



Applied
Mathematics

Semiclassical Analysis

Maciej Zworski

**Graduate Studies
in Mathematics**

Volume 138



American Mathematical Society

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American Mathematical Society
Providence, Rhode Island

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2010 *Mathematics Subject Classification*. Primary 35Q40, 81Q20, 35S05, 35S30, 35P20, 81S10.

For additional information and updates on this book, visit
www.ams.org/bookpages/gsm-138

Library of Congress Cataloging-in-Publication Data

Zworski, Maciej.

Semiclassical analysis / Maciej Zworski.

p. cm. — (Graduate studies in mathematics ; v. 138)

Includes bibliographical references and index.

ISBN 978-0-8218-8320-4 (alk. paper)

1. Quantum theory—Mathematics. 2. Differential equations, Partial. I. Title.

QC174.17.D54Z96 2012

515—dc23

2012010649

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10 9 8 7 6 5 4 3 2 1 17 16 15 14 13 12

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PREFACE

This book originated with a course I taught at UC Berkeley during the spring of 2003, with class notes taken by my colleague Lawrence C. Evans. Various versions of these notes have been available on-line as the *Evans-Zworski lecture notes on semiclassical analysis* and our original intention was to use them as the basis of a coauthored book. Craig Evans's contributions to the current manuscript can be recognized by anybody familiar with his popular partial differential equations (PDE) text [E]. In the end, the scope of the project and other commitments prevented Craig Evans from participating fully in the final stages of the effort, and he decided to withdraw from the responsibility of authorship, generously allowing me to make use of the contributions he had already made. I and my readers owe him a great debt, for this book would never have appeared without his participation.

Semiclassical analysis provides PDE techniques based on the *classical-quantum* (particle-wave) correspondence. These techniques include such well-known tools as geometric optics and the Wentzel–Kramers–Brillouin (WKB) approximation. Examples of problems studied in this subject are high energy eigenvalue asymptotics or effective dynamics for solutions of evolution equations. From the mathematical point of view, semiclassical analysis is a branch of *microlocal analysis* which, broadly speaking, applies *harmonic analysis* and *symplectic geometry* to the study of linear and non-linear PDE.

The book is intended to be a graduate level text introducing readers to semiclassical and microlocal methods in PDE. It is augmented in later chapters with many specialized advanced topics. Readers are expected to have reasonable familiarity with standard PDE theory (as recounted, for example, in Parts I and II of [E]), as well as a basic understanding of linear functional analysis. On occasion familiarity with differential forms will also prove useful.

Several excellent treatments of semiclassical analysis have appeared recently. The book **[D-S]** by Dimassi and Sjöstrand starts with the WKB-method, develops the general semiclassical calculus, and then provides high-tech spectral asymptotics. Martinez **[M]** provides a systematic development of FBI transform techniques, with applications to microlocal exponential estimates and to propagation estimates. This text is intended as a more elementary, but much broader, introduction. Except for the general symbol calculus, for which we followed Chapter 7 of **[D-S]**, there is little overlap with these other two texts or with the influential books by Helffer **[He]** and by Robert **[R]**. Guillemin and Sternberg **[G-St1]** offer yet another perspective on the subject, very much complementary to that given here. Their notes concentrate on global and functorial aspects of semiclassical analysis, in particular on the theory of Fourier integral operators and on trace formulas.

The approach to semiclassical analysis presented here is influenced by my long collaboration with Johannes Sjöstrand. I would like to thank him for sharing his philosophy and insights over the years. I first learned microlocal analysis from Richard Melrose, Victor Guillemin, and Gunther Uhlmann, and it is a pleasure to acknowledge my debt to them. Discussions of semiclassical physics and chemistry with Stéphane Nonnenmacher, Paul Brumer, William H. Miller, and Robert Littlejohn have been enjoyable and valuable. They have added a lot to my appreciation of the subject.

I am especially grateful to Stéphane Nonnenmacher, Semyon Dyatlov, Claude Zuily, Oran Gannot, Xi Chen, Hans Christianson, Jeff Galkowski, Justin Holmer, Long Jin, Gordon Linoff, and Steve Zelditch for their very careful reading of the earlier versions of this book and for their many valuable comments and corrections.

My thanks also go to Faye Yeager for typing the original lecture notes and to Jonathan Dorfman for $\text{\TeX}$ advice. Stephen Moye at the AMS provided fantastic help on deeper $\text{\TeX}$ issues and Arlene O'Sean's excellent copyediting removed many errors and inconsistencies.

I will maintain on my website at the UC Berkeley Mathematics Department <http://math.berkeley.edu/~zworski> a list of errata and corrections, as well as at the American Mathematical Society's website www.ams.org/bookpages/gsm-138. Please let me know about any errors you find.

I have been supported by NSF grants during the writing of this book, most recently by NSF grant DMS-0654436.

Maciej Zworski

Chapter 1

INTRODUCTION

- 1.1 Basic themes
- 1.2 Classical and quantum mechanics
- 1.3 Overview
- 1.4 Notes

1.1. BASIC THEMES

A major goal of this book is understanding the relationships between dynamical systems and the behavior of solutions to various linear partial differential equations (PDE) and pseudodifferential equations containing a small positive parameter h . We outline in this opening section some of the mathematical issues and challenges.

1.1.1. PDE with small parameters. The principal realm of motivation is quantum mechanics, in which case we informally understand h as related to *Planck's constant*. With this interpretation in mind, we break down our basic task into these two subquestions:

- (i) How and to what extent do *classical dynamics* determine the behavior as $h \rightarrow 0$ of solutions to *Schrödinger's equation*

$$ih\partial_t u = -h^2\Delta u + Vu$$

and the related *Schrödinger eigenvalue equation*

$$-h^2\Delta u + Vu = Eu?$$

The name “semiclassical” comes from this interpretation.

(ii) Conversely, given various mathematical objects associated with classical mechanics, for instance symplectic transformations, how can we profitably “quantize” them?

In fact the techniques of semiclassical analysis apply in many other settings and for many other sorts of PDE. For example we will later study the *damped wave equation*

$$(1.1.1) \quad \partial_t^2 u + a \partial_t u - \Delta u = 0$$

for large times. A rescaling in time will introduce the requisite small parameter h .

1.1.2. Basic techniques. We will construct, mostly in Chapters 2–4, 8–9, and 14, a wide variety of mathematical tools to address these issues, among them:

- the apparatus of symplectic geometry (to record succinctly the behavior of classical dynamical systems);
- the Fourier transform (to display dependence upon both the position variables x and the momentum variables ξ);
- stationary phase (to describe asymptotics as $h \rightarrow 0$ of various expressions involving rescaled Fourier transforms); and
- pseudodifferential operators (to localize or, as is said in the trade, to *microlocalize* functional behavior in phase space).

1.1.3. Microlocal analysis. There is a close relation between asymptotic properties of PDE with a small parameter and regularity of solutions to PDE. Asymptotic properties of $\hat{u}(\xi)$ as $1/|\xi| =: h \rightarrow 0$ are related to C^∞ regularity of u . For instance, we will see in Chapter 12 how to obtain results about *propagation of singularities* for general classes of equations. Answering questions about propagation of singularities has been one of the motivations of *microlocal analysis*, and most of the techniques presented in this book, such as pseudodifferential operators, come from that subject. Roughly speaking, in standard microlocal analysis $1/|\partial_x|$ plays the role of h . These ideas are behind the study of the damped wave equation (1.1.1).

Some techniques developed for pure PDE questions, such as local solvability, have acquired a new life when translated to the semiclassical setting. An example is the study of pseudospectra of nonselfadjoint operators; see Chapter 12. Another example is the connection between tunneling and

unique continuation. These were developed independently in physics and in mathematics and are unified nicely by semiclassical Carleman estimates; see Chapter 7.

1.1.4. Other directions. This book is devoted to semiclassical analysis as a branch of linear PDE theory. The ideas explored here are useful in other areas. One is the study of *quantum maps* where symplectic transformations on compact manifolds are quantized to give matrices. The semiclassical parameter is then related to the size of the matrix. These are popular models in physics partly due to the relative ease of numerical computations; see Haake [**Hak**] and references in Chapter 13 of this text. Many other *large N limit* problems enjoy semiclassical interpretation, in the sense of connecting analysis to geometry. In this book we present one example: a semiclassical proof of Quillen's Theorem (Theorem 13.18) which is related to Hilbert's 17th problem.

Semiclassical concepts also appear in the study of nonlinear PDE. One direction is provided by nonlinear equations with an asymptotic parameter which in some physically motivated problems plays a role similar to h in Section 1.1.1 above. One natural equation is the Gross-Pitaevskii nonlinear Schrödinger equation; see for instance the book by Carles [**Car**]. An example of a numerical study is given in Potter [**Po**] where a semiclassical approximation is used to describe solitons in an external field.

Another set of microlocal methods useful in nonlinear PDE is provided by the *paradifferential calculus* of Bony, Coifman, and Meyer; see for instance Métivier [**Me**], and for a brief introduction see Bényi–Maldonado–Naibo [**B-M-N**]. The semiclassical parameter appears in the Littlewood-Paley decomposition just as it does in Chapter 7, while the pseudodifferential classes are more exotic than the ones considered in Chapter 4.

1.2. CLASSICAL AND QUANTUM MECHANICS

We introduce and foreshadow a bit about quantum and classical correspondences.

1.2.1. Observables. We can think of a given function $a : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}$, $a = a(x, \xi)$, as a *classical observable* on phase space, where as above x denotes *position* and ξ denotes *momentum*. We usually call a a *symbol*.

Let $h > 0$ be given. We will associate with the observable a a corresponding *quantum observable* $a^w(x, hD)$, an operator defined by the formula

$$a^w(x, hD)u(x) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} a\left(\frac{x+y}{2}, \xi\right) u(y) d\xi dy$$

for appropriate smooth functions u . This is *Weyl's quantization formula*, and $a^w(x, hD)$ is a *pseudodifferential operator*.

One major task will be to understand how the analytic properties of the symbol a dictate the functional analytic properties of its quantization $a^w(x, hD)$. We will in fact build up a *symbol calculus*, meaning systematic rules for manipulating pseudodifferential operators.

1.2.2. Dynamics. We will be concerned as well with the evolution in time of classical particles and quantum states.

Classical evolution. Our most important example will concern the symbol

$$p(x, \xi) := |\xi|^2 + V(x),$$

corresponding to the phase space flow

$$\begin{cases} \dot{x} = 2\xi \\ \dot{\xi} = -\partial V, \end{cases}$$

where $\dot{} = \partial_t$. We generalize by introducing the arbitrary Hamiltonian $p : \mathbb{R}^{2n} \rightarrow \mathbb{R}$, $p = p(x, \xi)$, and the corresponding *Hamiltonian dynamics*

$$(1.2.1) \quad \begin{cases} \dot{x} = \partial_\xi p(x, \xi) \\ \dot{\xi} = -\partial_x p(x, \xi). \end{cases}$$

It is instructive to change our viewpoint somewhat, by writing

$$\varphi_t = \exp(tH_p)$$

for the solution of (1.2.1), where

$$H_p q := \{p, q\} = \langle \partial_\xi p, \partial_x q \rangle - \langle \partial_x p, \partial_\xi q \rangle$$

is the *Poisson bracket*. Select a symbol a and define

$$(1.2.2) \quad a_t(x, \xi) := a(\varphi_t(x, \xi)).$$

Then

$$(1.2.3) \quad \dot{a}_t = \{p, a_t\},$$

and this equation tells us how the symbol evolves in time, as dictated by the classical dynamics (1.2.1).

Quantum evolution. We can quantize the foregoing by putting

$$P = p^w(x, hD), \quad A = a^w(x, hD)$$

and defining

$$(1.2.4) \quad A(t) := F^{-1}(t)AF(t)$$

for $F(t) := e^{-\frac{itP}{h}}$. The operator $A(t)$ represents, according to the so-called *Heisenberg picture* of quantum mechanics, the evolution of the quantum observable A under the flow (1.2.1). Then we have the evolution equation

$$(1.2.5) \quad \partial_t A(t) = \frac{i}{h}[P, A(t)],$$

an obvious analogue of (1.2.3). Here then is a basic principle we will later work out in some detail: *an assertion about Hamiltonian dynamics, and so the Poisson bracket $\{\cdot, \cdot\}$, will involve at the quantum level the commutator $[\cdot, \cdot]$.*

REMARK: h and $\hbar$. In this book h denotes a dimensionless parameter and is consequently not immediately to be identified with the dimensional physical quantity

$$\hbar = \text{Planck's constant}/2\pi = 1.05457 \times 10^{-34} \text{joule-sec.}$$

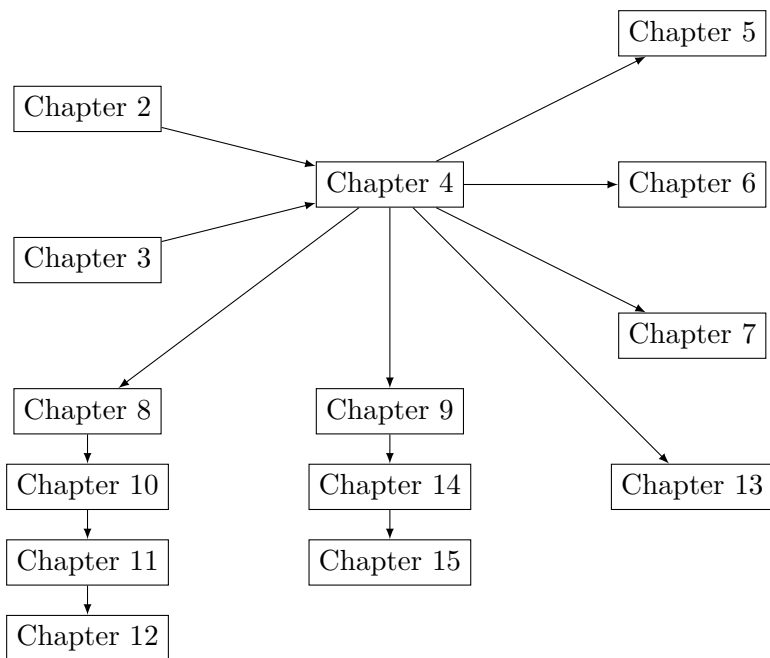
As the example of the damped wave equation (1.1.1) shows, the use of $h \rightarrow 0$ asymptotics is not restricted to problems motivated by quantum mechanics.

□

1.3. OVERVIEW

Chapters 2–4 develop the basic machinery, followed by applications to partial differential equations in Chapters 5–7. We develop more advanced theory and applications in Chapters 8–13, and in Chapters 14 and 15 we discuss semiclassical analysis on manifolds.

The following diagram indicates the dependencies of the chapters and may help in selective reading of the book:



Here is a quick overview of the book, with some of the highlights:

Chapter 2: We start with a quick introduction to symplectic analysis and geometry and their implications for classical Hamiltonian dynamical systems.

Chapter 3: This chapter provides the basics of the Fourier transform and derives also important *stationary phase* asymptotic estimates for the oscillatory integral

$$I_h := \int_{\mathbb{R}^n} e^{\frac{i\varphi}{h}} a \, dx$$

of the sort

$$I_h = (2\pi h)^{n/2} |\det \partial^2 \varphi(x_0)|^{-1/2} e^{\frac{i\pi}{4} \operatorname{sgn} \partial^2 \varphi(x_0)} e^{\frac{i\varphi(x_0)}{h}} a(x_0) + O\left(h^{\frac{n+2}{2}}\right)$$

as $h \rightarrow 0$, provided the gradient of the phase φ vanishes only at the point x_0 .

Chapter 4: Next we introduce the *Weyl quantization* $a^w(x, hD)$ of the symbol $a(x, \xi)$ and work out various properties, chief among them the composition formula

$$a^w(x, hD)b^w(x, hD) = c^w(x, hD),$$

where the symbol $c := a \# b$ is computed explicitly in terms of a and b . We will prove as well the sharp *Gårding inequality*, learn when a^w is a bounded operator on L^2 , etc.

Chapter 5: This part of the book introduces semiclassical defect measures and uses them to derive decay estimates for the damped wave equation (1.1.1), where $a \geq 0$ on the flat torus $\mathbb{T}^n$. A theorem of Rauch and Taylor provides a beautiful example of classical/quantum correspondence: the waves decay exponentially if all classical trajectories within a certain fixed time intersect the region where positive damping occurs.

Chapter 6: In Chapter 6 we begin our study of the eigenvalue problem

$$P(h)u(h) = E(h)u(h),$$

for the operator

$$P(h) := -h^2\Delta + V(x).$$

We prove *Weyl's Law* for the asymptotic distributions of eigenvalues as $h \rightarrow 0$, stating for all $a < b$ that

$$\#\{E(h) \mid a \leq E(h) \leq b\} = \frac{1}{(2\pi h)^n} (|\{a \leq |\xi|^2 + V(x) \leq b\}| + o(1))$$

as $h \rightarrow 0$. Our proof is a semiclassical analogue of the classical Dirichlet–Neumann bracketing argument of Courant.

Chapter 7: Chapter 7 deepens our study of eigenfunctions, first establishing an exponential vanishing theorem in the “classically forbidden” region. We derive as well a *Carleman-type* estimate: if $u(h)$ is an eigenfunction of a Schrödinger operator, then for any open set $U \subset \subset \mathbb{R}^n$,

$$\|u(h)\|_{L^2(U)} \geq e^{-c/h} \|u(h)\|_{L^2(\mathbb{R}^n)}.$$

This provides a quantitative estimate for quantum mechanical *tunneling*.

We also present a self-contained “semiclassical” derivation of interior Schauder estimates for the Laplacian.

Chapter 8: We return in Chapter 8 to the symbol calculus, first proving the semiclassical version of *Beals's Theorem*, characterizing pseudodifferential operators. As an application we show how quantization commutes with exponentiation at the level of order functions and then use these insights to define useful generalized Sobolev spaces. This chapter also introduces wavefront sets and the notion of microlocality.

Chapter 9: We next introduce the useful formalism of half-densities and use them to see how changing variables in a symbol affects the Weyl quantization. This motivates our introducing the new class of *Kohn–Nirenberg symbols*, which behave well under coordinate changes and are consequently useful later when we investigate the semiclassical calculus on manifolds.

Chapter 10: Chapter 10 discusses the local construction of propagators, using solutions of Hamilton–Jacobi PDE to build phase functions for *Fourier*

integral operators. Applications include the semiclassical Strichartz estimates and L^p bounds on eigenfunction clusters.

Chapter 11: This chapter proves *Egorov's Theorem*, characterizing propagators for bounded time intervals in terms of the classical dynamics applied to symbols, up to $O(h)$ error terms. We then employ Egorov's Theorem to quantize linear and nonlinear symplectic mappings and conclude the chapter by showing that Egorov's Theorem is in fact valid until times of order $\log(h^{-1})$, the so-called *Ehrenfest time*.

Chapter 12: Chapter 12 illustrates how methods from Chapter 11 provide elegant and useful normal forms of differential and pseudodifferential operators. Among the applications, we build quasimodes for certain nonnormal operators and discuss the implications for pseudospectra.

Chapter 13: We consider the question of how close semiclassical quantization can get to multiplication. This leads to an alternative presentation of the semiclassical calculus based on Toeplitz quantization acting on spaces of holomorphic functions. The FBI–Bargmann transform intertwines the quantization of Chapter 4 with the quantization by operators acting on holomorphic functions.

Chapter 14: Chapter 14 briefly discusses general manifolds and modifications to the symbol calculus to cover pseudodifferential operators on manifolds. Chapter 9 provides the change of variables formulas we need to work with coordinate patches.

Chapter 15: This chapter concerns the quantum implications of ergodicity for underlying dynamical systems on manifolds. A key assertion is that if the underlying dynamical system satisfies an appropriate ergodic condition, then

$$h^n \sum_{a \leq E_j \leq b} \left| \langle Au_j, u_j \rangle - \int_{\{a \leq p \leq b\}} \sigma(A) dx d\xi \right|^2 \rightarrow 0$$

as $h \rightarrow 0$, for a wide class of pseudodifferential operators A . In this expression the classical observable $\sigma(A)$ is the *symbol* of A .

Appendices: Appendix A records our notation in one convenient location, and Appendix B is a very quick review of differential forms. Appendix C collects various useful functional analysis theorems (with selected proofs). Appendix D discusses Fredholm operators within the framework of Grushin problems.

1.4. NOTES

The book by Griffiths [**G**] provides a nice elementary introduction to quantum mechanics, and Hannabuss [**Ha**] is a good mathematical text. For a modern physical perspective, consult Gutzwiller [**Gut**], Haake [**Hak**], Heller–Tomsovic [**H-T**], Miller [**Mi**], or Stöckmann [**Sto**].

Part 1

BASIC THEORY

SYMPLECTIC GEOMETRY AND ANALYSIS

- 2.1 Flows
- 2.2 Symplectic structure on $\mathbb{R}^{2n}$
- 2.3 Symplectic mappings
- 2.4 Hamiltonian vector fields
- 2.5 Lagrangian submanifolds
- 2.6 Notes

We provide in this chapter a quick discussion of the symplectic geometric structure on $\mathbb{R}^{2n} = \mathbb{R}^n \times \mathbb{R}^n$ and its interplay with Hamiltonian dynamics. These will be important for our later goal of understanding interrelationships between dynamics and PDE.

The reader may wish to first review our basic notation and also the theory of differential forms, set forth, respectively, in Appendices A and B.

2.1. FLOWS

Let $V : \mathbb{R}^N \rightarrow \mathbb{R}^N$ denote a smooth vector field. Fix a point $z \in \mathbb{R}^N$ and solve the ordinary differential equations (ODE)

$$(2.1.1) \quad \begin{cases} \dot{w} = V(w) & (t \in \mathbb{R}) \\ w(0) = z, \end{cases}$$

where $\dot{} = \partial_t$ and $w = w(t) = w(t, z)$, the latter notation used when we wish to display the dependence of the solution on the initial condition. We assume that the solution of the flow (2.1.1) exists for each z and is unique for all times $t \in \mathbb{R}$.

NOTATION. We define

$$\varphi_t z := w(t, z)$$

and will often also write

$$(2.1.2) \quad \varphi_t =: \exp(tV).$$

We call $\{\varphi_t\}_{t \in \mathbb{R}}$ the *flow map* or the *exponential map* generated by V .

The following are standard assertions from the theory of ordinary differential equations:

LEMMA 2.1 (Properties of flow map).

- (i) $\varphi_0 z = z$ for all $z \in \mathbb{R}^N$.
- (ii) $\varphi_{t+s} = \varphi_t \varphi_s$ for all $s, t \in \mathbb{R}$.
- (iii) For each time $t \in \mathbb{R}$, the mapping $\varphi_t : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a diffeomorphism, with

$$(\varphi_t)^{-1} = \varphi_{-t}.$$

2.2. SYMPLECTIC STRUCTURE ON $\mathbb{R}^{2n}$

We henceforth specialize to the even-dimensional space $\mathbb{R}^N = \mathbb{R}^{2n} = \mathbb{R}^n \times \mathbb{R}^n$.

NOTATION. We refine our previous notation and henceforth denote an element of $\mathbb{R}^{2n}$ as

$$z = (x, \xi)$$

and interpret $x \in \mathbb{R}^n$ as denoting *position*, $\xi \in \mathbb{R}^n$ as *momentum*. We will likewise write

$$w = (y, \eta)$$

for another typical point of $\mathbb{R}^{2n}$.

We let $\langle \cdot, \cdot \rangle$ denote the usual inner product on $\mathbb{R}^n$, and then define this new pairing on $\mathbb{R}^{2n}$:

DEFINITION. Given $z = (x, \xi)$, $w = (y, \eta)$ in $\mathbb{R}^{2n}$, define their *symplectic product*

$$(2.2.1) \quad \sigma(z, w) := \langle \xi, y \rangle - \langle x, \eta \rangle.$$

It is convenient to recast σ , employing a very useful matrix:

DEFINITION. We introduce the $2n \times 2n$ matrix

$$(2.2.2) \quad J := \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}.$$

LEMMA 2.2 (Properties of J and σ).

(i) *We have*

$$(2.2.3) \quad \sigma(z, w) = \langle Jz, w \rangle$$

for all $z, w \in \mathbb{R}^{2n}$, where $\langle \cdot, \cdot \rangle$ now means the standard inner product on $\mathbb{R}^{2n}$.

(ii) *The bilinear form σ is antisymmetric:*

$$(2.2.4) \quad \sigma(z, w) = -\sigma(w, z)$$

and nondegenerate:

$$(2.2.5) \quad \text{if } \sigma(z, w) = 0 \text{ for all } w, \text{ then } z = 0.$$

(iii) *Also*

$$(2.2.6) \quad J^2 = -I, \quad J^T = -J = J^{-1}.$$

We leave the simple proofs to the reader.

We now bring in the terminology of differential forms, reviewed in Appendix B:

NOTATION. We introduce for $x = (x_1, \dots, x_n)$ and $\xi = (\xi_1, \dots, \xi_n)$ the 1-forms dx_j and $d\xi_j$ for $j = 1, \dots, n$ and then write

$$(2.2.7) \quad \sigma = d\xi \wedge dx = \sum_{j=1}^n d\xi_j \wedge dx_j.$$

Observe also that

$$(2.2.8) \quad \sigma = d\omega \quad \text{for } \omega := \xi dx = \sum_{j=1}^n \xi_j dx_j.$$

Since $d^2 = 0$, it follows that

$$(2.2.9) \quad d\sigma = 0.$$

2.3. SYMPLECTIC MAPPINGS

Suppose next that $U, V \subset \mathbb{R}^{2n}$ are open sets and

$$\kappa : U \rightarrow V$$

is a smooth mapping. We will write

$$\kappa(x, \xi) = (y, \eta) = (y(x, \xi), \eta(x, \xi)).$$

DEFINITION. We call κ a *symplectic mapping*, or a *symplectomorphism*, provided

$$(2.3.1) \quad \kappa^* \sigma = \sigma.$$

Here the *pull-back* $\kappa^* \sigma$ of the symplectic product σ is defined by

$$(\kappa^* \sigma)(z, w) := \sigma(\kappa_*(z), \kappa_*(w)),$$

κ_* denoting the *push-forward* of vectors; see Appendix B.

NOTATION. We will usually write (2.3.1) in the more suggestive notation

$$(2.3.2) \quad d\eta \wedge dy = d\xi \wedge dx.$$

The ensuing sequence of examples will clarify the meaning of this notation:

EXAMPLE 1: Linear symplectic mappings. Suppose $\kappa : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ is linear:

$$\kappa(x, \xi) = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x \\ \xi \end{pmatrix} = (Ax + B\xi, Cx + D\xi) = (y, \eta),$$

where A, B, C, D are $n \times n$ matrices.

THEOREM 2.3 (Symplectic matrices). *The linear mapping κ is symplectic if and only if the matrix*

$$K := \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

satisfies

$$(2.3.3) \quad K^T J K = J.$$

In particular the linear mapping $(x, \xi) \mapsto (\xi, -x)$ determined by J is symplectic.

DEFINITION. We call a $2n \times 2n$ matrix K *symplectic* if (2.3.3) holds.

Proof. Let us compute

$$\begin{aligned} d\eta \wedge dy &= (Cdx + Dd\xi) \wedge (Adx + Bd\xi) \\ &= A^T C dx \wedge dx + B^T D d\xi \wedge d\xi + (A^T D - C^T B) d\xi \wedge dx \\ &= d\xi \wedge dx \end{aligned}$$

if and only if

$$(2.3.4) \quad A^T C \text{ and } B^T D \text{ are symmetric, } A^T D - C^T B = I.$$

Furthermore,

$$\begin{aligned} K^T J K &= \begin{pmatrix} A^T & C^T \\ B^T & D^T \end{pmatrix} \begin{pmatrix} O & I \\ -I & O \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} \\ &= \begin{pmatrix} A^T C - C^T A & A^T D - C^T B \\ B^T C - D^T A & B^T D - D^T B \end{pmatrix} \\ &= J \end{aligned}$$

if and only if (2.3.4) holds. □

We record some useful observations:

THEOREM 2.4 (More on symplectic matrices).

(i) *The product of two symplectic matrices is symplectic.*

(ii) *If K is a symplectic matrix, then*

$$(2.3.5) \quad \sigma(Kz, Kw) = \sigma(z, w) \quad (z, w \in \mathbb{R}^{2n}).$$

(iii) *A matrix K is symplectic if and only if*

$$(2.3.6) \quad K \text{ is invertible, } K^{-1} = JK^T J^T.$$

(iv) *If*

$$A^T J + JA = 0,$$

then

$$K_t := \exp(tA) \text{ is symplectic}$$

for each $t \in \mathbb{R}$.

Proof. Assertions (i), (ii), and (iii) follow directly from the definitions and the fact that $J^T = -J = J^{-1}$. To prove (iv), write $W_t := K_t^T J K_t - J$ and compute

$$\partial_t W_t = A^T W_t + W_t A + A^T J + JA = A^T W_t + W_t A.$$

Since $W_0 = 0$, we deduce from uniqueness that $W_t = 0$ for all $t \in \mathbb{R}$. □

EXAMPLE 2: Nonlinear symplectic mappings. Assume next that $\kappa : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ is nonlinear:

$$\kappa(x, \xi) = (y, \eta)$$

for smooth functions $y = y(x, \xi), \eta = \eta(x, \xi)$. Its linearization is the $2n \times 2n$ matrix

$$\partial\kappa = \begin{pmatrix} \partial_x y & \partial_\xi y \\ \partial_x \eta & \partial_\xi \eta \end{pmatrix}.$$

THEOREM 2.5 (Symplectic transformations). *The mapping κ is symplectic if and only if the matrix $\partial\kappa$ is symplectic at each point.*

Proof. We have

$$d\eta \wedge dy = (Cdx + Dd\xi) \wedge (Adx + Bd\xi)$$

for

$$A := \partial_x y, \quad B := \partial_\xi y, \quad C := \partial_x \eta, \quad D := \partial_\xi \eta.$$

Consequently, as in the previous proof, we have $d\eta \wedge dy = d\xi \wedge dx$ if and only if (2.3.4) is valid, which in turn is so if and only if $\partial\kappa$ is a symplectic matrix. $\square$

EXAMPLE 3: Lifting diffeomorphisms. Let

$$\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

be a diffeomorphism on $\mathbb{R}^n$, with nondegenerate Jacobian matrix $\partial\gamma = \partial_x \gamma$. We propose to extend γ to a symplectomorphism

$$\kappa : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$$

having the form

$$(2.3.7) \quad \kappa(x, \xi) = (\gamma(x), \eta(x, \xi)) = (y, \eta),$$

by “lifting” γ to variables ξ .

THEOREM 2.6 (Extending to a symplectic mapping). *The transformation (2.3.7) is symplectic if*

$$(2.3.8) \quad \eta(x, \xi) := [\partial\gamma(x)^{-1}]^T \xi.$$

Proof. As the statement suggests, it will be easier to look for ξ as a function of x and η . We compute

$$dy = A dx, \quad d\xi = E dx + F d\eta,$$

for

$$A := \partial_x y, \quad E := \partial_x \xi, \quad F := \partial_\eta \xi.$$

Therefore

$$d\eta \wedge dy = d\eta \wedge (A dx)$$

and

$$d\xi \wedge dx = (Edx \wedge Fd\eta) \wedge dx = Edx \wedge dx + d\eta \wedge F^T dx.$$

We would like to construct $\xi = \xi(x, \eta)$ so that

$$A = F^T \quad \text{and} \quad E \text{ is symmetric,}$$

the latter condition implying that $Edx \wedge dx = 0$. To do so, let us define

$$\xi(x, \eta) := (\partial\gamma)^T \eta.$$

Then clearly $F^T = A$, and $E = E^T = \partial^2\gamma$, as required. $\square$

EXAMPLE 4: Generating functions. Our next example demonstrates that we can, locally at least, build a symplectic transformation from a real-valued *generating function*.

Suppose $\varphi : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, $\varphi = \varphi(x, y)$, is smooth. Assume also that

$$(2.3.9) \quad \det(\partial_{xy}^2 \varphi(x_0, y_0)) \neq 0.$$

Define

$$(2.3.10) \quad \xi = \partial_x \varphi, \quad \eta = -\partial_y \varphi$$

and observe that the Implicit Function Theorem implies (y, η) is a smooth function of (x, ξ) near $(x_0, \partial_x \varphi(x_0, y_0))$.

THEOREM 2.7 (Generating functions and symplectic maps). *The mapping κ implicitly defined by*

$$(2.3.11) \quad (x, \partial_x \varphi(x, y)) \mapsto (y, -\partial_y \varphi(x, y))$$

is a symplectomorphism near (x_0, ξ_0) .

A simple example is $\varphi(x, y) = \langle x, y \rangle$, which generates the linear symplectic mapping represented by the matrix J .

Proof. We compute

$$\begin{aligned} d\eta \wedge dy &= d(-\partial_y \varphi) \wedge dy \\ &= [(-\partial_y^2 \varphi dy) \wedge dy] + [(-\partial_{xy}^2 \varphi dx) \wedge dy] \\ &= -(\partial_{xy}^2 \varphi) dx \wedge dy, \end{aligned}$$

since $\partial_y^2 \varphi$ is symmetric. Likewise,

$$\begin{aligned} d\xi \wedge dx &= d(\partial_x \varphi) \wedge dx \\ &= [(\partial_x^2 \varphi dx) \wedge dx] + [(\partial_{xy}^2 \varphi dy) \wedge dx] \\ &= -(\partial_{xy}^2 \varphi) dx \wedge dy = d\eta \wedge dy. \end{aligned} \quad \square$$

Section 2.5 will generalize this example and provide more geometric insight.

2.4. HAMILTONIAN VECTOR FIELDS

We will later be very concerned with dynamics compatible with the symplectic structure introduced earlier. These dynamics are generated by Hamiltonian vector fields:

DEFINITION. Given $f \in C^\infty(\mathbb{R}^{2n})$, we define the corresponding *Hamiltonian vector field*, H_f , by requiring

$$(2.4.1) \quad \sigma(z, H_f) = df(z) \quad \text{for all } z = (x, \xi).$$

So

$$(2.4.2) \quad H_f = \langle \partial_\xi f, \partial_x \rangle - \langle \partial_x f, \partial_\xi \rangle = \sum_{j=1}^n (f_{\xi_j} \partial_{x_j} - f_{x_j} \partial_{\xi_j}).$$

Another way of writing H_f uses the contraction $\lrcorner$ defined in Appendix B:

LEMMA 2.8 (Differentials and Hamiltonian vector fields). *We have*

$$(2.4.3) \quad df = -(H_f \lrcorner \sigma).$$

Proof. This follows directly from the definition, as we can calculate for each z that

$$(H_f \lrcorner \sigma)(z) = \sigma(H_f, z) = -\sigma(z, H_f) = -df(z). \quad \square$$

DEFINITION. If $f, g \in C^\infty(\mathbb{R}^{2n})$, we define their *Poisson bracket*

$$(2.4.4) \quad \{f, g\} := H_f g = \sigma(\partial f, \partial g).$$

That is,

$$(2.4.5) \quad \{f, g\} = \langle \partial_\xi f, \partial_x g \rangle - \langle \partial_x f, \partial_\xi g \rangle = \sum_{j=1}^n (f_{\xi_j} g_{x_j} - f_{x_j} g_{\xi_j}).$$

LEMMA 2.9 (Brackets, commutators).

(i) *We have Jacobi's identity*

$$(2.4.6) \quad \{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$$

for all functions $f, g, h \in C^\infty(\mathbb{R}^{2n})$.

(ii) *Furthermore,*

$$(2.4.7) \quad H_{\{f, g\}} = [H_f, H_g].$$

Proof. A direct calculation verifies assertion (i), and we observe that

$$H_{\{f,g\}}h = [H_f, H_g]h$$

is a rewriting of (2.4.6). $\square$

REMARK: Another derivation of Jacobi's identity. An alternative proof of (2.4.6) follows, this illustrating the essential property that $d\sigma = 0$. Theorem B.1 provides the identity

$$\begin{aligned} 0 &= d\sigma(H_f, H_g, H_h) \\ (2.4.8) \quad &= H_f\sigma(H_g, H_h) + H_g\sigma(H_h, H_f) + H_h\sigma(H_f, H_g) \\ &\quad - \sigma([H_f, H_g], H_h) - \sigma([H_g, H_h], H_f) - \sigma([H_h, H_f], H_g). \end{aligned}$$

Now (2.4.4) implies

$$H_f\sigma(H_g, H_h) = \{f, \{g, h\}\}$$

and

$$\begin{aligned} \sigma([H_f, H_g], H_h) &= [H_f, H_g]h = H_fH_g h - H_gH_f h \\ &= \{f, \{g, h\}\} - \{g, \{f, h\}\}. \end{aligned}$$

Similar identities hold for other terms. Substituting into (2.4.8) gives Jacobi's identity. $\square$

THEOREM 2.10 (Jacobi's Theorem). *If κ is a symplectomorphism, then*

$$(2.4.9) \quad H_f = \kappa_*(H_{\kappa^*f}).$$

In other words, the pull-back of a Hamiltonian vector field generated by f ,

$$(2.4.10) \quad \kappa^*H_f := (\kappa^{-1})_*H_f,$$

is the Hamiltonian vector field generated by the pull-back of f .

Proof. Using the notation of (2.4.10),

$$\begin{aligned} \kappa^*(H_f) \lrcorner \sigma &= \kappa^*(H_f) \lrcorner \kappa^* \sigma = \kappa^*(H_f \lrcorner \sigma) \\ &= -\kappa^*(df) = -d(\kappa^*f) \\ &= H_{\kappa^*f} \lrcorner \sigma. \end{aligned}$$

Since σ is nondegenerate, (2.4.9) follows. $\square$

EXAMPLE. Define $\kappa = J$, so that $\kappa(x, \xi) = (\xi, -x)$. We have $\kappa^*f(x, \xi) = f(\xi, -x)$, and therefore

$$H_{\kappa^*f} = \langle \partial_x f(\xi, -x), \partial_x \rangle + \langle \partial_\xi f(\xi, -x), \partial_\xi \rangle.$$

Then

$$\kappa^* H_f = \langle \partial_\xi f(\xi, -x), \partial_\xi \rangle - \langle \partial_x f(\xi, -x), -\partial_x \rangle = H_{\kappa^* f}. \quad \square$$

THEOREM 2.11 (Hamiltonian flows as symplectomorphisms). *If f is smooth, then for each time t , the mapping*

$$(x, \xi) \mapsto \varphi_t(x, \xi) = \exp(tH_f)$$

is a symplectomorphism.

Proof. According to Cartan's formula (Theorem B.3), we have

$$\partial_t(\varphi_t^* \sigma) = \mathcal{L}_{H_f} \sigma = d(H_f \lrcorner \sigma) + (H_f \lrcorner d\sigma).$$

Since $d\sigma = 0$, it follows that

$$\partial_t(\varphi_t^* \sigma) = d(-df) = -d^2 f = 0.$$

Thus $(\varphi_t)^* \sigma = \sigma$ for all times t . $\square$

The next result shows that locally all nondegenerate closed 2-forms are equivalent to the standard symplectic form σ on $\mathbb{R}^{2n}$.

THEOREM 2.12 (Darboux's Theorem). *Let U be a neighborhood of (x_0, ξ_0) and suppose η is a nondegenerate 2-form defined on U , satisfying*

$$d\eta = 0.$$

Then near (x_0, ξ_0) there exists a diffeomorphism κ such that

$$(2.4.11) \quad \kappa^* \eta = \sigma.$$

INTERPRETATION. A symplectic structure on $\mathbb{R}^{2n}$ is determined by a choice of a nondegenerate, closed 2-form η . Darboux's Theorem states that all symplectic structures are identical locally, in the sense that all are equivalent to that given by σ . This is a dramatic contrast to Riemannian geometry: there are no local invariants in symplectic geometry. $\square$

Proof. 1. Let us assume $(x_0, \xi_0) = (0, 0)$. We first need a linear mapping L so that

$$L^* \eta(0, 0) = \sigma(0, 0).$$

This means that we find a basis $\{e_k, f_k\}_{k=1}^n$ of $\mathbb{R}^{2n}$ such that

$$\eta(f_l, e_k) = \delta_{kl}, \quad \eta(e_k, e_l) = 0, \quad \eta(f_k, f_l) = 0$$

for all $1 \leq k, l \leq n$. Then if $u = \sum_{i=1}^n x_i e_i + \xi_i f_i$, $v = \sum_{j=1}^n y_j e_j + \eta_j f_j$, we

have

$$\begin{aligned}\eta(u, v) &= \sum_{i,j=1}^n x_i y_j \eta(e_i, e_j) + \xi_i \eta_j \eta(f_i, f_j) + x_i \eta_j \sigma(e_i, f_j) + \xi_i y_j \sigma(f_i, e_j) \\ &= \langle \xi, y \rangle - \langle x, \eta \rangle = \sigma((x, \xi), (y, \eta)).\end{aligned}$$

We leave finding L as a linear algebra exercise.

2. Next, define $\eta_t := t\eta + (1-t)\sigma$ for $0 \leq t \leq 1$. Our intention is to find κ_t so that $\kappa_t^* \eta_t = \sigma$ near $(0, 0)$; then $\kappa := \kappa_1$ solves our problem. We will construct κ_t by solving the flow

$$(2.4.12) \quad \begin{cases} \dot{w} = V_t(w) & (0 \leq t \leq 1) \\ w(0) = z \end{cases}$$

and setting $\kappa_t := \varphi_t$.

For this to work, we must design the vector fields V_t in (2.4.12) so that

$$\partial_t(\kappa_t^* \eta_t) = 0.$$

Let us therefore calculate

$$\begin{aligned}\partial_t(\kappa_t^* \eta_t) &= \kappa_t^* (\partial_t \eta_t) + \kappa_t^* \mathcal{L}_{V_t} \eta_t \\ &= \kappa_t^* [(\eta - \sigma) + d(V_t \lrcorner \eta_t) + V_t \lrcorner d\eta_t],\end{aligned}$$

where we used Cartan's formula, Theorem B.3. Now $d\eta_t = td\eta + (1-t)d\sigma$, and hence $(d/dt)(\kappa_t^* \eta_t) = 0$ provided

$$(2.4.13) \quad (\eta - \sigma) + d(V_t \lrcorner \eta_t) = 0.$$

3. According to Poincaré's Lemma (Theorem B.4), we can write

$$\eta - \sigma = d\alpha \quad \text{near } (0, 0).$$

So (2.4.13) will hold if

$$(2.4.14) \quad V_t \lrcorner \eta_t = -\alpha \quad (0 \leq t \leq 1).$$

Since $\eta = \sigma$ at $(0, 0)$, $\eta_t = \sigma$ at $(0, 0)$. In particular, η_t is nondegenerate for $0 \leq t \leq 1$ in a neighborhood of $(0, 0)$, and hence we can solve (2.4.13) for the vector field V_t . $\square$

2.5. LAGRANGIAN SUBMANIFOLDS

This section provides some further geometric interpretations of generating functions, introduced earlier in Example 4 in Section 2.3.

DEFINITION. A *Lagrangian submanifold* Λ in $\mathbb{R}^{2n}$ is an n -dimensional submanifold for which

$$(2.5.1) \quad \sigma|_{\Lambda} = 0.$$

The meaning of (2.5.1) is that $\sigma(u) = 0$ for each point $z \in \Lambda$ and for all $u = (u_1, u_2)$ with $u_1, u_2 \in T_z(\Lambda)$, the tangent space to Λ at z .

THEOREM 2.13 (Lagrangian submanifolds). *Let Λ be a Lagrangian submanifold of $\mathbb{R}^{2n}$. Then each point $z \in \Lambda$ lies in a relatively open neighborhood $U \subset \Lambda$ within which*

$$(2.5.2) \quad \omega|_{\Lambda} = d\varphi$$

for some smooth function $\varphi : U \rightarrow \mathbb{R}$.

Proof. Given $z \in \Lambda$, we find a relatively open neighborhood $U \subset \Lambda$ and a smooth diffeomorphism $\gamma : U \rightarrow V$, where $V = B^0(0, 1)$ is the open unit ball in $\mathbb{R}^n$. Then $\rho := \gamma^{-1}$ pulls back $\omega|_{\Lambda}$ to the 1-form $\alpha := \rho^*(\omega|_{\Lambda})$, defined on V .

According to (2.5.1), we have

$$d\alpha = d(\rho^*\omega|_{\Lambda}) = \rho^*(d\omega|_{\Lambda}) = \rho^*(\sigma|_{\Lambda}) = 0$$

within V . Poincaré's Lemma (Theorem B.5) therefore implies $\alpha = d\psi$ for some smooth function $\psi : V \rightarrow \mathbb{R}$. Set $\varphi := \psi \circ \gamma = \gamma^*\psi$. Then

$$d\varphi = d(\gamma^*\psi) = \gamma^*d\psi = \gamma^*\alpha = \omega|_{\Lambda}. \quad \square$$

We show next that a Lagrangian submanifold is locally determined by the graph of a generating function of appropriate coordinates:

THEOREM 2.14 (Generating functions for Lagrangian submanifolds).

Suppose that $\Lambda \subset \mathbb{R}^n \times \mathbb{R}^n$ is a smooth Lagrangian submanifold and that $(x_0, \xi_0) \in \Lambda$. Then there exist a neighborhood $U \subset \mathbb{R}^n \times \mathbb{R}^n$ of (x_0, ξ_0) , a splitting of coordinates

$$(2.5.3) \quad x = (x', x''), \quad \xi = (\xi', \xi''),$$

where $k \in \{0, \dots, n\}$ and $x', \xi' \in \mathbb{R}^k, x'', \xi'' \in \mathbb{R}^{n-k}$, and a smooth function

$$(2.5.4) \quad \varphi = \varphi(x', \xi'')$$

such that

$$(2.5.5) \quad \Lambda \cap U = \{(x', -\partial_{\xi''}\varphi; \partial_{x'}\varphi, \xi'') \mid x' \in \mathbb{R}^k, \xi'' \in \mathbb{R}^{n-k}\} \cap U.$$

We call $\varphi = \varphi(x', \xi'')$ a local *generating function* of Λ near (x_0, ξ_0) .

Proof. 1. Let $V \subset \mathbb{R}^n$ be a coordinate chart for a neighborhood of (x_0, ξ_0) in Λ :

$$\rho : V \rightarrow \Lambda \subset \mathbb{R}^{2n},$$

with $\rho(0) = (x_0, \xi_0)$. The Jacobian $\partial\rho(0)$ has full rank and hence has n independent rows. We choose n such rows and call the corresponding coordinates $x' \in \mathbb{R}^k$ and $\xi'' \in \mathbb{R}^{n-k}$.

2. Define $p : \mathbb{R}^{2n} \rightarrow \mathbb{R}^k \times \mathbb{R}^{n-k}$ by

$$p(x, \xi) := (x', \xi'').$$

Our choice of the coordinates (x', ξ'') means that

$$p \circ \rho : V \rightarrow \mathbb{R}^k \times \mathbb{R}^{n-k}$$

has an invertible Jacobian at $0 \in \mathbb{R}^n$. Hence the Implicit Function Theorem implies $p \circ \gamma$ is invertible in a neighborhood of 0. This means that we can use (x', ξ'') as local coordinates on Λ , and so there exists a neighborhood $U \subset \mathbb{R}^{2n}$ of (x_0, ξ_0) and smooth functions

$$f : \mathbb{R}^k \times \mathbb{R}^{n-k} \rightarrow \mathbb{R}^{n-k}, \quad g : \mathbb{R}^k \times \mathbb{R}^{n-k} \rightarrow \mathbb{R}^k$$

such that

$$\Lambda \cap U = \{(x', f(x', \xi''), g(x', \xi''), \xi'') \mid (x', \xi'') \in \mathbb{R}^k \times \mathbb{R}^{n-k}\} \cap U.$$

3. Recalling that $\omega = \xi dx$, we use Theorem 2.13 to see that for some function $\psi = \psi(x', \xi'')$,

$$\begin{aligned} \omega|_{\Lambda} &= \langle g, dx' \rangle + \langle \xi'', \partial_{x'} f dx' + \partial_{\xi''} f d\xi'' \rangle \\ &= \langle g + (\partial_{x'} f)^T \xi'', dx' \rangle + \langle (\partial_{\xi''} f)^T \xi'', d\xi'' \rangle \\ &= \langle \partial_{x'} \psi, dx' \rangle + \langle \partial_{\xi''} \psi, d\xi'' \rangle. \end{aligned}$$

That is,

$$\psi_{x'} = g + (\partial_{x'} f)^T \xi'' = g + \partial_{x'} \langle f, \xi'' \rangle, \quad \psi_{\xi''} = (\partial_{\xi''} f)^T \xi''.$$

If we put

$$\varphi(x', \xi'') := \psi(x', \xi'') - \langle f(x', \xi''), \xi'' \rangle,$$

then

$$f = -\partial_{\xi''} \varphi, \quad g = \partial_{x'} \varphi. \quad \square$$

EXAMPLES.

(i) The simplest case is $k = n$, for which

$$\Lambda \cap U = \{(x, \partial\varphi(x)) : x \in \mathbb{R}^n\}.$$

Then (2.5.2) reads $\omega|_{\Lambda} = d\varphi = \partial\varphi dx$.

(ii) Theorem 2.14 generalizes Theorem 2.7. To see this, consider the *twisted graph* of κ :

$$(2.5.6) \quad \Lambda_{\kappa} = \{(x, y, \xi, -\eta) \mid (x, \xi) = \kappa(y, \eta), (y, \eta) \in \mathbb{R}^{2n}\}.$$

We readily check that Λ_κ is a Lagrangian submanifold of $\mathbb{R}^{2n} \times \mathbb{R}^{2n}$, with the symplectic form $\sigma = d\eta \wedge dy + d\xi \wedge dx$.

If the map

$$(x, y, \xi, \eta) \mapsto (y, x)$$

has a nonvanishing differential on Λ_κ , we can employ (x, y) as coordinates in Theorem 2.14:

$$\Lambda_\kappa = \{(x, y, \partial_x \varphi(x, y), \partial_y \varphi(x, y)) \mid x, y \in \mathbb{R}^n\}.$$

Then (2.5.6) shows that this is equivalent to (2.3.11).

(iii) Another interesting class of generating functions for symplectic maps appears when

$$(x, y, \xi, \eta) \mapsto (x, \eta)$$

has nonvanishing differential on Λ_κ . Then

$$\begin{aligned} \Lambda_\kappa &= \{(x, -\partial_\eta \varphi(x, \eta), \partial_x \varphi(x, \eta), \eta) \mid x, \eta \in \mathbb{R}^n\} \\ (2.5.7) \quad &= \{(x, \partial_\eta \psi(x, \eta), \partial_x \psi(x, \eta), -\eta) \mid x, \eta \in \mathbb{R}^n\} \end{aligned}$$

for $\psi(x, \eta) := \varphi(x, -\eta)$. This means that

$$\kappa(\partial_\eta \psi(x, \eta), \eta) = (x, \partial_x \psi(x, \eta)).$$

□

2.6. NOTES

The proof of Theorem 2.12 is from Moser [Mo]; see also Cannas da Silva [CdS]. A PDE-oriented introduction to symplectic geometry may be found in Hörmander [H3, Chapter 21].

In Greek, the word “symplectic” means “intertwined”. This is consistent with Example 4, since the generating function $\varphi = \varphi(x, y)$ is a function of a mixture of half of the original variables (x, ξ) and half of the new variables (y, η) . “Symplectic” can also be interpreted as “complex”, mathematical usage due to H. Weyl who renamed “line complex group” the “symplectic group”; see Cannas da Silva [CdS].

FOURIER TRANSFORM, STATIONARY PHASE

- 3.1 Fourier transform on $\mathcal{S}$
- 3.2 Fourier transform on $\mathcal{S}'$
- 3.3 Semiclassical Fourier transform
- 3.4 Stationary phase in one dimension
- 3.5 Stationary phase in higher dimensions
- 3.6 Oscillatory integrals
- 3.7 Notes

We discuss in this chapter how to define the Fourier transform $\mathcal{F}$ and its inverse $\mathcal{F}^{-1}$ on various classes of smooth functions and nonsmooth distributions. We also introduce the rescaled semiclassical transforms $\mathcal{F}_h, \mathcal{F}_h^{-1}$ depending on the small parameter h and develop stationary phase asymptotics to help us understand various formulas involving $\mathcal{F}_h$ in the limit as $h \rightarrow 0$.

Be warned that our use of the symbols “ D ” and “ D^α ” differs from that in the textbook [E].

3.1. FOURIER TRANSFORM ON $\mathcal{S}$

We begin by defining and investigating the Fourier transform of smooth functions that decay rapidly as $|x| \rightarrow \infty$.

DEFINITIONS.

(i) The *Schwartz space* is

$$(3.1.1) \quad \mathcal{S} = \mathcal{S}(\mathbb{R}^n) \\ := \{ \varphi \in C^\infty(\mathbb{R}^n) \mid \sup_{\mathbb{R}^n} |x^\alpha \partial^\beta \varphi| < \infty \text{ for all multiindices } \alpha, \beta \}.$$

(ii) For each pair of multiindices α, β and each $\varphi \in \mathcal{S}$, we define the *seminorm*

$$(3.1.2) \quad |\varphi|_{\alpha, \beta} := \sup_{\mathbb{R}^n} |x^\alpha \partial^\beta \varphi|.$$

(iii) We say that

$$\varphi_j \rightarrow \varphi \quad \text{in } \mathcal{S}$$

provided

$$|\varphi_j - \varphi|_{\alpha, \beta} \rightarrow 0$$

for all multiindices α, β .

In words, the Schwartz space consists of functions which are smooth and which, together with all their derivatives, decay faster than any power of $|x|^{-1}$.

DEFINITION. If $\varphi \in \mathcal{S}$, define the *Fourier transform*

$$(3.1.3) \quad \mathcal{F}\varphi(\xi) = \hat{\varphi}(\xi) := \int_{\mathbb{R}^n} e^{-i\langle x, \xi \rangle} \varphi(x) dx \quad (\xi \in \mathbb{R}^n).$$

The reader is warned that many other texts use slightly different definitions, entailing normalizing factors involving π .

EXAMPLE: Exponential of a real quadratic form.

THEOREM 3.1 (Transform of a real exponential). *Let Q be a real, symmetric, positive definite $n \times n$ matrix. Then*

$$(3.1.4) \quad \mathcal{F}(e^{-\frac{1}{2}\langle Qx, x \rangle}) = \frac{(2\pi)^{n/2}}{(\det Q)^{1/2}} e^{-\frac{1}{2}\langle Q^{-1}\xi, \xi \rangle}.$$

Proof. Let us calculate

$$\begin{aligned}\mathcal{F}(e^{-\frac{1}{2}\langle Qx, x \rangle}) &= \int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle Qx, x \rangle - i\langle x, \xi \rangle} dx \\ &= \int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle Q(x+iQ^{-1}\xi), x+iQ^{-1}\xi \rangle} e^{-\frac{1}{2}\langle Q^{-1}\xi, \xi \rangle} dx \\ &= e^{-\frac{1}{2}\langle Q^{-1}\xi, \xi \rangle} \int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle Qy, y \rangle} dy.\end{aligned}$$

We compute the last integral by making an orthogonal change of variables that converts Q into the diagonal form $\text{diag}(\lambda_1, \dots, \lambda_n)$. Then

$$\begin{aligned}\int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle Qy, y \rangle} dy &= \int_{\mathbb{R}^n} e^{-\frac{1}{2}\sum_{k=1}^n \lambda_k w_k^2} dw = \prod_{k=1}^n \int_{-\infty}^{\infty} e^{-\frac{\lambda_k}{2} w^2} dw \\ &= \prod_{k=1}^n \frac{2^{1/2}}{\lambda_k^{1/2}} \int_{-\infty}^{\infty} e^{-y^2} dy \\ &= \frac{(2\pi)^{n/2}}{(\lambda_1 \cdots \lambda_n)^{1/2}} = \frac{(2\pi)^{n/2}}{(\det Q)^{1/2}}. \quad \square\end{aligned}$$

The Fourier transform $\mathcal{F}$ lets us move from position variables x to momentum variables ξ , and we need to catalog how it converts various algebraic and analytic expressions in x into related expressions in ξ :

THEOREM 3.2 (Properties of the Fourier transform).

(i) *The mapping $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$ is an isomorphism.*

(ii) *We have the Fourier inversion formula*

$$(3.1.5) \quad \mathcal{F}^{-1} = \frac{1}{(2\pi)^n} R\mathcal{F},$$

where $Rf(x) := f(-x)$. In other words,

$$(3.1.6) \quad \mathcal{F}^{-1}\psi(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\langle x, \xi \rangle} \psi(\xi) d\xi;$$

and therefore

$$(3.1.7) \quad \varphi(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\langle x, \xi \rangle} \hat{\varphi}(\xi) d\xi.$$

(iii) *In addition,*

$$(3.1.8) \quad D_\xi^\alpha(\mathcal{F}\varphi) = \mathcal{F}((-x)^\alpha \varphi)$$

and

$$(3.1.9) \quad \mathcal{F}(D_x^\alpha \varphi) = \xi^\alpha \mathcal{F}\varphi.$$

(iv) Furthermore,

$$(3.1.10) \quad \mathcal{F}(\varphi\psi) = \frac{1}{(2\pi)^n} \mathcal{F}(\varphi) * \mathcal{F}(\psi).$$

REMARKS. (i) In these formulas we employ the notation from Appendix A:

$$D^\alpha = \frac{1}{i^{|\alpha|}} \partial^\alpha.$$

In particular,

$$D_x^\alpha e^{-i\langle x, \xi \rangle} = (-\xi)^\alpha e^{-i\langle x, \xi \rangle}, \quad D_\xi^\alpha e^{-i\langle x, \xi \rangle} = (-x)^\alpha e^{-i\langle x, \xi \rangle}.$$

(ii) We will later interpret the Fourier inversion formula (3.1.6) as saying that

$$(3.1.11) \quad \delta_y(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\langle x-y, \xi \rangle} d\xi \quad \text{in the sense of distributions,}$$

with $\delta_y = \delta(\cdot - y)$ denoting the Dirac measure. $\square$

Proof. 1. Let us calculate for $\varphi \in \mathcal{S}$ that

$$\begin{aligned} D_\xi^\alpha (\mathcal{F}\varphi) &= D_\xi^\alpha \int_{\mathbb{R}^n} e^{-i\langle x, \xi \rangle} \varphi(x) dx \\ &= \int_{\mathbb{R}^n} e^{-i\langle x, \xi \rangle} (-x)^\alpha \varphi(x) dx = \mathcal{F}((-x)^\alpha \varphi). \end{aligned}$$

Likewise,

$$\begin{aligned} \mathcal{F}(D_x^\alpha \varphi) &= \int_{\mathbb{R}^n} e^{-i\langle x, \xi \rangle} D_x^\alpha \varphi dx = (-1)^{|\alpha|} \int_{\mathbb{R}^n} D_x^\alpha (e^{-i\langle x, \xi \rangle}) \varphi dx \\ &= (-1)^{|\alpha|} \int_{\mathbb{R}^n} (-\xi)^\alpha e^{-i\langle x, \xi \rangle} \varphi dx = \xi^\alpha (\mathcal{F}\varphi). \end{aligned}$$

This proves (iii).

2. Recall from Appendix A the useful notation $\langle x \rangle = (1 + |x|^2)^{\frac{1}{2}}$. Then for all multiindices α, β , we have

$$\begin{aligned} \sup_\xi |\xi^\beta D_\xi^\alpha \hat{\varphi}| &= \sup_\xi |\xi^\beta \mathcal{F}((-x)^\alpha \varphi)| \\ &= \sup_\xi |\mathcal{F}(D_x^\beta((-x)^\alpha \varphi))| \\ &= \sup_\xi \left| \int_{\mathbb{R}^n} e^{-i\langle x, \xi \rangle} \frac{1}{\langle x \rangle^{n+1}} \langle x \rangle^{n+1} D_x^\beta((-x)^\alpha \varphi) dx \right| \\ &\leq \sup_x |\langle x \rangle^{n+1} D_x^\beta((-x)^\alpha \varphi)| \int_{\mathbb{R}^n} \langle x \rangle^{-n-1} dx < \infty. \end{aligned}$$

Hence $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{S}$, and a similar calculation shows that $\varphi_j \rightarrow \varphi$ in $\mathcal{S}$ implies $\mathcal{F}(\varphi_j) \rightarrow \mathcal{F}(\varphi)$.

3. To show that $\mathcal{F}$ is invertible, note that

$$\begin{aligned} R\mathcal{F}F D_{x_j} &= R\mathcal{F}M_{\xi_j}\mathcal{F} \\ &= R(-D_{x_j})\mathcal{F}\mathcal{F} \\ &= D_{x_j}R\mathcal{F}\mathcal{F}, \end{aligned}$$

where M_{ξ_j} denotes multiplication by ξ_j . Thus $R\mathcal{F}\mathcal{F}$ commutes with D_{x_j} and it likewise commutes with the multiplication operators M_{x_j} . According to Lemma 3.3, stated and proved below, $R\mathcal{F}\mathcal{F}$ is a multiple of the identity operator:

$$(3.1.12) \quad R\mathcal{F}\mathcal{F} = cI.$$

From the example above, we know that

$$\mathcal{F}(e^{-\frac{|x|^2}{2}}) = (2\pi)^{n/2} e^{-\frac{|\xi|^2}{2}}.$$

Thus $\mathcal{F}(e^{-\frac{|\xi|^2}{2}}) = (2\pi)^{n/2} e^{-\frac{|x|^2}{2}}$. Consequently $c = (2\pi)^n$, and hence

$$\mathcal{F}^{-1} = \frac{1}{(2\pi)^n} R\mathcal{F}.$$

4. Finally, since

$$\varphi(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\langle x, \xi \rangle} \hat{\varphi}(\xi) d\xi, \quad \psi(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\langle x, \eta \rangle} \hat{\psi}(\eta) d\eta,$$

we have

$$\begin{aligned} \varphi\psi &= \frac{1}{(2\pi)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i\langle x, \xi + \eta \rangle} \hat{\varphi}(\xi) \hat{\psi}(\eta) d\xi d\eta \\ &= \frac{1}{(2\pi)^{2n}} \int_{\mathbb{R}^n} e^{i\langle x, \rho \rangle} \left(\int_{\mathbb{R}^n} \hat{\varphi}(\xi) \hat{\psi}(\rho - \xi) d\rho \right) d\xi \\ &= \frac{1}{(2\pi)^n} \mathcal{F}^{-1}(\hat{\varphi} * \hat{\psi}). \end{aligned}$$

But $\varphi\psi = \mathcal{F}^{-1}\mathcal{F}(\varphi\psi)$, and so assertion (iv) of Theorem 3.2 follows. $\square$

LEMMA 3.3 (Commutativity). *Let M_f be the multiplication operator, so that $M_fg = fg$. Suppose that $L : \mathcal{S} \rightarrow \mathcal{S}$ is linear and that*

$$(3.1.13) \quad LM_{x_j} = M_{x_j}L, \quad LD_{x_j} = D_{x_j}L,$$

$j = 1, \dots, n$. Then

$$L = cI$$

for some constant c , where I denotes the identity operator.

Proof. 1. Choose $\varphi \in \mathcal{S}$, fix $y \in \mathbb{R}^n$, and write

$$\varphi(x) - \varphi(y) = \sum_{j=1}^n (x_j - y_j) \psi_j(x)$$

for

$$\psi_j(x) := \int_0^1 \varphi_{x_j}(y + t(x - y)) dt.$$

Since typically $\psi_j \notin \mathcal{S}$, we select a smooth function χ with compact support such that $\chi \equiv 1$ for x near y . Write

$$\varphi_j(x) := \chi(x) \psi_j(x) + \frac{(x_j - y_j)}{|x - y|^2} (1 - \chi(x)) \varphi(x).$$

Then

$$(3.1.14) \quad \varphi(x) - \varphi(y) = \sum_{j=1}^n (x_j - y_j) \varphi_j(x)$$

with $\varphi_j \in \mathcal{S}$.

2. We claim next that if $\varphi(y) = 0$, then $L\varphi(y) = 0$. This follows from (3.1.14) since

$$L\varphi(x) = \sum_{j=1}^n (x_j - y_j) L\varphi_j = 0$$

at $x = y$.

Therefore $L\varphi(x) = c(x)\varphi(x)$ for some function c . Taking $\varphi(x) = e^{-|x|^2}$, we deduce that $c \in C^\infty$. Finally, since L commutes with differentiation, we conclude that c must be a constant. $\square$

THEOREM 3.4 (Integral identities). *If $\varphi, \psi \in \mathcal{S}$, then*

$$(3.1.15) \quad \int_{\mathbb{R}^n} \hat{\varphi} \psi \, dx = \int_{\mathbb{R}^n} \varphi \hat{\psi} \, dy$$

and

$$(3.1.16) \quad \int_{\mathbb{R}^n} \varphi \bar{\psi} \, dx = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{\varphi} \bar{\hat{\psi}} \, d\xi.$$

In particular,

$$(3.1.17) \quad \|\varphi\|_{L^2}^2 = \frac{1}{(2\pi)^n} \|\hat{\varphi}\|_{L^2}^2.$$

Proof. Note first that

$$\begin{aligned} \int_{\mathbb{R}^n} \hat{\varphi} \psi \, dx &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} e^{-i\langle x, y \rangle} \varphi(y) \, dy \right) \psi(x) \, dx \\ &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} e^{-i\langle y, x \rangle} \psi(x) \, dx \right) \varphi(y) \, dy = \int_{\mathbb{R}^n} \hat{\psi} \varphi \, dy. \end{aligned}$$

Replace ψ by $\bar{\hat{\psi}}$ in (3.1.15):

$$\int_{\mathbb{R}^n} \hat{\varphi} \bar{\hat{\psi}} \, d\xi = \int_{\mathbb{R}^n} \varphi (\bar{\hat{\psi}})^\wedge \, dx.$$

But $\bar{\hat{\psi}} = \int_{\mathbb{R}^n} e^{i\langle x, \xi \rangle} \bar{\psi}(x) \, dx = (2\pi)^n \mathcal{F}^{-1}(\bar{\psi})$ and so $(\bar{\hat{\psi}})^\wedge = (2\pi)^n \bar{\psi}$. $\square$

We now record some elementary estimates that we will need later:

LEMMA 3.5 (Useful estimates).

(i) *We have the bounds*

$$(3.1.18) \quad \|\hat{u}\|_{L^\infty} \leq \|u\|_{L^1}$$

and

$$(3.1.19) \quad \|u\|_{L^\infty} \leq \frac{1}{(2\pi)^n} \|\hat{u}\|_{L^1}.$$

(ii) *There exists a constant C such that*

$$(3.1.20) \quad \|\hat{u}\|_{L^1} \leq C \max_{|\alpha| \leq n+1} \|\partial^\alpha u\|_{L^1}.$$

Proof. Estimates (3.1.18) and (3.1.19) follow easily from (3.1.3) and (3.1.7). Furthermore,

$$\begin{aligned} \|\hat{u}\|_{L^1} &= \int_{\mathbb{R}^n} |\hat{u}| \langle \xi \rangle^{n+1} \langle \xi \rangle^{-n-1} \, d\xi \leq C \|\hat{u} \langle \xi \rangle^{n+1}\|_{L^\infty} \\ &\leq C \sup_{|\alpha| \leq n+1} \|\xi^\alpha \hat{u}\|_{L^\infty} = C \sup_{|\alpha| \leq n+1} \|(\partial^\alpha u)^\wedge\|_{L^\infty} \leq C \sup_{|\alpha| \leq n+1} \|\partial^\alpha u\|_{L^1}. \end{aligned}$$

This proves (3.1.20). $\square$

EXAMPLE 1: Solving a PDE. Consider the initial-value problem

$$(3.1.21) \quad \begin{cases} \partial_t u = x \partial_y u + \partial_x^2 u & \text{on } \mathbb{R}^2 \times (0, \infty) \\ u = \delta_{(x_0, y_0)} & \text{on } \mathbb{R}^2 \times \{t = 0\}. \end{cases}$$

Let $\hat{u} := \mathcal{F}u$ denote the Fourier transform of u in the variables x, y (but not in t). Then

$$(\partial_t + \eta \partial_\xi) \hat{u} = -\xi^2 \hat{u}.$$

This is a linear first-order PDE that we can solve by the method of characteristics ([E, Section 3.2]):

$$\begin{aligned}\hat{u}(t, \xi + t\eta, \eta) &= \hat{u}(0, \xi, \eta)e^{-\int_0^t (\xi + s\eta)^2 ds} \\ &= \hat{u}(0, \xi, \eta)e^{-\xi^2 t - \xi\eta t^2 - \frac{\eta^2 t^3}{3}} \\ &= \hat{u}(0, \xi, \eta)e^{-\frac{1}{2}\langle B_t(\xi, \eta), (\xi, \eta) \rangle},\end{aligned}$$

for

$$B_t := \begin{pmatrix} 2t & t^2 \\ t^2 & 2t^3/3 \end{pmatrix}.$$

Furthermore, $\hat{u}(0, \xi, \eta) = \hat{\delta}_{(x_0, y_0)}$.

Taking the inverse Fourier transform, $\mathcal{F}^{-1}$, we find that

$$\begin{aligned}u(t, x, y - tx) &= \delta_{(x_0, y)} * \mathcal{F}^{-1}(e^{-\frac{1}{2}\langle B_t(\xi, \eta), (\xi, \eta) \rangle}) \\ &= \frac{\sqrt{3}}{2\pi t^3} \exp\left(-\frac{(x - x_0)^2}{t} + \frac{3(x - x_0)(y - y_0)}{t^2} - \frac{3(y - y_0)^2}{t^3}\right);\end{aligned}$$

and hence

$$u(t, x, y) = \frac{\sqrt{3}}{2\pi t^3} e^{-\Phi(t, x, x_0, y - y_0)},$$

where

$$\Phi(t, x, x_0, y) = \frac{(x - x_0)^2}{t} - \frac{3(x - x_0)(y + tx)}{t^2} + \frac{3(y + tx)^2}{t^3}. \quad \square$$

EXAMPLE 2: Building almost analytic extensions. Let

$$\bar{\partial}_z := \frac{1}{2}(\partial_x + i\partial_y)$$

for $z = x + iy$ denote the Cauchy–Riemann operator, and remember that g is analytic provided $\bar{\partial}_z g \equiv 0$ in the complex plane $\mathbb{C}$.

A function $f \in \mathcal{S}(\mathbb{R})$ need not be the restriction to $\mathbb{R}$ of an analytic function in $\mathbb{C}$. But we can build an extension $\tilde{f}$ that is *almost analytic* in the sense that $\bar{\partial}_z \tilde{f}$ vanishes to infinite order near the real axis. We will use this almost analytic extension later.

For the construction below, select a function χ such that

$$\chi \in C_c^\infty((-1, 1)), \text{ with } \chi \equiv 1 \text{ on } [-1/2, 1/2].$$

THEOREM 3.6 (Almost analytic extension). *If $f \in \mathcal{S}(\mathbb{R})$, then*

$$(3.1.22) \quad \tilde{f}(z) := \frac{1}{2\pi} \chi(y) \int_{\mathbb{R}} \chi(y\xi) \hat{f}(\xi) e^{i\xi(x+iy)} d\xi$$

is an almost analytic extension of f to the complex plane. This means

$$\tilde{f} \in C^\infty(\mathbb{C}), \quad \tilde{f}|_{\mathbb{R}} = f, \quad \text{spt } \tilde{f} \subset \{z \mid |\text{Im } z| \leq 1\}$$

and

$$(3.1.23) \quad \bar{\partial}_z \tilde{f}(z) = O(|\text{Im } z|^\infty).$$

The notation (3.1.23) means that

$$|\bar{\partial}_z \tilde{f}(z)| \leq C_N |\text{Im } z|^N$$

for each N and $z \in \mathbb{C}$.

Proof. 1. The Fourier inversion formula shows that $\tilde{f} = f$ on $\mathbb{R}$ and the term $\chi(y)$ restricts the support of $\tilde{f}$ to $\{|\text{Im } z| \leq 1\}$.

2. Let

$$F(z) := \int_{\mathbb{R}} \chi(y\xi) \hat{f}(\xi) e^{i\xi(x+iy)} d\xi.$$

We calculate

$$\begin{aligned} \bar{\partial}_z F &= i \int_{\mathbb{R}} \xi \chi'(y\xi) \hat{f}(\xi) e^{i\xi(x+iy)} d\xi \\ &= y^N i \int_{\mathbb{R}} \frac{\chi'(t)}{t^N} \Big|_{t=y\xi} \xi^{N+1} \hat{f}(\xi) e^{i\xi(x+iy)} d\xi \\ &= O(|y|^N) \|\xi^{N+1} \hat{f}\|_{L^1} \sup_{t \in \mathbb{R}} \frac{|\chi'(t)| e^{-t}}{t^N}. \end{aligned}$$

Since $\chi(t) \equiv 1$ near $t = 0$ and $\hat{f} \in \mathcal{S}$, the right-hand side is bounded for any N . Thus

$$|\bar{\partial}_z F(x+iy)| \leq C_N |y|^N$$

for each N , and therefore (3.1.23) holds. $\square$

3.2. FOURIER TRANSFORM ON $\mathcal{S}'$

Next we extend the Fourier transform to $\mathcal{S}'$, the dual space of $\mathcal{S}$. We will then be able to study the Fourier transforms of various important, but nonsmooth, expressions.

DEFINITIONS.

(i) We write $\mathcal{S}' = \mathcal{S}'(\mathbb{R}^n)$ for the space of *tempered distributions*, which is the dual of $\mathcal{S}$. That is, $u \in \mathcal{S}'$ provided $u : \mathcal{S} \rightarrow \mathbb{C}$ is linear and $\varphi_j \rightarrow \varphi$ in $\mathcal{S}$ implies $u(\varphi_j) \rightarrow u(\varphi)$.

(ii) We say

$$u_j \rightarrow u \quad \text{in } \mathcal{S}'$$

if

$$u_j(\varphi) \rightarrow u(\varphi) \quad \text{for all } \varphi \in \mathcal{S}.$$

DEFINITION. If $u \in \mathcal{S}'$, we define

$$D^\alpha u, x^\alpha u, \mathcal{F}u \in \mathcal{S}'$$

by the rules

$$(3.2.1) \quad \begin{aligned} D^\alpha u(\varphi) &:= (-1)^{|\alpha|} u(D^\alpha \varphi), \\ (x^\alpha u)(\varphi) &:= u(x^\alpha \varphi), \\ (\mathcal{F}u)(\varphi) &:= u(\mathcal{F}\varphi) \end{aligned}$$

for $\varphi \in \mathcal{S}$.

EXAMPLE 1: Dirac measure. It follows from the definitions that

$$\hat{\delta}_0(\varphi) = \delta_0(\hat{\varphi}) = \hat{\varphi}(0) = \int_{\mathbb{R}^n} \varphi \, dx.$$

We interpret this calculation as saying that

$$\hat{\delta}_0 \equiv 1. \quad \square$$

EXAMPLE 2: Exponential of an imaginary quadratic form.

DEFINITION. The *signature* of a real, symmetric, nonsingular matrix Q is

$$(3.2.2) \quad \begin{aligned} \text{sgn } Q &:= \text{number of positive eigenvalues of } Q \\ &\quad - \text{number of negative eigenvalues of } Q. \end{aligned}$$

THEOREM 3.7 (Transform of an imaginary exponential). *Let Q be a real, symmetric, nonsingular $n \times n$ matrix. Then*

$$(3.2.3) \quad \mathcal{F} \left(e^{\frac{i}{2} \langle Qx, x \rangle} \right) = \frac{(2\pi)^{n/2} e^{\frac{i\pi}{4} \text{sgn}(Q)}}{|\det Q|^{1/2}} e^{-\frac{i}{2} \langle Q^{-1}\xi, \xi \rangle}.$$

Compare this carefully with the earlier formula (3.1.4). The extra *phase shift* term $e^{\frac{i\pi}{4} \text{sgn } Q}$ in (3.2.3) arises from the complex exponential.

Proof. 1. Let $\epsilon > 0$, $Q_\epsilon := Q + \epsilon iI$. Then

$$\begin{aligned} \mathcal{F} \left(e^{\frac{i}{2} \langle Q_\epsilon x, x \rangle} \right) &= \int_{\mathbb{R}^n} e^{\frac{i}{2} \langle Q_\epsilon x, x \rangle - i \langle x, \xi \rangle} \, dx \\ &= \int_{\mathbb{R}^n} e^{\frac{i}{2} \langle Q_\epsilon (x - Q_\epsilon^{-1} \xi), x - Q_\epsilon^{-1} \xi \rangle} e^{-\frac{i}{2} \langle Q_\epsilon^{-1} \xi, \xi \rangle} \, dx \\ &= e^{-\frac{i}{2} \langle Q_\epsilon^{-1} \xi, \xi \rangle} \int_{\mathbb{R}^n} e^{\frac{i}{2} \langle Q_\epsilon y, y \rangle} \, dy. \end{aligned}$$

Now change variables, to write Q in the form $\text{diag}(\lambda_1, \dots, \lambda_n)$, with $\lambda_1, \dots, \lambda_r > 0$ and $\lambda_{r+1}, \dots, \lambda_n < 0$. Then

$$\int_{\mathbb{R}^n} e^{\frac{i}{2} \langle Q_\epsilon y, y \rangle} dy = \int_{\mathbb{R}^n} e^{\sum_{k=1}^n \frac{1}{2} (i\lambda_k - \epsilon) w_k^2} dw = \prod_{k=1}^n \int_{-\infty}^{\infty} e^{\frac{1}{2} (i\lambda_k - \epsilon) w^2} dw.$$

2. If $1 \leq k \leq r$, then $\lambda_k > 0$ and we set $z = (\epsilon - i\lambda_k)^{1/2} w$, and we take the branch of the square root so that $\text{Im}(\epsilon - i\lambda_k)^{1/2} < 0$. Then

$$\int_{-\infty}^{\infty} e^{\frac{1}{2} (i\lambda_k - \epsilon) w^2} dw = \frac{1}{(\epsilon - i\lambda_k)^{1/2}} \int_{\Gamma_k} e^{-\frac{1}{2} z^2} dz,$$

for the contour Γ_k shown in Figure 3.2.1.

Since

$$\exp\left(-\frac{1}{2} z^2\right) = \exp\left((y^2 - x^2)/2 - ixy\right)$$

and since $x^2 > y^2$ on Γ_k , we can deform Γ_k into the real axis.

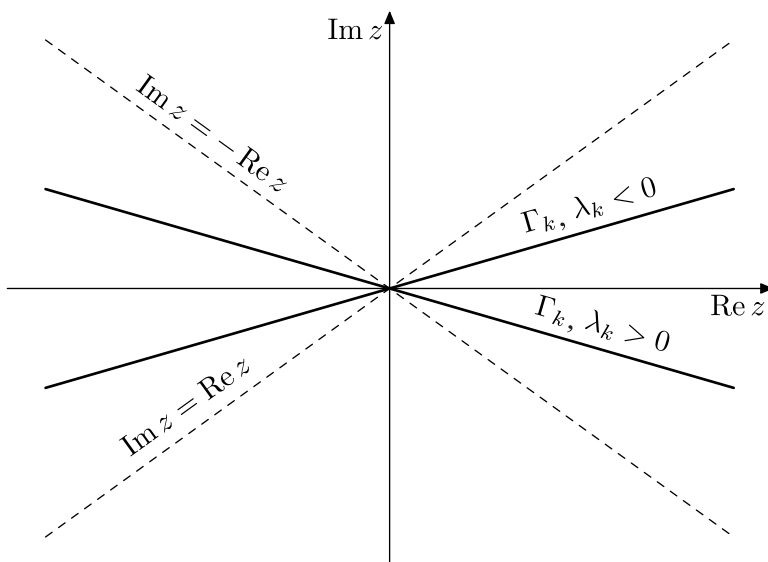


Figure 3.2.1. The contours used in the proof of Theorem 3.7.

Hence

$$\int_{\Gamma_k} e^{-\frac{1}{2} z^2} dz = \int_{-\infty}^{\infty} e^{-\frac{1}{2} x^2} dx = \sqrt{2\pi}.$$

Thus

$$\prod_{k=1}^r \int_{-\infty}^{\infty} e^{\frac{1}{2} (i\lambda_k - \epsilon) w^2} dw = (2\pi)^{r/2} \prod_{k=1}^r \frac{1}{(\epsilon - i\lambda_k)^{1/2}}.$$

Also for $1 \leq k \leq r$,

$$\lim_{\epsilon \rightarrow 0^+} \frac{1}{(\epsilon - i\lambda_k)^{1/2}} = \frac{1}{(-i)^{1/2} \lambda_k^{1/2}} = \frac{e^{\frac{i\pi}{4}}}{\lambda_k^{1/2}},$$

since we take the branch of the square root with $(-i)^{1/2} = e^{-i\pi/4}$.

3. Similarly for $r+1 \leq k \leq n$, we set $z = (\epsilon - i\lambda_k)^{1/2}w$ but now take the branch of the square root with $\text{Im}(\epsilon - i\lambda_k)^{1/2} > 0$. Hence

$$\prod_{k=r+1}^n \int_{-\infty}^{\infty} e^{\frac{1}{2}(i\lambda_k - \epsilon)w^2} dw = (2\pi)^{\frac{n-r}{2}} \prod_{k=r+1}^n \frac{1}{(\epsilon - i\lambda_k)^{1/2}};$$

and for $r+1 \leq k \leq n$,

$$\lim_{\epsilon \rightarrow 0^+} \frac{1}{(\epsilon - i\lambda_k)^{1/2}} = \frac{1}{(-i\lambda_k)^{1/2}} = \frac{e^{-\frac{i\pi}{4}}}{|\lambda_k|^{1/2}},$$

since we take the branch of the square root with $i^{1/2} = e^{\frac{i\pi}{4}}$.

4. Combining the foregoing calculations gives us

$$\begin{aligned} \mathcal{F}\left(e^{\frac{i}{2}\langle Qx, x \rangle}\right) &= \lim_{\epsilon \rightarrow 0} \mathcal{F}\left(e^{\frac{i}{2}\langle Q_\epsilon x, x \rangle}\right) \\ &= e^{-\frac{i}{2}\langle Q^{-1}\xi, \xi \rangle} \frac{(2\pi)^{n/2} e^{\frac{i\pi}{4}(r-(n-r))}}{|\lambda_1 \lambda_2 \dots \lambda_n|^{1/2}} \\ &= e^{-\frac{i}{2}\langle Q^{-1}\xi, \xi \rangle} \frac{(2\pi)^{n/2} e^{\frac{i\pi}{4} \text{sgn } Q}}{|\det Q|^{1/2}}. \quad \square \end{aligned}$$

3.3. SEMICLASSICAL FOURIER TRANSFORM

Now we introduce the small semiclassical parameter $h > 0$ and use it to rescale the Fourier transform:

DEFINITION. The *semiclassical Fourier transform* for $h > 0$ is

$$(3.3.1) \quad \mathcal{F}_h \varphi(\xi) := \int_{\mathbb{R}^n} e^{-\frac{i}{h}\langle x, \xi \rangle} \varphi(x) dx$$

and its inverse is

$$(3.3.2) \quad \mathcal{F}_h^{-1} \psi(x) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x, \xi \rangle} \psi(\xi) d\xi.$$

Consequently

$$(3.3.3) \quad \delta_{\{y=x\}} = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} d\xi \quad \text{in } \mathcal{S}'.$$

This is a rescaled version of (3.1.11).

Here are some useful formulas:

THEOREM 3.8 (Properties of $\mathcal{F}_h$). *We have*

$$(3.3.4) \quad (hD_\xi)^\alpha \mathcal{F}_h \varphi = \mathcal{F}_h((-x)^\alpha \varphi);$$

$$(3.3.5) \quad \mathcal{F}_h((hD_x)^\alpha \varphi) = \xi^\alpha \mathcal{F}_h \varphi;$$

and

$$(3.3.6) \quad \|\varphi\|_{L^2} = \frac{1}{(2\pi h)^{n/2}} \|\mathcal{F}_h \varphi\|_{L^2}.$$

Next we present a scaled version of the uncertainty principle.

THEOREM 3.9 (Uncertainty principle). *We have*

$$(3.3.7) \quad \frac{h}{2} \|f\|_{L^2} \|\mathcal{F}_h f\|_{L^2} \leq \|x_j f\|_{L^2} \|\xi_j \mathcal{F}_h f\|_{L^2} \quad (j = 1, \dots, n).$$

Proof. To see this, note first that

$$\xi_j \mathcal{F}_h f(\xi) = \mathcal{F}_h(hD_{x_j} f).$$

Also observe that

$$[x_j, hD_{x_j}]f = \frac{h}{i} [\langle x_j, \partial_{x_j} f \rangle - \partial_{x_j}(x_j f)] = ihf.$$

Thus

$$\begin{aligned} \|x_j f\|_{L^2} \|\xi_j \mathcal{F}_h f\|_{L^2} &= \|x_j f\|_{L^2} \|\mathcal{F}_h(hD_{x_j} f)\|_{L^2} \\ &= (2\pi h)^{n/2} \|x_j f\|_{L^2} \|hD_{x_j} f\|_{L^2} \\ &\geq (2\pi h)^{n/2} |\langle hD_{x_j} f, x_j f \rangle| \\ &\geq (2\pi h)^{n/2} |\operatorname{Im} \langle hD_{x_j} f, x_j f \rangle| \\ &= \frac{(2\pi h)^{n/2}}{2} |\langle [x_j, hD_{x_j}]f, f \rangle| \\ &= \frac{(2\pi h)^{n/2}}{2} h \|f\|_{L^2}^2 \\ &= \frac{h}{2} \|f\|_{L^2} \|\mathcal{F}_h f\|_{L^2}. \quad \square \end{aligned}$$

INTERPRETATION: Localization in phase space. The uncertainty principle estimate (3.3.7) shows that a function cannot be arbitrarily localized simultaneously in position x and momentum ξ . To see this, suppose

$$1 = \|f\|_{L^2} = \frac{1}{(2\pi h)^{n/2}} \|\mathcal{F}_h f\|_{L^2}.$$

A measure of concentration of f at $x = 0$ is provided by the quantities $\|x_j f\|_{L^2}$ for $j = 1, \dots, n$. For instance, take

$$f(x) = h^{-|\rho|/2} \psi\left(\frac{x_1}{h^{\rho_1}}, \dots, \frac{x_n}{h^{\rho_n}}\right)$$

for $0 \leq \rho_j \leq 1$ and $|\rho| = \rho_1 + \dots + \rho_n$, where $\psi \in \mathcal{S}$, $\|\psi\|_{L^2} = 1$. Then f is localized to the region $C_h(\epsilon) := [-h^{\rho_1-\epsilon}, h^{\rho_1-\epsilon}] \times \dots \times [-h^{\rho_n-\epsilon}, h^{\rho_n-\epsilon}]$ in the x variables, in the sense that for any $\epsilon > 0$,

$$\int_{\mathbb{R}^n \setminus C_h(\epsilon)} |f|^2 dx = O(h^\infty).$$

We can use the norms of $x_j f$ to make the localization statement more precise:

$$\|x_j f\|_{L^2} \simeq h^{\rho_j} \quad (j = 1, \dots, n).$$

Now

$$\mathcal{F}_h f(\xi) = h^{|\rho|/2} \widehat{\psi}\left(\frac{\xi_1}{h^{1-\rho_1}}, \dots, \frac{\xi_n}{h^{1-\rho_n}}\right),$$

$\widehat{\psi}$ denoting the usual Fourier transform. The localization in momentum ξ is expressed by

$$\frac{1}{(2\pi h)^{n/2}} \|\xi_j \mathcal{F}_h f\|_{L^2} \simeq h^{1-\rho_j}.$$

Hence localization in x_j is matched by delocalization in ξ_j :

$$\|x_j f\|_{L^2} \frac{1}{(2\pi h)^{n/2}} \|\xi_j \mathcal{F}_h f\|_{L^2} \simeq h^{\rho_j} \times h^{1-\rho_j} = h.$$

The inequality (3.3.7) also reveals the linkage of localization of the *symplectically conjugate* variables, x_j , ξ_j , satisfying $\{x_j, \xi_j\} = 1$; and the proof shows that this symplectic property is apparent in the noncommutativity of x_j and hD_{x_j} . This correspondence will be central in what follows; see for instance (4.3.11) in Theorem 4.12. $\square$

3.4. STATIONARY PHASE IN ONE DIMENSION

Understanding expressions like the right-hand side of (3.3.1) in the limit $h \rightarrow 0$ requires our studying integral expressions with rapidly oscillating integrands. We begin with the one-dimensional case.

DEFINITION. Given functions $a \in C_c^\infty(\mathbb{R})$, $\varphi \in C^\infty(\mathbb{R})$, we define for $h > 0$ the *oscillatory integral*

$$(3.4.1) \quad I_h = I_h(a, \varphi) := \int_{-\infty}^{\infty} e^{\frac{i\varphi}{h}} a dx.$$

We put

$$(3.4.2) \quad K := \text{spt}(a).$$

LEMMA 3.10 (Rapid decay). *If $\varphi' \neq 0$ on K , then*

$$(3.4.3) \quad I_h = O(h^\infty) \quad \text{as } h \rightarrow 0.$$

NOTATION. As explained in Appendix A, the identity (3.4.3) means that for each positive integer N , there exists a constant C_N such that

$$|I_h| \leq C_N h^N \quad \text{for all } 0 < h \leq 1.$$

Proof. We will integrate by parts N times. For this, observe that the operator

$$L := \frac{h}{i} \frac{1}{\varphi'} \partial_x$$

is defined on K , since $\varphi' \neq 0$ there. Notice also that

$$L \left(e^{\frac{i\varphi}{h}} \right) = e^{\frac{i\varphi}{h}}.$$

Hence $L^N(e^{i\varphi/h}) = e^{i\varphi/h}$, for $N = 1, 2, \dots$. Consequently

$$|I_h| = \left| \int_{-\infty}^{\infty} L^N \left(e^{\frac{i\varphi}{h}} \right) a \, dx \right| = \left| \int_{-\infty}^{\infty} e^{i\varphi/h} (L^*)^N a \, dx \right|,$$

L^* denoting the adjoint of L . Since a is smooth,

$$L^* a = -\frac{h}{i} \partial_x \left(\frac{a}{\varphi'} \right)$$

is of size h . We deduce that $|I_h| \leq C_N h^N$. $\square$

Suppose next that φ' vanishes at some point within $K = \text{spt}(a)$, in which case the oscillatory integral is no longer of order h^∞ . We instead want to expand I_h in an asymptotic expansion in powers of h :

THEOREM 3.11 (Stationary phase). *Let $a \in C_c^\infty(\mathbb{R})$. Suppose that $x_0 \in K = \text{spt}(a)$ and*

$$(3.4.4) \quad \varphi'(x_0) = 0, \quad \varphi''(x_0) \neq 0.$$

Assume further that $\varphi'(x) \neq 0$ on $K - \{x_0\}$.

(i) *There exist for each $k = 0, 1, \dots$ differential operators $A_{2k}(x, D)$, of order less than or equal to $2k$, such that for all N*

$$(3.4.5) \quad \left| I_h - \left(\sum_{k=0}^{N-1} A_{2k}(x, D) a(x_0) h^{k+\frac{1}{2}} \right) e^{\frac{i}{h}\varphi(x_0)} \right| \leq C_N h^{N+\frac{1}{2}} \sum_{0 \leq m \leq 2N+2} \sup_{\mathbb{R}} |a^{(m)}|.$$

The constant C_N also depends on the set K .

(ii) *In particular,*

$$(3.4.6) \quad A_0 = (2\pi)^{1/2} |\varphi''(x_0)|^{-1/2} e^{\frac{i\pi}{4} \operatorname{sgn} \varphi''(x_0)};$$

and consequently

$$(3.4.7) \quad I_h = (2\pi h)^{1/2} |\varphi''(x_0)|^{-1/2} e^{\frac{i\pi}{4} \operatorname{sgn} \varphi''(x_0)} e^{\frac{i\varphi(x_0)}{h}} a(x_0) + O(h^{3/2})$$

as $h \rightarrow 0$.

NOTATION. We will sometimes write (3.4.5) in the form

$$(3.4.8) \quad I_h \sim e^{\frac{i}{h}\varphi(x_0)} \sum_{k=0}^{\infty} A_{2k}(x, D) a(x_0) h^{k+\frac{1}{2}}.$$

See Section 4.4.2 for this notation.

We present two proofs of this important theorem. The second proof is more complicated but provides us with explicit expressions for the terms of the expansion (3.4.8); see (3.4.11).

First proof of Theorem 3.11. 1. We may without loss of generality assume $x_0 = 0$, $\varphi(0) = 0$. Then $\varphi(x) = \frac{1}{2}\psi(x)x^2$, for

$$\psi(x) := 2 \int_0^1 (1-t) \varphi''(tx) dt.$$

Notice that $\psi(0) = \varphi''(0) \neq 0$. We change variables by writing

$$y := |\psi(x)|^{1/2} x$$

for x near 0. Thus

$$\partial_y x = |\varphi''(0)|^{-1/2} \quad \text{at } x = y = 0.$$

Now select a smooth function $\chi : \mathbb{R} \rightarrow \mathbb{R}$ such that $0 \leq \chi \leq 1$, $\chi \equiv 1$ near 0, and $\operatorname{sgn} \varphi''(x) = \operatorname{sgn} \varphi''(0) \neq 0$ on the support of χ . Then Lemma 3.10 implies

$$\begin{aligned} I_h &= \int_{-\infty}^{\infty} e^{i\varphi(x)/h} \chi(x) a(x) dx + \int_{-\infty}^{\infty} e^{i\varphi(x)/h} (1 - \chi(x)) a(x) dx \\ &= \int_{-\infty}^{\infty} e^{\frac{i\epsilon}{2h} y^2} u(y) dy + O(h^\infty), \end{aligned}$$

for $\epsilon := \operatorname{sgn} \varphi''(0) = \pm 1$, $u(y) := \chi(x(y)) a(x(y)) |\det \partial_y x|$.

2. The Fourier transform formula (3.2.3) tells us that

$$\mathcal{F} \left(e^{-\frac{i\epsilon y^2}{2h}} \right) = (2\pi h)^{1/2} e^{-\frac{i\pi\epsilon}{4}} e^{\frac{i\epsilon h \xi^2}{2}}.$$

Applying (3.1.16), we see that consequently

$$I_h = \left(\frac{h}{2\pi}\right)^{1/2} e^{\frac{i\pi\epsilon}{4}} \int_{-\infty}^{\infty} e^{-\frac{i\epsilon h \xi^2}{2}} \hat{u}(\xi) d\xi + O(h^\infty).$$

The advantage is that the small parameter h , and not h^{-1} , occurs in the exponential.

3. Next, write

$$J(h, u) := \int_{-\infty}^{\infty} e^{-\frac{i\epsilon h \xi^2}{2}} \hat{u}(\xi) d\xi, \quad J(0, u) = 2\pi u(0).$$

Then

$$\partial_h J(h, u) = \int_{-\infty}^{\infty} e^{-\frac{i\epsilon h \xi^2}{2}} \left(\frac{\epsilon \xi^2}{2i}\right) \hat{u}(\xi) d\xi = J(h, Pu)$$

for $P := (\epsilon/2i)\partial^2$. Continuing, we discover

$$\partial_h^k J(h, u) = J(h, P^k u).$$

Therefore

$$J(h, u) = \sum_{k=0}^{N-1} \frac{h^k}{k!} J(0, P^k u) + \frac{h^N}{N!} R_N(h, u),$$

for the remainder term

$$R_N(h, u) := N \int_0^1 (1-t)^{N-1} J(th, P^N u) dt.$$

Thus Lemma 3.5 implies

$$|R_N| \leq C_N \|\widehat{P^N u}\|_{L^1} \leq C_N \sum_{0 \leq k \leq 2} \sup_{\mathbb{R}} |\partial^k (P^N u)|.$$

4. Since the definition of J gives

$$h^k J(0, P^k u) = h^2 P^k u(0) = (h/2i)^k u^{(2k)}(0)$$

and since $u = \chi(x(y))a(x(y))|\det \partial_y x|$, the expansion follows. $\square$

The second proof of stationary phase asymptotics will employ a quantitative version of Lemma 3.10:

LEMMA 3.12 (More on rapid decay). *Suppose that $a \in C_c^\infty(\mathbb{R})$ and that $\varphi \in C^\infty(\mathbb{R})$. For each positive integer m , there exists a constant C_m depending also on $\text{spt } a$ such that*

$$(3.4.9) \quad \left| \int_{-\infty}^{\infty} e^{i\varphi/h} a dx \right| \leq C_m h^m \sum_{0 \leq k \leq m} \sup_{\mathbb{R}} (|a^{(k)}| |\varphi'|^{k-2m}).$$

This inequality will be useful at points where φ' is small, provided $a^{(m)}$ is also small.

Proof. The proof is an induction on m , the case $m = 0$ being obvious. Assume the assertion for $m - 1$. Then

$$\begin{aligned} \int_{-\infty}^{\infty} e^{i\varphi/h} a \, dx &= \frac{h}{i} \int_{-\infty}^{\infty} \left(e^{i\varphi/h} \right)' \frac{a}{\varphi'} dx \\ &= -\frac{h}{i} \int_{-\infty}^{\infty} e^{i\varphi/h} \left(\frac{a}{\varphi'} \right)' dx = -\frac{h}{i} \int_{-\infty}^{\infty} e^{i\varphi/h} \tilde{a} \, dx, \end{aligned}$$

for

$$\tilde{a} := (a/\varphi')'.$$

Observe that

$$|\tilde{a}^{(k)}| = |(a/\varphi')^{(k+1)}| \leq C \sum_{0 \leq j \leq k+1} |a^{(j)}| |\varphi'|^{j-k-2}.$$

The induction hypothesis therefore implies

$$\begin{aligned} \left| \int_{-\infty}^{\infty} e^{i\varphi(x)/h} a \, dx \right| &\leq h \left| \int_{-\infty}^{\infty} e^{i\varphi(x)/h} \tilde{a} \, dx \right| \\ &\leq h C_{m-1} h^{m-1} \sum_{0 \leq k \leq m-1} \sup_{\mathbb{R}} (|\tilde{a}^{(k)}| |\varphi'|^{k-2(m-1)}) \\ &\leq C_m h^m \sum_{0 \leq j \leq m} \sup_{\mathbb{R}} (|a^{(j)}| |\varphi'|^{j-2m}). \quad \square \end{aligned}$$

Second proof of Theorem 3.11. 1. As before, we may assume $x_0 = 0$, $\varphi(0) = \varphi'(0) = 0$, $\varphi''(0) \neq 0$. To find the expansion in h of our integral

$$I_h = \int_{-\infty}^{\infty} e^{i\varphi/h} a \, dx,$$

we write

$$\varphi_s(x) := \varphi''(0)x^2/2 + sg(x)$$

for $0 \leq s \leq 1$, where

$$g(x) := \varphi(x) - \varphi''(0)x^2/2.$$

Then $\varphi = \varphi_1$ and $g = O(x^3)$ as $x \rightarrow 0$. Furthermore,

$$\varphi'_s(x) = \varphi''(0)x + O(x^2),$$

and therefore

$$|x| \leq |\varphi''(0)|^{-1} |\varphi'_s(x) + O(x^2)| \leq 2|\varphi''(0)|^{-1} |\varphi'(x)|$$

for sufficiently small x . Consequently, using a cutoff function χ as in the first proof, we may assume that

$$(3.4.10) \quad \frac{x}{\varphi'_s(x)} \text{ is bounded on } K = \text{spt}(a).$$

2. We also write

$$I_h(s) := \int_{-\infty}^{\infty} e^{i\varphi_s/h} a \, dx.$$

Let us calculate

$$\frac{d^{2m}}{ds^{2m}} I_h(s) = (i/h)^{2m} \int_{-\infty}^{\infty} e^{i\varphi_s/h} g^{2m} a \, dx.$$

Lemma 3.12, with $3m$ replacing m , implies

$$|I_h^{(2m)}(s)| \leq \frac{C}{h^{2m}} h^{3m} \sum_{0 \leq k \leq 3m} \sup_{\mathbb{R}} (|(ag^{2m})^{(k)}| |\varphi'_s|^{k-6m}).$$

Now the amplitude ag^{2m} vanishes to order $6m$ at $x = 0$. Consequently, for each $0 \leq k \leq 3m$ we recall (3.4.10) to estimate

$$|(ag^{2m})^{(k)}| |\varphi'_s|^{k-6m} \leq C|x|^{6m-k}|x|^{k-6m} \leq C.$$

Therefore

$$|I_h^{(2m)}(s)| \leq Mh^m.$$

It follows that

$$\begin{aligned} I_h &= I_h(1) = \sum_{l=0}^{2m-1} I_h^{(l)}(0)/l! + \frac{1}{(2m-1)!} \int_0^1 (1-s)^{2m-1} I_h^{(2m)}(s) \, ds \\ &= \sum_{l=0}^{2m-1} I_h^{(l)}(0)/l! + O(h^m). \end{aligned}$$

3. It remains to compute the expansions in h of the terms

$$I_h^{(l)}(0) = (i/h)^l \int_{-\infty}^{\infty} e^{i\varphi_0/h} g^l a \, dx$$

for $l = 0, \dots, 2m-1$. But this follows as in the first proof, since the phase $\varphi_0(x) = \varphi''(0)x^2/2$ is purely quadratic. Up to constants, the terms in the expansion are

$$h^{\frac{1}{2}+k-l} (g^l a)^{(2k)}(0)$$

for $l < 2m$ and $k = 0, 1, \dots$

This at first looks discouraging because of $-l$ in the power of h . Recall however that $g = O(x^3)$ near 0; so that $(g^l a)^{(2k)}(0) = 0$ unless $2k \geq 3l$. Also, if $k - l = j$, then

$$3j = 3k - 3l \geq k, \quad 2j = 2k - 2l \geq l.$$

This means that there are at most finitely many values of k and l in the expansion corresponding to the term $h^{\frac{1}{2}+j} = h^{\frac{1}{2}+k-l}$. $\square$

REMARK. This second proof avoids the Morse Lemma (see Theorem 3.15 below), but at some considerable technical expense. However this proof in fact provides the explicit expansion

$$(3.4.11) \quad \int_{\mathbb{R}} e^{i\varphi/h} a \, dx \sim e^{\frac{i\pi}{4} \operatorname{sgn} \varphi''(x_0)} \left(\frac{2\pi h}{|\varphi''(0)|} \right)^{\frac{1}{2}} \\ \times \sum_{k=0}^{\infty} \sum_{l=0}^{\infty} \left(\frac{h}{2i\varphi''(0)} \right)^k \frac{1}{l!} \frac{1}{k!} \frac{d^{2k}}{dx^{2k}} ((i/h)^l g^l a)(0).$$

More complicated, but in principle explicit, expansions can likewise be derived in higher dimensions as well. $\square$

3.5. STATIONARY PHASE IN HIGHER DIMENSIONS

We turn next to n -dimensional integrals.

DEFINITION. We now introduce the oscillatory integral

$$(3.5.1) \quad I_h = I_h(a, \varphi) = \int_{\mathbb{R}^n} e^{i\varphi/h} a \, dx,$$

where $a \in C_c^\infty(\mathbb{R}^n)$, $\varphi \in C^\infty(\mathbb{R}^n)$ are real-valued. As before, we also write

$$(3.5.2) \quad K := \operatorname{spt}(a).$$

3.5.1. Quadratic phase function. We begin with the case of a quadratic phase

$$\varphi(x) = \frac{1}{2} \langle Qx, x \rangle,$$

where Q is a nonsingular, symmetric, real matrix.

THEOREM 3.13 (Quadratic phase asymptotics). *For each positive integer N , we have the expansion*

$$(3.5.3) \quad I_h = (2\pi h)^{\frac{n}{2}} \frac{e^{\frac{i\pi}{4} \operatorname{sgn} Q}}{|\det Q|^{\frac{1}{2}}} \left(\sum_{k=0}^{N-1} \frac{h^k}{k!} \left(\frac{\langle Q^{-1}D, D \rangle}{2i} \right)^k a(0) + O(h^N) \right).$$

Proof. 1. The Fourier transform formulas (3.2.3) and (3.1.16) imply

$$I_h = \left(\frac{h}{2\pi} \right)^{n/2} \frac{e^{\frac{i\pi}{4} \operatorname{sgn} Q}}{|\det Q|^{\frac{1}{2}}} \int_{\mathbb{R}^n} e^{-\frac{ih}{2} \langle Q^{-1}\xi, \xi \rangle} \hat{a}(\xi) \, d\xi.$$

Write

$$J(h, a) := \int_{\mathbb{R}^n} e^{-\frac{ih}{2}\langle Q^{-1}\xi, \xi \rangle} \hat{a}(\xi) d\xi;$$

then

$$\partial_h J(h, a) = \int_{\mathbb{R}^n} e^{-\frac{ih}{2}\langle Q^{-1}\xi, \xi \rangle} \left(-\frac{i}{2} \langle Q^{-1}\xi, \xi \rangle \hat{a}(\xi) \right) d\xi = J(h, Pa)$$

for

$$P := -\frac{i}{2} \langle Q^{-1}D, D \rangle.$$

Therefore

$$J(h, a) = \sum_{k=0}^{N-1} \frac{h^k}{k!} J(0, P^k a) + \frac{h^N}{N!} R_N(h, a),$$

for the remainder term

$$R_N(h, a) := N \int_0^1 (1-t)^{N-1} J(th, P^N a) dt.$$

2. Now (3.1.7) gives

$$J(0, P^k a) = \int_{\mathbb{R}^n} \left(-\frac{i}{2} \langle Q^{-1}\xi, \xi \rangle \right)^k \hat{a}(\xi) d\xi = (2\pi)^n P^k a(0).$$

Furthermore, Lemma 3.5(ii) implies

$$|R_N| \leq C_N \|\widehat{P^N a}\|_{L^1} \leq C_N \sup_{|\alpha| \leq 2N+n+1} |\partial^\alpha a|. \quad \square$$

3.5.2. General phase function. Assume next that the phase φ is an arbitrary smooth function.

LEMMA 3.14 (Rapid decay again). *If $\partial\varphi \neq 0$ on K , then*

$$I_h = O(h^\infty).$$

In particular, for each positive integer N

$$(3.5.4) \quad |I_h| \leq Ch^N \sum_{|\alpha| \leq N} \sup_{\mathbb{R}^n} |\partial^\alpha a|,$$

where C depends upon only K and n .

Proof. Define the operator

$$L := \frac{h}{i} \frac{1}{|\partial\varphi|^2} \langle \partial\varphi, \partial \rangle$$

for $x \in K$, and observe that

$$L \left(e^{i\varphi/h} \right) = e^{i\varphi/h}.$$

Hence $L^N(e^{i\varphi/h}) = e^{i\varphi/h}$, and consequently

$$|I_h| = \left| \int_{\mathbb{R}^n} L^N(e^{i\varphi/h}) a \, dx \right| = \left| \int_{\mathbb{R}^n} e^{i\varphi/h} (L^*)^N a \, dx \right| \leq Ch^N. \quad \square$$

DEFINITION. We say $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ has a *nondegenerate critical point* at x_0 if

$$\partial\varphi(x_0) = 0, \quad \det \partial^2\varphi(x_0) \neq 0.$$

We also write

$$\begin{aligned} \operatorname{sgn} \partial^2\varphi(x_0) &:= \text{number of positive eigenvalues of } \partial^2\varphi(x_0) \\ &\quad - \text{number of negative eigenvalues of } \partial^2\varphi(x_0). \end{aligned}$$

Next we change variables locally to convert the phase function φ into a quadratic form:

THEOREM 3.15 (Morse Lemma). *Let $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ be smooth, with a nondegenerate critical point at x_0 . Then there exist neighborhoods U of 0 and V of x_0 and a diffeomorphism*

$$\gamma : V \rightarrow U$$

such that

$$(3.5.5) \quad (\varphi \circ \gamma^{-1})(x) = \varphi(x_0) + \frac{1}{2}(x_1^2 + \cdots + x_r^2 - x_{r+1}^2 - \cdots - x_n^2),$$

where r is the number of positive eigenvalues of $\partial^2\varphi(x_0)$.

Proof. 1. As usual, we suppose $x_0 = 0$, $\varphi(0) = 0$. After a linear change of variables, we have

$$\varphi(x) = \frac{1}{2}(x_1^2 + \cdots + x_r^2 - x_{r+1}^2 - \cdots - x_n^2) + O(|x|^3);$$

and so the problem is to design a further change of variables that removes the cubic and higher terms.

2. Now

$$\varphi(x) = \int_0^1 (1-t) \partial_t^2 \varphi(tx) \, dt = \frac{1}{2} \langle x, Q(x)x \rangle,$$

where

$$Q(0) = \partial^2\varphi(0) = \begin{pmatrix} I_r & O \\ O & -I_{n-r} \end{pmatrix}.$$

In this expression the upper identity matrix is $r \times r$ and the lower identity matrix is $(n-r) \times (n-r)$. We want to find a smooth mapping A from $\mathbb{R}^n$ to $GL(n, \mathbb{R})$ such that

$$(3.5.6) \quad \langle A(x)x, Q(0)A(x)x \rangle = \langle x, Q(x)x \rangle.$$

Then

$$\gamma(x) = A(x)x$$

is the desired change of variable.

Formula (3.5.6) will hold provided

$$(3.5.7) \quad A^T(x)Q(0)A(x) = Q(x).$$

Let $F : \mathbb{M}^{n \times n} \rightarrow \mathbb{S}^{n \times n}$ be defined by

$$F(A) = A^T Q(0) A.$$

We want to find a *right inverse* $G : \mathbb{S}^{n \times n} \rightarrow \mathbb{M}^{n \times n}$, so that

$$FG = I \quad \text{near } Q(0).$$

Then

$$A(x) := G(Q(x))$$

will solve (3.5.7).

3. We will apply a version of the Inverse Function Theorem (Theorem C.2). To do so, it suffices to find $B \in L(\mathbb{S}^{n \times n}, \mathbb{M}^{n \times n})$ such that

$$\partial F(I)B = I.$$

Now

$$\partial F(I)(C) = C^T Q(0) + Q(0)C.$$

Define

$$B(D) := \frac{1}{2} Q(0)^{-1} D$$

for $D \in \mathbb{S}^{n \times n}$. Then

$$\begin{aligned} \partial F(I)B(D) &= \frac{1}{2} \partial F(I)(Q^{-1}(0)D) \\ &= \frac{1}{2} [(Q(0)^{-1}D)^T Q(0) + Q(0)(Q(0)^{-1}D)] \\ &= D. \end{aligned}$$

□

Given now a general phase function φ , we apply the Morse Lemma to convert locally to a quadratic phase for which the asymptotics provided by Theorem 3.13 apply:

THEOREM 3.16 (Stationary phase asymptotics). *Assume that $a \in C_c^\infty(\mathbb{R}^n)$. Suppose $x_0 \in K$ and*

$$\partial \varphi(x_0) = 0, \quad \det \partial^2 \varphi(x_0) \neq 0.$$

Assume further that $\partial \varphi(x) \neq 0$ on $K - \{x_0\}$.

(i) Then there exist for $k = 0, 1, \dots$ differential operators $A_{2k}(x, D)$ of order less than or equal to $2k$, such that for each N

$$(3.5.8) \quad \left| I_h - \left(\sum_{k=0}^{N-1} A_{2k}(x, D) a(x_0) h^{k+\frac{n}{2}} \right) e^{\frac{i\varphi(x_0)}{h}} \right| \leq C_N h^{N+\frac{n}{2}} \sum_{|\alpha| \leq 2N+n+1} \sup_{\mathbb{R}^n} |\partial^\alpha a|.$$

(ii) In particular,

$$(3.5.9) \quad A_0 = (2\pi)^{n/2} |\det \partial^2 \varphi(x_0)|^{-1/2} e^{\frac{i\pi}{4} \operatorname{sgn} \partial^2 \varphi(x_0)};$$

and therefore

$$(3.5.10) \quad I_h = (2\pi h)^{n/2} |\det \partial^2 \varphi(x_0)|^{-1/2} e^{\frac{i\pi}{4} \operatorname{sgn} \partial^2 \varphi(x_0)} e^{\frac{i\varphi(x_0)}{h}} a(x_0) + O\left(h^{\frac{n+2}{2}}\right)$$

as $h \rightarrow 0$.

Proof. Without loss of generality $x_0 = 0$, $\varphi(0) = \partial\varphi(0) = 0$. Introducing a cutoff function χ and applying the Morse Lemma, Theorem 3.15, and then Lemma 3.14, we can write

$$I_h = \int_{\mathbb{R}^n} e^{i\varphi(x)/h} a \, dx = \int_{\mathbb{R}^n} e^{\frac{i}{2h} \langle Qx, x \rangle} u \, dx + O(h^\infty),$$

where

$$Q = \begin{pmatrix} I_r & O \\ O & -I_{n-r} \end{pmatrix},$$

and

$$u(x) := a(\kappa^{-1}(x)) |\det \partial \kappa^{-1}(x)|, \quad |\det \partial \kappa^{-1}(0)| = |\det \partial^2 \varphi(0)|^{-\frac{1}{2}}.$$

Note that $\operatorname{sgn} Q = \operatorname{sgn} \partial^2 \varphi(x_0)$ and $|\det Q| = 1$. We invoke Theorem 3.13 to finish the proof. $\square$

3.5.3. Important Examples. In Chapter 4 we will consider the asymptotic behavior of various expressions involving the Fourier transform. These involve the particular phase function

$$\varphi(x, y) = \langle x, y \rangle$$

on $\mathbb{R}^n \times \mathbb{R}^n$, corresponding to the Euclidean inner product. We will also encounter important applications with the phase

$$\varphi(z, w) = \sigma(z, w) = \langle Jz, w \rangle$$

on $\mathbb{R}^{2n} \times \mathbb{R}^{2n}$, corresponding to the symplectic structure. We therefore work out in this section the stationary phase expansions corresponding to these special cases.

THEOREM 3.17 (Important phase functions).

(i) Assume that $a \in C_c^\infty(\mathbb{R}^{2n})$. Then for each positive integer N ,

$$(3.5.11) \quad \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{\frac{i}{h}\langle x, y \rangle} a(x, y) dx dy \\ = (2\pi h)^n \left(\sum_{k=0}^{N-1} \frac{h^k}{k!} \left(\frac{\langle D_x, D_y \rangle}{i} \right)^k a(0, 0) + O(h^N) \right)$$

as $h \rightarrow 0$.

(ii) Assume that $a \in C_c^\infty(\mathbb{R}^{4n})$. Then for each positive integer N ,

$$(3.5.12) \quad \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{\frac{i}{h}\sigma(z, w)} a(z, w) dz dw \\ = (2\pi h)^{2n} \left(\sum_{k=0}^{N-1} \frac{h^k}{k!} \left(\frac{\sigma(D_x, D_\xi, D_y, D_\eta)}{i} \right)^k a(0, 0) + O(h^N) \right),$$

where $z = (x, \xi)$, $w = (y, \eta)$, and

$$\sigma(D_x, D_\xi, D_y, D_\eta) := \langle D_\xi, D_y \rangle - \langle D_x, D_\eta \rangle.$$

Proof. 1. We write (x, y) to denote a typical point of $\mathbb{R}^{2n}$ and let

$$Q := \begin{pmatrix} O & I \\ I & O \end{pmatrix}.$$

Then Q is symmetric, $Q^{-1} = Q$, $|\det Q| = 1$, $\operatorname{sgn}(Q) = 0$, and $Q(x, y) = (y, x)$. Consequently $\frac{1}{2}\langle Q(x, y), (x, y) \rangle = \langle x, y \rangle$.

Furthermore, since $D = (D_x, D_y)$,

$$\frac{1}{2}\langle Q^{-1}D, D \rangle = \langle D_x, D_y \rangle.$$

Hence Theorem 3.13 gives (3.5.11).

2. We write (z, w) to denote a typical point of $\mathbb{R}^{4n}$, where $z = (x, \xi)$, $w = (y, \eta)$. Set

$$Q := \begin{pmatrix} O & -J \\ J & O \end{pmatrix}.$$

Then Q is symmetric, $Q^{-1} = Q$, $|\det Q| = 1$, $\operatorname{sgn}(Q) = 0$, and $Q(z, w) = (-Jw, Jz)$. Consequently

$$\frac{1}{2}\langle Q(z, w), (z, w) \rangle = \langle Jz, w \rangle = \sigma(z, w).$$

We have $D = (D_z, D_w) = (D_x, D_\xi, D_y, D_\eta)$, and therefore

$$\frac{1}{2}\langle Q^{-1}D, D \rangle = \sigma(D_x, D_\xi, D_y, D_\eta).$$

Theorem 3.13 now provides us with the expansion (3.5.12). $\square$

3.6. OSCILLATORY INTEGRALS

So far we have considered oscillatory integrals which were absolutely convergent. In this section we study oscillatory integrals which do not necessarily converge but that define elements of $\mathcal{S}'$. To simplify notation, we put $h = 1$ as we are not concerned with asymptotic expansions but with making sense of integrals which do not converge.

EXAMPLE. If we put

$$u(x) := \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\langle x, \xi \rangle} d\xi,$$

then the integral does not make sense in the classical sense. It defines an element $u \in \mathcal{S}'(\mathbb{R}^n)$ as follows. For $\psi \in \mathcal{S}(\mathbb{R}^n)$ we put

$$u(\psi) := \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\langle x, \xi \rangle} \psi(x) dx d\xi.$$

The integral in x defines a Schwartz function in ξ . The Fourier inversion formula then shows that

$$u(\psi) = \psi(0), \quad \text{that is, } u(x) = \delta_0(x). \quad \square$$

THEOREM 3.18 (Distributions defined by oscillatory integrals).

Suppose that $\varphi = \varphi(w, z)$, $(w, z) \in \mathbb{R}^k \times \mathbb{R}^m$, is a real-valued quadratic function of w and z satisfying

$$(3.6.1) \quad |\partial_{w,z}\varphi| \geq |w + Kz|/C,$$

for some $K \in \mathbb{M}^{k \times m}$.

If $a \in C^\infty(\mathbb{R}^k \times \mathbb{R}^m)$ satisfies

$$(3.6.2) \quad |\partial^\alpha a(w, z)| \leq C_\alpha \langle w \rangle^M \langle z \rangle^M, \quad w \in \mathbb{R}^k, \quad z \in \mathbb{R}^m, \quad \alpha \in \mathbb{N}^{k+m},$$

for some fixed M , then the expression

$$(3.6.3) \quad u(z) = \int_{\mathbb{R}^k} e^{i\varphi(w,z)} a(w, z) dw$$

defines a unique $u \in \mathcal{S}'(\mathbb{R}^m)$ by

$$(3.6.4) \quad u(\psi) = \lim_{\epsilon \rightarrow 0+} \int_{\mathbb{R}^{k+m}} e^{i\varphi(w,z)} a(w,z) \psi(z) \chi(\epsilon w) dw dz,$$

$\psi \in \mathcal{S}(\mathbb{R}^k)$, where $\chi \in C_c^\infty(\mathbb{R}^m)$ is any function equal to 1 in a neighborhood of $0 \in \mathbb{R}^m$.

REMARK. (i) We can see the condition on φ more explicitly by writing

$$\varphi(w, z) = \frac{1}{2} \langle Aw, w \rangle + \langle Bw, z \rangle + \frac{1}{2} \langle Cz, z \rangle,$$

where $A \in \mathbb{S}^{k \times k}$, $B \in \mathbb{M}^{p \times k}$, $C \in \mathbb{S}^{p \times p}$. The assumption (3.6.1) then means that

$$(3.6.5) \quad |Aw + B^T z|^2 + |Bw + Cz|^2 \geq c|w + Kz|^2,$$

for some K . Hence (3.6.1) holds if A is invertible with $K = A^{-1}B^T$. When $k = m$ and B is invertible, then we get (3.6.1) with $K = B^{-1}C$.

(ii) When $|\partial_w \psi(w, z)| \geq |w|/C$ holds, we can conclude that

$$u(z) := \lim_{\epsilon \rightarrow 0+} \int_{\mathbb{R}^m} e^{i\varphi(w,z)} a(w,z) \chi(\epsilon w) dw$$

defines a smooth function of z . This follows from the proof below. □

Proof. 1. Because of the cut-off χ the integral defining $u(\psi)$ in (3.6.4) is well defined. We define the differential operator

$$L := \frac{1 + \langle \partial_z \varphi, D_z \rangle + \langle \partial_w \varphi, D_w \rangle}{1 + |\partial_z \varphi|^2 + |\partial_w \varphi|^2},$$

so that

$$Le^{i\varphi(z,w)} = e^{i\varphi(z,w)}.$$

Denoting by L^T the transpose of the operator L and using the assumption (3.6.2), we obtain

$$\begin{aligned} |(L^T)^N (a(z, w) \psi(z))| &\leq C_N \frac{\langle z \rangle^{M+N} \langle w \rangle^{M+N}}{\langle w + Kz \rangle^{2N}} \langle z \rangle^{-4N} \sup_{|\alpha| \leq N} |\langle z \rangle^{4N} \partial^\alpha \psi| \\ &\leq C'_N \langle z \rangle^{M-N} \langle w \rangle^{M-N} \sup_{|\alpha| \leq N} |\langle z \rangle^{4N} \partial^\alpha \psi|, \end{aligned}$$

since

$$\begin{aligned} \langle w \rangle &\leq 1 + |w| \leq 1 + |Kz| + |w + Kz| \\ &\leq (1 + \|K\|)(1 + |z|)(1 + |w + Kz|) \\ &\leq 2(1 + \|K\|) \langle z \rangle \langle w + Kz \rangle. \end{aligned}$$

2. For $N > M + \max(k, m)$,

$$\begin{aligned} u(\psi) &= \lim_{\epsilon \rightarrow 0+} \int_{\mathbb{R}^{k+m}} e^{i\varphi(w,z)} (L^T)^N (a(w, z)\psi(z)\chi(\epsilon w)) dw dz \\ &= \int_{\mathbb{R}^{k+m}} e^{i\varphi(w,z)} (L^T)^N (a(w, z)\psi(z)) dw dz, \end{aligned}$$

where the last integrand is bounded by $C\langle w \rangle^{-n-1}\langle z \rangle^{-k-1}$. Also,

$$|u(\psi)| \leq C \sup_{|\alpha| \leq N} |\langle z \rangle^{4N} \partial^\alpha \psi|,$$

which means that u defines an element of $\mathcal{S}'$. The proof also shows that the definition is independent of the choice of $\chi \in C_c^\infty(\mathbb{R}^m)$ as long $\chi(\epsilon w) \rightarrow 1$ as $\epsilon \rightarrow 0$. $\square$

NOTATION. Although the integral (3.6.3) does not make sense when a is not integrable in w , we write (3.6.3) for the tempered distribution defined by (3.6.4). The formal integral can then be manipulated as one would a convergent integral. In particular we can integrate by parts formally. $\square$

EXAMPLE. We illustrate this by a simple example in dimension one:

$$(3.6.6) \quad u(x) = \int_{\mathbb{R}} x e^{ix\xi} d\xi = \int_{\mathbb{R}} D_\xi(e^{ix\xi}) dx = 0,$$

by “integration by parts”. The rigorous interpretation is to consider $u(x)$ as an element of $\mathcal{S}'$ in which case we should consider, with $\psi \in \mathcal{S}$,

$$\begin{aligned} u(\psi) &:= \lim_{\epsilon \rightarrow 0+} \int_{\mathbb{R}} \int_{\mathbb{R}} x e^{ix\xi} \psi(x) \chi(\epsilon \xi) dx d\xi \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}} \frac{x}{(1+|\xi|^2)^k} e^{ix\xi} (I - \Delta_x)^k \psi(x) dx d\xi. \end{aligned}$$

The integral over $\mathbb{R}^2$ is now convergent and we can apply integration by parts in ξ . This shows that $u = 0$ as we have quickly seen in (3.6.6). Arguments like this will now be presented directly with the above understanding of what they mean rigorously.

3.7. NOTES

Good references are Friedlander–Joshi [F-J] and Hörmander [H1]. For more on the uncertainty principle, see Fefferman [Fe2]. The PDE example in Section 3.1 is from [H1, Section 7.6], and the second proof of the one-dimensional stationary phase is a variant of [H1, Section 7.7].

For an immediate application of the Fourier transform methods to PDE go directly to Section 7.5, which depends only on the results of this chapter.

SEMICLASSICAL QUANTIZATION

- 4.1 Definitions
- 4.2 Quantization formulas
- 4.3 Composition, asymptotic expansions
- 4.4 Symbol classes
- 4.5 Operators on L^2
- 4.6 Compactness
- 4.7 Inverses, Gårding inequalities
- 4.8 Notes

The Fourier transform and its inverse allow us to move at will between the position x and momentum ξ variables, but what we really want is to deal with both sets of variables simultaneously. This chapter therefore introduces the quantization of symbols, that is, of appropriate functions of both x and ξ . The resulting operators applied to functions entail information in the full (x, ξ) phase space, and particular choices of the symbol will later prove very useful, allowing us for example to “localize” in phase space.

The plan is to introduce quantization and then to work out the resulting *symbol calculus*, meaning the systematic rules for manipulating symbols and their associated operators. We will also establish criteria for the L^2 -boundedness, compactness, and approximate positivity of operators in terms of their symbols.

4.1. DEFINITIONS

NOTATION. For this section we take $h > 0$ and $a \in \mathcal{S}(\mathbb{R}^{2n})$, $a = a(x, \xi)$. We hereafter call a a *symbol*.

4.1.1. Quantization rules. To *quantize* a symbol means to associate with it an h -dependent linear operator acting on functions $u = u(x)$. There are several standard ways to do so:

DEFINITIONS.

(i) We define the *Weyl quantization* to be the operator $a^w(x, hD)$ acting on $u \in \mathcal{S}$ by the formula

$$(4.1.1) \quad a^w(x, hD)u(x) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} a\left(\frac{x+y}{2}, \xi\right) u(y) dy d\xi.$$

(ii) We also define the *standard quantization*

$$(4.1.2) \quad a(x, hD)u(x) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} a(x, \xi) u(y) dy d\xi$$

for $u \in \mathcal{S}$.

(iii) More generally, for $u \in \mathcal{S}$ and $0 \leq t \leq 1$, we set

$$(4.1.3) \quad \begin{aligned} \text{Op}_t(a)u(x) \\ := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} a(tx + (1-t)y, \xi) u(y) dy d\xi. \end{aligned}$$

Hence

$$(4.1.4) \quad \text{Op}_{\frac{1}{2}}(a) = a^w(x, hD), \quad \text{Op}_1(a) = a(x, hD).$$

We hereafter refer to any operator of the form $\text{Op}_t(a)$ as a *semiclassical pseudodifferential operator*. From the physical point of view the symbols should be thought of as *classical observables* and the corresponding operators as *quantum observables*; recall Section 1.2.

REMARKS. (i) Observe that

$$(4.1.5) \quad a(x, hD)u = \mathcal{F}_h^{-1}(a(x, \cdot)\mathcal{F}_h u(\cdot)).$$

This simple expression makes most of the subsequent calculations much easier for the standard quantization, as opposed to the Weyl quantization. However the latter has many better properties and will be our principal concern.

(ii) We will only rarely be directly interested in the operators Op_t for $t \neq \frac{1}{2}, 1$, but they will prove useful for interpolating between the Weyl and standard quantizations. $\square$

EXAMPLES.

(i) If $a(x, \xi) = \xi^\alpha$, then

$$(4.1.6) \quad \text{Op}_t(a)u = (hD)^\alpha u \quad (0 \leq t \leq 1).$$

(ii) If $a(x, \xi) = \sum_{|\alpha| \leq N} a_\alpha(x) \xi^\alpha$,

$$(4.1.7) \quad a(x, hD) = \sum_{|\alpha| \leq N} a_\alpha(x) (hD)^\alpha u.$$

(iii) If $a(x, \xi) = \langle x, \xi \rangle$, then

$$\text{Op}_t(a)u = (1-t)\langle hD, xu \rangle + t\langle x, hDu \rangle \quad (0 \leq t \leq 1).$$

In particular,

$$(4.1.8) \quad \langle x, hD \rangle^w u = \frac{h}{2} \langle D, xu \rangle + \frac{h}{2} \langle x, Du \rangle.$$

The formulas above follow straightforwardly from the definitions. We will provide many more explicit quantization formulas in the next section. $\square$

REMARK: Rescaling in h . It is often convenient to rescale to the case $h = 1$ by changing to the new variables

$$(4.1.9) \quad \tilde{x} := h^{-\frac{1}{2}}x, \quad \tilde{y} := h^{-\frac{1}{2}}y, \quad \tilde{\xi} := h^{-\frac{1}{2}}\xi.$$

Then

$$\begin{aligned} a^w(x, hD)u(x) &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a\left(\frac{x+y}{2}, \xi\right) e^{\frac{i}{h}\langle x-y, \xi \rangle} u(y) dy d\xi \\ &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a_h\left(\frac{\tilde{x}+\tilde{y}}{2}, \tilde{\xi}\right) e^{i\langle \tilde{x}-\tilde{y}, \tilde{\xi} \rangle} \tilde{u}(\tilde{y}) d\tilde{y} d\tilde{\xi}; \end{aligned}$$

and therefore

$$(4.1.10) \quad a^w(x, hD)u(x) = a_h^w(\tilde{x}, D)\tilde{u}(\tilde{x}),$$

for

$$(4.1.11) \quad \tilde{u}(\tilde{x}) := u(x) = u(h^{\frac{1}{2}}\tilde{x}), \quad a_h(\tilde{x}, \tilde{\xi}) := a(x, \xi) = a(h^{\frac{1}{2}}\tilde{x}, h^{\frac{1}{2}}\tilde{\xi}).$$

We call (4.1.9)–(4.1.11) the *standard rescaling*. $\square$

4.1.2. Quantization on $\mathcal{S}$ and $\mathcal{S}'$. To begin our careful study of pseudodifferential operators, we must first see how the quantizations act on the Schwartz space and on tempered distributions:

THEOREM 4.1 (Schwartz class symbols). *Assume $a \in \mathcal{S}$.*

(i) *Then for each $0 \leq t \leq 1$, $\text{Op}_t(a)$ can be defined as an operator mapping $\mathcal{S}'$ to $\mathcal{S}$; and furthermore*

$$\text{Op}_t(a) : \mathcal{S}' \rightarrow \mathcal{S}$$

is continuous.

(ii) *The formal adjoint is given by*

$$(4.1.12) \quad \text{Op}_t(a)^* = \text{Op}_{1-t}(\bar{a}) \quad (0 \leq t \leq 1);$$

and in particular the Weyl quantization of a real symbol is formally self-adjoint:

$$(4.1.13) \quad a^w(x, hD)^* = a^w(x, hD) \quad \text{if } a \text{ is real.}$$

We will later learn that for a very general class of symbols a , $a^w(x, hD)$ is bounded on L^2 , in which case $a^w(x, hD)$ is selfadjoint provided a is real.

Proof. 1. We have

$$\text{Op}_t(a)u(x) = \int_{\mathbb{R}^n} K_t(x, y)u(y) dy$$

for the kernel

$$\begin{aligned} K_t(x, y) &:= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} a(tx + (1-t)y, \xi) d\xi \\ &= \mathcal{F}_h^{-1}(a(tx + (1-t)y, \cdot))(x - y). \end{aligned}$$

Thus $K_t \in \mathcal{S}$, and so $\text{Op}_t(a)u(x) = u(K_t(x, \cdot))$ maps $\mathcal{S}'$ continuously into $\mathcal{S}$.

2. The kernel of $\text{Op}_t(a)^*$ is $K_t^*(x, y) := \overline{K}_t(y, x) = \overline{K}_{1-t}(x, y)$, which is the kernel of $\text{Op}_{1-t}(\bar{a})$. $\square$

We next observe that the formulas (4.1.1)–(4.1.3) make sense if a is merely a distribution:

THEOREM 4.2 (Distributional symbols). *If $a \in \mathcal{S}'$, then $\text{Op}_t(a)$ can be defined as an operator mapping $\mathcal{S}$ to $\mathcal{S}'$; and furthermore*

$$\text{Op}_t(a) : \mathcal{S} \rightarrow \mathcal{S}' \quad (0 \leq t \leq 1)$$

is continuous.

Proof. The formula for the distributional kernel K_t of $\text{Op}_t(a)$ given in the proof of Theorem 4.1 can be interpreted in the distributional sense if $a \in \mathcal{S}'$. This shows that $K_t \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^n)$. Hence $\text{Op}_t(a)$ is well defined as an operator from $\mathcal{S}$ to $\mathcal{S}'$. So if $u, v \in \mathcal{S}$, then

$$(\text{Op}_t(a)u)(v) := K_t(v \otimes u), \quad (v \times u)(x, y) := v(x)u(y). \quad \square$$

4.2. QUANTIZATION FORMULAS

Exact computations for quantization can be carried out only in certain cases, but these are important. For future reference, we collect in this section various explicit calculations of $\text{Op}_t(a)$ and especially $a^w(x, hD)$.

4.2.1. Symbols depending only on x . A first simple, but not entirely trivial, case is when a does not depend upon ξ :

THEOREM 4.3 (Quantizing symbols of x only). *If $a(x, \xi) = a(x)$, then*

$$(4.2.1) \quad \text{Op}_t(a) = M_a \quad (0 \leq t \leq 1), \quad M_a u(x) := a(x)u(x).$$

Proof. Let $u \in \mathcal{S}$ and compute the derivative with respect to t :

$$\begin{aligned} \partial_t \text{Op}_t(a)u &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} \langle \partial a(tx + (1-t)y), x-y \rangle u(y) dy d\xi \\ &= \frac{h}{i(2\pi h)^n} \int_{\mathbb{R}^n} \text{div}_\xi \left(\int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} \partial a(tx + (1-t)y) u(y) dy \right) d\xi \\ &= \frac{h}{i(2\pi h)^n} \int_{\mathbb{R}^n} \text{div}_\xi \left(e^{\frac{i}{h}\langle x, \xi \rangle} \hat{\alpha}(\xi) \right) d\xi \end{aligned}$$

for $\alpha(y) := \partial a(tx + (1-t)y)u(y)$. Here “div” denotes the divergence. Since $\hat{\alpha}(\xi) \rightarrow 0$ rapidly as $|\xi| \rightarrow \infty$, the last expression vanishes. Consequently for all $0 \leq t \leq 1$, $\text{Op}_t(a)u = \text{Op}_1(a)u = au$. $\square$

4.2.2. Linear symbols. The formulas (4.1.6) and (4.2.1) immediately imply

THEOREM 4.4 (Quantizing linear symbols). *Let l be a linear symbol of the form*

$$(4.2.2) \quad l(x, \xi) := \langle x^*, x \rangle + \langle \xi^*, \xi \rangle$$

for $(x^*, \xi^*) \in \mathbb{R}^{2n}$. Then

$$(4.2.3) \quad \text{Op}_t(l) = \langle x^*, x \rangle + \langle \xi^*, hD \rangle \quad (0 \leq t \leq 1).$$

NOTATION. In view of this result, we hereafter write

$$(4.2.4) \quad l(x, hD) = l^w(x, hD) = \langle x^*, x \rangle + \langle \xi^*, hD \rangle.$$

We can also compute explicitly the quantization of symbols linear in ξ , but nonlinear in x :

THEOREM 4.5 (Symbols linear in ξ). *Assume that*

$$c = (c_1(x), \dots, c_n(x))$$

does not depend on ξ . Then

$$(4.2.5) \quad \langle c, hD \rangle^w = \frac{h}{2} \sum_{j=1}^n (D_{x_j} c_j + c_j D_{x_j}).$$

The notation means $(D_{x_j} c_j)u = D_{x_j}(c_j u)$.

Proof. We calculate that

$$\begin{aligned} \langle c, hD \rangle^w u &= \frac{1}{(2\pi h)^n} \sum_{j=1}^n \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} c_j \left(\frac{x+y}{2} \right) \xi_j e^{\frac{i}{h} \langle x-y, \xi \rangle} u(y) d\xi dy \\ &= -\frac{1}{(2\pi h)^n} \sum_{j=1}^n \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} c_j \left(\frac{x+y}{2} \right) h D_{y_j} \left(e^{\frac{i}{h} \langle x-y, \xi \rangle} \right) u(y) d\xi dy \\ &= \frac{1}{(2\pi h)^n} \sum_{j=1}^n \frac{h}{2i} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} (\partial_{x_j} c_j) \left(\frac{x+y}{2} \right) e^{\frac{i}{h} \langle x-y, \xi \rangle} u(y) d\xi dy \\ &\quad + \frac{1}{(2\pi h)^n} \sum_{j=1}^n \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} c_j \left(\frac{x+y}{2}, \xi \right) e^{\frac{i}{h} \langle x-y, \xi \rangle} h D_{x_j} u(y) d\xi dy \\ &= \frac{h}{2i} \sum_{j=1}^n (\partial_{x_j} c_j)^w u + h \sum_{j=1}^n (c_j)^w D_{x_j} u \\ &= -\frac{ih}{2} \sum_{j=1}^n (\partial_{x_j} c_j) u + h \sum_{j=1}^n c_j D_{x_j} u, \end{aligned}$$

according (4.2.1). Consequently,

$$\langle c, hD \rangle^w = h \sum_{j=1}^n (c_j D_{x_j} - \frac{i}{2} \partial_{x_j} c_j) = \frac{h}{2} \sum_{j=1}^n (D_{x_j} c_j + c_j D_{x_j}). \quad \square$$

EXAMPLE. The case $c(x) = x$ gives

$$\langle x, hD \rangle^w = \frac{h}{2} \sum_{j=1}^n (D_{x_j} x_j + x_j D_{x_j}),$$

in agreement with our previous calculation (4.1.8). $\square$

4.2.3. Commutators. The Weyl quantizations of derivatives of a symbol can be characterized as appropriate commutators:

THEOREM 4.6 (Commutators and derivatives).

$$(4.2.6) \quad (D_{x_j} a)^w = [D_{x_j}, a^w]$$

and

$$(4.2.7) \quad h(D_{\xi_j} a)^w = -[x_j, a^w]$$

for $j = 1, \dots, n$.

Proof. We compute for $u \in \mathcal{S}$ that

$$\begin{aligned} (D_{x_j} a)^w u &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} D_{x_j} a \left(\frac{x+y}{2}, \xi \right) e^{\frac{i}{h} \langle x-y, \xi \rangle} u(y) d\xi dy \\ &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} (D_{x_j} + D_{y_j}) \left(a \left(\frac{x+y}{2}, \xi \right) \right) e^{\frac{i}{h} \langle x-y, \xi \rangle} u(y) d\xi dy \\ &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} D_{x_j} \left(a \left(\frac{x+y}{2}, \xi \right) \right) e^{\frac{i}{h} \langle x-y, \xi \rangle} u(y) d\xi dy \\ &\quad + \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a \left(\frac{x+y}{2}, \xi \right) e^{\frac{i}{h} \langle x-y, \xi \rangle} \left(\frac{\xi_j}{h} - D_{y_j} \right) u(y) d\xi dy \\ &= D_{x_j} (a^w u) - a^w (D_{x_j} u) \\ &= [D_{x_j}, a^w] u. \end{aligned}$$

Similarly,

$$\begin{aligned} h(D_{\xi_j} a)^w u &= \frac{h}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} D_{\xi_j} a \left(\frac{x+y}{2}, \xi \right) e^{\frac{i}{h} \langle x-y, \xi \rangle} u(y) d\xi dy \\ &= -\frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a \left(\frac{x+y}{2}, \xi \right) h D_{\xi_j} \left(e^{\frac{i}{h} \langle x-y, \xi \rangle} \right) u(y) d\xi dy \\ &= -\frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a \left(\frac{x+y}{2}, \xi \right) e^{\frac{i}{h} \langle x-y, \xi \rangle} (x_j - y_j) u(y) d\xi dy \\ &= -[x_j, a^w] u. \end{aligned} \quad \square$$

4.2.4. Exponentials of linear symbols. We will later need the Weyl quantization of complex exponentials of linear symbols:

THEOREM 4.7 (Quantizing exponentials of linear symbols).

(i) *For each linear symbol l of the form (4.2.2) we have the identity*

$$(4.2.8) \quad \left(e^{\frac{i}{h}l}\right)^w(x, hD) = e^{\frac{i}{h}l(x, hD)},$$

where

$$(4.2.9) \quad e^{\frac{i}{h}l(x, hD)}u(x) := e^{\frac{i}{h}\langle x^*, x \rangle + \frac{i}{2h}\langle x^*, \xi^* \rangle}u(x + \xi^*).$$

(ii) *If $l, m \in \mathbb{R}^{2n}$, then*

$$(4.2.10) \quad e^{\frac{i}{h}l(x, hD)}e^{\frac{i}{h}m(x, hD)} = e^{\frac{i}{2h}\sigma(l, m)}e^{\frac{i}{h}(l+m)(x, hD)},$$

where σ is the symplectic form in $\mathbb{R}^n \times \mathbb{R}^n$ given by (2.2.1).

Proof. 1. Consider for $u \in \mathcal{S}$ the PDE

$$\begin{cases} ih\partial_t v + l(x, hD)v = 0 & (t \in \mathbb{R}) \\ v(0) = u. \end{cases}$$

Its unique solution is denoted by

$$v(x, t) = e^{\frac{it}{h}l(x, hD)}u,$$

this formula defining the operators $e^{\frac{it}{h}l(x, hD)}$ for $t \in \mathbb{R}$. But we can check by a direct calculation using (4.2.4) that

$$v(x, t) = e^{\frac{it}{h}\langle x^*, x \rangle + \frac{it^2}{2h}\langle x^*, \xi^* \rangle}u(x + t\xi^*);$$

and therefore (4.2.9) holds.

2. Furthermore,

$$\begin{aligned} (e^{\frac{il}{h}})^w u &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} e^{\frac{i}{h}(\langle \xi^*, \xi \rangle + \langle x^*, \frac{x+y}{2} \rangle)} u(y) dy d\xi \\ &= \frac{e^{\frac{i}{2h}\langle x^*, x \rangle}}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y+\xi^*, \xi \rangle} \left(e^{\frac{i}{2h}\langle x^*, y \rangle} u(y) \right) dy d\xi \\ &= \frac{e^{\frac{i}{2h}\langle x^*, x \rangle}}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} \left(e^{\frac{i}{2h}\langle x^*, y+\xi^* \rangle} u(y + \xi^*) \right) dy d\xi \\ &= e^{\frac{i}{h}\langle x^*, x \rangle + \frac{i}{2h}\langle x^*, \xi^* \rangle} u(x + \xi^*), \end{aligned}$$

since

$$\delta_{\{y=x\}} = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} d\xi \quad \text{in } \mathcal{S}'$$

according to (3.3.3). This proves (4.2.8).

3. Suppose $l(x, \xi) = \langle x_1^*, x \rangle + \langle \xi_1^*, \xi \rangle$ and $m(x, \xi) = \langle x_2^*, x \rangle + \langle \xi_2^*, \xi \rangle$. According to (4.2.9),

$$e^{\frac{i}{h}m(x, hD)}u(x) = e^{\frac{i}{h}\langle x_2^*, x \rangle + \frac{i}{2h}\langle x_2^*, \xi_2^* \rangle}u(x + \xi_2^*);$$

and consequently

$$\begin{aligned} e^{\frac{i}{h}l(x, hD)}e^{\frac{i}{h}m(x, hD)}u(x) \\ = e^{\frac{i}{h}\langle x_1^*, x \rangle + \frac{i}{2h}\langle x_1^*, \xi_1^* \rangle}e^{\frac{i}{h}\langle x_2^*, x + \xi_1^* \rangle + \frac{i}{2h}\langle x_2^*, \xi_2^* \rangle}u(x + \xi_1^* + \xi_2^*). \end{aligned}$$

Furthermore, (4.2.9) also implies that

$$e^{\frac{i}{h}(l+m)(x, hD)}u(x) = e^{\frac{i}{h}\langle x_1^* + x_2^*, x \rangle + \frac{i}{2h}\langle x_1^* + x_2^*, \xi_1^* + \xi_2^* \rangle}u(x + \xi_1^* + \xi_2^*).$$

Using the formula above, we therefore compute

$$e^{\frac{i}{h}(l+m)(x, hD)}u(x) = e^{\frac{i}{2h}(\langle x_1^*, \xi_2^* \rangle - \langle x_2^*, \xi_1^* \rangle)}e^{\frac{i}{h}l(x, hD)}e^{\frac{i}{h}m(x, hD)}u(x).$$

This confirms (4.2.10), since $\sigma(l, m) = \langle \xi_1^*, x_2^* \rangle - \langle x_1^*, \xi_2^* \rangle$. $\square$

4.2.5. Exponentials of quadratic symbols. We next record some important integral representation formulas for the quantization of certain quadratic exponentials:

THEOREM 4.8 (Quantizing quadratic exponentials).

(i) Let Q denote a nonsingular, symmetric, $n \times n$ matrix. Then

$$(4.2.11) \quad e^{\frac{ih}{2}\langle QD, D \rangle}u(x) = \frac{|\det Q|^{-\frac{1}{2}}}{(2\pi h)^{\frac{n}{2}}}e^{\frac{i\pi}{4}\operatorname{sgn} Q} \int_{\mathbb{R}^n} e^{-\frac{i}{2h}\langle Q^{-1}y, y \rangle}u(x + y) dy$$

for $u \in \mathcal{S}(\mathbb{R}^n)$.

(ii) In particular, if $u \in \mathcal{S}(\mathbb{R}^{2n})$, $u = u(x, y)$, then

$$\begin{aligned} (4.2.12) \quad e^{ih\langle D_x, D_y \rangle}u(x, y) \\ = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-\frac{i}{h}\langle x_1, y_1 \rangle}u(x + x_1, y + y_1) dx_1 dy_1. \end{aligned}$$

(iii) Suppose that $u \in \mathcal{S}(\mathbb{R}^{4n})$, $u = u(z, w)$. Then

$$\begin{aligned} (4.2.13) \quad e^{ih\sigma(D_z, D_w)}u(z, w) \\ = \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{-\frac{i}{h}\sigma(z_1, w_1)}u(z + z_1, w + w_1) dz_1 dw_1. \end{aligned}$$

Proof. 1. Observe first that Theorem 3.7 gives

$$\begin{aligned} \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle w, \xi \rangle} e^{\frac{i}{2h}\langle Q\xi, \xi \rangle} d\xi &= \mathcal{F}_h^{-1}(e^{\frac{i}{2h}\langle Q\xi, \xi \rangle})(w) \\ &= \frac{|\det Q|^{-\frac{1}{2}}}{(2\pi h)^{\frac{n}{2}}} e^{\frac{i\pi}{4} \operatorname{sgn} Q} e^{-\frac{i}{2h}\langle Q^{-1}w, w \rangle}. \end{aligned}$$

Therefore

$$\begin{aligned} e^{\frac{ih}{2}\langle QD, D \rangle} u(x) &= e^{\frac{i}{2h}\langle QhD, hD \rangle} u(x) \\ &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} e^{\frac{i}{2h}\langle Q\xi, \xi \rangle} u(y) dy d\xi \\ &= \frac{|\det Q|^{-\frac{1}{2}}}{(2\pi h)^{\frac{n}{2}}} e^{\frac{i\pi}{4} \operatorname{sgn} Q} \int_{\mathbb{R}^n} e^{-\frac{i}{2h}\langle Q^{-1}(x-y), x-y \rangle} u(y) dy \\ &= \frac{|\det Q|^{-\frac{1}{2}}}{(2\pi h)^{\frac{n}{2}}} e^{\frac{i\pi}{4} \operatorname{sgn} Q} \int_{\mathbb{R}^n} e^{-\frac{i}{2h}\langle Q^{-1}y, y \rangle} u(x+y) dy. \end{aligned}$$

2. Assertion (4.2.12) is a special case of (4.2.11), obtained by replacing n by $2n$ and taking

$$Q := \begin{pmatrix} O & I \\ I & O \end{pmatrix}.$$

See the proof of Theorem 3.17(i).

3. Similarly, assertion (4.2.13) is a special case of (4.2.11) obtained by replacing n by $4n$ and taking

$$Q := \begin{pmatrix} O & -J \\ J & O \end{pmatrix}.$$

See the proof of Theorem 3.17(ii). □

4.2.6. Conjugation by Fourier transform.

THEOREM 4.9 (Conjugation and Fourier transform). *We have*

$$(4.2.14) \quad \mathcal{F}_h^{-1} a^w(x, hD) \mathcal{F}_h = a^w(hD, -x).$$

Note that $\tilde{a}(x, \xi) := a(\xi, -x)$ is the pull-back of a under J . In Section 11.3 we will generalize this insight and in particular interpret (4.2.14) as saying that *the semiclassical Fourier transform $\mathcal{F}_h$ quantizes the linear symplectic mapping J .*

Proof. We observe that the Schwartz kernel of $\mathcal{F}_h^{-1}a^w\mathcal{F}_h$ is

$$K_h(x, y) = \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\langle x', x \rangle + \langle x' - y', \zeta \rangle - \langle y', y \rangle)} a\left(\frac{x' + y'}{2}, \zeta\right) dy' dx' d\zeta.$$

The change of variables $x' = x'$, $z = \frac{x' + y'}{2}$ shows

$$K_h(x, y) = \frac{1}{(2\pi h)^{2n}} \frac{1}{2^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\Phi(x', z, \zeta, y, x)} a(z, \zeta) dx' dz d\zeta,$$

where

$$\Phi(x', z, \zeta, y, x) := 2(\langle x', \zeta + \frac{x+y}{2} \rangle - \langle z, y + \zeta \rangle).$$

Also note that

$$\frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{2i}{h}\langle x', \zeta + \frac{x+y}{2} \rangle} dx' = 2^n \delta\left(\zeta + \frac{x+y}{2}\right).$$

Hence

$$K_h(x, y) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x - y, z \rangle} a\left(z, -\left(\frac{x+y}{2}\right)\right) dz,$$

the Schwartz kernel of $\tilde{a}^w(x, hD)$ for $\tilde{a}(x, \xi) := a(\xi, -x)$. $\square$

4.3. COMPOSITION, ASYMPTOTIC EXPANSIONS

We now commence a careful study of the properties of the quantized operators defined in Section 4.1, especially the Weyl quantization.

4.3.1. Composing symbols. We next establish the fundamental formula

$$a^w b^w = (a \# b)^w,$$

along with a recipe for computing the new symbol $a \# b$. The plan is to represent the Weyl quantization of a general symbol in terms of the quantizations of complex exponentials of linear symbols. Remember that linear symbols have the form

$$l(x, \xi) := \langle x^*, x \rangle + \langle \xi^*, \xi \rangle$$

for $(x^*, \xi^*) \in \mathbb{R}^{2n}$. To simplify the calculations, *we will sometimes identify this linear symbol l with the point (x^*, ξ^*) .*

LEMMA 4.10 (Fourier decomposition of a^w).

(i) *Define*

$$\hat{a}(l) := \int_{\mathbb{R}^{2n}} e^{-\frac{i}{h}l(x, \xi)} a(x, \xi) dx d\xi$$

for $a \in \mathcal{S}$ and $l \in \mathbb{R}^{2n}$. Then

$$(4.3.1) \quad a^w(x, hD) = \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} \hat{a}(l) e^{\frac{i}{h} l(x, hD)} dl.$$

(ii) If $a \in \mathcal{S}'$, then the decomposition formula (4.3.1) holds in the sense of tempered distributions.

Assertion (ii) means that if $u, v \in \mathcal{S}$, then $\langle e^{\frac{i}{h} l(x, hD)} u, v \rangle \in \mathcal{S}$ as a function of $l = (x^*, \xi^*) \in \mathbb{R}^{2n}$ and

$$(4.3.2) \quad \langle a^w(x, hD) u, v \rangle = \frac{1}{(2\pi h)^{2n}} \hat{a} \left(\langle e^{\frac{i}{h} (\cdot)(x, hD)} u, v \rangle \right).$$

Proof. 1. For $a \in \mathcal{S}$, the Fourier inversion formula implies

$$(4.3.3) \quad a(x, \xi) = \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} e^{\frac{i}{h} l(x, \xi)} \hat{a}(l) dl;$$

and therefore (4.3.1) follows from Theorem 4.7.

2. To see the validity of (4.3.2) for $a \in \mathcal{S}'$, we only need to check that

$$\langle e^{\frac{i}{h} l(x, hD)} u, v \rangle = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h} (l(x+y/2, \xi) + \langle x-y, \xi \rangle)} u(y) \bar{v}(x) dy d\xi dx$$

lies in $\mathcal{S}$ as a function of l . We leave the verification as an exercise. $\square$

NOTATION. We will hereafter write

$$(4.3.4) \quad A(D) := \frac{1}{2} \sigma(D_x, D_\xi, D_y, D_\eta).$$

Now we show for the Weyl quantization that *the product of two pseudodifferential operators is a pseudodifferential operator.*

THEOREM 4.11 (Composition for Weyl quantization).

(i) Suppose that $a, b \in \mathcal{S}$. Then

$$(4.3.5) \quad a^w(x, hD) b^w(x, hD) = (a \# b)^w(x, hD)$$

for the symbol

$$(4.3.6) \quad a \# b(x, \xi) := e^{ihA(D)} (a(x, \xi) b(y, \eta)) \Big|_{\substack{y=x \\ \eta=\xi}}.$$

(ii) We have the integral representation formula

$$(4.3.7) \quad a \# b(x, \xi) = \frac{1}{(\pi h)^{2n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{-\frac{2i}{h} \sigma(w_1, w_2)} a(z + w_1) b(z + w_2) dw_1 dw_2,$$

where $z = (x, \xi)$.

Proof. 1. We have the representation formula (4.3.1) and likewise

$$b^w(x, hD) = \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} \hat{b}(m) e^{\frac{i}{h} m(x, hD)} dm.$$

Theorem 4.7(ii) lets us next compute

$$\begin{aligned} a^w(x, hD) b^w(x, hD) &= \frac{1}{(2\pi h)^{4n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} \hat{a}(l) \hat{b}(m) e^{\frac{i}{h} l(x, hD)} e^{\frac{i}{h} m(x, hD)} dm dl \\ &= \frac{1}{(2\pi h)^{4n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} \hat{a}(l) \hat{b}(m) e^{\frac{i}{2h} \sigma(l, m)} e^{\frac{i}{h} (l+m)(x, hD)} dm dl \\ &= \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} \hat{c}_1(r) e^{\frac{i}{h} r(x, hD)} dr \end{aligned}$$

for

$$(4.3.8) \quad \hat{c}_1(r) := \frac{1}{(2\pi h)^{2n}} \int_{\{l+m=r\}} \hat{a}(l) \hat{b}(m) e^{\frac{i\sigma(l, m)}{2h}} dl.$$

To get this, we changed variables by setting $r = m + l$.

2. We will show that $c_1 = c$, where $\hat{c}_1$ is defined by (4.3.8) and c is defined by the right-hand side of (4.3.6). We first simplify notation by writing $z = (x, \xi)$, $w = (y, \eta)$. Then

$$c(z) = e^{\frac{i\hbar}{2} \sigma(D_z, D_w)} a(z) b(w)|_{w=z} = e^{\frac{i}{2h} \sigma(hD_z, hD_w)} a(z) b(w)|_{w=z}$$

and

$$\begin{aligned} a(z) &= \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} e^{\frac{i}{h} l(z)} \hat{a}(l) dl, \\ b(w) &= \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} e^{\frac{i}{h} m(w)} \hat{b}(m) dm. \end{aligned}$$

Furthermore, since $l(z) = \langle l, z \rangle$ and $m(w) = \langle m, w \rangle$, we have

$$e^{\frac{i}{2h} \sigma(hD_z, hD_w)} e^{\frac{i}{h} (l(z) + m(w))} = e^{\frac{i}{h} (l(z) + m(w)) + \frac{i}{2h} \sigma(l, m)}.$$

Consequently

$$\begin{aligned} c(z) &= \frac{1}{(2\pi h)^{4n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{\frac{i}{2h} \sigma(hD_z, hD_w)} e^{\frac{i}{h} (l(z) + m(w))} \Big|_{z=w} \hat{a}(l) \hat{b}(m) dldm \\ &= \frac{1}{(2\pi h)^{4n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{\frac{i}{h} (l(z) + m(z)) + \frac{i}{2h} \sigma(l, m)} \hat{a}(l) \hat{b}(m) dldm. \end{aligned}$$

The semiclassical Fourier transform of c is therefore

$$\frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} \left(\frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^n} e^{\frac{i}{h} (l+m-r)(z)} dz \right) e^{\frac{i}{2h} \sigma(l, m)} \hat{a}(l) \hat{b}(m) dldm.$$

According to (3.3.3), the term inside the parentheses is $\delta_{\{l+m=r\}}$ in $\mathcal{S}'$. Thus, in view of (4.3.8), the foregoing equals

$$\frac{1}{(2\pi h)^{2n}} \int_{\{l+m=r\}} e^{\frac{i}{2h}\sigma(l,m)} \hat{a}(l) \hat{b}(m) dl = \hat{c}_1(r).$$

3. Formula (4.3.7) follows from Theorem 4.8(iii), with $h/2$ replacing h . $\square$

4.3.2. Asymptotics. We next apply the stationary phase to derive a useful asymptotic expansion of $a\#b$. Remember the definition (4.3.4) of the operator $A(D)$.

THEOREM 4.12 (Semiclassical expansions). *Assume $a, b \in \mathcal{S}$.*

(i) *We have for $N = 0, 1, \dots$,*

$$(4.3.9) \quad a\#b(x, \xi) = \sum_{k=0}^N \frac{i^k h^k}{k!} A(D)^k (a(x, \xi) b(y, \eta)) \Big|_{\substack{y=x \\ \eta=\xi}} + O_{\mathcal{S}}(h^{N+1})$$

as $h \rightarrow 0$, where $A(D)$ is given by (4.3.4).

(ii) *In particular,*

$$(4.3.10) \quad a\#b = ab + \frac{h}{2i} \{a, b\} + O_{\mathcal{S}}(h^2)$$

and

$$(4.3.11) \quad [a^w(x, hD), b^w(x, hD)] = \frac{h}{i} \{a, b\}^w(x, hD) + O_{\mathcal{S}}(h^3).$$

(iii) *If $\text{spt}(a) \cap \text{spt}(b) = \emptyset$, then*

$$(4.3.12) \quad a\#b = O_{\mathcal{S}}(h^\infty).$$

REMARKS. (i) The important formula (4.3.11) is an instance of the classical/quantum correspondence principle mentioned in Chapter 1, that an assertion about classical dynamics, and so the Poisson bracket $\{\cdot, \cdot\}$, entails at the quantum level the commutator $[\cdot, \cdot]$.

(ii) Formula (4.3.11) also shows that *the commutator of two pseudodifferential operators is of order h* . The error term $O(h^3)$ is specific to the Weyl quantization. Using Theorem 4.14, we see that

$$[a(x, hD), b(x, hD)] = \frac{h}{i} \{a, b\}(x, hD) + O_{\mathcal{S}}(h^2).$$

(iii) The notation $\varphi = O_{\mathcal{S}}(h^k)$ means that for all multiindices α, β

$$|\varphi|_{\alpha, \beta} := \sup_{\mathbb{R}^n} |x^\alpha \partial^\beta \varphi| \leq C_{\alpha, \beta} h^k$$

as $h \rightarrow 0$. □

Proof. 1. To prove (4.3.9), we apply the stationary phase Theorem 3.17(ii), with $h/2$ replacing h and $-\sigma$ replacing σ , to the integral formula (4.3.7). However, some care is needed to show that remainders are in $O_{\mathcal{S}}(h^{N+1})$.

For that we proceed directly as in the proof of the stationary phase theorem. In (4.3.6) we write

$$e^{ihA(D)} = \sum_{k=0}^N \frac{i^k h^k}{k!} A(D)^k + \frac{i^{N+1} h^{N+1}}{N!} \int_0^1 (1-t)^N e^{ithA(D)} A(D)^{N+1} dt.$$

The Fourier multiplier $e^{ithA(D)} A(D)^{N+1}$ is defined by multiplication by $\mathcal{F}^{-1} e^{itA(\zeta)} A(\zeta)^{N+1} \mathcal{F}$ where $\mathcal{F}$ is the (usual) Fourier transform, $\zeta \in \mathbb{R}^{4n}$. Hence it maps $\mathcal{S}(\mathbb{R}^{4n}) \rightarrow \mathcal{S}(\mathbb{R}^{4n})$ uniformly in h and t proving the estimate on the remainder in (4.3.9).

2. Next, compute

$$\begin{aligned} a \# b &= ab + ihA(D)(a(x, \xi)b(y, \eta)) \Big|_{\eta=\xi}^{y=x} + O(h^2) \\ &= ab + \frac{ih}{2} (\langle D_\xi a, D_y b \rangle - \langle D_x a, D_\eta b \rangle) \Big|_{\eta=\xi}^{y=x} + O(h^2) \\ &= ab + \frac{h}{2i} (\langle \partial_\xi a, \partial_x b \rangle - \langle \partial_x a, \partial_\xi b \rangle) + O(h^2) \\ &= ab + \frac{h}{2i} \{a, b\} + O(h^2). \end{aligned}$$

Similarly,

$$\begin{aligned} [a^w, b^w] &= a^w b^w - b^w a^w = (a \# b - b \# a)^w \\ &= \left(ab + \frac{h}{2i} \{a, b\} + \frac{1}{2} h^2 A(D)^2 (ab) \Big|_{\eta=\xi}^{y=x} + O_{\mathcal{S}}(h^3) \right)^w \\ &\quad - \left(ba + \frac{h}{2i} \{b, a\} + \frac{1}{2} h^2 A(D)^2 (ba) \Big|_{\eta=\xi}^{y=x} + O_{\mathcal{S}}(h^3) \right)^w \\ &= \frac{h}{i} \{a, b\}^w + O_{\mathcal{S}}(h^3)^w, \end{aligned}$$

since

$$(A(D)^2(a(x, \xi)b(y, \eta)) - A(D)^2(b(x, \xi)a(y, \eta))) \Big|_{\eta=\xi}^{y=x} = 0.$$

3. If $\text{spt}(a) \cap \text{spt}(b) = \emptyset$, each term in the expansion (4.3.9) vanishes. □

EXAMPLE: Symbols linear in ξ . Let $a = c_j(x)$ and $b = \xi_j$. Then

$$a^w b^w = (a \# b)^w = (ab)^w + \frac{h}{2i} \{a, b\}^w,$$

since $D^\alpha b = 0$ for $|\alpha| \geq 2$. Summing $j = 1, \dots, n$, we see that

$$\langle c, hD \rangle^w = h \sum_{j=1}^n (c_j D_{x_j} - \frac{i}{2} \partial_{x_j} c_j) = \frac{h}{2} \sum_{j=1}^n (D_{x_j} c_j + c_j D_{x_j}),$$

where $c = (c_1, \dots, c_n)$. This agrees with our previous calculation (4.2.5). $\square$

4.3.3. Transforming between different quantizations. We record an interesting conversion formula:

THEOREM 4.13 (Changing quantizations). *If*

$$A = \text{Op}_t(a_t) \quad (0 \leq t \leq 1),$$

then

$$(4.3.13) \quad a_t(x, \xi) = e^{i(t-s)h\langle D_x, D_\xi \rangle} a_s(x, \xi).$$

Proof. The decomposition formula (4.3.1) implies

$$\text{Op}_t(a_t) = \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^{2n}} \hat{a}_t(l) \text{Op}_t(e^{\frac{i}{h}l}) dl.$$

Denoting the Fourier transform used there by $\mathcal{F}_h$, we have

$$\mathcal{F}_h \left(e^{i(t-s)h\langle D_x, D_\xi \rangle} a_s(x, \xi) \right) (l) = e^{\frac{i}{h}(t-s)\langle x^*, \xi^* \rangle} \mathcal{F}_h a_s(l);$$

and as before we identify $l = (x^*, \xi^*) \in \mathbb{R}^{2n}$ with the linear function $l(x, \xi) = \langle x^*, x \rangle + \langle \xi^*, \xi \rangle$.

The theorem is a consequence of the identity

$$\text{Op}_t \left(e^{\frac{i}{h}l(x, \xi)} \right) = e^{\frac{i}{h}(s-t)\langle x^*, \xi^* \rangle} \text{Op}_s \left(e^{\frac{i}{h}l(x, \xi)} \right),$$

which can be checked by calculations similar to those in the proof of Theorem 4.7:

$$\text{Op}_t \left(e^{\frac{i}{h}l(x, \xi)} \right) u(x) = e^{\frac{i}{h}\langle x, x^* \rangle + \frac{i}{h}(1-t)\langle x^*, \xi^* \rangle} u(x + \xi^*). \quad \square$$

4.3.4. Standard quantization. Next we replace Weyl ($t = \frac{1}{2}$) by standard ($t = 1$) quantization in our formulas. The proofs are simpler.

THEOREM 4.14 (Formulas for standard quantization).

(i) Let $a, b \in \mathcal{S}$. Then

$$a(x, hD)b(x, hD) = c(x, hD)$$

for the symbol

$$(4.3.14) \quad c(x, \xi) = e^{ih\langle D_\xi, D_y \rangle} (a(x, \xi)b(y, \eta)) \Big|_{\substack{y=x \\ \eta=\xi}}.$$

(ii) We have the integral representation formula

$$(4.3.15) \quad c(x, \xi) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-\frac{i}{h}\langle x_1, \xi_1 \rangle} a(x, \xi + \xi_1) b(x + x_1, \xi) dx_1 d\xi_1.$$

(iii) For each $N = 0, 1, \dots$,

$$(4.3.16) \quad c(x, \xi) = \sum_{k=0}^N \frac{h^k}{k!} (i\langle D_\xi, D_y \rangle)^k (a(x, \xi)b(y, \eta)) \Big|_{\substack{y=x \\ \eta=\xi}} + O_{\mathcal{S}}(h^{N+1})$$

as $h \rightarrow 0$.

(iv) If $a \in \mathcal{S}$, then

$$a(x, hD)^* = b(x, hD),$$

for

$$(4.3.17) \quad b(x, \xi) := e^{ih\langle D_x, D_\xi \rangle} \bar{a}(x, \xi).$$

Proof. 1. Let $u \in \mathcal{S}$. Then

$$\begin{aligned} & a(x, hD)b(x, hD)u(x) \\ &= \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\langle x, \eta \rangle + \langle y, \xi - \eta \rangle)} a(x, \eta) b(y, \xi) \hat{u}(\xi) dy d\eta d\xi \\ &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} c(x, \xi) e^{\frac{i}{h}\langle x, \xi \rangle} \hat{u}(\xi) d\xi \\ &= c(x, hD)u(x), \end{aligned}$$

for

$$c(x, \xi) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-\frac{i}{h}\langle x-y, \xi-\eta \rangle} a(x, \eta) b(y, \xi) dy d\eta.$$

Change variables by putting $x_1 = y - x$, $\xi_1 = \eta - \xi$ to rewrite c in the form (4.3.15). Then (4.3.14) is a consequence of Theorem 4.8(ii). Finally,

the stationary phase Theorem 3.17(i) provides the asymptotic expansion (4.3.16). $\square$

2. We recall from (4.1.12) that $a(x, hD)^* = \text{Op}_1(a)^* = \text{Op}_0(\bar{a})$. Now invoke (4.3.13), to write

$$\text{Op}_0(\bar{a}) = \text{Op}_1(b),$$

the symbol b defined by (4.3.17). $\square$

4.4. SYMBOL CLASSES

We next extend our calculus to symbols which can depend on the parameter h and which can have varied behavior, in terms of growth and decay, as $(x, \xi) \rightarrow \infty$.

4.4.1. Order functions and symbol classes.

DEFINITION. A measurable function $m : \mathbb{R}^{2n} \rightarrow (0, \infty)$ is called an *order function* if there exist constants C, N such that

$$(4.4.1) \quad m(w) \leq C \langle z - w \rangle^N m(z)$$

for all $w, z \in \mathbb{R}^{2n}$.

EXAMPLES. Standard examples are

$$m(z) \equiv 1, \quad m(z) = \langle z \rangle = (1 + |z|^2)^{1/2}.$$

We also check that for any $a, b \in \mathbb{R}$

$$m(z) = \langle x \rangle^a \langle \xi \rangle^b$$

are order functions, where $z = (x, \xi)$. Observe also that if m_1, m_2 are order functions, so is $m_1 m_2$. $\square$

DEFINITIONS.

(i) Given an order function m on $\mathbb{R}^{2n}$, we define the corresponding class of symbols:

$$(4.4.2) \quad S(m) := \{a \in C^\infty \mid \text{for each multiindex } \alpha \\ \text{there exists a constant } C_\alpha \text{ so that } |\partial^\alpha a| \leq C_\alpha m\}.$$

(ii) We as well define

$$(4.4.3) \quad S_\delta(m) := \{a \in C^\infty \mid |\partial^\alpha a| \leq C_\alpha h^{-\delta|\alpha|} m \text{ for all multiindices } \alpha\}.$$

REMARKS. (i) Symbols $a = a(x, \xi)$ in $S(m)$ are allowed to depend upon h , although this dependence is usually not reflected in our notation. Symbols in $S_\delta(m)$ can likewise depend on h , although again our notation will mostly not show this explicitly.

(ii) The space of Schwartz functions, $\mathcal{S} \subset S(m)$, is dense in $S(m)$ in the topology of $S(\langle x, \xi \rangle^\epsilon m)$ for any $\epsilon > 0$. We leave this as an exercise to the reader, noting that this is similar to the density of $\mathcal{S}(\mathbb{R}^n) \subset L^\infty(\mathbb{R}^n)$ in the topology of $\langle x \rangle^\epsilon L^\infty(\mathbb{R}^n)$ for any $\epsilon > 0$.

(iii) If $a \in S(m)$ or $a \in S_\delta(m)$ depends on h , we require that the constants C_α in the definitions be uniform for $0 < h \leq h_0$ for some number $h_0 > 0$.

(iv) The spaces $S_\delta(m)$ will appear naturally in later applications, for example the sharp Gårding inequality (Section 4.7) and the Ehrenfest time theorem (Section 11.4). The index $\delta > 0$ allows for increasing singularity of the higher derivatives. $\square$

NOTATION. If the order function is the constant function $m \equiv 1$, we will usually not write it. Thus $S := S(1)$, $S_\delta := S_\delta(1)$; that is,

$$(4.4.4) \quad S = \{a \in C^\infty(\mathbb{R}^{2n}) \mid |\partial^\alpha a| \leq C_\alpha \text{ for all } \alpha\},$$

$$(4.4.5) \quad S_\delta = \{a \in C^\infty(\mathbb{R}^{2n}) \mid |\partial^\alpha a| \leq C_\alpha h^{-\delta|\alpha|} \text{ for all } \alpha\}.$$

REMARK: Critical and subcritical values of δ . Note that if $a \in S_\delta$, then

$$(4.4.6) \quad |\partial^\alpha a_h| = h^{\frac{|\alpha|}{2}} |\partial^\alpha a| \leq C_\alpha h^{|\alpha|(\frac{1}{2}-\delta)}$$

for each multiindex α , where a_h is given by the standard rescaling (4.1.11). If $\delta > \frac{1}{2}$, the last term is unbounded as $h \rightarrow 0$; and consequently we will henceforth always assume

$$0 \leq \delta \leq \frac{1}{2}.$$

We also see that *the case $\delta = \frac{1}{2}$ is critical* in that we do not then get decay as $h \rightarrow 0$ for the terms on the right-hand side of (4.4.6) when $|\alpha| > 0$. $\square$

4.4.2. Asymptotic series. Next we consider infinite sums of terms in various symbol classes.

DEFINITION. Let $a_j \in S_\delta(m)$ for $j = 0, 1, \dots$. We say that $a \in S_\delta(m)$ is *asymptotic* to $\sum_{j=0}^\infty h^j a_j$ and write

$$(4.4.7) \quad a \sim \sum_{j=0}^\infty h^j a_j \quad \text{in } S_\delta(m),$$

provided that for each $N = 1, 2, \dots$

$$(4.4.8) \quad a - \sum_{j=0}^{N-1} h^j a_j = O_{S_\delta(m)}(h^N).$$

REMARKS. (i) The notation (4.4.8) means that

$$\left| \partial^\alpha \left(a - \sum_{j=0}^{N-1} h^j a_j \right) \right| \leq C_{\alpha,N} h^{N-\delta|\alpha|} m$$

for all multiindices α .

(ii) Observe that the formal series $\sum_{j=0}^{\infty} h^j a_j$ need not converge for any $h > 0$. We are requiring rather in (4.4.8) that for each N , the difference $a - \sum_{j=0}^{N-1} h^j a_j$ and its derivatives vanish at appropriate rates as $h \rightarrow 0$.

(iii) If the expansion (4.4.7) holds, we call a_0 the *principal symbol* of a . $\square$

Perhaps surprisingly, we can always construct such an asymptotic sum of symbols:

THEOREM 4.15 (Borel's Theorem).

(i) Assume $a_j \in S_\delta(m)$ for $j = 0, 1, \dots$. Then there exists a symbol $a \in S_\delta(m)$ such that

$$(4.4.9) \quad a \sim \sum_{j=0}^{\infty} h^j a_j \quad \text{in } S_\delta(m).$$

(ii) If also $\hat{a} \sim \sum_{j=0}^{\infty} h^j a_j$, then

$$(4.4.10) \quad a - \hat{a} = O_{S(m)}(h^\infty).$$

Proof. 1. Choose a C^∞ function χ such that $0 \leq \chi \leq 1$, $\chi \equiv 1$ on $[0, 1]$, and $\chi \equiv 0$ on $[2, \infty)$. Define

$$(4.4.11) \quad a := \sum_{j=0}^{\infty} h^j \chi(\lambda_j h) a_j,$$

where the sequence $\lambda_j \rightarrow \infty$ must be selected. Since $\lambda_j \rightarrow \infty$, there are for each $h > 0$ at most finitely many nonzero terms in the sum (4.4.11).

2. Now for each multiindex α , with $|\alpha| \leq j$, we have

$$(4.4.12) \quad \begin{aligned} h^j \chi(\lambda_j h) |\partial^\alpha a_j| &\leq C_{j,\alpha} h^{j-\delta|\alpha|} \chi(\lambda_j h) m \\ &= C_{j,\alpha} h^{j-\delta|\alpha|} \chi(\lambda_j h) \frac{\lambda_j h}{\lambda_j h} m \\ &\leq 2C_{j,\alpha} \frac{h^{j-1-\delta|\alpha|}}{\lambda_j} m \\ &\leq h^{j-1-\delta|\alpha|} 2^{-j} m \end{aligned}$$

if λ_j is selected sufficiently large. We can accomplish this for all j and multiindices α with $|\alpha| \leq j$. We may also assume that $\lambda_{j+1} \geq \lambda_j$, for all j .

3. We have

$$a - \sum_{j=0}^N h^j a_j = \sum_{j=N+1}^{\infty} h^j a_j \chi(\lambda_j h) + \sum_{j=0}^N h^j a_j (\chi(\lambda_j h) - 1).$$

Fix any multiindex α . Then taking $N \geq |\alpha|$, we have

$$\begin{aligned} \left| \partial^\alpha \left(a - \sum_{j=0}^N h^j a_j \right) \right| &\leq \sum_{j=N+1}^{\infty} h^j |(\partial^\alpha a_j) \chi(\lambda_j h)| + \sum_{j=0}^N h^j |\partial^\alpha a_j| (1 - \chi(\lambda_j h)) \\ &=: A + B. \end{aligned}$$

According to estimate (4.4.12),

$$A \leq \sum_{j=N+1}^{\infty} h^{j-1-\delta|\alpha|} 2^{-j} m \leq h^{N-\delta|\alpha|} m.$$

Also,

$$B \leq \sum_{j=0}^N C_{\alpha,j} h^{j-\delta|\alpha|} m (1 - \chi(\lambda_j h)).$$

Since $\chi \equiv 1$ on $[0, 1]$, $B = 0$ if $0 < h \leq \lambda_N^{-1}$. If $\lambda_N^{-1} \leq h \leq 1$, we have $1 \leq \lambda_N h$ and hence

$$B \leq m \sum_{j=0}^N C_{\alpha,j} h^{-\delta|\alpha|} \leq m \sum_{j=0}^N C_{\alpha,j} \lambda_N^N h^{N-\delta|\alpha|} = m \tilde{C}_{\alpha,N} h^{N-\delta|\alpha|}.$$

Thus

$$\left| \partial^\alpha \left(a - \sum_{j=0}^N h^j a_j \right) \right| \leq C_{\alpha,N} h^{N-\delta|\alpha|} m$$

if $N \geq |\alpha|$. Therefore, for any N

$$\left| \partial^\alpha \left(a - \sum_{j=0}^{N-1} h^j a_j \right) \right| \leq C_{\alpha,N} h^{N-\delta|\alpha|} m. \quad \square$$

4.4.3. Quantization. Next we discuss the Weyl quantization of symbols in the class $S_\delta(m)$. The mapping properties in the next theorem concern a fixed value of h and the main point is their validity for general order functions m .

THEOREM 4.16 (Quantizing general symbols). *If $a \in S_\delta(m)$, then*

$$(4.4.13) \quad a^w(x, hD) : \mathcal{S} \rightarrow \mathcal{S}$$

and

$$(4.4.14) \quad a^w(x, hD) : \mathcal{S}' \rightarrow \mathcal{S}'$$

are continuous linear transformations.

Proof. 1. We take $h = 1$ for simplicity; so that

$$a^w(x, D)u(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i\langle x-y, \xi \rangle} a\left(\frac{x+y}{2}, \xi\right) u(y) dy d\xi$$

for $u \in \mathcal{S}$. Observe next that $L_1 e^{i\langle x-y, \xi \rangle} = e^{i\langle x-y, \xi \rangle}$, where

$$L_1 := \frac{1 - \langle \xi, D_y \rangle}{1 + |\xi|^2},$$

and $L_2 e^{i\langle x-y, \xi \rangle} = e^{i\langle x-y, \xi \rangle}$ for

$$L_2 := \frac{1 + \langle x-y, D_\xi \rangle}{1 + |x-y|^2}.$$

We employ these operators and an integration by parts argument to show that $a^w(x, D) : \mathcal{S} \rightarrow L^\infty$. We first integrate by parts in the x integral using L_1 to obtain decay in ξ , which then justifies integration by parts using L_2 .

Furthermore,

$$x_j a^w(x, D)u = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} (D_{\xi_j} + y_j) e^{i\langle x-y, \xi \rangle} a\left(\frac{x+y}{2}, \xi\right) u(y) dy d\xi.$$

We can again integrate by parts to conclude that $x^\alpha a^w(x, D) : \mathcal{S} \rightarrow L^\infty$ for each multinomial x^α .

2. Using the Fourier conjugation formula (4.2.14), we see that

$$D^\beta a^w(x, D) = \mathcal{F}^{-1} \langle (\xi^\beta a^w(-D, \xi)) \rangle \mathcal{F}.$$

Now step 1 implies $\langle x \rangle^{n+1} x^\beta a^w(x, D) : \mathcal{S} \rightarrow L^\infty$ for all β . Hence

$$\partial^\beta a^w(x, D) : \mathcal{S} \rightarrow \mathcal{F}^{-1}(\langle \xi \rangle^{-n-1} L^\infty) \subset L^\infty,$$

according to Lemma 3.5. We similarly show that $x^\alpha \partial^\beta a^w(x, D) : \mathcal{S} \rightarrow L^\infty$ for all multiindices α, β . This proves (4.4.13). The continuity statement easily follows from similar arguments: if all seminorms of $u_j \in \mathcal{S}$ tend to 0 with j , so do the seminorms of $a^w(x, D)u_j$.

3. To establish (4.4.14), we note that if $u, v \in \mathcal{S}$, then we have the distributional pairing

$$(a^w(x, D)u)(v) = u(\tilde{a}^w(x, D)v),$$

where

$$\tilde{a}(x, \xi) = a(x, -\xi) \in S(m).$$

According to (4.4.13) we have

$$\tilde{a}^w(x, D)v \in \mathcal{S},$$

and this means that $a^w(x, D)u$ is well defined for $u \in \mathcal{S}'$. The continuity of a^w on $\mathcal{S}'$ follows from the continuity of (4.4.13) and the definition of the topology on $\mathcal{S}'$ in Section 3.2. $\square$

4.4.4. Semiclassical expansions in S_δ . Next we need to reexamine some of our earlier asymptotic expansions, deriving improved estimates on the error terms. The following theorem will let us translate results derived for $a \in \mathcal{S}$ in Section 4.2 into assertions for $a \in S_\delta(m)$.

THEOREM 4.17 (Semiclassical expansions in S_δ). *Let Q be a symmetric, nonsingular matrix.*

(i) *If $0 \leq \delta \leq \frac{1}{2}$, then the operator $e^{\frac{ih}{2}\langle QD, D \rangle} : \mathcal{S} \rightarrow \mathcal{S}$ extends uniquely to an operator*

$$e^{\frac{ih}{2}\langle QD, D \rangle} : S_\delta(m) \rightarrow S_\delta(m).$$

(ii) *If $0 \leq \delta < \frac{1}{2}$, we furthermore have for each symbol $a \in S_\delta(m)$ the expansion*

$$(4.4.15) \quad e^{\frac{ih}{2}\langle QD, D \rangle} a \sim \sum_{k=0}^{\infty} \frac{h^k}{k!} \left(i \frac{\langle QD, D \rangle}{2} \right)^k a \quad \text{in } S_\delta(m).$$

Proof. 1. First, let $0 \leq \delta < \frac{1}{2}$ and $a \in S_\delta(m)$. Theorem 4.8(i) showed that for $a \in \mathcal{S}$,

$$e^{\frac{ih}{2}\langle QD, D \rangle} a(z) = \frac{|\det Q|^{-\frac{1}{2}}}{(2\pi h)^n} e^{\frac{i\pi}{4} \operatorname{sgn} Q} \int_{\mathbb{R}^{2n}} e^{\frac{i}{h}\varphi(w)} a(z+w) dw$$

for the quadratic phase

$$\varphi(w) := -\frac{1}{2}\langle Q^{-1}w, w \rangle.$$

Using Theorem 3.18 this integral defines an element of $\mathcal{S}'$ for any $a \in S_\delta(m)$, as the condition (3.6.2) holds.

Let $\chi : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function with $\chi \equiv 1$ on $B(0, 1)$, $\chi \equiv 0$ on $\mathbb{R}^n \setminus B(0, 2)$. Then

$$\begin{aligned} e^{\frac{i\hbar}{2}\langle QD, D \rangle} a(z) &= \frac{C}{h^n} \int_{\mathbb{R}^{2n}} e^{\frac{i\varphi(w)}{h}} a(z-w) dw \\ &= \frac{C}{h^n} \int_{\mathbb{R}^{2n}} e^{\frac{i\varphi(w)}{h}} \chi(w) a(z-w) dw \\ &\quad + \frac{C}{h^n} \int_{\mathbb{R}^{2n}} e^{\frac{i\varphi(w)}{h}} (1-\chi(w)) a(z-w) dw \\ &=: A + B, \end{aligned}$$

for the constant

$$C := \frac{|\det Q|^{-\frac{1}{2}}}{(2\pi)^n} e^{\frac{i\pi}{4} \operatorname{sgn} Q}.$$

2. *Estimate of A.* Since $\chi(w)a(z-w)$ has compact support, the method of stationary phase, Theorem 4.8, gives

$$A \sim \sum_{k=0}^{\infty} \frac{h^k}{k!} \left(\frac{i}{2} \langle QD, D \rangle \right)^k a(z).$$

Furthermore, if $|w| \leq 2$, we have $m(z-w) \leq Cm(z)$. The remainder estimate in (3.5.8) and the expansion above show that

$$\begin{aligned} |\partial^\alpha A(z)| &\leq C_0 h^{-|\alpha|\delta} m(z) + C_1 h^N \sup_{\substack{0 \leq \beta \leq 2N+n+1 \\ |w| \leq 2}} |\partial^{\alpha+\beta} a(z+w)| \\ &\leq C_2 h^{-|\alpha|\delta} m(z). \end{aligned}$$

Hence $A \in S_\delta(m)$.

3. *Estimate of B.* Let

$$L := \frac{\langle \partial\varphi, hD \rangle}{|\partial\varphi|^2};$$

then $Le^{i\varphi/h} = e^{i\varphi/h}$. Since $|\partial\varphi(w)| \geq \gamma|w|$ for some positive constant γ , the operator L has smooth coefficients on the support of $1-\chi$ and

$$|(L^*)^M((1-\chi)a)| \leq C_M \frac{h^M}{\langle w \rangle^M} \sup_{|\alpha| \leq M} |\partial^\alpha a(z-w)|.$$

Consequently,

$$\begin{aligned} |B| &= \frac{C}{h^n} \left| \int_{\mathbb{R}^{2n}} \left(L^M e^{i\varphi/h} \right) (1-\chi(w)) a(z-w) dw \right| \\ &= \frac{C}{h^n} \left| \int_{\mathbb{R}^{2n}} e^{i\varphi/h} (L^*)^M((1-\chi)a) dw \right| \end{aligned}$$

$$\begin{aligned}
&\leq Ch^{M-n} \int_{\mathbb{R}^{2n}} \langle w \rangle^{-M} \sup_{|\alpha| \leq M} |\partial^\alpha a(z-w)| dw \\
&\leq Ch^{M-n-\delta M} \int_{\mathbb{R}^{2n}} \langle w \rangle^{N-M} m(z) dw \\
&= Ch^{M-n-\delta M} m(z),
\end{aligned}$$

provided $M > 2n + N$. The number N is from the definition (4.4.1) of the order function m .

We similarly also check the higher derivatives, to conclude that $B = O_{S_\delta(m)}(h^\infty)$.

4. Now assume $\delta = 1/2$. In this case we can rescale by setting

$$\tilde{w} = wh^{-1/2}.$$

Then

$$e^{\frac{i}{2}h\langle QD, D \rangle} a(z) = C \int_{\mathbb{R}^{2n}} e^{i\varphi(\tilde{w})} a(z - \tilde{w}h^{1/2}) d\tilde{w}.$$

We use $\chi = \chi(\tilde{w})$ to break the integral into two pieces A and B , as above. Then

$$|\partial^\alpha A| \leq C \sup_{|\tilde{w}| \leq 2} |\partial^\alpha a(z + h^{1/2}\tilde{w})| \leq h^{-|\alpha|/2} m.$$

Furthermore

$$|\partial^\alpha B| \leq C_\alpha h^{-|\alpha|/2} m$$

for each k and α . We leave the verification to the reader. $\square$

REMARKS. (i) Observe that since we can always rescale to the case $h = 1$, there cannot exist an expansion like (4.4.15) for $\delta = 1/2$.

(ii) Theorem 4.17 is interesting and nontrivial for $h = 1$. It then states that if $|\partial^\alpha a| \leq C_\alpha m$ for some order function m and all multiindices α , then

$$\partial^\alpha \left(e^{i\langle QD, D \rangle/2} a \right) = O(m)$$

for all α . $\square$

Recall from Theorem 4.11 that for $a, b \in \mathcal{S}$,

$$a^w(x, hD)b^w(x, hD) = (a \# b)^w(x, hD),$$

where $a \# b$ is defined by (4.3.6).

THEOREM 4.18 (Symbol class of $a \# b$).

(i) If $a \in S_\delta(m_1)$ and $b \in S_\delta(m_2)$, then

$$(4.4.16) \quad a \# b \in S_\delta(m_1 m_2),$$

and

$$(4.4.17) \quad a^w(x, hD)b^w(x, hD) = (a\#b)^w(x, hD)$$

as operators mapping $\mathcal{S}$ to $\mathcal{S}$.

(ii) Furthermore,

$$(4.4.18) \quad a\#b = ab + \frac{i}{2h}\{a, b\} + O_{S_\delta(m_1m_2)}(h^{1-2\delta}),$$

and

$$(4.4.19) \quad [a^w(x, hD), b^w(x, hD)] = \frac{h}{i}\{a, b\}^w(x, hD) + O_{S_\delta(m_1m_2)}(h^{3(1-2\delta)}).$$

Proof. 1. Clearly

$$c(z, w) := a(z)b(w) \in S_\delta(m_1(z)m_2(w))$$

in $\mathbb{R}^{4n}$. If we put $D = (D_x, D_\xi, D_y, D_\eta)$ and $\langle QD, D \rangle = \sigma(D_x, D_\xi; D_y, D_\eta)$ for $z = (x, \xi)$ and $w = (y, \eta)$, then Theorem 4.17 implies

$$e^{\frac{ih}{2}\langle QD, D \rangle} c \in S_\delta(m_1(z)m_2(w)).$$

Then (4.4.16) and (4.4.17) follow, since (4.3.6) and (4.3.4) say

$$a\#b(z) = e^{ihA(D)}c(z, z) = e^{\frac{ih}{2}\langle QD, D \rangle}c(z, z).$$

The second statement of assertion (i) follows from the density of $\mathcal{S}$ in $S_\delta(m)$.

2. We leave the verification of (4.4.18) as an exercise. $\square$

4.4.5. More useful formulas. We next describe how to obtain the symbol from the operator in the particularly nice case of the standard quantization ($t = 1$). This is called *oscillatory testing*.

THEOREM 4.19 (Constructing the symbol from the operator). Suppose $a \in S_\delta(m)$. Then

$$(4.4.20) \quad a(x, \xi) = e^{-\frac{i}{h}\langle x, \xi \rangle} a(x, hD) (e^{\frac{i}{h}\langle \cdot, \xi \rangle}).$$

Proof. For $a \in \mathcal{S}$ we verify this formula using the inverse Fourier transform:

$$\begin{aligned} \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(x, \eta) e^{\frac{i}{h}\langle x, \eta - \xi \rangle} e^{-\frac{i}{h}\langle y, \eta - \xi \rangle} dy d\xi \\ = \int_{\mathbb{R}^n} a(x, \eta) \delta_0(\xi - \eta) e^{\frac{i}{h}\langle x, \eta - \xi \rangle} d\eta = a(x, \xi). \end{aligned}$$

Approximation of a by elements of $\mathcal{S}$ concludes the proof. $\square$

Finally, we record the following interesting fact. Suppose m is an order function on $\mathbb{R}^{2n}$ and put

$$\tilde{m}(x, y, \xi) := m(x, \xi) + m(y, \xi);$$

this is an order function on $\mathbb{R}^{3n}$.

THEOREM 4.20 (Another transformation formula). *Suppose that $0 \leq \delta \leq 1/2$ and $\tilde{a} \in S_\delta(\tilde{m})$. Define*

$$(4.4.21) \quad Au(x) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \tilde{a}(x, y, \xi) e^{\frac{i}{h}\langle x-y, \xi \rangle} u(y) dy d\xi$$

for $u \in \mathcal{S}(\mathbb{R}^n)$.

Then

$$A = a^w(x, hD)$$

for the symbol $a \in S_\delta(m)$ given by

$$(4.4.22) \quad a(x, \xi) = e^{ih\langle D_z, D_\xi \rangle} \tilde{a}\left(x - \frac{z}{2}, x + \frac{z}{2}, \xi\right) \Big|_{z=0}.$$

Proof. We outline the idea. The equality of the Schwartz kernels of A and $a^w(x, hD)$ implies

$$a(x, \xi) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \tilde{a}(x - z/2, x + z/2, \zeta + \xi) e^{-\frac{i}{h}\langle \zeta, z \rangle} dz d\zeta$$

for $\tilde{a} \in \mathcal{S}$. We then use (4.2.12) to obtain (4.4.22). Theorem 4.17 implies the validity for $\tilde{a} \in S_\delta(\tilde{m})$. $\square$

REMARK: Pseudolocality of pseudodifferential operators. In all the arguments it was essential that away from the diagonal $x - y$ integration by parts using

$$\frac{\langle x - y, hD_\xi \rangle}{|x - y|^2} e^{\frac{i}{h}\langle x-y, \xi \rangle} = e^{\frac{i}{h}\langle x-y, \xi \rangle}$$

produces negligible terms. In (4.4.21) the properties of $\tilde{a}$ near $x = y$ contribute to the nonnegligible (that is, not $O(h^\infty)$) properties of the operator A . As we will see in many places in this book, that reflects the *pseudolocal* nature of pseudodifferential operators: $Au(x)$ is determined, modulo negligible terms, by $u(y)$ for y close to x . Differential operators are completely local in the strong sense that for $P(x, hD) = \sum_{|\alpha| \leq m} a_\alpha(x) (hD)^\alpha$, $\text{spt } Pu \subset \text{spt } u$.

4.5. OPERATORS ON L^2

Thus far our symbol calculus has produced operators acting on either the Schwartz space $\mathcal{S}$ or its dual space $\mathcal{S}'$. But for applications we would also like to handle functions in more convenient spaces, most notably L^2 .

4.5.1. Symbols in $\mathcal{S}$. We first observe that if $a \in \mathcal{S}$, then $a^w(x, hD)$ is in fact a bounded linear operator on L^2 :

THEOREM 4.21 (L^2 boundedness for symbols in $\mathcal{S}$). *If the symbol a belongs to $\mathcal{S}$, then*

$$a^w(x, hD) : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$$

is bounded independently of h .

Proof. 1. We recall from the proof of Theorem 4.1 that

$$(4.5.1) \quad a^w(x, hD)u(x) = \int_{\mathbb{R}^n} K(x, y)u(y) dy$$

for the kernel

$$K(x, y) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle x-y, \xi \rangle} a\left(\frac{x+y}{2}, \xi\right) d\xi = \mathcal{F}_h^{-1}\left(a\left(\frac{x+y}{2}, \cdot\right)\right)(x-y).$$

Since $a \in \mathcal{S}$, we have

$$(4.5.2) \quad C_1 := \sup_x \int_{\mathbb{R}^n} |K(x, y)| dy < \infty, \quad C_2 := \sup_y \int_{\mathbb{R}^n} |K(x, y)| dx < \infty.$$

2. We estimate for $u \in L^2$ that

$$\begin{aligned} \|a^w u\|_{L^2}^2 &\leq \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |K(x, y)| |K(x, z)| |u(y)| |u(z)| dx dy dz \\ &\leq \frac{1}{2} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |K(x, y)| |K(x, z)| (|u(y)|^2 + |u(z)|^2) dx dy dz. \end{aligned}$$

Now

$$\begin{aligned} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |K(x, y)| |K(x, z)| |u(y)|^2 dx dy dz \\ \leq C_1 \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} |K(x, y)| |u(y)|^2 dx dy \\ \leq C_1 C_2 \int_{\mathbb{R}^n} |u(y)|^2 dy; \end{aligned}$$

and a similar estimate holds with the roles of y and z reversed. Thus

$$\|a^w u\|_{L^2} \leq (C_1 C_2)^{\frac{1}{2}} \|u\|_{L^2}. \quad \square$$

The second part of this proof is a special case of *Schur's inequality*.

4.5.2. Symbols in S and S_δ . It is important for applications that we extend the foregoing to a wider class of symbols. Our next task therefore is showing that if $a \in S_\delta$ for some $0 \leq \delta \leq \frac{1}{2}$, then $a^w(x, hD)$ extends to become a bounded linear operator acting upon L^2 . This is much harder than the calculations above for $a \in \mathcal{S}$.

For the time being, we take $h = 1$.

Preliminaries. We select $\chi \in C_c^\infty(\mathbb{R}^{2n})$ such that $0 \leq \chi \leq 1$, $\chi \equiv 0$ on $\mathbb{R}^{2n} \setminus B(0, 2)$, and

$$\sum_{\alpha \in \mathbb{Z}^{2n}} \chi_\alpha \equiv 1,$$

where $\chi_\alpha := \chi(\cdot - \alpha)$ denotes χ shifted by the lattice point $\alpha \in \mathbb{Z}^{2n}$. Write

$$(4.5.3) \quad a_\alpha := \chi_\alpha a;$$

then

$$a = \sum_{\alpha \in \mathbb{Z}^{2n}} a_\alpha.$$

We also define

$$(4.5.4) \quad b_{\alpha\beta} := \bar{a}_\alpha \# a_\beta \quad (\alpha, \beta \in \mathbb{Z}^{2n}).$$

THEOREM 4.22 (Decay of mixed terms).

(i) For each N and each multiindex γ , we have the estimate

$$(4.5.5) \quad |\partial^\gamma b_{\alpha\beta}(z)| \leq C_{\gamma,N} \langle \alpha - \beta \rangle^{-N} \langle z - \frac{\alpha + \beta}{2} \rangle^{-N}$$

for $z = (x, \xi) \in \mathbb{R}^{2n}$.

(ii) For each N , there exists a constant C_N such that

$$(4.5.6) \quad \|b_{\alpha\beta}^w(x, D)\|_{L^2 \rightarrow L^2} \leq C_N \langle \alpha - \beta \rangle^{-N}$$

for all $\alpha, \beta \in \mathbb{Z}^{2n}$.

Proof. 1. We can rewrite formula (4.3.7) to read

$$b_{\alpha\beta}(z) = \frac{1}{\pi^{2n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{i\varphi(w_1, w_2)} \bar{a}_\alpha(z - w_1) a_\beta(z - w_2) dw_1 dw_2,$$

for $\varphi(w_1, w_2) = -2\sigma(w_1, w_2)$.

Select $\zeta : \mathbb{R}^{4n} \rightarrow \mathbb{R}$ such that

$$0 \leq \zeta \leq 1, \quad \zeta \equiv 1 \text{ on } B(0, 1), \quad \zeta \equiv 0 \text{ on } \mathbb{R}^{4n} \setminus B(0, 2).$$

Then

$$\begin{aligned} b_{\alpha\beta}(z) &= \frac{1}{\pi^{2n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{i\varphi} \zeta(w) \bar{a}_\alpha(z - w_1) a_\beta(z - w_2) dw_1 dw_2 \\ &\quad + \frac{1}{\pi^{2n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{i\varphi} (1 - \zeta(w)) \bar{a}_\alpha(z - w_1) a_\beta(z - w_2) dw_1 dw_2 \\ &=: A + B. \end{aligned}$$

2. *Estimate of A.* We have

$$|A| \leq C \iint_{\{|w| \leq 2\}} |\bar{a}_\alpha(z - w_1)| |a_\beta(z - w_2)| dw_1 dw_2,$$

for $w = (w_1, w_2)$. The integrand equals

$$\chi(z - w_1 - \alpha) \chi(z - w_2 - \beta) |a(z - w_1)| |a(z - w_2)|$$

and thus vanishes unless

$$|z - w_1 - \alpha| \leq 2 \quad \text{and} \quad |z - w_2 - \beta| \leq 2.$$

But then

$$|\alpha - \beta| \leq 4 + |w_1| + |w_2| \leq 8$$

and

$$\left| z - \frac{\alpha + \beta}{2} \right| \leq 4 + |w_1| + |w_2| \leq 8.$$

Hence

$$|A| \leq C_N \langle \alpha - \beta \rangle^{-N} \langle z - \frac{\alpha + \beta}{2} \rangle^{-N}$$

for any N . Similarly, for each multiindex γ we can estimate

$$(4.5.7) \quad |\partial^\gamma A| \leq C_{N,\gamma} \langle \alpha - \beta \rangle^{-N} \langle z - \frac{\alpha + \beta}{2} \rangle^{-N}.$$

3. *Estimate of B.* If $w_1 = (x_1, \xi_1)$ and $w_2 = (x_2, \xi_2)$, then

$$\partial_{w_1} \sigma(w_1, w_2) = (-\xi_2, x_2), \quad \partial_{w_2} \sigma(w_1, w_2) = (\xi_1, -x_1).$$

Hence,

$$|\partial \varphi(w)| = 2|w|$$

and $Le^{i\varphi} = e^{i\varphi}$, for

$$L := \frac{\langle \partial \varphi, D \rangle}{|\partial \varphi|^2}.$$

Since the integrand of B vanishes unless $|w| \geq 1$, the usual argument based on integration by parts shows that

$$|B| \leq C_M \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} \langle w \rangle^{-M} \bar{c}_\alpha(z - w_1) c_\beta(z - w_2) dw_1 dw_2$$

for appropriate functions c_α, c_β , with $\text{spt}(c_\alpha) \subset B(\alpha, 2)$, $\text{spt}(c_\beta) \subset B(\beta, 2)$.

Thus the integrand vanishes unless

$$\langle \alpha - \beta \rangle \leq C \langle w \rangle, \quad \langle z - \frac{\alpha + \beta}{2} \rangle \leq C \langle w \rangle.$$

Hence

$$\begin{aligned} |B| &\leq C_M \langle \alpha - \beta \rangle^{-N} \langle z - \frac{\alpha + \beta}{2} \rangle^{-N} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} \langle w \rangle^{2N-M} dw_1 dw_2 \\ &\leq C_M \langle \alpha - \beta \rangle^{-N} \langle z - \frac{\alpha + \beta}{2} \rangle^{-N} \end{aligned}$$

if M is large enough. Likewise,

$$(4.5.8) \quad |\partial^\gamma B| \leq C_{N,\gamma} \langle \alpha - \beta \rangle^{-N} \langle z - \frac{\alpha + \beta}{2} \rangle^{-N}.$$

This proves (4.5.5).

4. Recall next that

$$a^w(x, D) = \frac{1}{(2\pi)^{2n}} \int_{\mathbb{R}^{2n}} \hat{a}(l) e^{il(x,D)} dl$$

and that, owing to (4.2.9), $e^{il(x,D)}$ is a unitary operator on L^2 . Consequently

$$\|a^w(x, D)\|_{L^2 \rightarrow L^2} \leq C \int_{\mathbb{R}^{2n}} |\hat{a}(l)| dl.$$

Therefore we can estimate

$$\begin{aligned} \|b_{\alpha\beta}^w(x, D)\|_{L^2 \rightarrow L^2} &\leq C \|\hat{b}_{\alpha\beta}\|_{L^1} \leq C \|\langle \xi \rangle^{2n+1} \hat{b}_{\alpha\beta}\|_{L^\infty} \\ &\leq C \sup_{|\gamma| \leq 2n+1} \|\widehat{D^\gamma b_{\alpha\beta}}\|_{L^\infty} \\ &\leq C \sup_{|\gamma| \leq 2n+1} \|D^\gamma b_{\alpha\beta}\|_{L^1} \\ &\leq C \sup_{|\gamma| \leq 2n+1} \|\langle z \rangle^{2n+1} D^\gamma b_{\alpha\beta}\|_{L^\infty} \\ &\leq C \langle \alpha - \beta \rangle^{-N}, \end{aligned}$$

according to (4.5.5) □

THEOREM 4.23 (L^2 boundedness for symbols in S).

(i) *If the symbol a belongs to S , then*

$$a^w(x, D) : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$$

is bounded, with the estimate

$$(4.5.9) \quad \|a^w(x, D)\|_{L^2 \rightarrow L^2} \leq C \sum_{|\alpha| \leq Mn} \sup_{\mathbb{R}^n} |\partial^\alpha a|,$$

where M is a universal constant.

(ii) *Furthermore, if $a \in S_\delta$ for some $0 \leq \delta \leq 1/2$, then*

$$(4.5.10) \quad \|a^w(x, hD)\|_{L^2 \rightarrow L^2} \leq C \sum_{|\alpha| \leq Mn} h^{|\alpha|/2} \sup_{\mathbb{R}^n} |\partial^\alpha a|.$$

REMARK. Theorem 13.13 will provide a more precise bound than (4.5.10).

Proof. 1. We have $b_{\alpha\beta}^w(x, D) = A_\alpha^* A_\beta$, where $A_\alpha := a_\alpha^w(x, D)$. Thus Theorem 4.22(ii) asserts that

$$\|A_\alpha^* A_\beta\|_{L^2 \rightarrow L^2} \leq C \langle \alpha - \beta \rangle^{-N}.$$

Therefore

$$\sup_\alpha \sum_\beta \|A_\alpha A_\beta^*\|^{1/2} \leq C \sum_\beta \langle \alpha - \beta \rangle^{-N/2} \leq C;$$

and similarly

$$\sup_\alpha \sum_\beta \|A_\alpha^* A_\beta\|^{1/2} \leq C.$$

Since $a^w(x, D) = \sum_\alpha A_\alpha$, we can apply the Cotlar–Stein Theorem (Theorem C.5). The constants in the estimates in Theorem 4.22 depend only on a finite number of derivatives of a , growing linearly with the dimension. That proves (4.5.9).

2. Estimate (4.5.10) follows from a rescaling, the details of which for $\delta = 0$ we will provide later in the proof of Theorem 5.1. $\square$

As a first application, we record the useful

THEOREM 4.24 (Composition and multiplication). *Suppose that $a, b \in S_\delta$ for $0 \leq \delta < \frac{1}{2}$.*

Then

$$(4.5.11) \quad (ab)^w(x, hD) = a^w(x, hD)b^w(x, hD) + O_{L^2 \rightarrow L^2}(h^{1-2\delta})$$

as $h \rightarrow 0$.

Proof. In light of (4.4.18), we have

$$a \# b - ab = O_{S_\delta}(h^{1-2\delta}).$$

Hence Theorem 4.23 implies

$$a^w b^w - (ab)^w = (a \# b - ab)^w = O_{L^2 \rightarrow L^2}(h^{1-2\delta}). \quad \square$$

For the borderline case $\delta = \frac{1}{2}$, we have this assertion:

THEOREM 4.25 (Disjoint supports). *Suppose that $a, b \in S_{\frac{1}{2}}$ and that*

$$(4.5.12) \quad \text{dist}(\text{spt}(a), \text{spt}(b)) \geq \gamma > 0$$

for some constant γ , independent of h .

Then

$$(4.5.13) \quad a^w(x, hD)b^w(x, hD) = O_{L^2 \rightarrow L^2}(h^\infty).$$

Proof. Remember from (4.3.7) that

$$a\#b(z) = \frac{1}{(h\pi)^{2n}} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} e^{\frac{i}{h}\varphi(w_1, w_2)} a(z - w_1) b(z - w_2) dw_1 dw_2,$$

for $z = (x, \xi)$ and $\varphi(w_1, w_2) = -2\sigma(w_1, w_2)$.

We proceed as in the proof of Theorem 4.22: $|\partial\varphi| = 2|w|$ and thus the operator

$$L := \frac{\langle \partial\varphi, hD \rangle}{|\partial\varphi|^2}$$

has smooth coefficients on the support of $a(z - w_1)b(z - w_2)$. From our assumption that $a, b \in S_{\frac{1}{2}}$, we see that

$$(L^*)^M(a(z - w_1)b(z - w_2)) = O(h^{\frac{M}{2}} \langle w \rangle^{-M}).$$

Hence, for M sufficiently large we obtain

$$\begin{aligned} |a\#b(z)| &\leq Ch^{\frac{M}{2}-2n} \int_{\mathbb{R}^{2n}} \int_{\mathbb{R}^{2n}} (1 + |w_1|^2 + |w_2|^2)^{-M/2} dw_1 dw_2 \\ &= O(h^{\frac{M}{2}-2n}), \end{aligned}$$

with similar estimates for derivatives. We conclude that $a\#b = O_S(h^\infty)$. Its quantization is therefore bounded on L^2 , with norm of order $O(h^\infty)$. $\square$

4.6. COMPACTNESS

In this section we modify the proof of Theorem 4.23 to show that if $a \in S(m)$ and if m goes to zero as $(x, \xi) \rightarrow \infty$, then $a^w(x, D)$ is a compact operator on L^2 .

A first observation follows:

LEMMA 4.26 (Schwartz symbols and compactness). *Suppose that $a \in \mathcal{S}$. Then*

$$a^w(x, D) : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n) \text{ is a compact operator.}$$

Proof. 1. The Schwartz kernel of $c^w(x, D)$ is given by

$$K(x, y) := \frac{1}{(2\pi)^n} \int a\left(\frac{x+y}{2}, \xi\right) e^{i\langle \xi, x-y \rangle} d\xi,$$

and thus $K \in \mathcal{S}(\mathbb{R}^n \times \mathbb{R}^n)$. Hence, for any α and β ,

(4.6.1)

$$\begin{aligned} \sup_{x \in \mathbb{R}^n} |x^\alpha \partial_x^\beta (c^w(x, D)u)| &\leq \sup_{(x, y) \in \mathbb{R}^{2n}} |x^\alpha \partial_x^\beta \langle y \rangle^N K(x, y)| \int_{\mathbb{R}^n} \langle y \rangle^{-N} |u(y)| dy \\ &\leq C_{\alpha\beta} \|u\|_{L^2}, \end{aligned}$$

where for the last estimate we applied the Cauchy–Schwartz inequality and took $N > n/2$.

2. Let $\mathcal{F} \subset L^2(\mathbb{R}^n)$ be a bounded set. We need to find $\{f_p\}_{p=1}^\infty \subset \mathcal{F}$ such that $c^w(x, D)f_p$ converges in L^2 . It is enough to show that

$$g_p(x) := \langle x \rangle^N c^w(x, D)f_p(x), \quad N > n/2,$$

converges in $L^\infty(\mathbb{R}^n)$:

$$\|c^w(x, D)f_p - c^w(x, D)f_k\|_{L^2} \leq \|\langle x \rangle^{-N}\|_{L^2} \|g_p - g_k\|_{L^\infty}.$$

3. To find the sequence $\{g_p\}_{p=1}^\infty$, we use the standard Ascoli–Arzelà argument, which we recall for the reader’s convenience.

Let $\mathbb{Q}^n = \{x_k\}_{k=1}^\infty$ be the set of rational points in $\mathbb{R}^n$ (or any other countable dense set). The estimate (4.6.1) shows that for any $f \in \mathcal{F}$,

$$g(x) := \langle x \rangle^N c^w(x, D)f(x)$$

is uniformly bounded. The standard diagonal argument shows that there exists a sequence $\{f_p\}_{p=1}^\infty \subset \mathcal{F}$ such that for all $x_k \in \mathbb{Q}^n$,

$$g_p(x_k) := \langle x_k \rangle^N c^w(x, D)f_p(x_k) \rightarrow b_k \in \mathbb{C}, \quad p \rightarrow \infty.$$

(In the diagonal argument we first choose $\{f_p^1\}_{\ell=1}^\infty \subset \mathcal{F}$ with convergence at x_1 and then inductively a subsequence $\{f_p^k\}_{p=1}^\infty \subset \{f_p^{k-1}\}_{p=1}^\infty$, such that $\langle x \rangle^N c^w(x, D)f_p^k(x)$ converges at x_k . We then put $f_p := f_p^p$.)

4. To show that g_p converges in L^∞ , we show that it is a Cauchy sequence. Estimate (4.6.1) shows that for some M ,

$$|\partial g_p(x)| \leq M/3, \quad \langle x \rangle |g_p(x)| \leq M/2.$$

Fix $\epsilon > 0$ and choose R such that $M/R < \epsilon$. Then choose $\{y_p\}_{p=1}^P \subset \mathbb{Q}^n$ such that

$$B(0, R) \subset \bigcup_{p=1}^P B(y_p, \epsilon/M).$$

Each sequence $g_\ell(y_p)$ is a Cauchy sequence and hence there exists K such that

$$(4.6.2) \quad |g_k(y_p) - g_\ell(y_p)| < \epsilon/3, \quad k, \ell > K, \quad 1 \leq p \leq P.$$

We now estimate

$$\|g_k - g_\ell\|_{L^\infty} \leq \max \left(\sup_{|x| < R} |g_k(x) - g_\ell(x)|, \sup_{|x| \geq R} |g_k(x) - g_\ell(x)| \right).$$

The choice for R shows that

$$\sup_{|x| \geq R} |g_k(x) - g_\ell(x)| \leq 2R^{-1} \sup_{|x| \geq R} \langle x \rangle |g_\ell(x)| < \epsilon.$$

On the other hand for $x \in B(0, R)$ we choose y_p such that $|x - y_p| < \epsilon/M$. Using (4.6.2), it follows that for $k, \ell > K$,

$$\begin{aligned} |g_k(x) - g_\ell(x)| &\leq |g_k(y_p) - g_\ell(y_p)| + |g_k(x) - g_k(y_p)| + |g_\ell(x) - g_\ell(y_p)| \\ &< \epsilon/3 + (\sup |\partial g_k| + \sup |\partial g_\ell|) |x - y_p| < \epsilon. \end{aligned}$$

This gives the Cauchy property in L^∞ and g_p converges to some $g \in L^\infty$. The sequence $c^w(x, D)f_k$ then converges to $\langle x \rangle^{-N}g$ in L^2 . $\square$

Next we revisit Theorem 4.22 for general symbol classes. Recall that a_α and $b_{\alpha\beta}$ are defined in (4.5.4).

THEOREM 4.27 (Decay of mixed terms for general symbols). *Suppose that $a \in S(m)$. Then for each N , there exists a constant C_N such that*

$$(4.6.3) \quad \|b_{\alpha\beta}^w(x, D)\|_{L^2 \rightarrow L^2} \leq C_N m(\alpha) m(\beta) \langle \alpha - \beta \rangle^{-N}$$

for all $\alpha, \beta \in \mathbb{Z}^{2n}$.

Proof. We observe that

$$\begin{aligned} |\partial^\gamma a_\alpha(w)| &= |\partial^\gamma (\chi(w - \alpha) a(w))| \\ (4.6.4) \quad &\leq C_\gamma \sup_{|\rho| \leq |\gamma|} |\partial^\rho \chi(w - \alpha)| m(w) \\ &\leq C_\gamma m(\alpha), \end{aligned}$$

since the support of $\chi(w - \alpha)$ is contained in $|w - \alpha| \leq 2$.

Given (4.6.4), the proof of (4.5.5) now shows that

$$(4.6.5) \quad |\partial^\gamma b_{\alpha\beta}(z)| \leq C_{\gamma, N} m(\alpha) m(\beta) \langle \alpha - \beta \rangle^{-N} \langle z - \frac{\alpha + \beta}{2} \rangle^{-N}$$

for all N, γ and $z = (x, \xi) \in \mathbb{R}^{2n}$. We now apply (4.6.5) in the same way that (4.5.5) was employed in the proof of Theorem 4.22. This gives (4.6.3). $\square$

THEOREM 4.28 (Compactness for decaying order functions). *Suppose $a \in S(m)$ and*

$$(4.6.6) \quad \lim_{(x, \xi) \rightarrow \infty} m = 0.$$

Then

$$(4.6.7) \quad a^w(x, D) : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n) \text{ is a compact operator.}$$

REMARK. The same assertion holds for the other quantizations of a . A converse is also true: if for every $a \in S(m)$, $a^w(x, D)$ is compact, then (4.6.6) holds. $\square$

Proof. 1. We recall the notation of the proof of Theorem 4.23,

$$A := a^w(x, D), \quad A_\alpha := a_\alpha^w(x, D),$$

and define

$$A_M := \sum_{|\alpha| < M} A_\alpha.$$

According to Lemma 4.26, A_M is a finite sum of compact operators and hence is compact.

2. The space of compact operators is closed in operator norm topology, and hence to prove that A is compact, it suffices to show that

$$(4.6.8) \quad \lim_{M \rightarrow \infty} \|A - A_M\|_{L^2 \rightarrow L^2} = 0.$$

We write $A - A_M = \sum_{|\alpha| \geq M} A_\alpha$ and use the Cotlar-Stein Theorem (Theorem C.5) to estimate:

$$\|A - A_M\| \leq \max \left(\sup_{|\alpha| \geq M} \sum_{|\beta| \geq M} \|A_\alpha A_\beta^*\|^{1/2}, \sup_{|\alpha| \geq M} \sum_{|\beta| \geq M} \|A_\alpha^* A_\beta\|^{1/2} \right).$$

Since $A_\alpha^* A_\beta = (\bar{a}_\alpha \# a_\beta)^w = b_{\alpha\beta}^w$, we can apply Theorem 4.27 to obtain

$$\begin{aligned} \sup_{|\alpha| \geq M} \sum_{|\beta| \geq M} \|A_\alpha^* A_\beta\|^{1/2} &\leq C_N \sup_{|\alpha| \geq M} \sum_{|\beta| \geq M} \sqrt{m(\alpha)m(\beta)} \langle \alpha - \beta \rangle^{-N/2} \\ &\leq C \sup_{|\alpha| \geq M} m(\alpha). \end{aligned}$$

A similar estimate applies in the case of $A_\alpha A_\beta^*$. Therefore

$$\|A - A_M\| \leq C \sup_{|\alpha| \geq M} m(\alpha) \rightarrow 0$$

as $M \rightarrow \infty$, thanks to our hypothesis (4.6.6) on m . This shows (4.6.8). $\square$

4.7. INVERSES, GÄRDING INEQUALITIES

At this stage we have constructed in appropriate generality the quantizations $a^w(x, hD)$ of various symbols a . We turn therefore to the practical problem of understanding how the algebraic and analytic behavior of the function a dictates properties of the corresponding quantized operators.

4.7.1. Inverses. Suppose in particular that $a : \mathbb{R}^{2n} \rightarrow \mathbb{C}$ is nonvanishing and so is pointwise invertible. Can we draw the same conclusion about $a^w(x, hD)$?

DEFINITIONS. (i) We say the symbol a is *elliptic* if there exists a constant $\gamma > 0$, independent of h , such that

$$(4.7.1) \quad |a| \geq \gamma > 0 \quad \text{on } \mathbb{R}^{2n}.$$

(ii) More generally, a is *elliptic in $S(m)$* if for some constant $\gamma > 0$

$$|a| \geq \gamma m.$$

THEOREM 4.29 (Inverses for elliptic symbols). Assume that $a \in S_\delta(m)$ for some $0 \leq \delta < \frac{1}{2}$ and that a is elliptic in $S(m)$.

(i) If $m \geq 1$, there exist $h_0 > 0$ and $C > 0$ such that

$$(4.7.2) \quad \|a^w(x, hD)u\|_{L^2} \geq C\|u\|_{L^2}$$

for all $u \in \mathcal{S}$ and $0 < h < h_0$.

(ii) If $m = 1$, there exists $h_0 > 0$, such that

$$a^w(x, hD)^{-1}$$

exists as a bounded linear operator on $L^2(\mathbb{R}^n)$ for $0 < h \leq h_0$.

Proof. 1. Let $b := 1/a$, $b \in S_\delta(1/m)$. Then (4.4.18) gives

$$a \# b = 1 + r_1, \quad \text{with } r_1 \in h^{1-2\delta} S_\delta.$$

Likewise

$$b \# a = 1 + r_2, \quad \text{with } r_2 \in h^{1-2\delta} S_\delta.$$

Hence if $A := a^w(x, hD)$, $B := b^w(x, hD)$, $R_1 := r_1^w(x, hD)$, and $R_2 := r_2^w(x, hD)$, we have

$$\begin{aligned} AB &= I + R_1, \\ BA &= I + R_2, \end{aligned}$$

with

$$\|R_1\|_{L^2 \rightarrow L^2}, \|R_2\|_{L^2 \rightarrow L^2} = O(h^{1-2\delta}) \leq \frac{1}{2}$$

if $0 < h \leq h_0$ and h_0 is small enough.

2. When $m = 1$, $A = a^w(x, hD)$ has an approximate left inverse and an approximate right inverse. Then applying Theorem C.3, we deduce that A^{-1} exists.

3. If $m \geq 1$, we see that for $u \in \mathcal{S}$

$$\|u\|_{L^2} = \|(I + R_2)^{-1}b^w(x, hD)a^w(x, hD)u\|_{L^2} \leq C\|a^w(x, hD)u\|_{L^2},$$

since $b \in S(1/m) \subset S(1)$ is bounded on L^2 , according to Theorem 4.23. $\square$

4.7.2. Gårding inequalities. We suppose next that a is real-valued and nonnegative and ask the consequences for $a^w(x, hD)$.

THEOREM 4.30 (Easy Gårding inequality). *Assume a is a real-valued symbol in S and*

$$(4.7.3) \quad a \geq \gamma > 0 \quad \text{on } \mathbb{R}^{2n}.$$

Then for each $\epsilon > 0$ there exists $h_0 = h_0(\epsilon) > 0$ such that

$$(4.7.4) \quad \langle a^w(x, hD)u, u \rangle \geq (\gamma - \epsilon)\|u\|_{L^2(\mathbb{R}^n)}^2$$

for all $0 < h \leq h_0$ and $u \in L^2(\mathbb{R}^n)$.

Proof. We will show that

$$(4.7.5) \quad (a - \lambda)^{-1} \in S \quad \text{if } \lambda < \gamma - \epsilon.$$

Indeed if $b := (a - \lambda)^{-1}$, then

$$(a - \lambda)\#b = 1 + \frac{h}{2i}\{a - \lambda, b\} + O_S(h^2) = 1 + O_S(h^2),$$

the bracket term vanishing since b is a function of $a - \lambda$. Therefore

$$(a^w(x, hD) - \lambda)b^w(x, hD) = I + O_{L^2 \rightarrow L^2}(h^2),$$

and so $b^w(x, hD)$ is an approximate right inverse of $a^w(x, hD) - \lambda$. Likewise $b^w(x, hD)$ is an approximate left inverse.

Hence Theorem C.3 implies $a^w(x, hD) - \lambda$ is invertible for each $\lambda < \gamma - \epsilon$. Consequently,

$$\text{Spec}(a^w(x, hD)) \subset [\gamma - \epsilon, \infty).$$

Then according to Theorem C.8,

$$\langle a^w(x, hD)u, u \rangle \geq (\gamma - \epsilon)\|u\|_{L^2}^2$$

for all $u \in L^2$. $\square$

To improve the preceding estimate, we will need a simple calculus inequality:

LEMMA 4.31 (Gradient estimate). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 , with*

$$|\partial^2 f| \leq A.$$

Suppose also that $f \geq 0$. Then

$$|\partial f| \leq (2Af)^{1/2}.$$

Proof. By Taylor's Theorem,

$$f(x+y) = f(x) + \langle \partial f(x), y \rangle + \int_0^1 (1-t) \langle \partial^2 f(x+ty)y, y \rangle dt.$$

Let $y = -\lambda \partial f(x)$, $\lambda > 0$ to be selected. Then since $f \geq 0$, we have

$$\begin{aligned} \lambda |\partial f(x)|^2 &\leq f(x) + \lambda^2 \int_0^1 (1-t) \langle \partial^2 f(x - \lambda t \partial f(x)) \partial f(x), \partial f(x) \rangle dt \\ &\leq f(x) + \frac{\lambda^2}{2} A |\partial f(x)|^2. \end{aligned}$$

Putting $\lambda = 1/A$, we conclude that $|\partial f(x)|^2 \leq 2Af(x)$. $\square$

We next improve Theorem 4.30:

THEOREM 4.32 (Sharp Gårding inequality). *Assume $a \in S$ and*

$$(4.7.6) \quad a \geq 0 \quad \text{on } \mathbb{R}^{2n}.$$

Then there exist constants $C \geq 0$ and $h_0 > 0$ such that

$$(4.7.7) \quad \langle a^w(x, hD)u, u \rangle \geq -Ch \|u\|_{L^2(\mathbb{R}^n)}^2$$

for all $0 < h < h_0$ and $u \in L^2(\mathbb{R}^n)$.

REMARK. The estimate (4.7.7) is in fact true for each quantization $\text{Op}_t(a)$ ($0 \leq t \leq 1$). For the Weyl quantization, the stronger *Fefferman–Phong inequality* holds:

$$\langle a^w(x, hD)u, u \rangle \geq -Ch^2 \|u\|_{L^2(\mathbb{R}^n)}^2$$

for $0 < h \leq h_0$, $u \in L^2(\mathbb{R}^n)$. $\square$

Proof. 1. Our goal is to show that if we fix $\tilde{h}$ sufficiently small and write

$$(4.7.8) \quad \lambda = h/\tilde{h},$$

then

$$(4.7.9) \quad h(a + \lambda)^{-1} \in \tilde{h}S_{1/2},$$

with estimates independent of $\tilde{h}$. We can then argue as in the proof of Theorem 4.30.

In our notation

$$b \in \tilde{h}S_{1/2}$$

means

$$|\partial^\alpha b| \leq C_\alpha h^{-|\alpha|/2} \tilde{h}$$

for all multiindices α , with C_α independent of h and $\tilde{h}$.

2. We first claim that

$$(4.7.10) \quad \partial^\alpha (a + \lambda)^{-1} = (a + \lambda)^{-1} \sum_{k=1}^{|\alpha|} \sum_{\substack{\alpha = \beta^1 + \dots + \beta^k \\ |\beta^j| \geq 1}} C_{\beta^1, \dots, \beta^k} \prod_{j=1}^k \left((a + \lambda)^{-1} \partial^{\beta^j} a \right),$$

for appropriate constants $C_{\beta^1, \dots, \beta^k}$. To see this, observe that when we compute $\partial^\alpha (a + \lambda)^{-1}$, a typical term involves k differentiations of $(a + \lambda)^{-1}$ with the remaining derivatives falling on a . For each $k \leq |\alpha|$ we partition α into multiindices $\beta^1, \dots, \beta^k$, each of which corresponds to one derivative falling on $(a + \lambda)^{-1}$ and the remaining derivatives falling on a . Summing over k gives (4.7.10).

3. Lemma 4.31 implies that

$$\lambda^{1/2} |\partial a| \leq C \lambda^{1/2} a^{1/2} \leq C(\lambda + a).$$

Hence for $|\beta| = 1$

$$(4.7.11) \quad |\partial^\beta a| (a + \lambda)^{-1} \leq C \lambda^{-1/2};$$

and furthermore

$$(4.7.12) \quad |\partial^\beta a| (a + \lambda)^{-1} \leq C \lambda^{-1}$$

if $|\beta| \geq 2$, since $a \in S$. Consequently, for each partition $\alpha = \beta^1 + \dots + \beta^k$ and $0 < \lambda \leq 1$:

$$\left| \prod_{j=1}^k (a + \lambda)^{-1} \partial^{\beta_j} a \right| \leq C \prod_{|\beta_j| \geq 2} \lambda^{-1} \prod_{|\beta_j| = 1} \lambda^{-1/2} \leq C \prod_{j=1}^k \lambda^{-\frac{|\beta_j|}{2}} = C \lambda^{-\frac{|\alpha|}{2}}.$$

Therefore

$$(4.7.13) \quad |\partial^\alpha (a + \lambda)^{-1}| \leq C_\alpha (a + \lambda)^{-1} \lambda^{-\frac{|\alpha|}{2}}.$$

Because $\lambda = h/\tilde{h}$, this implies

$$(a + \lambda)^{-1} \in \frac{\tilde{h}}{h} S_{1/2};$$

that is,

$$(4.7.14) \quad h(a + \lambda)^{-1} \in \tilde{h} S_{1/2},$$

with estimates independent of λ .

4. Since $a + \lambda \in S \subset S_{\frac{1}{2}}$, we can define $(a + \lambda)\#b$, for $b = (a + \lambda)^{-1}$. Using Taylor's formula, we compute

$$\begin{aligned} (a + \lambda)\#b(z) &= e^{ihA(D)}(a(z) + \lambda)b(w)\Big|_{w=z} \\ &= 1 + \int_0^1 (1-t)e^{ithA(D)}(ihA(D))^2(a(z) + \lambda)b(w)\Big|_{w=z} dt \\ &=: 1 + r(z), \end{aligned}$$

where we used $\{a + \lambda, (a + \lambda)^{-1}\} = 0$.

Now according to (4.7.14), $hb \in \tilde{h}S_{1/2}$ and so $h^2\partial^\alpha b \in \tilde{h}S_{1/2}$ for $|\alpha| = 2$. An application of $e^{ihA(D)}$ preserves the symbol class $\tilde{h}S_{1/2}$. Consequently,

$$\|r^w(x, hD)\|_{L^2 \rightarrow L^2} \leq C\tilde{h} \leq \frac{1}{2}$$

if $\tilde{h}$ is now fixed small enough. Thus $b^w(x, hD)$ is an approximate right inverse of $a^w(x, hD) + \lambda$ and is similarly an approximate left inverse.

5. So $(a^w(x, hD) + \lambda)^{-1}$ exists. Likewise $(a^w(x, hD) + \gamma + \lambda)^{-1}$ exists for all $\gamma \geq 0$. Therefore

$$\text{Spec}(a^w(x, hD)) \subset [-\lambda, \infty).$$

Then according to Theorem C.8,

$$\langle a^w(x, hD)u, u \rangle \geq -\lambda\|u\|_{L^2}^2$$

for all $u \in L^2$. Since $\lambda = h/\tilde{h}$, this inequality finishes the proof. $\square$

REMARK: More on rescaling. The rescaling (4.1.9) can be generalized to

$$(4.7.15) \quad \tilde{x} := (\tilde{h}/h)^{\frac{1}{2}}x, \quad \tilde{y} := (\tilde{h}/h)^{\frac{1}{2}}y, \quad \tilde{\xi} := (\tilde{h}/h)^{\frac{1}{2}}\xi.$$

Then the calculation which lead to (4.1.10) gives

$$(4.7.16) \quad a^w(x, hD)u(x) = a_h^w(\tilde{x}, \tilde{h}D)\tilde{u}(\tilde{x}),$$

for

$$\tilde{u}(\tilde{x}) := u((h/\tilde{h})^{\frac{1}{2}}\tilde{x}), \quad a_h(\tilde{x}, \tilde{\xi}) := a((h/\tilde{h})^{\frac{1}{2}}\tilde{x}, (h/\tilde{h})^{\frac{1}{2}}\tilde{\xi}).$$

We have thus rescaled from the h -semiclassical calculus to the $\tilde{h}$ -semiclassical calculus.

Note in particular that if

$$\partial^\alpha a = O((\tilde{h}/h)^{-|\alpha|/2}),$$

then $\tilde{a} \in S$. The bound (4.7.13) is precisely an estimate of this type. It is essential in the proof of Theorem 4.32 that if $a \in S$, then $\partial^\alpha \tilde{a} = O((h/\tilde{h})^{|\alpha|/2})$; that is, the derivative improves. $\square$

4.8. NOTES

Our presentation of semiclassical calculus is based upon Dimassi–Sjöstrand [D-S, Chapter 7]. See also Martinez [M] for the Fefferman–Córdoba [C-F] proof of the sharp Gårding inequality. The argument presented here followed the proof of [D-S, Theorem 7.12].

Good introductions to the theory of pseudodifferential operators include Alinhac–Gérard [A-G], Grigis–Sjöstrand [G-S], Martinez [M], and Saint Raymond [SR]. A major treatise is Hörmander [H1]–[H4].

Part 2

APPLICATIONS TO PARTIAL DIFFERENTIAL EQUATIONS

Chapter 5

SEMICLASSICAL DEFECT MEASURES

- 5.1 Construction, examples
- 5.2 Defect measures and PDE
- 5.3 Damped wave equation
- 5.4 Notes

One way to understand limits as $h \rightarrow 0$ of a collection of functions $u = \{u(h)\}_{0 < h \leq h_0}$ bounded in L^2 is to construct corresponding *semiclassical defect measures* μ , which record the limiting behavior of certain quadratic forms acting on $u(h)$. If in addition these functions solve certain operator equations or PDE, we can deduce various properties of the measure μ and thereby indirectly recover information about asymptotics as $h \rightarrow 0$ of the functions $u(h)$.

5.1. CONSTRUCTION, EXAMPLES

In the first two sections of this chapter, we consider a collection of functions $u = \{u(h)\}_{0 < h \leq h_0}$ that is bounded in $L^2(\mathbb{R}^n)$:

$$(5.1.1) \quad \sup_{0 < h \leq h_0} \|u(h)\|_{L^2} < \infty.$$

For the time being, we do not assume that $u(h)$ solves any particular PDE or pseudodifferential equation.

THEOREM 5.1 (An operator norm bound). *Suppose $a \in S$. Then*

$$(5.1.2) \quad \|a^w(x, hD)\|_{L^2 \rightarrow L^2} \leq C \sup_{\mathbb{R}^{2n}} |a| + O(h^{\frac{1}{2}})$$

as $h \rightarrow 0$.

REMARK. Theorem 13.13 will provide a more precise bound than (5.1.2).

Proof. We showed earlier in Theorem 4.23 that if $a \in S$ and $h = 1$, then

$$(5.1.3) \quad \|a^w(x, D)\|_{L^2 \rightarrow L^2} \leq C \sup_{|\alpha| \leq Mn} \|\partial^\alpha a\|_{L^\infty}.$$

Suppose now that $a \in S$ and $u \in \mathcal{S}$. We rescale according to (4.1.9) but now write

$$\tilde{u}(\tilde{x}) := h^{\frac{n}{4}} u(x) = h^{\frac{n}{4}} u(h^{\frac{1}{2}} \tilde{x}).$$

This is different from the standard rescaling (4.1.11) of u , the advantage being that $u \mapsto \tilde{u}$ is now a unitary transformation of L^2 : $\|u\|_{L^2} = \|\tilde{u}\|_{L^2}$. Then

$$\begin{aligned} a^w(x, hD)u(x) &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a\left(\frac{x+y}{2}, \xi\right) e^{\frac{i}{h}\langle x-y, \xi \rangle} u(y) dy d\xi \\ (5.1.4) \quad &= \frac{h^{-\frac{n}{4}}}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a_h\left(\frac{\tilde{x}+\tilde{y}}{2}, \tilde{\xi}\right) e^{i\langle \tilde{x}-\tilde{y}, \tilde{\xi} \rangle} \tilde{u}(\tilde{y}) d\tilde{y} d\tilde{\xi} \\ &= h^{-\frac{n}{4}} a_h^w(\tilde{x}, D)\tilde{u}(\tilde{x}) \end{aligned}$$

for a_h as in the standard rescaling (4.1.11). We hence deduce from (5.1.4) and (5.1.3) that

$$\begin{aligned} \|a^w(x, hD)u\|_{L^2} &= \|a_h^w(\tilde{x}, D)\tilde{u}\|_{L^2} \\ &\leq \|a_h^w\|_{L^2 \rightarrow L^2} \|\tilde{u}\|_{L^2} \\ &\leq C \sup_{|\alpha| \leq Mn} |\partial^\alpha a_h| \|u\|_{L^2} \\ &\leq C \sup_{|\alpha| \leq Mn} h^{\frac{|\alpha|}{2}} |\partial^\alpha a| \|u\|_{L^2}. \end{aligned}$$

This implies (5.1.2). □

THEOREM 5.2 (Existence of defect measure). *There exists a non-negative Radon measure μ on $\mathbb{R}^{2n}$ and a sequence $h_j \rightarrow 0$ such that*

$$(5.1.5) \quad \langle a^w(x, h_j D)u(h_j), u(h_j) \rangle \rightarrow \int_{\mathbb{R}^{2n}} a(x, \xi) d\mu$$

for each symbol $a \in C_c^\infty(\mathbb{R}^{2n})$.

DEFINITION. We call μ a *semiclassical defect measure* associated with the family $u = \{u(h)\}_{0 < h \leq h_0}$.

Proof. 1. Choose $\{a_k\}_{k=0}^\infty \subset C_c^\infty(\mathbb{R}^{2n})$ to be dense in $C_c(\mathbb{R}^{2n})$. Select a sequence $h_j^1 \rightarrow 0$ such that

$$\langle a_1^w(x, h_j^1 D) u(h_j^1), u(h_j^1) \rangle \rightarrow \alpha_1.$$

Next choose a further subsequence $\{h_j^2\} \subset \{h_j^1\}$ such that

$$\langle a_2^w(x, h_j^2 D) u(h_j^2), u(h_j^2) \rangle \rightarrow \alpha_2.$$

Continue at the k^{th} step extracting a subsequence $\{h_j^k\} \subset \{h_j^{k-1}\}$ such that

$$\langle a_k^w(x, h_j^k D) u(h_j^k), u(h_j^k) \rangle \rightarrow \alpha_k.$$

A standard diagonal argument shows that for $h_j := h_j^j \rightarrow 0$ we have

$$\langle a_k^w(x, h_j D) u(h_j), u(h_j) \rangle \rightarrow \alpha_k$$

for all $k = 1, \dots$

2. Define $\Phi(a_k) := \alpha_k$. Owing to Theorem 5.1, we see for each k that

$$(5.1.6) \quad \begin{aligned} |\Phi(a_k)| &= |\alpha_k| = \lim_{h_j \rightarrow 0} |\langle a_k^w u(h_j), u(h_j) \rangle| \\ &\leq C \limsup_{h_j \rightarrow 0} \|a_k^w\|_{L^2 \rightarrow L^2} \leq C \sup_{\mathbb{R}^{2n}} |a_k|. \end{aligned}$$

The mapping Φ is bounded, linear, and densely defined. Hence it uniquely extends to a bounded linear functional on $C_c(\mathbb{R}^{2n})$, with the estimate

$$|\Phi(a)| \leq C \sup_{\mathbb{R}^{2n}} |a|$$

for all $a \in C_c(\mathbb{R}^{2n})$. The Riesz Representation Theorem (see e.g. [E-G, Section 1.8]) therefore implies the existence of a (possibly complex-valued) Radon measure on $\mathbb{R}^{2n}$ such that

$$\Phi(a) = \int_{\mathbb{R}^{2n}} a(x, \xi) d\mu. \quad \square$$

3. To show that μ is real and nonnegative, we must demonstrate that $a \geq 0$ implies

$$\int_{\mathbb{R}^{2n}} a d\mu \geq 0.$$

Now when $a \geq 0$, the sharp Gårding inequality, Theorem 4.32, implies

$$a^w(x, hD) \geq -Ch;$$

that is,

$$\langle a^w(x, hD) u(h), u(h) \rangle \geq -Ch \|u(h)\|_{L^2}^2$$

for sufficiently small $h > 0$. Let $h = h_j \rightarrow 0$, to deduce

$$\int_{\mathbb{R}^{2n}} a \, d\mu = \lim_{h_j \rightarrow 0} \langle a^w(x, h_j D) u(h_j), u(h_j) \rangle \geq 0.$$

□

REMARK. Theorem 5.2 is also valid if we replace the Weyl quantization $a^w = \text{Op}_{1/2}(a)$ by $\text{Op}_t(a)$ for any $0 \leq t \leq 1$. This is so since according to (4.3.13), we can write $\text{Op}_t(a) = a_t^w(x, hD)$ with $a_t = a + O_S(h)$. □

EXAMPLE 1: Coherent states. Fix a point (x_0, ξ_0) and define the corresponding *coherent state*

$$u(h)(x) := (\pi h)^{-\frac{n}{4}} e^{\frac{i}{h} \langle x - x_0, \xi_0 \rangle - \frac{1}{2h} |x - x_0|^2},$$

where we have normalized so that $\|u(h)\|_{L^2} = 1$. Then there exists precisely one associated semiclassical defect measure, namely

$$\mu := \delta_{(x_0, \xi_0)}.$$

To confirm this statement, we use the remark above, take $t = 1$ in the quantization, and calculate

$$\begin{aligned} & \langle a(x, hD) u(h), u(h) \rangle \\ &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(x, \xi) e^{\frac{i}{h} \langle x - y, \xi \rangle} u(h)(y) \overline{u(h)}(x) \, dy d\xi dx \\ &= \frac{2^{\frac{n}{2}}}{(2\pi h)^{\frac{3n}{2}}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(x, \xi) e^{\frac{i}{h} (\langle x - y, \xi \rangle + \langle y - x_0, \xi_0 \rangle - \langle x - x_0, \xi_0 \rangle)} \\ & \quad \times e^{-\frac{1}{2h} (|y - x_0|^2 + |x - x_0|^2)} \, dy d\xi dx \\ &= \frac{2^{\frac{n}{2}}}{(2\pi h)^{\frac{3n}{2}}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(x, \xi) e^{\frac{i}{h} \langle x - y, \xi - \xi_0 \rangle} \\ & \quad \times e^{-\frac{1}{2h} (|y - x_0|^2 + |x - x_0|^2)} \, dy d\xi dx. \end{aligned}$$

For each fixed x and ξ , the integral in y is

$$\begin{aligned} \int_{\mathbb{R}^n} e^{\frac{i}{h} \langle x - y, \xi - \xi_0 \rangle} e^{-\frac{1}{2h} |y - x_0|^2} \, dy &= e^{\frac{i}{h} \langle x - x_0, \xi - \xi_0 \rangle} \int_{\mathbb{R}^n} e^{-\frac{i}{h} \langle y, \xi - \xi_0 \rangle} e^{-\frac{1}{2h} |y|^2} \, dy \\ &= e^{\frac{i}{h} \langle x - x_0, \xi - \xi_0 \rangle} \mathcal{F} \left(e^{-\frac{1}{2h} |y|^2} \right) \left(\frac{\xi - \xi_0}{h} \right) \\ &= (2\pi h)^{\frac{n}{2}} e^{\frac{i}{h} \langle x - x_0, \xi - \xi_0 \rangle} e^{-\frac{1}{2h} |\xi - \xi_0|^2}, \end{aligned}$$

where we used formula (3.1.4) for the last equality. Therefore

$$\begin{aligned}
 & \langle a(x, hD)u(h), u(h) \rangle \\
 &= \frac{2^{\frac{n}{2}}}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(x, \xi) e^{\frac{i}{h} \langle x-x_0, \xi-\xi_0 \rangle} e^{-\frac{1}{2h}(|x-x_0|^2 + |\xi-\xi_0|^2)} dx d\xi \\
 &= a(x_0, \xi_0) \frac{2^{\frac{n}{2}}}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h} \langle x-x_0, \xi-\xi_0 \rangle} e^{-\frac{1}{2h}(|x-x_0|^2 + |\xi-\xi_0|^2)} dx d\xi + o(1) \\
 &= Ca(x_0, \xi_0) + o(1),
 \end{aligned}$$

for the constant

$$C := \frac{2^{\frac{n}{2}}}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i \langle x, \xi \rangle} e^{-\frac{1}{2}(|x|^2 + |\xi|^2)} dx d\xi.$$

Taking $a \equiv 1$ and recalling that $\|u(h)\|_{L^2} = 1$, we deduce that $C = 1$. $\square$

EXAMPLE 2: Stationary phase and defect measures. For our next example, take

$$u(h)(x) := e^{\frac{i\varphi(x)}{h}} b(x),$$

where $\varphi, b \in C^\infty$ and $\|b\|_{L^2} = 1$. Then

$$\begin{aligned}
 & \langle a(x, hD)u(h), u(h) \rangle \\
 &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(x, \xi) e^{\frac{i}{h}(\langle x-y, \xi \rangle + \varphi(y) - \varphi(x))} b(y) \overline{b(x)} dy d\xi dx.
 \end{aligned}$$

We assume $a \in C_c^\infty(\mathbb{R}^{2n})$ and apply the stationary phase. For a given value of x , define

$$\phi(y, \xi) := \langle x - y, \xi \rangle + \varphi(y) - \varphi(x).$$

Then

$$\partial_y \phi = \partial \varphi(y) - \xi, \quad \partial_\xi \phi = x - y;$$

and the Hessian matrix of ϕ is

$$\partial^2 \phi = \begin{pmatrix} \partial^2 \varphi & -I \\ -I & O \end{pmatrix}.$$

The signature of a matrix is integer-valued and consequently is invariant if we move along a curve of nonsingular matrices. Since

$$\operatorname{sgn} \begin{pmatrix} O & -I \\ -I & O \end{pmatrix} = 0,$$

it follows that

$$\operatorname{sgn} \begin{pmatrix} t\partial^2 \varphi & -I \\ -I & O \end{pmatrix} = 0$$

for $0 \leq t \leq 1$; and therefore

$$\operatorname{sgn}(\partial^2 \phi) = 0.$$

In addition, $|\det \partial^2 \phi| = 1$. Thus as $h \rightarrow 0$, the stationary phase asymptotic expression (3.5.10) implies

$$\langle a^w(x, hD)u(h), u(h) \rangle \rightarrow \int_{\mathbb{R}^n} a(x, \partial\varphi(x)) |b(x)|^2 dx = \int_{\mathbb{R}^{2n}} a(x, \xi) d\mu$$

for the semiclassical defect measure

$$\mu := |b(x)|^2 \delta_{\{\xi = \partial\varphi(x)\}} dx,$$

dx denoting the n -dimensional Lebesgue measure in the x variables. $\square$

5.2. DEFECT MEASURES AND PDE

We now assume more about the family $\{u(h)\}_{0 < h \leq h_0}$, namely that each function $u(h)$ is an approximate solution of an equation involving the operator $P(h) = p^w(x, hD)$ for some real symbol $p \in S(\langle \xi \rangle^k)$ satisfying

$$(5.2.1) \quad |p| \geq c \langle \xi \rangle^k \quad \text{if } |\xi| \geq R$$

for constants $k, c, R > 0$. We also assume that p is *independent of h* .

5.2.1. Properties of semiclassical defect measures. First, let us suppose $P(h)u(h)$ vanishes up to an $o(1)$ error term and see what we can conclude about a corresponding semiclassical defect measure μ .

THEOREM 5.3 (Support of defect measure). *Suppose that $u(h)$ satisfies*

$$(5.2.2) \quad \begin{cases} \|P(h)u(h)\|_{L^2} = o(1) & \text{as } h \rightarrow 0 \\ \|u(h)\|_{L^2} = 1. \end{cases}$$

Then

$$(5.2.3) \quad \text{spt}(\mu) \subset p^{-1}(0)$$

for any semiclassical defect measure μ associated with the family $\{u(h)\}$.

INTERPRETATION. We sometimes call $p^{-1}(0)$ the *characteristic variety* or *zero energy surface* of the symbol p . We understand (5.2.3) as saying that in the semiclassical limit $h \rightarrow 0$, all of the “mass” of the approximate solution $u(h)$ coalesces in phase space onto this set.

See Theorem 6.4 for a somewhat related assertion. $\square$

Proof. 1. Select $a \in C_c^\infty(\mathbb{R}^{2n})$ such that $\text{spt}(a) \cap p^{-1}(0) = \emptyset$. We must show

$$\int_{\mathbb{R}^{2n}} a d\mu = 0.$$

To do so, first select $q \in S(\langle \xi \rangle^k)$ such that $\text{spt}(a) \cap \text{spt}(q) = \emptyset$ and

$$|p + iq| \geq \delta \langle \xi \rangle^k > 0 \quad \text{on } \mathbb{R}^{2n}$$

for some $\delta > 0$. We can for instance choose a function $q \in C^\infty$ that is equal to one on $p^{-1}(0)$ and then modify it near the compact support of a .

2. Write $Q(h) := q^w(x, hD)$. Then Theorem 4.29 ensures us that for small enough h the operator $\langle hD \rangle^{-k}(P(h) + iQ(h))$ is invertible on L^2 .

Next, put $A(h) := a^w(x, hD)$. We observe that

$$\frac{ap}{p + iq} - a = -i \frac{aq}{p + iq}.$$

Since a and q have disjoint support, Theorems 4.24 and 4.25 imply

$$\|A(h)(P(h) + iQ(h))^{-1}P(h) - A(h)\|_{L^2 \rightarrow L^2} = O(h).$$

Therefore (5.2.2) implies

$$\|A(h)u(h)\|_{L^2} = o(1);$$

and thus

$$\langle A(h)u(h), u(h) \rangle \rightarrow 0.$$

But also

$$\langle A(h_j)u(h_j), u(h_j) \rangle = \langle a^w(x, h_j D)u(h_j), u(h_j) \rangle \rightarrow \int_{\mathbb{R}^{2n}} a \, d\mu. \quad \square$$

Now we make the stronger assumption that the error term in (5.2.2) is of order $o(h)$:

THEOREM 5.4 (Flow invariance). *Assume*

$$(5.2.4) \quad \begin{cases} \|P(h)u(h)\|_{L^2} = o(h) & \text{as } h \rightarrow 0 \\ \|u(h)\|_{L^2} = 1. \end{cases}$$

Then

$$(5.2.5) \quad \int_{\mathbb{R}^{2n}} \{p, a\} \, d\mu = 0$$

for all $a \in C_c^\infty(\mathbb{R}^{2n})$.

INTERPRETATION. Let φ_t be the flow generated by the Hamiltonian vector field H_p . Then

$$\partial_t \int_{\mathbb{R}^{2n}} \varphi_t^* a \, d\mu = \int_{\mathbb{R}^{2n}} (H_p a)(\varphi_t) \, d\mu = \int_{\mathbb{R}^{2n}} \{p, a\} \, d\mu.$$

Consequently (5.2.5) asserts that *the semiclassical defect measure μ is invariant under the Hamiltonian flow generated by p* . $\square$

The proof illustrates one of the basic principles mentioned in Chapter 1, that an assertion about Hamiltonian dynamics involving the Poisson bracket corresponds to a commutator argument at the quantum level.

Proof. Since p is real, $P(h) = p^w(x, hD)$ is selfadjoint on L^2 . Select a as above and write $A(h) = a^w(x, hD)$. Recall that $A(h) = A(h)^*$. Then

$$\begin{aligned} \langle [P(h), A(h)]u(h), u(h) \rangle &= \langle (P(h)A(h) - A(h)P(h))u(h), u(h) \rangle \\ &= \langle A(h)u(h), P(h)u(h) \rangle \\ &\quad - \langle P(h)u(h), A(h)u(h) \rangle \\ &= o(h) \end{aligned}$$

as $h \rightarrow 0$.

Note that even though p may not have compact support, $\{p, a\}$ does. Consequently the formula (4.4.19) and Theorem 4.23 imply

$$[P(h), A(h)] = \frac{h}{i} \{p, a\}^w(x, hD) + O_{L^2 \rightarrow L^2}(h^2).$$

Hence

$$\langle [P(h), A(h)]u(h), u(h) \rangle = \frac{h}{i} \langle \{p, a\}^w u(h), u(h) \rangle + o(h).$$

Divide by $h > 0$ and let $h = h_j \rightarrow 0$:

$$\int_{\mathbb{R}^{2n}} \{p, a\} d\mu = 0.$$

□

5.3. DAMPED WAVE EQUATION

This section illustrates applications of semiclassical defect measures to PDE theory. We study a damped wave equation on the torus $\mathbb{T}^n$ and learn when a damping term localized in space can induce exponential energy decay.

5.3.1. Quantization and semiclassical defect measures on the torus.

First of all, we need to extend the foregoing theory to the torus $\mathbb{T}^n$, which is the simplest compact manifold for which we can consider semiclassical quantization. We defer the general study of quantization on manifolds until Chapter 13.

NOTATION.

- (i) We identify the *torus* $\mathbb{T}^n$ with the fundamental domain

$$\mathbb{T}^n \simeq \{x \mid 0 \leq x_i < 1, 1 \leq i \leq n\} \subset \mathbb{R}^n.$$

(ii) We likewise identify functions on $\mathbb{T}^n$ with periodic functions on $\mathbb{R}^n$:

$$(5.3.1) \quad u(x+k) = u(x) \quad (k \in \mathbb{Z}^n).$$

(iii) Symbols a on $\mathbb{T}^n \times \mathbb{R}^n$ are similarly identified with symbols a on $\mathbb{R}^{2n}$ that are periodic in x :

$$(5.3.2) \quad a(x+k, \xi) = a(x, \xi) \quad (k \in \mathbb{Z}^n).$$

Operators obtained by quantizing such symbols satisfy

$$(a^w(x, hD)u)(x+k) = (a^w(x, hD)u(\cdot+k))(x)$$

and hence preserve periodicity. We note here that a periodic function u identified with $u \in L^2(\mathbb{T}^n)$ is an element of $\mathcal{S}'$. Thus the action of a^w on u is well defined.

A key fact is that these operators are also bounded on $L^2(\mathbb{T}^n)$:

THEOREM 5.5 (L^2 estimates on the torus). *Suppose that $a \in S$ satisfies the periodicity condition (5.3.2).*

Then for u satisfying (5.3.1), we have the estimate

$$(5.3.3) \quad \int_{\mathbb{T}^n} |a^w(x, hD)u|^2 dx \leq C \int_{\mathbb{T}^n} |u|^2 dx.$$

Proof. 1. We use the periodicity to write

$$a^w(x, hD)u(x) = \sum_{k \in \mathbb{Z}^n} A_k u(x), \quad x \in \mathbb{T}^n,$$

for

$$A_k u(x) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a\left(\frac{x+y-k}{2}, \xi\right) e^{\frac{i}{h}\langle x-y+k, \xi \rangle} u(y) dy d\xi.$$

Letting $\mathbf{1}_{\mathbb{T}^n}$ denote the characteristic function of $\mathbb{T}^n \subset \mathbb{R}^n$ and observing that

$$a\left(\frac{x+y-k}{2}, \xi\right) = a\left(\frac{x+k+y}{2} - k, \xi\right) = a\left(\frac{x+k+y}{2}, \xi\right),$$

we can write

$$(5.3.4) \quad A_k = \mathbf{1}_{\mathbb{T}^n} T_{-k} a^w \mathbf{1}_{\mathbb{T}^n}, \quad T_{-k} v(x) := v(x+k).$$

2. We next claim that for $|k| > 2$

$$(5.3.5) \quad \|A_k\|_{L^2(\mathbb{T}^n) \rightarrow L^2(\mathbb{T}^n)} = O(h^\infty \langle k \rangle^{-\infty}).$$

To prove this, we note that for $|k| > 2$

$$e^{\frac{i}{h}\langle x-y+k, \xi \rangle} = h^{2N} |x-y+k|^{-2N} |D_\xi|^{2N} e^{\frac{i}{h}\langle x-y+k, \xi \rangle}$$

for $x, y \in \mathbb{T}^n$. Hence

$$A_k = \mathbf{1}_{\mathbb{T}^n} T_{-k} \tilde{A}_k \mathbf{1}_{\mathbb{T}^n},$$

where

$$\tilde{A}_k v(x) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \tilde{a}_k(x, y, \xi) e^{\frac{i}{h} \langle x-y, \xi \rangle} v(y) dy d\xi$$

for

$$\tilde{a}_k(x, y, \xi) := \chi(x-k)\chi(y)h^{2N}|x-y|^{-2N}(|D_\xi|^{2N}a)\left(\frac{x+y}{2}, \xi\right).$$

Here $\chi \in C_c^\infty(\mathbb{R}^n)$ is equal to 1 near $\mathbb{T}^n$. Theorem 4.20 then shows that

$$\tilde{A}_k = b_k^w(x, hD)$$

for a symbol $b_k \in S$, seminorms of which in S are bounded independently of k . Theorem 4.23 therefore implies (5.3.5).

3. Since (5.3.4) shows L^2 boundedness for any fixed k , the estimate (5.3.3) follows. $\square$

Using Theorem 5.5, we can now mimic the theory developed earlier. We assume that $P = p^w(x, hD)$ where $p \in S(\langle \xi \rangle^m)$ is periodic in x :

$$p(x+k, \xi) = p(x, \xi), \quad k \in \mathbb{Z}^n.$$

Hence $P : C^\infty(\mathbb{T}^n) \rightarrow C^\infty(\mathbb{T}^n)$.

THEOREM 5.6 (Defect measures on the torus). *Let $\{u(h)\}_{0 < h < h_0} \subset L^2(\mathbb{T}^n)$ be a family of functions with $\|u(h)\|_{L^2} = 1$.*

(i) *There exists a nonnegative Radon measure μ on $\mathbb{T}^n \times \mathbb{R}^n$ and a sequence $h_j \rightarrow 0$ such that*

$$(5.3.6) \quad \langle a^w(x, h_j D) u(h_j), u(h_j) \rangle \rightarrow \int_{\mathbb{T}^n \times \mathbb{R}^n} a(x, \xi) d\mu$$

for each symbol $a \in C_c^\infty(\mathbb{T}^n \times \mathbb{R}^n)$.

(ii) *If $\|P(h)u(h)\|_{L^2} = o(1)$, then*

$$\text{spt}(\mu) \subset p^{-1}(0).$$

(iii) *If $\|P(h)u(h)\|_{L^2} = o(h)$, then*

$$\int_{\mathbb{R}^{2n}} \{p, a\} d\mu = 0$$

for all $a \in C_c^\infty(\mathbb{T}^n \times \mathbb{R}^n)$.

5.3.2. A damped wave equation. In this section $\mathbb{T}^n$ denotes the flat n -dimensional torus. We consider now the initial-value problem

$$(5.3.7) \quad \begin{cases} (\partial_t^2 + a\partial_t - \Delta)u = 0 & \text{on } \mathbb{T}^n \times \{t > 0\} \\ u = 0, \quad u_t = f & \text{on } \mathbb{T}^n \times \{t = 0\}, \end{cases}$$

in which the smooth function $a = a(x)$ is nonnegative and thus represents a damping mechanism, as we will see. For simplicity we assume that f and hence u are real-valued.

DEFINITION. The *energy* at time t is

$$E(t) := \frac{1}{2} \int_{\mathbb{T}^n} (\partial_t u)^2 + |\partial_x u|^2 dx.$$

LEMMA 5.7 (Elementary energy estimates).

- (i) If $a \equiv 0$, $t \mapsto E(t)$ is constant.
- (ii) If $a \geq 0$, $t \mapsto E(t)$ is nonincreasing.

Proof. These assertions follow easily from the calculation

$$\begin{aligned} E'(t) &= \int_{\mathbb{T}^n} \partial_t u \partial_t^2 u + \langle \partial_x u, \partial_{xt}^2 u \rangle dx \\ &= \int_{\mathbb{T}^n} \partial_t u (\partial_t^2 u - \Delta u) dx = - \int_{\mathbb{T}^n} a (\partial_t u)^2 dx \leq 0. \quad \square \end{aligned}$$

Our eventual goal is to show that if the support of the damping term a is large enough, then we have exponential energy decay for our solution of the wave equation (5.3.7). Here is the key assumption:

Dynamical hypothesis.

$$(5.3.8) \quad \begin{cases} \text{There exists a time } T > 0 \text{ such that any} \\ \text{trajectory of the Hamiltonian vector field of} \\ p(x, \xi) = |\xi|^2, \text{ starting at time 0 with } |\xi| = 1, \\ \text{intersects the set } \{a > 0\} \text{ by the time } T. \end{cases}$$

Equivalently, for each initial point $z = (x, \xi) \in \mathbb{T}^n \times \mathbb{R}^n$, with $|\xi| = 1$, we have

$$\langle a \rangle_T = \oint_0^T a(x + t\xi) dt := \frac{1}{T} \int_0^T a(x + t\xi) dt > 0,$$

the slash through the first integral denoting an average.

5.3.3. Resolvent estimates.

MOTIVATION. Since the damping term a in general depends upon x , we cannot use the Fourier transform (or Fourier series) in x to solve (5.3.7). Instead we take a partial Fourier transform in t :

$$(5.3.9) \quad \check{u}(x, \tau) := \int_0^\infty e^{it\tau} u(x, t) dt \quad (\operatorname{Im} \tau > 0).$$

We note that $|e^{it\tau}| = e^{-t\operatorname{Im} \tau}$ so that the integrand decays for $\operatorname{Im} \tau > 0$.

The partial Fourier transform satisfies

$$\begin{aligned} \Delta \check{u}(\cdot, \tau) &= \int_0^\infty e^{it\tau} \Delta u dt = \int_0^\infty e^{it\tau} (\partial_t^2 u + a \partial_t u) dt \\ &= \int_0^\infty ((i\tau)^2 - ai\tau) e^{-it\tau} u dt - f = -(\tau^2 + ai\tau) \check{u} - f. \end{aligned}$$

Consequently,

$$(5.3.10) \quad P(\tau) \check{u} := (-\Delta - i\tau a - \tau^2) \check{u} = f.$$

Now take τ to be complex and define

$$(5.3.11) \quad P(z, h) := -h^2 \Delta - i\sqrt{z} h a - z = h^2 P(h^{-1} \sqrt{z})$$

for the rescaled variable

$$(5.3.12) \quad z = \tau^2 h^2.$$

For $\operatorname{Re} z > 0$ we choose the branch of the square root so that $\operatorname{Re} \sqrt{z} > 0$.

Then (5.3.10) reads

$$P(z, h) \check{u} = h^2 f;$$

and so, if $P(z, h)$ is invertible,

$$(5.3.13) \quad \check{u} = h^2 P(z, h)^{-1} f.$$

We therefore need to study the inverse of the operator $P(z, h)$, in particular for complex z . □

We start with a general result about the inverse of $P(\tau)$:

THEOREM 5.8 (Meromorphy of the resolvent). *The operator $P(\tau)$ is invertible except at a discrete set of points in $\mathbb{C}$. More precisely,*

$$P(\tau)^{-1} : L^2(\mathbb{T}^n) \rightarrow H^2(\mathbb{T}^n)$$

is a meromorphic family of operators with poles of finite rank. It has no poles for $\tau \in \mathbb{R} \setminus \{0\}$, a simple pole at $\tau = 0$, and is holomorphic for $\operatorname{Im} \tau > 0$ and for $\operatorname{Im} \tau < -\|a\|_\infty$.

We will invoke the theory of Fredholm operators, as developed in Appendix D.

Proof. 1. We first note that the pseudodifferential calculus and Theorem 5.5 show that $(-h^2\Delta + 1)^{-1} : L^2 \rightarrow H^2$ exists. That means that for $\tau_0 = is$ and large $|s|$, we have $(-\Delta - \tau_0^2)^{-1} : L^2 \rightarrow H^2$.

Consequently,

$$P(\tau) = (-\Delta - \tau_0^2)(I + (-\Delta - \tau_0^2)^{-1}(\tau_0^2 - \tau^2 - i\tau a(x)));$$

and the existence of the inverse of $P(\tau)$ is equivalent to the invertibility of

$$I + K(\tau)$$

on $L^2(\mathbb{T}^2)$, where

$$K(\tau) := (-\Delta - \tau_0^2)^{-1}(\tau_0^2 - \tau^2 - i\tau a(x)).$$

2. Theorem 4.28 shows that $K(\tau)$ is compact and hence $I + K(\tau)$ is a holomorphic family of Fredholm operators. If $|\tau_0| \gg 1$, then

$$\|K(\tau_0)\|_{L^2 \rightarrow H^2} < 1,$$

and therefore $I + K(\tau_0)$ is invertible. Theorem D.4 gives the meromorphy of $\tau \mapsto P(\tau)^{-1}$.

For $\text{Im } \tau > 0$ we use the inequality $a \geq 0$ to see that

$$\begin{aligned} \|P(\tau)u\|_{L^2}\|u\|_{L^2} &\geq \max(|\text{Im}\langle P(\tau)u, u \rangle|, |\text{Re}\langle P(\tau)u, u \rangle|) \\ &\geq \max(2|\text{Re } \tau|\text{Im } \tau, (\text{Im } \tau)^2 - (\text{Re } \tau)^2)\|u\|_{L^2}^2 \\ &\geq \frac{1}{2}(\text{Im } \tau)^2\|u\|_{L^2}^2. \end{aligned}$$

This proves invertibility, since the meromorphy implies that the Fredholm alternative holds. A similar estimate is valid for $\text{Im } \tau < -\|a\|_{L^\infty}$.

3. Finally we exclude the possibility of a real nonzero τ satisfying

$$(5.3.14) \quad (-\Delta - \tau^2 - i\tau a)u = 0$$

for some $u \neq 0$. Multiplying by $\bar{u}$, integrating, and taking the imaginary part shows that

$$\int_{\mathbb{T}^n} a|u|^2 dx = 0.$$

Since $a \geq 0$, this implies that $u \equiv 0$ on $\text{spt } a$. Hence $(-\Delta - \tau^2)u = 0$. But this is impossible owing to unique continuation results which we will prove in Section 7.2, since $\text{spt } a$ has a nonempty interior. The Fredholm alternative now guarantees that $P(\tau)^{-1}$ has no pole on the real axis. $\square$

With these general considerations in hand, we next utilize the dynamical hypothesis (5.3.8) to derive some subtle resolvent estimates:

THEOREM 5.9 (Resolvent bounds). *Under the dynamical assumption (5.3.8), there exist constants $\alpha, C, h_0 > 0$ such that*

$$(5.3.15) \quad \|P(z, h)^{-1}\|_{L^2 \rightarrow L^2} \leq \frac{C}{h}$$

for

$$(5.3.16) \quad |\operatorname{Im} z| \leq \alpha h, \quad |z - 1| \leq \alpha, \quad 0 < h \leq h_0.$$

Proof. 1. It is enough to show that there exists a constant C such that

$$\|u\|_{L^2} \leq \frac{C}{h} \|P(z, h)u\|_{L^2}$$

for all $u \in L^2$, provided z and h satisfy (5.3.16).

We argue by contradiction. If the assertion were false, then for $m = 1, 2, \dots$ there would exist $z_m \in \mathbb{C}$, $0 < h_m \leq 1/m$, and functions u_m in L^2 such that

$$\|P(z_m, h_m)u_m\|_{L^2} \leq \frac{h_m}{m} \|u_m\|_{L^2}, \quad |\operatorname{Im} z_m| \leq \frac{h_m}{m}, \quad |z_m - 1| \leq \frac{1}{m}.$$

We may assume $\|u_m\|_{L^2} = 1$. Then

$$(5.3.17) \quad P(z_m, h_m)u_m = o(h_m).$$

Also,

$$(5.3.18) \quad z_m \rightarrow 1, \quad \operatorname{Im}(z_m) = o(h_m).$$

2. Let μ be a semiclassical defect measure associated with $\{u_m\}_{m=1}^\infty$. Then Theorem 5.3 implies for the symbol $p := |\xi|^2 - 1$ that

$$\operatorname{spt}(\mu) \subset p^{-1}(0) = \{|\xi|^2 = 1\}.$$

But $\langle u_m, u_m \rangle = 1$, and so

$$(5.3.19) \quad \int_{\mathbb{T}^n \times \mathbb{R}^n} d\mu = 1.$$

We will derive a contradiction to this.

3. Hereafter write $P_m := P(z_m, h_m)$. Then

$$\begin{aligned} P_m &= -h_m^2 \Delta - i\sqrt{z_m} h_m a - z_m, \\ P_m^* &= -h_m^2 \Delta + i\sqrt{\bar{z}_m} h_m a - \bar{z}_m; \end{aligned}$$

and therefore

$$(5.3.20) \quad P_m - P_m^* = -i(\sqrt{z_m} + \sqrt{\bar{z}_m}) h_m a - z_m + \bar{z}_m = -2i h_m a + o(h_m),$$

since (5.3.18) implies that $\sqrt{z_m} + \sqrt{\bar{z}_m} = 2 + o(1)$ and that $-z_m + \bar{z}_m = -2i\text{Im}(z_m) = o(h_m)$.

Now select $b \in C_c^\infty(\mathbb{T}^n \times \mathbb{R}^n)$ and set $B_m := b^w(x, h_m D)$. Then $B_m = B_m^*$. Using (5.3.17) and (5.3.20), we calculate that

$$\begin{aligned} o(h_m) &= 2i \text{Im} \langle B_m P_m u_m, u_m \rangle = \langle B_m P_m u_m, u_m \rangle - \langle u_m, B_m P_m u_m \rangle \\ &= \langle (B_m P_m - P_m^* B_m) u_m, u_m \rangle \\ &= \langle [B_m, P_m] u_m, u_m \rangle \\ &\quad + \langle (P_m - P_m^*) B_m u_m, u_m \rangle \\ &= \frac{h_m}{i} \langle \{b, p\}^w u_m, u_m \rangle \\ &\quad - 2h_m i \langle (ab)^w u_m, u_m \rangle + o(h_m). \end{aligned}$$

Divide by h_m/i and let $h_m \rightarrow 0$, through a subsequence if necessary, to discover that

$$(5.3.21) \quad \int_{\mathbb{T}^n \times \mathbb{R}^n} \{p, b\} + 2ab \, d\mu = 0.$$

We will build a function b so that $\{p, b\} + 2ab > 0$ on $\text{spt}(\mu)$. This will imply $\int_{\mathbb{T}^n \times \mathbb{R}^n} d\mu = 0$, a contradiction to (5.3.19).

4. For $(x, \xi) \in \mathbb{T}^n \times \mathbb{R}^n$, with $|\xi| = 1$, define

$$c(x, \xi) := \int_0^T (T-t) a(x + \xi t) \, dt,$$

where T is the time from the dynamical hypothesis (5.3.8). Hence

$$\begin{aligned} \langle \xi, \partial_x c \rangle &= \int_0^T (T-t) \langle \xi, \partial_x a(x + \xi t) \rangle \, dt \\ &= \int_0^T (T-t) \partial_t a(x + \xi t) \, dt \\ &= \int_0^T a(x + \xi t) \, dt - a(x) \\ &= \langle a \rangle_T - a. \end{aligned}$$

Let

$$b := e^c \chi(p),$$

where $\chi \in C_c^\infty(\mathbb{R})$ is equal to 1 near 0. Then

$$\langle \xi, \partial_x b \rangle = e^c \langle \xi, \partial_x c \rangle \chi(p) = e^c \langle a \rangle_T \chi(p) - a e^c \chi(p)$$

since $H_p(\chi(p)) = 0$. Consequently

$$\{p, b\} + 2ab = 2\langle \xi, \partial_x b \rangle + 2ab = 2e^c \langle a \rangle_T \chi(p) > 0 \quad \text{on } p^{-1}(0),$$

as desired. □

5.3.4. Energy decay.

THEOREM 5.10 (Exponential energy decay). *Assume the dynamic hypothesis (5.3.8) and suppose u solves the wave equation with damping (5.3.7).*

Then there exist constants $C, \beta > 0$ such that

$$(5.3.22) \quad E(t) \leq C e^{-\beta t} \|f\|_{L^2} \quad \text{for all times } t > 0.$$

MOTIVATION. The following calculations are based upon this idea: to get decay estimates of g on the positive real axis, we estimate $\check{g}(\tau) := \hat{g}(-\tau)$ in a complex strip $|\operatorname{Im} z| \leq \alpha$. Then if $\beta < \alpha$,

$$\widehat{e^{\beta t} g}(-\tau) = \int_{-\infty}^{\infty} e^{\beta t} g(t) e^{it\tau} dt = \int_{-\infty}^{\infty} g(t) e^{it(\tau - i\beta)} dt = \check{g}(\tau - i\beta).$$

Hence our L^2 estimate of $\check{g}(\tau - i\beta)$ will imply exponential decay of $g(t)$ for $t \rightarrow \infty$. $\square$

Proof. 1. Recall from (5.3.11), (5.3.12) that

$$P(\tau) = h^{-2} P(z, h) \quad \text{for } \tau^2 = h^{-2} z.$$

First we assert that there exists $\gamma > 0$ such that

$$(5.3.23) \quad \|P(\tau)^{-1}\|_{L^2 \rightarrow H^1} \leq C \quad \text{for } |\operatorname{Im} \tau| \leq \gamma, \quad |\tau| > 1/\gamma,$$

where, here and below, $L^2 = L^2(\mathbb{T}^n)$, $H^k = H^k(\mathbb{T}^n)$.

To prove (5.3.23), we note that provided the inequalities (5.3.16) hold, then

$$\|h^2 \Delta P(z, h)^{-1} u\|_{L^2} = \|(i\sqrt{z} h a + z) P(z, h)^{-1} u + u\|_{L^2} \leq \frac{C}{h} \|u\|_{L^2},$$

the last inequality holding according to Theorem 5.9. Since

$$\|u\|_{H^2} \leq C \|\Delta u\|_{L^2} + \|u\|_{L^2},$$

we conclude that

$$(5.3.24) \quad \|h^2 P(z, h)^{-1} u\|_{H^2} \leq \frac{C}{h} \|u\|_{L^2}.$$

We now put

$$z = h^2 \tau^2, \quad \tau = \lambda + i\mu, \quad h = \frac{1}{\lambda},$$

so that

$$|\operatorname{Im} z| \leq \alpha h, \quad |z - 1| \leq \alpha \quad \text{if } |\mu| \leq \gamma, \quad |\lambda| > 1/\gamma,$$

for some sufficiently small γ , and so the inequalities (5.3.16) hold. Hence for $|\mu| \leq \gamma$ and $|\lambda| > 1/\gamma$, (5.3.24) implies

$$(5.3.25) \quad \|P(\tau)^{-1} u\|_{H^2} \leq C |\tau| \|u\|_{L^2}.$$

A rescaled version of (5.3.15) reads

$$\|P(\tau)^{-1}u\|_{L^2} \leq \frac{C}{|\tau|}\|u\|_{L^2}.$$

Since $\|v\|_{H^1(\mathbb{T}^n)} \leq \|v\|_{L^2(\mathbb{T}^n)}^{\frac{1}{2}}\|v\|_{H^2(\mathbb{T}^n)}^{\frac{1}{2}}$, this and (5.3.25) give

$$\|P(\tau)^{-1}u\|_{H^1} \leq C\|u\|_{L^2}$$

for $|\operatorname{Im} \tau| \leq \gamma$ and $|\tau| > 1/\gamma$.

This proves (5.3.23) except for a bounded range of τ 's. Theorem 5.8 shows that the set of poles of $P(\tau)^{-1}$ is discrete and there are no poles on the real axis other than $\tau = 0$. We conclude that there exists $\beta_0 > 0$ such that

$$(5.3.26) \quad \sup_{\tau \in \mathbb{R}} \|P(\tau - i\beta)^{-1}\|_{L^2 \rightarrow H^1} \leq C\beta, \quad 0 < \beta < \beta_0.$$

2. Next select $\chi : \mathbb{R} \rightarrow \mathbb{R}$, $\chi = \chi(t)$, such that

$$0 \leq \chi \leq 1, \quad \chi \equiv 1 \text{ on } [1, \infty), \quad \chi \equiv 0 \text{ on } (-\infty, 0).$$

Then if $u_1 := \chi u$, we have

$$(5.3.27) \quad (\partial_t^2 + a\partial_t - \Delta)u_1 = g_1,$$

for

$$(5.3.28) \quad g_1 := \chi''u + 2\chi'\partial_t u + a(x)\chi'u.$$

Note that $u_1(t) = 0$ for $t \leq 0$, and observe also that the support of g_1 lies within $\mathbb{T}^n \times [0, 1]$. Furthermore, using energy estimates in Lemma 5.7, we see that

$$(5.3.29) \quad \begin{aligned} &\|g_1\|_{L^2(\mathbb{R}_+; L^2)} \\ &\leq C(\|u\|_{L^2((0,1); L^2)} + \|\partial_t u\|_{L^2((0,1); L^2)}) \leq C\|f\|_{L^2}. \end{aligned}$$

Now take the Fourier transform of (5.3.27) in time, as in (5.3.9),

$$P(\tau)\check{u}_1(\tau) = \check{g}_1(\tau), \quad \check{w}(\tau, x) := \hat{w}(-\tau, x).$$

Then

$$(5.3.30) \quad \check{u}_1(\tau) = P(\tau)^{-1}\check{g}_1(\tau),$$

where, in principle, we allow the left-hand side to have a pole at $\tau = 0$.

3. We now deduce exponential decay. For that, let us fix β satisfying the condition in (5.3.26). Noting that u_1 is supported in $t > 0$, we use

Plancherel's Theorem to compute

$$\begin{aligned}
 \|e^{\beta t} u_1\|_{L^2(\mathbb{R}_+; H^1)} &= (2\pi)^{-\frac{1}{2}} \|\widehat{e^{\beta t} u_1}\|_{L^2(\mathbb{R}; H^1)} \\
 &= (2\pi)^{-\frac{1}{2}} \|\check{u}_1(\cdot - i\beta)\|_{L^2(\mathbb{R}; H^1)} \\
 &= (2\pi)^{-\frac{1}{2}} \|P(\cdot - i\beta)^{-1} \check{g}_1(\cdot - i\beta)\|_{L^2(\mathbb{R}; H^1)} \\
 &\leq C \|\check{g}_1(\cdot - i\beta)\|_{L^2(\mathbb{R}; L^2)}.
 \end{aligned}$$

Since g_1 is compactly supported in t , we also see that

$$\check{g}_1(\tau - i\beta) = \widehat{e^{\beta t} g_1}(-\tau);$$

and hence

$$\begin{aligned}
 \|e^{\beta t} u_1\|_{L^2(\mathbb{R}_+; H^1)} &\leq C' \|e^{\beta t} g_1\|_{L^2(\mathbb{R}; L^2)} \\
 &\leq C'' \|g_1\|_{L^2(\mathbb{R}; L^2)} \leq C \|f\|_{L^2}.
 \end{aligned}$$

Since $u_1 = \chi u$, it follows that

$$(5.3.31) \quad \|e^{\beta t} u\|_{L^2((1, \infty); H^1)} \leq C \|f\|_{L^2}.$$

4. Finally, fix $T > 3$ and

$$\chi_T := \chi(t - T + 1),$$

where χ is as in step 2. Let $u_2 = \chi_T u$. Then

$$(5.3.32) \quad (\partial_t^2 + a\partial_t - \Delta)u_2 = g_2,$$

for

$$(5.3.33) \quad g_2 := \chi_T'' u + 2\chi_T' \partial_t u + a\chi_T' u.$$

Therefore $\text{spt}(g_2) \subset \mathbb{T}^n \times (T - 1, T)$.

Define

$$E_2(t) := \frac{1}{2} \int_{\mathbb{T}^n} (\partial_t u_2)^2 + |\partial_x u_2|^2 dx,$$

where we recall that we assumed, without loss of generality, that u is real-valued.

Modifying the calculations in the proof of Lemma 5.7, we use (5.3.32) and (5.3.33) to compute

$$\begin{aligned}
 E_2'(t) &= \int_{\mathbb{T}^n} (\partial_t u_2 \partial_t^2 u_2 + \langle \partial_x u_2, \partial_{x_t}^2 u_2 \rangle) dx \\
 &= \int_{\mathbb{T}^n} \partial_t u_2 (\partial_t^2 u_2 - \Delta u_2) dx \\
 &= - \int_{\mathbb{T}^n} a(\partial_t u_2)^2 dx + \int_{\mathbb{T}^n} \partial_t u_2 g_2 dx
 \end{aligned}$$

$$\begin{aligned}
&\leq C \int_{\mathbb{T}^n} |\partial_t u_2| (|\partial_t u| + |u|) dx \\
&\leq CE_2(t) + C \int_{\mathbb{T}^n} (u^2 + (\partial_t u)^2) dx.
\end{aligned}$$

Since $E_2(T-1) = 0$ and $E_2(T) = E(T)$, Gronwall's inequality ([**E**, Appendix B]) implies that

$$(5.3.34) \quad E(T) \leq C \left(\|u\|_{L^2((T-1,T);L^2)}^2 + \|\partial_t u\|_{L^2((T-1,T);L^2)}^2 \right).$$

5. We need to control the right-hand term in (5.3.34). For this, select $\chi_1 : \mathbb{R} \rightarrow \mathbb{R}$, such that

$$\begin{cases} 0 \leq \chi_1 \leq 1 \\ \chi_1 \equiv 0 \text{ for } t \leq T-2 \text{ and } t \geq T+1 \\ \chi_1 \equiv 1 \text{ for } T-1 \leq t \leq T. \end{cases}$$

We multiply the wave equation (5.3.7) by $\chi_1(t)^2 u(x, t)$ and integrate by parts to find

$$\begin{aligned}
0 &= \int_{T-2}^{T+1} \int_{\mathbb{T}^n} \chi_1^2 u (\partial_t^2 u + a \partial_t u - \Delta u) dx dt \\
&= \int_{T-2}^{T+1} \int_{\mathbb{T}^n} -\chi_1^2 (\partial_t u)^2 - 2\chi_1 \chi_1' u \partial_t u + \chi_1^2 a u \partial_t u + \chi_1^2 |\partial_x u|^2 dx dt.
\end{aligned}$$

From this identity we derive the estimate

$$\|\partial_t u\|_{L^2((T-1,T);L^2)} \leq C \|u\|_{L^2((T-2,T+1);H^1)}.$$

This, (5.3.34), and (5.3.31) therefore imply

$$E(T) \leq C \|u\|_{L^2((T-2,T+1);H^1)}^2 \leq C e^{-2\beta T} \|f\|_{L^2}^2,$$

as asserted. □

5.4. NOTES

Semiclassical defect measures were introduced independently in Gérard [**Ge**] and Lions–Paul [**L-P**]; see also Tartar [**T**] and Gérard–Leichtnam [**Ge-Le**].

Theorem 5.10 is due to Rauch–Taylor [**R-T**], but the proof here follows Lebeau [**L**]. To pass from the resolvent estimate to the energy decay, we also use some ideas of Morawetz, explained to us by N. Burq. The results extend with no particular difficulty if $\mathbb{T}^n$ is replaced by a general compact Riemannian manifold; see Theorem 14.4.

EIGENVALUES AND EIGENFUNCTIONS

- 6.1 The harmonic oscillator
- 6.2 Symbols and eigenfunctions
- 6.3 Spectrum and resolvents
- 6.4 Weyl's Law
- 6.5 Notes

In this chapter we are given the *potential* $V : \mathbb{R}^n \rightarrow \mathbb{R}$ and investigate how the symbol

$$(6.0.1) \quad p(x, \xi) = |\xi|^2 + V(x)$$

provides interesting information about the corresponding operator

$$(6.0.2) \quad P(h) := P(x, hD) = -h^2 \Delta + V.$$

We will focus mostly upon learning how p controls the asymptotic distribution of the eigenvalues of $P(h)$ in the semiclassical limit $h \rightarrow 0$.

6.1. THE HARMONIC OSCILLATOR

We investigate first the simplest case of a quadratic potential and, to simplify even more, begin in one dimension. So suppose that $n = 1$, $h = 1$, and $V(x) = x^2$. Thus we start with the *one-dimensional quantum harmonic oscillator*, meaning the operator

$$(6.1.1) \quad P_0 := -\partial^2 + x^2, \quad \text{where } \partial = \frac{d}{dx}.$$

6.1.1. Eigenvalues and eigenfunctions of P_0 . We can as follows employ certain auxiliary first-order differential operators to compute explicitly the eigenvalues and eigenfunctions for P_0 .

NOTATION. Let us write

$$(6.1.2) \quad A_+ := D_x + ix, \quad A_- := D_x - ix,$$

where $D_x = \frac{1}{i}\partial$, and call A_+ the *creation operator* and A_- the *annihilation operator*. (This terminology is from particle physics.)

LEMMA 6.1 (Properties of $A_{\pm}$). *The creation and annihilation operators satisfy these identities:*

$$(6.1.3) \quad A_+^* = A_-, \quad A_-^* = A_+,$$

$$(6.1.4) \quad P_0 = A_+A_- + 1 = A_-A_+ - 1.$$

Proof. It is easy to check that $D_x^* = D_x$ and $(ix)^* = -ix$. Furthermore,

$$\begin{aligned} A_+A_-u &= (D_x + ix)(D_x - ix)u \\ &= \left(\frac{1}{i}\partial_x + ix\right)\left(\frac{1}{i}u_x - ixu\right) \\ &= -u_{xx} - (xu)_x + xu_x + x^2u \\ &= -u_{xx} - u - xu_x + xu_x + x^2u \\ &= P_0u - u; \end{aligned}$$

and similarly,

$$\begin{aligned} A_-A_+u &= (D_x - ix)(D_x + ix)u \\ &= \left(\frac{1}{i}\partial_x - ix\right)\left(\frac{1}{i}u_x + ixu\right) \\ &= -u_{xx} + (xu)_x - xu_x + x^2u \\ &= P_0u + u. \end{aligned}$$

□

We can use $A_{\pm}$ to find all the eigenvalues and eigenfunctions of P_0 :

THEOREM 6.2 (Eigenvalues and eigenfunctions for the harmonic oscillator).

(i) *We have*

$$\langle P_0u, u \rangle \geq \|u\|_{L^2}^2$$

for all $u \in C_c^\infty(\mathbb{R}^n)$. That is,

$$P_0 \geq 1.$$

(ii) *The function*

$$v_0 =: e^{-\frac{x^2}{2}}$$

is an eigenfunction corresponding to the smallest eigenvalue 1.

(iii) *Set*

$$v_n := A_+^n v_0$$

for $n = 1, 2, \dots$. Then

$$(6.1.5) \quad P_0 v_n = (2n + 1)v_n.$$

(iv) *Define the normalized eigenfunctions*

$$u_n := \frac{v_n}{\|v_n\|_{L^2}}.$$

Then

$$(6.1.6) \quad u_n(x) = H_n(x) e^{-\frac{x^2}{2}}$$

where $H_n(x) = c_n x^n + \dots + c_0$ ($c_n \neq 0$) is a polynomial of degree n .

(v) *We have*

$$\langle u_n, u_m \rangle = \delta_{nm} \quad (n, m \in \mathbb{N});$$

and furthermore, the collection of eigenfunctions $\{u_n\}_{n=0}^\infty$ is complete in $L^2(\mathbb{R}^n)$.

REMARKS. (i) The functions H_n mentioned in assertion (iv) are the *Hermite polynomials*.

(ii) The completeness in assertion (v) shows that we have found *all* the eigenvalues of the harmonic oscillator.

Proof. 1. Note that

$$[D_x, x]u = \frac{1}{i}(xu)_x - \frac{x}{i}u_x = \frac{u}{i},$$

and consequently $i[D_x, x] = 1$. Therefore

$$\begin{aligned} \|u\|_{L^2}^2 &= \langle i[D_x, x]u, u \rangle \leq 2\|xu\|_{L^2}\|D_x u\|_{L^2} \\ &\leq \|xu\|_{L^2}^2 + \|D_x u\|_{L^2}^2 = \langle P_0 u, u \rangle. \end{aligned}$$

Next, observe that

$$A_- v_0 = \frac{1}{i} \left(e^{-\frac{x^2}{2}} \right)_x - ix e^{-\frac{x^2}{2}} = 0,$$

so that $P_0 v_0 = (A_+ A_- + 1)v_0 = v_0$.

2. We can further calculate that

$$\begin{aligned}
 P_0 v_n &= (A_+ A_- + 1) A_+ v_{n-1} \\
 &= A_+ (A_- A_+ - 1) v_{n-1} + 2 A_+ v_{n-1} \\
 &= A_+ P_0 v_{n-1} + 2 A_+ v_{n-1} \\
 &= (2n - 1) A_+ v_{n-1} + 2 A_+ v_{n-1} \quad (\text{by induction}) \\
 &= (2n + 1) v_n.
 \end{aligned}$$

The form (6.1.6) of v_n, u_n follows by induction.

3. Also note that

$$\begin{aligned}
 [A_-, A_+] &= A_- A_+ - A_+ A_- \\
 &= (P_0 + 1) - (P_0 - 1) = 2.
 \end{aligned}$$

Hence if $m > n$,

$$\begin{aligned}
 \langle v_n, v_m \rangle &= \langle A_+^n v_0, A_+^m v_0 \rangle \\
 &= \langle A_-^m A_+^n v_0, v_0 \rangle \quad (\text{since } A_- = A_+^*) \\
 &= \langle A_-^{m-1} (A_+ A_- + 2) A_+^{n-1} v_0, v_0 \rangle.
 \end{aligned}$$

After finitely many steps, the foregoing equals

$$\langle (\dots) A_- v_0, v_0 \rangle = 0,$$

since $A_- v_0 = 0$. Alternatively, we can simply note that $\langle P_0 v_n, v_m \rangle = \langle v_n, P_0 v_m \rangle$, $P_0 v_k = (2k + 1) v_k$, $k = m, n$.

4. Finally, we demonstrate that the collection of eigenfunctions that we have found spans L^2 . Suppose $\langle u_n, g \rangle = 0$ for $n = 0, 1, 2, \dots$; we must show that $g \equiv 0$.

Now since $H_n(x) = c_n x^n + \dots$, with $c_n \neq 0$, we have

$$\int_{-\infty}^{\infty} g(x) e^{-\frac{x^2}{2}} p(x) dx = 0$$

for each polynomial p . Hence

$$\int_{-\infty}^{\infty} g(x) e^{-\frac{x^2}{2}} e^{-ix\xi} dx = \int_{-\infty}^{\infty} g(x) e^{-\frac{x^2}{2}} \sum_{k=0}^{\infty} \frac{(-ix\xi)^k}{k!} dx;$$

and so $\mathcal{F} \left(g e^{-\frac{x^2}{2}} \right) \equiv 0$. This implies $g e^{-\frac{x^2}{2}} \equiv 0$ and consequently $g \equiv 0$. $\square$

6.1.2. Higher dimensions, rescaling. Suppose now that $n > 1$, and write

$$(6.1.7) \quad P_0 := -\Delta + |x|^2;$$

this is the n -dimensional quantum harmonic oscillator. We also define

$$u_\alpha(x) := \prod_{j=1}^n u_{\alpha_j}(x_j) = \prod_{j=1}^n H_{\alpha_j}(x_j) e^{-\frac{|x|^2}{2}}$$

for each multiindex $\alpha = (\alpha_1, \dots, \alpha_n)$. Then

$$P_0 u_\alpha = (-\Delta + |x|^2) u_\alpha = (2|\alpha| + n) u_\alpha,$$

for $|\alpha| = \alpha_1 + \dots + \alpha_n$. Hence u_α is an eigenfunction of P_0 corresponding to the eigenvalue $2|\alpha| + n$.

We next restore the parameter $h > 0$ by setting

$$(6.1.8) \quad P_0(h) := -h^2 \Delta + |x|^2,$$

$$(6.1.9) \quad u_\alpha(h)(x) := h^{-\frac{n}{4}} \prod_{j=1}^n H_{\alpha_j} \left(\frac{x_j}{\sqrt{h}} \right) e^{-\frac{|x|^2}{2h}},$$

and

$$(6.1.10) \quad E_\alpha(h) := (2|\alpha| + n)h.$$

Then

$$P_0(h) u_\alpha(h) = E_\alpha(h) u_\alpha(h);$$

and upon reindexing, we can write these eigenfunction equations as

$$(6.1.11) \quad P_0(h) u_j(h) = E_j(h) u_j(h) \quad (j = 1, \dots).$$

6.1.3. Asymptotic distribution of eigenvalues. With these explicit formulas in hand, we can study the behavior in the semiclassical limit of the eigenvalues $E(h)$ of the harmonic oscillator:

THEOREM 6.3 (Weyl's Law for the harmonic oscillator). *Assume that $0 \leq a < b < \infty$. Then*

$$(6.1.12) \quad \begin{aligned} \#\{E(h) \mid a \leq E(h) \leq b\} \\ = \frac{1}{(2\pi h)^n} (|\{a \leq |\xi|^2 + |x|^2 \leq b\}| + o(1)) \end{aligned}$$

as $h \rightarrow 0$.

Proof. 1. We may assume that $a = 0$. Since $E(h) = (2|\alpha| + n)h$ for some multiindex α according to (6.1.10), we have

$$\begin{aligned}\#\{E(h) \mid 0 \leq E(h) \leq b\} &= \#\left\{\alpha \mid 0 \leq 2|\alpha| + n \leq \frac{b}{h}\right\} \\ &= \#\{\alpha \mid \alpha_1 + \cdots + \alpha_n \leq R\},\end{aligned}$$

for $R := (b - nh)/2h$. Therefore

$$\begin{aligned}\#\{E(h) \mid 0 \leq E(h) \leq b\} &= |\{x \mid x_i \geq 0, x_1 + \cdots + x_n \leq R\}| + o(R^n) \\ &= \frac{1}{n!}R^n + o(R^n) \quad \text{as } R \rightarrow \infty \\ &= \frac{1}{n!}\left(\frac{b}{2h}\right)^n + o(h^{-n}) \quad \text{as } h \rightarrow 0.\end{aligned}$$

Note that the volume of $\{x \mid x_i \geq 0, x_1 + \cdots + x_n \leq 1\}$ is $(n!)^{-1}$.

2. Next we observe that $|\{|\xi|^2 + |x|^2 \leq b\}| = \alpha(2n)b^n$, where $\alpha(k) := \pi^{\frac{k}{2}}(\Gamma(\frac{k}{2} + 1))^{-1}$ is the volume of the unit ball in $\mathbb{R}^k$. Setting $k = 2n$, we compute that $\alpha(2n) = \pi^n(n!)^{-1}$. Hence

$$\begin{aligned}\#\{E(h) \mid 0 \leq E(h) \leq b\} &= \frac{1}{n!}\left(\frac{b}{2h}\right)^n + o(h^{-n}) \\ &= \frac{1}{(2\pi h)^n}|\{|\xi|^2 + |x|^2 \leq b\}| + o(h^{-n}). \quad \square\end{aligned}$$

6.2. SYMBOLS AND EIGENFUNCTIONS

For this section, we return to the general symbol (6.0.1) and the quantized operator (6.0.2). We assume that the potential $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth and satisfies the growth conditions:

$$(6.2.1) \quad \begin{cases} |\partial^\alpha V(x)| \leq C_\alpha \langle x \rangle^k & \text{for each multiindex } \alpha \\ V(x) \geq c \langle x \rangle^k & \text{for } |x| \geq R, \end{cases}$$

for appropriate constants $k, c, C_\alpha, R > 0$.

Our plan in the next section is to employ our detailed knowledge about the eigenvalues of the harmonic oscillator (6.1.8) to estimate the asymptotics of the eigenvalues of $P(h)$.

6.2.1. Concentration in phase space. First, we make the important observation that in the semiclassical limit the eigenfunctions $u(h)$ “are concentrated in phase space” on the energy surface $\{|\xi|^2 + V(x) = E\}$. (This assertion is somewhat related to the earlier Theorem 5.3.)

THEOREM 6.4 (h^∞ estimates). *Suppose that $u(h) \in L^2(\mathbb{R}^n)$ solves*

$$(6.2.2) \quad P(h)u(h) = E(h)u(h).$$

Assume as well that $a \in S$ is a symbol satisfying

$$\{|\xi|^2 + V(x) = E\} \cap \text{spt}(a) = \emptyset.$$

Then if

$$|E(h) - E| < \delta$$

for some sufficiently small $\delta > 0$, we have the estimate

$$(6.2.3) \quad \|a^w(x, hD)u(h)\|_{L^2} = O(h^\infty)\|u(h)\|_{L^2}.$$

Proof. 1. The set $K := \{|\xi|^2 + V(x) = E\} \subset \mathbb{R}^{2n}$ is compact. Hence there exists $\chi \in C_c^\infty(\mathbb{R}^{2n})$ such that

$$0 \leq \chi \leq 1, \quad \chi \equiv 1 \text{ on } K, \quad \chi \equiv 0 \text{ on } \text{spt}(a).$$

Define the symbol

$$b := |\xi|^2 + V(x) - E(h) + i\chi = p - E(h) + i\chi$$

and the order function

$$m := \langle \xi \rangle^2 + \langle x \rangle^k.$$

Therefore if $|E(h) - E|$ is small enough,

$$|b| \geq \gamma m \quad \text{on } \mathbb{R}^{2n}$$

for some constant $\gamma > 0$. Consequently $b \in S(m)$, with $b^{-1} \in S(m^{-1})$.

2. Thus there exist $c \in S(m^{-1})$, $r_1, r_2 \in S$ such that

$$\begin{cases} b^w(x, hD)c^w(x, hD) = I + r_1^w(x, hD) \\ c^w(x, hD)b^w(x, hD) = I + r_2^w(x, hD), \end{cases}$$

where $r_1^w(x, hD), r_2^w(x, hD)$ are $O(h^\infty)$. Then

$$(6.2.4) \quad a^w(x, hD)c^w(x, hD)b^w(x, hD) = a^w(x, hD) + O(h^\infty),$$

and

$$(6.2.5) \quad b^w(x, hD) = P(h) - E(h) + i\chi^w(x, hD).$$

Furthermore

$$a^w(x, hD)c^w(x, hD)\chi^w(x, hD) = O(h^\infty),$$

since $\text{spt}(a) \cap \text{spt}(\chi) = \emptyset$. Since $P(h)u(h) = E(h)u(h)$, (6.2.4) and (6.2.5) imply that

$$\begin{aligned} a^w(x, hD)u &= a^w(x, hD)c^w(x, hD)(P(h) - E(h) + i\chi^w)u + O(h^\infty) \\ &= O(h^\infty). \end{aligned} \quad \square$$

For the next result, we temporarily return to the case of the quantum harmonic oscillator, developing some sharper estimates:

THEOREM 6.5 (Improved estimates for the harmonic oscillator).

Suppose that $u(h) \in L^2(\mathbb{R}^n)$ is an eigenfunction of the harmonic oscillator:

$$(6.2.6) \quad P_0(h)u(h) = E(h)u(h).$$

Assume also that $a \in C_c^\infty$.

Then there exists $E_0 > 0$, depending only on the support of a , such that for $E(h) > E_0$,

$$(6.2.7) \quad \|a^w(x, hD)u(h)\|_{L^2} = O\left(\left(\frac{h}{E(h)}\right)^\infty\right) \|u(h)\|_{L^2}.$$

The precise form of the right-hand side of (6.2.7) will later let us handle eigenvalues $E(h) \rightarrow \infty$.

Proof. 1. We rescale the harmonic oscillator so that we can work near a fixed energy level E . Set

$$y := \frac{x}{E^{\frac{1}{2}}}, \quad \tilde{h} := \frac{h}{E}, \quad E(\tilde{h}) := \frac{E(h)}{E},$$

where we choose E so that $|E(h) - E| \leq E/4$. Then put

$$P_0(h) := -h^2 \Delta_x + |x|^2, \quad P_0(\tilde{h}) := -\tilde{h}^2 \Delta_y + |y|^2,$$

whence

$$P_0(h) - E(h) = E(P(\tilde{h}) - \tilde{E}(\tilde{h})).$$

We next introduce the unitary transformation

$$U\tilde{u}(y) := E^{\frac{n}{4}}\tilde{u}(E^{\frac{1}{2}}y).$$

Then

$$UP_0(h)U^{-1} = EP_0(\tilde{h});$$

and more generally

$$Ub^w(x, hD)U^{-1} = \tilde{b}^w(y, \tilde{h}D), \quad \tilde{b}(y, \eta) := b(E^{\frac{1}{2}}y, E^{\frac{1}{2}}\eta).$$

We will denote the symbol classes defined using $\tilde{h}$ by the symbol $\tilde{S}_\delta$.

2. We now apply Theorem 6.4. If

$$(P_0(\tilde{h}) - E(\tilde{h}))\tilde{u}(\tilde{h}) = 0, \quad |E(\tilde{h}) - 1| < \delta,$$

and $\tilde{b}(y, \eta) \in \tilde{S}$ has its support contained in $\{|y|^2 + |\eta|^2 \leq 1/2\}$, then

$$\|\tilde{b}^w(y, \tilde{h}D)\tilde{u}(\tilde{h})\|_{L^2} = O(\tilde{h}^\infty)\|\tilde{u}(\tilde{h})\|_{L^2}.$$

Translated to the original h and x as above, this assertion provides us with the bound

$$(6.2.8) \quad \|b^w(x, hD)u(h)\|_{L^2} = O((h/E)^\infty)\|u(h)\|_{L^2},$$

for

$$b(x, \xi) = \tilde{b}(E^{-1/2}x, E^{-1/2}\xi) \in S.$$

Note that $\text{spt}(b) \subset \{|x|^2 + |\xi|^2 \leq E/2\}$.

3. We now assume that $\tilde{b} = 1$ in $\{|y|^2 + |\eta|^2 \leq 1/4\}$. That corresponds to $b = 1$ in $\{|x|^2 + |\xi|^2 \leq E/4\}$. In view of (6.2.8), we only need to show that for

$$a \in C^\infty(\mathbb{R}^{2n}), \quad \text{spt}(a) \subset \{|x|^2 + |\xi|^2 \leq R^2\},$$

we have

$$\|(a^w(x, hD)(1 - b^w(x, hD))\|_{L^2 \rightarrow L^2} = O((h/E)^\infty),$$

for E large enough, where b is as in (6.2.8). That is the same as showing

$$(6.2.9) \quad \|\tilde{a}^w(y, \tilde{h}D)(1 - \tilde{b}^w(y, \tilde{h}D))\|_{L^2 \rightarrow L^2} = O(\tilde{h}^\infty),$$

for

$$\tilde{a}(y, \eta) = a(E^{\frac{1}{2}}y, E^{\frac{1}{2}}\eta).$$

We first observe that $E = h/\tilde{h} < 1/\tilde{h}$ and hence

$$\tilde{a} \in \tilde{S}_{\frac{1}{2}}.$$

Since the support of $\tilde{a}$ is contained in $\{|y|^2 + |\eta|^2 \leq R^2/E\}$, we see that for E large enough,

$$\text{dist}(\text{spt}(\tilde{a}), \text{spt}(1 - \tilde{b})) \geq 1/C > 0,$$

uniformly in $\tilde{h}$. The estimate (6.2.9) is now a consequence of Theorem 4.25. $\square$

6.2.2. Projections. We next study how projections onto the span of various eigenfunctions of the harmonic oscillator $P_0(h)$ are related to our symbol calculus.

THEOREM 6.6 (Projections and symbols). *Suppose for the symbol $a \in S$ that*

$$\text{spt}(a) \subset \{|\xi|^2 + |x|^2 < R\}.$$

Let

$$\Pi := \text{projection in } L^2 \text{ onto}$$

$$\text{span}\{u(h) \mid P_0(h)u(h) = E(h)u(h), E(h) \leq R\}.$$

Then

$$(6.2.10) \quad a^w(x, hD)(I - \Pi) = O_{L^2 \rightarrow L^2}(h^\infty)$$

and

$$(6.2.11) \quad (I - \Pi)a^w(x, hD) = O_{L^2 \rightarrow L^2}(h^\infty).$$

Proof. First of all, observe that

$$(I - \Pi) = \sum_{E_j(h) > R} u_j(h) \otimes u_j(h),$$

meaning that

$$(I - \Pi)u = \sum_{E_j(h) > R} \langle u_j(h), u \rangle u_j(h).$$

Therefore

$$a^w(x, hD)(I - \Pi) = \sum_{E_j(h) > R} (a^w(x, hD)u_j(h)) \otimes u_j(h);$$

and so

$$(6.2.12) \quad \begin{aligned} \|a^w(x, hD)(I - \Pi)\|_{L^2 \rightarrow L^2}^2 &\leq \sum_{E_j(h) > R} \|a^w(x, hD)u_j(h)\|_{L^2}^2 \\ &\leq \left(\sum_{R < E_j(h) \leq E_1} + \sum_{E_1 < E_j(h)} \right) \|a^w(x, hD)u_j(h)\|_{L^2}^2 \\ &:= A + B, \end{aligned}$$

where $E_1 = \max(E_0, R)$ with E_0 given in Theorem 6.5. We recall that E_0 depends only on $\text{spt}(a)$.

For the term A we can use Theorems 6.3 and 6.4:

$$\begin{aligned} A &\leq \#\{E_j(h) : R < E_j(h) < E_0\} \max_{R < E_j(h) < E_0} \|a^w(x, hD)u_j(h)\|^2 \\ &\leq Ch^{-n}O(h^\infty) = O(h^\infty). \end{aligned}$$

(Here we noted that the assumptions of Theorem 6.4 are satisfied for any E in $R \leq E \leq E_0$, and hence $\|a^w(x, hD)u_j(h)\| = O(h^\infty)$ holds for $|E_j(h) - E| < \delta_E$, with any of these E 's. A finite covering of $[R, E_0]$ provides a uniform estimate.)

Next, observe that Weyl's Law for the harmonic oscillator, Theorem 6.3, implies that

$$E_j(h) \geq \gamma j^{\frac{1}{n}} h$$

for some constant $\gamma > 0$. Then according to Theorem 6.5, for each $M < N$ and for $E_j(h) > E_0$, we have

$$\begin{aligned} \|a^w(x, hD)u_j(h)\|_{L^2}^2 &\leq C_N \left(\frac{h}{E_j(h)} \right)^N \\ &\leq C_N h^M \left(\frac{h}{E_j(h)} \right)^{N-M} \\ &\leq Ch^M j^{-\frac{N-M}{n}}. \end{aligned}$$

Consequently, if we fix $N - M \geq 2n$, we obtain

$$\begin{aligned} B &= \sum_{E_j(h) > E_0} \|a^w(x, hD)u_j(h)\|_{L^2}^2 \\ &\leq Ch^M \sum_{j \geq 1} j^{-2} \leq C'h^M. \end{aligned}$$

Since M is arbitrary, we obtain $B = O(h^\infty)$. This proves (6.2.10), and the proof of (6.2.11) is similar. $\square$

6.3. SPECTRUM AND RESOLVENTS

We next show that the spectrum of $P(h)$ consists entirely of eigenvalues.

THEOREM 6.7 (Resolvents and spectrum).

(i) *There exists a constant $h_0 > 0$ such that if $0 < h \leq h_0$, then the resolvent*

$$(P(h) - i)^{-1} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$$

is a compact operator.

(ii) *The mapping $z \mapsto (P(h) - z)^{-1}$ is meromorphic, with real and simple poles.*

(iii) *The spectrum of $P(h)$ is discrete.*

(iv) *Furthermore, there exists an orthonormal basis of $L^2(\mathbb{R}^n)$ comprised of eigenfunctions $\{u_j(h)\}_{j=1}^\infty$:*

$$(6.3.1) \quad P(h)u_j(h) = E_j(h)u_j(h) \quad (j = 1, 2, \dots).$$

Proof. 1. Let

$$m(x, \xi) := 1 + |\xi|^2 + |x|^k.$$

Then $p \in S(m)$, $C|p - i| \geq m$, and $P(h) = p^w(x, hD)$.

For h small enough, $m^w(x, hD)$ has a right inverse,

$$m^w(x, hD)^{-1} := (1/m)^w(x, hD)(I + hr^w(x, hD))^{-1},$$

$$r = (m\#(1/m) - 1)/h \in S.$$

We can therefore define the Hilbert space:

$$(6.3.2) \quad \mathcal{H} := \{u \in \mathcal{S}' \mid (I - h^2\Delta + \langle x \rangle^k)u \in L^2\} = m^w(x, hD)^{-1}L^2.$$

For small h , the inverse

$$(P(h) - i)^{-1} : L^2 \rightarrow \mathcal{H}$$

is bounded. Theorem 4.28 shows that $m(x, hD)^{-1} : L^2 \rightarrow L^2$ is compact and hence $\{u \in \mathcal{H} : \|u\|_{\mathcal{H}} \leq 1\}$ is compact in L^2 . That means that the inclusion of $\mathcal{H}$ in L^2 is compact and consequently $(P(h) - i)^{-1} : L^2 \rightarrow L^2$ is a compact operator.

2. We now write

$$P(h) - z = (I - K(z, h))(P(h) - i),$$

for

$$K(z, h) := (z - i)(P(h) - i)^{-1}.$$

Since $I - K(-i, h) = I$ and $K(z, h)$ is compact, Theorem D.4 shows that $z \rightarrow (I - K(z, h))^{-1}$ is a meromorphic family of operators, with poles of finite rank. Consequently,

$$(P(h) - z)^{-1} = (P(h) - i)^{-1}(I - K(z, h))^{-1}$$

is a meromorphic family of compact operators from L^2 to L^2 .

3. Since the poles of $(P(h) - z)^{-1}$ are discrete, there exists $\lambda \in \mathbb{R}$ for which $P(h) - \lambda : \mathcal{H} \rightarrow L^2$ is invertible. Hence for any $v_j \in L^2$, there exists $u_j \in \mathcal{H}$ such that $(P(h) - \lambda)u_j = v_j$ and

$$\begin{aligned} \langle (P(h) - \lambda)^{-1}v_1, v_2 \rangle &= \langle (P(h) - \lambda)^{-1}(P(h) - \lambda)u_1, (P(h) - \lambda)u_2 \rangle \\ &= \langle u_1, (P(h) - \lambda)u_2 \rangle. \end{aligned}$$

We integrate by parts, to find

$$\langle (P(h) - \lambda)^{-1}v_1, v_2 \rangle = \langle (P(h) - \lambda)u_1, u_2 \rangle = \langle v_1, (P(h) - \lambda)^{-1}v_2 \rangle.$$

Hence $(P(h) - \lambda)^{-1}$ is selfadjoint.

4. We now apply part (v) of Theorem C.7 to obtain an orthonormal set $\{u_j(h)\}_{j=1}^J$ and a sequence of real numbers $\{E_j(h)\}_{j=1}^J$ such that

$$(6.3.3) \quad (P(h) - \lambda)^{-1}v = \sum_{j=1}^J (E_j(h) - \lambda)^{-1}u_j(h)\langle v, u_j(h) \rangle,$$

for all $v \in L^2$, where either $J \in \mathbb{N}$ or else $J = \infty$.

5. Taking $v = u_j$ and applying $P(h) - \lambda$ to both sides of (6.3.3), we deduce that $P(h)u_j(h) = E_j(h)u_j$. Applying $P(h) - \lambda$ to both sides of (6.3.3) for an arbitrary $v \in L^2$, we discover that

$$v = \sum_{j=1}^J u_j(h) \langle v, u_j(h) \rangle.$$

Consequently the eigenfunctions $\{u_j(h)\}_{j=1}^J$ form a complete orthonormal set, and in particular $J = \infty$. $\square$

REMARK: Eigenfunctions in $\mathcal{S}$. Using (6.3.1) and the fact that $V \in C^\infty$, we can apply Theorem 7.1 iteratively to conclude that $u \in H_h^l(\mathbb{R}^n)$ for all l and in particular that $u \in C^\infty(\mathbb{R}^n)$. Similarly we can use $V(x) \geq c\langle x \rangle^k - C$ to obtain $\langle x \rangle^N u \in H_h^l(\mathbb{R}^n)$. Putting this together, we deduce that

$$(6.3.4) \quad u_j(h) \in \mathcal{S},$$

with seminorms depending on h . $\square$

REMARK: An alternative proof of meromorphy. To illustrate further the semiclassical calculus, we provide a different proof of the meromorphy of $z \mapsto (P - z)^{-1}$ for h small.

1. Let $|z| \leq E$, where E is fixed; and as before let $P_0(h) = -h^2\Delta + |x|^2$ be the harmonic oscillator. As in Theorem 6.6 define

$$\Pi := \text{projection in } L^2 \text{ onto} \\ \text{span}\{u \mid P_0(h)u = E(h)u \text{ for } E(h) \leq R + 1\}.$$

Now suppose that $\text{spt}(a) \subset \{|x|^2 + |\xi|^2 \leq R\}$. Owing to Theorem 6.6, we have

$$a^w(x, hD) - a^w(x, hD)\Pi = O_{L^2 \rightarrow L^2}(h^\infty)$$

and

$$a^w(x, hD) - \Pi a^w(x, hD) = O_{L^2 \rightarrow L^2}(h^\infty).$$

2. Fix $R > 0$ so large that

$$\{|\xi|^2 + V(x) \leq E\} \subset \{|x|^2 + |\xi|^2 < R\}.$$

Select $\chi \in C^\infty(\mathbb{R}^{2n})$ with $\text{spt}(\chi) \subset \{|x|^2 + |\xi|^2 \leq R\}$ so that

$$|\xi|^2 + V(x) - z + \chi \geq \gamma m$$

for $m = \langle \xi \rangle^2 + \langle x \rangle^k$ and all $|z| \leq E$. Then $\chi = \Pi\chi\Pi + O(h^\infty)$. Recall that the symbolic calculus guarantees that $P(h) - z + \chi$ is invertible if h is small enough. Consequently, so is $P(h) - z + \Pi\chi\Pi$, since the two operators differ by an $O(h^\infty)$ term.

3. Now write

$$P(h) - z = P(h) - z + \Pi\chi\Pi - \Pi\chi\Pi.$$

Consequently

$$P(h) - z = (P(h) - z + \Pi\chi\Pi)(I - (P(h) - z + \Pi\chi\Pi)^{-1}\Pi\chi\Pi).$$

Note that $\Pi\chi\Pi$ is an operator of finite rank. So Theorem D.4 asserts that the family of operators

$$(I - (P(h) - z + \Pi\chi\Pi)^{-1}\Pi\chi\Pi)^{-1}$$

is meromorphic in z . It follows that $(P(h) - z)^{-1}$ is meromorphic on L^2 . The poles are the eigenvalues, and the symmetry of $P(h)$ implies that these eigenvalues are real. $\square$

6.4. WEYL'S LAW

Here is the main result of this chapter:

THEOREM 6.8 (Weyl's Law). *Suppose that V satisfies the conditions of (6.2.1) and that the $E(h)$ are the eigenvalues of $P(h) = -h^2\Delta + V(x)$.*

Then for each $a < b$, we have

$$\begin{aligned} (6.4.1) \quad \# \{E(h) \mid a \leq E(h) \leq b\} \\ = \frac{1}{(2\pi h)^n} (|\{a \leq |\xi|^2 + V(x) \leq b\}| + o(1)) \end{aligned}$$

as $h \rightarrow 0$. Here $|\{a \leq |\xi|^2 + V(x) \leq b\}|$ denotes the volume of the set of (x, ξ) such that $a \leq |\xi|^2 + V(x) \leq b$.

INTERPRETATION: Density of states. The uncertainty principle proved in Theorem 3.9 tells us the eigenfunctions cannot be arbitrarily localized in phase space. Roughly speaking, their “tightest localization” is to balls of radius $O(h^{\frac{1}{2}})$, so that $\Delta x_j \Delta \xi_j \simeq h$ for $j = 1, \dots, n$ in compliance with the uncertainty principle. Such a localized state “takes up a volume of at least order $O(h^n)$ ”; and since the eigenfunctions are orthogonal, they also “take up different volumes in any region of phase space.” Therefore their total number is at most approximately the volume of the region times h^{-n} .

Consequently in a region of volume $|\{a \leq |\xi|^2 + V(x) \leq b\}|$ in $\mathbb{R}^{2n}$ we could plausibly expect to find at most $O(h^{-n})$ states, and Weyl's Law tells us that the eigenfunctions in fact pack the region at this maximum density. $\square$

These heuristics in fact motivate the following proof.

Proof. 1. Let

$$N(\lambda) = \#\{E(h) \mid E(h) \leq \lambda\}.$$

Select $\chi \in C_c^\infty(\mathbb{R}^{2n})$ so that

$$\chi \equiv 1 \text{ on } \{p \leq \lambda + \epsilon\}, \quad \chi \equiv 0 \text{ on } \{p \geq \lambda + 2\epsilon\}.$$

Then for M large enough

$$a := p + M\chi - \lambda \geq \gamma_\epsilon m,$$

for $m = \langle \xi \rangle^2 + \langle x \rangle^m$ and some constant $\gamma_\epsilon > 0$. Hence a is elliptic; and so for small $h > 0$, $a^w(x, hD)$ is invertible.

2. *Claim #1:* We have

$$(6.4.2) \quad \langle (P(h) + M\chi^w(x, hD) - \lambda)u, u \rangle \geq \gamma \|u\|_{L^2}^2, \quad u \in \mathcal{H},$$

for some $\gamma > 0$. Here $\mathcal{H}$ is given in (6.3.2). Theorem C.14 shows that $\mathcal{H}$ is the domain of $P(h)$.

To prove (6.4.2), take $b \in S(m^{1/2})$ so that $b^2 = a$. Then $b^2 = b\#b + r_0$, where $r_0 \in hS(m)$. We also recall from the proof of Theorem 4.29 that the right inverse $b^w(x, hD)^{-1}$ exists and

$$b^w(x, hD)^{-1} r_0^w(x, hD) b^w(x, hD)^{-1} = O_{L^2 \rightarrow L^2}(h).$$

Thus

$$\begin{aligned} a^w(x, hD) &= b^w(x, hD) b^w(x, hD) + r_0^w(x, hD) \\ &= b^w(x, hD) (1 + b^w(x, hD)^{-1} r_0^w(x, hD) b^w(x, hD)^{-1}) b^w(x, hD) \\ &= b^w(x, hD) (1 + O_{L^2 \rightarrow L^2}(h)) b^w(x, hD). \end{aligned}$$

Hence for sufficiently small $h > 0$,

$$\begin{aligned} \langle (P(h) + M\chi^w - \lambda)u, u \rangle &= \langle a^w(x, hD)u, u \rangle \\ &\geq \|b^w(x, hD)u\|_{L^2}^2 (1 - O(h)) \\ &\geq \gamma \|u\|_{L^2}^2, \end{aligned}$$

for some $\gamma > 0$, in view of (4.7.2). This proves (6.4.2).

3. *Claim #2:* For each $\delta > 0$, there exists a bounded linear operator Q such that

$$(6.4.3) \quad \chi^w(x, hD) = Q + O_{L^2 \rightarrow L^2}(h^\infty)$$

and

$$(6.4.4) \quad \text{rank}(Q) \leq \frac{1}{(2\pi h)^n} (|\{p \leq \lambda + 2\epsilon\}| + \delta).$$

To prove this, cover the set $\{p \leq \lambda + 2\epsilon\}$ with balls

$$B_j := B((x_j, \xi_j), r_j^2) \quad (j = 1, \dots, N)$$

such that

$$\sum_{j=1}^N |B_j| \leq |\{p \leq \lambda + 2\epsilon\}| + \frac{\delta}{2}.$$

We then define the “shifted” harmonic oscillator

$$P_j(h) := |hD_x - \xi_j|^2 + |x - x_j|^2$$

and set

$$\begin{aligned} \Pi &:= \text{orthogonal projection in } L^2 \text{ onto } V, \text{ the span of} \\ &\quad \{u \mid P_j(h)u = E_j(h)u, \ E_j(h) \leq r_j, \ j = 1, \dots, N\}. \end{aligned}$$

We now claim that

$$(6.4.5) \quad (I - \Pi)\chi^w(x, hD) = O_{L^2 \rightarrow L^2}(h^\infty).$$

To see this, let $\chi = \sum_{j=1}^N \chi_j$, where $\text{spt } \chi_j \subset B((x_j, \xi_j), r_j^2)$, and put

$$\begin{aligned} \Pi_j &:= \text{orthogonal projection in } L^2 \text{ onto the span of} \\ &\quad \{u \mid P_j(h)u = E_j(h)u, \ E_j(h) \leq r_j\}. \end{aligned}$$

Theorem 6.6 shows that $(I - \Pi_j)\chi_j^w(x, hD) = O(h^\infty)$. We note that $\Pi\Pi_j = \Pi_j$ and hence

$$\begin{aligned} (I - \Pi)\chi^w(x, hD) &= \sum_{j=1}^N (I - \Pi)\chi_j^w(x, hD) \\ &= \sum_{j=1}^N (I - \Pi)(I - \Pi_j)\chi_j^w(x, hD) \\ &= O_{L^2 \rightarrow L^2}(h^\infty). \end{aligned}$$

This proves (6.4.5).

It now follows that

$$\chi^w(x, hD) = \Pi\chi^w(x, hD) + (I - \Pi)\chi^w(x, hD) = Q + O(h^\infty)$$

for

$$Q := \Pi\chi^w(x, hD).$$

Clearly Q has finite rank, since

$$\begin{aligned} \text{rank } Q &= \dim(\text{image of } Q) \leq \dim(\text{image of } \Pi) \\ &\leq \sum_{j=1}^N \#\{E_j(h) \mid E_j(h) \leq r_j\} \\ &= \frac{1}{(2\pi h)^n} \left(\sum_{j=1}^N |B_j| + o(1) \right), \end{aligned}$$

according to Weyl's Law for the harmonic oscillator, Theorem 6.3. Consequently

$$(6.4.6) \quad \text{rank } Q \leq \frac{1}{(2\pi h)^n} \left(|\{p \leq \lambda + 2\epsilon\}| + \frac{\delta}{2} + o(1) \right).$$

This proves Claim #2.

4. We next employ Claims #1 and #2 and Theorem C.15. We have

$$\begin{aligned} \langle P(h)u, u \rangle &\geq (\lambda + \gamma) \|u\|_{L^2}^2 - M \langle Qu, u \rangle - O(h^\infty) \|u\|_{L^2}^2 \\ &\geq \lambda \|u\|_{L^2}^2 - M \langle Qu, u \rangle, \end{aligned}$$

where the rank of Q is bound by (6.4.6). Theorem C.15(i) implies then that

$$N(\lambda) \leq \frac{1}{(2\pi h)^n} (|\{p \leq \lambda + 2\epsilon\}| + \delta + o(1)).$$

This holds for all $\epsilon, \delta > 0$, and so

$$(6.4.7) \quad N(\lambda) \leq \frac{1}{(2\pi h)^n} (|\{p \leq \lambda\}| + o(1))$$

as $h \rightarrow 0$.

5. We must prove the opposite inequality.

Claim #3: Suppose $B_j = B((x_j, \xi_j), r_j^2) \subset \{p < \lambda\}$ and put

$$V_j := \text{span}\{u \mid P_j(h)u = E_j(h)u, \ E_j(h) \leq r_j\}.$$

We claim that for $u \in V_j$,

$$(6.4.8) \quad \langle P(h)u, u \rangle \leq (\lambda + \epsilon + O(h^\infty)) \|u\|_{L^2}^2.$$

To prove this claim, select a symbol $a \in C_c^\infty(\mathbb{R}^{2n})$, with

$$a \equiv 1 \text{ on } \{p \leq \lambda\}, \quad \text{spt}(a) \subset \{p \leq \lambda + \frac{\epsilon}{2}\}.$$

Let $c := 1 - a$. Then $u - a^w(x, hD)u = c^w(x, hD)u = O(h^\infty)$ according to Theorem 6.6, since $\text{spt}(1 - a) \cap B_j = \emptyset$.

Define $b^w := P(h)a^w(x, hD)$. Now $p \in S(m)$ and $a \in S(m^{-1})$. Thus $b = pa + O(h) \in S$ and so b^w is bounded in L^2 . Observe also that $b \leq \lambda + \frac{\epsilon}{2}$, and so

$$b^w(x, hD) \leq \lambda + \frac{3\epsilon}{4}.$$

Therefore

$$\langle P(h)a^w(x, hD)u, u \rangle = \langle b^w(x, hD)u, u \rangle \leq \left(\lambda + \frac{3\epsilon}{4} \right) \|u\|_{L^2}^2.$$

Since $a^w(x, hD)u = u + O(h^\infty)$, we deduce that

$$\langle P(h)u, u \rangle \leq (\lambda + \epsilon + O(h^\infty)) \|u\|_{L^2}^2.$$

This proves Claim #3.

6. Now find disjoint balls $B_j \subset \{p < \lambda\}$ such that

$$|\{p < \lambda\}| \leq \sum_{j=1}^N |B_j| + \delta$$

and denote

$$V = V_1 + V_2 + \cdots + V_N.$$

The spaces V_i and V_j , $i \neq j$, are not orthogonal; but because B_i and B_j are disjoint, we see, as in Theorem 6.6, that

$$(6.4.9) \quad \langle u, v \rangle = O(h^\infty) \|u\| \|v\|$$

if $u \in V_i, v \in V_j$, and $i \neq j$. Since each V_j has an orthonormal basis of eigenvectors, (6.4.8) holds for $u \in V_j$. The approximate orthogonality (6.4.9) then gives

$$\langle Pu, u \rangle \leq (\lambda + \delta) \|u\|_{L^2}^2$$

for all $u \in V$. Also, (6.4.9) and Theorem 6.3 imply that for h small enough

$$\begin{aligned} \dim V &= \sum_{j=1}^N \dim V_j \\ &= \sum_{j=1}^N \#\{E_j(h) \leq r_j\} \\ &= \frac{1}{(2\pi h)^n} \left(\sum_{j=1}^N |B_j| + o(1) \right) \\ &\geq \frac{1}{(2\pi h)^n} (|\{p < \lambda\}| - \delta + o(1)). \end{aligned}$$

Then according to Theorem C.15(ii),

$$N(\lambda) \geq \frac{1}{(2\pi h)^n} (|\{p < \lambda\}| - \delta + o(1)).$$

□

6.5. NOTES

The proof of Weyl asymptotics is a semiclassical version of the classical Dirichlet–Neumann bracketing proof for the bounded domains.

In Chapter 12 we will present a more general form of Weyl’s Law, proved using a functional calculus of pseudodifferential operators. That proof leads to many further improvements, as discussed in Dimassi–Sjöstrand [D-S]. The proof using min-max principle comparisons with the harmonic oscillator has the advantage of providing upper bounds for the number of eigenvalues of nonselfadjoint operators.

ESTIMATES FOR SOLUTIONS OF PDE

- 7.1 Classically forbidden regions
- 7.2 Tunneling
- 7.3 Order of vanishing
- 7.4 L^∞ estimates for quasimodes
- 7.5 Schauder estimates
- 7.6 Notes

We continue our study of the semiclassical behavior of eigenfunctions:

$$(7.0.1) \quad P(h)u(h) = E(h)u(h)$$

for the operator

$$P(h) = -h^2\Delta + V(x).$$

We first demonstrate that if $E(h)$ is close to the energy level E , then $u(h)$ is exponentially small within the *classically forbidden region*

$$(7.0.2) \quad \{V > E\} := \{x \in \mathbb{R}^n \mid V(x) > E\}.$$

Then we show conversely that in any open set the L^2 norm of $u(h)$ is bounded from below by a quantity exponentially small in h . We discuss as well the order of vanishing of eigenfunctions and conclude with a semiclassical derivation of the interior Schauder estimates.

7.1. CLASSICALLY FORBIDDEN REGIONS

First we appropriately rescale the norm for the standard Sobolev spaces $H^k(U)$, as defined and discussed for example in [E, Chapter 5]. Note in particular that we allow for complex-valued functions. The modified norms differ from the usual ones by the introduction of appropriate powers of h :

DEFINITION. Let $U \subset \mathbb{R}^n$ be an open set, $h > 0$, and $k \in \mathbb{N}$. The *semiclassical norm* of a function $u \in H^k(U)$ is

$$(7.1.1) \quad \|u\|_{H_h^k(U)} := \left(\sum_{|\alpha| \leq k} \int_U |(hD)^\alpha u|^2 dx \right)^{1/2}.$$

This semiclassical, rescaled form of a standard elliptic estimate will be very useful:

THEOREM 7.1 (H_h^2 estimates). *Write*

$$Q(h) := -h^2 \Delta + \langle a, hDu \rangle + b,$$

where the coefficients a, b are smooth and possibly complex-valued. Assume also that $U \subset \subset W \subset \mathbb{R}^n$ are open sets.

There exists a constant C such that

$$(7.1.2) \quad \|u\|_{H_h^2(U)} \leq C(\|Q(h)u\|_{L^2(W)} + \|u\|_{L^2(W)})$$

for all $u \in C^\infty(W)$.

Proof. 1. Let $\chi \in C_c^\infty(W)$, $\chi \equiv 1$ on U . We multiply $Q(h)u$ by $\chi^2 \bar{u}$ and integrate by parts:

$$\begin{aligned} \operatorname{Re} \int_W Q(h)u \bar{u} \chi^2 dx &= \int_W h^2 \langle \partial(\chi^2 \bar{u}), \partial u \rangle + \operatorname{Re} \langle a, hDu \rangle \chi^2 \bar{u} \\ &\quad + \operatorname{Re} b |u|^2 \chi^2 dx \\ &\geq \int_W \chi^2 |hDu|^2 - C \chi |hDu| |u| - C |u|^2 dx \\ &\geq \frac{1}{2} \int_W \chi^2 |hDu|^2 dx - C \int_W |u|^2 dx. \end{aligned}$$

Therefore

$$\int_U |hDu|^2 dx \leq C \int_W |Q(h)u|^2 + |u|^2 dx.$$

2. Also note that if $u \in C_c^\infty(W)$, an integration by parts gives

$$\begin{aligned} \int_W |\Delta u|^2 dx &= \sum_{j,k=1}^n \int_W \partial_{x_j}^2 u \partial_{x_k}^2 \bar{u} dx \\ &= \sum_{j,k=1}^n \int_W |\partial_{x_j x_k} u|^2 dx = \int_W |D^2 u|^2 dx. \end{aligned}$$

We now proceed as above, multiplying $Q(h)u$ by $-\chi^2 h^2 \Delta \bar{u}$ and integrating by parts, eventually to derive the inequality

$$\int_U |(hD)^2 u|^2 dx \leq C \int_W |Q(h)u|^2 + |u|^2 dx. \quad \square$$

Before turning again to eigenfunctions, we present the following general estimates. Our primary tool will be properly designed conjugations of the operator $P(h)$.

DEFINITION Given $\varphi \in C^\infty(\mathbb{R}^n)$, we define the *conjugation* of $P(h)$ by $e^{\varphi/h}$:

$$(7.1.3) \quad P_\varphi(h) := e^{\varphi/h} P e^{-\varphi/h}.$$

LEMMA 7.2 (Symbol of conjugation). *We have*

$$(7.1.4) \quad P_\varphi(h) = p_\varphi^w(x, hD)$$

for the symbol

$$(7.1.5) \quad p_\varphi(x, \xi) := \langle \xi + i\partial\varphi(x), \xi + i\partial\varphi(x) \rangle + V(x).$$

Proof. We calculate for functions $u \in C^\infty(\mathbb{R}^n)$ that

$$\begin{aligned} P_\varphi(h)u &= e^{\varphi/h} (-h^2 \Delta + V)(e^{-\varphi/h} u) \\ &= -h^2 \Delta u + 2h \langle \partial\varphi, \partial u \rangle - |\partial\varphi|^2 u + Vu + h \Delta \varphi u. \end{aligned}$$

But Theorem 4.5 shows that

$$p_\varphi^w(x, hD)u = -h^2 \Delta u + i(\langle \partial\varphi, hDu \rangle + \langle hD, \partial\varphi u \rangle) - |\partial\varphi|^2 u + Vu,$$

and this proves (7.1.4). $\square$

We now turn to estimates valid within the classically forbidden region $\{V > E\}$:

THEOREM 7.3 (Exponential estimate from above). *Suppose that U is an open set such that*

$$U \subset\subset \{V > E\}.$$

Then for each open set $W \supset\supset U$ and for each λ near E , there exist constants $h_0, \delta, C > 0$, such that

$$(7.1.6) \quad \|u\|_{L^2(U)} \leq C e^{-\delta/h} \|u\|_{L^2(W)} + C \|(P(h) - \lambda)u\|_{L^2(W)}$$

for $u \in C_c^\infty(\mathbb{R}^n)$ and $0 < h \leq h_0$.

We call (7.1.6) an *Agmon* or *Agmon–Lithner estimate*.

Proof. 1. Select $\psi, \varphi \in C_c^\infty(W)$ such that $0 \leq \psi, \varphi \leq 1$, $\psi \equiv 1$ on U , and $\varphi \equiv 1$ on $\text{spt}(\psi)$. We may assume as well that $W \subset\subset \{V > E\}$.

As in Lemma 7.2, we observe that the symbol of

$$A(h) := e^{\delta\psi/h} (P(h) - \lambda) e^{-\delta\psi/h}$$

is

$$\langle \xi + i\delta\partial\psi, \xi + i\delta\partial\psi \rangle + V - \lambda.$$

Now for λ close to E , $x \in W$, and δ sufficiently small, we have

$$(7.1.7) \quad \begin{aligned} & |\langle \xi + i\delta\partial\psi, \xi + i\delta\partial\psi \rangle + V - \lambda|^2 \\ & \geq |\xi^2 + V - \delta^2|\partial\psi|^2 - \lambda|^2 \geq \gamma^2 \langle \xi \rangle^4 > 0 \end{aligned}$$

for some positive constant γ .

2. Let φ_1 have the same properties as φ and let it be equal to one on $\text{spt} \varphi$. The lower bound (7.1.7) implies that

$$B := \langle hD \rangle^{-2} (\varphi_1 A(h)^* A(h) \varphi_1 - \gamma^2 \varphi_1^2) \langle hD \rangle^{-2} = b^w(x, hD)$$

for a symbol $b \in S$ such that $b \geq -Ch$.

According to the sharp Gårding inequality, Theorem 4.32, we see that

$$\langle Bv, v \rangle \geq -Ch \|v\|_{L^2}^2$$

provided $\delta > 0$ is sufficiently small. Putting $v = \langle hD \rangle^2 \varphi w$ in this inequality gives

$$\|A(h)\varphi w\|_{L^2}^2 \geq \gamma^2 \|\varphi w\|_{L^2}^2 - Ch \|\langle hD \rangle^2 \varphi w\|_{L^2}^2$$

for $w \in H_{\text{loc}}^2(\mathbb{R}^n)$.

3. Theorem 7.1 applied with W such that $\text{spt} \varphi \subset\subset W$ shows that

$$\|\langle hD \rangle^2 \varphi w\|_{L^2} \leq C \|\varphi w\|_{H_h^2} \leq C (\|\varphi w\|_{L^2} + \|A(h)\varphi w\|_{L^2}).$$

Hence for h small enough, we have

$$\|A(h)\varphi w\|_{L^2} \geq \frac{\gamma}{2} \|\varphi w\|_{L^2}$$

when $w \in H_{h,\text{loc}}^2(\mathbb{R}^n)$. We put $w = e^{\delta\psi/h}u$ in this inequality, to learn that

$$\begin{aligned}\|e^{\delta\psi/h}\varphi u\|_{L^2} &\leq C\|A(h)(e^{\delta\psi/h}\varphi u)\|_{L^2} = C\|e^{\delta\psi/h}(P(h) - \lambda)\varphi u\|_{L^2} \\ &\leq C\|e^{\delta\psi/h}\varphi(P(h) - \lambda)u\|_{L^2} + C\|e^{\delta\psi/h}[P(h), \varphi]u\|_{L^2}\end{aligned}$$

for $u \in C_c^\infty(\mathbb{R}^n)$.

4. Next is the key observation that since $\varphi \equiv 1$ on $\text{spt}(\psi)$, we have $\psi \equiv 0$ on $\text{spt}[P(h), \varphi]u$. Choosing $W_1 \subset\subset W$ such that $\text{spt} \varphi \subset W$, we can apply Theorem 7.1, which gives

$$\begin{aligned}\|e^{\delta\psi/h}[P(h), \varphi]u\|_{L^2} &= \|[P(h), \varphi]u\|_{L^2} \leq C(\|hD_x u\|_{L^2(W_1)} + \|u\|_{L^2(W_1)}) \\ &\leq C\|u\|_{L^2(W)} + C\|(P(h) - \lambda)u\|_{L^2(W)}.\end{aligned}$$

Combining these estimates, we conclude that

$$\begin{aligned}e^{\delta/h}\|u\|_{L^2(U)} &\leq \|e^{\delta\psi/h}\varphi u\|_{L^2} \\ &\leq C\|u\|_{L^2(W)} + C(e^{\delta/h} + 1)\|(P(h) - \lambda)u\|_{L^2(W)}.\end{aligned}$$

Mutlplying both sides by $e^{-\delta/h}$ gives the estimate (7.1.6). $\square$

Specializing to eigenfunctions, we deduce

THEOREM 7.4 (Exponential decay estimates). *Suppose that*

$$U \subset\subset \{V > E\}$$

and that $u(h) \in L^2(\mathbb{R}^n)$ solves

$$P(h)u(h) = E(h)u(h),$$

where

$$E(h) \rightarrow E \quad \text{as } h \rightarrow 0.$$

Then there exists a constant $\delta > 0$ such that

$$(7.1.8) \quad \|u(h)\|_{L^2(U)} \leq e^{-\delta/h}\|u(h)\|_{L^2(\mathbb{R}^n)}$$

for all sufficiently small $h > 0$.

7.2. TUNNELING

In this section we assume $u(h)$ solves the eigenvalue problem (7.0.1) and we suppose concerning the potential V that

$$(7.2.1) \quad \begin{cases} |\partial^\alpha V(x)| \leq C_\alpha \langle x \rangle^k & \text{for each multiindex } \alpha \\ V(x) \geq c \langle x \rangle^k & \text{for } |x| \geq R \end{cases}$$

for constants $k, c, R, C_\alpha > 0$.

In Section 7.1 we showed that $u(h)$ is exponentially small in the physically forbidden region. In this section we will show that it can never be smaller than this: for small $h > 0$ and any bounded, open subset U of $\mathbb{R}^n$, we have the lower bound

$$\|u\|_{L^2(U)} \geq e^{-\frac{C}{h}} \|u\|_{L^2(\mathbb{R}^n)}.$$

This is a mathematical version of quantum mechanical “tunneling into the physically forbidden region” and is closely related to unique continuation phenomena for solutions of second-order elliptic equations.

DEFINITION. Hörmander’s *hypoellipticity condition* is the requirement for the symbol p_φ , defined by (7.1.5), that

$$(7.2.2) \quad \text{if } p_\varphi = 0, \quad \text{then } i\{p_\varphi, \bar{p}_\varphi\} > 0.$$

Observe that for any complex function $q = q(x, \xi)$,

$$i\{q, \bar{q}\} = i\{\operatorname{Re} q + i\operatorname{Im} q, \operatorname{Re} q - i\operatorname{Im} q\} = 2\{\operatorname{Re} q, \operatorname{Im} q\}.$$

Hence the expression $i\{p_\varphi, \bar{p}_\varphi\}$ is real.

THEOREM 7.5 (L^2 estimate for $P_\varphi(h)$). *Let $W \subset\subset \mathbb{R}^n$ be an open set, and assume that Hörmander’s hypoellipticity condition (7.2.2) is valid within $\bar{W}$.*

Then there exists a constant C such that

$$(7.2.3) \quad h^{1/2} \|u\|_{L^2(W)} \leq C \|P_\varphi(h)u\|_{L^2(W)}$$

for all $u \in C_c^\infty(W)$, provided $0 < h \leq h_0$ with $h_0 > 0$ sufficiently small.

Proof. We calculate

$$\begin{aligned} \|P_\varphi(h)u\|_{L^2}^2 &= \langle P_\varphi(h)u, P_\varphi(h)u \rangle = \langle P_\varphi^*(h)P_\varphi u, u \rangle \\ &= \langle P_\varphi(h)P_\varphi^*(h)u, u \rangle + \langle [P_\varphi^*(h), P_\varphi(h)]u, u \rangle \\ &= \|P_\varphi^*(h)u\|_{L^2}^2 + \langle [P_\varphi^*(h), P_\varphi(h)]u, u \rangle. \end{aligned}$$

The idea will be to use the positivity of the second term on the right-hand side wherever $P_\varphi^*(h)$ fails to be elliptic. More precisely, for any $M > 1$ and h small enough the calculation above gives

$$\begin{aligned} \|P_\varphi(h)u\|_{L^2}^2 &\geq Mh \|P_\varphi^*(h)u\|_{L^2}^2 + \langle [P_\varphi^*(h), P_\varphi(h)]u, u \rangle \\ &= h\langle (M|p_\varphi|^2 + i\{p_\varphi, \bar{p}_\varphi\})^\omega u, u \rangle - O(h^2) \|u\|_{H_h^2}^2, \end{aligned}$$

the last term resulting from estimates of the lower-order terms in $\bar{p}_\varphi \# p_\varphi$ and the commutator. Hörmander’s hypoellipticity condition (7.2.2) implies for M large enough that

$$M|p_\varphi(x, \xi)|^2 + i\{p_\varphi, \bar{p}_\varphi\}(x, \xi) \geq \gamma^2 \langle \xi \rangle^4 > 0$$

for $x \in \bar{W}$. Then the sharp Gårding inequality (Theorem 4.32), applied as in the proof of Theorem 7.3, and Theorem 7.1 show us that

$$\|P_\varphi(h)u\|_{L^2}^2 \geq Ch\|u\|_{L^2}^2 - O(h^2)(\|P_\varphi(h)u\|_{L^2}^2 + \|u\|_{L^2}^2). \quad \square$$

Next we carefully design a weight φ , to ensure that $P_\varphi(h)$ satisfies the hypothesis of Theorem 7.5.

LEMMA 7.6 (Constructing a weight). *Let $0 < r < R$. There exists a positive nonincreasing radial function $\varphi \in C^\infty(\mathbb{R}^n)$ such that p_φ satisfies the Hörmander hypoellipticity condition (7.2.2) in $B(0, R) \setminus B(0, r)$.*

Proof. 1. Recall that

$$p_\varphi = |\xi|^2 + 2i\langle \xi, \partial\varphi \rangle - |\partial\varphi|^2 + V - E.$$

So $p_\varphi = 0$ implies both

$$(7.2.4) \quad |\xi|^2 - |\partial\varphi|^2 + V - E = 0$$

and

$$(7.2.5) \quad \langle \xi, \partial\varphi \rangle = 0.$$

Furthermore,

$$\begin{aligned} \frac{i}{2}\{p_\varphi, \overline{p_\varphi}\} &= \{\operatorname{Re} p_\varphi, \operatorname{Im} p_\varphi\} \\ &= \langle \partial_\xi(|\xi|^2 - |\partial\varphi|^2 + V - E), 2\partial_x \langle \xi, \partial\varphi \rangle \rangle \\ (7.2.6) \quad &\quad - \langle \partial_x(|\xi|^2 - |\partial\varphi|^2 + V - E), 2\partial_\xi \langle \xi, \partial\varphi \rangle \rangle \\ &= 4\langle \partial^2 \varphi \xi, \xi \rangle + 4\langle \partial^2 \varphi \partial\varphi, \partial\varphi \rangle - 2\langle \partial V, \partial\varphi \rangle. \end{aligned}$$

2. Now assume that

$$\varphi = e^{\lambda\psi},$$

where $\lambda > 0$ will be selected and $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ is positive and radial, $\psi = \psi(|x|)$. Then

$$\partial\varphi = \lambda\partial\psi e^{\lambda\psi}$$

and

$$\partial^2 \varphi = (\lambda^2 \partial\psi \otimes \partial\psi + \lambda \partial^2 \psi) e^{\lambda\psi}.$$

Hence

$$\langle \partial^2 \varphi \xi, \xi \rangle = (\lambda^2 \langle \partial\psi, \xi \rangle^2 + \lambda \langle \partial^2 \psi \xi, \xi \rangle) e^{\lambda\psi} = \lambda \langle \partial^2 \psi \xi, \xi \rangle e^{\lambda\psi},$$

since (7.2.5) implies $\langle \partial\psi, \xi \rangle = 0$. Also

$$\langle \partial^2 \varphi \partial\varphi, \partial\varphi \rangle = \lambda^4 |\partial\psi|^4 e^{3\lambda\psi} + \lambda^3 \langle \partial^2 \psi \partial\psi, \partial\psi \rangle e^{3\lambda\psi},$$

and

$$\langle \partial V, \partial\varphi \rangle = \lambda \langle \partial V, \partial\psi \rangle e^{\lambda\psi}.$$

According to (7.2.6), we have

$$(7.2.7) \quad \frac{i}{2}\{p_\varphi, \overline{p_\varphi}\} = 4\lambda\langle\partial^2\psi\xi, \xi\rangle e^{\lambda\psi} + 4\lambda^4|\partial\psi|^4 e^{3\lambda\psi} \\ + 4\lambda^3\langle\partial^2\psi\partial\psi, \partial\psi\rangle e^{3\lambda\psi} - 2\lambda\langle\partial V, \partial\psi\rangle e^{\lambda\psi}.$$

3. Now take

$$\psi := \mu - |x|,$$

for a constant μ so large that $\psi \geq 1$ on $B(0, R)$. Then φ is radial and nonincreasing. Furthermore

$$|\partial\psi| = 1, \quad |\partial^2\psi| \leq C \quad \text{on } B(0, R) \setminus B(0, r).$$

Owing to (7.2.4) we have

$$|\xi|^2 \leq C + |\partial\varphi|^2 \leq C + C\lambda^2 e^{2\lambda\psi} \quad \text{on } B(0, R) \setminus B(0, r).$$

Inserting these estimates into (7.2.7), we compute

$$\frac{i}{2}\{p_\varphi, \overline{p_\varphi}\} \geq 2\lambda^4 e^{3\lambda\psi} - C\lambda^3 e^{3\lambda\psi} - C \geq 1,$$

in $B(0, R) \setminus B(0, r)$, if λ is selected large enough.

Finally, we modify ψ within $B(0, r)$ to obtain a smooth function on $B(0, R)$. $\square$

THEOREM 7.7 (Exponential estimate from below). *Let $a < b$ and suppose $U \subset\subset \mathbb{R}^n$ is an open set.*

There exist constants $C, h_0 > 0$ such that if $u(h)$ solves

$$P(h)u = E(h)u(h) \quad \text{in } \mathbb{R}^n$$

for $E(h) \in [a, b]$ and $0 < h \leq h_0$, then

$$(7.2.8) \quad \|u(h)\|_{L^2(U)} \geq e^{-\frac{C}{h}} \|u(h)\|_{L^2(\mathbb{R}^n)}.$$

We call (7.2.8) a *Carleman estimate*.

REMARK. The condition (7.2.1) can easily be relaxed to allow $k = 0$. In that case we have to assume that $V(x) - b \geq c$ for $|x| \geq R$. $\square$

Proof. 1. We may assume without loss of generality that $U = B(0, 3r)$ for some $0 < r < \frac{1}{3}$. Select $R > 1$ so large that

$$p(x, \xi) - \lambda = |\xi|^2 + V(x) - \lambda \geq |\xi|^2 + \langle x \rangle^k / C$$

for $|x| \geq R$ and $a \leq \lambda \leq b$. Since $p - E(h)$ is therefore elliptic on $\mathbb{R}^n \setminus B(0, R)$, we have the estimate

$$(7.2.9) \quad \|v\|_{L^2(\mathbb{R}^n \setminus B(0, R))} \leq C\|(P(h) - E(h))v\|_{L^2(\mathbb{R}^n \setminus B(0, R))}$$

for all $v \in C_c^\infty(\mathbb{R}^n \setminus B(0, R))$ and for h small enough. In fact, we can consider $P(h) + V_0(x) - \lambda$ where $\text{spt } V_0 \subset B(0, R)$ which is elliptic everywhere in $S(\langle x \rangle^2 + \langle x \rangle^k)$. Then for all $v \in C_c^\infty(\mathbb{R}^n)$ and for $a \leq \lambda \leq b$, $\|(P(h) + V_0(x) - \lambda)v\|_{L^2} \geq \|v\|_{L^2}$ by Theorem 4.29. The term V_0 disappears when $v \in C_c^\infty(\mathbb{R}^n \setminus B(0, R))$.

2. Select two radial functions $\chi_1, \chi_2 : \mathbb{R}^n \rightarrow \mathbb{R}$ such that $0 \leq \chi_1 \leq 1$ and

$$\begin{cases} \chi_1 \equiv 0 & \text{on } B(0, r) \\ \chi_1 \equiv 1 & \text{on } B(0, R+2) \setminus B(0, 2r) \\ \chi_1 \equiv 0 & \text{on } \mathbb{R}^{2n} \setminus B(0, R+3) \end{cases}$$

and such that $0 \leq \chi_2 \leq 1$ and

$$\begin{cases} \chi_2 \equiv 0 & \text{on } B(0, R) \\ \chi_2 \equiv 1 & \text{on } \mathbb{R}^{2n} \setminus B(0, R+1). \end{cases}$$

Applying (7.2.9) to $v = \chi_2 u$ gives

$$\|\chi_2 u\|_{L^2} \leq C\|(P(h) - E(h))(\chi_2 u)\|_{L^2} = C\|[P(h), \chi_2]u\|_{L^2}.$$

Now

$$[P(h), \chi_2]u = -h^2 u \Delta \chi_2 - 2h^2 \langle \partial \chi_2, \partial u \rangle,$$

and consequently $[P(h), \chi_2]u$ is supported within $B(0, R+1) \setminus B(0, R)$. Hence Theorem 7.1 implies

$$\begin{aligned} \|[P(h), \chi_2]u\|_{L^2} &\leq Ch\|u\|_{H_h^1(B(0, R+1) \setminus B(0, R))} \\ &\leq Ch(\|(P(h) - E(h))u\|_{L^2(B(0, R+2) \setminus B(0, R-1))} \\ &\quad + \|u\|_{L^2(B(0, R+2) \setminus B(0, R-1))}) \\ &\leq Ch\|u\|_{L^2(B(0, R+2) \setminus B(0, R-1))} \\ &\leq Ch\|\chi_1 u\|_{L^2}. \end{aligned}$$

Therefore

$$(7.2.10) \quad \|\chi_2 u\|_{L^2} \leq Ch\|\chi_1 u\|_{L^2}.$$

3. Next apply Theorem 7.5:

$$h^{1/2}\|e^{\frac{\varphi}{h}}\chi_1 u\|_{L^2} \leq C\|e^{\frac{\varphi}{h}}(P(h) - E(h))(\chi_1 u)\|_{L^2} = C\|e^{\frac{\varphi}{h}}[P(h), \chi_1]u\|_{L^2}.$$

Now $[P(h), \chi_1]$ is supported within the union of $B(0, 2r) \setminus B(0, r)$ and $B(0, R+3) \setminus B(0, R+2)$. Since φ is nonincreasing, we therefore have

$$\begin{aligned} \|e^{\frac{\varphi}{h}}[P(h), \chi_1]u\|_{L^2} &\leq Che^{\frac{\varphi(R+2)}{h}}\|\chi_2 u\|_{H_h^1(B(0, R+3) \setminus B(0, R+2))} \\ &\quad + Che^{\frac{\varphi(0)}{h}}\|u\|_{H_h^1(B(0, 2r))}. \end{aligned}$$

The right-hand sides can be estimated by Theorem 7.1. This gives

$$(7.2.11) \quad \|e^{\frac{\varphi}{h}} \chi_1 u\|_{L^2} \leq Ch^{1/2} e^{\frac{\varphi(R+2)}{h}} \|\chi_2 u\|_{L^2} + Ch^{1/2} e^{\frac{\varphi(0)}{h}} \|u\|_{L^2(U)}.$$

4. Put $A = \varphi(R+2)$. We observe that

$$e^{\frac{2A}{h}} \chi_1^2 \leq 2(e^{\frac{2\varphi}{h}} \chi_1^2 + e^{\frac{2A}{h}} \chi_2^2).$$

Hence multiplying (7.2.10) by $e^{A/h}$ gives

$$\|e^{\frac{A}{h}} \chi_2 u\|_{L^2} \leq Ch \left(\|e^{\frac{\varphi}{h}} \chi_1 u\|_{L^2} + \|e^{\frac{A}{h}} \chi_2 u\|_{L^2} \right).$$

Adding this to (7.2.11) results in

$$\begin{aligned} & \|e^{\frac{A}{h}} \chi_2 u\|_{L^2} + \|e^{\frac{\varphi}{h}} \chi_1 u\|_{L^2} \\ & \leq Ch \|e^{\frac{\varphi}{h}} \chi_1 u\|_{L^2} + Ch^{1/2} \|e^{\frac{A}{h}} \chi_2 u\|_{L^2} + Ch^{1/2} e^{\frac{\varphi(0)}{h}} \|u\|_{L^2(U)}. \end{aligned}$$

Take $0 < h \leq h_0$, for h_0 sufficiently small, to deduce that

$$\|e^{\frac{A}{h}} \chi_2 u\|_{L^2} + \|e^{\frac{\varphi}{h}} \chi_1 u\|_{L^2} \leq Ch^{1/2} e^{\frac{\varphi(0)}{h}} \|u\|_{L^2(U)}.$$

This gives, upon our using the fact that $\varphi > 0$, that

$$\|\chi_2 u\|_{L^2} + \|\chi_1 u\|_{L^2} \leq Ch^{1/2} e^{\frac{\varphi(0)}{h}} \|u\|_{L^2(U)}.$$

Since $\chi_1^2 + \chi_2^2 \geq 1/2$ on $\mathbb{R}^n \setminus B(0, 2r) \supset \mathbb{R}^n - U$, the theorem follows. $\square$

7.3. ORDER OF VANISHING

Assume, as usual, that

$$(7.3.1) \quad P(h)u(h) = E(h)u(h),$$

where

$$a \leq E(h) \leq b$$

for given a, b . We intend in this section to estimate the order of vanishing of $u(h)$ at any given point.

DEFINITION. We say a function u *vanishes to order* N at the point x_0 if

$$u(x) = O(|x - x_0|^N) \quad \text{as } x \rightarrow x_0.$$

We will consider potentials V that are analytic in x and, to avoid technical difficulties, make a strong assumption on the growth of derivatives:

$$(7.3.2) \quad \begin{cases} C + V(x) \geq c \langle x \rangle^m \\ |\partial^\alpha V(x)| \leq C^{1+|\alpha|} |\alpha|^{|\alpha|} \langle x \rangle^m \end{cases}$$

for constants $C, c, m > 0$ and all multiindices α .

We first establish the analyticity of solutions to (7.3.1) in a semiclassical quantitative way. This provides an h -dependent estimate on the derivatives.

THEOREM 7.8 (H_h^k estimates). *There exists a constant C such that for any positive integer k :*

$$(7.3.3) \quad \|u(h)\|_{H_h^k(\mathbb{R}^n)} \leq C^k (1 + kh)^k \|u(h)\|_{L^2(\mathbb{R}^n)}.$$

Proof. 1. By adding a constant to V , we can assume without loss of generality that $V(x) \geq c\langle x \rangle^m$.

The theorem will follow from the following stronger estimate, which we will prove by induction:

$$(7.3.4) \quad \begin{aligned} \|\langle x \rangle^{m/2} (hD)^\alpha u(h)\|_{L^2} + \|(h\partial)(hD)^\alpha u(h)\|_{L^2} \\ \leq C_2^{k+2} (1 + kh)^{k+1} \|u(h)\|_{L^2} \end{aligned}$$

for $|\alpha| = k$.

2. To prove (7.3.4), we observe first that by multiplying (7.3.1) by $\bar{u}(h)$ and integrating by parts, estimate (7.3.4) holds for $|\alpha| = 0$.

Next, note that

$$\begin{aligned} & \|V^{\frac{1}{2}}(hD)^\alpha u(h)\|_{L^2}^2 + \|(h\partial)(hD)^\alpha u(h)\|_{L^2}^2 \\ &= \langle (-h^2\Delta + V - E(h))(hD)^\alpha u(h), (hD)^\alpha u(h) \rangle + E(h) \|(hD)^\alpha u(h)\|_{L^2}^2 \\ &= \langle V^{-\frac{1}{2}}[V, (hD)^\alpha]u(h), V^{\frac{1}{2}}(hD)^\alpha u(h) \rangle + E(h) \|(hD)^\alpha u(h)\|_{L^2}^2 \\ &\leq \|V^{-\frac{1}{2}}[V, (hD)^\alpha]u(h)\|_{L^2}^2 + \frac{1}{4} \|V^{\frac{1}{2}}(hD)^\alpha u(h)\|_{L^2}^2 + E(h) \|(hD)^\alpha u(h)\|_{L^2}^2. \end{aligned}$$

Hence

$$(7.3.5) \quad \begin{aligned} & \frac{1}{2} \|V^{\frac{1}{2}}(hD)^\alpha u(h)\|_{L^2}^2 + \|(h\partial)(hD)^\alpha u(h)\|_{L^2}^2 \\ & \leq \|V^{-\frac{1}{2}}[V, (hD)^\alpha]u(h)\|_{L^2}^2 + E(h) \|(hD)^\alpha u(h)\|_{L^2}^2. \end{aligned}$$

3. We can now expand the commutator and use (7.3.2) to deduce that for $|\alpha| = k$ we have

$$(7.3.6) \quad \begin{aligned} & \|V^{-\frac{1}{2}}[V, (hD)^\alpha]u(h)\|_{L^2} \\ & \leq C_0^{\frac{3}{2}} \sum_{l=0}^{k-1} \binom{k}{l} C_0^{k-l} (h(k-l))^{k-l} \sup_{|\beta|=l} \|\langle x \rangle^{m/2} (hD)^\beta u(h)\|_{L^2}. \end{aligned}$$

This follows from the Leibnitz rule

$$\partial^k(wv) = \sum_{l=0}^k \binom{k}{l} \partial^{k-l} w \partial^l v.$$

4. We prove (7.3.4) by induction and thus assume that (7.3.4) is valid for $|\alpha| < k$. Now Stirling's formula implies

$$\binom{k}{l} \leq C \frac{k^k}{l^l (k-l)^{k-l}}.$$

Hence, in view of (7.3.5) and (7.3.6), it is enough to show that there exists a constant C_2 such that

$$\begin{aligned} C_0^{\frac{3}{2}} \sum_{l=0}^{k-1} h^{k-l} k^k l^{-l} C_0^{k-l} C_2^{l+2} (1+lh)^{l+1} + C_2^{k+1} (1+hk)^k \\ \leq C_2^{k+2} (1+hk)^{k+1}. \end{aligned}$$

We rewrite this estimate as

$$\begin{aligned} C_0^{\frac{3}{2}} \sum_{l=1}^{k-1} \left(\frac{C_0}{C_2} \right)^{k-l} (hl)^{-l} (1+hl)^l (1+hl) + C_2^{-1} (hk)^{-k} (1+hk)^k \\ \leq (hk)^{-k} (1+hk)^k (1+hk). \end{aligned}$$

Since we can choose C_2 to be large and since we can estimate the $(1+hl)$ factor in the sum by $(1+hk)$, this will follow once we show that for ϵ small enough,

$$\sum_{l=0}^{k-1} \epsilon^{k-l} a_l \leq a_k \quad \text{for } a_l := (1 + (hl)^{-1})^l.$$

This is true by induction if a_{k-1}/a_k is bounded: the induction hypothesis and $a_{k-1} \leq C_3 a_k$ imply

$$\sum_{l=0}^{k-1} \epsilon^{k-l} a_l = \epsilon \sum_{l=0}^{k-2} \epsilon^{k-1-l} a_l + \epsilon a_{k-1} \leq 2\epsilon a_{k-1} \leq 2C_3 \epsilon a_k,$$

and we need $\epsilon \leq 1/(2C_3)$.

For our $a_k = (1 + (hk)^{-1})^k$,

$$\begin{aligned} \frac{a_{k-1}}{a_k} &= \left(\frac{1 + (h(k-1))^{-1}}{1 + (hk)^{-1}} \right)^{k-1} (1 + (hk)^{-1})^{-1} \\ &= \left(1 + \frac{1}{(k-1)(1+hk)} \right)^{k-1} (1 + (hk)^{-1})^{-1} \\ &\leq \exp \left(\frac{1}{1+hk} \right) \frac{hk}{1+hk} < 1. \end{aligned}$$

This completes the proof of (7.3.4), a stronger statement than (7.3.3). $\square$

THEOREM 7.9 (Semiclassical estimate on vanishing order). *Suppose that $u(h) \in L^2$ solves (7.3.1) for $a \leq E(h) \leq b$. Assume also that V is a real analytic potential satisfying (7.3.2). Take K to be a compact subset of $\mathbb{R}^n$.*

There exists a constant C such that if $u(h)$ vanishes to order N at a point $x_0 \in K$, we have the estimate

$$(7.3.7) \quad N \leq Ch^{-1}.$$

Proof. 1. Assume that $\|u(h)\|_{L^2} = 1$ and that $u(h)$ vanishes to order N at a point $x_0 \in K$.

Then $D^\alpha u(h)(x_0) = 0$ for $|\alpha| < N$ and Taylor's formula shows that

$$(7.3.8) \quad |u(h)(x)| \leq \frac{\epsilon^N}{N!} \sup_{|\alpha|=N} \sup_{y \in \mathbb{R}^n} |D^\alpha u(h)(y)| \quad \text{for } |x - x_0| < \epsilon.$$

Lemma 3.5 and Theorem 7.8 allow us to estimate the derivatives. If

$$M = N + n, \quad |\alpha| = N,$$

then

$$\sup_{y \in \mathbb{R}^n} |D^\alpha u(h)(y)| \leq \|u(h)\|_{H^M} \leq h^{-M} \|u(h)\|_{H_h^M} \leq h^{-M} C_1^M (1 + hM)^M.$$

Inserting this into (7.3.8) and using Stirling's formula, we deduce that for $|x - x_0| < \epsilon$,

$$\begin{aligned} |u(h)(x)| &\leq C \left(\frac{e\epsilon}{N} \right)^N \left(\frac{C_1}{h} \right)^M (1 + hM)^M \\ &\leq N^n \left(\frac{C_1 \epsilon^{1/n} e}{Nh} \right)^M (1 + hM)^M. \end{aligned}$$

2. If we put

$$A := Mh, \quad \frac{1}{K} := C_1 \epsilon^{1/n} e < 1,$$

then

$$\begin{aligned} |u(h)(x)| &\leq (Ah^{-1})^n (KA)^{-Ah^{-1}} (1 + A)^{Ah^{-1}} \\ &= (Ah^{-1})^n (1 + 1/A)^{Ah^{-1}} \exp(-Ah^{-1} \log K) \\ &\leq \exp(-Ah^{-1}(\log K - 1 - nh(\log(1/h) + \log A)/A)). \end{aligned}$$

We can assume that A is large, as otherwise

$$N < M = Ah^{-1},$$

in which case the conclusion of the theorem holds.

Hence, taking ϵ small enough, depending only on C_1 and n ,

$$|u(h)(x)| \leq \exp(-\alpha Ah^{-1}),$$

for $|x - x_0| < \epsilon$ and for a fixed $\alpha > 0$ depending on ϵ .

3. It follows that

$$\int_{\{|x-x_0|<\epsilon\}} |u(h)|^2 dx \leq C_2 \epsilon^n e^{-2\alpha A/h},$$

uniformly in h . But according to Theorem 7.7,

$$\int_{\{|x-x_0|<\epsilon\}} |u(h)|^2 dx > e^{-C_3/h}.$$

Consequently $A = Mh = (N + n)h$ is bounded, and this means that $N \leq Ch^{-1}$, as claimed. $\square$

EXAMPLE: Optimal order of vanishing. Theorem 7.9 is optimal in the semiclassical limit, meaning as regards the dependence on h in estimate (7.3.7).

We can see this by considering the harmonic oscillator in dimension $n = 2$. In polar coordinates (r, θ) the harmonic oscillator for $h = 1$ takes the form

$$P_0 = r^{-2}((rD_r)^2 + D_\theta^2 + r^4).$$

The eigenspace corresponding to the eigenvalue $2k + 2$ has dimension $k + 1$, corresponding to the number of multiindices $\alpha = (\alpha_1, \alpha_2)$, with $|\alpha| = \alpha_1 + \alpha_2 = k$. Separating variables, we look for eigenfunctions of the form

$$u = u_{km}(r)e^{im\theta}.$$

Then

$$r^{-2}((rD_r)^2 + m^2 + r^4 - (2k + 2)r^2)u_{km}(r) = 0.$$

Since the number of linearly independent eigenfunctions is $k + 1$, there must be a solution for some integer $m > k/2$. Near $r = 0$, we have the asymptotics $u_{km} \simeq r^{\pm m}$, and the case $u_{km} \simeq r^{-m}$ is impossible since $u \in L^2$. Therefore $u \simeq r^m$ has to vanish to order m .

Rescaling to the semiclassical case, we see that for the eigenvalue $E(h) = (2k + 2)h \simeq 1$ we have an eigenfunction vanishing to order $\simeq 1/h$. $\square$

7.4. L^∞ ESTIMATES FOR QUASIMODES

Next we show how a natural frequency localization condition on approximate solutions to pseudodifferential equations implies h -dependent L^∞ bounds. As an application we will provide bounds on eigenfunction clusters for compact Riemannian manifolds.

7.4.1. Quasimodes. We consider again in this section families of approximate solutions $\{u(h)\}_{0 < h \leq h_0}$ of $P(h)u(h) = 0$, where $P(h)$ is an h -dependent differential or pseudodifferential operator.

NOTATION. We will hereafter informally refer to approximate solutions of $P(h)u = 0$ as *quasimodes*.

We will also informally classify quasimodes according to the size of $P(h)u$ as $h \rightarrow 0$. The crudest condition is

$$(7.4.1) \quad P(h)u(h) = O_{L^2}(h)$$

and $u(h) = O_{L^2}(1)$. This bound is valid for eigenvalue clusters; see Theorem 7.13. Finer quasimodes require that

$$(7.4.2) \quad P(h)u(h) = o_{L^2}(h),$$

and the finest quasimodes are defined by conditions such as

$$(7.4.3) \quad P(h)u(h) = O_{\mathcal{S}}(h^\infty).$$

REMARK: Quasimodes and localization. In certain cases, quasimodes can be localized. For instance, if (7.4.1) holds and $\chi \in \mathcal{S}$, then

$$(7.4.4) \quad \begin{aligned} P(h)\chi^w(x, hD)u \\ = \chi^w(x, hD)P(h)u + [P(h), \chi^w(x, hD)]u = O_{L^2}(h), \end{aligned}$$

since we gain a factor h from the commutator, as noted after Theorem 4.24. This means that the condition (7.4.1) can be achieved without relying on global properties of the operator $P(h)$.

On the other hand, we saw in Theorem 5.4 that the finer quasimodes satisfying (7.4.2) have semiclassical defect measure invariant under the flow of p . Their construction requires global considerations. $\square$

7.4.2. Preliminary estimates. We now pose a basic question: What pointwise bounds are valid for approximate solutions of $P(h)u(h) = 0$ in the semiclassical limit $h \rightarrow 0$?

We start with the following variant of Lemma 3.5:

LEMMA 7.10 (A simple L^∞ estimate). *Suppose that $\{u(h)\}_{0 < h \leq h_0}$ is bounded in $L^2(\mathbb{R}^n)$ and there exists $\psi \in C_c^\infty(\mathbb{R}^n)$ such that*

$$(7.4.5) \quad \|(1 - \psi(hD))u(h)\|_{H_h^k(\mathbb{R}^n)} = O(h^\infty)\|u(h)\|_{L^2(\mathbb{R}^n)}$$

for all k .

Then

$$(7.4.6) \quad \|u(h)\|_{L^\infty(\mathbb{R}^n)} \leq Ch^{-n/2}\|u(h)\|_{L^2(\mathbb{R}^n)}.$$

We regard (7.4.5) as a *frequency localization* condition.

Proof. We can assume that $\|u(h)\|_{L^2} = 1$. Then condition (7.4.5) and Lemma 3.5 imply

$$\|(1 - \psi(hD))u(h)\|_{L^\infty} = O(h^\infty).$$

To estimate $\|\psi(hD)u\|_{L^\infty}$, we use (3.3.2):

$$\|\psi(hD)u\|_{L^\infty} \leq \frac{1}{(2\pi h)^n} \|\psi\|_{L^2} \|\mathcal{F}_h u\|_{L^2} = \frac{1}{(2\pi h)^{n/2}} \|\psi\|_{L^2} \|u\|_{L^2}. \quad \square$$

We will later need the following

LEMMA 7.11 (A simple L^2 estimate). *Suppose that $\{a_t\}_{t \in \mathbb{R}} \subset S$ is a family of real-valued symbols, depending continuously on t , and suppose that*

$$\begin{cases} (hD_t + a_t^w(x, hD))u = f & (t \in \mathbb{R}) \\ u(\cdot, 0) = u_0. \end{cases}$$

Then

$$(7.4.7) \quad \|u(\cdot, t)\|_{L^2(\mathbb{R}^n)} \leq \frac{\sqrt{|t|}}{h} \|f\|_{L^2(\mathbb{R} \times \mathbb{R}^n)} + \|u_0\|_{L^2(\mathbb{R}^n)}$$

for each $t \in \mathbb{R}$.

Proof. Since $a_t^w(x, hD)$ is a family of bounded operators on $L^2(\mathbb{R}^n)$, the existence of solutions follows from the existence theory for linear ordinary differential equations in the variable t .

Suppose first that $f \equiv 0$. Then

$$\begin{aligned} \frac{1}{2} \partial_t \|u(\cdot, t)\|_{L^2(\mathbb{R}^n)}^2 &= \operatorname{Re} \langle \partial_t u(\cdot, t), u(\cdot, t) \rangle_{L^2(\mathbb{R}^n)} \\ &= \frac{1}{h} \operatorname{Re} \langle i a_t^w u(\cdot, t), u(\cdot, t) \rangle = 0. \end{aligned}$$

Thus, if we set $E(t)u_0 := u(t)$, we have

$$\|E(t)u_0\|_{L^2(\mathbb{R}^n)} = \|u_0\|_{L^2(\mathbb{R}^n)}.$$

If $f \neq 0$, Duhamel's formula gives

$$u(\cdot, t) = E(t)u_0 + \frac{i}{h} \int_0^t E(t-s)f(\cdot, s) ds.$$

Hence

$$\|u(\cdot, t)\|_{L^2(\mathbb{R}^n)} \leq \|u_0\|_{L^2(\mathbb{R}^n)} + \frac{1}{h} \int_0^t \|f(\cdot, s)\|_{L^2(\mathbb{R}^n)} ds.$$

The estimate (7.4.7) is an immediate consequence. $\square$

7.4.3. Nondegeneracy, localization, and L^∞ bounds. Let us now suppose m is an order function and $K \subset \mathbb{R}^{2n}$ is a compact set. We will suppose that a real-valued symbol $p \in S(m)$ satisfies the nondegeneracy condition:

$$(7.4.8) \quad \partial_\xi p \neq 0 \quad \text{on } \{p = 0\} \cap K.$$

We will also need this localization condition:

DEFINITION. Let K be a compact subset of $\mathbb{R}^{2n}$. We say that a family of functions $\{u(h)\}_{0 < h \leq h_0}$ is *localized to K* if there exists a smooth function χ with

$$(7.4.9) \quad \text{spt}(\chi) \subseteq K$$

for which

$$(7.4.10) \quad (I - \chi^w(x, hD))u(h) = O_{\mathcal{S}}(h^\infty).$$

The interpretation is that $u(h)$ is “negligible as $h \rightarrow 0$ ” outside of K .

THEOREM 7.12 (L^∞ bounds for quasimodes). *Assume p satisfies the nondegeneracy condition (7.4.8). Suppose further that $u = \{u(h)\}_{0 < h \leq h_0}$ is bounded in $L^2(\mathbb{R}^n)$ and localized to K .*

Then if

$$(7.4.11) \quad \|p^w(x, hD)u(h)\|_{L^2(\mathbb{R}^n)} = O(h)\|u(h)\|_{L^2(\mathbb{R}^n)},$$

we have the estimate

$$(7.4.12) \quad \|u(h)\|_{L^\infty(K)} \leq Ch^{-(n-1)/2}\|u(h)\|_{L^2(\mathbb{R}^n)}.$$

So if $u(h)$ is a quasimode in the sense of the estimate (7.4.11), we can improve the earlier L^∞ estimate (7.4.6) by a factor of $h^{\frac{1}{2}}$. The localization condition (7.4.10) is not very restrictive in view of (7.4.4).

Proof. 1. We can assume that $\|u(h)\| = 1$. In light of (7.4.10) it is enough to prove the theorem for $u(h)$ replaced by $\chi^w u(h)$, $\text{spt } \chi \subset K \subset \subset \mathbb{R}^{2n}$. For every $\rho_0 = (x_0, \xi_0) \in K$ there exists an open neighborhood V_{ρ_0} of ρ_0 such that

$$(7.4.13) \quad \text{either } |p(\rho)| \geq c > 0 \text{ or } |\partial_\xi p(\rho)| \geq c > 0 \text{ for } \rho \in V_{\rho_0}.$$

Then

$$K \subset \bigcup_{j=1}^N V_{\rho_j},$$

and we can take a partition of unity associated to this cover, $\{\varphi_j\}_{j=1}^N$, $\text{spt } \varphi_j \subset V_{\rho_j}$. We have

$$\chi = \sum_{j=1}^N \chi_j, \quad \chi_j := \varphi_j \chi.$$

Recalling (7.4.4), we see that we only need to prove (7.4.12) for u replaced by $\chi_j^w(x, hD)u$, where (7.4.13) holds on the support of χ_j .

To simplify notation, we now drop the index j and assume that (7.4.13) holds on a neighborhood of the support of χ .

2. Suppose that $p \neq 0$ on the support of χ . Then $q = \tilde{\chi}/p \in C_c^\infty(\mathbb{R}^{2n})$, for some $\tilde{\chi} \in C_c^\infty(\mathbb{R}^n)$ equal to 1 on $\text{spt } \chi$. The composition formula in Theorem 4.18 shows that

$$q^w(x, hD)p^w(x, hD) = \tilde{\chi}^w(x, hD) + hr^w(x, hD)$$

for $r \in S$. Hence $P(h)\chi^w u(h) = O_{L^2}(h)$ implies that $\chi^w u(h) = O_{L^2}(h)$. Lemma 7.10 then shows that

$$\|\chi^w u(h)\|_{L^\infty} \leq Chh^{-n/2} \leq Ch^{-(n-1)/2}.$$

3. Now suppose that the other $|\partial_\xi p| \neq 0$ on the support of χ . By applying a linear change of variables we can assume that $p_{\xi_1} > 0$ there. The Implicit Function Theorem shows that in a neighborhood of $\text{spt } \chi$, we have

$$(7.4.14) \quad p(x, \xi) = e(x, \xi)(\xi_1 - a(x, \xi')),$$

where $\xi = (\xi_1, \xi')$ and $e > 0$. We extend e arbitrarily to a symbol in S , with $e \geq \gamma > 0$, and extend a as well to a real-valued symbol in S . Then

$$\begin{aligned} e^w(x, hD)(hD_{x_1} - a(x, hD_{x'}))\chi^w u(h) &= P(h)(\chi^w u(h)) + O_{L^2}(h) \\ &= O_{L^2}(h). \end{aligned}$$

Since e^w is elliptic, Theorem 4.29 shows that

$$(7.4.15) \quad (hD_{x_1} - a(x, hD_{x'}))\chi^w u(h) = O_{L^2}(h).$$

4. Lemma 7.11 and (7.4.15) show that

$$(7.4.16) \quad \|(\chi^w u)(x_1, \cdot)\|_{L^2(\mathbb{R}^{n-1})} = O(1).$$

We now apply Lemma 7.10 in x' variables only, that is, with $n-1$ replacing n and t replacing x_1 . To justify this step, we need to verify the hypothesis (7.4.5): for some $\psi \in C_c^\infty(\mathbb{R}^{n-1})$

$$(7.4.17) \quad \|(1 - \psi(hD'))\chi^w u(h)(x_1, \cdot)\|_{H_h^k(\mathbb{R}^{n-1})} = O(h^\infty),$$

uniformly in x_1 , for all k . We choose $\psi \in C_c^\infty$ so that $\psi(\xi') = 1$ if $(x_1, x'; \xi_1 \xi')$ is in a neighborhood of $\text{spt } \chi$. Then, by the composition formula (4.3.5)

$$(1 - \psi(hD'))\chi^w(x, hD) = c^w(x, hD), \quad c \in h^\infty \mathcal{S}(\mathbb{R}^{2n}),$$

and $c^w(x, hD) = O_{L^2 \rightarrow \mathcal{S}}(h^\infty)$. This implies (7.4.17). $\square$

REMARKS. (i) The bound provided by Theorem 7.12 is already optimal in the simplest case in which the assumptions are satisfied: $p(x, \xi) = \xi_1$. Indeed, write $x = (x_1, x')$ and let $\varphi \in C_c^\infty(\mathbb{R})$ and $\chi \in C_c^\infty(\mathbb{R}^{n-1})$. Then

$$u(h) := h^{-(n-1)/2} \varphi(x_1) \chi(x'/h)$$

satisfies $\|u(h)\|_{L^2} = O(1)$,

$$P(h)u(h) = hD_{x_1}u(h) = O_{L^2}(h),$$

and for any nontrivial choices of φ and χ , $\|u(h)\|_{L^\infty} \simeq h^{-(n-1)/2}$.

(ii) That the nondegeneracy condition (7.4.8) is in general necessary is shown by another simple example. Let $p(x, \xi) = x_1$ and

$$u(h) = h^{-n/2} \varphi(x_1/h) \chi(x'/h).$$

Then $\|u(h)\|_{L^2} = O(1)$,

$$P(h)u(h) = hh^{-n/2} (t\varphi(t))|_{t=x_1/h} \chi(x'/h) = O_{L^2}(h),$$

and $\|u(h)\|_{L^\infty} \simeq h^{-n/2}$. This is the general bound of Lemma 7.10. $\square$

7.4.4. Bounds for spectral clusters. Next we provide L^∞ bounds on “spectral clusters”, that is, linear combinations of eigenfunctions for the Laplace–Beltrami operator on a compact manifold. (The proof requires the material presented in Chapter 14.)

Suppose that M is an n -dimensional compact Riemannian manifold and let Δ_g be its Laplace–Beltrami operator. Assume that

$$0 = \lambda_0 < \lambda_1 \leq \dots \leq \lambda_j \rightarrow \infty$$

are the eigenvalues of $-\Delta_g$ and that

$$-\Delta_g \varphi_j = \lambda_j \varphi_j \quad (j = 0, 1, \dots)$$

are a corresponding orthonormal basis of eigenfunctions.

THEOREM 7.13 (L^∞ bounds on eigenfunction clusters).

(i) *There exists a constant C such that for any choices of constants $c_j \in \mathbb{C}$, we have the inequality*

$$(7.4.18) \quad \|u_\mu\|_{L^\infty(M)} \leq C \mu^{(n-1)/2} \|u_\mu\|_{L^2(M)}$$

where

$$u_\mu := \sum_{\mu \leq \sqrt{\lambda_j} \leq \mu+1} c_j \varphi_j$$

and $\mu \geq 1$.

(ii) In particular,

$$(7.4.19) \quad \|\varphi_j\|_{L^\infty(M)} \leq C \lambda_j^{(n-1)/4} \|\varphi_j\|_{L^2(M)}.$$

Proof. Put $h = 1/\mu$, $P(h) := -h^2 \Delta_g - 1$, and $u(h) = u_\mu$. Then the assumption (7.4.8) holds everywhere. Also

$$\begin{aligned} \|P(h)u(h)\|_{L^2} &= \left\| \sum_{\mu \leq \sqrt{\lambda_j} \leq \mu+1} c_j (h^2 \lambda_j - 1) \varphi_j \right\|_{L^2} \\ &= \left(\sum_{\mu \leq \sqrt{\lambda_j} \leq \mu+1} |c_j|^2 (h^2 \lambda_j - 1)^2 \|\varphi_j\|_{L^2}^2 \right)^{\frac{1}{2}} \\ &\leq 2h \|u(h)\|_{L^2}. \end{aligned}$$

Thus (7.4.11) holds.

To obtain the frequency localization condition, we first note that, from the Spectral Theorem (Theorem C.10) and Theorem 14.7 we see that for $\psi \in C_c^\infty(\mathbb{R})$ satisfying $\psi(t) \equiv 1$ for $|t| \leq 2$,

$$\begin{aligned} \psi(-h^2 \Delta_g) u(h) &= \sum_j \psi(h^2 \lambda_j) \langle u(h), \varphi_j \rangle_{L^2(M)} \varphi_j \\ &= \sum_{\mu \leq \sqrt{\lambda_j} \leq \mu+1} \psi\left(\frac{\lambda_j}{\mu^2}\right) c_j \varphi_j \\ &= u(h). \end{aligned}$$

Hence, on a compact manifold, (7.4.5) follows from Theorem 14.9. $\square$

7.5. SCHAUDER ESTIMATES

All of the estimates provided so far in this and previous chapters have been based upon calculations in L^2 , related Sobolev spaces, and L^∞ . We next show that we can in fact use Fourier methods to characterize spaces of Hölder continuous functions and as application provide a semiclassical proof of the interior Schauder estimates for the Laplacian. This section is independent of the results of Chapter 4 and can be read immediately after Chapter 3.

7.5.1. Littlewood–Paley decomposition. This important tool is a decomposition of a given function into components with controlled Fourier frequencies. We start with the following lemma.

LEMMA 7.14 (Dyadic partition of unity). *There exist functions $\psi_0 \in C_c^\infty(\mathbb{R})$ and $\psi \in C_c^\infty(\mathbb{R} \setminus \{0\})$ such that*

$$(7.5.1) \quad 1 = \psi_0(|x|) + \sum_{j=0}^{\infty} \psi(2^{-j}|x|)$$

for each point $x \in \mathbb{R}^n$.

Proof. Select $\varphi_0 \in C_c^\infty((-1, 1))$ so that $0 \leq \varphi_0 \leq 1$ and $\varphi_0(\rho) = 1$ for $|\rho| \leq 1/2$. Then

$$\varphi_1(\rho) := \sum_{j=-\infty}^{\infty} \varphi_0(\rho - j)$$

is a smooth positive function satisfying $\varphi_1(\rho) \geq 1$ and $\varphi_1(\rho - j) = \varphi_1(\rho)$ for all $j \in \mathbb{Z}$. Hence for $\varphi := \varphi_0/\varphi_1$ we have

$$\sum_{j=-\infty}^{\infty} \varphi(\rho - j) = 1$$

for all $\rho \in \mathbb{R}$. Now define

$$\psi(r) := \varphi(\log r / \log 2).$$

Then $\psi \in C_c^\infty(1/2, 2)$ and

$$\sum_{j=-\infty}^{\infty} \psi(2^{-j}r) = 1$$

for each $r > 0$.

We next put

$$\psi_0(r) := 1 - \sum_{j=0}^{\infty} \psi(2^{-j}r).$$

Notice that $\psi_0(r) = 1$ for $r < 1/2$ and $\psi_0(r) = 0$ for $r > 1$. Hence $\psi_0 \in C_c^\infty([0, \infty))$. $\square$

We will hereafter identify ψ_0 and ψ with smooth radial functions on $\mathbb{R}^n$: $\psi_0(x) = \psi_0(|x|)$ and $\psi(x) = \psi(|x|)$.

DEFINITION. The *Littlewood–Paley decomposition* of $u \in \mathcal{S}'(\mathbb{R}^n)$ is

$$(7.5.2) \quad u = \psi_0(D)u + \sum_{j=0}^{\infty} \psi(2^{-j}D)u,$$

the functions ψ_0, ψ from (7.5.1).

The terms in the decomposition (7.5.2) are localized near momenta comparable to 2^j . We can therefore think of $h \simeq 2^{-j}$ as the relevant semiclassical parameter.

We record for future reference some useful estimates:

THEOREM 7.15 (Multiplier estimates).

(i) For each $\chi \in C_c^\infty(\mathbb{R}^n)$, we have

$$(7.5.3) \quad \|\chi(hD)u\|_{L^\infty(\mathbb{R}^n)} \leq Ch^{-\frac{n}{p}} \|u\|_{L^p(\mathbb{R}^n)}, \quad 1 \leq p \leq \infty.$$

(ii) Furthermore,

$$(7.5.4) \quad \|\chi(hD)\|_{L^p(\mathbb{R}^n) \rightarrow L^p(\mathbb{R}^n)} \leq \frac{1}{(2\pi)^n} \|\widehat{\chi}\|_{L^1(\mathbb{R}^n)}$$

for $1 \leq p \leq \infty$.

(iii) Suppose $\varphi \in \mathcal{S}(\mathbb{R}^n)$ and $\chi, \tilde{\chi} \in C_c^\infty(\mathbb{R}^n)$, with $\tilde{\chi} \equiv 1$ on a neighborhood of $\text{spt } \chi$. Then

$$(7.5.5) \quad \|\chi(hD)\varphi(1 - \tilde{\chi}(hD))\|_{L^p(\mathbb{R}^n) \rightarrow L^q(\mathbb{R}^n)} = O(h^\infty),$$

for all $1 \leq p, q \leq \infty$.

Proof. 1. We have

$$(7.5.6) \quad \begin{aligned} \chi(hD)u(x) &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \chi(\xi) e^{\frac{i}{h}\langle x-y, \xi \rangle} u(y) d\xi dy \\ &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \hat{\chi}\left(\frac{x-y}{h}\right) u(y) dy. \end{aligned}$$

Thus

$$|\chi(hD)u(x)| \leq \frac{1}{(2\pi)^n} \|\widehat{\chi}\|_q \|u(x - h \cdot)\|_p = C_\chi h^{-\frac{n}{p}} \|u\|_p,$$

where $\frac{1}{q} + \frac{1}{p} = 1$, $1 \leq p \leq \infty$.

2. The bound (7.5.4) follows from (7.5.6) and Young's inequality

$$\|f * g\|_{L^p(\mathbb{R}^n)} \leq \|f\|_{L^p(\mathbb{R}^n)} \|g\|_{L^1(\mathbb{R}^n)}.$$

3. The estimate (7.5.5) is an immediate consequence of the composition rule for pseudodifferential operators (Section 4.3), which shows that

$$\chi(hD)\varphi(1 - \tilde{\chi}(hD)) = O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty).$$

To give a direct proof, we write the operator in (7.5.5) using an integral kernel:

$$K_h u(x) := \chi(hD)\varphi(1 - \tilde{\chi}(hD))u(x) = \int_{\mathbb{R}^n} K_h(x, y) u(y) dy,$$

where

$$\begin{aligned} K_h(x, y) &= \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \chi(\xi) \varphi(z) (1 - \tilde{\chi}(\eta)) e^{\frac{i}{h}(\langle x, \xi \rangle - \langle y, \eta \rangle + \langle \eta - \xi, z \rangle)} dz d\xi d\eta \\ &= \frac{1}{(2\pi h)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \chi(\xi) (1 - \tilde{\chi}(\eta)) \hat{\varphi}\left(\frac{\eta - \xi}{h}\right) e^{\frac{i}{h}(\langle x, \xi \rangle - \langle y, \eta \rangle)} d\xi d\eta. \end{aligned}$$

4. Fix N . Since $\text{spt } \chi \cap \text{spt}(1 - \tilde{\chi}) = \emptyset$ and since $\varphi \in \mathcal{S}$, we see that on the support of the integrand

$$(\partial^\alpha \hat{\varphi})\left(\frac{\eta - \xi}{h}\right) = O(h^N \langle \xi - \eta \rangle^{-N})$$

for each multiindex α . Now

$$(1 - h^2 \Delta_\xi)^N (1 - h^2 \Delta_\eta)^N e^{\frac{i}{h}(\langle x, \xi \rangle - \langle y, \eta \rangle)} = \langle x \rangle^{2N} \langle y \rangle^{2N} e^{\frac{i}{h}(\langle x, \xi \rangle - \langle y, \eta \rangle)},$$

and so integration by parts shows that

$$|K_h(x, y)| \leq C_N h^N \langle x \rangle^{-N} \langle y \rangle^{-N}.$$

Then, for $1 \leq q < \infty$,

$$\|K_h u\|_{L^q}^q \leq C_N h^{Nq} \int_{\mathbb{R}^n} \langle x \rangle^{-Nq} \left(\int_{\mathbb{R}^n} \langle y \rangle^{-N} |u(y)| dy \right)^q dx \leq C_N h^{Nq} \|u\|_{L^p}^q,$$

where we used Hölder's inequality to get the last estimate. □

7.5.2. Hölder continuity. We now show how the Littlewood–Paley decomposition (7.5.2) provides a characterization of Hölder continuous functions.

Let $U \subset \mathbb{R}^n$ be an open set. We write

$$(7.5.7) \quad \|u\|_{C^{k,\gamma}(\bar{U})} := \max_{|\alpha| \leq k} \|\partial^\alpha u\|_{L^\infty(U)} + \max_{|\alpha|=k} \sup_{\substack{x \neq y \\ x, y \in U}} \frac{|\partial^\alpha u(x) - \partial^\alpha u(y)|}{|x - y|^\gamma}$$

for $k = 0, 1, \dots$ and $0 < \gamma \leq 1$.

THEOREM 7.16 (Characterization of Hölder spaces). *Suppose $u \in L^p(\mathbb{R}^n)$ for some $1 \leq p \leq \infty$.*

Then for $k = 0, 1, \dots$ and $0 < \gamma < 1$, we have

$$(7.5.8) \quad u \in C^{k,\gamma}(\mathbb{R}^n)$$

if and only if

$$(7.5.9) \quad \|\chi(hD)u\|_{L^\infty(\mathbb{R}^n)} \leq C_\chi h^{k+\gamma}$$

for each $\chi \in C_c^\infty(\mathbb{R}^n \setminus \{0\})$ and all $0 < h < 1$.

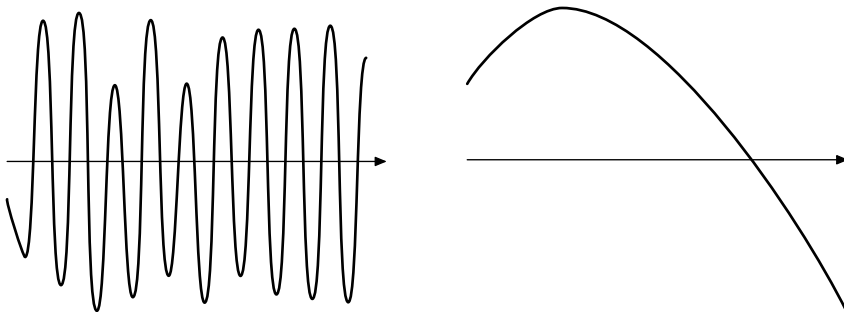


Figure 7.5.1. In the highly oscillatory regime, $h < r$, the oscillation of the function is effectively given by twice the supremum norm, as used in (7.5.16). When oscillations are low, $h \geq r$, an effective bound is obtained by using the supremum of the derivative, as in (7.5.17).

When we assert here that $u \in L^p$ in fact belongs to $C^{k,\gamma}$, we mean that there exists a function $\bar{u} \in C^{k,\gamma}$ such that $u = \bar{u}$ almost everywhere. We call $\bar{u}$ a *version* of u .

Proof. In the proof we can assume that $k = 0$, as the modification in the case of higher derivatives is straightforward.

1. We start with the easier implication that (7.5.8) implies (7.5.9). For this, we use (7.5.6) to write

$$\chi(hD)u(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{\chi}(y) u(x - hy) dy.$$

Since $\chi(0) = 0$ and $\chi \in \mathcal{S}$, we have $\int_{\mathbb{R}^n} \hat{\chi}(y) dy = 0$; and hence

$$\chi(hD)u(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{\chi}(y) (u(x - hy) - u(x)) dy.$$

Now $|\hat{\chi}(y)| \leq C_N(1 + |y|^2)^{-N}$; and so since $u \in C^{0,\gamma}(\mathbb{R}^n)$, we obtain

$$\begin{aligned} |\chi(hD)u(x)| &\leq \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} |\hat{\chi}(y)| |u(x - hy) - u(x)| dy \\ &\leq C \int_{\mathbb{R}^n} (1 + |y|^2)^{-N} |yh|^\gamma dy \leq Ch^\gamma. \end{aligned}$$

Consequently,

$$\|\chi(hD)u\|_{L^\infty} \leq Ch^\gamma;$$

this is (7.5.9) for $k = 0$.

2. To prove the opposite implication, let us write

$$(7.5.10) \quad \Lambda_\gamma(u) := \sup_{0 < h < 1} h^{-\gamma} \left(\|\psi(hD)u\|_{L^\infty} + \max_{1 \leq k \leq n} \|\psi_k(hD)u\|_{L^\infty} \right),$$

where ψ is as in (7.5.1) and

$$\psi_k(\xi) := \xi_k \psi(\xi).$$

We are assuming now that $\Lambda_\gamma(u)$ is finite.

Let us estimate

$$\begin{aligned} \|u\|_{L^\infty} &\leq \|\psi_0(D)u\|_{L^\infty} + \sum_{j=0}^{\infty} \|\psi(2^{-j}D)u\|_{L^\infty} \\ &\leq \|\psi_0(D)u\|_{L^\infty} + \Lambda_\gamma(u) \sum_{j=0}^{\infty} 2^{-j\gamma} \\ &= \|\psi_0(D)u\|_{L^\infty} + (2^\gamma - 1)^{-1} \Lambda_\gamma(u). \end{aligned}$$

Since

$$(7.5.11) \quad \psi_0(hD)u(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \widehat{\psi}_0(x-y)u(y) dy,$$

we see that

$$(7.5.12) \quad \|u\|_{L^\infty} \leq C(\|u\|_{L^p} + \Lambda_\gamma(u)).$$

3. We next claim that

$$(7.5.13) \quad |u(x) - u(y)| \leq C(\|u\|_{L^p} + \Lambda_\gamma(u))r^\alpha$$

if $|x - y| \leq r$. In view of the Littlewood–Paley decomposition (7.5.2) it is enough to show that

$$(7.5.14) \quad |\psi_0(D)u(x) - \psi_0(D)u(y)| \leq Cr^\gamma \|u\|_{L^p}$$

and that

$$(7.5.15) \quad \sum_{j=0}^{\infty} |\psi(2^{-j}D)u(x) - \psi(2^{-j}D)u(y)| \leq Cr^\gamma \Lambda_\gamma(u).$$

Inequality (7.5.14) is immediate from the integral representation (7.5.11) and the smoothness of $\widehat{\psi}$.

4. To establish (7.5.15), we exploit different estimates in regions with different rates of oscillations. The high oscillation estimate is simply

$$(7.5.16) \quad \|\psi(hD)u(x) - \psi(hD)u(y)\| \leq 2\|\psi(hD)u\|_{L^\infty} \leq 2h^\gamma \Lambda_\gamma(u);$$

and for $\psi_k(\xi) = \xi_k \psi(\xi)$ as above, the low oscillation estimate has the form

$$\begin{aligned} \|\psi(hD)u(x) - \psi(hD)u(y)\| &\leq rC \max_{1 \leq k \leq n} \|D_{x_k} \psi(hD)u\|_{L^\infty} \\ (7.5.17) \quad &\leq rh^{-1}C \max_{1 \leq k \leq n} \|\psi_k(hD)u\|_{L^\infty} \\ &\leq rh^{-1+\gamma}C\Lambda_\gamma(u). \end{aligned}$$

To prove (7.5.15), we divide up the sum and use (7.5.17) for $h = 2^{-j} \geq r$ and (7.5.16) for $h = 2^{-j} < r$:

$$\begin{aligned} & \sum_{j=0}^{\infty} |\psi(2^{-j}D)u(x) - \psi(2^{-j}D)u(y)| \\ &= \left(\sum_{2^{-j} \geq r} + \sum_{2^{-j} < r} \right) |\psi(2^{-j}D)u(x) - \psi(2^{-j}D)u(y)| \\ &\leq \Lambda_{\gamma}(u)C \left(r \sum_{2^j \leq r^{-1}} 2^{j(1-\gamma)} + \sum_{2^j > r^{-1}} 2^{-j\gamma} \right) \leq C\Lambda_{\gamma}(u)r^{\gamma}. \end{aligned}$$

This establishes (7.5.15). $\square$

The proof provides the useful estimate that for $u \in L^p(\mathbb{R}^n)$,

$$\begin{aligned} (7.5.18) \quad \frac{1}{C} (\|u\|_{L^{\infty}(\mathbb{R}^n)} + \Lambda_{\gamma}(u)) &\leq \|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \\ &\leq C_p (\|u\|_{L^p(\mathbb{R}^n)} + \Lambda_{\gamma}(u)). \end{aligned}$$

As an immediate application, we have

THEOREM 7.17 (Morrey's inequality). *If $u \in L^p(\mathbb{R}^n)$ and $\partial u \in L^p(\mathbb{R}^n)$ for some $p > n$, then $u \in C^{0,\gamma}(\mathbb{R}^n)$ for $\gamma = 1 - \frac{n}{p}$.*

Furthermore,

$$(7.5.19) \quad \|u\|_{C^{0,\gamma}(\mathbb{R}^n)} \leq C (\|u\|_{L^p(\mathbb{R}^n)} + \|\partial u\|_{L^p(\mathbb{R}^n)}).$$

The constant C depends on n and p only.

For a real variables proof see [E, Section 5.6.2].

Proof. For ψ as in (7.5.10) we write

$$\psi(\xi) = \sum_{k=1}^n \tilde{\psi}_k(\xi) \xi_k, \quad \tilde{\psi}_k(\xi) := \frac{\xi_k}{|\xi|^2} \psi(\xi).$$

Then $\tilde{\psi}_k \in C_c^{\infty}(\mathbb{R}^n \setminus \{0\})$. Estimate (7.5.3) now implies

$$\begin{aligned} \|\psi(hD)u\|_{\infty} &\leq \sum_{k=1}^n \|\tilde{\psi}_k(hD)hD_k u\|_{\infty} \\ &\leq \sum_{k=1}^n h^{1-\frac{n}{p}} \|D_k u\|_{L^p} \\ &\leq Ch^{1-\frac{n}{p}} \|\partial u\|_{L^p}. \end{aligned}$$

Similar estimates are valid for all functions ψ_k in (7.5.10). Hence

$$\Lambda_{1-\frac{n}{p}}(u) \leq C \|\partial u\|_{L^p}.$$

□

7.5.3. Schauder estimates. In this section we derive an important estimate for distributional solutions u of the PDE

$$(7.5.20) \quad -\Delta u = f \quad \text{in } U.$$

Here $U \subset \mathbb{R}^n$ is an open set.

THEOREM 7.18 (Interior Schauder estimate). *Suppose that $u \in L^1(U)$ solves (7.5.20), where $f \in C^{k,\gamma}(\bar{U})$ for some $0 < \gamma < 1$ and $k \in \mathbb{N}$.*

Then for each compactly contained open subset $V \subset\subset U$, we have

$$u \in C^{k+2,\gamma}(\bar{V}),$$

with the estimate

$$(7.5.21) \quad \|u\|_{C^{k+2,\gamma}(\bar{V})} \leq C \left(\|f\|_{C^{k,\gamma}(\bar{U})} + \|u\|_{L^1(U)} \right).$$

REMARK. To illustrate the idea of the proof, we first assume that $u \in L^1(\mathbb{R}^n)$ and that $-\Delta u = f$ in $\mathbb{R}^n$, with $f \in C^{k,\gamma}(\mathbb{R}^n)$. In view of (7.5.18) we need to show for each $\chi \in C_c^\infty(\mathbb{R}^n \setminus \{0\})$ that

$$(7.5.22) \quad \|\chi(hD)u\|_{L^\infty} \leq C_\chi h^{k+2+\gamma}$$

for all $0 < h < 1$. But this follows from our writing the equation in a way consistent with the semiclassical viewpoint of this text:

$$(7.5.23) \quad -h^2 \Delta u = h^2 f.$$

From this it follows that

$$(7.5.24) \quad \chi(hD)u = \chi_0(hD)h^2 f,$$

where

$$\chi_0(\xi) := \frac{\chi(\xi)}{|\xi|^2} \in C_c^\infty(\mathbb{R}^n \setminus \{0\}).$$

We now apply Theorem 7.16 to f .

□

Proof. 1. Let $\{\psi_j\}_{j=1}^N$ be a sequence of functions satisfying

$$\text{spt } \psi_1 \subset\subset U, \quad \text{spt } \psi_{j+1} \subset\subset \{\psi_j = 1\}, \quad V \subset\subset \{\psi_N = 1\}$$

for $j = 1, \dots, N-1$. We will select N later.

Then

$$(7.5.25) \quad -h^2 \Delta(\psi_{j+1}u) = -[h^2 \Delta, \psi_{j+1}] \psi_j u + \psi_{j+1} h^2 f.$$

The commutator is given by

$$-[h^2\Delta, \psi_{j+1}] = h(h\Delta\psi_{j+1} + 2hD \cdot D\psi_{j+1}),$$

where we use the notation $(D \cdot D\varphi)u = D \cdot ((D\varphi)u)$.

2. We now assert that for any functions $\chi, \tilde{\chi} \in C_c^\infty(\mathbb{R}^n \setminus \{0\})$ with $\tilde{\chi} \equiv 1$ on $\text{spt } \chi$, we have the estimate

$$(7.5.26) \quad \|\chi(hD)\psi_{j+1}u\|_{L^\infty} \leq Ch\|\tilde{\chi}(hD)\psi_ju\|_{L^\infty} \\ + Ch^2\|\tilde{\chi}(hD)(\psi_{j+1}f)\|_{L^\infty} + O(h^\infty)(\|\psi_ju\|_{L^1} + \|f\|_{L^\infty}).$$

To prove (7.5.26), we use the notation of (7.5.24) and (7.5.25) and obtain

$$(7.5.27) \quad \chi(hD)\psi_{j+1}u = \chi_0(hD)(-[h^2\Delta, \psi_{j+1}]\psi_ju) + \chi_0(hD)(\psi_{j+1}h^2f).$$

Since $\text{spt } \chi_0 = \text{spt } \chi$, Theorem 7.15 shows that

$$(7.5.28) \quad \|\chi_0(hD)(\psi_{j+1}h^2f)\|_{L^\infty} = h^2\|\chi_0(hD)\tilde{\chi}(hD)(\psi_{j+1}f)\|_{L^\infty} \\ \leq Ch^2\|\tilde{\chi}(hD)(\psi_{j+1}f)\|_{L^\infty}.$$

We expand the first term on the right-hand side of (7.5.27) using the expression for the commutator,

$$h^2\chi_0(hD)((\Delta\psi_{j+1})\psi_ju) + 2h\sum_{k=1}^n\chi_0(hD)(hD_{x_k}((D_{x_k}\psi_{j+1})\psi_ju)),$$

and apply (7.5.5) from Theorem 7.15 to both terms. The first terms is estimated as

$$\|h^2\chi_0(hD)(\Delta\psi_{j+1})\psi_ju\|_{L^\infty} = h^2\|\chi_0(hD)(\Delta\psi_{j+1})\tilde{\chi}(hD)(\psi_ju)\|_{L^\infty} \\ + O(h^\infty)\|\psi_ju\|_{L^1} \\ \leq Ch^2\|\tilde{\chi}(hD)(\psi_ju)\|_{L^\infty} + O(h^\infty)\|\psi_ju\|_{L^1}.$$

For the second term write

$$\varphi_k(x) := D_{x_k}\psi_j(x), \quad \chi_k(\xi) := \xi_k\chi_0(\xi),$$

so that

$$h\|\chi_k(hD)\varphi_k\psi_ju\|_{L^\infty} = h\|\chi_k(hD)\varphi_k\tilde{\chi}(hD)\psi_ju\|_{L^\infty} + O(h^\infty)\|\psi_ju\|_{L^1} \\ \leq Ch\|\tilde{\chi}(hD)(\psi_ju)\|_{L^\infty} + O(h^\infty)\|\psi_ju\|_{L^1}.$$

Combining the foregoing estimates proves (7.5.26).

3. From (7.5.3) we see that

$$\|\chi(hD)\psi_1u\|_{L^\infty} \leq Ch^{-n}\|u\|_{L^1(U)}.$$

Then (7.5.26) applied inductively shows that for $1 \leq j \leq N - 1$,

$$\|\chi(hD)\psi_{j+1}u\|_{L^\infty} \leq C_j \left(h^{j-n}\|u\|_{L^1(U)} + h^{2+k+\gamma}\|f\|_{C^{k,\gamma}(\bar{U})} \right),$$

where we used (7.5.18) to estimate $\|\tilde{\chi}(hD)f\|_\infty$.

Now we select

$$N > n + 2 + k + \gamma$$

and obtain

$$\|\chi(hD)\psi_N u\|_{L^\infty} \leq Ch^{2+k+\gamma} \left(\|u\|_{L^1(U)} + \|f\|_{C^{k,\gamma}(\bar{U})} \right).$$

According to Theorem 7.16, this shows that $\psi_N u \in C^{2+k,\gamma}(\mathbb{R}^n)$. $\square$

7.6. NOTES

Agmon estimates play a crucial role in the analysis of the spectra of multiple well potentials and of the Witten complex, as discussed in Dimassi–Sjöstrand [D-S, Chapter 6]. Here we followed an argument of Nakamura [N].

The presentation of Carleman estimates in Section 7.2 is based on discussions with N. Burq and D. Tataru, and Burq suggested the estimates for the order of vanishing. For an approach based on Almgren’s monotonicity formula and for references see Han–Lin [HaLi].

[K-T-Z] presents the semiclassical pointwise bounds reproduced here. The estimate (7.4.18) is essentially optimal, whereas the optimality of the bound on individual eigenfunctions (7.4.19) is rare. See [S-Z] for a recent discussion.

H. Smith suggested the application to Schauder estimates. For an in-depth discussion of Schauder estimates based on real analysis methods, see Gilbarg–Trudinger [G-T], and for developments of Littlewood–Paley techniques, consult Stein [St].

Part 3

ADVANCED THEORY AND APPLICATIONS

MORE ON THE SYMBOL CALCULUS

- 8.1 Beals's Theorem
- 8.2 Real exponentiation of operators
- 8.3 Generalized Sobolev spaces
- 8.4 Wavefront sets, essential support, and microlocality
- 8.5 Notes

This chapter collects various more advanced topics concerning the symbol calculus. Subsequent chapters will provide many applications.

8.1. BEALS'S THEOREM

We present next a semiclassical version of Beals's Theorem, a characterization of pseudodifferential operators in terms of h -dependent bounds on commutators. This theorem answers a fundamental question: *When can a given linear operator be represented using the symbol calculus?*

We start with $h = 1$:

THEOREM 8.1 (Estimating a symbol by operator norms). *There exist constants $C, M > 0$ such that*

$$(8.1.1) \quad \|b\|_{L^\infty(\mathbb{R}^{2n})} \leq C \sum_{|\gamma| \leq M} \|(\partial^\gamma b)^w(x, D)\|_{L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)}$$

for all $b \in \mathcal{S}'$.

Proof. 1. We will first consider the classical quantization

$$b(x, D)u(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} b(x, \xi) e^{i\langle x, \xi \rangle} \hat{u}(\xi) d\xi,$$

where by the integration we mean the Fourier transform in $\mathcal{S}'$. We will choose a fixed $\chi \in \mathcal{S}(\mathbb{R}^{2n})$, $|\chi(x, \xi)| = 1$ near $(0, 0)$, and prove the estimate

$$(8.1.2) \quad \|\chi b\|_{L^\infty} \leq C_\chi \sum_{|\gamma| \leq 2n+1} \|(\partial^\gamma b)(x, D)\|_{L^2 \rightarrow L^2},$$

assuming that the norms on the right-hand side are finite. Since $|\chi(x, \xi)| = 1$ in a neighborhood of $(0, 0)$, we have

$$\begin{aligned} \|b\|_{L^\infty(\mathbb{R}^{2n})} &\leq \sup_{(y, \eta) \in \mathbb{R}^{2n}} \|\chi(\cdot - y, \cdot - \eta)b\|_{L^\infty} \\ (8.1.3) \quad &\leq \sup_{(y, \eta) \in \mathbb{R}^{2n}} \|\chi b(\cdot + y, \cdot + \eta)\|_{L^\infty} \\ &\leq C_\gamma \sup_{(y, \eta) \in \mathbb{R}^{2n}} C_\chi \sum_{|\gamma| \leq 2n+1} \|(\partial^\gamma b)(x + y, D + \eta)\|_{L^2 \rightarrow L^2}. \end{aligned}$$

We now observe that

$$\begin{aligned} b(x + y, D + \eta)u &= \frac{1}{(2\pi)^n} \int b(x + y, \zeta + \eta) e^{i\langle x - z, \zeta \rangle} u(z) dz d\zeta \\ &= \frac{e^{-i\langle x, \eta \rangle}}{(2\pi)^n} \int b(x + y, \xi) e^{i\langle x + y - w, \xi \rangle} e^{i\langle w - y, \eta \rangle} u(w - y) dw d\xi \\ &= Ub(x, D)U^*u, \end{aligned}$$

where

$$Uv(x) := e^{-i\langle \eta, x \rangle} v(x + y), \quad U^*u(w) = e^{i\langle w - y, \eta \rangle} u(w - y).$$

The same unitary equivalence is valid for $\partial^\gamma b$.

Since U is a unitary operator, it follows that

$$\begin{aligned} \|(\partial^\gamma b)(x + y, D + \eta)\|_{L^2 \rightarrow L^2} &= \|U(\partial^\gamma b)(x, D)U^*\|_{L^2 \rightarrow L^2} \\ &= \|(\partial^\gamma b)(x, D)\|_{L^2 \rightarrow L^2}. \end{aligned}$$

Using this, (8.1.2), and (8.1.3), prove the theorem for the classical quantization $b \mapsto b(x, D)$.

2. To construct $\chi \in \mathcal{S}$ for which (8.1.2) holds, let $\varphi = \varphi(x)$, $\hat{\psi} = \hat{\psi}(\xi)$ be functions in $\mathcal{S}$, equal to 1 near 0. We put

$$\chi(x, \xi) := \overline{\varphi}(x) \hat{\psi}(\xi) e^{i\langle x, \xi \rangle}.$$

We consider the Fourier transform in $\mathbb{R}^n \times \mathbb{R}^n$,

$$(x^*, \xi^*) \mapsto \mathcal{F}(b\chi)(x^*, \xi^*) = \mathcal{F}(b\overline{\varphi}\hat{\psi}e^{i\langle x, \xi \rangle})(x^*, \xi^*),$$

as a function of the dual variables $(x^*, \xi^*) \in \mathbb{R}^{2n}$. We have

$$\begin{aligned} |\mathcal{F}(b\bar{\varphi}\hat{\psi}e^{i\langle x, \xi \rangle})(0, 0)| &= \left| \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} b(x, \xi) \bar{\varphi}(x) \hat{\psi}(\xi) e^{i\langle x, \xi \rangle} dx d\xi \right| \\ &= (2\pi)^n |\langle b(x, D)\psi, \varphi \rangle| \\ &\leq (2\pi)^n \|b(x, D)\|_{L^2 \rightarrow L^2} \|\varphi\|_{L^2} \|\psi\|_{L^2}. \end{aligned}$$

Fix $(x^*, \xi^*) \in \mathbb{R}^{2n}$ and rewrite this inequality with $\varphi(x)e^{i\langle x^*, x \rangle}$ replacing $\varphi(x)$ and $\hat{\psi}(\xi)e^{-i\langle \xi^*, \xi \rangle}$ replacing $\hat{\psi}(\xi)$, a procedure which does not change the L^2 norms. It follows that

$$(8.1.4) \quad \frac{1}{(2\pi)^n} |\mathcal{F}(b\bar{\varphi}(x)\hat{\psi}(\xi)e^{i\langle x, \xi \rangle})(x^*, \xi^*)| \leq \|b(x, D)\|_{L^2 \rightarrow L^2} \|\varphi\|_{L^2} \|\psi\|_{L^2}.$$

That is,

$$(8.1.5) \quad \|\mathcal{F}(\chi b)\|_{L^\infty} \leq C \|b(x, D)\|_{L^2 \rightarrow L^2}.$$

3. Next, we assert that

$$(8.1.6) \quad |\mathcal{F}(\chi b)(x^*, \xi^*)| \leq C \langle (x^*, \xi^*) \rangle^{-2n-1} \sum_{|\alpha| \leq 2n+1} \|(\partial^\alpha b)(x, D)\|_{L^2 \rightarrow L^2}.$$

To see this, compute

$$\begin{aligned} (x^*)^\alpha (\xi^*)^\beta \mathcal{F}(\chi b)(x^*, \xi^*) &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} (x^*)^\alpha (\xi^*)^\beta e^{-i(\langle x^*, x \rangle + \langle \xi^*, \xi \rangle)} \chi b(x, \xi) dx d\xi \\ &= (-1)^{|\alpha|+|\beta|} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} D_x^\alpha D_\xi^\beta \left(e^{-i(\langle x^*, x \rangle + \langle \xi^*, \xi \rangle)} \right) \chi b dx d\xi \\ &= \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{-i(\langle x^*, x \rangle + \langle \xi^*, \xi \rangle)} D_x^\alpha D_\xi^\beta (\chi b) dx d\xi. \end{aligned}$$

Summing absolute values of the left-hand side over all (α, β) with $|\alpha| + |\beta| \leq 2n + 1$ and using the estimate (8.1.5), we obtain the bound

$$\begin{aligned} \|\langle (x^*, \xi^*) \rangle^{2n+1} \mathcal{F}(\chi b)\|_{L^\infty} &\leq C \sum_{|\alpha|+|\beta| \leq 2n+1} \|\mathcal{F}(D_x^\alpha D_\xi^\beta (\chi b))\|_{L^\infty} \\ &\leq C \sum_{|\gamma| \leq 2n+1} \|(\partial^\gamma b)(x, hD)\|_{L^2 \rightarrow L^2}. \end{aligned}$$

This gives (8.1.6). We now recall the estimate (3.1.19) to obtain

$$\|\chi b\|_{L^\infty} \leq C \|\mathcal{F}(\chi b)\|_{L^1} \leq C \sum_{|\gamma| \leq 2n+1} \|(\partial^\gamma b)(x, D)\|_{L^2 \rightarrow L^2}.$$

Hence we have proved (8.1.2).

4. As shown in step 1 of the proof, this implies the desired inequality (8.1.1), except that we used the classical ($t = 1$) and not the Weyl ($t = 1/2$) quantization. To remedy this, recall from Theorem 4.13 that if

$$\tilde{b} := e^{\frac{i}{2}\langle D_x, D_\xi \rangle} b,$$

then

$$b^w(x, D) = \tilde{b}^w(x, D), \quad (\partial^\alpha b)^w(x, D) = (\partial^\alpha \tilde{b})^w(x, D).$$

The continuity statement in Theorem 4.17 shows that

$$\|b\|_{L^\infty} \leq C \sum_{|\alpha| \leq K} \|\partial^\alpha \tilde{b}\|_{L^\infty}$$

and reduces the argument to the classical quantization. $\square$

The following notation will be useful, if slightly odd looking, in expressions involving multiple commutators:

NOTATION. If A, B are operators, we henceforth write

$$(8.1.7) \quad \text{ad}_B A := [B, A];$$

“ad” is called the *adjoint action*.

Easy calculations show

LEMMA 8.2 (Properties of ad). *The adjoint action ad satisfies the derivation property*

$$(8.1.8) \quad \text{ad}_A(BC) = (\text{ad}_A B)C + B(\text{ad}_A C)$$

and therefore

$$(8.1.9) \quad \text{ad}_A B = -B(\text{ad}_A B^{-1})B.$$

Remember that we identify a pair $(x^*, \xi^*) \in \mathbb{R}^{2n}$ with the linear operator $l(x, \xi) = \langle x^*, x \rangle + \langle \xi^*, \xi \rangle$. Recall also from Theorem 4.4 that

$$l^w(x, hD) = l(x, hD) = \langle x^*, x \rangle + \langle \xi^*, hD \rangle.$$

THEOREM 8.3 (Semiclassical Beals Theorem). *Let $A : \mathcal{S} \rightarrow \mathcal{S}'$ be a continuous linear operator. Then*

$$(i) \quad A = a^w(x, hD) \text{ for a symbol } a \in S$$

if and only if

$$(ii) \text{ for all } N = 0, 1, 2, \dots \text{ and all linear functions } l_1, \dots, l_N, \text{ we have}$$

$$(8.1.10) \quad \|\text{ad}_{l_1(x, hD)} \cdots \text{ad}_{l_N(x, hD)} A\|_{L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)} = O(h^N).$$

Proof. 1. That (i) implies (ii) follows from the symbol calculus developed in Chapter 4. Indeed, $\|A\|_{L^2 \rightarrow L^2} = O(1)$ according to Theorem 4.23, and formula (4.3.11) shows that each commutator with $l_j(x, hD)$ yields a bounded operator of order h . Observe that although $l_j \notin S$, we can still apply the composition formula since $\partial^\alpha l_j \in S$ for $|\alpha| \geq 1$.

2. That (ii) implies (i) is harder to prove, and we will first prove the implication for $h = 1$.

The Schwartz Kernel Theorem (Theorem C.1) asserts that we can write

$$(8.1.11) \quad Au(x) = \int_{\mathbb{R}^n} K_A(x, y) u(y) dy$$

for $K_A \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^n)$. We call K_A the *kernel* of A .

We now claim that if we define $a \in \mathcal{S}'(\mathbb{R}^{2n})$ by

$$(8.1.12) \quad a(x, \xi) := \int_{\mathbb{R}^n} e^{-i\langle w, \xi \rangle} K_A\left(x + \frac{w}{2}, x - \frac{w}{2}\right) dw,$$

then

$$(8.1.13) \quad K_A(x, y) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} a\left(\frac{x+y}{2}, \xi\right) e^{i\langle x-y, \xi \rangle} d\xi,$$

where the integrals are shorthand for the Fourier transforms defined on $\mathcal{S}'$. To confirm this, using (8.1.12) and the Fourier inversion formula, we calculate that

$$\begin{aligned} & \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} a\left(\frac{x+y}{2}, \xi\right) e^{i\langle x-y, \xi \rangle} d\xi \\ &= \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{i\langle x-y-w, \xi \rangle} K_A\left(\frac{x+y}{2} + \frac{w}{2}, \frac{x+y}{2} - \frac{w}{2}\right) dw d\xi \\ &= \int_{\mathbb{R}^n} \delta(x - y - w) K_A\left(\frac{x+y}{2} + \frac{w}{2}, \frac{x+y}{2} - \frac{w}{2}\right) dw \\ &= K_A(x, y). \end{aligned}$$

In view of (8.1.11) and (8.1.13), we see that $A = a^w(x, D)$, for a defined by (8.1.12).

3. Now we must show that a belongs to the symbol class S ; that is,

$$(8.1.14) \quad \sup_{\mathbb{R}^{2n}} |\partial^\alpha a| \leq C_\alpha$$

for each multiindex α .

To do so, we will make use of our hypothesis (8.1.10) with $l = x_j, \xi_j$, that is, with $l(x, D) = x_j, D_j$. We recall the commutator formulas (4.2.7)

and (4.2.6), which imply for $j = 1, \dots, n$ that

$$(8.1.15) \quad \begin{cases} \text{ad}_{x_j} A = [x_j, a^w] = -(D_{\xi_j} a)^w \\ \text{ad}_{D_{x_j}} A = [D_{x_j}, a^w] = (D_{x_j} a)^w. \end{cases}$$

This and the hypothesis (8.1.10) with $h = 1$ imply that

$$\|(\partial^\alpha a)^w\|_{L^2 \rightarrow L^2} \leq C_\alpha,$$

for all multiindices α . The estimate (8.1.14) now follows from Theorem 8.1.

4. Next we convert the case with arbitrary h to the case of $h = 1$ by rescaling (4.1.9). For this, define

$$U_h u(x) := h^{n/4} u(h^{1/2} x)$$

and check that $U_h : L^2 \rightarrow L^2$ is unitary. Then

$$U_h a^w(x, hD) U_h^{-1} = a^w(h^{1/2} x, h^{1/2} D) = a_h^w(x, D)$$

for

$$(8.1.16) \quad a_h(x, \xi) := a(h^{1/2} x, h^{1/2} \xi).$$

Our hypothesis (8.1.10) is invariant under conjugation by U_h and is consequently equivalent to

$$(8.1.17) \quad \text{ad}_{l_1(h^{1/2} x, h^{1/2} D)} \cdots \text{ad}_{l_N(h^{1/2} x, h^{1/2} D)} a_h^w = O_{L^2 \rightarrow L^2}(h^N).$$

But since l_j is linear, $l_j(h^{1/2} x, h^{1/2} D) = h^{1/2} l(x, D)$. Thus (8.1.17) is equivalent to

$$(8.1.18) \quad \text{ad}_{l_1(x, D)} \cdots \text{ad}_{l_N(x, D)} a_h^w = O_{L^2 \rightarrow L^2}(h^{N/2}).$$

Taking $l_k(x, \xi) = x_j$ or ξ_j , it follows from (8.1.18) that

$$(8.1.19) \quad \|(\partial^\beta a_h)^w\|_{L^2 \rightarrow L^2} \leq C h^{\frac{|\beta|}{2}}$$

for all multiindices β .

5. Finally, we claim that

$$(8.1.20) \quad |\partial^\alpha a_h| \leq C_\alpha h^{|\alpha|/2} \quad \text{for each multiindex } \alpha.$$

But this follows from Theorem 8.1, owing to estimate (8.1.19):

$$\|\partial^\alpha a_h\|_{L^\infty} \leq C \sum_{|\beta| \leq M} \|(\partial^{\alpha+\beta} a_h)^w\|_{L^2 \rightarrow L^2} \leq C_\alpha h^{|\alpha|}.$$

Recalling (8.1.16), we rescale to derive the desired inequality (8.1.14). $\square$

REMARK: Beals's Theorem for S_δ . Similar arguments show that

$$(8.1.21) \quad A = a^w(x, hD) \quad \text{for a symbol } a \in S_\delta$$

if and only if

$$(8.1.22) \quad \|\mathrm{ad}_{l_1(x,hD)} \cdots \mathrm{ad}_{l_N(x,hD)} A\|_{L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)} = O(h^{N(1-\delta)})$$

for all $N = 0, 1, 2, \dots$ and all linear functions $l_1, \dots, l_N$. $\square$

EXAMPLE: Resolvents as pseudodifferential operators. Suppose $a \in S$ is real-valued, so that $A = a^w(x, hD)$ is a selfadjoint operator on L^2 . If λ does not lie in the spectrum of A , the resolvent $B = (A + \lambda)^{-1}$ is a bounded operator on L^2 . Can we represent B as a pseudodifferential operator?

To see that we can, first calculate using (8.1.9) that

$$\mathrm{ad}_{l(x,hD)} B = -B(\mathrm{ad}_{l(x,hD)}(A + \lambda))B = -B(\mathrm{ad}_{l(x,hD)} A)B$$

for each linear l . Therefore

$$\|\mathrm{ad}_{l(x,hD)} B\|_{L^2 \rightarrow L^2} \leq C \|\mathrm{ad}_{l(x,hD)} A\|_{L^2 \rightarrow L^2} = O(h),$$

according to (8.1.10). A similar computation shows that for each N

$$\|\mathrm{ad}_{l_1(x,hD)} \cdots \mathrm{ad}_{l_N(x,hD)} B\|_{L^2 \rightarrow L^2} = O(h^N),$$

and so the assumptions of Beals's Theorem are satisfied. Consequently $B = (A + \lambda)^{-1} = b^w(x, hD)$ for some symbol $b \in S$. $\square$

Many more applications will appear later, particularly in Chapter 11.

8.2. REAL EXPONENTIATION OF OPERATORS

We devote this section and the next to building certain generalized Sobolev type spaces $H_h(m)$ compatible with a given order function m . These spaces are useful since for each symbol $a \in S(m)$, its quantization $a^w(x, hD)$ is well behaved on $H_h(m)$.

As a preliminary step in constructing the generalized Sobolev spaces, we consider in this section families of operators which give real exponentials of certain pseudodifferential operators.

We henceforth assume $m = m(x, \xi)$ is an order function. Set

$$(8.2.1) \quad g := \log m.$$

We also suppose that

$$(8.2.2) \quad |\partial^\alpha g| \leq C_\alpha \quad \text{for all multiindices } |\alpha| \geq 1.$$

Then

$$(8.2.3) \quad e^{tg} = m^t \in S(m^t) \quad (t \in \mathbb{R}).$$

In Section 8.3 we will discuss how to find order functions m for which these conditions hold.

LEMMA 8.4 (Inverting exponentials). *Consider*

$$U(t) := (\exp tg)^w(x, D)$$

as a mapping from $\mathcal{S}$ to itself. There exists $t_0 > 0$ such that the operator $U(t)$ is invertible for $|t| < t_0$ and

$$(8.2.4) \quad U(t)^{-1} = b_t^w(x, D)$$

for a symbol

$$(8.2.5) \quad b_t \in S(m^{-t}).$$

Proof. 1. Owing to (8.2.3), $U(t)$ is the quantization of an element of $S(m^t)$.

We assert that

$$(8.2.6) \quad U(-t)U(t) = I + e_t^w(x, D) \quad \text{for a symbol } e_t \in S.$$

To see this, we employ the composition formula (4.3.6) with $h = 1$, to write

$$\begin{aligned} (8.2.7) \quad e_t(x, \xi) &= e^{iA(D)}(e^{-tg(x^1, \xi^1) + tg(x^2, \xi^2)})|_{x^1, x^2=x, \xi^1, \xi^2=\xi} - 1 \\ &= \int_0^1 \frac{d}{ds} e^{isA(D)}(e^{-tg(x^1, \xi^1) + tg(x^2, \xi^2)})|_{x^1, x^2=x, \xi^1, \xi^2=\xi} ds \\ &= \int_0^1 e^{isA(D)} iA(D)(e^{-tg(x^1, \xi^1) + tg(x^2, \xi^2)})|_{x^1, x^2=x, \xi^1, \xi^2=\xi} ds \\ &= \frac{it}{2} \int_0^1 e^{isA(D)} \left(F e^{-tg(x^1, \xi^1) + tg(x^2, \xi^2)} \right) |_{x^1, x^2=x, \xi^1, \xi^2=\xi} ds, \end{aligned}$$

where $A(D) = \frac{1}{2}\sigma(D_{x^1}, D_{\xi^1}; D_{x^2}, D_{\xi^2})$ and

$$F = \partial_{x^1} g(x^1, \xi^1) \cdot \partial_{\xi^2} g(x^2, \xi^2) - \partial_{\xi^1} g(x^1, \xi^1) \cdot \partial_{x^2} g(x^2, \xi^2).$$

Our assumptions imply that $F \in S$ and that

$$\exp(-tg(x^1, \xi^1) + tg(x^2, \xi^2)) \in S(\tilde{m}^t)$$

for $\tilde{m}(x^1, x^2, \xi^1, \xi^2) := m(x^2, \xi^2)/m(x^1, \xi^1)$. Thus Theorem 4.17 shows that

$$e^{isA(D)} : S(\tilde{m}^t) \rightarrow S(\tilde{m}^t).$$

Furthermore the restriction to $x^1 = x^2$, $\xi^1 = \xi^2$ shows that $e_t \in S$, since $\tilde{m}(x^1, x^1, \xi^1, \xi^1) \equiv 1$. This proves (8.2.6).

2. It follows from (8.2.7) that

$$e_t = t\tilde{e}_t$$

for $\tilde{e}_t \in S$. Therefore Theorem 4.23 implies

$$\|e_t^w(x, D)\|_{L^2 \rightarrow L^2} = O(t),$$

and so $I + e_t^w(x, D)$ is invertible for $|t|$ small enough. Then the application of Beals's Theorem (Theorem 8.3) to resolvents presented on page 177 implies

$$(I + e_t^w(x, D))^{-1} = c_t^w(x, D)$$

for a symbol $c_t \in S$. Hence $b_t = c_t \# \exp(-tg(x, \xi)) \in S(m^{-t})$, according to Theorem 4.18. $\square$

We record the following for later reference:

LEMMA 8.5 (Solving an operator equation). *Suppose that*

$$C(t) = c_t^w(x, D),$$

where the symbols $c_t \in S$ depend continuously on t for $|t| \leq t_0$. Assume also that $q \in S$.

Then the equation

$$(8.2.8) \quad \begin{cases} (\partial_t + C(t))Q(t) = 0 \\ Q(0) = q^w(x, D) \end{cases}$$

has a unique solution $Q(t) : \mathcal{S} \rightarrow \mathcal{S}$ given by

$$Q(t) = q_t^w(x, D),$$

the symbols $q_t \in S$ depending continuously on t for $|t| \leq t_0$.

Proof. 1. The Picard Theorem for ODE shows that there exists a unique solution $Q(t)$ that is bounded on L^2 .

2. Next we assert that for any choice of l_j 's and any N

$$(8.2.9) \quad \text{ad}_{l_1(x,D)} \cdots \text{ad}_{l_N(x,D)} Q(t) : L^2 \rightarrow L^2.$$

We prove this by induction on N . Observe from the derivation property (8.1.8) of ad_l that

$$(8.2.10) \quad \begin{aligned} \text{ad}_{l_1(x,D)} \cdots \text{ad}_{l_N(x,D)} (C(t)Q(t)) \\ = C(t) \text{ad}_{l_1(x,D)} \cdots \text{ad}_{l_N(x,D)} Q(t) - R(t), \end{aligned}$$

where $R(t)$ is the sum of terms of the form

$$A^k(t) \text{ad}_{l_1(x,D)} \cdots \text{ad}_{l_k(x,D)} Q(t)$$

with $k < N$, for $A^k(t) = (a_t^k)^w$ and symbols $a_t^k \in S$ depending continuously on t . Then the induction hypothesis implies $R(t)$ is bounded on L^2 .

Now

$$\partial_t (\text{ad}_{l_1(x,D)} \cdots \text{ad}_{l_N(x,D)} Q(t)) + \text{ad}_{l_1(x,D)} \cdots \text{ad}_{l_N(x,D)} (C(t)Q(t)) = 0,$$

and consequently

$$(\partial_t + C(t)) (\text{ad}_{l_1(x,D)} \cdots \text{ad}_{l_N(x,D)} Q(t)) = R(t).$$

Since $R(t)$ is bounded on L^2 and the assertion (8.2.9) is clearly valid at $t = 0$, it also holds for all $|t| < t_0$.

3. In view of (8.2.9) and Beals's Theorem for $h = 1$, the unique solution bounded on L^2 is a pseudodifferential operator and hence maps $\mathcal{S}$ to $\mathcal{S} \subset L^2$. As such, it is also unique. $\square$

Our next theorem identifies $\exp(tg^w(x, hD))$ as the quantization of an element of $S(m^t)$.

THEOREM 8.6 (Exponentials and order functions). *Assume for the order function m and for $g = \log m$ that conditions (8.2.1) and (8.2.2) hold.*

(i) *Then the equation*

$$(8.2.11) \quad \begin{cases} \partial_t B(t) = g^w(x, hD) B(t) \\ B(0) = I \end{cases}$$

has a unique solution $B(t) : \mathcal{S} \rightarrow \mathcal{S}$ for $t \in \mathbb{R}$.

(ii) *Furthermore, we have*

$$(8.2.12) \quad B(t) = b_t^w(x, hD)$$

for a symbol

$$(8.2.13) \quad b_t \in S(m^t).$$

As we have seen in Theorem 4.7, quantization and exponentiation commute for linear symbols. This is certainly not true for nonlinear symbols, but according to (ii) it is in a certain sense valid *at the level of order functions*.

Proof. Using the rescaling given in (4.1.9), we only need to prove the result for the case $h = 1$.

1. To begin, let us assume that a solution of (8.2.11) exists, with $B(t) : \mathcal{S} \rightarrow \mathcal{S}$. We assert that

$$(8.2.14) \quad \partial_t (U(-t)B(t)) = V(t)B(t)$$

in the notation of Lemma 8.4, where

$$(8.2.15) \quad V(t) = a_t^w(x, D) \quad \text{for } a_t \in S(m^{-t}).$$

In fact

$$(8.2.16) \quad \partial_t U(-t) = -(g \exp(-tg))^w(x, D)$$

and

$$(8.2.17) \quad U(-t)g^w(x, D) = (\exp(-tg)\#g)^w(x, D).$$

Hence (8.2.14) holds with

$$V(t) = (\exp(-tg)\#g - (g \exp(-tg)))^w(x, D).$$

2. To analyze $V(t)$, we note that

$$\exp(iA(D)) = 1 + i \int_0^1 \exp(isA(D))A(D) ds,$$

as in (8.2.7). Consequently (4.3.6) gives

$$\begin{aligned} & \exp(-tg)\#g - \exp(-tg)g \\ &= i \int_0^1 \exp(isA(D))A(D) \left(e^{-tg(x^1, \xi^1)} g(x^2, \xi^2) \right) |_{x^1=x^2=x, \xi^1=\xi^2=\xi} ds. \end{aligned}$$

From the hypothesis on g we see that $A(D) \exp(-tg(x^1, \xi^1))g(x^2, \xi^2)$ is a sum of terms of the form $a(x^1, \xi^1)b(x^2, \xi^2)$, where $a \in S(m^{-t})$ and $b \in S$. The continuity of $\exp(iA(D))$ on the spaces of symbols in Theorem 4.17 now gives (8.2.15).

3. Set

$$C(t) := -V(t)U(-t)^{-1}.$$

Then Lemma 8.4 implies $C(t) = c_t^w$ where $c_t \in S$. The symbolic calculus shows that c_t depends smoothly on t and

$$(\partial_t + C(t))(U(-t)B(t)) = 0.$$

4. The existence part of Lemma 8.5 implies that $B(t) = U(-t)^{-1}Q(t)$ and $Q(0) = I$. This shows that $B(t)$ exists and that it is unique. Since $Q(t)$ quantizes $q_t \in S$, Lemma 8.4 gives the statement of Theorem 8.6 for small times. Because the solution of (8.2.11) has the group property $B(t)B(s) = B(t+s)$, the assertion for small times and the pseudodifferential calculus imply the assertion for all times $t \in \mathbb{R}$. $\square$

REMARK: Real and complex exponentials. The foregoing Lemmas 8.4 and 8.5 and Theorem 8.6 concern real exponential expressions arising from operator dynamics of the form

$$(\partial_t + C(t))Q(t) = 0.$$

Quantum dynamics like

$$(hD_t + C(t))Q(t) = 0$$

yield instead complex exponential expressions, and these we will study more in Chapters 10, 11, and 14. $\square$

8.3. GENERALIZED SOBOLEV SPACES

8.3.1. Sobolev spaces compatible with symbols. The quantization of real exponentials developed in the previous section now allows us to define generalized Sobolev spaces $H_h(m)$ on which operators with symbols in $S(m)$ naturally act.

We first record

LEMMA 8.7 (Logarithms of order functions).

(i) Suppose that m is an order function and that

$$(8.3.1) \quad m \in S(m).$$

Then

$$(8.3.2) \quad m^{-1} \in S(m^{-1}),$$

and $g = \log m$ satisfies the assumptions (8.2.1) and (8.2.2).

(ii) Given an arbitrary order function m , there exists another order function $\tilde{m}$ such that

$$S(m) = S(\tilde{m})$$

and

$$\tilde{m} \in S(\tilde{m}).$$

Proof. 1. The statement (8.3.2) follows from the formula (4.7.10) applied with $a = m$ and $\lambda = 0$. That g satisfies (8.2.1) and (8.2.2) follows from (8.3.2).

2. For an arbitrary order function m define $\tilde{m} := m * \eta$, where $\eta \in C_c^\infty(\mathbb{R}^{2n})$, $\eta \geq 0$, $\int \eta dw = 1$. According to the definition (4.4.1) of an order function,

$$C^{-1}\langle w \rangle^{-N} \leq \frac{m(z-w)}{m(z)} \leq C\langle w \rangle^N$$

for all $w, z \in \mathbb{R}^{2n}$. Consequently $C^{-1}m \leq \tilde{m} \leq Cm$ and

$$|\partial^\alpha \tilde{m}| \leq C_\alpha m \leq C_\alpha \tilde{m}$$

for all multiindices α . Hence $S(m) = S(\tilde{m})$ and $\tilde{m} \in S(\tilde{m})$. □

Hereafter m denotes an order function satisfying $m \in S(m)$ and, as above, set $g := \log m$. Recall from Theorems 4.16 and 8.6 that $\exp(\pm g^w) : \mathcal{S}' \rightarrow \mathcal{S}'$.

DEFINITIONS. (i) We define the *generalized Sobolev space associated to m* as

$$(8.3.3) \quad \begin{aligned} H_h(m) &:= \{u \in \mathcal{S}'(\mathbb{R}^n) \mid \exp(g^w(x, hD))u \in L^2(\mathbb{R}^n)\} \\ &= \exp(-g^w(x, hD))L^2(\mathbb{R}^n) \subset \mathcal{S}'. \end{aligned}$$

(ii) The norm on $H_h(m)$ is defined by

$$(8.3.4) \quad \|u\|_{H_h(m)} := \|\exp(g^w(x, hD))u\|_{L^2}.$$

When m and thus g are functions of both x and ξ , we sometimes call $H_h(m)$ a *microlocally weighted space*.

EXAMPLES.

(i) If $m = \langle \xi \rangle^s$ for $s \in \mathbb{R}$, then

$$(8.3.5) \quad \begin{aligned} H_h(m) &= H_h^s(\mathbb{R}^n) = \{u \in L^2(\mathbb{R}^n) \mid (1 + |\xi|^2)^{s/2} \mathcal{F}_h u \in L^2(\mathbb{R}^n)\}, \\ \|u\|_{H_h^s}^2 &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \langle \xi \rangle^{2s} |\mathcal{F}_h u(\xi)|^2 d\xi, \end{aligned}$$

are the usual semiclassical Sobolev spaces.

(ii) When m depends only on x , the space $H_h(m)$ corresponds to changing Lebesgue measure dx in the definition of $L^2(\mathbb{R}^n)$ to $\exp(2g(x))dx$. So

$$(8.3.6) \quad \|u\|_{H_h(m(x))} = \|u\|_{L^2(\exp(2g(x))dx)}.$$

In particular,

$$(8.3.7) \quad H_h(m) = L^2(\mathbb{R}^n) \quad \text{if } m \equiv 1.$$

(iii) If m depends only on ξ , then the measure is changed on the semiclassical Fourier transform side to $\exp(2g(\xi))d\xi$:

$$(8.3.8) \quad \|u\|_{H_h(m(\xi))} = (2\pi h)^{-\frac{n}{2}} \|\mathcal{F}_h u\|_{L^2(\exp(2g(\xi))d\xi)},$$

where the prefactor is explained by Theorem 3.8 □

THEOREM 8.8 (Properties of $H_h(m)$ spaces).

(i) Suppose that $m \in S(m)$, $\tilde{m} \in S(\tilde{m})$ are two order functions satisfying

$$c^{-1}m \leq \tilde{m} \leq cm,$$

where $c > 0$. Then

$$(8.3.9) \quad H_h(m) = H_h(\tilde{m})$$

and

$$(8.3.10) \quad C^{-1}\|u\|_{H_h(m)} \leq \|u\|_{H_h(\tilde{m})} \leq C\|u\|_{H_h(m)}$$

for a constant $C > 0$ and all $u \in H_h(m)$.

(ii) We can use the L^2 inner product to identify the dual space of $H_h(m)$ with $H_h(1/m)$:

$$(8.3.11) \quad (H_h(m))' = H_h(1/m).$$

REMARKS. (i) So given any order function m , we can define

$$H_h(m) := H_h(\tilde{m}),$$

where $\tilde{m}$ is any order function satisfying $S(m) = S(\tilde{m})$ and $\tilde{m} \in S(\tilde{m})$.

(ii) The precise identification abbreviated by (8.3.11) will be explained in the proof. $\square$

Proof. 1. Let $g = \log m$ and $\tilde{g} = \log \tilde{m}$. To prove (8.3.10), we note that Theorem 8.6 implies

$$\exp(g^w(x, hD)) \exp(-\tilde{g}^w(x, hD)) = a^w(x, hD)$$

for a symbol $a \in S$. By Theorem 4.23, $a^w(x, hD) = O_{L^2 \rightarrow L^2}(1)$; so that

$$\|u\|_{H_h(m)} = \|e^{g^w} u\|_{L^2} = \|a^w e^{\tilde{g}^w} u\|_{L^2} \leq C \|e^{\tilde{g}^w} u\|_{L^2} = C \|u\|_{H_h(\tilde{m})}.$$

This proves the first inequality in (8.3.10) and the second one follows as m and $\tilde{m}$ are exchangeable.

2. The definition shows that $\exp(\pm g^w) : H_h(m^{\pm 1}) \rightarrow L^2$ are Hilbert space isometries. Since L^2 is its own dual under the pairing $u(v) = \int_{\mathbb{R}^n} v \bar{u} dx$, we identify the dual of $H_h(m)$ with $H_h(1/m)$ using these isometries. Explicitly, if $v \in H_h(m)$ and $u \in H_h(1/m)$, then

$$u(v) = \langle v, u \rangle = \langle e^{g^w} v, e^{-g^w} u \rangle.$$

$\square$

THEOREM 8.9 (Generalized Sobolev spaces and Schwartz space).

For each fixed $h > 0$, we have

$$(8.3.12) \quad \mathcal{S} = \bigcap_{m \in \mathcal{M}} H_h(m), \quad \mathcal{S}' = \bigcup_{m \in \mathcal{M}} H_h(m),$$

where $\mathcal{M}$ denotes the set of all order functions on $\mathbb{R}^{2n}$.

We also see from (8.3.12) that

$$(\mathcal{S}')' = \mathcal{S},$$

a standard result in functional analysis.

Proof. 1. If $g = \log m$ for $m \in S(m)$, then Theorem 8.6 shows that $\exp(g^w) = a^w$ for a symbol $a \in S(m)$. Hence Theorem 4.16 implies that if $u \in \mathcal{S}$, then $\exp(g^w)u \in \mathcal{S} \subset L^2$. Consequently $\mathcal{S} \subseteq H_h(m)$ for all m and consequently $\mathcal{S} \subseteq \bigcap_{m \in \mathcal{M}} H_h(m)$.

2. Next, put

$$m(x, \xi) = \langle x \rangle^{|\beta|} \langle \xi \rangle^{2n+|\alpha|}, \quad g := \log m.$$

Then

$$\begin{aligned} \sup_{x \in \mathbb{R}^n} |D_x^\alpha x^\beta u| &\leq Ch^{-2n-|\alpha|} \|(I - h^2 \Delta)^n (hD_x)^\alpha x^\beta u\|_{L^2} \\ (8.3.13) \quad &\leq Ch^{-2n-|\alpha|} \|(I - h^2 \Delta)^n (hD_x)^\alpha x^\beta e^{-g^w} e^{g^w} u\|_{L^2} \\ &\leq C_1 h^{-2n-|\alpha|} \|e^{g^w} u\|_{L^2} \\ &= C_1 h^{-2n-|\alpha|} \|u\|_{H_h(m)}. \end{aligned}$$

The last inequality holds since

$$(I - h^2 \Delta)^n (hD_x)^\alpha x^\beta = b^w$$

for a symbol $b \in S(m)$; and $e^{-g^w} = c^w$ for a symbol $c \in S(1/m)$. Consequently their composition is bounded on L^2 . This proves that $\mathcal{S} \supseteq \bigcap_{m \in \mathcal{M}} H_h(m)$. $\square$

Next we see that the quantizations of symbols in $S(m)$ act naturally on generalized Sobolev spaces:

THEOREM 8.10 (Pseudodifferential operators acting on generalized Sobolev spaces). *Suppose that m_1 and m_2 are two order functions and that $a \in S(m_1)$.*

(i) *Then*

$$(8.3.14) \quad a^w(x, hD) : H_h(m_2) \rightarrow H_h(m_2/m_1)$$

is a bounded operator, with norm bounded independently of h .

(ii) *In particular,*

$$(8.3.15) \quad a^w(x, hD) : H_h(m_1) \rightarrow L^2(\mathbb{R}^n)$$

is a bounded operator.

(iii) If

$$\lim_{(x,\xi) \rightarrow \infty} m_1 = 0,$$

then

$$(8.3.16) \quad a^w(x, hD) : H_h(m_2) \rightarrow H_h(m_2)$$

is a compact operator.

Proof. 1. Following Theorem 8.8, we can take $m_j \in S(m_j)$. Lemma 8.7 also implies that $m_2/m_1 \in S(m_2/m_1)$. We restrict ourselves to the case $h = 1$ as we can again use the rescaling (4.1.9).

2. In view of the definition of $H(m) = H_1(m)$, the theorem is equivalent to showing the boundedness of

$$(8.3.17) \quad A := e^{-g_1^w(x,D)+g_2^w(x,D)} a^w(x, D) e^{-g_2^w(x,D)}$$

on L^2 , where $g_j := \log m_j$. Theorem 8.6 tells us that

$$e^{-g_2^w(x,D)} = b^w(x, D), \quad e^{-g_1^w(x,D)+g_2^w(x,D)} = c^w(x, D),$$

for symbols $b \in S(1/m_2), c \in S(m_2/m_1)$. Hence the composition rule in Theorem 4.11 implies

$$e^{-g_1^w+g_2^w} a^w e^{-g_2^w} = c^w a^w b^w = \tilde{a}^w,$$

where

$$\tilde{a} \in S(m_2/m_1 \times m_1 \times 1/m_2) = S.$$

So Theorem 4.23 implies $A = \tilde{a}^w(x, D)$ is bounded on L^2 .

3. Assertion (ii) is equivalent to our showing that

$$B := e^{g_2^w(x,D)} a^w(x, D) e^{-g_2^w(x,D)}$$

is a compact operator on L^2 . As above, we observe that $B = b^w(x, D)$ for a symbol $b \in S(m_1)$. We then apply Theorem 4.28. $\square$

8.3.2. Application: Estimates for eigenfunctions. The next theorem provides a general regularity assertion for L^2 -eigenfunctions of pseudodifferential operators.

Let $m \geq 1$ be an order function. Suppose that $a \in S(m)$ is real and that

$$C + a \geq cm$$

for constants $C \geq 0, c > 0$.

THEOREM 8.11 (Eigenfunctions and $H_h(m)$). Assume there exist $h_0 > 0$ and constants $\alpha < \beta$ such that for $0 < h < h_0$ we have

$$a^w(x, hD)u(h) = E(h)u(h),$$

with $u(h) \in L^2(\mathbb{R}^n)$ and $\alpha \leq E(h) \leq \beta$.

Then there exist $0 \leq h_1 \leq h_0$ and constants C_k , such that

$$(8.3.18) \quad \|u(h)\|_{H_h(m^k)} \leq C_k \|u(h)\|_{L^2(\mathbb{R}^n)},$$

for all $k = 0, 1, \dots$ and $0 < h < h_1$.

Proof. Replacing a by $a + C$ if necessary, we may assume that $a \geq cm$ and $0 \notin [\alpha, \beta]$.

Hence for $h < h_1$, with h_1 small enough, we have $a^w(x, hD)^{-1} = b^w(x, hD)$ for $b \in S(1/m)$. This implies that $H_h(m^k) = (b^w(x, hD))^k L^2$. Since

$$u(h) = E(h)^k (b^w(x, hD))^k u(h),$$

we obtain the estimate (8.3.18). $\square$

8.4. WAVEFRONT SETS, ESSENTIAL SUPPORT, AND MICROLOCALITY

We introduce in this section some precise ways to describe asymptotic properties of families of functions and operators in phase space as $h \rightarrow 0$.

8.4.1. Tempered functions and operators, localization. We begin by identifying some convenient classes of h -dependent distributions, which can deteriorate as $h \rightarrow 0$ but for which we have some uniform control:

DEFINITION. We call $u = \{u(h)\}_{0 < h \leq h_0} \subset \mathcal{S}'$ a *tempered family of distributions* if there exist an order function m and a constant $N \geq 0$ such that

$$(8.4.1) \quad \|u(h)\|_{H_h(m)} = O(h^{-N}).$$

From (8.3.12) we know that each $u(h)$ belongs to $H_h(m)$ for some m . The purpose of the definition is to have m and N independent of h .

We likewise introduce some useful classes of operators:

DEFINITION. A family $T = \{T(h)\}_{0 < h \leq h_0}$ of linear operators

$$T(h) : \mathcal{S} \rightarrow \mathcal{S}'$$

is called *tempered* if there exists $N \geq 0$ such that for any order function m

$$(8.4.2) \quad \|T(h)\|_{H_h(m) \rightarrow H_h(m \langle (x, \xi) \rangle^{-N})} = O(h^{-N}).$$

The point is that although the operators $T(h)$ may become increasingly singular as $h \rightarrow 0$, they do so at a controlled rate.

REMARK. Most of the operators we encounter in this book are tempered. According to Theorem 8.10, if $a \in h^k S(m)$, for any $k \in \mathbb{R}$ and any order function m , then $T = a^w(x, hD)$ is tempered. Theorem 10.1 will show that the operators discussed later in Theorems 11.1 and 11.5 are also tempered. The precise definitions of tempered functions and operators vary in the literature, and we have taken formulations most convenient for our purposes. $\square$

Since tempered operators deteriorate only algebraically as $h \rightarrow 0$, they behave well under composition with $O(h^\infty)$ operators:

LEMMA 8.12 (Tempered operators and h^∞). *If T is tempered and if $a \in h^\infty \mathcal{S}$, then*

$$(8.4.3) \quad \begin{aligned} Ta^w(x, hD) &= O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty), \\ a^w(x, hD)T &= O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty). \end{aligned}$$

This follows from Theorem 8.10.

We next introduce the very useful concept of asymptotic localization in phase space:

DEFINITION. We call a tempered family $u = \{u(h)\}_{0 < h \leq h_0}$ *localized in phase space* if there exists a function $\chi \in C_c^\infty(\mathbb{R}^{2n})$ such that

$$(8.4.4) \quad (I - \chi^w(x, hD))u(h) = O_{\mathcal{S}}(h^\infty).$$

The interpretation is that u is negligible off the support of χ as $h \rightarrow 0$.

8.4.2. Semiclassical wavefront sets. We next introduce the semiclassical wavefront set $\text{WF}_h(u)$ for a family u of tempered distribution. This wavefront set records where u is asymptotically localized in phase space as $h \rightarrow 0$. Being localized to a neighborhood of x_0 means that away from that neighborhood u is $O(h^\infty)$. Being localized near a frequency ξ_0 means the same thing for the semiclassical Fourier transform of u .

We now want to understand what simultaneous localization means.

DEFINITION. The *semiclassical wavefront set*

$$\text{WF}_h(u)$$

of a tempered family $u = \{u(h)\}_{0 < h \leq h_0}$ is the *complement* of the set of points $(x_0, \xi_0) \in \mathbb{R}^{2n}$ for which there exists a symbol $a \in S$ such that, for some constant $\gamma > 0$ independent of h ,

$$(8.4.5) \quad |a(x_0, \xi_0)| > \gamma$$

and

$$(8.4.6) \quad \|a^w(x, hD)u(h)\|_{H_h(m)} = O(h^\infty),$$

where the order function m is the same as in (8.4.1).

Theorem 8.13 below will show that definition is independent of the $H_h(m)$ norm used in (8.4.6).

REMARKS. (i) The name “wavefront set” is motivated by the example

$$u(h) = e^{i\varphi(x)/h}$$

with $\partial\varphi \neq 0$. Then

$$\mathrm{WF}_h(u) = \{(x, \partial\varphi(x)) \mid x \in \mathbb{R}^n\},$$

which is indeed the wavefront of the highly oscillatory wave. For example, if $\varphi(x) = \langle x, \omega \rangle$, with $|\omega| = 1$, then u is a plane wave with wavefront $\mathbb{R}^n \times \{\omega\}$.

(ii) The wavefront set provides a crude but robust measure of semiclassical regularity of a function. For instance a smooth function u not depending on h has $\mathrm{WF}_h(u) = \mathrm{spt} \, u \times \{0\}$. Observe also that if $\mathrm{WF}_h(u) = K \times \{0\}$ and $u = \chi^w(hD)u + O_{\mathcal{S}}(h^\infty)$, $\chi \in C_c^\infty(\mathbb{R}^n)$, that is, u is localized in frequency, we cannot conclude that the partial derivatives $\partial^\alpha u$ are uniformly bounded in h on K , only that $(h\partial)^\alpha u = o(1)$ as $h \rightarrow 0$. Away from K we have the estimate $\partial^\alpha u = O(h^\infty)$.

(iii) An alternative definition of the wavefront set is this: $(x_0, \xi_0) \notin \mathrm{WF}_h(u)$ if there exist $\varphi, \psi \in C_c^\infty$ such that $\varphi = 1$ near x_0 , $\psi = 1$ near ξ_0 , and

$$\psi \mathcal{F}_h(\varphi u) = O(h^\infty);$$

see Theorem 8.13 below. Because of the compactly supported localizations, the O term can be taken in any form one wishes: L^2, L^∞, C^∞ . They are all equivalent.

EXAMPLES.

(i) Fix a point (x_0, ξ_0) and recall from Section 5.1 the corresponding coherent state:

$$u(h)(x) := (\pi h)^{-\frac{n}{4}} e^{\frac{i}{h} \langle x - x_0, \xi_0 \rangle - \frac{1}{2h} |x - x_0|^2}.$$

Then

$$\mathrm{WF}_h(u) = \{(x_0, \xi_0)\},$$

as can be seen from the argument in Example 1 of Section 5.1.

(ii) Suppose that $b \in \mathcal{S}(\mathbb{R}^n)$ and $\varphi \in C^\infty(\mathbb{R}^n)$ is real. Then

$$\mathrm{WF}_h(b e^{\frac{i}{h}\varphi}) = \{(x, \partial_x \varphi(x)) \mid x \in \mathrm{spt}(b)\}.$$

This corresponds to Example 2 in Section 5.1.

(iii) This example generalizes the first two. Suppose that $b \in S(\mathbb{R}^n)$ and that $\varphi \in C^\infty(\mathbb{R}^n)$ is complex, satisfying

$$\operatorname{Im} \varphi(x) \geq 0, \quad \operatorname{Im} \varphi(x) \geq c\langle x \rangle^2 - C, \quad \varphi \in S(\langle x \rangle^2).$$

Then

$$\operatorname{WF}_h(b(x)e^{\frac{i}{h}\varphi(x)}) = \{(x, \partial_x \varphi(x)) \mid x \in \operatorname{spt}(b), \operatorname{Im} \varphi(x) = 0\}.$$

The verification is an exercise.

(iv) We have, for $x \in \mathbb{R}^n$ and a fixed vector $\omega \in \mathbb{R}^n$,

$$(8.4.7) \quad \operatorname{WF}_h(\exp(i\langle x, \omega \rangle)/h^\alpha) = \begin{cases} \mathbb{R}^n \times \{0\}, & \alpha < 1, \\ \mathbb{R}^n \times \{\omega\}, & \alpha = 1, \\ \emptyset, & \alpha > 1. \end{cases}$$

This reflects the fact that the semiclassical wavefront picks up the oscillations at wave length h . The slower oscillations are smoother and the higher oscillations push the wavefront set to ∞ .

(v) Suppose that $f \in L^2(\mathbb{R})$ and that $\operatorname{spt} f$ is compact. Define

$$u(h)(x) := f\left(x - \frac{1}{h}\right).$$

Then

$$\operatorname{WF}_h(u) = \emptyset.$$

This simple example shows that our wavefront set measures behavior only in compact sets in phase space and may not be sensitive to C^∞ regularity.

(vi) The Fourier conjugation formula (4.2.14) implies

$$\operatorname{WF}_h(\mathcal{F}_h u) = \{(x, \xi) \mid (-\xi, x) \in \operatorname{WF}_h(u)\}.$$

(vii) Example (ii) states that if $u \in \mathcal{S}$ and u independent of h , then

$$\operatorname{spt}(u) \times \{0\} = \operatorname{WF}_h(u).$$

More generally, if $u \in \mathcal{S}'$ is independent of h , we have

$$\operatorname{spt}(u) \times \{0\} \subseteq \operatorname{WF}_h(u).$$

REMARKS: Classical wavefront set. (i) For readers familiar with Hörmander's classical wavefront sets $\operatorname{WF}(u)$ ([H1, Chapter 8]), we suggest checking the following statement:

(8.4.8) If $u \in \mathcal{S}'$ is independent of h , then

$$\operatorname{WF}_h(u) = (\operatorname{spt}(u) \times \{0\}) \cup \operatorname{WF}(u).$$

(ii) There is no unique way of designing a wavefront set which simultaneously captures C^∞ and semiclassical properties. To illustrate these complications, suppose $u \in L^2(\mathbb{R})$ and $|\hat{u}(\xi)| \leq C\langle\xi\rangle^{-\epsilon}$ for some $\epsilon > 0$. Then

$$\mathrm{WF}_h(u(x) \exp(ie^{1/h}x)) = \emptyset.$$

But $\mathrm{WF}(u \exp(ie^{1/h}x)) = \mathrm{WF}(u)$ for every h , and this is nonempty if u is not a smooth function. Compare this example to (8.4.7). $\square$

Next we show that we can replace the given operator $a^w(x, hD)$ in the definition of wavefront set with others:

THEOREM 8.13 (Localization and wavefront sets). *Suppose that u is a tempered family of functions and*

$$(x_0, \xi_0) \notin \mathrm{WF}_h(u).$$

Then for any $b \in C_c^\infty(\mathbb{R}^{2n})$ with support sufficiently close to (x_0, ξ_0) , we have

$$(8.4.9) \quad \|b^w(x, hD)u(h)\|_{L^2} = O(h^\infty).$$

Proof. 1. Suppose $a \in S$, $a(x_0, \xi_0) \neq 0$. There exists $\chi \in C^\infty(\mathbb{R}^{2n})$ supported near (x_0, ξ_0) such that

$$|\chi(x, \xi)(a(x, \xi) - a(x_0, \xi_0)) + a(x_0, \xi_0)| \geq \gamma > 0$$

for $(x, \xi) \in \mathbb{R}^{2n}$. Then according to Theorem 4.29 there exists $c \in S$ for which

$$c^w(x, hD)(\chi^w(x, hD)a^w(x, hD) + a(x_0, \xi_0)(I - \chi^w(x, hD))) = I,$$

provided that h is small enough.

2. Now consider

$$(8.4.10) \quad \begin{aligned} b^w(x, hD)u(h) &= b^w(x, hD)c^w(x, hD)\chi^w(x, hD)a^w(x, hD)u(h) \\ &\quad + a(x_0, \xi_0)b^w(x, hD)c^w(x, hD)(I - \chi^w(x, hD))u(h). \end{aligned}$$

If we choose a to be the symbol appearing in (8.4.6), then the first term on the right-hand side is bounded by $O(h^\infty)$ in L^2 . If the support of b is sufficiently close to (x_0, ξ_0) , then $\mathrm{spt}(b) \cap \mathrm{spt}(1 - \chi) = \emptyset$ and according to Theorem 4.12,

$$\|b^w(x, hD)c^w(x, hD)(I - \chi^w(x, hD))\|_{\mathcal{S}' \rightarrow \mathcal{S}} = O(h^\infty).$$

Hence the second term in (8.4.10) is also $O(h^\infty)$ in L^2 . $\square$

In Chapter 9 we will study how pseudodifferential operators vary under coordinate changes. Anticipating Theorem 9.3, we record here the invariance property of the wavefront set:

THEOREM 8.14 (Invariance of wavefront sets). *Suppose $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a diffeomorphism equal to the identity outside of a compact set. Suppose also that u is a tempered family of functions.*

Then

$$(8.4.11) \quad \text{WF}_h(\gamma^*u) = \{(x, (\partial\gamma(x))^T\xi) \mid (\gamma(x), \xi) \in \text{WF}_h(u)\}.$$

8.4.3. Essential support. Since compactness of the support of a symbol is not preserved under operations such as composition, we introduce the more flexible notion of the essential support. It will be useful for studying phenomena localized in phase space.

DEFINITIONS.

(i) Let $a = \{a(x, \xi, h)\}_{0 < h \leq h_0} \subset \mathcal{S}$ be an h -dependent family of symbols, with $\mathcal{S}$ seminorms bounded uniformly in h .

We say that a has a compact *essential support* if there exists a compact set $K \subset \mathbb{R}^{2n}$ such that

$$(8.4.12) \quad \text{spt } \chi \cap K = \emptyset \quad \text{implies} \quad \chi a \in h^\infty \mathcal{S}$$

for all $\chi \in S$.

(ii) The smallest such compact set $K \subset \mathbb{R}^n \times \mathbb{R}^n$ is called the *essential support* of a , denoted

$$\text{ess-spt}(a).$$

The essential support $\text{ess-spt}(a)$ is defined only when we know that a has compact essential support.

REMARK. The assumption on the family of symbols a can be replaced by $a = \{a(x, \xi, h)\}_{0 < h \leq h_0} \subset S$, with $\partial^\alpha a = O(1)$ independently of h . The condition (8.4.12) then provides bounds on the $\mathcal{S}$ seminorms, as we can take $\chi = 1$ outside a compact set. $\square$

THEOREM 8.15 (Wavefront sets under pseudodifferential operators).

(i) *Suppose that the family $a = \{a(x, \xi, h)\}_{0 < h \leq h_0} \subset S(m)$ for some order function m and that $u = \{u(h)\}_{0 < h \leq h_0}$ is tempered.*

Then

$$(8.4.13) \quad \text{WF}_h(a^w u) \subseteq \text{WF}_h(u).$$

(ii) If $a = \{a(x, \xi, h)\}_{0 < h \leq h_0} \subset \mathcal{S}$ has compact essential support, then

$$(8.4.14) \quad \text{WF}_h(a^w u) \subseteq \text{WF}_h(u) \cap \text{ess-spt}(a).$$

Proof. For part (i) we need to show that

$$\text{if } (x_0, \xi_0) \notin \text{WF}_h(u), \text{ then } (x_0, \xi_0) \notin \text{WF}_h(a^w u).$$

Similarly for part (ii), we need the same conclusion when $(x_0, \xi_0) \notin \text{ess-spt}(a)$.

1. First suppose that $(x_0, \xi_0) \notin \text{WF}_h(u)$. Choose $b \in C_c^\infty(\mathbb{R}^{2n})$, with $b(x_0, \xi_0) \neq 0$ and $b^w(x, hD)u(h) = O_{L^2}(h^\infty)$. The existence of such b follows from Theorem 8.13. The composition formula in Theorem 4.11 shows that

$$b^w(x, hD)a^w(x, hD, h) = c^w(x, hD, h) + r^w(x, hD, h),$$

where $\text{spt } c \subset \text{spt } b$ and $r \in h^\infty \mathcal{S}$. Theorem 8.13 implies that

$$\begin{aligned} \|b^w(x, hD)a^w(x, hD, h)u(h)\|_{L^2} &\leq \|c^w(x, hD, h)u(h)\| + O(h^\infty)\|u\|_{L^2} \\ &= O(h^\infty). \end{aligned}$$

This shows that $(x_0, \xi_0) \notin \text{WF}_h(a^w u)$.

2. Now assume that a has compact essential support and that $(x_0, \xi_0) \notin \text{ess-spt}(a)$. If the support of a symbol b is sufficiently close to (x_0, ξ_0) , (8.4.12) implies that $b^w(x, hD)a^w(x, hD, h) = c^w(x, hD, h)$, where $c \in h^\infty \mathcal{S}$. Consequently

$$\|b^w(x, hD)a^w(x, hD, h)u(h)\|_{L^2} = O(h^\infty).$$

□

THEOREM 8.16 (Alternative characterization of $\text{ess-spt}(a)$). Suppose that $a = \{a(x, \xi, h)\}_{0 < h \leq h_0} \subset \mathcal{S}$ has compact essential support.

Then

$$(8.4.15) \quad (x, \xi) \notin \text{ess-spt}(a)$$

if and only if

$$(8.4.16) \quad (x, \xi) \notin \text{WF}_h(a^w u)$$

for each family $u = \{u(h)\}_{0 < h \leq h_0}$ bounded in $L^2(\mathbb{R}^n)$.

Proof. To see that if $(x_0, \xi_0) \notin \text{WF}_h(a^w u)$ for all u , then $(x_0, \xi_0) \notin \text{ess-spt}(a)$, use the family of coherent states from example (i) above. The opposite implication is given by (8.4.14). □

This alternative definition gives a quick proof of the following invariance statement:

THEOREM 8.17 (Invariance of essential support). *Suppose $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a diffeomorphism equal to the identity outside of a compact set. Suppose that $a = \{a(x, \xi, h)\}_{0 < h \leq h_0}$ is a family with compact essential support. Write $A = a^w(x, hD)$, acting on functions or on half-densities.*

If

$$(\gamma^{-1})^* A \gamma^* = a_\gamma^w(x, hD),$$

then

$$(8.4.17) \quad \text{ess-spt}(a_\gamma) = \{(x, (\partial\gamma(x))^T \xi) \mid (\gamma(x), \xi) \in \text{ess-spt}(a)\}.$$

Proof. We employ Theorems 8.14 and 8.16:

$$(x, \xi) \notin \text{ess-spt}(a_\gamma)$$

if and only if

$$(x, \xi) \notin \text{WF}_h((\gamma^*)^{-1} A \gamma^* u) \quad \text{for all } u$$

if and only if

$$(\gamma(x), (\partial\gamma(x))^T \xi) \notin \text{WF}_h(A \gamma^* u) \quad \text{for all } u$$

if and only if

$$(\gamma(x), (\partial\gamma(x))^T \xi) \notin \text{WF}_h(Av) \quad \text{for all } v$$

if and only if

$$(\gamma(x), (\partial\gamma(x))^T \xi) \notin \text{ess-spt}(a).$$

Here the families of functions u and v are assumed to satisfy (8.4.1). $\square$

REMARK. Operators $A = a^w(x, hD, h)$ where a has compact essential support can be characterized as pseudodifferential operators such that

$$(8.4.18) \quad (1 - \chi^w(x, hD))A = O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty)$$

for some $\chi \in C_c^\infty(\mathbb{R}^{2n})$. This follows from the composition formula since $(1 - \chi)\#a$ satisfies (8.4.12) with $K = \text{spt } \chi \subset\subset \mathbb{R}^{2n}$. $\square$

The remark above motivates the following

DEFINITION. If $A = a^w(x, hD, h)$ and a has a compact essential support, then

$$(8.4.19) \quad \text{WF}_h(A) := \text{ess-spt}(a).$$

Roughly speaking, $\text{WF}_h(A)$ describes the compact set in the phase space $\mathbb{R}^{2n}$ on which the operator A is localized.

8.4.4. Wavefront sets of localized functions. We conclude this section by defining a stronger notion of the wavefront set for a family of functions $u = \{u(h)\}_{0 < h \leq h_0} \subset L^2(\mathbb{R}^n)$ that is tempered and localized in phase space.

DEFINITION. We define

$$(8.4.20) \quad \text{WF}_h^*(u) := \bigcap \{ \text{spt } \chi \mid \chi, (1 - \chi^w(x, hD))u = O_{\mathcal{S}}(h^\infty) \}.$$

This notation assumes that the localization condition (8.4.4) holds for some χ .

We note that $\text{WF}_h^*(Au)$ is defined for any operator A for which $\text{WF}_h(A)$ is defined. We leave it as an exercise to check that if $\text{WF}_h^*(u)$ is defined, then

$$(8.4.21) \quad \text{WF}_h^*(u) = \text{WF}_h(u).$$

Hence WF_h^* serves only as a useful shorthand notation indicating that the localization (8.4.4) holds.

8.4.5. Microlocality. Finally, we introduce the concept of microlocal identity of operators:

DEFINITIONS. Let U, V be open, bounded subsets of $\mathbb{R}^{2n}$.

(i) Given two tempered operators T and S , we say that

$$(8.4.22) \quad T \equiv S \quad \text{microlocally on } U \times V$$

if there exist open sets $\tilde{U} \supset U$ and $\tilde{V} \supset V$ such that

$$A(T - S)B = O_{\mathcal{S} \rightarrow \mathcal{S}}(h^\infty)$$

for all A, B with $\text{WF}_h(A) \subset \tilde{V}, \text{WF}_h(B) \subset \tilde{U}$.

(ii) In particular, we say

$$(8.4.23) \quad T \equiv I \quad \text{microlocally near } U \times V$$

if there exists an open set $\tilde{U} \supset U$ such that

$$A - TA = A - AT = O_{\mathcal{S} \rightarrow \mathcal{S}}(h^\infty)$$

for all A with $\text{WF}_h(A) \subset \tilde{U}$.

(iii) We say that T is *microlocally invertible* near $U \times V$ if there exists an operator S such that $TS \equiv I$ and $ST \equiv I$ microlocally near $V \times V$ and $U \times U$, respectively. When confusion is unlikely, we will write

$$S = T^{-1}$$

and call S a *microlocal inverse* of T .

EXAMPLE. Suppose that $a \in S$ and $a(x_0, \xi_0) \neq 0$. Then $a^w(x, hD)$ is microlocally invertible in a neighborhood of (x_0, ξ_0) . $\square$

INTERPRETATION. The notion of microlocality gives us a quick way to refer to phenomena occurring in phase space. Owing to the uncertainty principle, localization in phase space is more subtle than localization in space, but in many ways it has the same properties.

As shown in Theorem 8.15, pseudodifferential operators do not “move things in phase space”. That however will not be so for other natural operators introduced later, in Chapters 10 and 11. We can best and most conveniently understand their properties geometrically using the notion of microlocality. It will be particularly useful in Chapter 12 when we study microlocal normal forms. $\square$

8.5. NOTES

Our presentation of Beals’s Theorem follows Dimassi–Sjöstrand [D-S, Chapter 8] and Helffer–Sjöstrand [H-S].

Section 8.2 provides a special case of a general result in Bony–Chemin [B-C, Théoreme 6.4]. The self-contained proof in the simple case considered here comes from the appendix to [S-Z3]. See [D-S-Z], [S-Z3] for examples of conjugation techniques and Martinez [M] for a different perspective based on the FBI transform.

The semiclassical wavefront set is an analogue of the usual wavefront set in microlocal analysis (see Hörmander [H1, Chapter 8], [H3, Chapter 18], and Grigis–Sjöstrand [G-S] and is closely related to the frequency set introduced in Guillemin–Sternberg [G-St]. See Alexandrova [A] for more about semiclassical wavefront sets.

CHANGING VARIABLES

- 9.1 Invariance, half-densities
- 9.2 Changing symbols
- 9.3 Invariant symbol classes
- 9.4 Notes

9.1. INVARIANCE, HALF-DENSITIES

9.1.1. Motivation, definitions. We begin with a general discussion concerning the invariance of various quantities under the change of variables

$$(9.1.1) \quad \tilde{x} = \gamma(x),$$

where $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a diffeomorphism.

Functions. Recall first that functions transform under (9.1.1) by pull-back. This means that we change u into a function $\tilde{u}$ of the new variables $\tilde{x}$ by the rule

$$(9.1.2) \quad \tilde{u}(\tilde{x}) = \tilde{u}(\gamma(x)) := u(x)$$

for $x \in \mathbb{R}^n$. In the notation of pull-backs *acting on functions*, this is written

$$(9.1.3) \quad \gamma^* \tilde{u} = \tilde{u} \circ \gamma = u.$$

Notice however that in general the integral of $|u|^2$ over a Borel set E is then not invariant:

$$\int_{\gamma(E)} |\tilde{u}(\tilde{x})|^2 d\tilde{x} \neq \int_E |u(x)|^2 dx.$$

We introduce the formalism of half-densities to fix this defect.

Half-densities. In quantum mechanics the probability of “finding our state in the set E ” is given by

$$\int_E |u(x)|^2 dx,$$

where dx is the Lebesgue density.

This probability density should be invariantly defined and so should not depend on the choice of coordinates x . This means that it is *not* the function u which should be defined invariantly but rather the density $|u|^2 dx$, or, up to the phase information, the *half-density*

$$(9.1.4) \quad u|dx|^{\frac{1}{2}}.$$

For half-densities we therefore demand that

$$(9.1.5) \quad \tilde{u}(\tilde{x})|d\tilde{x}|^{\frac{1}{2}} = u(x)|dx|^{\frac{1}{2}},$$

meaning that integrals of the squares should be invariantly defined. To accomplish this, we modify our earlier definitions (9.1.2), (9.1.3) as follows:

DEFINITION. The pull-back of γ acting on half-densities is defined by the rule

$$(9.1.6) \quad \gamma^* \left(\tilde{u}|d\tilde{x}|^{\frac{1}{2}} \right) = u|dx|^{\frac{1}{2}},$$

for

$$(9.1.7) \quad \tilde{u}(\tilde{x}) = \tilde{u}(\gamma(x)) := u(x)|\det(\partial\gamma(x))|^{-\frac{1}{2}}.$$

With this modified definition we have

$$\int_{\gamma(E)} |\tilde{u}(\tilde{x})|^2 d\tilde{x} = \int_E |u(x)|^2 dx$$

for all Borel sets $E \subseteq \mathbb{R}^n$.

We will use the shorter, informal, notation (9.1.5) in place of (9.1.6). This corresponds to writing

$$|d\tilde{x}|^{\frac{1}{2}} = |\partial\tilde{x}/\partial x|^{\frac{1}{2}}|dx|^{\frac{1}{2}} = |\det(\partial\gamma(x))|^{\frac{1}{2}}|dx|^{\frac{1}{2}}.$$

A rigorous interpretation in terms of line bundles will be given in Example 3 in Section 14.1.2.

We stress that when a specific density is chosen, such as the Lebesgue density on $\mathbb{R}^n$, then half-densities can be identified with functions. However,

the transformation rule under the change of variables is now (9.1.7) rather than (9.1.2).

NOTATION. We denote smooth half-densities by

$$C^\infty(\mathbb{R}^n; \Omega^{\frac{1}{2}}),$$

and the notation $\Omega^{\frac{1}{2}}$ will appear naturally in Example 3 of Section 14.1.2, where half-densities are defined as sections of a (trivial) line bundle. We can identify $C^\infty(\mathbb{R}^n; \Omega^{\frac{1}{2}})$ with $C^\infty(\mathbb{R}^n)$ using the Lebesgue density $|dx|$,

$$v = u(x)|dx|^{\frac{1}{2}},$$

paying attention to the action of γ^* different than in the case of functions.

Since for a half-density v , $|v|^2$ can be integrated in an invariant way, we can define

$$L^2(\mathbb{R}^n; \Omega^{\frac{1}{2}})$$

for the collection of half-densities on $\mathbb{R}^n$ satisfying

$$\|v\|_{L^2(\mathbb{R}^n; \Omega^{\frac{1}{2}})}^2 = \int_{\mathbb{R}^n} |v|^2 < \infty.$$

The pull-backs by diffeomorphisms act on these spaces:

$$(9.1.8) \quad \gamma^*, (\gamma^{-1})^* : L^2(\mathbb{R}^n; \Omega^{\frac{1}{2}}) \rightarrow L^2(\mathbb{R}^n; \Omega^{\frac{1}{2}}).$$

The space $L^2(\mathbb{R}^n; \Omega^{\frac{1}{2}})$ can be identified with $L^2(\mathbb{R}^n)$ by writing $v = u(x)|dx|^{\frac{1}{2}}$, $u \in L^2(\mathbb{R}^n)$. However, for the general diffeomorphism γ , γ^* defined by (9.1.3) does not map $L^2(\mathbb{R}^n)$ to itself.

DISCUSSION. The foregoing formalism is at first rather unintuitive but turns out later to be useful in the rigorous semiclassical calculus, in particular in the theory of Fourier integral operators, which we will touch upon later. Section 9.2 will show how the half-density viewpoint fits naturally within the Weyl calculus, and Section 10.2 will explain how half-densities simplify some related calculations for a propagator. $\square$

9.1.2. Operators on half-densities. We continue to suppose $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a smooth diffeomorphism equal to the identity outside a bounded set.

DEFINITION. If

$$A : C^\infty(\mathbb{R}^n; \Omega^{\frac{1}{2}}) \rightarrow C^\infty(\mathbb{R}^n; \Omega^{\frac{1}{2}})$$

or

$$A : L^2(\mathbb{R}^n; \Omega^{\frac{1}{2}}) \rightarrow L^2(\mathbb{R}^n; \Omega^{\frac{1}{2}}),$$

we define the *pull-back* by γ^{-1} of A to be the operator

$$(9.1.9) \quad \tilde{A} := (\gamma^{-1})^* A \gamma^*,$$

the pull-backs acting on half-densities.

In half-density notation, this is written

$$(9.1.10) \quad \tilde{A}(\tilde{u}|d\tilde{x}|^{\frac{1}{2}}) = A(u|dx|^{\frac{1}{2}}), \quad \tilde{x} = \gamma(x).$$

EXAMPLE. It is instructive to consider differential operators acting on half-densities. Let

$$P(x, hD) = \sum_{|\alpha| \leq m} a_\alpha(x) (hD)^\alpha.$$

If we identify $C^\infty(\mathbb{R}^n, \Omega^{\frac{1}{2}})$ with $C^\infty(\mathbb{R}^n)$, $v = u(x)|dx|^{\frac{1}{2}}$, we define

$$P(x, hD)v := (P(x, hD)u(x))|dx|^{\frac{1}{2}}.$$

Then

$$\begin{aligned} (\gamma^{-1})^* P(x, hD) \gamma^* v &= ((\gamma^{-1})^* P(x, hD) \gamma^*) (u(x)|dx|^{\frac{1}{2}}) \\ &= (P_\gamma(x, hD)u(x))|dx|^{\frac{1}{2}}, \end{aligned}$$

where $P_\gamma(x, hD)u(x)$ is given by

$$(P(y, hD_y)(u(\gamma(y))| \det \partial \gamma(y)|^{\frac{1}{2}})_{y=\gamma^{-1}(x)} | \det \partial \gamma(\gamma^{-1}(x))|^{-\frac{1}{2}}.$$

In the special case of $P(x, \xi) = \langle a, \xi \rangle$, $a \in \mathbb{C}^n$, a short calculation shows that

$$P_\gamma(x, hD) = p_\gamma^w(x, hD), \quad p_\gamma(\gamma(x), \xi) := \langle a, \partial \gamma(x)^T \xi \rangle.$$

We will see far-reaching generalizations of this statement in Theorems 9.3 and 9.10.

Half-densities also elegantly appear when we use operator kernels:

DEFINITION. Let $K \in \mathcal{S}(\mathbb{R}^{2n})$. Then the formal expression

$$K(x, y)|dx|^{\frac{1}{2}}|dy|^{\frac{1}{2}}$$

acts as an integral kernel, defining as follows a map on half-densities:

$$\begin{aligned} (9.1.11) \quad A(u|dx|^{\frac{1}{2}}) &= \int_{\mathbb{R}^n} K(x, y)|dx|^{\frac{1}{2}}|dy|^{\frac{1}{2}}u(y)|dy|^{\frac{1}{2}} \\ &:= \left(\int_{\mathbb{R}^n} K(x, y)u(y)dy \right) |dx|^{\frac{1}{2}}. \end{aligned}$$

THEOREM 9.1 (Operator kernels and half-densities). *Let K be the kernel for an operator A acting on half-densities, as above. Then the kernel for the pull-back $\tilde{A}$ is*

$$\tilde{K}(\tilde{x}, \tilde{y}) |d\tilde{x}|^{\frac{1}{2}} |d\tilde{y}|^{\frac{1}{2}},$$

where

$$(9.1.12) \quad \tilde{K}(\tilde{x}, \tilde{y}) = K(x, y) |\det \partial\gamma(x)|^{-\frac{1}{2}} |\det \partial\gamma(y)|^{-\frac{1}{2}}$$

for $\tilde{x} = \gamma(x)$, $\tilde{y} = \gamma(y)$.

Proof. To simplify notation, we will write $|\partial\gamma| := |\det \partial\gamma|$.

From (9.1.10) we have

$$\left(\int \tilde{K}(\tilde{x}, \tilde{y}) \tilde{u}(\tilde{y}) |d\tilde{y}| \right) |d\tilde{x}|^{\frac{1}{2}} = \left(\int K(x, y) u(y) |dy| \right) |dx|^{\frac{1}{2}},$$

where $\tilde{u}(\gamma(y)) = u(y) |\partial\gamma(y)|^{-\frac{1}{2}}$ and $|d\tilde{x}|^{\frac{1}{2}} = |\partial\gamma(x)|^{\frac{1}{2}} |dx|^{\frac{1}{2}}$. Hence

$$|\partial\gamma(x)|^{\frac{1}{2}} \int \tilde{K}(\tilde{x}, \tilde{y}) u(\gamma^{-1}(\tilde{y})) |\partial\gamma(\gamma^{-1}(\tilde{y}))|^{-\frac{1}{2}} d\tilde{y} = \int K(x, y) u(y) dy.$$

The change of variable in the integral on the right-hand side gives

$$|\partial\gamma(x)|^{\frac{1}{2}} \int \tilde{K}(\gamma(x), \gamma(y)) |\partial\gamma(y)|^{\frac{1}{2}} u(y) dy = \int K(x, y) u(y) dy.$$

Since this is valid for all u , we have

$$|\partial\gamma(x)|^{\frac{1}{2}} \tilde{K}(\gamma(x), \gamma(y)) |\partial\gamma(y)|^{\frac{1}{2}} = K(x, y),$$

and that is the same as (9.1.12). $\square$

REMARK: Numerical interpretation. The half-density formalism is also a surprisingly useful bookkeeping technique, appearing implicitly in some numerical schemes.

Suppose that an operator A is given by a symmetric integral kernel $K(x, y)$. A numerical implementation converts A into a matrix $A^N = ((K(x_i, x_j)))_{1 \leq i, j \leq N}$, which acts on vectors $v \in \mathbb{C}^N$ interpreted as $v = (u(x_i))_{1 \leq i \leq N}$. Integration of u involves multiplication of each entry by a weight $\mu_j > 0$ and summation:

$$\int u(x) dx \approx \sum_{j=1}^N \mu_j v_j.$$

Hence

$$(Au)(x_i) = \int K(x_i, y) u(y) dy \approx \sum_{j=1}^N A_{ij}^N \mu_j v_j.$$

In other words, the application of A is approximated by the matrix $A^N M$ where

$$M := \begin{pmatrix} \mu_1 & 0 & \dots & 0 \\ 0 & \mu_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \mu_N \end{pmatrix}.$$

What is lost is the symmetry of the matrix and this may be numerically inconvenient.

A simple remedy is to consider

$$(Au)(x_i)\mu_i^{\frac{1}{2}} \approx \sum_{j=1}^N (\mu_i^{\frac{1}{2}} A_{ij}^N \mu_j^{\frac{1}{2}}) (\mu_j^{\frac{1}{2}} v_j),$$

which acts on, and returns, discretized functions multiplied by $M^{\frac{1}{2}}$. The application of A corresponds to the *symmetric* matrix

$$M^{\frac{1}{2}} A^N M^{\frac{1}{2}}.$$

Since the weights correspond to integration, this is a discrete version of the half-density formalism. $\square$

9.1.3. Quantization and half-densities. We can also interpret the Weyl quantization of a symbol $a \in \mathcal{S}$ as an operator acting on half-densities:

DEFINITIONS.

(i) If $a \in \mathcal{S}$, we write

$$(9.1.13) \quad \begin{aligned} & K_a(x, y) |dx|^{\frac{1}{2}} |dy|^{\frac{1}{2}} \\ & := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} a\left(\frac{x+y}{2}, \xi\right) e^{\frac{i}{h}\langle x-y, \xi \rangle} d\xi |dx|^{\frac{1}{2}} |dy|^{\frac{1}{2}}. \end{aligned}$$

(ii) We define $a^w(x, hD)$ as an operator acting on half-densities by the formula

$$(9.1.14) \quad a^w(x, hD)(u|dx|^{\frac{1}{2}}) := \int_{\mathbb{R}^n} K_a(x, y) u(y) dy |dx|^{\frac{1}{2}}.$$

THEOREM 9.2 (Quantization acting on half-densities). *If $a \in \mathcal{S}$, then*

$$a^w(x, hD) : L^2(\mathbb{R}^n, \Omega^{\frac{1}{2}}) \rightarrow L^2(\mathbb{R}^n, \Omega^{\frac{1}{2}})$$

is a bounded operator.

This is shown by arguments as in Chapter 4, details of which we omit.

9.2. CHANGING SYMBOLS

In this section we investigate the invariance properties of quantization under changes of variables.

9.2.1. Changing variables and changing symbols. Next, let $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a smooth diffeomorphism which we assume to be the identity outside a bounded set. As above, we write $\tilde{x} = \gamma(x)$ and

$$\tilde{u}(\tilde{x})|d\tilde{x}|^{\frac{1}{2}} = u(x)|dx|^{\frac{1}{2}}.$$

THEOREM 9.3 (Quantization and change of variables). *Assume $a \in \mathcal{S}$.*

(i) *Then*

$$(9.2.1) \quad (\gamma^{-1})^* a^w(x, hD) \gamma^* = \tilde{a}^w(x, hD)$$

as operators on half-densities, for the symbol satisfying

$$(9.2.2) \quad \tilde{a}(\gamma(x), \xi) := a(x, \partial\gamma(x)^T \xi) + O_{\mathcal{S}}(h^2).$$

(ii) *Furthermore,*

$$(9.2.3) \quad (\gamma^{-1})^* a^w(x, hD) \gamma^* = \tilde{a}_1^w(x, hD)$$

as operators on functions, for the symbol satisfying

$$(9.2.4) \quad \tilde{a}_1(\gamma(x), \xi) := a(x, \partial\gamma(x)^T \xi) + O_{\mathcal{S}}(h).$$

DEFINITION. We call

$$(9.2.5) \quad b(x, \xi) := a(\gamma^{-1}(x), \partial\gamma(\gamma^{-1}(x))^T \xi)$$

the *pull-back* of the symbol a under the mapping γ^{-1} . (It is more precisely the pull-back by the lift of the diffeomorphism γ^{-1} to the cotangent bundle; recall Theorem 2.6.)

Theorem 9.3 states that the pull-back of the quantization of a by γ^{-1} is to leading order given by quantization of the pull-back of a by γ^{-1} in the sense of this definition.

DISCUSSION. Yet another motivation for half-densities is that assertion (i) for half-densities (with error term of order $O_{\mathcal{S}}(h^2)$) is more precise than the assertion (ii) for functions (with error term $O_{\mathcal{S}}(h)$).

Remember that the notation $\varphi = O_{\mathcal{S}}(h^k)$ means

$$|x^\alpha \xi^\beta \partial^\gamma \varphi| \leq C_{\alpha\beta\gamma} h^k$$

for all multiindices α, β, γ . □

Proof. 1. Write $A = a^w(x, hD)$. Since $a \in \mathcal{S}$, we have $K_a \in \mathcal{S}$. Take $\tilde{a} \in \mathcal{S}(\mathbb{R}^{2n})$ for which $(\gamma^{-1})^* A \gamma^* = \tilde{a}^w$.

Recall that

$$A(u|dx|^{\frac{1}{2}}) = \int_{\mathbb{R}^n} K_a(x, y) |dx|^{\frac{1}{2}} |dy|^{\frac{1}{2}} u(y) |dy|^{\frac{1}{2}}$$

for

$$K_a(x, y) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} a\left(\frac{x+y}{2}, \xi\right) e^{\frac{i}{h}\langle x-y, \xi \rangle} d\xi.$$

Likewise

$$\tilde{A}(\tilde{u}|d\tilde{x}|^{\frac{1}{2}}) = \int_{\mathbb{R}^n} K_{\tilde{a}}(\tilde{x}, \tilde{y}) |d\tilde{x}|^{\frac{1}{2}} |d\tilde{y}|^{\frac{1}{2}} \tilde{u}(\tilde{y}) |d\tilde{y}|^{\frac{1}{2}}$$

for

$$K_{\tilde{a}}(\tilde{x}, \tilde{y}) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \tilde{a}\left(\frac{\tilde{x}+\tilde{y}}{2}, \tilde{\xi}\right) e^{\frac{i}{h}\langle \tilde{x}-\tilde{y}, \tilde{\xi} \rangle} d\tilde{\xi}.$$

Since $\tilde{u}(\tilde{y})|d\tilde{y}|^{\frac{1}{2}} = u(y)|dy|^{\frac{1}{2}}$ and $d\tilde{y} = |\partial\gamma(y)|dy$, the invariance (9.1.10) implies

$$(9.2.6) \quad K_a(x, y) = K_{\tilde{a}}(\tilde{x}, \tilde{y}) |\partial\gamma(y)|^{\frac{1}{2}} |\partial\gamma(x)|^{\frac{1}{2}}, \quad \tilde{x} = \gamma(x), \quad \tilde{y} = \gamma(y).$$

Our goal is to find an expression for $\tilde{a}$ in terms of a , up to $O_{\mathcal{S}}(h^2)$ errors.

2. In what follows we can assume that $|x - y|$ in the integrand is small. To justify this, we write the integral kernel as

$$(9.2.7) \quad K_{\tilde{a}}(\tilde{x}, \tilde{y}) = K_{\tilde{a}}(\tilde{x}, \tilde{y}) \chi(\tilde{x} - \tilde{y}) + K_{\tilde{a}}(\tilde{x}, \tilde{y}) (1 - \chi)(\tilde{x} - \tilde{y}),$$

where $\chi \in C_c^\infty(\mathbb{R}^n)$ is equal to 1 near 0. In the second term integration by parts using

$$\frac{\langle x - y, hD_\xi \rangle}{|x - y|^2} e^{\frac{i}{h}\langle x-y, \xi \rangle} = e^{\frac{i}{h}\langle x-y, \xi \rangle}$$

shows that the corresponding operator is $O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty)$. Hence it is given by $b^w(x, hD)$ where $b = O_{\mathcal{S}}(h^\infty)$. Owing to Theorem 4.20 (see also the remark after the proof of Theorem 4.20), the first term in (9.2.7) gives an operator of the form $(\tilde{a} + O_{\mathcal{S}}(h^\infty))^w(x, hD)$.

These principles of *pseudolocality* stating that only contributions near the diagonal are significant should be kept in mind while reading the rest of the proof.

3. We need to compute $\tilde{a}$ in terms of a and γ , up to $O(h^2)$ error terms. Now

$$\begin{aligned} K_{\tilde{a}}(\tilde{x}, \tilde{y}) &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \tilde{a}\left(\frac{\tilde{x}+\tilde{y}}{2}, \tilde{\xi}\right) e^{\frac{i}{h}\langle \tilde{x}-\tilde{y}, \tilde{\xi} \rangle} d\tilde{\xi} \\ &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \tilde{a}\left(\frac{\gamma(x)+\gamma(y)}{2}, \tilde{\xi}\right) e^{\frac{i}{h}\langle \gamma(x)-\gamma(y), \tilde{\xi} \rangle} d\tilde{\xi}. \end{aligned}$$

Taylor's formula gives

$$\begin{aligned}\gamma(x) &= \gamma\left(\frac{x+y}{2}\right) + \partial\gamma\left(\frac{x+y}{2}\right)\left(\frac{x-y}{2}\right) + \frac{1}{8}\langle\partial^2\gamma\left(\frac{x+y}{2}\right)(x-y), x-y\rangle \\ &\quad + \langle F(x, y), x-y\rangle, \\ \gamma(y) &= \gamma\left(\frac{x+y}{2}\right) - \partial\gamma\left(\frac{x+y}{2}\right)\left(\frac{x-y}{2}\right) + \frac{1}{8}\langle\partial^2\gamma\left(\frac{x+y}{2}\right)(x-y), x-y\rangle \\ &\quad - \langle F(y, x), x-y\rangle,\end{aligned}$$

where $F(x, y)$ is a matrix satisfying $F(x, y) = O(|x-y|^2)$. Hence

$$(9.2.8) \quad \gamma(x) - \gamma(y) = g(x, y)(x-y),$$

where the matrix g satisfies

$$(9.2.9) \quad g(x, y) = \partial\gamma\left(\frac{x+y}{2}\right) + F(x, y) - F(y, x) = \partial\gamma\left(\frac{x+y}{2}\right) + O(|x-y|^2).$$

Also

$$(9.2.10) \quad \gamma(x) + \gamma(y) = 2\gamma\left(\frac{x+y}{2}\right) + O(|x-y|^2).$$

Let us also write

$$(9.2.11) \quad \tilde{\xi} = (g(x, y)^T)^{-1}\xi.$$

Substituting above, we deduce that

$$\begin{aligned}K_{\tilde{a}}(\tilde{x}, \tilde{y}) &= \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} [\tilde{a}\left(\gamma\left(\frac{x+y}{2}\right), (g(x, y)^T)^{-1}\xi\right) + O(|x-y|^2)] \\ &\quad \times e^{\frac{i}{h}\langle\gamma(x)-\gamma(y), (g(x, y)^T)^{-1}\xi\rangle} d\tilde{\xi}.\end{aligned}$$

We now use the “Kuranishi trick” to rewrite this expression as a pseudodifferential operator. First,

$$\begin{aligned}\langle\gamma(x) - \gamma(y), (g(x, y)^T)^{-1}\xi\rangle &= \langle g(x, y)^{-1}(\gamma(x) - \gamma(y)), \xi\rangle \\ &= \langle x - y, \xi\rangle,\end{aligned}$$

according to (9.2.8). Remembering also (9.2.9), we compute

$$K_{\tilde{a}}(\tilde{x}, \tilde{y}) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \left(\tilde{b}\left(\frac{x+y}{2}, \xi\right) + O(|x-y|^2)\right) e^{\frac{i}{h}\langle x-y, \xi\rangle} d\tilde{\xi},$$

where

$$(9.2.12) \quad \tilde{b}(x, \xi) := \tilde{a}(\gamma(x), (\partial\gamma(x)^T)^{-1}\xi).$$

Furthermore $d\tilde{\xi} = |\det g(x, y)|^{-1}d\xi$ and

$$\det g(x, y) = \det \partial\gamma\left(\frac{x+y}{2}\right) + O(|x-y|^2).$$

Also, we claim that

$$\left|\det \partial\gamma\left(\frac{x+y}{2}\right)\right|^2 = |\partial\gamma(x)| |\partial\gamma(y)| + O(|x-y|^2).$$

This identity is clear if we add a term $\langle A(\frac{x+y}{2}), x-y \rangle$ on the right-hand side. But the symmetry under switching x and y shows that $A \equiv 0$.

4. Finally, we observe that

$$(9.2.13) \quad (x-y)^\alpha e^{\frac{i}{h}\langle x-y, \xi \rangle} = (hD_\xi)^\alpha e^{\frac{i}{h}\langle x-y, \xi \rangle}.$$

Hence integrating by parts in the terms with $O(|x-y|^2)$ gives us terms of order $O(h^2)$. So

$$K_{\tilde{a}}(\tilde{x}, \tilde{y}) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \left(b\left(\frac{x+y}{2}, \xi\right) + O(h^2) \right) e^{\frac{i}{h}\langle x-y, \xi \rangle} d\xi \\ \times |\partial\gamma(x)|^{-1/2} |\partial\gamma(y)|^{-1/2}.$$

Comparing this with (9.2.6) and (9.2.12) proves (9.2.2).

5. When A acts on functions, then K_a has to transform as a density. In other words, we need to show that

$$(9.2.14) \quad K_a(x, y) = K_{a_1}(\tilde{x}, \tilde{y}) |\partial\gamma(y)|$$

implies (9.2.4).

Since

$$|\partial\gamma(y)| = |\partial\gamma(y)|^{1/2} |\partial\gamma(x)|^{1/2} + O(|x-y|),$$

we see from (9.2.13) that (9.2.14) follows from (9.2.6) with $a_1 = \tilde{a} + O(h)$. $\square$

9.3. INVARIANT SYMBOL CLASSES

We now introduce more general classes of symbols a than those discussed in Chapter 4 and explain their invariance under mappings $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

9.3.1. Classical symbols. To explain the need for these new classes of symbols, let m be an order function and recall the class $S(m)$ introduced in Chapter 4:

$$S(m) := \{a \in C^\infty(\mathbb{R}^{2n}) \mid |\partial^\alpha a| \leq C_\alpha m \text{ for all } \alpha\}.$$

In view of Theorem 9.3, for a symbol a in $S(m)$ to be invariant under γ , we would need for its pull-back by γ^{-1} , $b(\gamma(x), \xi) := a(x, \partial\gamma(x)^T \xi)$, that

$$(9.3.1) \quad |\partial^\alpha b| \leq C_\alpha m$$

for all multiindices α . But this bound is in general false except for a very restrictive class of order functions m ; that in particular would exclude all differential operators.

However an estimate of the type (9.3.1) would hold if differentiation in ξ improves the decay in ξ . This observation leads us to the following definition, in which we restrict ourselves to the order functions $\langle \xi \rangle^m$.

DEFINITIONS. (i) The *Kohn–Nirenberg symbols* for $m \in \mathbb{Z}$ are

$$(9.3.2) \quad S^m := \{a \in C^\infty(\mathbb{R}^{2n}) \mid |\partial_x^\alpha \partial_\xi^\beta a| \leq C_{\alpha\beta} \langle \xi \rangle^{m-|\beta|} \text{ for all } \alpha, \beta\}.$$

(ii) We also write

$$(9.3.3) \quad \Psi^m := \{a^w(x, hD) \mid a \in S^m\}$$

and

$$(9.3.4) \quad \Psi^{-\infty} := \bigcap_{m \in \mathbb{Z}} \Psi^m.$$

As for the classes introduced in Chapter 4, *symbols in S^m are allowed to depend upon h , although this dependence is usually not displayed in our notation.* If $a \in S^m$ depends on h , we require that the constants $C_{\alpha\beta}$ are uniform for $0 < h \leq h_0$.

THEOREM 9.4 (Invariance of S^m). *Assume $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a smooth diffeomorphism, with*

$$|\partial^\alpha \gamma|, |\partial^\alpha \gamma^{-1}| \leq C_\alpha$$

for all multiindices α . Then for each symbol $a \in S^m$, its pull-back

$$b(x, \xi) := a(\gamma^{-1}(x), \partial\gamma(\gamma^{-1}(x))^T \xi)$$

also belongs to S^m .

Proof. We have $b(x, \xi) = c(x, \partial\gamma(\gamma^{-1}(x))^T \xi)$ for $c(x, \xi) := a(\gamma^{-1}(x), \xi)$. Now $\partial_x^\alpha b$ has the form

$$\partial_x^\alpha b = \sum g_{\gamma\sigma\rho} (\partial_x^\gamma \partial_\xi^\sigma c) \xi^\rho,$$

the sum over multiindices satisfying $|\gamma| + |\sigma| \leq |\alpha|$ and $|\sigma| = |\rho|$. Therefore

$$\partial_x^\alpha \partial_\xi^\beta b = \sum g_{\gamma\sigma\rho\nu\kappa\lambda} (\partial_x^\gamma \partial_\xi^{\sigma+\nu-\kappa} c) \xi^{\rho-\lambda},$$

where $|\gamma| + |\sigma| \leq |\alpha|$, $|\sigma| = |\rho|$, $|\kappa| = |\lambda|$, $|\nu| = |\beta|$, $\nu \geq \kappa$, $\rho \geq \lambda$. Since clearly $c \in S^m$, we can estimate

$$|\partial_x^\alpha \partial_\xi^\beta b| \leq \sum C_{\gamma\sigma\rho\nu\kappa\lambda} \langle \xi \rangle^{m-|\sigma|-|\nu|+|\kappa|} \langle \xi \rangle^{|\rho|-|\lambda|} \leq C_{\alpha\beta} \langle \xi \rangle^{m-|\beta|}.$$

□

9.3.2. Symbol calculus for S^m . Since $S^m \subset S(\langle \xi \rangle^m)$, the results of Chapter 4 are all applicable; but due to the improvement under differentiation in ξ , there are many new features, important in the study of partial differential equations.

THEOREM 9.5 (Composition for Ψ^m).

(i) Assume $a \in S^{m_1}$ and $b \in S^{m_2}$. Then

$$a^w(x, hD)b^w(x, hD) = c^w(x, hD)$$

where the symbol $c \in S^{m_1+m_2}$ is given by (4.3.6).

(ii) Moreover,
(9.3.5)

$$c(x, \xi) = \sum_{k=0}^N \frac{i^k h^k}{k!} A(D)^k a(x, \xi) b(y, \eta)|_{x=\xi, y=\eta} + O_{S^{m_1+m_2-N-1}}(h^{N+1})$$

where $A(D) = \frac{1}{2}\sigma(D_x, D_\xi; D_y, D_\eta)$.

(iii) Similar statements hold for the usual quantization:

$$\begin{aligned} a(x, hD)b(x, hD) &= c_1(x, hD) \\ &= e^{ih\langle D_\xi, D_y \rangle} a(x, \xi) b(y, \eta)|_{y=x, \eta=\xi} \end{aligned}$$

and

$$c_1(x, \xi) = \sum_{|\alpha| \leq N} \frac{1}{\alpha!} \partial_\xi^\alpha a(x, \xi) (hD_x)^\alpha b(x, \xi) + O_{S^{m_1+m_2-N-1}}(h^{N+1}).$$

Proof. 1. Since $S^{m_j} \subset S(\langle \xi \rangle^{m_j})$, (4.3.6) follows from Theorem 4.11, but we must show that $c \in S^{m_1+m_2}$. Similarly, the expansion (9.3.5) is valid, but we need to prove the stated error bounds.

2. We first recall from (4.3.5) that

$$c(x, \xi) = \exp(hA(D))a(x, \xi)b(y, \eta)|_{x=y, \eta=\xi}$$

and also observe that

$$i^k h^k A(D)^k a(x, \xi) b(y, \eta)|_{x=\xi, y=\eta} \in h^k S^{m_1+m_2-k},$$

since the differential operator $A(D)^k$ entails k derivatives in the variables ξ, η and k derivatives in x, y . We claim that the remainder satisfies

$$\begin{aligned} (9.3.6) \quad c(x, \xi) - \sum_{k=0}^N \frac{i^k h^k}{k!} A(D)^k a(x, \xi) b(y, \eta)|_{x=\xi, y=\eta} \\ \in h^{N+1} S(\langle \xi \rangle^{m_1+m_2-N-1}). \end{aligned}$$

Since N is arbitrary, it will follow from (9.3.6) that

$$|\partial_x^\alpha \partial_\xi^\beta c(x, \xi)| \leq C_{\alpha, \beta} \langle \xi \rangle^{m_1 + m_2 - |\beta|},$$

uniformly for $0 < h \leq 1$; and consequently, $c \in S^{m_1 + m_2}$.

3. The left-hand side of (9.3.6) is a constant times

$$(9.3.7) \quad \int_0^1 (1-t)^N \exp(thA(D))(hA(D))^{N+1}(a(x, \xi)b(y, \eta))|_{x=y, \xi=\eta} dt.$$

As $a \in S^{m_1}$ and $b \in S^{m_2}$, we see that

$$(hA(D))^{N+1}(a(x, \xi)b(y, \eta)) \in h^{N+1} \sum_{k=0}^{N+1} S(\langle \xi \rangle^{m_1-k} \langle \eta \rangle^{m_2-N-1+k}).$$

Theorem 4.17 shows that

$$\exp(thA(D)) : S(\langle \xi \rangle^{m_1-k} \langle \eta \rangle^{m_2-N-1+k}) \rightarrow S(\langle \xi \rangle^{m_1-k} \langle \eta \rangle^{m_2-N-1+k}).$$

Observing that

$$S(\mathbb{R}^{4n}, \langle \xi \rangle^{m_1-k} \langle \eta \rangle^{m_2-N-1+k})|_{x=y, \xi=\eta} = S(\mathbb{R}^{2n}, \langle \xi \rangle^{m_1+m_2-N-1}),$$

we use the foregoing to estimate (9.3.7) and thereby establish the claim (9.3.6). $\square$

Let us also record the following useful result:

THEOREM 9.6 (Schwartz kernels of operators in Ψ^m).

(i) Suppose that $a \in S^m$ and that $K_a \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^n)$ is the Schwartz kernel of $a^w(x, hD)$. Then

$$(9.3.8) \quad K_a(x, y) \in C^\infty(\mathbb{R}^n \times \mathbb{R}^n \setminus \Delta)$$

for the diagonal $\Delta := \{(x, x) \mid x \in \mathbb{R}^n\}$.

(ii) Furthermore, we have the estimates

$$(9.3.9) \quad |(hD_x)^\alpha (hD_y)^\beta K_a(x, y)| \leq C_{N\alpha\beta} \left(\frac{h}{|x-y|} \right)^N$$

for $(x, y) \in \mathbb{R}^n \times \mathbb{R}^n \setminus \Delta$ and $N > |\alpha| + |\beta| + m + n$.

(iii) If K satisfies

$$(9.3.10) \quad |\partial_x^\alpha \partial_y^\beta K(x, y)| \leq C_{N\alpha\beta} \left(\frac{h}{\langle x-y \rangle} \right)^N$$

for all N , α , β , and $(x, y) \in \mathbb{R}^n \times \mathbb{R}^n$, then K is the Schwartz kernel of a^w for some symbol $a \in h^\infty S^{-\infty}$.

This illustrates one of the many advantages of symbol classes S^m , since we now have smoothness and rapid decay away from the diagonal.

Proof. 1. We can consider either the Weyl quantization or the standard quantization. This follows from Theorem 4.13, since

$$(9.3.11) \quad \exp(i(t-s)h\langle D_x, D_\xi \rangle) : S^m \rightarrow S^m,$$

as in the proof of Theorem 9.5. So for simplicity of notation, we take $a(x, hD)$.

2. First suppose that $a \in \mathcal{S}$. Then the kernel of $a(x, hD)$ is

$$K_a(x, y) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} a(x, \xi) e^{i\langle x-y, \xi \rangle/h} d\xi.$$

Since

$$(9.3.12) \quad (x-y)^\gamma K_a(x, y) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} (-hD_\xi)^\gamma a(x, \xi) e^{i\langle x-y, \xi \rangle/h} d\xi,$$

we have

$$|(x-y)^\gamma K_a(x, y)| \leq C_\gamma h^{|\gamma|} \|\langle \xi \rangle^{n+1} \partial_\xi^\gamma a\|_{L^\infty} \int_{\mathbb{R}^n} \langle \xi \rangle^{-n-1} d\xi.$$

Next observe that

$$\sup_{|\gamma|=N} |(x-y)^\gamma| \geq n^{-\frac{N}{2}} |x-y|^N,$$

and therefore

$$(9.3.13) \quad |K_a(x, y)| \leq C_N \left(\frac{h}{|x-y|} \right)^N \sup_{|\gamma|=N} \|\langle \xi \rangle^{n+1} \partial_\xi^\gamma a\|_{L^\infty}.$$

Going back to (9.3.12) gives a similar estimate for derivatives:

$$\begin{aligned} |(hD_x)^\alpha (hD_y)^\beta ((x-y)^\gamma K_a(x, y))| \\ \leq C_{N, \alpha \beta} h^N \sup_{|\rho| \leq |\alpha|, |\gamma|=N} \|\langle \xi \rangle^{n+1+|\alpha|+|\beta|} \partial_\xi^\gamma \partial_x^\rho a\|_{L^\infty}. \end{aligned}$$

Since

$$\begin{aligned} (hD_x)^\alpha (hD_y)^\beta ((x-y)^\gamma K_a(x, y)) &= (x-y)^\gamma (hD_x)^\alpha (hD_y)^\beta K_a(x, y) \\ &+ \sum_{\substack{|\alpha'|+|\beta'| \leq |\alpha|+|\beta| \\ |\gamma'| \leq |\gamma|}} c_{\alpha' \beta' \gamma'} (x-y)^{\gamma'} (hD_x)^{\alpha'} (hD_y)^{\beta'} K_a(x, y), \end{aligned}$$

induction on $|\alpha| + |\beta|$ shows that

$$\begin{aligned} (9.3.14) \quad |(hD_x)^\alpha (hD_y)^\beta K_a(x, y)| \\ \leq C_{N, \alpha \beta} \left(\frac{h}{|x-y|} \right)^N \sup_{|\rho| \leq |\alpha|, |\gamma|=N} \|\langle \xi \rangle^{n+1+|\alpha|+|\beta|} \partial_x^\rho \partial_\xi^\gamma a\|_{L^\infty}. \end{aligned}$$

If $a \in S^m$, we observe that seminorms appearing on the right-hand side of (9.3.14) are finite and bounded if $N > |\alpha| + |\beta| + m + n$. Approximation by symbols in $\mathcal{S}$ concludes the proof of (9.3.9).

3. Assertion (iii) follows from the inverse Fourier transform. Since

$$|\partial_x^\alpha \partial_z^\beta K_a(x, x - z)| \leq C_N \frac{h^N}{\langle z \rangle^N},$$

for all N and

$$a(x, \xi) = \int_{\mathbb{R}^n} K_a(x, x - z) e^{i\langle z, \xi \rangle / h} dz,$$

we have $a \in h^\infty S^{-\infty}$. □

As an immediate consequence we obtain the following

THEOREM 9.7 (More on disjoint support). *Let $b \in S^m$ and suppose $\varphi, \psi \in C_c^\infty(\mathbb{R}^n)$. If*

$$(9.3.15) \quad \text{spt}(\varphi) \cap \text{spt}(\psi) = \emptyset,$$

then

$$(9.3.16) \quad \|\varphi b^w(x, hD) \psi\|_{H_h^{-N}(\mathbb{R}^n) \rightarrow H_h^N(\mathbb{R}^n)} = O(h^\infty)$$

for all N .

9.3.3. Changing variables for S^m . Before restating Theorem 9.3 in this more general setting, we will discuss the usual quantization acting on functions.

LEMMA 9.8 (Changing variables and exponentials). *Suppose that $a \in \mathcal{S}$. Then*

$$(9.3.17) \quad (\gamma^{-1})^* a(x, hD) \gamma^* = a_\gamma(x, hD),$$

where

$$(9.3.18) \quad a_\gamma(\gamma(x), \eta) = e^{-\frac{i}{h}\langle \gamma(x), \eta \rangle} a(x, hD) e^{\frac{i}{h}\langle \gamma(\cdot), \eta \rangle}.$$

Proof. Theorem 4.19 shows that

$$a_\gamma(y, \eta) = e^{-\frac{i}{h}\langle y, \eta \rangle} a_\gamma(x, hD) e^{\frac{i}{h}\langle y, \eta \rangle} = e^{-\frac{i}{h}\langle y, \eta \rangle} (\gamma^{-1})^* a(x, hD) \gamma^* (e^{-\frac{i}{h}\langle \cdot, \eta \rangle}),$$

and this gives (9.3.18). □

THEOREM 9.9 (Changing variables. I). *Suppose that $a \in S^m$.*

(i) *Then the formula (9.3.17) defines a symbol $a_\gamma \in S^m$ for which (9.3.18) holds.*

(ii) *Moreover,*

$$(9.3.19) \quad a_\gamma(\gamma(x), \eta) = \sum_{|\alpha| \leq N} \frac{1}{\alpha!} \partial_\xi^\alpha a(x, \partial\gamma(x)^T \eta) (hD_y)^\alpha e^{\frac{i}{h} \langle \rho_x(y), \eta \rangle} \Big|_{y=x} + O_{S^{m-N-1}}(h^{N+1}),$$

where

$$\rho_x(y) = \gamma(y) - \gamma(x) - \partial\gamma(x)(y - x).$$

(iii) *In particular,*

$$(9.3.20) \quad a_\gamma(\gamma(x), \eta) = a(x, \partial\gamma(x)^T \eta) + O_{S^{m-1}}(h).$$

Proof. 1. We need to show that a_γ defined by (9.3.18) is in S^m . This will imply

$$(\gamma^{-1})^* a(x, hD) \gamma^* u = a_\gamma(x, hD) u$$

for $u(x) = \exp(i\langle x, \eta \rangle/h)$. The operator identity (9.3.17) will follow, since such functions u are dense in $\mathcal{S}'$.

2. Select a function $\chi \in C_c^\infty(\mathbb{R}, [0, 1])$ satisfying $\chi \equiv 1$ on $[-1, 1]$ and $\chi \equiv 0$ on $\mathbb{R} \setminus [-2, 2]$. We assert that

$$(9.3.21) \quad a_\gamma(\gamma(x), \eta) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(x, \xi) \chi_x(y) \chi_\eta(\xi) e^{i(\langle x-y, \xi \rangle + \langle \gamma(y) - \gamma(x), \eta \rangle)/h} dy d\xi + O_S(\langle \eta \rangle^{-\infty} h^\infty)$$

where

$$\chi_x(y) := \chi(x - y), \quad \chi_\eta(\xi) := \chi((\xi - \partial\gamma(x)^T \eta)/\langle \eta \rangle).$$

To prove this, observe that on the support of $1 - \chi_x(y) \chi_\eta(\xi)$ the phase is not stationary, since

$$d_{y, \xi}(\langle x - y, \xi \rangle + \langle \gamma(y) - \gamma(x), \eta \rangle) = 0$$

if and only if

$$x = y \quad \text{and} \quad \xi = \partial\gamma(x)^T \eta.$$

Consequently a standard integration-by-parts argument, which we leave to the reader, gives (9.3.21).

3. We rewrite the integral term in (9.3.21) as

$$\frac{1}{(2\pi\tilde{h})^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a_\eta(x, \tilde{\xi}) \chi_x(y) \tilde{\chi}_\eta(\tilde{\xi}) e^{i(\langle x-y, \tilde{\xi} \rangle + \langle \gamma(y) - \gamma(x), \eta / \langle \eta \rangle \rangle) / \tilde{h}} dy d\tilde{\xi}$$

for

$$a_\eta(x, \tilde{\xi}) := a(x, \langle \eta \rangle \tilde{\xi}), \quad \tilde{\chi}_\eta(\tilde{\xi}) := \chi_\eta(\tilde{\xi} \langle \eta \rangle), \quad \tilde{h} := h / \langle \eta \rangle.$$

The support of the integrand is contained in a fixed compact set, and

$$\langle \eta \rangle^{-m} a_\eta \chi_\eta \in S,$$

uniformly in h and η . Hence the stationary phase method (Theorem 3.16) yields an expansion in powers of $\tilde{h} = h / \langle \eta \rangle$. Computing the leading term, we easily check that (9.3.20) holds.

4. Since we will not use (9.3.19), we refer to Hörmander [H2, Theorem 18.1.17] for the proof. $\square$

The proof given in Section 9.2 can be adapted to the present setting using the integration by parts arguments from the proof of Theorem 9.6. This gives

THEOREM 9.10 (Changing variables. II). *Let $a \in S^m$.*

(i) *Then*

$$(9.3.22) \quad (\gamma^{-1})^* a^w(x, hD) \gamma^* = \tilde{a}^w(x, hD),$$

acting on half-densities, for

$$(9.3.23) \quad \tilde{a}(\gamma(x), \xi) := a(x, \partial\gamma(x)^T \xi) + O_{S^{m-2}}(h^2).$$

(ii) *Also,*

$$(\gamma^{-1})^* a^w(x, hD) \gamma^* = \tilde{a}_1^w(x, hD),$$

acting on functions, for

$$(9.3.24) \quad \tilde{a}_1(\gamma(x), \xi) := a(x, \partial\gamma(x)^T \xi) + O_{S^{m-1}}(h).$$

The invariance presented in the two theorems above leads to a definition of a symbol of a pseudodifferential operator:

DEFINITION. We define the *principal symbol* of the pseudodifferential operator $A = a^w(x, hD) \in \Psi^m$ to be the *equivalence class*:

$$(9.3.25) \quad \sigma(A) = \sigma_m(A) := a(x, \xi) \bmod (hS^{m-1}).$$

Theorem 9.9 shows that this is defined independently of the choice of coordinates, and Theorem 9.10 demonstrates that the principal symbol does not depend upon the particular quantization nor upon whether we act on functions or on half-densities. For the Weyl quantization and for action

on half-densities, the same theorem shows that the symbol can be defined $\text{mod}(h^2 S^{m-2})$.

9.3.4. Sharp Gårding inequality again. Theorem 4.32 provided a lower bound for quantizations of nonnegative symbols in the class S . A stronger inequality is valid for nonnegative symbols in S^m . The proof below is an instructive illustration of the interplay between the semiclassical and Kohn–Nirenberg (h -free) quantizations. It uses the Littlewood–Paley decomposition introduced in Section 7.5.1. The semiclassical Sobolev spaces appearing in the statement of the theorem were defined in (8.3.5).

THEOREM 9.11 (Sharp Gårding inequality for Ψ^m). *Suppose that $a \in S^m$ and that $a \geq 0$. Then, for some $C \geq 0$,*

$$(9.3.26) \quad \langle a^w(x, hD)u, u \rangle \geq -h \|u\|_{H_h^{\frac{m-1}{2}}}^2, \quad u \in \mathcal{S}.$$

Proof. 1. We recall from Lemma 7.14 that we can find χ_0 and χ such that

$$1 = \sum_{j=0}^{\infty} \chi_j(\xi), \quad \chi_0 \in C_c^\infty(\mathbb{R}^n, [0, 1]),$$

$$\chi_j(\xi) = \chi(2^{-j}\xi), \quad j \geq 1, \quad \chi \in C_c^\infty(\mathbb{R}^n \setminus \{0\}, [0, 1]).$$

Suppose $\tilde{\chi}_0 \in C_c^\infty(\mathbb{R}^n, [0, 1])$ and $\tilde{\chi} \in C_c^\infty(\mathbb{R}^n \setminus \{0\}, [0, 1])$ satisfy

$$\tilde{\chi}_0 = 1 \text{ on } \text{spt } \chi_0, \quad \tilde{\chi} = 1 \text{ on } \text{spt } \chi.$$

If $\tilde{\chi}_j(\xi) := \tilde{\chi}(2^{-j}\xi)$, $j \geq 1$, we claim that there exists C such that for $u \in \mathcal{S}$

$$(9.3.27) \quad \|u\|_{H_h^s}^2 / C \leq \sum_{j=0}^{\infty} 2^{2js} \|\tilde{\chi}_j(hD)u\|_{L^2}^2 \leq C \|u\|_{H_h^s}^2, \quad s \in \mathbb{R}.$$

To establish the claim, we recall from (8.3.5) that

$$(9.3.28) \quad \|u\|_{H_h^s}^2 = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \langle \xi \rangle^{2s} |\mathcal{F}_h u(\xi)|^2 d\xi.$$

On the other hand,

$$(9.3.29) \quad \sum_{j=0}^{\infty} 2^{2js} \|\tilde{\chi}_j(hD)u\|_{L^2}^2 = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \sum_{j=0}^{\infty} 2^{2js} |\tilde{\chi}_j(\xi)|^2 |\mathcal{F}_h u|^2 d\xi.$$

We have

$$\langle \xi \rangle^{2s} / C_0 \leq 2^{2js} \leq C_0 \langle \xi \rangle^{2s}, \quad \xi \in \text{spt } \tilde{\chi}_j.$$

Also, for any ξ , $\tilde{\chi}_j(\xi) \neq 0$ for at most J values of j , and it is equal to 1 for at least one j . Hence,

$$\langle \xi \rangle^{2s} / C_0 \leq \sum_{j=0}^{\infty} 2^{2js} |\tilde{\chi}_j(\xi)|^2 \leq JC_0 \langle \xi \rangle^{2s},$$

and (9.3.27) follows from (9.3.28) and (9.3.29).

2. Let us first assume that $m = 0$. We write

$$(9.3.30) \quad a(x, \xi) = \sum_{j=0}^{\infty} \chi_j(\xi) a(x, \xi) = \sum_{j=0}^{\infty} a_j(x, 2^{-j}\xi),$$

where $a_0(x, \xi) = \chi_0(\xi) a(x, \xi)$, $j = 0$, $a_j(x, \xi) = \chi(\xi) a(x, 2^j \xi)$ for $j \geq 1$. Since $\partial_x^\alpha \partial_\xi^\beta a = O(\langle \xi \rangle^{-|\beta|})$ and $0 \notin \text{spt } \chi$, we have, for $j \geq 1$,

$$|\partial_x^\alpha \partial_\xi^\beta a_j| \leq C_{\alpha\beta} 2^{j|\beta|} \max_{\xi \in \text{spt } \chi} \langle 2^j \xi \rangle^{-|\beta|} \leq \tilde{C}_{\alpha\beta},$$

and the estimate is obvious for $j = 0$. This means that $a_j \in S$ with semi-norms bounded independently of j .

3. Returning to (9.3.30), we see that

$$(9.3.31) \quad a^w(x, hD) = \sum_{j=0}^{\infty} a_j^w(x, h_j D), \quad h_j := 2^{-j}h.$$

The composition formula in Theorem 4.11 and the fact that $\tilde{\chi} = 1$ on the support of a_j show, for $j \geq 1$, that

$$(9.3.32) \quad \begin{aligned} \tilde{\chi}_j(hD) a_j^w(x, h_j D) \tilde{\chi}_j(hD) &= \tilde{\chi}(h_j D) a_j^w(x, h_j D) \tilde{\chi}(h_j D) \\ &= a_j^w(x, h_j D) + O_{\mathcal{S}' \rightarrow \mathcal{S}}(h_j^\infty), \end{aligned}$$

with the same statement valid for $j = 0$.

The sharp Gårding inequality of Theorem 4.32 gives

$$\langle a_j^w(x, h_j D) v, v \rangle \geq -Ch_j \|v\|_{L^2}^2, \quad v \in \mathcal{S},$$

where C is independent of j .

Combining this with (9.3.31) and (9.3.32), we have for $u \in \mathcal{S}$

$$\begin{aligned} \langle a^w u, u \rangle &= \sum_{j=0}^{\infty} \left(\langle a_j^w(x, h_j D) \tilde{\chi}_j(hD) u, \tilde{\chi}_j(hD) u \rangle - O(h^N 2^{-jN}) \|u\|_{H_h^{-\frac{1}{2}}} \right) \\ &\geq -Ch \sum_{j=0}^{\infty} 2^{-j} \|\tilde{\chi}_j(hD) u\|_{L^2}^2 - O(h^N) \|u\|_{H_h^{-\frac{1}{2}}}. \end{aligned}$$

From this and (9.3.27), we obtain (9.3.26) for $m = 0$.

4. To obtain the result for general m , we put $v = \langle hD \rangle^{m/2} u$, so that

$$\langle a^w(x, hD)u, u \rangle = \langle b^w(x, hD)v, v \rangle, \quad b := \langle \xi \rangle^{-m/2} \# a \# \langle \xi \rangle^{-m/2}.$$

We have

$$b \in S^0, \quad b(x, \xi) = \langle \xi \rangle^{-m/2} a(x, \xi) \langle \xi \rangle^{-m/2} + c(x, \xi), \quad c \in hS^{-1},$$

and hence, by the result for $m = 0$,

$$\begin{aligned} \langle a^w(x, hD)u, u \rangle &= \langle b^w(x, hD)v, v \rangle \\ &\geq -Ch \|v\|_{H_h^{-\frac{1}{2}}}^2 - Ch \langle c(x, hD)v, v \rangle \\ &\geq -C'h \|v\|_{H_h^{-\frac{1}{2}}}^2 \\ &= -C'h \|u\|_{H_h^{\frac{m-1}{2}}}^2. \end{aligned} \quad \square$$

REMARK. The inequality (9.3.26) is valid when the Weyl quantization is replaced by any other quantization, $\text{Op}_t(a)$, $0 \leq t$. It also remains valid for systems. For scalar symbols and for the Weyl quantization a stronger *Fefferman–Phong* inequality is valid:

$$\langle a^w(x, hD)u, u \rangle \geq -h^2 \|u\|_{H_h^{\frac{m-2}{2}}}^2, \quad u \in \mathcal{S}.$$

Consult [H3, Chapter 18.6] for more details and references.

9.3.5. Beals’s Theorem again.

THEOREM 9.12 (Beals’s Theorem for Ψ^m). *Let $A : \mathcal{S} \rightarrow \mathcal{S}'$ be a continuous linear operator. Then*

$$A \in \Psi^m$$

if and only if

$$\|\text{ad}_{x_{i_1}} \cdots \text{ad}_{x_{i_N}} \text{ad}_{D_{x_{j_1}}} \cdots \text{ad}_{D_{x_{j_M}}} A\|_{H_h^{l+m}(\mathbb{R}^n) \rightarrow H_h^{l+N}(\mathbb{R}^n)} = O(h^N)$$

for some l and all $N, M = 0, 1, 2, \dots$ and all sequences $i_k, j_l \in \{1, \dots, n\}$, $1 \leq k \leq N, 1 \leq l \leq M$.

See Grigis and Sjöstrand [G-S] for the modifications needed in the proof of Theorem 8.3.

9.4. NOTES

For more on invariance, half-densities, and a functorial point of view towards semiclassical analysis see Guillemin–Sternberg [**G-St1**]. J. Strain suggested the numerical interpretation of half-densities. The proof of symbol invariance is from the appendix to [**S-Z1**].

For $h = 1$, Ψ^m is the class of Kohn–Nirenberg pseudodifferential operators; see Hörmander [**H2**, Section 18.1] or Grigis–Sjöstrand [**G-S**] for a thorough presentation. This calculus of pseudodifferential operators goes back to the works of Bokobza–Unterberger, Kohn–Nirenberg, and Hörmander in the 1960s.

FOURIER INTEGRAL OPERATORS

- 10.1 Operator dynamics
- 10.2 An integral representation formula
- 10.3 Strichartz estimates
- 10.4 L^p estimates for quasimodes
- 10.5 Notes

Let m be an order function and suppose that $\{p_t\}_{t \in \mathbb{R}} \subset S(m)$ is a family of real-valued symbols that depend smoothly on $t \in \mathbb{R}$. Set

$$P(t) := p_t^w(x, hD).$$

We investigate in this chapter and the next the corresponding quantum operator dynamics:

$$(10.0.1) \quad \begin{cases} hD_t F(t) + P(t)F(t) = 0 & (t \in \mathbb{R}) \\ F(0) = I. \end{cases}$$

This chapter mostly concerns explicit representations of approximate solutions of (10.0.1) valid for small times $|t| < t_0$, whereas Chapter 11 addresses global properties of solutions for large times.

10.1. OPERATOR DYNAMICS

We start by identifying several circumstances under which the quantum dynamics (10.0.1) are uniquely solved by a family $\{F(t)\}_{t \in \mathbb{R}}$ of bounded, unitary operators on $L^2(\mathbb{R}^n)$.

10.1.1. Symbols in S . We first consider the case $m \equiv 1$:

THEOREM 10.1 (Quantum dynamics in S). (i) *If $\{p_t\}_{t \in \mathbb{R}} \subset S$ is a family of real-valued symbols that depend smoothly on t , then the operator equation (10.0.1) is uniquely solved by a family $\{F(t)\}_{t \in \mathbb{R}}$ of bounded, unitary operators on $L^2(\mathbb{R}^n)$.*

(ii) *Furthermore, for any order function m ,*

$$(10.1.1) \quad F(t) : H_h(m) \rightarrow H_h(m)$$

is a bounded linear operator, for each $t \in \mathbb{R}$.

Proof. 1. Since $p_t \in S$, Theorem 4.23 implies that the operator $P(t) = p_t^w(x, hD)$ is bounded from L^2 to L^2 , with

$$P(t)^* = P(t).$$

As $P(t)$ depends smoothly on t as an operator, equation (11.1.3) consequently has a unique solution. Furthermore,

$$(10.1.2) \quad \begin{cases} hD_t F(t)^* - F(t)^* P(t) = 0 \\ F(0)^* = I. \end{cases}$$

2. We assert that $F(t)$ is unitary on L^2 . To confirm this, using (10.0.1) and (10.1.2), let us calculate that

$$\begin{aligned} hD_t(F(t)^* F(t)) &= hD_t F(t)^* F(t) + F(t)^* hD_t F(t) \\ &= F(t)^* P(t) F(t) - F(t)^* P(t) F(t) = 0. \end{aligned}$$

Since $F(0)^* F(0)^* = I$, it follows that $F(t)^* F(t) \equiv I$. On the other hand,

$$\begin{aligned} hD_t(F(t) F(t)^* - I) &= -P(t) F(t) F(t)^* + F(t) P(t) F(t)^* \\ &= [F(t)^* F(t) - I, P(t)] \end{aligned}$$

with $F(0) F(0)^* - I = 0$. Since this equation for $F(t) F(t)^* - I$ is homogeneous, it follows that $F(t) F(t)^* \equiv I$.

3. To prove (10.1.1), we can assume according to Lemma 8.7 that $m \in S(m)$. We then define $g := \log m$, write $G := g^w(x, hD)$, and introduce the operator

$$F_G(t) := e^G F(t) e^{-G}.$$

Then

$$F_G(t) : H_h(1/m) \rightarrow H_h(1/m),$$

and in particular $F_G(t) : \mathcal{S} \rightarrow \mathcal{S}'$; see Theorem 8.9.

Now consider the equation satisfied by $F_G(t)$:

$$(10.1.3) \quad hD_t F_G(t) = -e^G P(t) F(t) e^{-G} = -P_G(t) F_G(t),$$

for $P_G(t) := e^G P(t) e^{-G}$. Theorem 8.6 implies

$$e^{\pm G} = c_{\pm}^w(x, hD)$$

for symbols $c_{\pm} \in S(m^{\pm 1})$; and hence $P_G(t) = b_t(x, hD)$ for a symbol $b_t \in S$, and $P_G : \mathcal{S}' \rightarrow \mathcal{S}'$. This means that (10.1.3) makes sense as an equation for $F_G(t) : \mathcal{S} \rightarrow \mathcal{S}'$.

Since $P_G(t)$ is bounded on L^2 and $F_G(0) = I$, the solution to equation (10.1.3) extends to a bounded operator on L^2 . Hence

$$\|F(t)\|_{H_h(m) \rightarrow H_h(m)} = \|F_G(t)\|_{L^2 \rightarrow L^2} \leq \exp \int_0^t \|P_G(s)\|_{L^2 \rightarrow L^2} ds < \infty. \quad \square$$

REMARKS. (i) According to (8.3.11), we can use the L^2 inner product to identify the dual space of $H_h(m)$ with $H_h(1/m)$. Hence (10.1.1) implies that

$$(10.1.4) \quad F(t)^{-1} = F(t)^* : H_h(m) \rightarrow H_h(m)$$

for all order functions m .

(ii) We recall from (8.3.12) that

$$\mathcal{S} = \bigcap_m H_h(m).$$

Then (10.1.1) and (10.1.4) imply

$$(10.1.5) \quad F(t), F(t)^{-1} : \mathcal{S} \rightarrow \mathcal{S}. \quad \square$$

10.1.2. Time-independent, elliptic symbols. Assume next that $m \geq 1$ is an order function and that the real, time-independent symbol p belongs to $S(m)$. We next establish under an ellipticity condition the selfadjointness of $P = p^w(x, hD)$ for small h and hence the solvability of (10.0.1) for all times.

THEOREM 10.2 (Ellipticity and selfadjointness). *Assume that m is an order function satisfying $m \geq 1$.*

Let $P = p^w(x, hD)$ for a real-valued symbol $p \in S(m)$, and assume p is elliptic in the sense that

$$C + p \geq cm$$

for constants $C \geq 0, c > 0$.

(i) *For h small enough, the operator P with the domain $H_h(m)$ is essentially selfadjoint on $L^2(\mathbb{R}^n)$.*

(ii) *Consequently,*

$$(10.1.6) \quad F(t) := e^{-itP/h} \quad (t \in \mathbb{R})$$

is a family of unitary operators on $L^2(\mathbb{R}^n)$ solving (10.0.1).

DEFINITION. We call $\{e^{-itP/h}\}_{t \in \mathbb{R}}$ the *propagators* generated by P .

Proof. 1. The operator P with domain $\mathcal{S}$ is symmetric, since $\langle Pu, v \rangle = \langle u, Pv \rangle$ for all $u, v \in \mathcal{S}$.

We claim that for h small the inverses

$$(10.1.7) \quad (P \pm i)^{-1} : L^2 \rightarrow H_h(m)$$

exist and are surjective. It then follows from Theorem C.12 that P with domain $H_h(m)$ is essentially selfadjoint on L^2 .

2. We have $(p \pm i)^{-1} \in S(1/m)$ and hence

$$(P \pm i)((p \pm i)^{-1})^w = (p \pm i)^w((p \pm i)^{-1})^w = I + h^2(r_{\pm}^R)^w,$$

for $r_{\pm}^R \in S$. Also

$$((p \pm i)^{-1})^w(P \pm i) = ((p \pm i)^{-1})^w(p \pm i)^w = I + h^2(r_{\pm}^L)^w$$

for $r_{\pm}^L \in S$. Since $(r_{\pm}^R)^w$ and $(r_{\pm}^L)^w$ are bounded on L^2 , uniformly with respect to h , Theorem C.3 implies $(P \pm i)^{-1}$ exist when h is small enough.

Furthermore,

$$\exp(g^w(x, hD))((p \pm i)^{-1})^w = a^w(x, hD)$$

for a symbol $a \in S$, and thus a^w is bounded on L^2 . Consequently

$$((p \pm i)^{-1})^w = O_{L^2 \rightarrow H_h(m)}(1).$$

Combining these two statements, we see that the operators $P \pm i : H_h(m) \rightarrow L^2$ have two-sided inverses, and consequently (10.1.7) holds.

3. The second assertion follows from Stone's Theorem (Theorem C.13).

□

10.1.3. Time-dependent elliptic symbols. For completeness we now consider the case of time-dependent elliptic symbols. In what follows we will never use this particular result, but it ties nicely into the previous two cases.

We assume that an order function satisfies

$$(10.1.8) \quad m \geq 1$$

and that a family of symbols, $p_t \in S(m)$, satisfies symbolic estimates uniformly in t :

$$(10.1.9) \quad |\partial_t^k \partial_{(x,\xi)}^\alpha p_t(x, \xi)| \leq C_{k\alpha} m(x, \xi),$$

with $C_{k\alpha}$ independent of time.

THEOREM 10.3. *Suppose that $P(t) := p_t(x, hD)$ where $\{p_t\}_{t \in \mathbb{R}} \subset S(m)$ satisfies (10.1.8) and (10.1.9). Assume p_t is uniformly elliptic:*

$$p_t \geq m/C - C,$$

for a time-independent constant C .

Then for h small enough, the operator equation (10.0.1) is uniquely solved by a family of invertible operators $\{F(t)\}_{t \in \mathbb{R}}$,

$$(10.1.10) \quad F(t), F(t)^{-1} \in C_{\text{loc}}(\mathbb{R}_t; L(H_h(m^k), H_h(m^k))),$$

for any $k \in \mathbb{Z}$.

In addition, the operator $F(t) : L^2 \rightarrow L^2$ is unitary.

Proof. 1. We will construct $F(t)$ by solving the Cauchy problem

$$(10.1.11) \quad \begin{aligned} hD_t u(t) + P(t)u(t) &= 0, & u(0) &= u_0 \in H_h(m^k), \\ u &\in C_{\text{loc}}([0, \infty); H_h(m^k)), \end{aligned}$$

and putting $F(t)u_0 := u(t)$.

The equation $(hD_t) + P(t) + \lambda w(t) = 0$, $w(0) = u_0$, is solved by $w(t) = e^{-i\lambda t/h} u(t)$. Since $m \geq 1$, $p_t + \lambda \in S(m)$ for any fixed λ , and we can replace $p_t + \lambda$.

Hence we can assume that p_t satisfies

$$p_t \geq m/C, \quad p_t \in S(m).$$

Once this holds and h is small enough,

$$P(t) : H_h(m^k) \rightarrow H_h(m^{k-1})$$

is invertible with the inverse

$$P(t)^{-1} : H_h(m^{k-1}) \rightarrow H_h(m^k).$$

It follows that as the norm on $H_h(m^k)$, $k \in \mathbb{Z}$, we can use $\|P(t)^k u\|_{L^2}$:

$$(10.1.12) \quad \|P(t)^k u\|_{L^2}/C_k \leq \|u\|_{H_h(m)} \leq C_k \|P(t)^k u\|_{L^2},$$

with constants independent of t .

2. Suppose that $v \in C_c^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n))$ and $v(t) := v(t, \cdot) \in \mathcal{S}(\mathbb{R}^n)$. Then

$$\begin{aligned} -\partial_t \|P(t)^{-k} v(t)\|_{L^2(\mathbb{R}^n)}^2 &= -\frac{2}{h} \operatorname{Re} \langle i h D_t P(t)^{-k} v(t), P(t)^{-k} v(t) \rangle_{L^2(\mathbb{R}^n)} \\ &= \frac{2}{h} \operatorname{Im} \langle (h D_t + P(t)) P(t)^{-k} v(t), P(t)^{-k} v(t) \rangle_{L^2(\mathbb{R}^n)}, \end{aligned}$$

since the reality of p_t implies that $\langle P(t)w, w \rangle_{L^2(\mathbb{R}^n)}$ is real for $w \in \mathcal{S}$.

As $P'(t) = (\partial_t p_t)^w(x, hD)$ and $\partial_t p_t \in S(m)$, the composition formula (4.4.16) and invertibility of $P(t)$ give

$$h D_t P(t)^{-k} = P(t)^{-k} h D_t + h a_k(t)^w(x, hD) P(t)^{-k},$$

where $a_k(t) \in S$ uniformly in t .

Hence, by Theorem 4.23,

$$h D_t P(t)^{-k} v(t) = P(t)^{-k} h D_t v(t) + h A_k(t) P(t)^{-k} v(t),$$

where $\|A_k\|_{L^2 \rightarrow L^2} \leq \Lambda_k$.

This gives us the bound

$$\begin{aligned} -\partial_t \|P(t)^{-k} v(t)\|_{L^2(\mathbb{R}^n)} &\leq \frac{1}{h} \|P(t)^{-k} (h D_t + P(t)) v(t)\|_{L^2(\mathbb{R}^n)} \\ &\quad + \Lambda_k \|P(t)^{-k} v(t)\|_{L^2(\mathbb{R}^n)}, \end{aligned}$$

or

$$-\partial_t \left(e^{\Lambda_k t} \|P(t)^{-k} v(t)\|_{L^2(\mathbb{R}^n)} \right) \leq \frac{e^{\Lambda_k t}}{h} \|P(t)^{-k} (h D_t + P(t)) v(t)\|_{L^2(\mathbb{R}^n)}.$$

Owing to the equivalence of norms (10.1.12), we now get

$$(10.1.13) \quad \|v(0)\|_{H_h(m^{-k})} \leq \frac{C_k^2}{h} \int_0^\infty e^{\Lambda_k t} \|(h D_t + P(t)) v(t)\|_{H_h(m^{-k})} dt, \\ v \in C_c^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n)).$$

3. The equation (10.1.11) is satisfied in the distributional sense if for v in (10.1.13),

$$(10.1.14) \quad \frac{1}{h} \int_0^\infty \langle u(t), (h D_t + P(t)) v(t) \rangle_{L^2(\mathbb{R}^n)} dt = i \langle u_0, v(0) \rangle_{L^2(\mathbb{R}^n)}.$$

From (10.1.13)

$$|\langle u_0, v(0) \rangle_{L^2(\mathbb{R}^n)}| \leq \frac{C_k^2}{h} \|u_0\|_{H_h(m^k)} \|(h D_t + P(t)) v\|_{L^1([0, \infty); H_h(m^{-k}), e^{\Lambda_k t} dt)}.$$

The Hahn–Banach Theorem then shows that

$$\frac{1}{h}(hD_t + P(t))v \mapsto \langle u_0, v(0) \rangle_{L^2(\mathbb{R}^n)}$$

extends to a linear functional

$$L^1([0, \infty); H_h(m^{-k}), e^{\Lambda_k t} dt) \rightarrow \mathbb{C},$$

bounded by $\|u_0\|_{H_h(m^k)}$. Hence there exists an element of the dual space, $u \in e^{\Lambda_k t} L^\infty([0, \infty); H_h(m^k))$, for which (10.1.14) holds. This proves the existence of u satisfying the equation in (10.1.11), with the map $u_0 \mapsto u$ continuous from $H_h(m^k) \rightarrow e^{\Lambda_k t} L^\infty([0, \infty); H_h(m^k))$.

4. We now need to improve the statement that $u \in L^\infty_{\text{loc}}([0, \infty); H_h(m^k))$ to the $u \in C_{\text{loc}}([0, \infty); H_h(m^k))$.

For that we note that if, without any uniformity with respect to h ,

$$w \in C([0, T], H_h(m^{k+1})) \quad \text{and} \quad \partial_t w \in C([0, T], H_h(m^k)),$$

then the argument presented in part 2 of the proof (applied to $\partial_t \|P(t)^k w(t)\|_{L^2}$) shows that

$$(10.1.15) \quad \sup_{0 \leq t \leq T} \|w(t)\|_{H_h(m^k)} \leq C_{k,T} \|w(0)\|_{H_h(m^k)} + \frac{C_{k,T}}{h} \int_0^T \|(hD_t + P(t))w(t)\|_{H_h(m^k)} dt.$$

If

$$(10.1.16) \quad u_0 \in \mathcal{S} \subset \bigcap_k H_h(m^k),$$

then the solution to (10.1.11) satisfies $u \in L^\infty_{\text{loc}}([0, \infty), H_h(m^\ell))$ for any ℓ . Hence, replacing k by $k+2$, we see that

$$hD_t u(t) = -P(t)u(t) \in L^\infty_{\text{loc}}([0, \infty), H_h(m^{k+1}))$$

so that

$$u(t) \in C_{\text{loc}}([0, \infty), H_h(m^{k+1})).$$

But then

$$hD_t u(t) = -P(t)u(t) \in C_{\text{loc}}([0, \infty), H_h(m^k)),$$

and we conclude that

$$u \in C^1_{\text{loc}}([0, \infty), H_h(m^k)) \cap C_{\text{loc}}([0, \infty), H_h(m^{k+1})),$$

for any k .

We can now find $u_0^p \in \mathcal{S}$ such that

$$\|u_0^\ell - u_0\|_{H_h(m^k)} \rightarrow 0, \quad \ell \rightarrow \infty.$$

We can apply (10.1.15) to the solutions to $(hD_t + P(t))u^k = 0$, $u^k(0) = u_0^k$, to obtain

$$\sup_{0 \leq t \leq T} \|u^{\ell'}(t) - u^{\ell}(t)\|_{H_h(m^k)} \leq C_{k,T} \|u_0^{\ell'} - u_0^{\ell}\|_{H_h(m^k)}.$$

Thus the u^{ℓ} form a Cauchy sequence in $C([0, T], H_h(m^k))$ and, by continuity established in part 3 of the proof, $u^{\ell} \rightarrow u$ in $L^{\infty}([0, T], H_h(m^k))$. Hence $u \in C([0, T], H_h(m^k))$ for any T , completing the proof of the existence of a solution satisfying (10.1.11).

5. To establish the uniqueness of the solution to (10.1.11), we suppose that $u \in C_{\text{loc}}([0, \infty), H_h(m^k))$ solves (10.1.11) with $u_0 = 0$. The equation shows that $\partial_t u \in C_{\text{loc}}([0, \infty), H_h(m^{k-1}))$ and hence we can apply (10.1.15) with k replaced by $k-1$. This shows that $u = 0$.

6. For the invertibility of $F(t)$ we note that we have in fact uniquely solved a more general problem than (10.1.11):

$$(10.1.17) \quad \begin{aligned} hD_t u(t) + P(t)u(t) &= 0, & u(r) &= u_1 \in H_h(m^k), \\ u &\in C_{\text{loc}}(\mathbb{R}; H_h(m^k)). \end{aligned}$$

The initial time 0 played no special role, neither did the direction of time, and the continuity at 0 allowed the extension to $\mathbb{R}$.

We define $F(t, r)u_1 := u(t)$, so that $F(r, r) = I$, on $H_h(m^k)$. Then

$$F(t) = F(t, 0) \quad \text{and} \quad F(t)^{-1} = F(0, t).$$

This proves that $F(t)$ is invertible and that the inverse has the same properties as $F(t)$.

7. To show unitarity on L^2 , we only need to show that $F(t)^*F(t) = I$. For that, we first check that $\partial_t \|F(t)u_0\|_{L^2} = 0$ for u_0 satisfying (10.1.16). Then an approximation argument presented in part 4 of the proof shows that $\|F(t)u_0\|_{L^2} = \|u_0\|_{L^2}$ for $u_0 \in L^2$. $\square$

10.2. AN INTEGRAL REPRESENTATION FORMULA

Since the propagators $F(t) := e^{-itP/h}$ given by Theorem 10.2 commute with P , we have

$$\begin{cases} (hD_t + P)F(t)u = 0 & (t \in \mathbb{R}) \\ F(0)u = u \end{cases}$$

for each $u \in L^2$.

Given an order function m , we next consider a local, time-dependent version of these dynamics:

$$(10.2.1) \quad \begin{cases} (hD_t + P(t))F(t)u = 0 & (-t_0 < t < t_0) \\ F(0)u = u \end{cases}$$

where $P(t) = p_t^w(x, hD)$ for a family of symbols $\{p_t\}_{|t| < t_0} \subset S(m)$.

Our goal is constructing for small times $|t| < t_0$, a microlocal representation of operators approximately solving (10.2.1). More precisely, given a point (x_0, ξ_0) , we will construct fairly explicit oscillatory integral operators $U(t)$ so that

$$(10.2.2) \quad \begin{cases} (hD_t + P(t))U(t)u = O_{L^2}(h^\infty) & (-t_0 < t < t_0) \\ U(0)u = u \end{cases}$$

for appropriately localized families of functions $u = \{u(h)\}_{0 < h \leq h_0}$.

10.2.1. A microlocal representation. To obtain the solution to (10.2.2) for $u \in L^2$, microlocalized near (x_0, ξ_0) we will construct an asymptotic oscillatory solutions, known as the WKB approximation (named after Wentzel, Kramers, and Brillouin):

$$(10.2.3) \quad \begin{cases} (hD_t + P(t)) \left(b(t, x, \eta) e^{\frac{i}{h}\varphi(t, x, \eta)} \right) = O(h^\infty) \\ b(0, x, \eta) = 1 \text{ near } (x_0, \xi_0), \quad \varphi(0, x, \eta) = \langle x, \eta \rangle, \end{cases}$$

where $b(0, x, \eta) \in C_c^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ and η is considered as a parameter. For u with $\text{WF}_h(u)$ in a small neighborhood of (x_0, ξ_0) ,

$$\iint b(0, x, \eta) e^{\frac{i}{h}\langle x-y, \eta \rangle} u(y) dy d\eta = b(0, x, hD)u = u + O_{L^2}(h^\infty).$$

Since we can apply the exact solution operator $F(t)$ to the error term $O_{L^2}(h^\infty)$, solving (10.2.3) gives a solution to (10.2.2).

THEOREM 10.4 (Oscillatory integral representation).

(i) *Given a point $(x_0, \xi_0) \in \mathbb{R}^{2n}$, there exist $t_0 > 0$, a phase function φ , and an amplitude b defined near (x_0, ξ_0) for times $|t| < t_0$ such that*

$$(10.2.4) \quad U(t)u(x) := \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\varphi(t, x, \eta) - \langle y, \eta \rangle)} b(t, x, \eta; h) u(y) dy d\eta$$

satisfies (10.2.2) for each family of functions $u = \{u(h)\}_{0 < h \leq h_0} \subset L^2(\mathbb{R}^n)$ for which $\text{WF}_h(u)$ lies within a small neighborhood of (x_0, ξ_0) .

(ii) In addition, if $A = a^w(x, hD)$ for $a \in S$ and $\text{WF}_h(A)$ is contained within some small neighborhood of (x_0, ξ_0) , we have

$$(10.2.5) \quad \begin{cases} (hD_t + P(t))U(t)A = O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty) & (-t_0 < t < t_0) \\ U(0)A = A + O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty). \end{cases}$$

In other words, $U(t)u$ solves (10.2.1) microlocally near (x_0, ξ_0) .

(iii) If $m \equiv 1$ and so $\{p_t\}_{t \in \mathbb{R}} \subset S$, then

$$(10.2.6) \quad U(t) = F(t) \quad \text{microlocally near } (x_0, \xi_0)$$

for times $|t| \leq t_0$, where $F(t)$ is the unitary operator given by Theorem 10.1.

DEFINITION. The right-hand side of (10.2.4) is called a *Fourier integral operator*.

The proof will appear after the following constructions of the phase and amplitude. (For the special case of a quadratic time-independent symbol p , see Lemma 11.8.) These constructions provide a *WKB approximation* for a general class of operators.

10.2.2. Construction of the phase function. We start by identifying the phase function φ for (10.2.4) as a local generating function (see Section 2.3) associated with the symplectomorphisms generated by the flow

$$(10.2.7) \quad \partial_t \kappa_t = H_{p_t}(\kappa_t)$$

on $\mathbb{R}^{2n}$.

We can assume that $(x_0, \xi_0) = (0, 0)$ and let U denote a bounded open set containing $(0, 0)$.

LEMMA 10.5 (Hamilton–Jacobi equation).

(i) If $t_0 > 0$ is small enough, there exists a smooth function $\varphi = \varphi(t, x, \eta)$ defined in $(-t_0, t_0) \times U \times U$, such that κ_t given by (10.2.7) locally satisfies

$$\kappa_t(y, \eta) = (x, \xi)$$

if and only if

$$(10.2.8) \quad \xi = \partial_x \varphi(t, x, \eta), \quad y = \partial_\eta \varphi(t, x, \eta).$$

(ii) Furthermore, $\varphi = \varphi(t, x, \eta)$ solves the Hamilton–Jacobi equation

$$(10.2.9) \quad \begin{cases} \partial_t \varphi + p_t(x, \partial_x \varphi) = 0 & (-t_0 \leq t \leq t_0) \\ \varphi(0, x, \eta) = \langle x, \eta \rangle, \end{cases}$$

for $(x, \eta) \in U \times U$.

The proof of the lemma is based on the methods of characteristics; see [E, Section 3.2.4]. In the remark after the proof we present a geometric argument based on the Poincaré Lemma (Theorem B.4).

Proof. 1. The definition (10.2.7) means that $\kappa_s(y, \eta) = (x, \xi)$ is equivalent to $(y, \eta) = (z(0), \zeta(0))$, $(x, \xi) = (z(s), \zeta(s))$, where

$$(10.2.10) \quad \frac{\partial z}{\partial s} = \frac{\partial p_s}{\partial \xi}(z, \zeta), \quad \frac{\partial \zeta}{\partial s} = -\frac{\partial p_s}{\partial x}(z, \zeta).$$

For (y, η) in a compact set in $\mathbb{R}^{2n}$, the map

$$(10.2.11) \quad (x, \xi, y, \eta) = (\kappa_s(y, \eta), (y, \eta)) \mapsto (x, \eta)$$

is surjective for times $|t| \leq t_0$, provided t_0 is small enough. Hence we can write

$$\xi = \xi(s, x, \eta), \quad y = y(s, x, \eta), \quad \kappa_s(y, \eta) = (x, \xi).$$

Since κ_s is symplectic, we have $\sum_{j=1}^n d\xi_j \wedge dx_j = \sum_{j=1}^n d\eta_j \wedge dy_j$. Hence, putting $\xi = \xi(s, x, \eta)$, $y = y(s, x, \eta)$ and comparing terms $d\eta_i \wedge dx_j$ on both sides shows that

$$(10.2.12) \quad \frac{\partial \xi_j}{\partial \eta_i} = \frac{\partial y_i}{\partial x_j},$$

that is, $\partial \xi / \partial \eta = (\partial y / \partial x)^T$.

2. The desired properties of $\varphi(t, x, \eta)$ imply that

$$\varphi(t, x, \eta) = \langle x, \eta \rangle - \int_0^t p_s(x, \partial_x \varphi(s, x, \eta)) ds, \quad \xi(t, x, \eta) = \partial_x \varphi(t, x, \eta).$$

This suggests the following formula for φ obtained using the implicitly defined function $(s, x, \eta) \mapsto \xi$:

$$(10.2.13) \quad \varphi(t, x, \eta) := \langle x, \eta \rangle - \int_0^t p_s(x, \xi(s, x, \eta)) ds.$$

We only need to verify (10.2.8) as then (10.2.9) follows by differentiation with respect to t .

3. We calculate

$$\partial_x \varphi(t, x, \eta) = \eta - \int_0^t (\partial_x p_s(x, \xi(s, x, \eta)) + \partial_x \xi \cdot \partial_\xi p_s(x, \xi(s, x, \eta))) ds,$$

where $\partial_x \xi \cdot \partial_\xi p_s$ means the matrix $\partial_x \xi$ applied to the vector $\partial_\xi p_s$.

Since $\zeta(s) = \xi(s, z(s), \eta)$, we have

$$\begin{aligned} \partial_x p_s(z(s), \xi(s, z(s), \eta)) &= -\partial_s \zeta(s) = \partial_s (\xi(s, z(s), \eta)) \\ &= -\partial_x \xi(s, z(s), \eta) - \partial_x \xi \cdot \partial_s z(s). \end{aligned}$$

Recalling from (10.2.10) that $\partial_\xi p_s(z(s), \xi(s, z(s), \eta)) = \partial_s z(s)$, we conclude that

$$\partial_x p_s + \partial_x \xi \partial_\xi p_s(z(s), \xi(s, z(s), \eta)) = -\partial_s \xi(s, z(s), \eta).$$

We now note that $z(s) = x$ and thus

$$\partial_x \varphi(t, x, \eta) = \eta + \int_0^t \partial_s \xi(s, x, \eta) ds = \xi(t, x, \eta),$$

as claimed.

4. It remains to show that $\partial_\eta \varphi(t, x, \eta) = y$ when $\kappa_t(y, \eta) = (x, \xi)$. From (10.2.12) we have

$$\begin{aligned} \partial_\eta \varphi(t, x, \eta) &= x - \int_0^t \partial_\xi \eta \partial_\xi p_s(x, \xi(s, x, \eta)) ds = x - \int_0^t \partial_z y \partial_s z(s) ds \\ &= x - \int_0^t \partial_s y(s, x, \eta) ds = y, \end{aligned}$$

as claimed in (10.2.8). $\square$

REMARK: A geometric construction of φ . We can also construct φ using general principles for finding generating functions of Lagrangian submanifolds; see Theorem 2.14.

1. We have already noted in (10.2.11) that $(x, \xi, y, \eta) \mapsto (x, \eta)$ is surjective on the graph of κ_t for $|t|$ small.

2. To show the existence of φ , consider the $(2n + 1)$ -dimensional submanifold of $\mathbb{R}^2 \times \mathbb{R}^{2n} \times \mathbb{R}^{2n}$

$$\Lambda := \{(t, p_t(x, \xi); \kappa_t(y, \eta); y, \eta) \mid t \in \mathbb{R}, (y, \eta) \in \mathbb{R}^{2n}\}.$$

We write a typical point of Λ as $(t, \tau, x, \xi, y, \eta)$.

We introduce now the 1-form

$$\nu := -\tau dt + \sum_{j=1}^n \xi_j dx_j + \sum_{j=1}^n y_j d\eta_j.$$

That κ_t is a symplectic implies $d\nu|_\Lambda = 0$. Here $d\nu|_\Lambda := \iota^* d\nu = d(\iota^* \nu)$, where $\iota: \Lambda \rightarrow \mathbb{R}^2 \times \mathbb{R}^{2n} \times \mathbb{R}^{2n}$ is the inclusion map; see Theorem B.2.

By Poincaré's Lemma (Theorem B.4), there exists a smooth function φ on Λ such that

$$d\varphi = \nu|_\Lambda.$$

In view of (10.2.11) we can use (t, x, η) as coordinates on

$$\Lambda \cap ((-t_0, t_0) \times U \times U);$$

and hence

$$-\tau dt + \sum_{j=1}^n \xi_j dx_j + \sum_{j=1}^n y_j d\eta_j = \partial_t \varphi dt + \sum_{j=1}^n \partial_{x_j} \varphi dx_j + \sum_{j=1}^n \partial_{\eta_j} \varphi d\eta_j.$$

Comparing terms on the two sides shows that (10.2.8) holds, as well as the Hamilton–Jacobi PDE. Since $\kappa_0(x, \eta) = (x, \eta)$, we see as well that $d\varphi|_{t=0} = d(\langle x, \eta \rangle)$. Since φ is determined up to an additive constant, we can therefore ensure that $\varphi = \langle x, \eta \rangle$ at $t = 0$. $\square$

10.2.3. Construction of the amplitude. The amplitude b in (10.2.4) must satisfy (10.2.3):

$$(hD_t + p_t^w(x, hD))(e^{i\varphi/h}b) = O_S(h^\infty),$$

and so

$$(10.2.14) \quad (\partial_t \varphi + hD_t + e^{-i\varphi/h} p_t^w e^{i\varphi/h})b = O_S(h^\infty),$$

for $(x, \eta) \in U, |t| \leq t_0$. In this equation η appears only as a parameter.

We will construct b in the form

$$(10.2.15) \quad b(t, x, \eta; h) \sim \sum_{k=0}^{\infty} h^k b_k(t, x, \eta).$$

Once all the terms b_k have been computed, Borel’s Theorem (Theorem 4.15) produces the amplitude b , defined up to terms in $O_S(h^\infty)$. We assume that b is supported in a neighborhood of U and the equation holds for $(x, \eta) \in U$.

Although the value of b_0 is not important in our applications, it is sometimes useful to know it:

THEOREM 10.6 (Calculation of b_0). *For $(x, \eta) \in U$ the leading term of the expansion (10.2.15) is*

$$(10.2.16) \quad b_0(t, x, \eta) = (\det \partial_{x\eta}^2 \varphi(t, x, \eta))^{\frac{1}{2}}.$$

Note that at $t = 0$, $\det \partial_{\eta x}^2 \varphi = 1$ and hence $\det \partial_{\eta x}^2 \varphi > 0$ for small $|t|$.

Proof. 1. Since we are solving the equations for $(x, \eta) \in U \times U$, a small neighborhood of $(0, 0)$, we can assume that φ is compactly supported. We then claim that for compactly supported $\varphi(x)$, we have

$$(10.2.17) \quad e^{-i\varphi/h} p_t^w(x, hD) e^{i\varphi/h} = q_t^w(x, hD),$$

where

$$(10.2.18) \quad q_t(x, \xi; h) = p_t(x, \partial_x \varphi + \xi) + O_{S(m)}(h^2).$$

To see this, write $\varphi(x) - \varphi(y) = \langle F(x, y), x - y \rangle$, where

$$F(x, y) = \partial_x \varphi\left(\frac{x+y}{2}\right) + O(|x - y|^2).$$

We check that

$$(10.2.19) \quad e^{-i\varphi/h} p_t^w e^{i\varphi/h} u = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} p_t\left(\frac{x+y}{2}, \xi + F(x, y)\right) e^{i\langle x-y, \xi \rangle/h} u(y) dy d\xi.$$

Hence

$$e^{-i\varphi/h} p_t^w e^{i\varphi/h} u = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} p_t\left(\frac{x+y}{2}, \xi + \partial_x \varphi\left(\frac{x+y}{2}\right)\right) + \langle E_t(x, y, \xi)(x - y), x - y \rangle e^{i\langle x-y, \xi \rangle/h} u(y) dy d\xi,$$

the entries of the matrix-valued function E_t belonging to $S(m)$. Integration by parts based on (9.2.13) and Theorem 4.20 gives (10.2.18).

2. Recalling from Lemma 10.5 that $\partial_t \varphi = -p_t(x, \partial_x \varphi)$, we deduce from (10.2.14) that for (x, η) in a neighborhood of U ,

$$(10.2.20) \quad (hD_t + f_t^w(x, hD, \eta))b(t, x, \eta) = O_S(h^2),$$

where

$$f_t(x, \xi) := p_t(x, \partial_x \varphi(t, x, \eta) + \xi) - p_t(x, \partial_x \varphi(t, x, \eta))$$

and where η is again considered as a parameter. So

$$(10.2.21) \quad f_t(x, \xi, \eta) = \sum_{j=1}^n \xi_j \partial_{\xi_j} p_t(x, \partial_x \varphi(t, x, \eta)) + e_t(x, \xi, \eta)$$

for a symbol e_t of the form $e_t(x, \xi, \eta) = \langle F_t(x, \xi, \eta)\xi, \xi \rangle$. Remember that if $g \in S$, then $(hD_x)^2 g \in O_S(h^2)$. Hence if g is compactly supported in x , we have $e_t^w g \in O_S(h^2)$.

Since $\partial_{\xi_j} p_t = \partial_{\xi_j} p_t(x, \partial_x \varphi)$ does not depend on ξ , (10.2.21) and the quantization formula (4.2.5) for symbols linear in ξ imply

$$f_t^w(x, hD, \eta)b = \frac{1}{2} \sum_{j=1}^n hD_{x_j} (\partial_{\xi_j} p_t b) + (\partial_{\xi_j} p_t) hD_{x_j} b + O_S(h^2).$$

Consequently the leading term in the expansion b_0 satisfies

$$hD_t b_0 + \frac{1}{2} \sum_{j=1}^n hD_{x_j} (\partial_{\xi_j} p_t b_0) + (\partial_{\xi_j} p_t) hD_{x_j} b_0 = 0.$$

Rewrite this equation as follows:

$$(10.2.22) \quad (\partial_t + V_t + \frac{1}{2} \operatorname{div} V_t) b_0 = 0$$

for

$$(10.2.23) \quad V_t := \sum_{j=1}^n \partial_{\xi_j} p_t(x, \partial_x \varphi) \partial_{x_j}.$$

We also require that $b_0(0, x, \eta) = 1$ in U .

3. Next observe from (10.2.22) that $\sigma := b_0^2$ solves

$$(10.2.24) \quad \partial_t \sigma + \operatorname{div}_x(\sigma \partial_{\xi} p_t(x, \partial_x \varphi)) = 0,$$

with $\sigma(0, x, \eta) = 1$ in U . We claim that a solution is

$$(10.2.25) \quad \sigma = \det \partial_{x\eta}^2 \varphi.$$

To see this, first differentiate the Hamilton–Jacobi equation (10.2.9) with respect to η :

$$\partial_t \partial_{\eta} \varphi + (\partial_{x\eta}^2 \varphi)^T \partial_{\xi} p_t = 0.$$

Apply the cofactor matrix $\operatorname{cof}(\partial_{x\eta}^2 \varphi) = \det \partial_{x\eta}^2 \varphi ((\partial_{x\eta}^2 \varphi)^T)^{-1}$, to deduce

$$\operatorname{cof}(\partial_{x\eta}^2 \varphi) \partial_t \partial_{\eta} \varphi + \sigma \partial_{\xi} p_t = 0,$$

where we defined σ by (10.2.25).

With $\operatorname{div}_x(\operatorname{cof}(\partial_{x\eta}^2 \varphi)) = 0$, as shown in [E, Section 8.1.4], we discover consequently that

$$\operatorname{tr}(\operatorname{cof}(\partial_{x\eta}^2 \varphi) \partial_t \partial_{x\eta}^2 \varphi) + \operatorname{div}_x(\sigma \partial_{\xi} p_t(x, \partial_x \varphi)) = 0.$$

On the other hand, (A.2.1) shows that

$$\partial_t \sigma = \partial_t \det \partial_{x\eta}^2 \varphi = \operatorname{tr}(\operatorname{cof}(\partial_{x\eta}^2 \varphi) \partial_t \partial_{x\eta}^2 \varphi).$$

This proves (10.2.24); and then (10.2.16) follows, since

$$b_0 = \sigma^{\frac{1}{2}} = (\det \partial_{x\eta}^2 \varphi)^{\frac{1}{2}},$$

with the $+$ sign determined by $b_0|_{t=0} = 1$ near $U \times U$. □

Proof of Theorem 10.4:

1. After constructing b_0 , we construct successive terms in the expansion of b by induction using

$$(\partial_t + V_t + \frac{1}{2} \operatorname{div} V_t) b_k = F(b_0, \dots, b_{k-1}),$$

where V_t is given by (10.2.23) and, in the notation of (10.2.17),

$$F(b_0, \dots, b_{k-1}) := \frac{1}{h^{k+1}} (hD_t + \partial_t \varphi + q_t^w(x, hD, \eta)) \left(\sum_{j=0}^{k-1} h^j b_j(t, x, \eta) \right) \in S.$$

Borel's Theorem (Theorem 4.15) then gives $b(t, x, \eta)$, which is supported within some neighborhood of U in the (x, η) variables. Thus (10.2.5) holds for A for which $\text{WF}_h(A) \subset U$; and so (10.2.2) is valid.

2. When $p_t \in S$, the operators $F(t)$ are well defined according to Theorem 10.1. We can now use Duhamel's formula ([E, Chapter 2]) to show that $U(t) = F(t)$ microlocally. For this, consider

$$\begin{cases} (hD_t + P(t))F(t, r) = 0 \\ F(r, r) = I. \end{cases}$$

If

$$\begin{cases} (hD_t U(t) + P(t)U(t))A = -E(t) \\ U(0)A = A - R, \end{cases}$$

where $E(t), R = O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty)$, then

$$\begin{aligned} F(t)A &= U(t)A + F(t)R + \int_0^t F(t, s)E(s) ds \\ &= U(t)A + O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty). \end{aligned}$$

□

REMARK: Half-densities, geometric interpretation of the amplitude. We provide an alternative derivation of the formula for b_0 given in (10.2.16) that yields an invariant interpretation of the evolution.

To understand the equation (10.2.22) geometrically, we consider $b_0(t, \cdot, \eta)$ as a function on

$$\Lambda_{t,\eta} := \{(x, \partial_x \varphi(t, x, \eta)) \mid x \in \mathbb{R}^n\}.$$

Although we write $x \in \mathbb{R}^n$, it should be understood that $\Lambda_{t,\eta}$ is defined for $(x, \eta) \in U$. Since $\Lambda_{0,\eta} = \{(y, \eta) \mid y \in \mathbb{R}^n\}$ and $\kappa_t(\partial_\eta \varphi(t, x, \eta), \eta) = (x, \partial_x \varphi(t, x, \eta))$, we have

$$\kappa_t(\Lambda_{0,\eta}) = \Lambda_{t,\eta}.$$

We parametrize $\Lambda_{t,\eta}$ by

$$(10.2.26) \quad x \mapsto (x, \partial_x \varphi(t, x, \eta))$$

and consider the function

$$\kappa_{t,\eta} := \pi \circ \kappa_t$$

mapping $\Lambda_{0,\eta} \rightarrow \mathbb{R}^n$, where $\pi(x, \xi) := x$. Then (10.2.7) implies

$$\partial_t \kappa_{t,\eta}^* u = \partial_t \kappa_t^* \pi^* u = \kappa_t^* (H_{p_t} \pi^* u) = \kappa_t^* \pi^* (\pi_* H_{p_t} u) = \kappa_{t,\eta}^* (V_t u),$$

V_t defined by (10.2.22). In other words,

$$\partial_t \kappa_{t,\eta}(x) = V_t(\kappa_{t,\eta}(x)).$$

We now introduce

$$a := u|dx|^{\frac{1}{2}},$$

a half-density on $\Lambda_{t,\eta}$, keeping in mind that $\Lambda_{t,\eta}$ is parametrized by $\mathbb{R}^n$ using (10.2.26). We claim that

$$(10.2.27) \quad \mathcal{L}_{V_t} a := \partial_t \kappa_{t,\eta}^* a = \kappa_{t,\eta}^* \left(\left(V_t u + \frac{1}{2} \operatorname{div} V_t u \right) |dx|^{\frac{1}{2}} \right).$$

To see this, we use (9.1.7) to compute

$$\kappa_{t,\eta}^* a = \kappa_{t,\eta}^* u (\det \partial \kappa_{t,\eta})^{\frac{1}{2}} |dx|^{\frac{1}{2}}.$$

Now (10.2.27) follows from (A.2.1):

$$(10.2.28) \quad \partial_t (\det \partial \kappa_{t,\eta})^{\frac{1}{2}} = \frac{1}{2} \kappa_{t,\eta}^* (\operatorname{div} V_t) (\det \partial \kappa_{t,\eta})^{\frac{1}{2}}.$$

So if we consider $b_0(t, x, \eta) |dx|^{\frac{1}{2}}$ as a half-density on $\Lambda_{t,\eta}$, then (10.2.22) becomes

$$\partial_t \kappa_{t,\eta}^* (b_0 |dx|^{\frac{1}{2}}) = (\partial_t + \mathcal{L}_{V_t}) \left(b_0(t, x, \eta) |dx|^{\frac{1}{2}} \right) = 0.$$

This is the same as

$$\kappa_{t,\eta}^* \left(b_0(t, x, \eta) |dx|^{\frac{1}{2}} \right) = |dx|^{\frac{1}{2}}$$

as half-densities on $\Lambda_{t,\eta}$ and $\Lambda_{0,\eta}$, with $\kappa_{t,\eta}(\Lambda_{0,\eta}) = \Lambda_{t,\eta}$. It follows that

$$\kappa_{t,\eta}^* b_0 = (\det \partial \kappa_{t,\eta})^{-\frac{1}{2}}.$$

We now recall that $\kappa_{t,\eta}^{-1}$ maps x to $\partial_\eta \varphi(t, x, \eta)$, and thus

$$\partial \kappa_{t,\eta}^{-1} = \partial_{x\eta}^2 \varphi.$$

This completes our alternative derivation of (10.2.16).

It is appealing that the amplitude, interpreted as a half-density, is invariant under the flow. When coordinates change, and in particular when we move to larger times at which (10.2.8) and (10.2.9) are no longer valid, the statement about the amplitude as a half-density remains simple. $\square$

10.3. STRICHARTZ ESTIMATES

In this section and the next we will employ Theorem 10.4 to obtain L^p bounds on weak quasimodes (7.4.1). These supplement the L^∞ estimates we derived earlier in Section 7.4.

10.3.1. Strichartz estimates. We first recall the following general result (see Keel–Tao [K-T] and the references given there):

THEOREM 10.7 (Strichartz estimates). *Let $\{U(t)\}_{t \in \mathbb{R}}$ be a family of linear operators satisfying the bounds*

$$(10.3.1) \quad \begin{cases} \|U(t)\|_{L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)} \leq C \\ \|U(t)U(s)^*\|_{L^1(\mathbb{R}^n) \rightarrow L^\infty(\mathbb{R}^n)} \leq Ch^{-\gamma}|t-s|^{-\sigma}, \end{cases}$$

for all $t, s \in \mathbb{R}$, where $C, \sigma, \gamma > 0$ are fixed constants.

Then we have the estimate

$$(10.3.2) \quad \left(\int_{\mathbb{R}} \|U(t)f\|_{L^q(\mathbb{R}^n)}^p dt \right)^{\frac{1}{p}} \leq Ch^{-\frac{\gamma}{p\sigma}} \|f\|_{L^2(\mathbb{R}^n)}$$

for every pair p, q satisfying

$$\frac{2}{p} + \frac{2\sigma}{q} = \sigma, \quad 2 \leq p \leq \infty, \quad 1 \leq q \leq \infty, \quad (p, q) \neq (2, \infty).$$

We will need this assertion only for $p = q$, and for the reader's convenience we present the proof for that case.

Proof (for $p = q$): 1. A rescaling in time converts $h = 1$, and inequality (10.3.2) for $p = q$ then reads

$$(10.3.3) \quad \|U(t)f\|_{L^p(\mathbb{R} \times \mathbb{R}^n)} \leq C\|f\|_{L^2(\mathbb{R}^n)}$$

where $p = \frac{2(1+\sigma)}{\sigma}$.

2. We apply the Riesz–Thorin Interpolation Theorem (Hörmander [H1, Theorem 7.1.12]) to the operator $U(t)U(s)^*$, for fixed times $t, s \in \mathbb{R}$. Interpolating between the two estimates provided by the hypothesis (10.3.1) gives

$$\|U(t)U(s)^*\|_{L^{p'} \rightarrow L^p} \leq C|t-s|^{-\sigma(2/p'-1)}$$

for $1 \leq p' \leq 2$, where $1/p + 1/p' = 1$. In particular, we have

$$(10.3.4) \quad \begin{aligned} & |\langle U(t)^*G(t, \cdot), U(s)^*F(s, \cdot) \rangle| \\ & \leq C|t-s|^{-\sigma(2/p'-1)} \|G(t, \cdot)\|_{L^{p'}} \|F(s, \cdot)\|_{L^{p'}} \end{aligned}$$

for functions $F = F(t, x)$, $G = G(t, x)$.

3. Next, we recall the Hardy–Littlewood–Sobolev inequality (Hörmander [H1, Theorem 4.5.3]), which says for $K_a(t) := |t|^{-1/a}$ and $1 < a < \infty$ that

$$(10.3.5) \quad \|K_a * u\|_{L^r(\mathbb{R})} \leq C\|u\|_{L^{p'}(\mathbb{R})}$$

for

$$\frac{1}{p} + \frac{1}{r} = \frac{1}{a}, \quad 1 < p' < r.$$

We apply (10.3.5) to (10.3.4) with $\frac{1}{a} = \sigma \left(\frac{2}{p'} - 1 \right)$ and $p = r$, in which case $p = \frac{2(1+\sigma)}{\sigma}$. The resulting inequality is

$$(10.3.6) \quad \left| \int_{\mathbb{R}} \int_{\mathbb{R}} \langle U(t)^* G(t, \cdot), U(s)^* F(s, \cdot) \rangle dt ds \right| \leq C \|G\|_{L^{p'}(\mathbb{R} \times \mathbb{R}^n)} \|F\|_{L^{p'}(\mathbb{R} \times \mathbb{R}^n)}.$$

Taking $F = G$, we deduce that

$$\left\| \int_{\mathbb{R}} U(t)^* G(t, x) dt \right\|_{L^2(\mathbb{R}^n)} \leq C \|G\|_{L^{p'}(\mathbb{R} \times \mathbb{R}^n)};$$

and therefore

$$\left| \int_{\mathbb{R} \times \mathbb{R}^n} U(t) f(x) G(t, x) dx dt \right| \leq C \|f\|_{L^2(\mathbb{R}^n)} \|G\|_{L^{p'}(\mathbb{R} \times \mathbb{R}^n)}$$

for all $G \in L^{p'}(\mathbb{R} \times \mathbb{R}^n)$. This implies (10.3.3). $\square$

10.3.2. Semiclassical Strichartz estimates. Let $\{a_t\}_{t \in \mathbb{R}}$ be a smooth family of real-valued symbols in S . We introduce at a point (t, x, ξ) the *nondegeneracy condition* that

$$(10.3.7) \quad \partial_{\xi}^2 a_t(x, \xi) \text{ is nonsingular.}$$

Put $A(t) := a_t^w(x, hD)$ and then consider the initial-value problem

$$(10.3.8) \quad \begin{cases} (hD_t + A(t))F(t, r) = 0 \\ F(r, r) = I \end{cases}$$

for some fixed time $r \in \mathbb{R}$.

THEOREM 10.8 (Semiclassical Strichartz estimates). *Select a function $\chi \in C_c^\infty(\mathbb{R}^{2k})$ and assume the nondegeneracy condition that (10.3.7) holds for all $(x, \xi) \in \text{spt}(\chi)$ and $t \in \mathbb{R}$.*

Then given $\psi \in C_c^\infty(\mathbb{R})$ with support sufficiently close to 0 and a compact interval $I \subset \mathbb{R}$, we have the estimate

$$(10.3.9) \quad \sup_{r \in I} \left(\int_{\mathbb{R}} \|U(t, r) f\|_{L^q(\mathbb{R}^k)}^p dt \right)^{\frac{1}{p}} \leq C h^{-\frac{1}{p}} \|f\|_{L^2(\mathbb{R}^k)},$$

for

$$U(t, r) := \psi(t - r) \chi^w(x, hD) F(t, r),$$

where

$$\frac{2}{p} + \frac{k}{q} = \frac{k}{2}, \quad 2 \leq p \leq \infty, \quad 1 \leq q \leq \infty, \quad (p, q) \neq (2, \infty).$$

Proof. 1. In view of Theorem 10.7 we need to show that

$$(10.3.10) \quad \|U(t, r)U(s, r)^*\|_{L^1 \rightarrow L^\infty} \leq Ch^{-k/2}|t - s|^{-k/2}$$

for $s, t \in \mathbb{R}$, with constants independent of $r \in I$. For simplicity we can put $r = 0$ in the argument and drop the dependence on r in U and F .

2. Since $a_t \in S$, Theorem 10.4 provides a microlocal description of $U(t)$ for small values of $|t|$. We have

$$U(t) = \tilde{U}(t) + E(t),$$

where

$$E(t) = O_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty)$$

and the Schwartz kernel of $\tilde{U}(t)$ is

$$(10.3.11) \quad \tilde{U}(t, x, y) = \frac{1}{(2\pi h)^k} \int_{\mathbb{R}^k} e^{\frac{i}{h}(\varphi(t, x, \eta) - \langle y, \eta \rangle)} \tilde{b}(t, y, x, \eta; h) d\eta,$$

for $\tilde{b} \in S \cap C_c^\infty(\mathbb{R}^{1+3k})$ and $\varphi = \varphi(t, x, \eta)$ solving the Hamilton–Jacobi equation

$$(10.3.12) \quad \begin{cases} \partial_t \varphi + a(t, x, \partial_x \varphi) = 0 \\ \varphi(0, x, \eta) = \langle x, \eta \rangle. \end{cases}$$

All of this is true since the local construction of Section 10.2 works on the support of χ .

3. Hence we need to prove (10.3.10) with U replaced by $\tilde{U}$, and for this we need an L^∞ bound on the Schwartz kernel of $W(t, s) := \tilde{U}(t)\tilde{U}(s)^*$, which is

$$W(t, s, x, y) = \frac{1}{(2\pi h)^{2k}} \int_{\mathbb{R}^k} \int_{\mathbb{R}^k} \int_{\mathbb{R}^k} e^{\frac{i}{h}(\varphi(t, x, \eta) - \varphi(s, y, \zeta) - \langle z, \eta - \zeta \rangle)} B dz d\zeta d\eta,$$

where $B = B(t, s, x, y, z, \eta, \zeta; h) \in S \cap C_c^\infty(\mathbb{R}^{2+6k})$.

The phase φ is nondegenerate in the (z, ζ) variables and is stationary at $\zeta = \eta$, $z = \partial_\zeta \varphi(s, y, \zeta)$. Hence Theorem 3.16 implies

$$W(t, s, x, y) = \frac{1}{(2\pi h)^k} \int_{\mathbb{R}^k} e^{\frac{i}{h}(\varphi(t, x, \eta) - \varphi(s, y, \eta))} B_1(t, s, x, y, \eta; h) d\eta,$$

where $B_1 \in S \cap C_c^\infty(\mathbb{R}^{2+3k})$. We now rewrite the phase:

$$\begin{aligned} \tilde{\varphi} &:= \varphi(t, x, \eta) - \varphi(s, y, \eta) \\ &= (t - s)(a(0, x, \eta) + O(|t| + |s|)) + \langle x - y, \eta + sF(s, x, y, \eta) \rangle \end{aligned}$$

for $F \in C^\infty(\mathbb{R}^{1+3k})$. Here we used (10.3.12) to write

$$\varphi(s, x, \eta) - \varphi(s, y, \eta) = \langle x - y, \eta \rangle + \langle x - y, sF(s, x, y, \eta) \rangle.$$

The phase is stationary when

$$\partial_\eta \tilde{\varphi} = (I + s\partial_\eta F)(x - y) + (t - s)(\partial_\eta a + O(|t| + |s|)) = 0.$$

In particular, for s small, $(I + s\partial_\eta F)$ is invertible, and hence having a stationary point implies

$$x - y = O(t - s).$$

The Hessian is

$$\begin{aligned} \partial_\eta^2 \tilde{\varphi} &= s(\partial_\eta^2 F)(x - y) + (t - s)(\partial_\eta^2 a + O(|t| + |s|)) \\ &= (t - s)(\partial_\eta^2 a + O(|t| + |s|)), \end{aligned}$$

where $\partial_\eta^2 a = \partial_\eta^2 a(0, x, \eta)$.

4. Hence for t and s sufficiently small, that is, for a suitable choice of the support of ψ in the definition of $U(t)$, the nondegeneracy assumption (10.3.7) implies that

$$\partial_\eta^2 \tilde{\varphi} = (t - s)\psi(x, y)$$

at the critical point. If $|t - s| > Mh$, where M is a large constant, we can use the stationary phase estimate in Theorem 3.16 to deduce that

$$|W(t, s, x, y)| \leq Ch^{-k/2}|t - s|^{-k/2}.$$

When $|t - s| < Mh$,

$$\begin{aligned} |W(t, s, x, y)| &\leq \frac{1}{(2\pi h)^{2k}} \int_{\mathbb{R}^{3k}} |B_1(t, s, x, y, z, \eta, \zeta; h)| dz d\zeta d\eta \\ &\leq Ch^{-k} \leq Ch^{-k/2}|t - s|^{-k/2}. \end{aligned}$$

The proof of (10.3.10) is complete. $\square$

We will later need this theorem:

THEOREM 10.9. *In the notation of Theorem 10.8, we have*

$$\begin{aligned} (10.3.13) \quad &\left\| \int_0^t U(t, s) \mathbf{1}_I(s) f(s, x) ds \right\|_{L^p(\mathbb{R}_t \times \mathbb{R}^k)} \\ &\leq Ch^{-1/p} \int_{\mathbb{R}} \|f(s, x)\|_{L^2(\mathbb{R}^k)} ds. \end{aligned}$$

Proof. We apply the integral version of Minkowski's inequality:

$$\begin{aligned} &\left\| \int_0^t U(t, s) \mathbf{1}_I(s) f(s, x) ds \right\|_{L^p(\mathbb{R}_t \times \mathbb{R}^k)} \\ &\leq C \int_{I \cap \mathbb{R}_+} \|\mathbf{1}_{[s, \infty)}(t) U(t, s) f(s, x)\|_{L^p(\mathbb{R}_t \times \mathbb{R}^k)} ds \\ &\leq C \int_{I \cap \mathbb{R}_+} \|U(t, s) f(s, x)\|_{L^p(\mathbb{R}_t \times \mathbb{R}^k)} ds. \end{aligned}$$

Now we use the estimate (10.3.9) with $p = q = 2(n+1)/(n-1)$:

$$\|U(t, s)f(s, x)\|_{L^p(\mathbb{R} \times \mathbb{R}^k)} \leq Ch^{-1/p} \|f(s, x)\|_{L^2(\mathbb{R}^k)}.$$

Since I is compact, (10.3.13) follows. $\square$

10.4. L^p ESTIMATES FOR QUASIMODES

This section mirrors Section 7.4, where we earlier derived h -dependent L^∞ bounds for appropriate families of quasimodes. Our goal here is finding analogous L^p bounds.

10.4.1. Nondegeneracy, localization, and L^p bounds. Let us suppose that $K \subset \mathbb{R}^{2n}$ is a compact set and that the real-valued symbol $p \in S(m)$ satisfies

$$(10.4.1) \quad \partial_\xi p \neq 0 \quad \text{on } \{p = 0\} \cap K.$$

Thus if $p(x_0, \xi_0) = 0$ and $(x_0, \xi_0) \in K$, then $\{\xi \mid p(x_0, \xi) = 0\}$ is a smooth hypersurface in $\mathbb{R}^n$ near ξ_0 . After a linear change of variables, we have $\partial_\xi p(x_0, \xi_0) = (\rho, 0, \dots, 0)$ where $\rho \neq 0$. Then near (x_0, ξ_0) ,

$$(10.4.2) \quad p(x, \xi) = e(x, \xi)(\xi_1 - a(x, \xi'))$$

where $\xi = (\xi_1, \xi')$ and $e(x, \xi) > 0$. We make the further assumption that for each (x_0, ξ_0) this surface has nondegenerate second fundamental form, meaning that

$$(10.4.3) \quad \partial_{\xi'}^2 a(x_0, \xi'_0) \text{ is nondegenerate.}$$

Remember from Section 7.4 what it means for a family of functions to be localized to a compact set K in $\mathbb{R}^{2n}$.

THEOREM 10.10 (L^p bounds for quasimodes). *Assume p satisfies the nondegeneracy conditions (10.4.1) and (10.4.3). Suppose further that $u = \{u(h)\}_{0 < h \leq h_0}$ is bounded in $L^2(\mathbb{R}^n)$ and is localized to K in the sense of (8.4.4).*

Then if

$$(10.4.4) \quad \|p^w(x, hD)u(h)\|_{L^2(\mathbb{R}^n)} = O(h) \|u(h)\|_{L^2(\mathbb{R}^n)},$$

we have the estimate

$$(10.4.5) \quad \|u(h)\|_{L^p(K)} \leq Ch^{-1/p} \|u(h)\|_{L^2(\mathbb{R}^n)}$$

for

$$p = \frac{2(n+1)}{n-1}.$$

REMARK. Estimates of the form (10.4.5) have proved useful in linear and nonlinear PDE theory. The improvement from estimate (10.4.5), as compared to interpolation using the L^∞ bound provided by Theorem 7.12, is most pronounced in lower dimensions; see Figure 10.4.1. $\square$

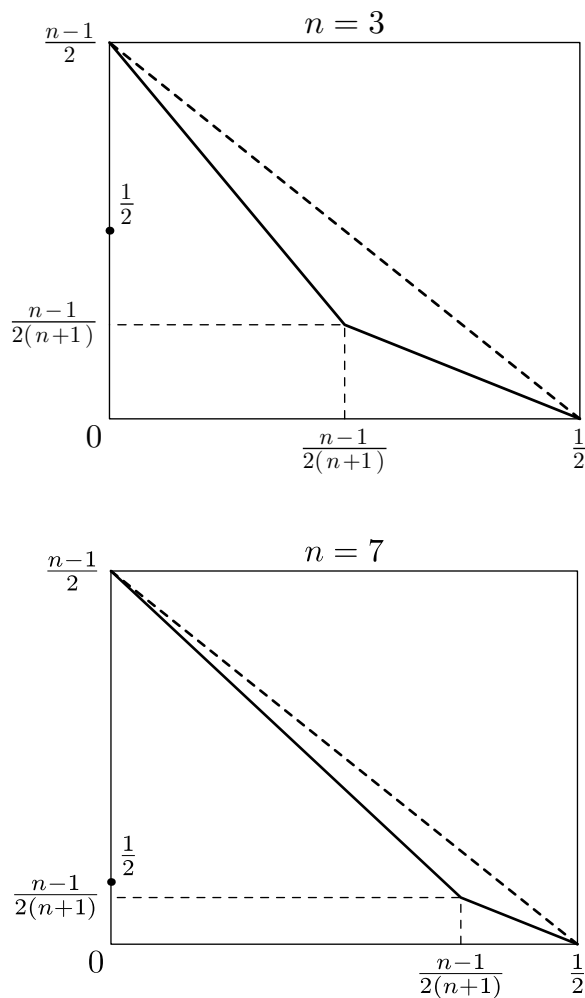


Figure 10.4.1. The horizontal axis is $1/p$ and the vertical axis is α in the estimate $\|u\|_{L^p} \leq Ch^{-\alpha} \|u\|_{L^2}$.

Proof. 1. We may assume $\|u(h)\|_{L^2} = 1$. We follow the same procedure as in the proof of Theorem 7.12. The partition of unity argument is the same as in that proof, and so is the estimate near points where $p \neq 0$. Hence we only need to consider neighborhoods of points where $p = 0$.

2. Let $\chi \in C_c^\infty(\mathbb{R}^{2n})$ be supported near a point where (10.4.2) holds. Since

$$\|p^w(x, hD)\chi^w(x, hD)u(h)\|_{L^2} = O(h)$$

and since e is elliptic, we conclude that

$$\|(hD_{x_1} - a(x, hD_{x'}))\chi^w(x, hD)u(h)\|_{L^2} = O(h)$$

for χ with sufficiently small support. Let

$$f(x_1, x') := (hD_{x_1} - a(x, hD_{x'}))\chi^w u(h).$$

If $\chi_0 \in C_c^\infty(\mathbb{R})$ has the property that $\chi_0(x_1) = 1$ if $x \in \pi(\text{spt } \chi)$, $\pi(y, \eta) := y$, then

$$\chi_0 f = f + O_{\mathcal{S}}(h^\infty).$$

Replacing $f(x_1, \cdot)$ by $\chi_0(x_1)f(x_1, \cdot)$ and using the Cauchy–Schwartz inequality gives

$$(10.4.6) \quad \int_{\mathbb{R}} \|f(x_1, \cdot)\|_{L^2(\mathbb{R}^{n-1})} dx_1 \leq C\|f\|_{L^2(\mathbb{R}^n)} + O(h^\infty) = O(h).$$

3. We now apply Theorem 10.8 with $t = x_1$ and x replaced by $x' \in \mathbb{R}^{n-1}$; that is, $k = n - 1$. The assumption (10.4.3) shows that $\partial_\xi^2 a$ is nondegenerate in the support of χ . We can choose ψ and χ in the definition of $U(t, s)$ in the statement of Theorem 10.8, so that

$$\chi^w(x, hD)u(x_1, x', h) = \frac{i}{h} \int_0^{x_1} U(t, s)f(s, x') ds + O_{\mathcal{S}}(h^\infty).$$

Let us choose $p = q$, $k = n - 1$ in (10.3.9); that is,

$$p = q = \frac{2(n+1)}{n-1}.$$

Then using (10.3.9), (10.4.6), and (10.3.13), we deduce that

$$\begin{aligned} \|\chi^w(x, hD)u\|_{L^p} &\leq \frac{1}{h} h^{-1/p} \int_{\mathbb{R}} \|f(s, \cdot)\|_{L^2(\mathbb{R}^{n-1})} ds + O(h^\infty) \\ &= O(h^{-1/p}). \end{aligned}$$

A partition of unity argument finishes the proof. $\square$

REMARKS. (i) The example in Remark (i) after Theorem 7.12 shows that the curvature condition (10.4.3) is in general necessary. In fact, if $P(h) = hD_{x_1}$ and $u(h) = h^{-(n-1)/2}\chi(x_1)\chi(x'/h)$, then for $p = 2(n+1)/(n-1)$,

$$\|u\|_{L^p} \simeq h^{(n-1)(1/p-1/2)} = h^{-(n-1)/(n+1)} \neq O(h^{-1/p}).$$

(ii) For the simplest case in which (10.4.3) holds, $p(x, \xi) = \xi_1 - \xi_2^2 - \cdots - \xi_n^2$, the estimate (10.4.5) is optimal. To see this, put

$$u(h) := h^{-(n-1)/4} \chi_0(x_1) \exp(-|x'|^2/2h),$$

where $x = (x_1, x')$, $\chi_0 \in C_c^\infty(\mathbb{R})$. Then

$$(-h^2 \Delta_{x'} + |x'|^2)u(h) = (n-1)h u(h),$$

$\|u(h)\|_{L^2} \simeq 1$, $|x'|^{2k}u(h) = O_{L^2}(h^k)$. Hence $P(h)u(h) = O_{L^2}(h)$, and

$$\|u(h)\|_{L^p(\mathbb{R}^n)} \simeq h^{(n-1)(2/p-1)/4} = h^{-1/p}$$

for $p = 2(n+1)/(n-1)$. □

10.4.2. Bounds for spectral clusters. Suppose that M is a compact Riemannian manifold of dimension n and let Δ_g be its Laplace–Beltrami operator. Assume

$$0 = \lambda_0 < \lambda_1 \leq \cdots \lambda_j \rightarrow \infty$$

is the complete set of eigenvalues of $-\Delta_g$ and

$$-\Delta_g \varphi_j = \lambda_j \varphi_j$$

are the corresponding eigenfunctions. Theorem 10.10 specialized to this setting gives the result due to Sogge:

THEOREM 10.11 (L^p bounds on eigenfunction clusters).

(i) *There exists a constant C such that for any choices of constants $c_j \in \mathbb{C}$, we have the inequality*

$$(10.4.7) \quad \|u_\mu\|_{L^p(M)} \leq C \mu^{\sigma(p)} \|u_\mu\|_{L^2(M)},$$

where $\mu \geq 1$,

$$u_\mu := \sum_{\mu \leq \sqrt{\lambda_j} \leq \mu+1} c_j \varphi_j,$$

and

$$(10.4.8) \quad \sigma(p) := \begin{cases} \frac{n-1}{2} \left(\frac{1}{2} - \frac{1}{p} \right) & \text{if } 2 \leq p \leq \frac{2(n+1)}{n-1} \\ \frac{n-1}{2} - \frac{n}{p} & \text{if } \frac{2(n+1)}{n-1} \leq p \leq \infty. \end{cases}$$

(ii) *In particular,*

$$(10.4.9) \quad \|\varphi_j\|_{L^p(M)} \leq C \lambda_j^{\sigma(p)/2} \|\varphi_j\|_{L^2(M)}.$$

Proof. 1. We argue as in the proof of Theorem 7.13 but need to check the curvature assumption (10.4.3). But at any point (x_0, ξ_0) , we have in suitable coordinates

$$p(x_0, \xi) = |\xi|^2 - 1, \quad \xi_0 = (1, 0, \dots, 0).$$

The hypersurface $p(x_0, \xi) = 0$ is the unit sphere in $\mathbb{R}^n$ and clearly has a nondegenerate second fundamental form.

2. Complex interpolation (see Hörmander [**H1**, Theorem 7.1.12]) between the estimate in Theorem 7.13, the trivial L^2 estimate, and the estimate in Theorem 10.10 gives the full result. $\square$

10.5. NOTES

Quantization of a general real-valued symbol $p \in S(m)$ might not be self-adjoint on $L^2(\mathbb{R}^n)$; see Dimassi–Sjöstrand [**D-S**].

The construction of $U(t)$ borrows from the presentations in Helffer–Sjöstrand [**H-S1**, Appendix A] and [**S-Z1**, Section 7]. For a more general problem of p depending on h we refer to [**S-Z1**, Section 7] and the references given there. Here we note that the proof works for $P(t) = p^w(t, x, hD) + h^2 p_2^w(t, x, hD)$ and that this class of operators acting on half-densities is invariant under changes of coordinates (see Theorem 9.3). See Duistermaat [**D**] and Hörmander [**H2**, Chapter 25] for more on Fourier integral operators. For a related modern perspective in molecular dynamics see Miller [**Mi**].

Semiclassical Strichartz estimates for $P = -h^2 \Delta_g - 1$ appeared in Burq–Gérard–Tzvetkov [**B-G-T**]. The adaptation of Sogge’s L^p estimates to the semiclassical setting is from [**K-T-Z**] and was inspired by discussions with N. Burq, H. Koch, C. Sogge, D. Tataru, and C. Zuily. See also Koch–Tataru [**Ko-T**] and Sogge [**S**].

QUANTUM AND CLASSICAL DYNAMICS

- 11.1 Egorov's Theorem
- 11.2 Quantizing symplectic mappings
- 11.3 Quantizing linear symplectic mappings
- 11.4 Egorov's Theorem for longer times
- 11.5 Notes

Chapter 10's discussion of Fourier integral operators provided an explicit, but only local in time, representation for solutions of quantum dynamics. This chapter switches attention to dynamics over longer time intervals and to the corresponding classical evolution.

11.1. EGOROV'S THEOREM

We assume for this section that $\{q_t\}_{0 \leq t \leq T}$ denotes a family of smooth symbols that vanish outside some fixed, bounded open set U_0 . When $q = q_t$ is independent of t , the flow

$$\varphi_t = \exp(tH_q)$$

induces a smooth family $\{\kappa_t\}_{0 \leq t \leq T}$ of symplectomorphisms of $\mathbb{R}^{2n}$, which equal the identity outside U_0 and satisfy $\partial_t \kappa_t = H_q = (\kappa_t)_* H_q$, where the last equality follows from Jacobi's Theorem (Theorem 2.10).

For q_t depending on t we define the flow of symplectomorphisms by demanding that

$$(11.1.1) \quad \begin{cases} \partial_t \kappa_t = (\kappa_t)_* H_{q_t} & (0 \leq t \leq T) \\ \kappa_0 = I. \end{cases}$$

REMARK. The statement (11.1.1) means that for each function $a \in C^\infty$, we have

$$(11.1.2) \quad \partial_t \kappa_t^* a = H_{q_t} \kappa_t^* a.$$

In fact,

$$\partial_t \kappa_t^* a = \langle da, \partial_t \kappa_t \rangle = \langle da, (\kappa_t)_* H_{q_t} \rangle = H_{q_t} \kappa_t^* a,$$

where $\langle \cdot, \cdot \rangle$ is the pairing of differential 1-forms and vector fields; see Section B.2 in Appendix B. Figure 11.1.1 and an explicit calculation in Section 11.3.1 below can also help in clarifying this. $\square$

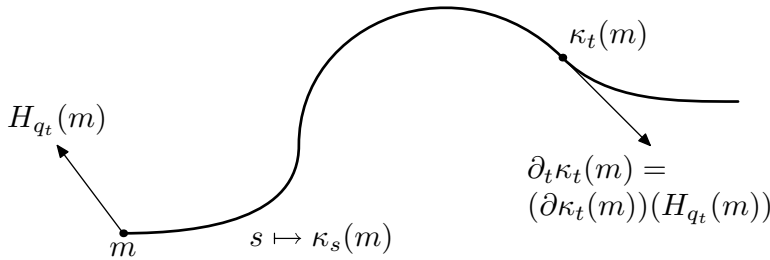


Figure 11.1.1. The relation between κ_t and H_{q_t} in (11.1.1): $\partial_t \kappa_t(m)$, $m \in \mathbb{R}^n \times \mathbb{R}^n$, is a tangent vector to the curve $s \mapsto \kappa_s(m)$, and it is equal to $(\kappa_t)_* H_{q_t}(\kappa_t(m)) = \partial \kappa_t(m)(H_{q_t}(m))$; see (B.2.1).

Write

$$Q(t) := q_t^w(x, hD).$$

Since $\{q_t\}_{0 \leq t \leq T} \subset S$, Theorem 10.1 implies that the operator equation

$$(11.1.3) \quad \begin{cases} hD_t F(t) + F(t)Q(t) = 0 & (0 \leq t \leq T) \\ F(0) = I \end{cases}$$

has a unique solution of unitary operators $\{F(t)\}_{0 \leq t \leq T}$ on $L^2(\mathbb{R}^n)$.

Observe for later use that (11.1.3) implies

$$(11.1.4) \quad hD_t(F(t)^{-1}) - Q(t)F(t)^{-1} = 0.$$

The choice of the convention in (11.1.3) is motivated by the agreement with the propagation of observables. If

$$A(t) := F(t)^{-1} A F(t),$$

for $A : L^2 \rightarrow L^2$, then

$$(11.1.5) \quad hD_t A(t) = [A(t), Q(t)],$$

or

$$\partial_t A(t) = \frac{i}{h} [Q(t), A(t)].$$

The evolution of observables following (11.1.5) is sometimes referred to as the *Heisenberg picture of quantum mechanics* as opposed to the *Schrödinger picture* in which the propagators $F(t)$ are considered directly.

THEOREM 11.1 (Egorov's Theorem). *Let m be an order function and suppose $a \in S(m)$.*

Then for $0 \leq t \leq T$, we have

$$(11.1.6) \quad F(t)^{-1} a^w(x, hD) F(t) = b_t^w(x, hD)$$

for a symbol $b_t \in S(m)$ having the form

$$(11.1.7) \quad b_t = \kappa_t^* a + O_S(h).$$

Theorem 10.1 asserts that $F(t) : H_h(m) \rightarrow H_h(m)$ and consequently the formula (11.1.6) makes sense on $H_h(m)$.

INTERPRETATION. Quantization given by the conjugation (11.1.6) with $F(t)$ follows the *Heisenberg picture* of quantum mechanics. The assertion is that the quantum evolution of $A = a^w(x, hD)$ is well approximated up to time T by the classical evolution, in the sense that the quantum operator is a quantization of the classical observable $\kappa_t^* a \in S(m)$, up to an error of order h in S . $\square$

Proof. 1. Write $A := a^w(x, hD)$ and define

$$(11.1.8) \quad B(t) := F(t)^{-1} A F(t).$$

Our aim is to prove

$$(11.1.9) \quad B(t) = b_t^w(x, hD)$$

for a symbol $b_t \in S(m)$ having the form (11.1.7).

2. To show this, we first note that $\kappa_t^* a \in S(m)$ since the vector field defining κ_t is smooth and has compact support. Hence we can define the pseudodifferential operators

$$B_0(t) := (\kappa_t^* a)^w(x, hD).$$

The dynamics (11.1.1) imply

$$\partial_t \kappa_t^* a = H_{q_t} \kappa_t^* a.$$

We can therefore calculate

$$\begin{aligned} hD_t B_0(t) &= \frac{h}{i} (\partial_t \kappa_t^* a)^w = \frac{h}{i} (H_{q_t} \kappa_t^* a)^w \\ &= \frac{h}{i} \{q_t, \kappa_t^* a\}^w = [Q(t), B_0(t)] + E(t), \end{aligned}$$

where

$$E(t) = e_t^w(x, hD) \quad \text{for } e_t \in h^2 \mathcal{S}.$$

To see this, remember that the expansion formulas in Theorem 4.12 show that e_t is computed in terms of expressions involving the product of q_t and $\kappa_t^* a$, thus has compact support and thus lies in $\mathcal{S}$. It follows from Theorem 4.21 that

$$E(t) = O_{L^2 \rightarrow L^2}(h^2).$$

Then the operator equations (11.1.3), (11.1.4) let us calculate

$$\begin{aligned} hD_t(F(t)B_0(t)F(t)^{-1}) &= hD_t F(t)B_0(t)F(t)^{-1} + F(t)hD_t B_0(t)F(t)^{-1} \\ &\quad + F(t)B_0(t)hD_t(F(t)^{-1}) \\ &= -F(t)Q(t)B_0(t)F(t)^{-1} + F(t)([Q(t), B_0(t)] + E(t))F(t)^{-1} \\ &\quad + F(t)B_0(t)Q(t)F(t)^{-1} \\ &= F(t)E(t)F(t)^{-1} = O_{L^2 \rightarrow L^2}(h^2). \end{aligned}$$

Integrating and dividing by h gives

$$\begin{aligned} F(t)B_0(t)F(t)^{-1} &= A + \frac{i}{h} \int_0^t F(s)E(s)F(s)^{-1} ds \\ &= A + O_{L^2 \rightarrow L^2}(h) \end{aligned}$$

for $0 \leq t \leq T$. Then according to (11.1.8)

$$B(t) := B_0(t) - \frac{i}{h} F(t)^{-1} \int_0^t F(s)E(s)F(s)^{-1} ds F(t),$$

so that

$$B_0(t) - B(t) = O_{L^2 \rightarrow L^2}(h).$$

3. We next construct families of pseudodifferential operators $B_k(t)$ so that for each $k = 1, 2, \dots$

$$(11.1.10) \quad \begin{cases} hD_t B_k(t) = [Q(t), B_k(t)] + E_k(t) \\ B_k(0) = A, \end{cases}$$

and

$$(11.1.11) \quad \begin{aligned} B_{k+1}(t) - B_k(t) &= O_{L^2 \rightarrow L^2}(h^{k+1}), \\ E_k(t) &= e_{k,t}^w(x, hD), \quad e_{k,t} \in h^{k+2}S. \end{aligned}$$

We proceed by induction. Assuming (11.1.11), define

$$c_{k+1,t} := \frac{i}{h}(\kappa_t)^* \int_0^t (\kappa_s^{-1})^* e_{k,s} ds.$$

Then

$$C_{k+1}(t) := c_{k+1,t}^w(x, hD) \in h^{k+1}\Psi.$$

Therefore

$$\begin{aligned} hD_t C_{k+1}(t) &= (hD_t c_{k+1,t})^w \\ &= \frac{h}{i} \left(\partial_t \left(\frac{i}{h}(\kappa_t)^* \int_0^t (\kappa_s^{-1})^* e_{k,s} ds \right) \right)^w \\ &= \left(\frac{h}{i} \left(H_{q_t} c_{k+1,t} + \frac{i}{h} e_{k,t} \right) \right)^w \\ &= [Q(t), C_{k+1}(t)] + E_k(t) - E_{k+1}(t), \end{aligned}$$

where $E_{k+1} = e_{k+1,t}^w$ for $e_{k+1,t} \in h^{k+3}S$. Hence

$$B_{k+1}(t) := B_k(t) - C_{k+1}(t)$$

satisfies (11.1.10) and (11.1.11) with k replaced by $k+1$.

4. To compare $B_k(t)$ with $B(t)$, we proceed as in step 1 above to obtain

$$hD_t (F(t)B_k(t)F(t)^{-1}) = F(t)E_k(t)F(t)^{-1}.$$

Integrating in t gives

$$(11.1.12) \quad \begin{aligned} B(t) - B_k(t) &= F(t)^{-1}(A - F(t)B_k(t)F(t)^{-1})F(t) \\ &= \frac{i}{h} \int_0^t F(t)^{-1}F(s)E_k(s)F(s)^{-1}F(t) ds. \end{aligned}$$

In particular,

$$(11.1.13) \quad B(t) = B_k(t) + O_{L^2 \rightarrow L^2}(h^{k+1}).$$

5. It remains to show that $B(t)$ is a pseudodifferential operator and that

$$B(t) - B_0(t) = O_{L^2 \rightarrow L^2}(h).$$

To do so, we invoke Beals's Theorem (Theorem 8.3) by showing that for any linear $l_1, \dots, l_M$, we have the estimate

$$(11.1.14) \quad \text{ad}_{l_1(x, hD)} \cdots \text{ad}_{l_M(x, hD)}(B(t) - B_0(t)) = O_{L^2 \rightarrow L^2}(h^{M+1}).$$

It suffices to show that

$$(11.1.15) \quad \text{ad}_{l_1(x,hD)} \cdots \text{ad}_{l_M(x,hD)}(B(t) - B_k(t)) = O_{L^2 \rightarrow L^2}(h^{k+1}).$$

Then in light of (11.1.13) it will follow that

$$\begin{aligned} \text{ad}_{l_1} \cdots \text{ad}_{l_M}(B(t) - B_0(t)) &= \text{ad}_{l_1} \cdots \text{ad}_{l_M}(B_k(t) - B_0(t)) \\ &\quad + \text{ad}_{l_1} \cdots \text{ad}_{l_M}(B(t) - B_k(t)) \\ &= O_{L^2 \rightarrow L^2}(h^{M+1}) + O_{L^2 \rightarrow L^2}(h^{k+1}) \\ &= O_{L^2 \rightarrow L^2}(h^{M+1}), \end{aligned}$$

if $k \geq M$.

6. To establish (11.1.15), we select a linear l and use (11.1.12):

$$\text{ad}_l(B(t) - B_k(t)) = \frac{i}{h} \int_0^t \text{ad}_l(F(t)^{-1}F(s)E_k(s)F(s)^{-1}F(t)) ds.$$

In view of (11.1.11), we have

$$\frac{i}{h} \text{ad}_l E_k(s) = O_{L^2 \rightarrow L^2}(h^{k+2}) = O_{L^2 \rightarrow L^2}(h^{k+1});$$

and a similar calculation holds for additional applications of ad_{l_j} .

Recalling the formulas (8.1.8), (8.1.9), we see that we now need to show that

$$(11.1.16) \quad \text{ad}_{l_1(x,hD)} \cdots \text{ad}_{l_M(x,hD)} F(t) = O_{L^2 \rightarrow L^2}(1)$$

for linear l_j on $\mathbb{R}^{2n}$. For this, observe from the derivation property (8.1.8) that $\text{ad}_l F(t)$ for a linear l satisfies

$$\begin{cases} hD_t(\text{ad}_l F(t)) + (\text{ad}_l F(t))Q(t) = -F(t)\text{ad}_l Q(t) \\ \text{ad}_l F(0) = 0. \end{cases}$$

Therefore

$$hD_t(\text{ad}_l F(t)F(t)^{-1}) = -F(t)\text{ad}_l Q(t)F(t)^{-1},$$

according to (11.1.4). Since $Q(t)$ is a pseudodifferential operator, we have $\text{ad}_l Q(t) = hQ_1(t)$, where $Q_1 = O_{L^2 \rightarrow L^2}(1)$. Hence

$$\text{ad}_l F(t) = i \int_0^t F(s)Q_1(s)F(s)^{-1}F(t) ds = O_{L^2 \rightarrow L^2}(1).$$

This argument can be iterated for additional applications of ad , thereby proving (11.1.16). $\square$

REMARKS. (i) The proof shows that if

$$a \sim \sum_{k=0}^{\infty} h^k a_k$$

for $a_k \in S(m)$, then

$$b_t \sim \kappa_t^* a_0 + \sum_{k=1}^{\infty} h^k b_t^k.$$

However, the higher-order terms are difficult to compute.

(ii) In Theorem 11.1 we assumed that q_t vanishes outside of a fixed compact set and that $a \in S(m)$. The proof applies directly when $q_t \in S$ and $a \in S$. We are then in the setting of Theorem 10.1 and the statements (11.1.6) and (11.1.7) are valid. $\square$

11.2. QUANTIZING SYMPLECTIC MAPPINGS

Now suppose we are given a symplectomorphism

$$\kappa : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n},$$

with $\kappa(0, 0) = (0, 0)$. Our next task is to *quantize* κ *locally*, which means to construct a unitary operator $F : L^2 \rightarrow L^2$ such that if $a \in S(m)$, then

$$F^{-1} a^w(x, hD) F = b^w(x, hD) \quad \text{microlocally near } (0, 0),$$

for a symbol $b \in S(m)$ of the form

$$b = \kappa^* a + O_S(h).$$

This assertion can be very useful in practice, since sometimes we can design κ so that $\kappa^* a$ is more tractable than a . See Chapter 12 for applications.

EXAMPLE. To illustrate this, consider $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$, a diffeomorphism defined near $0 \in \mathbb{R}^n$, $\gamma(0) = 0$. Then

$$Fu := \gamma^* u$$

is defined microlocally near $\{0\} \times \mathbb{R}^n$, since it is defined locally near 0. Consider

$$\kappa : (\gamma(x), \xi) \rightarrow (x, (\partial\gamma(x))^T \xi),$$

which is defined near $\{0\} \times \mathbb{R}^n$. Then the change of variables formula in Theorem 9.3 shows that

$$F^{-1} a^w(x, hD) F = b^w(x, hD) \quad \text{microlocally near } (0, \xi), \text{ for any } \xi \in \mathbb{R}^n,$$

where

$$b = \kappa^* a + O_S(h^2).$$

Since the statement is microlocal, that is, it involves localization to a compact subset of $\mathbb{R}^n \times \mathbb{R}^n$, the statement for $a \in \mathcal{S}$ in Theorem 9.3 is sufficient.

We can thus say that the pullback $u \mapsto \gamma^*u$ quantizes the symplectic extension of γ^{-1} . We recall that the symplectic extension of a diffeomorphism was defined in Theorem 2.6. $\square$

11.2.1. More on symplectic matrices. We will first of all require some more precise information about symplectic matrices. Remember from Example 1 in Section 2.3 that a real $2n \times 2n$ matrix K is symplectic provided

$$(11.2.1) \quad K^T J K = J.$$

THEOREM 11.2 (Decomposing symplectic matrices). *Let K be a symplectic matrix.*

Then we can write

$$(11.2.2) \quad K = \exp(B) \exp(A),$$

where A, B are real $2n \times 2n$ matrices satisfying

$$(11.2.3) \quad A^T = A, \quad A^T J + J A = 0,$$

and

$$(11.2.4) \quad B^T = -B, \quad B^T J + J B = 0.$$

REMARK. It is in general impossible to write an arbitrary symplectic matrix K as $\exp A$, with $A^T J + J A = 0$. We leave it as an exercise to show this for

$$K = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}.$$

$\square$

Proof. 1. Recall from (2.3.6) that K 's being symplectic is equivalent to

$$(11.2.5) \quad K^{-1} = J K^T J^T.$$

As K is invertible, we have the unique polar decomposition

$$K = QP,$$

where Q is orthogonal and P is symmetric, positive definite. From (11.2.5) we deduce that

$$(Q^T)^{-1}(P^T)^{-1} = (K^T)^{-1} = J Q J^T J P J^T,$$

whence the uniqueness of the polar decomposition implies

$$(Q^T)^{-1} = J Q J^T, \quad (P^T)^{-1} = J P J^T.$$

Consequently (2.3.6) implies that both Q and P are symplectic.

2. Since P is symmetric and positive definite, we can write $P = \exp A$, where $A = A^T$. We assert further that

$$AJ + JA = 0.$$

To see this, compute

$$\exp(-A) = P^{-1} = JP^T J^T = JPJ^T = J \exp(A) J^T = \exp(JAJ^T),$$

whence $-A = JAJ^T$.

3. We next identify $\mathbb{R}^{2n}$ with $\mathbb{C}^n$, using the relation $(x, \xi) = x + i\xi$. Since

$$\langle x + i\xi, x' + i\xi' \rangle_{\mathbb{C}^n} = \langle (x, \xi), (x', \xi') \rangle_{\mathbb{R}^{2n}} + i\sigma((x, \xi), (x', \xi')),$$

that Q is orthogonal and symplectic implies that it is unitary:

$$\begin{aligned} \langle Q(x + i\xi), Q(x' + i\xi') \rangle_{\mathbb{C}^n} &= \langle Q(x, \xi), Q(x', \xi') \rangle_{\mathbb{R}^{2n}} + i\sigma(Q(x, \xi), Q(x', \xi')) \\ &= \langle (x, \xi), (x', \xi') \rangle_{\mathbb{R}^{2n}} + i\sigma((x, \xi), (x', \xi')) \\ &= \langle x + i\xi, x' + i\xi' \rangle_{\mathbb{C}^n}. \end{aligned}$$

So Q is normal and thus diagonalizable on $\mathbb{C}^n$; there exists V so that $Q = V^{-1} \text{diag}(\lambda_1, \dots, \lambda_n) V$ for complex eigenvalues λ_k satisfying $|\lambda_k| = 1$ for $k = 1, \dots, n$. We can write

$$Q = \exp B$$

for $B = V^{-1} \text{diag}(\log \lambda_1, \dots, \log \lambda_n) V$. In particular $B = -B^*$ is anti-Hermitian on $\mathbb{C}^n$.

4. We note next that the action of Q on $\mathbb{C}^n$ is given by $Q_1 + iQ_2$, where Q_1 and Q_2 are real:

$$(Q_1 + iQ_2)(x + i\xi) = (Q_1x - Q_2\xi) + i(Q_2x + Q_1\xi).$$

Likewise the action of B is $B_1 + iB_2$ for real B_1 and B_2 . So the action of Q and B on $\mathbb{R}^{2n}$ is represented by the matrices

$$Q = \begin{pmatrix} Q_1 & -Q_2 \\ Q_2 & Q_1 \end{pmatrix}, \quad B = \begin{pmatrix} B_1 & -B_2 \\ B_2 & B_1 \end{pmatrix}.$$

Since B is anti-Hermitian acting on $\mathbb{C}^n$,

$$(11.2.6) \quad B_1^T = -B_1, \quad B_2^T = B_2;$$

and hence $B^T = -B$ acting on $\mathbb{R}^{2n}$. The properties (11.2.6) also imply that

$$-BJ + JB = 0. \quad \square$$

11.2.2. Deformation of symplectomorphisms. The plan is to locally embed κ into a one-parameter family of symplectomorphisms, generated by a family $\{q_t\}_{0 \leq t \leq 1}$ to which Egorov's Theorem (Theorem 11.1) applies.

So assume that $U_0 \subset \mathbb{R}^{2n} = \mathbb{R}^n \times \mathbb{R}^n$ is an open set *star-shaped* with respect to $(0, 0)$, meaning that

$$tU_0 := \{tw \mid w \in U_0\} \subseteq U_0$$

for all $0 \leq t \leq 1$. Suppose further that

$$\kappa : U_0 \rightarrow U_1 := \kappa(U_0)$$

is a symplectomorphism with $\kappa(0, 0) = (0, 0)$.

THEOREM 11.3 (Deforming symplectomorphisms. I). *There exists a continuous, piecewise smooth family*

$$\{\kappa_t\}_{0 \leq t \leq 1}$$

of symplectomorphisms $\kappa_t : U_0 \rightarrow U_t := \kappa_t(U_0)$ such that

$$(i) \quad \kappa_t(0, 0) = 0 \quad (0 \leq t \leq 1).$$

$$(ii) \quad \kappa_1 = \kappa, \quad \kappa_0 = I.$$

(iii) Also, there exists a family of smooth symbols $\{q_t\}_{0 \leq t \leq 1} \subset C^\infty(U_0)$, which is piecewise smooth in t , such that

$$(11.2.7) \quad \partial_t \kappa_t = (\kappa_t)_* H_{q_t} \quad (0 \leq t \leq 1).$$

Proof. 1. First assume that κ is linear, given by the symplectic matrix K . Theorem 11.2 asserts that we can write

$$K = \exp B \exp A$$

for matrices A, B satisfying

$$A^T J + JA = 0, \quad B^T J + JB = 0.$$

According to Theorem 2.4, $\exp(tA)$ and $\exp(tB)$ are both linear symplectomorphisms; and so also is their composition. A smooth deformation to the identity is now clear:

$$K_t := \exp(tB) \exp(tA) \quad (0 \leq t \leq 1).$$

2. For the general case that κ is nonlinear, set $K := \partial \kappa(0, 0)$. Then for $1/2 \leq t \leq 1$,

$$\kappa_t := K_{2-2t}^{-1} \circ \kappa$$

is a piecewise smooth family of symplectomorphisms satisfying

$$\kappa_1 = \kappa, \quad \partial \kappa_{1/2}(0, 0) = I.$$

For $0 \leq t \leq 1/2$, we set

$$\kappa_t(z) := \frac{1}{2t} \kappa_{1/2+t}(2tz).$$

3. Define $V_t := \partial_t \kappa_t$; we must show that

$$V_t = (\kappa_t)_* H_{q_t}$$

for some function q_t . According to Cartan's formula (Theorem B.3),

$$\mathcal{L}_{V_t} \sigma = V_t \lrcorner d\sigma + d(V_t \lrcorner \sigma).$$

But $\mathcal{L}_{V_t} \sigma = \partial_t \kappa_t^* \sigma = \frac{d}{dt} \sigma = 0$, since $\kappa_t^* \sigma = \sigma$. Furthermore, $d\sigma = 0$, and consequently $d(V_t \lrcorner \sigma) = 0$. Owing to Poincaré's Lemma (Theorem B.4), we have

$$\kappa_t^*(V_t \lrcorner \sigma) = dq_t$$

for a function q_t ; and this means that $V_t = (\kappa_t)_* H_{q_t}$. $\square$

Next we extend symplectomorphisms defined only locally to be globally defined on $\mathbb{R}^{2n}$. That will allow us to apply the deformation methods developed above.

THEOREM 11.4 (Deforming symplectomorphisms. II). *Suppose that*

$$\kappa : U_0 \rightarrow U_1$$

is a symplectomorphism and that κ extends as a symplectomorphism to an open set $\tilde{U}_0 \supset \supset U_0$, where $\tilde{U}_0$ is star-shaped with respect to $(0, 0)$.

(i) *Then κ extends to a symplectomorphism*

$$\tilde{\kappa} : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$$

which equals the identity outside a compact set.

(ii) *There exists a continuous, piecewise smooth family $\{\tilde{\kappa}_t\}_{0 \leq t \leq 1}$ of symplectomorphisms $\tilde{\kappa}_t : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ such that $\tilde{\kappa}_1 = \tilde{\kappa}$, $\tilde{\kappa}_0 = I$, and*

$$(11.2.8) \quad \partial_t \tilde{\kappa}_t = (\tilde{\kappa}_t)_* H_{q_t} \quad (0 \leq t \leq 1)$$

for a family of smooth compactly supported functions $\{q_t\}_{0 \leq t \leq 1} \subset C_c^\infty(\tilde{U}_0; \mathbb{R})$ that is piecewise smooth in t .

Proof. Let κ denote the extension of κ to the star-shaped neighborhood $\tilde{U}_0$. We then apply Theorem 11.3 to obtain a family of symbols $\tilde{q}_t \in C^\infty(\tilde{U}_0)$. Let $\chi \in C_c^\infty(U_0)$ satisfy $0 \leq \chi \leq 1$ with $\chi \equiv 1$ in U_0 , and put

$$q_t := \chi \tilde{q}_t \in C_c^\infty(U_0).$$

We then define $\tilde{\kappa}_t$ by (11.2.8). Since q_t and $\tilde{q}_t$ are the same within U_0 , we have $\kappa_t|_{U_0} = \tilde{\kappa}_t|_{U_0}$. In particular, $\kappa|_{U_0} = \tilde{\kappa}|_{U_0}$, $\tilde{\kappa} := \tilde{\kappa}_1$. Since q_t has compact support, $\tilde{\kappa}_t$ equals the identity outside a compact set. $\square$

11.2.3. Locally quantizing symplectomorphisms. We now apply Theorem 11.1 to the deformations provided by Theorem 11.4:

THEOREM 11.5 (Local quantization of symplectomorphisms). *Let $\kappa : U_0 \rightarrow U_1$ be a symplectomorphism satisfying the assumptions of Theorem 11.4.*

Then there exists a unitary operator

$$F : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$$

such that for all $a \in S(m)$, we have

$$(11.2.9) \quad F^{-1}a^w(x, hD)F = b^w(x, hD)$$

for a symbol $b \in S(m)$ satisfying

$$(11.2.10) \quad b|_{U_0} := \kappa^*(a|_{U_1}) + O_S(h).$$

Proof. According to Theorem 11.4, there exists a piecewise smooth family of symplectomorphisms $\kappa_t : U_0 \rightarrow U_t$ for $0 \leq t \leq 1$, such that $\kappa = \kappa_1$, $\kappa_0 = I$, and κ_t extends to $\tilde{\kappa}_t : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ satisfying

$$\partial_t \kappa_t = (\kappa_t)_* H_{q_t} \quad (0 \leq t \leq 1).$$

Furthermore, $q_t \in C_c^\infty(U)$ for some fixed $U_0 \subset \mathbb{R}^{2n}$.

We invoke Theorem 11.1 to obtain the family of unitary operators $\{F(t)\}_{0 \leq t \leq 1}$ and put $F = F(1)$. Then (11.1.7) shows that (11.2.9) and (11.2.10) hold. $\square$

11.2.4. Microlocal reformulation. It is useful to restate Theorem 11.5 without reference to the global properties of the operator F . Remember the notion of microlocality from Section 8.4.

THEOREM 11.6 (Local Egorov Theorem). *Let $\kappa : U_0 \rightarrow U_1$ be a symplectomorphism satisfying the assumptions of Theorem 11.4.*

Then there exists a unitary operator

$$F : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$$

such that for all $a \in S(m)$, we have

$$(11.2.11) \quad F^{-1}a^w(x, hD)F = b^w(x, hD) \quad \text{microlocally near } U_1 \times U_0,$$

where $b \in S(m)$ has the form

$$(11.2.12) \quad b := \kappa^* a + O_S(h).$$

In (11.2.12) we do not specify any neighborhoods, as we did in (11.2.10), since the statement needs to make sense only *microlocally* near $U_1 \times U_0$.

11.3. QUANTIZING LINEAR SYMPLECTIC MAPPINGS

Linear symplectic maps are not, strictly speaking, covered in Section 11.2. The problem is that the deformation given in the proof of Theorem 11.3 requires that κ be the identity outside some compact set. We therefore devote this section to a direct study of linear symplectic functions, deriving in particular an exact form of Egorov's Theorem.

11.3.1. Quantizing J . As motivation, we first discuss how to quantize J . The Fourier conjugation formula (4.2.14) shows that for $F = \mathcal{F}_h$ we have

$$(11.3.1) \quad F^{-1} a^w(x, hD) F = (J^* a)^w(x, hD),$$

and thus (11.1.7) holds without any error terms. Consequently we can say that *the semiclassical Fourier transform $F = \mathcal{F}_h$ quantizes J .*

Let us now reconsider this quantization of J from the perspective of deformations introduced in Section 11.2. Since $\kappa(x, \xi) = (\xi, -x)$, we can take

$$\kappa_t(x, \xi) = \left(\cos\left(\frac{t\pi}{2}\right) x + \sin\left(\frac{t\pi}{2}\right) \xi, -\sin\left(\frac{t\pi}{2}\right) x + \cos\left(\frac{t\pi}{2}\right) \xi \right)$$

for $0 \leq t \leq 1$. Then

$$(11.3.2) \quad \partial_t \kappa_t = (\kappa_t)_* H_q = H_q$$

for

$$(11.3.3) \quad q := \frac{\pi}{4}(|x|^2 + |\xi|^2).$$

The equation (11.3.2) means that $\partial_t \kappa_t(x, \xi)$, which is a vector tangent to the curve $s \mapsto \kappa_s(x, \xi)$ at $\kappa_t(x, \xi)$, is equal to $H_q(\kappa_t(x, \xi)) = \partial \kappa(x, \xi)(H_q(x, \xi))$. The last equality follows from the invariance of q under the flow; see Theorem 2.10.

THEOREM 11.7 (J quantized). *Suppose that an operator F is associated with the transformation J as in Theorem 11.5. That is, $F = F(1)$, where $F(t)$ solves (11.1.3) with $Q(t) = q^w(x, hD)$ and q is given by (11.3.3).*

Then

$$(11.3.4) \quad Fu(x) := \frac{e^{-\frac{\pi}{4}i}}{(2\pi h)^{\frac{n}{2}}} \int_{\mathbb{R}^n} e^{-\frac{i\langle x,y \rangle}{h}} u(y) dy = \frac{e^{-\frac{\pi}{4}i}}{(2\pi h)^{\frac{n}{2}}} \mathcal{F}_h u.$$

Proof. 1. Theorem 10.2 shows that $F(t) = \exp(-itQ/h)$ solves (11.1.3) and is unitary. Since $F(t)Q = QF(t)$ and

$$\mathcal{S} = \bigcap_{N \in \mathbb{N}} \langle Q \rangle^{-N} L^2,$$

we see that $F(t) : \mathcal{S} \rightarrow \mathcal{S}$.

2. To show that $F = F(1)$ verifies (11.3.4), we first show that for any order function m and any symbol $a \in S(m)$, we have

$$(11.3.5) \quad F^{-1}a^w(x, hD)F = a^w(-hD, x),$$

the operators acting on $\mathcal{S}$.

Write $A(t) := F(t)^{-1}a^w(x, hD)F(t)$. As in the proof of Theorem 11.1,

$$hD_t A(t) = \frac{\pi}{4}[-h^2 \Delta + |x|^2, A(t)].$$

Let $l = l(x, \xi)$ be a linear function on $\mathbb{R}^{2n}$ and consider as in Theorem 4.7 the exponential symbol

$$a_t(x, \xi) := \exp(i\kappa_t^* l(x, \xi)/h).$$

An explicit computation shows that

$$hD_t a_t^w(x, hD) = \frac{\pi}{4}[-h^2 \Delta + |x|^2, a_t^w(x, hD)].$$

Since (4.3.1) and (4.3.2) show that a Weyl operator is a superposition of exponentials of linear symbols, assertion (11.3.5) follows. This also shows that (11.3.5) remains valid for $a \in S(m)$.

3. Suppose now that $\tilde{F}$ is another unitary operator for which (11.3.5) holds. Then $\tilde{F} = cF$ for $c \in \mathbb{C}$, $|c| = 1$, as follows from applying Lemma 3.3 to $L = F^* \tilde{F}$. Since by Theorem 4.9 the Fourier transform satisfies (11.3.5) and since $(2\pi h)^{-n/2} \mathcal{F}_h$ is unitary, we deduce that

$$F = \frac{c}{(2\pi h)^{\frac{n}{2}}} \mathcal{F}_h.$$

4. Thus it remains to compute the constant c . For this, let us put $u_0 = \exp(-|x|^2/2)$ and consider the ODE

$$\begin{cases} hD_t u(t) = \frac{\pi}{4}(-h^2 \Delta + |x|^2)u(t) \\ u(0) = u_0. \end{cases}$$

Then $u(t) = F(t)^*u_0$. Since u_0 is the ground state of the harmonic oscillator and its eigenvalue is $\hbar$, we deduce that $u(t) = a(t)u_0$, where a solves the ODE

$$\begin{cases} \partial_t a = \frac{\pi i}{4} a \\ a(0) = 1. \end{cases}$$

Therefore $a(t) = \exp(\pi i t/4)$. Finally, we note that

$$e^{\pi i/4} u_0 = F(1)^* u_0 = \bar{c} (2\pi\hbar)^{-n/2} \mathcal{F}_h u_0 = \bar{c} u_0,$$

whence

$$c = \exp(-\pi i/4).$$

This proves (11.3.4) □

11.3.2. Quantizing linear symplectic mappings. Now suppose that $\kappa : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ is linear and symplectic and that it is given by the symplectic matrix K .

Preliminaries. To quantize K , we deform it to the identity. Assume first that K has the form

$$(11.3.6) \quad K = \exp A, \quad JA + A^T J = 0.$$

We write

$$A = \begin{pmatrix} A_1 & A_2 \\ A_3 & -A_1^T \end{pmatrix}, \quad A_2^T = A_2, \quad A_3^T = A_3.$$

We will define $F_K := F(1)$ where $F(t)$ is a family of unitary operators associated to $K_t := \exp(tA)$. To proceed in the spirit of Section 11.1, we want to find q_t such that

$$\partial_t K_t = K_t(H_{q_t}),$$

where $(K_t)_* = K_t$ since K_t is linear. We have $\partial_t K_t = \exp(tA)A$; and therefore

$$H_{q_t}(x, \xi) = (A_1 x + A_2 \xi) \partial_x + (A_3 x - A_1^T \xi) \partial_\xi.$$

Hence we can take $q_t = p(x, \xi)$ for

$$(11.3.7) \quad p(x, \xi) := \frac{1}{2} \langle A_2 \xi, \xi \rangle + \langle A_1 x, \xi \rangle - \frac{1}{2} \langle A_3 x, x \rangle.$$

LEMMA 11.8 (Propagators for quadratic Hamiltonians). *Suppose that*

$$P := p^w(x, hD),$$

where p is given by (11.3.7).

Then the equation

$$(11.3.8) \quad \begin{cases} hD_t F(t) + F(t)P = 0 & (t \in \mathbb{R}) \\ F(0) = I \end{cases}$$

is uniquely solved by a family $\{F(t)\}_{t \in \mathbb{R}}$ of bounded, unitary operators on $L^2(\mathbb{R}^n)$.

We remark that Theorem 10.2 cannot be invoked here, as P is not elliptic if the quadratic form $p(x, \xi)$ is not positive definite. Instead, motivated by Theorem 10.4, we propose an explicit formula for $F(t)$ for small values of t .

Proof. 1. Let us solve the Hamilton–Jacobi equation

$$\begin{cases} \partial_t \varphi + p(x, \partial_x \varphi) = 0 \\ \varphi(0, x, \eta) = \langle x, \eta \rangle. \end{cases}$$

As shown in the proof of Lemma 10.5, the solution $\varphi = \varphi(x, \eta)$ generates $K_t = \exp(tA)$ in the sense that

$$K_t : (\partial_\eta \varphi(x, \eta), \eta) \mapsto (x, \partial_x \varphi(x, \eta)).$$

If

$$K_t = \begin{pmatrix} A_t & B_t \\ C_t & D_t \end{pmatrix},$$

where

$$A_t = I + O(t), \quad D_t = I + O(t), \quad B_t, C_t = O(t),$$

then an explicit calculation shows

$$\varphi(t, x, \eta) = \frac{1}{2} \langle C_t A_t^{-1} x, x \rangle + \langle A_t^{-1} x, \eta \rangle - \frac{1}{2} \langle A_t^{-1} B_t \eta, \eta \rangle.$$

2. Motivated by the amplitude computation in Theorem 10.6, we define

$$F(t)u := \frac{(\det A_t)^{-\frac{1}{2}}}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\varphi(t, x, \eta) - \langle y, \eta \rangle)} u(y) dy d\eta$$

for $u \in \mathcal{S}$. The square root is well defined as A_t is close to the identity.

A calculation like that in the proof of Theorem 10.4 shows that the equation (11.3.8) holds exactly for small times t . We can now define $F(t) : \mathcal{S} \rightarrow \mathcal{S}$ for all times using the group property $F(t+s) = F(t)F(s)$. We similarly show that $F(t)^* = F(-t)$ is well defined on $\mathcal{S}$; and hence $F(t)^* F(t) = F(-t) F(t) = I$. $\square$

THEOREM 11.9 (Quantizing linear symplectic maps). *Let $K : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ be a linear symplectic mapping.*

(i) *There exists a unitary mapping $F_K : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ such that*

$$(11.3.9) \quad F_K^{-1} a^w(x, hD) F_K = (K^* a)^w(x, hD)$$

for all symbols $a \in S(m)$, m denoting any order function.

(ii) The operator F_K is unique up to a multiplicative factor. Hence for two linear symplectic maps K and L , we have

$$(11.3.10) \quad F_K F_L = c F_{KL}, \quad |c| = 1.$$

Remember that the pull-back of a under K is

$$K^* a(x, \xi) = a(K(x, \xi)).$$

INTERPRETATION. The quantization formula (11.3.9) for linear symplectic mappings generalizes (11.3.1), is a counterpart of the nonlinear formulas (11.2.9) and (11.2.11) from Section 11.2, and furthermore is exact since there are no error terms. $\square$

Proof. 1. We only need to consider the case $K = \exp A$, where $A^T J + J A = 0$, as Theorem 11.2 shows that general linear symplectic K can be written as a product of two such linear maps. Define $F_K = F(1)$ where $F(t)$ is provided by Lemma 11.8.

2. We need to show that

$$(11.3.11) \quad F(t)^{-1} a^w(x, hD) F(t) = (K_t^* a)^w(x, hD)$$

for $a \in S(m)$. We proceed as in the proof of Theorem 11.7 and consider

$$a(x, \xi) := \exp(il/h)$$

where $l = l(x, \xi)$ is real-valued and linear. We recall from Theorem 4.7 that

$$a^w(x, hD) = \exp(il(x, hD)/h).$$

This operator is bounded on L^2 and hence both sides of (11.3.11) are well defined. The superposition described in Lemma 4.10 and then Lemma 11.8 provide the general case.

3. Let $A(t) = F(t)^{-1} a^w(x, hD) F(t)$; so that

$$(11.3.12) \quad \begin{cases} hD_t A(t) = [P, A(t)] \\ A(0) = a^w(x, hD). \end{cases}$$

Define

$$a_t(x, \xi) := \exp(iK_t^* l(x, \xi)/h).$$

We claim that $a_t^w(x, hD)$ solves the same equation (11.3.12) as $A(t)$, and therefore

$$a_t^w(x, hD) = A(t).$$

Let us therefore calculate that

$$(11.3.13) \quad hD_t a_t^w = \left((\partial_t K_t^* l) e^{iK_t^* l/h} \right)^w = \left(H_p(K_t^* l) e^{iK_t^* l/h} \right)^w.$$

On the other hand, we see that

$$[P, a_t^w] = (p \# \exp(iK_t^* l/h) - \exp(iK_t^* l/h) \# p)^w.$$

Since p is quadratic, (4.3.6) and (4.3.9) show that

$$p \# \exp(iK_t^* l/h) - \exp(iK_t^* l/h) \# p = H_p(K_t^* l) e^{iK_t^* l/h}.$$

This is the right-hand side of (11.3.13).

4. To see the uniqueness up to a multiplicative constant, we apply Lemma 3.3 to $F_K \tilde{F}_K^*$ where both F_K and $\tilde{F}_K$ satisfy (11.3.11). The conclusion that $K \mapsto F_K$ gives a projective representation of the group of linear symplectic maps (that is, a representation up to a multiplicative constant) is immediate from the uniqueness. $\square$

11.3.3. An explicit formula. We next provide a representation for F_K in a special case.

THEOREM 11.10 (An explicit quantization rule). *Suppose that*

$$K = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

is symplectic and $\det A \neq 0$.

Then we have the formula

$$(11.3.14) \quad F_K u(x) = \frac{|\det A|^{-\frac{1}{2}}}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)} u(y) dy d\eta$$

for $u \in \mathcal{S}$, where

$$(11.3.15) \quad \varphi(x, \eta) := \frac{1}{2} \langle CA^{-1}x, x \rangle + \langle A^{-1}x, \eta \rangle - \frac{1}{2} \langle A^{-1}B\eta, \eta \rangle.$$

Proof. 1. Let Λ_K be the twisted graph of K :

$$\Lambda_K := \{(x, y, \xi, -\eta) \mid (x, \xi) = K(y, \eta)\}.$$

The condition $\det A \neq 0$ means that linear map $\Lambda_K \rightarrow \mathbb{R}^n \times \mathbb{R}^n$ given by

$$(11.3.16) \quad (x, y, \xi, \eta) \mapsto (x, \eta)$$

is surjective:

$$(Ay + B\eta, Cy + D\eta, x, -\eta) \mapsto (Ay + B\eta, -\eta).$$

From (2.5.7) we see that there exists a generating function $\varphi = \varphi(x, \eta)$ such that K is implicitly given by

$$(x, \partial_x \varphi(x, \eta)) \mapsto (\partial_\eta \varphi(y, \eta), \eta).$$

By following the proof of Theorem 2.14 or by direct inspection we see that φ is given by (11.3.15), up to an additive constant.

2. As suggested by the proof of Theorem 10.4, we arrive at the definition (11.3.14). It remains to check that F_K is unitary and that

$$(11.3.17) \quad a^w(x, hD)F_K = F_K(K^*a)^w(x, hD).$$

3. The unitarity of F_K follows once we calculate the Schwartz kernel of $F_K^*F_K$:

$$\begin{aligned} (F_K^*F_K)(x, y) &= \frac{\det A}{(2\pi h)^{2n}} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\langle y, \eta \rangle - \varphi(x, \eta) + \varphi(z, \zeta) - \langle y, \zeta \rangle)} dz d\zeta d\eta \\ &= \frac{\det A}{(2\pi h)^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\varphi(y, \eta) - \varphi(x, \eta))} d\eta \\ &= \frac{\det A}{(2\pi h)^n} e^{\frac{i}{2h}(\langle Cy, y \rangle - \langle Cx, x \rangle)} \int_{\mathbb{R}^n} e^{\frac{i}{h}\langle \eta, A^{-1}(x-y) \rangle} d\eta \\ &= \delta(x - y), \end{aligned}$$

where in the last integral (interpreted in the $\mathcal{S}'$ sense) we made a change of variables $\eta = A\xi$.

4. To obtain (11.3.17), we check it first for the linear function

$$a(x, \xi) = l(x, \xi) = \langle x^*, x \rangle + \langle \xi^*, \xi \rangle.$$

Using the properties of A, B, C, D , we see that

$$\begin{aligned} xe^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)} &= AhD_\eta e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)} + (Ay + B\eta)e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)}, \\ hD_x e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)} &= ChD_\eta e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)} + (Cy + D\eta)e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)}. \end{aligned}$$

Since the variable η appears only in the phase, the terms with D_η disappear after integration. Also for any $n \times n$ matrix M ,

$$\begin{aligned} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} (M\eta) e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)} u(y) dy d\eta \\ = \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)} (MhD_y) u(y) dy d\eta. \end{aligned}$$

Hence

$$\begin{aligned} l(x, hD)F_K u &= \frac{|\det A|^{-\frac{1}{2}}}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}(\varphi(x, \eta) - \langle y, \eta \rangle)} (K^*l)(y, hD_y) u(y) dy d\eta \\ &= F_K(K^*l)(x, hD)u. \end{aligned}$$

This implies that

$$e^{il(x, hD)/h} F_K = F_K e^{iK^*l(x, hD)/h}.$$

The decomposition formula in Theorem 4.7 shows that (11.3.11) is therefore valid for all $a \in S(m)$. $\square$

REMARK. The construction in the proof of Lemma 11.8 is a special case of the formula (11.3.14). A representation of $F(t)$ as an oscillatory integral for larger times would involve linear symplectic transformations for which the map (11.3.16) is no longer surjective. The change of the generating function of Λ_K produces phase factors of the form $\exp(im\pi/4)$. This leads to the study of the metaplectic group and of the Maslov index. See the references in Section 11.5 for more. $\square$

11.4. EGOROV'S THEOREM FOR LONGER TIMES

In Section 11.1 we proved Egorov's Theorem relating the quantum and classical evolutions of time-dependent Hamiltonians. In this section we show that for time-independent Hamiltonians Egorov's Theorem (Theorem 11.1) in fact holds for times

$$0 \leq t \leq C \log h^{-1}.$$

That means that the classical/quantum correspondence described by (11.4.8) and (11.4.9) below is valid up to times comparable to $\log(h^{-1})$, known as the *Ehrenfest time*.

11.4.1. Estimates for flows. Suppose for the following that $m \geq 1$ is an order function and $p \in S(m)$. We assume as well that p is *independent of h* and that

$$C + p \geq cm$$

for constants $c > 0, C \geq 0$.

We introduce the corresponding Hamiltonian flow

$$\varphi_t = \exp tH_p$$

and finally define

$$(11.4.1) \quad \Gamma_R := \sup_{\{(x,\xi)|p_0 \leq R\}} \{|\partial^2 p(x,\xi)|\}$$

for $R > 0$.

LEMMA 11.11 (Exponential estimates for flows). *For each $\gamma_0 > \Gamma_R$ and each point (x, ξ) satisfying $p(x, \xi) \leq R$ we have the estimate*

$$(11.4.2) \quad |\partial^\alpha \varphi_t(x, \xi)| \leq C_\alpha e^{\gamma_0 |\alpha| t}, \quad \alpha \in \mathbb{N}^{2n}, \quad |\alpha| > 0.$$

Proof. 1. The set $\{p \leq R\}$ is preserved by φ_t . Consequently on this set

$$|\partial^\alpha p| \leq C_\alpha m \leq C_\alpha(1 + p) \leq C_\alpha$$

for all multiindices α .

2. The proof of (11.4.2) is an induction on $|\alpha|$. For $\alpha = 0$, we have a weaker estimate on $|\varphi_t(x, \xi)| \leq Ct$, which is valid since

$$(11.4.3) \quad \frac{d}{dt} \varphi_t = H_p(\varphi_t)$$

and H_p is bounded on $\{p \leq R\}$.

3. Now assume $|\alpha| = \ell$ and suppose the estimate (11.4.2) is valid for all multiindices of order less than or equal to $\ell - 1$. We differentiate (11.4.3), to find

$$(11.4.4) \quad \frac{d}{dt} (\partial^\alpha \varphi_t) = \partial H_p(\varphi_t) \partial^\alpha \varphi_t + \rho_t,$$

where ρ_t is a sum of terms having the form

$$g \partial^{\alpha_1} \varphi_t \cdots \partial^{\alpha_k} \varphi_t$$

for $\alpha_1 + \cdots + \alpha_k = \alpha$ and $0 < |\alpha_j| < |\alpha| = \ell$ ($j = 1, \dots, k$). The induction hypothesis implies

$$|\rho_t| \leq C e^{\gamma_0 |\alpha| |t|}.$$

From (11.4.4) it follows that

$$\frac{d}{dt} |\partial^\alpha \varphi_t| \leq \Gamma_R |\partial^\alpha \varphi_t| + C e^{\gamma_0 |\alpha| |t|}.$$

We use Gronwall's inequality to finish the proof of (11.4.2). □

REMARK. The flows can also be estimated using a bound on Lyapunov exponents:

$$(11.4.5) \quad \tilde{\Gamma}_R := \lim_{t \rightarrow \infty} \frac{1}{t} \sup_{p_0 \leq R} \log |\partial \varphi_t|.$$

The limit exists since the function $t \mapsto \sup_{p \leq R} \log |\partial \varphi_t|$ is subadditive. (A function f is *subadditive* if $f(t+s) \leq f(t) + f(s)$, in which case $\lim_{t \rightarrow \infty} f(t)/t$ exists.)

Then Lemma 11.11 holds for $\gamma_0 > \tilde{\Gamma}_R$, as shown in Anantharaman–Nonnenmacher [A–N, Section 5.2] and Dyatlov–Guillarmou [Dy–Gu, Appendix A]. A striking example is provided by $p = |\xi|^2 + |x|^2$, for which $\Gamma_R = 2$ whereas $\tilde{\Gamma}_R = 0$. However, Γ_R is much easier to compute. □

11.4.2. Egorov's Theorem for long times. We will hereafter write

$$(11.4.6) \quad F(t) := e^{-itP/h} \quad (t \in \mathbb{R})$$

to denote the propagators generated by the operator $P = p^w(x, hD)$, as discussed in Theorem 10.2. We will also use the notation that

$$b \in S_\delta(m^{-\infty})$$

provided

$$|\partial^\alpha b| \leq C_{\alpha N} h^{-\delta|\alpha|} m^{-N}$$

for all multiindices α and all $N \geq 0$.

THEOREM 11.12 (Egorov's Theorem up to Ehrenfest time). *Suppose $a \in S$ and*

$$\text{spt } a \subset \{p \leq R\}$$

for some $R > 0$.

For any $\gamma > \Gamma_R$, $T \geq 0$, and $\delta \in [0, 1/2)$, if

$$(11.4.7) \quad |t| \leq T + \frac{\delta}{\gamma} \log(h^{-1}),$$

then

$$(11.4.8) \quad F(t)^{-1} a^w(x, hD) F(t) = b_t^w(x, hD)$$

for a symbol $b_t \in S_\delta(m^{-\infty})$ having the form

$$(11.4.9) \quad b_t = \varphi_t^* a + O_{S_\delta(m^{-\infty})}(h^{2-3\delta}).$$

INTERPRETATION. The statement $b_t - \varphi_t^* a = O(h^{2-3\delta})$ means that the quantum evolution of $a^w(x, hD)$ given by the conjugation with $\exp(-itP/h)$ is well approximated by the classical evolution up to the time $\delta/\gamma \log(h^{-1})$. Until that time we also know that the quantum evolved operator is a quantization of the (slightly exotic) symbol b_t .

We also note that when $\delta = 0$, we have an improved estimate valid up to any fixed time T . $\square$

REMARKS. (i) The theorem is also valid for $P = p^w(x, hD)$ for a real-valued symbol p having the form

$$p = p_0 + h^2 p_1,$$

where $p_0, p_1 \in S(m)$ and p_0 satisfies the bound $C + p_0 \geq cm$.

(ii) When $m = 1$, we can drop the assumption on the support of a and consider any $a \in S$.

Both assertions follow from the proof below. $\square$

We will mimic the proof of Theorem 11.1, paying careful attention to the dependence on t .

Proof. 1. Select γ_0 so that

$$\gamma > \gamma_0 > \Gamma_R.$$

We always assume that the time t satisfies

$$(11.4.10) \quad 0 \leq t \leq T + \frac{\delta}{\gamma} \log \frac{1}{h},$$

as the case of negative times is the same. Consequently,

$$(11.4.11) \quad t^k e^{\gamma_0 q t} \leq C_{kp} h^{-q\delta}$$

for all $k, q \geq 0$,

As noted previously, $|\partial^\alpha p| \leq C_\alpha$ within $\{p \leq R\}$. Since $\text{spt } a \subset \{p \leq R\}$, Lemma 11.11 shows that

$$(11.4.12) \quad |\partial^\alpha \varphi_t^* a| \leq C_\alpha e^{\gamma_0 |\alpha| t}.$$

In particular, we have $\text{spt } \varphi_t^* a \subset \{p \leq R\}$ and

$$(11.4.13) \quad \varphi_t^* a \in S_\delta.$$

2. Put $A := a^w(x, hD)$ and define

$$B(t) := F(t)^{-1} A F(t).$$

As useful approximations, we construct a family of pseudodifferential operators

$$B_k(t) = b_{k,t}^w(x, hD), \quad E_k(t) = e_{k,t}^w(x, hD) \quad (k = 0, 1, \dots)$$

such that

$$(11.4.14) \quad \begin{cases} hD_t B_k(t) = [P, B_k(t)] + E_k(t) + \mathcal{O}_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty) \\ B_k(0) = A, \end{cases}$$

where

$$(11.4.15) \quad \begin{cases} \partial^\alpha (b_{k,t} - b_{k-1,t}) = O(t^k h^{k(2-3\delta)} e^{\gamma_0 |\alpha| t}) \\ \partial^\alpha e_{k,t} = O(t^k h^{1+(k+1)(2-3\delta)} e^{\gamma_0 |\alpha| t}) \\ \text{spt } e_{k,t} \subset \{p \leq R\}, \end{cases}$$

with the convention that $b_{-1,t} \equiv 0$.

3. We will build $B_k(t)$ and $E_k(t)$ by induction. For $k = 0$, define

$$B_0(t) := (\varphi_t^* a)^w(x, hD).$$

Then using the definition of the flow (11.4.3), we calculate that

$$\begin{aligned} hD_t B_0(t) &= \frac{h}{i} (\partial_t \varphi_t^* a)^w = \frac{h}{i} (H_p \varphi_t^* a)^w \\ &= \frac{h}{i} \{p, \varphi_t^* a\}^w \\ &= [P, B_0(t)] + E_0(t) + \mathcal{O}_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty). \end{aligned}$$

In this expression

$$P = p^w(x, hD), \quad E_0(t) = e_{0,t}^w(x, hD)$$

for a symbol

$$e_{0,t}(x, \xi) \sim 2 \sum_{l=1}^{\infty} \frac{h^{2l+1}}{(2l+1)!} \left(\frac{i}{2} \sigma(D_z, D_w) \right)^{2l+1} p(z) \varphi_t^* a(w)|_{z=w=(x,\xi)}$$

satisfying $\text{spt } e_{0,t} \subset \{p \leq R\}$. We used the composition formula (4.3.9) to define $e_{0,t}$. This is justified in view of (11.4.12) and the assumption (11.4.10). According to (11.4.13) each term a_l in the expansion belongs $h^{(2l+1)(1-\delta)} S_\delta$ and so we can apply Theorem 4.15. Since we also have

$$\partial^\alpha a_l = O(h^{2l+1} e^{\gamma_0 |\alpha| t}),$$

it follows that

$$\partial^\alpha e_{0,t} = O(h^3 e^{\gamma_0(3+|\alpha|)t}) = O(h^{3-3\delta} e^{\gamma_0 |\alpha| t}).$$

This confirms (11.4.14) and (11.4.15) for $k = 0$.

4. Next, assume by induction that (11.4.14) and (11.4.15) hold for some k , and put

$$c_{k+1,t} := \frac{i}{h} \int_0^t \varphi_{t-s}^* e_{k,s} ds, \quad C_{k+1}(t) := c_{k+1,t}^w(x, hD).$$

Lemma 11.11 and (11.4.15) imply

$$\begin{aligned} (11.4.16) \quad |\partial^\alpha c_{k+1,t}| &\leq C_\alpha h^{(k+1)(2-3\delta)} \int_0^t s^k e^{\gamma_0 |\alpha| s} ds \\ &\leq C_\alpha t^{(k+1)(2-3\delta)} h^{(k+1)/2} e^{\gamma_0 |\alpha| t}. \end{aligned}$$

Therefore

$$\begin{aligned} hD_t C_{k+1}(t) &= (hD_t c_{k+1,t}^w(x, hD))^w \\ &= \frac{h}{i} \left(\partial_t \left(\frac{i}{h} \int_0^t \varphi_{t-s}^* e_{k,s} ds \right) \right)^w \\ &= \left(\frac{h}{i} \left(H_p c_{k+1,t} + \frac{i}{h} e_{k,t} \right) \right)^w \\ &= [P, C_{k+1}(t)] + E_k(t) - E_{k+1}(t) + \mathcal{O}_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty), \end{aligned}$$

where

$$E_{k+1} = e_{k+1,t}^w(x, hD),$$

with $\text{spt } e_{k+1,t} \subset \{p \leq R\}$ and

$$e_{k+1,t}(x, \xi) \sim 2 \sum_{l=1}^{\infty} \frac{h^{2l+1}}{(2l+1)!} \left(\frac{i}{2} \sigma(D_z, D_w) \right)^{2l+1} p(z) c_{k+1,t}(w)|_{z=w=(x,\xi)}.$$

In view of (11.4.16) and (11.4.11) we obtain

$$\begin{aligned}\partial^\alpha e_{k+1,t} &= O(t^{k+1} h^{(k+1)(2-3\delta)+3} e^{\gamma_0(3+|\alpha|)t}) \\ &= O(t^{k+1} h^{1+(k+2)(2-3\delta)} e^{\gamma_0|\alpha|t}).\end{aligned}$$

Then

$$B_{k+1}(t) := B_k(t) - C_{k+1}(t)$$

where $B_{k+1}(t) = b_{k+1,t}^w(x, hD)$ for a symbol $b_{k+1,t} \in S_\delta$ provides the operator satisfying (11.4.14) and (11.4.15), with k replaced by $k+1$.

5. We next compare $B_k(t)$ with $B(t)$. To do so, we proceed again as in the proof of Theorem 11.1. A calculation from that proof gives

$$hD_t(F(t)B_k(t)F(t)^{-1}) = F(t)(E_k(t) + \mathcal{O}_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty))F(t)^{-1}.$$

Integrating in t and conjugating by $F(t)^{-1}$, we see that

$$\begin{aligned}(11.4.17) \quad B(t) - B_k(t) &= F(t)^{-1}(A - F(t)B_k(t)F(t)^{-1})F(t) \\ &= \frac{i}{h} \int_0^t F(s-t)(E_k(s) + \mathcal{O}_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty))F(t-s)ds \\ &= O_{L^2 \rightarrow L^2} \left(h^{(k+1)(2-3\delta)} t^{k+1} \right).\end{aligned}$$

In the last estimate we used the unitarity of $F(t)$ and the bound on the L^2 norm in terms of derivatives of $e_{k,t}$ given in part (ii) of Theorem 4.23.

6. Next, we assert that for any linear $l_1, \dots, l_M$, we have

$$(11.4.18) \quad \text{ad}_{l_1} \cdots \text{ad}_{l_M}(B(t) - B_k(t)) = O_{L^2 \rightarrow L^2}(h^{(k+1)/2}).$$

As we will see in step 9 of the proof, the exact power of h is not important, so long as it goes to infinity with k .

To establish (11.4.18), use (11.4.17). We have

$$\begin{aligned}\text{ad}_l(B(t) - B_k(t)) &= \frac{i}{h} \int_0^t \text{ad}_l(F(s-t)E_k(s)F(t-s)) ds \\ &\quad + \frac{i}{h} \int_0^t \text{ad}_l(F(s-t)\mathcal{O}_{\mathcal{S}' \rightarrow \mathcal{S}}(h^\infty)F(t-s)) ds,\end{aligned}$$

and a similar expression is valid for $M > 1$.

We will focus upon the first term on the right-hand side, the second term being handled by the same methods. Since $E_k(s) = e_{k,s}^w$ for a symbol $e_{k,s} \in S_\delta$ with $\text{spt } e_{k,s} \subset \{p \leq R\}$, we can write

$$E_k(s) = (P+i)^{-N} \tilde{E}_k(s) (P+i)^{-N}, \quad \tilde{E}_k(s) := (P+i)^N E_k(s) (P+i)^N,$$

where

$$\tilde{E}_k(s) = \tilde{e}_{k,s}^w(x, hD), \quad \tilde{e}_{k,s} \in h^{1+(k+1)(2-3\delta)} S_\delta.$$

Since $\delta < 1/2$, the remark near the end of Section 8.1 shows that

$$\frac{i}{h} \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} \widetilde{E}_k(s) = O_{L^2 \rightarrow L^2}(h^{(k+1)(2-3\delta)+M(1-\delta)}).$$

Recalling (8.1.8) and (8.1.9), to obtain (11.4.18) we only need to show that

$$(11.4.19) \quad \begin{aligned} \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} ((P+i)^{-N} F(t)) &= O_{L^2 \rightarrow L^2}(t^M), \\ \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} ((P+i)^{-N} F(t)^{-1}) &= O_{L^2 \rightarrow L^2}(t^M) \end{aligned}$$

for l_j linear on $\mathbb{R}^{2n}$ and for $N \geq 2M$. (Recall the assumption (11.4.10) which gives the control of powers of t .)

7. To make the argument clear, we first consider the case of $M = 1$ and look at the equation satisfied by $\operatorname{ad}_l((P+i)^{-N} F(t))$:

$$(11.4.20) \quad \begin{aligned} hD_t (\operatorname{ad}_l((P+i)^{-N} F(t))) &= \operatorname{ad}_l((P+i)^{-N} hD_t F(t)) \\ &= -\operatorname{ad}_l(F(t)(P+i)^{-N} P) \\ &= -\operatorname{ad}_l(F(t)(P+i)^{-N}) P - F(t)(P+i)^{-N} \operatorname{ad}_l P, \end{aligned}$$

with $\operatorname{ad}_l(F(0)(P+i)^{-N}) = \operatorname{ad}_l(P+i)^{-N} = O_{L^2 \rightarrow L^2}(h)$. We also note that for $N \geq 2$, the pseudodifferential calculus gives

$$\begin{aligned} Q_N &:= (P+i)^{-N} \operatorname{ad}_l P \\ &= (P+i)^{-1} \operatorname{ad}_l((P+i)^{-N+1} P) + (P+i)^{-1} (\operatorname{ad}_l(P+i)^{-N+1}) P \\ &= O_{L^2 \rightarrow L^2}(h). \end{aligned}$$

Applying Duhamel's formula to (11.4.20) gives

$$\begin{aligned} \operatorname{ad}_l((P+i)^{-N} F(t)) &= F(t) \operatorname{ad}_l((P+i)^{-N} F(0)) - \frac{i}{h} \int_0^t F(s) Q_N F(t-s) ds \\ &= O_{L^2 \rightarrow L^2}(t). \end{aligned}$$

This argument applies to $F(t)^{-1}$ and shows (11.4.19) for $M = 1$.

8. To check the first claim in (11.4.19) for $M > 1$, we modify it to

$$(11.4.21) \quad \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} ((P+i)^{-N} F(t)) (P+i)^K = O_{L^2 \rightarrow L^2}(t^M)$$

for $N \geq 2M + K$ and proceed by induction on M . We note that (11.4.21) holds for $M = 0$ and $N \geq K$.

Put

$$S(t) := \operatorname{ad}_{l_{M+1}} \cdots \operatorname{ad}_{l_1} ((P+i)^{-N} F(t)) (P+i)^K.$$

Then, arguing as in step 7, we get

$$hD_t S(t) = S(t)P + R(t),$$

with

$$\begin{aligned} R(t) &= \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} ((P+i)^{-N} F(t)) (P+i)^{K+2} (P+i)^{-2} \operatorname{ad}_{l_{M+1}} P \\ &= O_{L^2 \rightarrow L^2}(ht^M). \end{aligned}$$

We note that, as required by the induction hypothesis, we need to take

$$N \geq 2M + K + 2 = 2(M+1) + K.$$

As in step 7 above an application of the Duhamel formula gives (11.4.21).

9. It remains to show that $B(t)$ is a pseudodifferential operator, and to do so, we invoke the remark after the proof of Beals's Theorem (Theorem 8.3) and hence need to show for any linear $l_1, \dots, l_M$ that

$$(11.4.22) \quad \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} B(t) = O_{L^2 \rightarrow L^2}(h^{M(1-\delta)}).$$

Using (11.4.18), we have

$$\begin{aligned} \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} B(t) &= \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} B_k(t) + \operatorname{ad}_{l_M} \cdots \operatorname{ad}_{l_1} (B(t) - B_k(t)) \\ &= O_{L^2 \rightarrow L^2}(h^{M(1-\delta)}) + O_{L^2 \rightarrow L^2}(h^{(k+1)/2}) \\ &= O_{L^2 \rightarrow L^2}(h^{M(1-\delta)}), \end{aligned}$$

if $(k+1)/2 \geq M$.

Retracing our steps, we see that $B(t) = b_t^w(x, hD)$, where $b_t \in S_\delta$. Since the arguments in step 6 above also apply to $(P+i)^N B(t)$ for any N , we see furthermore that $b_t \in S_\delta(m^{-\infty})$. $\square$

11.5. NOTES

The definition of the quantization of symplectomorphisms using deformation follows the *Heisenberg picture of quantum mechanics*. The essence of the proof of Theorem 11.1 comes from Taylor [Ta, §Section 8.1]. For a semiclassical version, consult Christiansen [Chr, Section 3]. See Bony [B] and the references therein for generalizations of Theorem 11.1, allowing the symbols $\{q_t\}_{0 \leq t \leq T}$ to belong to more general classes.

See Dimassi–Sjöstrand [D-S, Appendix to Chapter 7] for another self-contained presentation of semiclassical quantization of linear symplectic relations and Folland [F, Chapter 4] for more on this and for references. For the discussion of the Maslov index we refer to Hörmander [H3, Section 21.6], Guillemin–Sternberg [G-St], [G-St1], and Lion–Vergne [L-V]. See Robert [R1] for the closely related theory of coherent states.

The term *Ehrenfest time* has origins in the physics literature (Chirikov [Ch], Zaslavsky[Za]), and a standard reference is Bouzouina–Robert [B-R]. The precise dependence upon the dynamics of the constant in front of $\log(1/h)$ is not clear. In [B-R] a finer version is given for a class of Schrödinger operators. An interesting application to scattering theory is Bony–Burq–Ramond [B-B-R]. For a physics perspective and references, see Zurek [Zu].

NORMAL FORMS

- 12.1 Overview
- 12.2 Normal forms: real symbols
- 12.3 Propagation of singularities
- 12.4 Normal forms: complex symbols
- 12.5 Quasimodes, pseudospectra
- 12.6 Notes

12.1. OVERVIEW

We devote this chapter to deriving microlocal *normal forms* for principal type operators, that is, operators P whose symbols p satisfy

$$(12.1.1) \quad \partial p \neq 0 \quad \text{on } \{p = 0\}.$$

The main endeavor will be to find ways to microlocally convert such operators P into much simpler forms, either hD_{x_1} for real p (Section 12.2) or $hD_{x_1} \pm ix_1$ for complex p (Section 12.4).

MOTIVATION: WKB method. It is remarkable that entire classes of differential and pseudodifferential equations can be microlocally transformed into the same simple transport PDE $hD_{x_1}u = 0$, but we should understand that this is meaningful only for highly oscillatory solutions in the limit $h \rightarrow 0$, with the transformations having good mapping properties on L^2 .

We can provide some motivation by discussing briefly Wentzel–Kramers–Brillouin (WKB) approximation for solutions of a linear PDE, say of the

form

$$(12.1.2) \quad P(h)u(h) = \sum_{|\alpha| \leq N} a_\alpha(x)(hD)^\alpha u(h) = 0.$$

Let us look for a solution having the so-called *WKB form*

$$(12.1.3) \quad u(h) = ae^{\frac{i}{h}\varphi},$$

where φ is independent of h and where the amplitude a has the expansion

$$(12.1.4) \quad a \sim \sum_{k=0}^{\infty} a_k(x)h^k.$$

If we formally plug (12.1.3) and (12.1.4) into (12.1.2) and compare like powers of h , we discover that φ solves the nonlinear PDE

$$(12.1.5) \quad p(x, \partial\varphi) = 0$$

for

$$(12.1.6) \quad p(x, \xi) = \sum_{|\alpha| \leq N} a_\alpha(x)\xi^\alpha.$$

As explained for instance in [E, Section 3.2], we can locally study the PDE (12.1.5) by introducing the *characteristic equations*

$$\begin{cases} \dot{x} = \partial_\xi p(x, \xi) \\ \dot{\xi} = -\partial_x p(x, \xi), \end{cases}$$

which are of course the Hamiltonian dynamics generated by the symbol p .

Now it is certainly plausible that we can sometimes change variables to make this ODE flow into translation in the x_1 -direction, dynamics corresponding to the operator D_{x_1} . Consequently it may be reasonable to expect that the PDE (12.1.2) is, in this WKB setting at least, somehow “equivalent” to the linear transport equation $hD_{x_1}u = 0$ as $h \rightarrow 0$.

This motivation for the theory of normal forms is at best only heuristic, since we may not in general be able to write the solution of (12.1.2) in the simple WKB form (12.1.3). (But remember that in Section 10.2 we have represented solutions of evolution equations using a Fourier integral operator (10.2.4), which is a superposition of WKB states.) $\square$

An example will further motivate the later theory:

EXAMPLE: Helmholtz’s equation. Let us consider *Helmholtz’s equation*

$$(12.1.7) \quad (-h^2\Delta - 1)u(h) = 0,$$

the symbol of which is $p = |\xi|^2 - 1$.

Changing variables. We introduce the change of variables $\xi = \gamma(\eta)$, defined near $\eta = 0$ by

$$\eta_1 = |\xi|^2 - 1, \quad \eta_2 = \xi_2, \quad \dots, \quad \eta_n = \xi_n.$$

Proceeding as in Theorem 2.6, we extend this to a symplectic transformation

$$(x, \xi) \mapsto (y, \eta)$$

by putting $y = (\partial\gamma(\eta))^T x$. That is,

$$y_1 = x_1/2\xi_1, \quad y_2 = x_2 - x_1\xi_2/\xi_1, \quad \dots, \quad y_n = x_n - x_1\xi_n/\xi_1.$$

This is well defined near any point $(x_0, \xi_0) \in \mathbb{R}^{2n}$ at which $\xi_1 \neq 0$. The symplectic transformation κ given by the above transformation then gives

$$\kappa^* p = \eta_1.$$

In this sense κ “simplifies” the symbol p near (x_0, ξ_0) .

Quantizing κ . We next quantize κ^{-1} and thereby “simplify” Helmholtz’s equation. We accomplish this by defining the operator

$$Tu := \mathcal{F}_h^{-1}(\chi\gamma^*(\mathcal{F}_h u)),$$

where $u \in L^2$ and $\chi \in C_c^\infty(B(0, 1/2))$ equals one near 0. We check directly that for functions u with $\mathcal{F}_h u$ supported in a sufficiently small neighborhood of ξ_0 , we have

$$\begin{aligned} hD_{y_1} Tu &= \mathcal{F}_h^{-1}(\chi\eta_1\gamma^*(\mathcal{F}_h u)) \\ &= \mathcal{F}_h^{-1}(\chi\gamma^*((|\xi|^2 - 1)\mathcal{F}_h u)) \\ &= T((-h^2\Delta - 1)u). \end{aligned}$$

So T has “intertwined” the Helmholtz operator and the much simpler operator hD_{y_1} .

We stress that this simplification holds only locally in phase space but nevertheless, as we will see, allows us to draw interesting conclusions. $\square$

The point of the general theory of normal forms is that the approach described above for constant coefficient equations works for operators with nonconstant coefficients.

12.2. NORMAL FORMS: REAL SYMBOLS

In this section we show that microlocally near a point an operator P quantizing a real symbol with a nonvanishing differential is equivalent to hD_{x_1} . This is important since we can then transplant various mathematical objects associated to P to those for hD_{x_1} , which are much easier to study.

12.2.1. More symplectic geometry. To apply the local theory of quantized symplectic transformations from Chapter 11 to the study of semiclassical operators, we will need two results from symplectic geometry. The first is a stronger form of Darboux's Theorem (Theorem 2.12):

THEOREM 12.1 (Variant of Darboux's Theorem). *Let A and B be two subsets of $\{1, \dots, n\}$, and suppose that*

$$p_j(x, \xi) \quad (j \in A), \quad q_k(x, \xi) \quad (k \in B)$$

are smooth, real-valued functions defined in a neighborhood of $(0, 0) \in \mathbb{R}^{2n}$, with linearly independent gradients at $(0, 0)$.

If

$$(12.2.1) \quad \begin{aligned} \{q_i, q_j\} &= 0 \quad (i, j \in A), & \{p_k, p_l\} &= 0 \quad (k, l \in B), \\ \{p_k, q_j\} &= \delta_{kj} \quad (j \in A, k \in B), \end{aligned}$$

then there exists a symplectomorphism κ , locally defined near $(0, 0)$, such that $\kappa(0, 0) = (0, 0)$ and

$$(12.2.2) \quad \kappa^* q_j = x_j \quad (j \in A), \quad \kappa^* p_k = \xi_k \quad (k \in B).$$

We will also need

THEOREM 12.2 (Symplectic integrating factor). *Let p and q be smooth, real-valued functions defined near $(0, 0) \in \mathbb{R}^{2n}$, satisfying*

$$(12.2.3) \quad p(0, 0) = q(0, 0) = 0, \quad \{p, q\}(0, 0) > 0.$$

Then there exists a smooth, positive function u for which

$$(12.2.4) \quad \{up, uq\} \equiv 1$$

in a neighborhood of $(0, 0)$.

Consult Hörmander [H2, Theorem 21.1.6] and [H2, Lemma 21.3.4] for proofs.

12.2.2. Symbols of real principal type.

Now let m be an order function and set $P = p^w(x, hD)$, where

$$p(x, \xi) \sim \sum_{k=0}^{\infty} h^k p_k(x, \xi)$$

for symbols $p_j \in S(m)$.

DEFINITIONS.

(i) The operator P is of *real principal type* if p_0 is real-valued and

$$(12.2.5) \quad \partial p_0 \neq 0 \quad \text{on } \{p_0 = 0\}.$$

(ii) If the principal symbol p_0 is real-valued and satisfies

$$(12.2.6) \quad p_0(0,0) = 0, \quad \partial p_0(0,0) \neq 0,$$

we say that P is of *real principal type at the point* $(0,0)$.

THEOREM 12.3 (Normal form for real principal type operators).

Suppose that $P = p^w(x, hD)$ is of real principal type at $(0,0)$.

Then there exist

(i) a local canonical transformation κ defined near $(0,0)$, such that $\kappa(0,0) = (0,0)$ and

$$(12.2.7) \quad \kappa^* p_0 = \xi_1;$$

and

(ii) an operator T , quantizing κ^{-1} in the sense of Theorem 11.6, such that

$$(12.2.8) \quad T^{-1} \text{ exists microlocally near } ((0,0), (0,0))$$

and

$$(12.2.9) \quad TPT^{-1} = hD_{x_1} \quad \text{microlocally near } ((0,0), (0,0)).$$

Proof. 1. Theorem 12.1 with $A = \emptyset$ and $B = \{1\}$ provides κ satisfying (12.2.7) near $(0,0)$. By taking a sufficiently small neighborhood of $(0,0)$, we can assume that the assumptions of Theorem 11.5 are satisfied, and consequently we have a quantization T_0 of κ^{-1} . Thus

$$(12.2.10) \quad T_0 P T_0^{-1} = hD_{x_1} + E \quad \text{microlocally near } (0,0),$$

where $E = e^w$ for a symbol $e \in hS$.

2. We want to transform away the term E in (12.2.10). To do so, we look for a symbol $a \in S$ so that a is elliptic at $(0,0)$ and

$$(12.2.11) \quad hD_{x_1} + E = A hD_{x_1} A^{-1} \quad \text{microlocally near } (0,0),$$

where $A := a^w$. This is the same as solving

$$(12.2.12) \quad [hD_{x_1}, A] + EA = 0 \quad \text{microlocally near } (0,0).$$

Since $P = p_0^w + hp_1^w + h^2p_2^w + \cdots$, the Remark (i) after the proof of Theorem 11.1 shows that

$$e(x, \xi) = he_0(x, \xi) + h^2e_1(x, \xi) + \cdots.$$

We can solve the differential equation

$$\frac{1}{i} \{ \xi_1, a_0 \} + e_0 a_0 = 0$$

near $(0, 0)$ for $a_0 \in S$ such that $a_0(0, 0) \neq 0$. Define $A_0 := a_0^w$; then

$$[hD_{x_1}, A_0] + EA_0 = r_0^w$$

for a symbol $r_0 \in h^2S$.

3. We now inductively find $A_k = a_k^w$ for $a_k \in S$, satisfying

$$[hD_{x_1}, A_0 + hA_1 + \cdots + h^N A_N] + E(A_0 + hA_1 + \cdots + h^N A_N) = r_N^w,$$

where $r_N \in h^{N+2}S$. We invoke Borel's Theorem (Theorem 4.15) to put

$$a \sim \sum_{k=0}^{\infty} h^k a_k, \quad A = a^w \sim \sum_{k=0}^{\infty} h^k A_k.$$

Then A solves (12.2.12), and a is elliptic near $(0, 0)$. Finally, define

$$T := A^{-1}T_0.$$

According to (12.2.10) and (12.2.11), this operator verifies (12.2.9). $\square$

12.2.3. L^2 estimates and principal type. Here is a first application of Theorem 12.3:

THEOREM 12.4 (Principal type and estimates). *Suppose that $P = p^w$ is of real principal type at $(0, 0)$. Assume for $u = u(h) \in L^2(\mathbb{R}^n)$ that $\text{WF}_h^*(u)$ is contained in a small neighborhood of $(0, 0)$. Then we have the estimate*

$$(12.2.13) \quad \|u\|_{L^2(\mathbb{R}^n)} \leq \frac{C}{h} \|Pu\|_{L^2(\mathbb{R}^n)}.$$

Proof. Let T be the operator given in Theorem 12.3 and let T^{-1} be its microlocal inverse near $(0, 0)$. The localization property of u implies

$$\begin{aligned} \|Pu\|_{L^2} &= \|T^{-1}(hD_{x_1})Tu\|_{L^2} + O(h^\infty)\|u\|_{L^2} \\ &\geq c\|TT^{-1}hD_{x_1}Tu\|_{L^2} + O(h^\infty)\|u\|_{L^2} \\ &= c\|hD_{x_1}Tu\|_{L^2} + O(h^\infty)\|u\|_{L^2}. \end{aligned}$$

Localization of u and the local Egorov Theorem (Theorem 11.6) imply that

$$Tu = \psi Tu + O_{\mathcal{S}}(h^\infty)\|u\|_{L^2}$$

for $\psi \in C_c^\infty(\mathbb{R}^n)$. Also

$$v(x) = \frac{i}{h} \int_{-\infty}^{x_1} hD_{x_1} v(y, x') dy,$$

where $x = (x_1, x')$, for $v := \psi Tu$. Therefore

$$\|v\|_{L^2} \leq \frac{C}{h} \|hD_{x_1} v\|_{L^2};$$

and hence

$$\begin{aligned}\|Pu\|_{L^2} &\geq ch\|Tu\|_{L^2} + O(h^\infty)\|u\|_{L^2} \\ &\geq ch\|T^{-1}T\chi^w u\|_{L^2} + O(h^\infty)\|u\|_{L^2} \\ &\geq ch\|\chi^w u\|_{L^2} + O(h^\infty)\|u\|_{L^2}.\end{aligned}$$

This proves (12.2.13), since $\text{WF}_h^*(u)$ lies within a small neighborhood of $(0, 0)$. $\square$

REMARK. The operators used in the microlocal transformations have good mapping properties on L^2 -based spaces, such as our generalized Sobolev spaces $H_h(m)$. However, as we have seen in Sections 10.3 and 10.4, the mapping properties on L^p spaces are complicated and depend on finer geometric aspects of the canonical transformation. Hence the normal forms cannot be directly used for the study of such phenomena as dispersions or multiplicative properties of solutions. $\square$

12.3. PROPAGATION OF SINGULARITIES

12.3.1. Propagation of wavefront sets. As a further application of Theorem 12.3 we next prove a propagation theorem that refines our earlier Theorem 5.4.

Suppose that $P(h) = p^w(x, hD)$ is an operator of real principal type and let $p_0(x, \xi)$ be its principal symbol:

$$p \sim \sum_{k=0}^{\infty} h^k p_k,$$

where $p, p_k \in S(m)$ for some order function m .

THEOREM 12.5 (Flow invariance of wavefront sets). *Assume that*

$$\partial p \neq 0 \quad \text{on } \{p = 0\}.$$

Then for each tempered family $u = \{u(h)\}_{0 < h < 1} \subset H_h(m)$ solving

$$P(h)u(h) = f(h),$$

where $f = \{f(h)\}_{0 < h < 1} \subset L^2(\mathbb{R}^n)$, we have

(12.3.1) $\text{WF}_h(u) \setminus \text{WF}_h(f)$ *is invariant under the Hamiltonian flow of p .*

More precisely, the theorem asserts that if $(x_0, \xi_0) \in \text{WF}_h(u) \setminus \text{WF}_h(f)$ and if $\exp(tH_p)(x_0, \xi_0) \notin \text{WF}_h(f)$ for $t \in (a, b)$, where $a < 0 < b$, then $\exp(tH_p)(x_0, \xi_0) \in \text{WF}_h(u)$ for all $t \in (a, b)$.

Proof. 1. Suppose that $(x_0, \xi_0) \in \text{WF}_h(u)$ and that $\exp([a, b]H_p) \notin \text{WF}_h(f)$. We need to show that the set

$$G := \{t \in (a, b) \mid \exp(tH_p)(x_0, \xi_0) \in \text{WF}_h(u)\}$$

is both open and relatively closed.

Since $\text{WF}_h(u)$ is closed and $t \mapsto \exp(tH_p)(x_0, \xi_0)$ is continuous, G is clearly relatively closed.

2. To see that G is open, we invoke Theorem 12.3. Let T microlocally quantize the symplectomorphism κ given in Theorem 12.3, so that

$$TPT^{-1} = hD_{x_1}$$

microlocally near $((0, 0), (0, 0))$. The definition of the semiclassical wavefront set in Section 8.4 then implies

$$(12.3.2) \quad \text{WF}_h(Tu) \cap V = \kappa(\text{WF}_h(u) \cap V),$$

when V is a small neighborhood of $(0, 0)$.

We therefore need to check that if

$$(12.3.3) \quad hD_{x_1}v = O_{L^2}(h^\infty)$$

microlocally near $(0, 0)$ and if $(0, 0; 0, 0) \in \text{WF}_h(v)$, then $(t, 0; 0, 0) \in \text{WF}_h(v)$ for small $|t|$. Here and hereafter we write $(x, \xi) = (x_1, x'; \xi_1, \xi')$ for $x', \xi' \in \mathbb{R}^{n-1}$.

3. The equation (12.3.3) shows that

$$(12.3.4) \quad v(x_1, x') = v(0, x') + O_{L^2}(h^\infty)$$

for $|x_1| < \epsilon$.

Then if $(t, 0; 0, 0) \notin \text{WF}_h(v)$, it follows that $(0, 0; 0, 0) \notin \text{WF}_h(v)$ for small $|t|$. To see this, suppose that $a \in C_c^\infty$, $a(t, 0; 0, 0) \neq 0$, and

$$a^w(x, hD)v = O_{L^2}(h^\infty).$$

Define $b(x, \xi) := a(x_1 + t, x', \xi)$. Then $b(0, 0) \neq 0$ if $|t|$ is small enough. If we put $U_t w := w(x_1 + t, x)$, then (12.3.4) implies

$$\begin{aligned} b^w(x, hD)v &= U_t a^w(x, hD)U_{-t}v = U_t a^w(x, hD)v + O_{L^2}(h^\infty) \\ &= O_{L^2}(h^\infty). \end{aligned}$$

□

REMARK: Propagation of C^∞ singularities. Using Theorem 12.5 and the relation (8.4.8) between the semiclassical wavefront set and the wavefront set measuring C^∞ singularities, we can recover the now classical theorem of Hörmander on propagation of singularities for operators of real principal type.

Suppose that

$$P(x, D) := \sum_{|\alpha| \leq m} a_\alpha(x) D^\alpha$$

is a differential operator with coefficients $a_\alpha \in C^\infty$ and that $u \in \mathcal{S}'$ solves

$$P(x, D)u = f.$$

We reformulate this semiclassically by putting

$$\tilde{P}(x, hD) := h^m P(x, D) = \sum_{|\alpha| \leq m} a_\alpha(x) h^{m-|\alpha|} (hD)^\alpha.$$

In the notation of Theorem 12.5, $\tilde{P}(x, hD)$ has the principal symbol

$$p(x, \xi) = \sum_{|\alpha|=m} a_\alpha(x) \xi^\alpha.$$

This is the same as the classical principal symbol obtained by taking the highest-order derivatives. If we assume that

$$dp|_{p^{-1}(0)} \neq 0 \quad \text{in a conic neighborhood of } (x_0, \xi_0) \in \mathbb{R}^{2n} \setminus 0,$$

then (8.4.8) and Theorem 12.5 show that in a conic neighborhood of (x_0, ξ_0)

$$\text{WF}(u) \setminus \text{WF}(f)$$

is invariant under the flow of H_p .

EXAMPLE. Suppose that $u \in \mathcal{S}'$ solves the inhomogeneous wave equation in $\mathbb{R} \times \mathbb{R}^n$:

$$(\partial_t^2 - \Delta)u = f$$

where $f \in \mathcal{S}'$. Then

$$\text{WF}(u) \setminus \text{WF}(f) \subset \{(t, x, \tau, \xi) \mid \tau^2 = |\xi|^2\}$$

is invariant under the flow

$$\dot{t} = 1, \quad \dot{x} = \frac{\xi}{|\xi|}, \quad \dot{\tau} = 0, \quad \dot{\xi} = 0.$$

So singularities of solutions to the wave equation, as measured by the wave-front set, propagate along light rays. $\square$

12.4. NORMAL FORMS: COMPLEX SYMBOLS

Motivated by the study of pseudospectra in Section 12.5 below, we now consider the case of complex p . In that case we need a stronger nondegeneracy condition stating that $\operatorname{Re} \partial p$ and $\operatorname{Im} \partial p$ are independent. Once that is true, then $p = 0$ is a codimension two submanifold and it can have different symplectic properties, determining possible normal forms.

Operators of complex principal type. Assume as before that $P = p^w(x, hD)$ has the symbol

$$p \sim \sum_{k=0}^{\infty} h^k p_k$$

with $p_j \in S(m)$. We now allow $p(x, \xi)$ to be *complex-valued*, and we still say that P is of *principal type* at $(0, 0)$ provided

$$(12.4.1) \quad p_0(0, 0) = 0, \quad \partial p_0(0, 0) \neq 0.$$

THEOREM 12.6 (Normal form for the complex symplectic case).

Suppose that $P = p^w(x, hD; h)$ is a semiclassical principal type operator at $(0, 0)$, with principal symbol p_0 satisfying

$$(12.4.2) \quad p_0(0, 0) = 0, \quad \pm\{\operatorname{Re} p_0, \operatorname{Im} p_0\}(0, 0) > 0.$$

Then there exist

(i) *a local canonical transformation κ defined near $(0, 0)$ and a smooth function u such that $\kappa(0, 0) = (0, 0)$, $u(0, 0) > 0$, and*

$$\kappa^*(\xi_1 \pm ix_1) = up_0,$$

and

(ii) *an operator T , quantizing κ in the sense of Theorem 11.6, and a pseudodifferential operator A , elliptic at $(0, 0)$, such that*

$$(12.4.3) \quad T^{-1} \text{ exists microlocally near } ((0, 0), (0, 0))$$

and

$$(12.4.4) \quad TPT^{-1} = A(hD_{x_1} \pm ix_1) \text{ microlocally near } ((0, 0), (0, 0)).$$

INTERPRETATION. If $\partial(\operatorname{Re} p_0)$ and $\partial(\operatorname{Im} p_0)$ are linearly independent, then the submanifold $p^{-1}(0) \subset \mathbb{R}^{2n}$, where P is not elliptic, has codimension two, as opposed to codimension one in the real-valued case. The symplectic form restricted to that submanifold is nondegenerate if

$$\pm\{\operatorname{Re} p_0, \operatorname{Im} p_0\} > 0.$$

Under this assumption a combination of Theorems 12.1 and 12.2 shows that there exists a canonical transformation κ , defined near $(0, 0)$, and a smooth positive function u such that

$$\kappa^*(\xi_1 \pm ix_1) = up_0.$$

That is, after a multiplication by a function we obtain the symbol of the creation or annihilation operator for the harmonic oscillator in the (x_1, ξ_1) variables. $\square$

Proof. 1. Let us assume that

$$\{\operatorname{Re} p_0, \operatorname{Im} p_0\} > 0.$$

As noted above, using Theorems 12.1 and 12.2, we can find a smooth function u , with $u(0, 0) > 0$, and a local canonical transformation κ such that $\kappa(0, 0) = (0, 0)$ and $\kappa^*(\xi_1 + ix_1) = up_0$.

Quantizing as before, we obtain an operator T_0 satisfying

$$(12.4.5) \quad T_0 P T_0^{-1} = Q(hD_{x_1} + ix_1 + E)$$

microlocally near $(0, 0)$, where $Q = q^w$ for a function q satisfying

$$\kappa^* q = 1/u$$

and $E = e^w$ for some $e \in hS$. As in the proof of Theorem 12.3, we have

$$e = he_0(x, \xi) + h^2 e_1(x, \xi) + \cdots$$

2. To remove the term E from (12.4.5), we now need to find pseudodifferential operators B and C , elliptic at $(0, 0)$, such that

$$(12.4.6) \quad (hD_{x_1} + ix_1 + E)B = C(hD_{x_1} + ix_1) \quad \text{microlocally.}$$

Note the difference between this and (12.2.11), where we were able to take the same operator on both sides.

We will find the symbols of B and C by computing successive terms in their expansions:

$$b \sim \sum_{k=0}^{\infty} h^k b_k, \quad c \sim \sum_{k=0}^{\infty} h^k c_k.$$

We use Theorem 4.11 to write

$$(hD_{x_1} + ix_1 + E)B - C(hD_{x_1} + ix_1) = r^w,$$

where

$$r = \sum_{k=0}^{\infty} h^k r_k$$

with

$$\begin{aligned} r_0 &= (\xi_1 + ix_1)(b_0 - c_0), \\ r_1 &= (\xi_1 + ix_1)(b_1 - c_1) + e_0 b_0 + \{\xi_1 + ix_1, b_0 + c_0\}/2i. \end{aligned}$$

3. We intend to select b and c so that $r_j \equiv 0$ for all j . For $r_0 = 0$ we simply need $b_0 = c_0$. To force $r_1 = 0$, we must solve

$$-i(\partial_{x_1} - i\partial_{\xi_1})b_0 + e_0 b_0 + (\xi_1 + ix_1)(b_1 - c_1) = 0.$$

We first find b_0 such that

$$(12.4.7) \quad \begin{cases} -i(\partial_{x_1} - i\partial_{\xi_1})b_0 + e_0 b_0 = O(x_1^\infty) \\ b_0|_{x_1=0} = 1. \end{cases}$$

The notation means that the left-hand side of the first equation vanishes to infinite order at $x_1 = 0$. We compute $\partial_{x_1}^{k+1} b_0|_{x_1=0}$ inductively:

$$\partial_{x_1}^{k+1} b_0 = i \left(\partial_{\xi_1} \partial_{x_1}^k b_0 - \partial_{x_1}^k (e_0 b_0) \right).$$

We see that the derivatives $\partial_{x_1}^k e_0|_{x_1=0}$ therefore determine $\partial_{x_1}^k b_0|_{x_1=0}$. Then Borel's Theorem (Theorem 4.15) produces a smooth function b_0 with these prescribed derivatives, and that b_0 therefore solves (12.4.7).

With $b_0 = c_0$ chosen as above, we see that

$$t_1 := (-i(\partial_{x_1} - i\partial_{\xi_1})b_0 + e_0 b_0)/(\xi_1 + ix_1)$$

is a smooth function, since the numerator vanishes to infinite order on the zero set of the denominator. So if we put

$$(12.4.8) \quad c_1 = b_1 + t_1,$$

then $r_1 = 0$.

4. Now using (12.4.8) and the same calculation as before, we see that

$$r_3 = (\xi_1 + ix_1)(b_2 - c_2) + e_0 b_1 - i\{\xi_1 + ix_1, b_1\} + \tilde{r}_3,$$

where $\tilde{r}_3$ depends only on $b_0 = c_0$, t_1 , and e . Hence $\tilde{r}_3$ is already determined. We proceed as in step 3 and first solve

$$\begin{cases} -i(\partial_{x_1} - i\partial_{\xi_1})b_1 + e_0 b_1 + \tilde{r}_3 = O(x_1^\infty) \\ b_1|_{x=1} = 0. \end{cases}$$

This determines b_1 and hence c_1 . We continue in the same way to determine b_2 and thus c_2 . An iteration of the argument completes the construction of b and c , for which (12.4.6) holds microlocally near $(0, 0)$.

5. Finally, we put $T = B^{-1}T_0$, where B^{-1} is the microlocal inverse of B near $(0, 0)$, and $A = B^{-1}QC$. $\square$

Applications of Theorem 12.6 appear in Section 12.5. The next example shows that other phenomena occur when symbols are complex but the assumptions of the previous theorem do not hold.

EXAMPLE: Heat equation. Now consider the semiclassical heat equation

$$(12.4.9) \quad h\partial_t u - h^2 \Delta u = 0.$$

The symbol, $p = \tau - i\xi^2$, is complex,

$$p = \{\tau = 0, \xi = 0\} \subset \mathbb{R}^{2n+2},$$

and for any $\chi \in C_c^\infty(\mathbb{R}^{n+1})$,

$$\text{WF}_h(\chi u) \subset \mathbb{R}^{n+1} \times \{(0, 0)\}.$$

The conclusion about the wavefront set corresponds to the fact that solutions to (12.4.9) have no *strong* oscillations in h .

Using the fact that $\partial \text{Re } p \neq 0$ and $\text{Im } p \leq 0$, we can prove the following propagation assertion. Suppose that u is tempered and solves

$$h\partial_t u - h^2 \Delta u = f.$$

Then

$$\text{WF}_h(u) \setminus \text{WF}_h(f) \subset \{p = 0\}$$

is closed under *backward time propagation*. This means that if $(t, x, 0, 0) \in \text{WF}_h(u)$ and $(t - s, x, 0, 0) \notin \text{WF}_h(f)$ for $0 \leq s \leq T$, then $(t - T, x, 0, 0) \in \text{WF}_h(u)$.

A similar statement can be made for the initial-value problem and we illustrate it with a simple example. Let u solve

$$(12.4.10) \quad \begin{cases} h\partial_t u - h^2 \Delta u = 0 & (t > 0) \\ u = g & (t = 0), \end{cases}$$

where

$$g = e^{\frac{i}{h}\varphi} b$$

for smooth functions b, φ , the phase φ being real. A tempered solution to (12.4.10) is (see [E, Section 2.2]):

$$u(t, x) = \frac{1}{(4\pi ht)^{n/2}} \int_{\mathbb{R}^n} e^{\frac{|x-y|^2}{4ht} + \frac{i}{h}\varphi(y)} b(y) dy.$$

For $t > \epsilon > 0$ we have the pointwise estimate

$$(12.4.11) \quad (hD_x)^\alpha (hD_t)^k u(t, x) = O(h^{|\alpha|/2 + k/2}).$$

This shows that the set $\tau = \xi = 0$ contains the wavefront set of u . When $x \notin \text{spt } b$, we also have

$$(12.4.12) \quad (hD_x)^\alpha (hD_t)^k u(t, x) = O(h^\infty).$$

When $\partial\varphi(x) \neq 0$, integration by parts based upon the identity

$$2t \frac{\langle 2t\partial\varphi(y) - i(x-y), hD_y \rangle}{4t^2|\varphi'(y)|^2 + |x-y|^2} e^{\frac{|x-y|^2}{4ht} + \frac{i}{h}\varphi(y)} = e^{\frac{|x-y|^2}{4ht} + \frac{i}{h}\varphi(y)}$$

shows that (12.4.12) also holds. It follows therefore that

$$\text{WF}_h(u) \cap \{t > 0\} = ((0, \infty) \times \{x \in \text{spt } b \mid \partial\varphi(x) = 0\}) \times \{(0, 0)\}.$$

We note that the statement that $\text{WF}_h(u)$ is contained in the zero section means that we do not have oscillations on the scale h but not that the solution is actually smooth uniformly with respect to h ; see (12.4.11). $\square$

This example again shows that the existence of solutions to $Pu = 0$ that are highly oscillatory as $h \rightarrow 0$ comes from $\{p = 0\} \neq \emptyset$. The heat equation example also shows that the situation is richer for complex symbols, as we can have different conditions on $\text{Re } p$ and $\text{Im } p$.

12.5. QUASIMODES, PSEUDOSPECTRA

We present in this section an application of Theorem 12.6 to the construction of quasimodes for nonnormal operators.

12.5.1. Quasimodes and eigenvalues. Recall from Theorem 6.7 that if $P(h) = -h^2\Delta + V$ and V is real-valued, satisfying

$$(12.5.1) \quad \begin{cases} |\partial^\alpha V(x)| \leq C_\alpha \langle x \rangle^k & \text{for each multiindex } \alpha \\ V(x) \geq c \langle x \rangle^k & \text{for } |x| \geq R, \end{cases}$$

then the spectrum of $P(h)$ is discrete.

Since V is real and consequently $P(h)$ is selfadjoint, the approximate location of eigenvalues can be found using quasimodes, that is, approximate eigenfunctions:

THEOREM 12.7 (Quasimodes and eigenvalues). *Suppose under the foregoing assumptions that*

$$(12.5.2) \quad \|(P(h) - z(h))u(h)\|_{L^2} = O(h^\infty), \quad \|u(h)\|_{L^2} = 1.$$

Then there exist an eigenvalue $E(h)$ and eigenfunction $v(h)$ such that

$$(12.5.3) \quad \begin{cases} (P(h) - E(h))v(h) = 0, & \|v(h)\|_{L^2} = 1 \\ |E(h) - z(h)| = O(h^\infty). \end{cases}$$

INTERPRETATION. This means that if we can construct a quasimode solving (12.5.2), then the approximate eigenvalue $z(h)$ is in fact close to a true eigenvalue $E(h)$. However the quasimode $u(h)$ need not be near a true eigenfunction $v(h)$. $\square$

Proof. If $z(h) \notin \text{Spec}(P(h))$, we can use (C.2.17) to compute

$$\begin{aligned} 1 &= \|u(h)\|_{L^2} = \|(P(h) - z(h))^{-1}(P(h) - z(h))u(h)\|_{L^2} \\ &\leq \|(P(h) - z(h))^{-1}\| \| (P(h) - z(h))u(h) \|_{L^2} \\ &= O(h^\infty)/\text{dist}(z(h), \text{Spec}(P(h))). \end{aligned}$$

As noted above, $\text{Spec}(P(h))$ consists solely of eigenvalues. $\square$

12.5.2. Quasimodes for nonnormal operators. But Theorem 12.7 is in general false for nonnormal operators P , meaning those for which the commutator $[P^*, P]$ does not vanish. In particular, if V is complex and $p = |\xi|^2 + V(x)$, the symbol of this commutator is

$$(12.5.4) \quad \frac{1}{i} \{\bar{p}, p\} = 2\{\text{Re } p, \text{Im } p\};$$

and when this is nonzero, we are in the situation discussed in Theorem 12.6.

This discussion leads us to

THEOREM 12.8 (Constructing quasimodes). *Suppose*

$$P(h) = -h^2\Delta + V$$

and

$$(12.5.5) \quad z_0 = \xi_0^2 + V(x_0), \quad \text{Im}\langle \xi_0, \partial V(x_0) \rangle \neq 0.$$

(i) *Then there exists a family of functions $u(h) \in C_c^\infty(\mathbb{R}^n)$ such that*

$$(12.5.6) \quad \|(P(h) - z_0)u(h)\|_{L^2} = O(h^\infty), \quad \|u(h)\|_{L^2} = 1.$$

(ii) *Moreover, changing the sign of ξ_0 if necessary,*

$$(12.5.7) \quad \text{WF}_h(u(h)) = \{(x_0, \xi_0)\}, \quad \text{Im}\langle \xi_0, \partial V(x_0) \rangle < 0.$$

INTERPRETATION. The point is that, although we can solve the quasimode equation (12.5.6) under the hypotheses of (12.5.5), the point z_0 may in fact be far from the spectrum of $P(h)$. We informally say that z_0 belongs to the *pseudospectrum* of $P(h)$. $\square$

Proof. Our function $u(h)$ will be constructed with support near x_0 , and therefore with no loss of generality we can assume that V is compactly

supported. By changing the sign of ξ_0 if necessary, but without changing z_0 , we can also assume that

$$\{\operatorname{Re} p, \operatorname{Im} p\}(x_0, \xi_0) = 2\operatorname{Im}\langle \xi_0, \partial V(x_0) \rangle < 0.$$

Now according to Theorem 12.6, $P - z_0$ is microlocally conjugate to $A(hD_{x_1} - ix_1)$ near $((x_0, \xi_0), (0, 0))$. Let

$$u_0(x, h) := \exp(-|x|^2/2h),$$

so that

$$(hD_{x_1} - ix_1)u_0(h) = 0, \quad \operatorname{WF}_h(u_0(h)) = \{(0, 0)\}.$$

Following the notation of Theorem 12.6, we define $u(h) := T^{-1}u_0(h)$. Then $\operatorname{WF}_h(u(h)) = \{(x_0, \xi_0)\}$ and

$$(P(h) - z_0)u(h) = T^{-1}A(hD_{x_1} - ix_1)T(T^{-1}u_0) = 0 \quad \text{microlocally.} \quad \square$$

EXAMPLE 1. If $p(x, \xi) = |\xi|^2 + V(x)$, the potential V satisfies (12.5.1), and

$$\{p(x, \xi) \mid (x, \xi) \in \mathbb{R}^{2n}\} \neq \mathbb{C},$$

then the operator $P(h)$ has a discrete spectrum. This follows from the proof of Theorem 6.7, once we have found a point z at which $P(h) - z$ is elliptic. But this is so for any z not in the set of values of $p(x, \xi)$. However, the hypotheses of Theorem 12.8 hold in a dense open subset of the interior of the closure of the range of p . $\square$

EXAMPLE 2. It is also clear that more general operators can be considered. As a simple one-dimensional example, take

$$P(h) = (hD_x)^2 + ihD_x + x^2$$

with

$$p(x, \xi) = \xi^2 + i\xi + x^2, \quad \{\operatorname{Re} p, \operatorname{Im} p\} = -2x.$$

Then there exists a quasimode corresponding to any point in the interior of the range of p , namely $\{z \mid \operatorname{Re} z \geq (\operatorname{Im} z)^2\}$. This set is thus contained in the pseudospectrum of $P(h)$. But since

$$e^{x/2h}P(h)e^{-x/2h} = (hD)^2 + x^2 + \frac{1}{4},$$

$P(h)$ has the discrete spectrum $\{1/4 + (2n+1)h \mid n \in \mathbb{N}\}$; recall Theorem 6.2. $\square$

12.6. NOTES

See Hörmander [H2, Theorem 21.1.6 and Lemma 21.3.4] for proofs of the theorems cited in Section 12.2. Theorem 12.3 is a semiclassical analogue of the standard result of Duistermaat–Hörmander; see [H2, Proposition 26.1.3']. Theorem 12.6 is a semiclassical adaptation of a microlocal result of Duistermaat–Sjöstrand [Du-Sj]; see [H2, Proposition 26.3.1]. For another take on the wave equation with highly oscillatory initial data, see Hörmander [H2, Section 12.2].

We consider the case in which $\{p = 0\}$ is symplectic, that is, the symplectic form restricted to $\{p = 0\}$ is nondegenerate. The opposite case is the involutive case in which $\{\operatorname{Re} p, \operatorname{Im} p\}|_{p^{-1}(0)} = 0$. The normal form in that case is given by the Cauchy–Riemann operator; consult Hörmander [H4, Section 26.2]. When $n = 2$, then the operator is elliptic, but in higher dimensions we have interesting propagation phenomena.

Davies [Da] proved Theorem 12.8 in one dimension and [Zw] explained the link with Hörmander’s bracket condition (12.4.1). See Trefethen–Embree [T-E] for background on quasimodes and pseudospectra and for further references; and see [D-S-Z] for more on semiclassical pseudospectra.

THE FBI TRANSFORM

- 13.1 Motivation
- 13.2 Complex analysis
- 13.3 FBI transforms and Bergman kernels
- 13.4 Quantization and Toeplitz operators
- 13.5 Applications
- 13.6 Notes

13.1. MOTIVATION

In this chapter we will present a different perspective on the semiclassical quantization of Chapter 4. Starting with $a = a(x, \xi) \in S$, a function on phase space, we have defined an operator $a^w(x, hD)$ acting on $L^2(\mathbb{R}^n)$. This procedure is associated to a classical observable a and a quantum observable a^w . In the semiclassical limit, a^w shares many properties with multiplication by a . For instance in Theorem 4.29 we saw that if $1/a \in S$, then $(a^w)^{-1}$ exists. In fact, as pointed out in Section 8.4, this statement is microlocal; that is, nonvanishing of a near (x_0, ξ_0) implies microlocal invertibility of a^w near that point.

How close can one get to “quantization as multiplication”? The fact that quantum observables may not commute provides an obvious limitation. We recall from Theorem 3.9 that the uncertainty principle is a consequence of noncommutativity.

In Chapter 4 the space on which the quantum observables acted, the space of wave functions, was given by $L^2(\mathbb{R}^n)$. Instead we can consider as

the space of wave functions the Hilbert space of L^2 weighted holomorphic functions,

$$H_\Phi(\mathbb{C}^n) := \{f \in L_\Phi^2(\mathbb{C}^n) : \partial_{\bar{z}_j} f \equiv 0, j = 1, \dots, n\},$$

$$L_\Phi^2(\mathbb{C}^n) := \left\{ f : \mathbb{C}^n \rightarrow \mathbb{C} : \int_{\mathbb{C}^n} |f(z)|^2 e^{-2\Phi(z)/h} dm(z) < \infty \right\},$$

where $\Phi : \mathbb{C}^n \rightarrow \mathbb{R}$ is a strictly plurisubharmonic quadratic function, such as

$$(13.1.1) \quad \Phi(z) = \frac{1}{2} |\operatorname{Im} z|^2.$$

(See Section 13.2 for precise definitions.) The Hilbert space norm on L_Φ^2 is

$$\|u\|_{L_\Phi^2} := \left(\int_{\mathbb{C}^n} |f(z)|^2 e^{-2\Phi(z)/h} dm(z) \right)^{\frac{1}{2}}$$

and H_Φ is a closed subspace of L_Φ^2 .

We associate to Φ a natural real symplectic form on $\mathbb{C}^n$

$$\sigma_\Phi := \frac{2}{i} \sum_{i,j=1}^n \frac{\partial^2 \Phi}{\partial z_i \partial \bar{z}_j} dz_i \wedge d\bar{z}_j,$$

and we can consider $\mathbb{C}^n$ with this symplectic form as classical phase space. A function $q \in L^\infty(\mathbb{C}^n)$, a classical observable, is then quantized as

$$\Pi_\Phi M_q \Pi_\Phi : H_\Phi(\mathbb{C}^n) \rightarrow H_\Phi(\mathbb{C}^n),$$

where $M_q : u \mapsto qu$ is the multiplication operator and

$$\Pi_\Phi : L_\Phi^2(\mathbb{C}^n) \rightarrow H_\Phi(\mathbb{C}^n)$$

is the orthogonal projector.

We consider $\Pi_\Phi M_q \Pi_\Phi$ as a quantum observable and call it the *Toeplitz quantization* of the classical observable q . We note that it is defined for q that are merely bounded and that a stronger version of the sharp Gårding inequality is immediate:

$$\text{if } q \geq 0, \text{ then } \Pi_\Phi M_q \Pi_\Phi \geq 0.$$

The transform which makes this approach equivalent to the approach in Chapter 4 is called by different names depending on the context: Bargmann transform, Gabor transform, Segal–Bargmann transform, wave packet transform, and it is closely related to the Wigner transform. In microlocal analysis it is normally called the Fourier–Bros–Iagolnitzer (FBI) transform as it was used by physicists Bros and Iagolnitzer, and also Iagolnitzer and Stapp, to

characterize analytic wavefront sets. For Φ given in (13.1.1) this transform is defined as follows:

$$(13.1.2) \quad \mathcal{T}_\varphi u(z) := \frac{2^{n/4}}{(2\pi h)^{3n/4}} \int_{\mathbb{R}^n} e^{\frac{i}{h}\varphi(z,x)} u(x) dx,$$

$$\varphi(z, x) := i(z - x)^2/2.$$

The relation between Φ and φ is given by

$$\Phi(z) = \max_{y \in \mathbb{R}^n} -\operatorname{Im} \varphi(z, y).$$

We will see that

$$\mathcal{T}_\varphi : L^2(\mathbb{R}^n) \rightarrow H_\Phi(\mathbb{C}^n)$$

is a unitary operator and therefore

$$\Pi_\Phi = \mathcal{T}_\varphi \mathcal{T}_\varphi^*,$$

where the adjoint is taken in the sense of L^2_Φ . The operator Π_Φ is the orthogonal projector from L^2_Φ onto H_Φ and is called the Bergman projector. Its Schwartz kernel is called the Bergman kernel.

The equivalence between Weyl quantization, $a \rightarrow a^w$, and Toeplitz quantization, $q \rightarrow \Pi_\Phi M_q \Pi_\Phi$, is given using the FBI transform. There exists a map

$$\Sigma : S(\mathbb{C}^n) \rightarrow S(T^*\mathbb{R}^n), \quad \Sigma : q \mapsto a = \Sigma(q),$$

such that

$$(13.1.3) \quad \mathcal{T}_\varphi a^w(x, hD) \mathcal{T}_\varphi^* = \Pi_\Phi M_q \Pi_\Phi.$$

This map is invertible only in an asymptotic sense as $h \rightarrow 0$. That means that for $a \in S(T^*\mathbb{R}^n)$ we can find $q \in S(\mathbb{C}^n)$ such that (13.1.3) holds up to terms $O_{H_\Phi \rightarrow H_\Phi}(h^\infty)$.

The asymptotic expansion relates q to the symbol a in a geometric way, modulo $O(h)$ errors. For Φ given by (13.1.1) that relation is given as follows:

$$q(z) = a(\operatorname{Re} z, -\operatorname{Im} z) + O_S(h).$$

We will also show that

$$\mathcal{T}_\varphi a^w(x, hD) = a(\operatorname{Re} z, -\operatorname{Im} z) \mathcal{T}_\varphi + O_{L^2(\mathbb{R}^n) \rightarrow L^2_\Phi(\mathbb{C}^n)}(\sqrt{h}),$$

which means that the action of pseudodifferential operators on the FBI transform side equals the multiplication by the symbol up to $\sqrt{h}$ errors.

13.2. COMPLEX ANALYSIS

In this section we will present a few facts about analysis on $\mathbb{C}^k$. We stress that none of the finer aspects of the theory play a role in our presentation.

13.2.1. Complex differential forms. The complex vector space $\mathbb{C}^k$ can be identified with $\mathbb{R}^k \times \mathbb{R}^k$ using the identification

$$z = x + iy \in \mathbb{C}^k, \quad x, y \in \mathbb{R}^k.$$

It can then be considered a real vector space and we should be careful to distinguish between real and complex linear transformations.

We recall complex conjugation $\bar{z} = x - iy$ and consider differential forms

$$dz_j = dx_j + i dy_j, \quad d\bar{z}_j = dx_j - i dy_j.$$

The complex differentiation operators and the Cauchy–Riemann operators are defined by

$$\frac{\partial}{\partial z_j} = \frac{1}{2} \left(\frac{\partial}{\partial x_j} - i \frac{\partial}{\partial y_j} \right), \quad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2} \left(\frac{\partial}{\partial x_j} + i \frac{\partial}{\partial y_j} \right),$$

respectively. A differential of a function considered as a 1-form can be written as

$$df = \sum_{j=1}^k \frac{\partial f}{\partial z_j} dz_j + \frac{\partial f}{\partial \bar{z}_j} d\bar{z}_j.$$

A function $f : \mathbb{C}^k \rightarrow \mathbb{C}$ is *holomorphic* if $\partial_{\bar{z}_j} f \equiv 0$. It follows that $f(z)$ is holomorphic in each of the variables z_j , in the sense of complex function theory. This then implies that near any $z_0 \in \mathbb{C}^k$, f is given by a convergent Taylor series,

$$f(z) = \sum_{\alpha \in \mathbb{N}^k} \frac{(z - z_0)^\alpha}{\alpha!} \partial_z^\alpha f(z_0).$$

Contour integrals are defined by integrals over k -dimensional submanifolds of $\mathbb{C}^k \simeq \mathbb{R}^{2k}$ but we will only consider a very special case. Suppose that $k = 2n$, $(z, \zeta) \in \mathbb{C}^n \times \mathbb{C}^n$ and that

$$\Gamma := \{(z, F(z)) : z \in \mathbb{C}^n\},$$

where $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is a smooth map.

Then

$$(13.2.1) \quad \iint_{\Gamma} f(z, \zeta) dz \wedge d\zeta := \int_{\mathbb{C}^n} f(z, F(z)) \det \left(\frac{\partial F}{\partial \bar{z}} \right) dz \wedge d\bar{z},$$

where

$$dz \wedge d\bar{z} = \left(\frac{2}{i} \right)^n dm(z)$$

is a multiple of the Lebesgue measure. (We identify measures with highest degree forms by requiring that the basis $(dx_1, \dots, dx_n, dy_1, \dots, dy_n)$ is positively oriented.) In this notation

$$dz = dz_1 \wedge \dots \wedge dz_n, \quad d\bar{z} = d\bar{z}_1 \wedge \dots \wedge d\bar{z}_n.$$

For (13.2.1) to make sense, we assume that $f(z, F(z)) \det(\partial_{\bar{z}} F)$ is Lebesgue integrable on $\mathbb{C}^n \simeq \mathbb{R}^{2n}$.

Now suppose that $\Gamma_j = \{(z, F_j(z))\}$ are two contours which are equal outside of a compact set,

$$F_j : \mathbb{C}^n \rightarrow \mathbb{C}^n, \quad F_1(z) = F_2(z), \quad |z| \geq R.$$

We define

$$\Omega := \{(z, tF_1(z) + (1-t)F_2(z)) : z \in \mathbb{C}^n, 0 \leq t \leq 1\},$$

so that by Stokes's Theorem,

$$\begin{aligned} \iint_{\Gamma_1} f(z, \zeta) dz \wedge d\zeta - \iint_{\Gamma_2} f(z, \zeta) dz \wedge d\zeta &= \int_{\Omega} df(z, \zeta) \wedge dz \wedge d\zeta \\ &= \sum_{j=1}^n \int_{\Omega} (\partial_{\bar{z}_j} f d\bar{z}_j + \partial_{\bar{\zeta}_j} f d\bar{\zeta}_j) \wedge dz_1 \wedge \cdots \wedge dz_n \wedge d\zeta_1 \wedge \cdots \wedge d\zeta_n. \end{aligned}$$

(Formally speaking, one should apply Stokes's Theorem on $[0, 1] \times \mathbb{C}^n$ to the pull-back of $f(z, \zeta) dz \wedge d\zeta$ under the map $(t, z) \mapsto (z, tF_1(z) + (1-t)F_2(z))$, whose image is Ω . This also provides the orientation on Ω .) In particular, for holomorphic f we have

$$(13.2.2) \quad \iint_{\Gamma_1} f(z, \zeta) dz \wedge d\zeta = \iint_{\Gamma_2} f(z, \zeta) dz \wedge d\zeta.$$

When $f(z, \zeta)$ has sufficiently strong decay properties, we can extend this result to the case of Γ_1 and Γ_2 disjoint near infinity. That follows the same arguments as in complex analysis in one variable.

13.2.2. Quadratic forms. In Chapter 4 real quadratic forms such as the symplectic form $\sigma(D_z, D_w)$ played an important role. In this chapter complex-valued forms on $\mathbb{R}^k$ and $\mathbb{C}^k$ will be important.

DEFINITION. The complex bilinear inner product on $\mathbb{C}^k$ is defined by

$$\langle z, z' \rangle := \sum_{j=1}^k z_j z'_j.$$

We stress that this is *not* the Hermitian inner product on $\mathbb{C}^k$.

In our applications we will typically deal with holomorphic quadratic forms on $\mathbb{C}^n \times \mathbb{C}^n$ and real quadratic forms on $\mathbb{C}^n$:

DEFINITION. (i) A holomorphic quadratic form on $\mathbb{C}^N$ is a function of the form

$$f(z) = \frac{1}{2} \langle Cz, z \rangle,$$

where $C \in M^{N \times N}(\mathbb{C})$ satisfies $C^T = C$. In particular, if we consider the product decomposition $\mathbb{C}^{n+k} \simeq \mathbb{C}^n \times \mathbb{C}^k$, then a holomorphic quadratic form on $\mathbb{C}^n \times \mathbb{C}^k$ is a function $\varphi = \varphi(z, w)$, $z \in \mathbb{C}^n$, $w \in \mathbb{C}^k$ given by

$$(13.2.3) \quad \varphi(z, w) := \frac{1}{2} \langle Az, z \rangle + \langle Bz, w \rangle + \frac{1}{2} \langle Dw, w \rangle,$$

where $A \in \mathbb{M}^{n \times n}(\mathbb{C})$, $B \in \mathbb{M}^{k \times n}$, $D \in \mathbb{M}^{k \times k}$ satisfy

$$A^T = A, \quad D^T = D.$$

(ii) A real quadratic form on $\mathbb{C}^n$, $\Phi = \Phi(z)$, is a function

$$(13.2.4) \quad \Phi(z) := \frac{1}{2} \langle A_0 z, z \rangle + \langle B_0 z, \bar{z} \rangle + \frac{1}{2} \langle D_0 \bar{z}, \bar{z} \rangle, \quad x, y \in \mathbb{R}^n,$$

where the complex $n \times n$ matrices, A_0, B_0, D_0 , satisfy

$$A_0^T = A_0 = \bar{D}_0, \quad B_0 = \bar{B}_0^T.$$

Here $\overline{(a_{ij})}_{1 \leq i, j \leq n} = (\bar{a}_{ij})_{1 \leq i, j \leq n}$.

DEFINITION. A k -dimensional $\mathbb{R}$ -linear subspace $\Lambda \subset \mathbb{C}^k$ is called *totally real* if $i\Lambda \cap \Lambda = \{0\}$.

DISCUSSION. For a totally real subspace Λ there exists a complex linear change of variables γ such that $\gamma(\Lambda) = \mathbb{R}^k$. Indeed, if $e_1, \dots, e_k$ is a real basis of Λ , then it is a complex basis of $\mathbb{C}^k$. We then apply the complex linear change of variables mapping $e_1, \dots, e_k$ to the canonical basis of $\mathbb{C}^k$. This shows that a holomorphic function on $\mathbb{C}^k$ is determined by its restriction to Λ . $\square$

EXAMPLE. The subspace $\Lambda = \{(z, \bar{z}) : z \in \mathbb{C}^n\}$ is a totally real subspace of $\mathbb{C}^n \times \mathbb{C}^n$. We see that for $\gamma(z, w) = (z + w, i(z - w))$, $\gamma(\Lambda) = \mathbb{R}^{2n}$.

LEMMA 13.1 (Properties of Φ). Let Φ be a real quadratic form on $\mathbb{C}^n$.

(i) There exists a unique holomorphic quadratic form Ψ on $\mathbb{C}^n \times \mathbb{C}^n$ such that

$$(13.2.5) \quad \Psi(z, \bar{z}) = \Phi(z).$$

It satisfies

$$(13.2.6) \quad \Psi(w, \bar{z}) = \overline{\Psi(z, \bar{w})}.$$

(ii) We have

$$(13.2.7) \quad \Phi(z) - \Phi(w) = 2 \operatorname{Re} \left\langle \partial_z \Phi \left(\frac{z+w}{2} \right), z-w \right\rangle.$$

Proof. 1. Substituting $\bar{z} = w$ into (13.2.4) gives an expression for $\Psi(z, w)$. Since $\Lambda = \{(z, \bar{z}) : z \in \mathbb{C}^n\}$ is totally real, Ψ is the unique holomorphic function extending Φ from Λ to $\mathbb{C}^n \times \mathbb{C}^n$.

2. The holomorphic function $\overline{\Psi(\bar{v}, \bar{w})}$ restricted to $w = \bar{v}$, $v \in \mathbb{C}^n$, is equal to $\Psi(w, v)$. Hence it is equal to $\Psi(w, v)$ and (13.2.6) follows by putting $v = \bar{z}$.

3. As $\partial_z \Phi$ is an $\mathbb{R}$ -linear function on $\mathbb{C}^n$, the right-hand side of (13.2.7) is given by

$$\operatorname{Re}(\langle \partial_z \Phi(z), z \rangle - \langle \partial_w \Phi(w), w \rangle + \langle \partial_w \Phi(w), z \rangle - \langle \partial_z \Phi(z), w \rangle).$$

From the fact that $\Phi(z) = \Psi(z, \bar{z}) \in \mathbb{R}$, we see that

$$\begin{aligned} \operatorname{Re}(\partial_z \Phi(z), z) &= \frac{1}{2} (\langle \partial_z \Psi(z, \bar{z}), z \rangle + \langle \partial_{\bar{z}} \Psi(z, \bar{z}), \bar{z} \rangle) \\ &= \Psi(z, \bar{z}) = \Phi(z). \end{aligned}$$

Hence it remains to show that

$$\operatorname{Re}(\langle \partial_w \Phi(w), z \rangle - \langle \partial_z \Phi(z), w \rangle) = 0,$$

and this follows from (13.2.4) and the properties of A_0 and B_0 . $\square$

In Chapter 3 we have expressions for the integrals of $\exp(iq(x, y)/h)$ where q was a real quadratic form. In this chapter we will need results about such integrals where $q(x, y)$ is complex-valued and $\operatorname{Im} \partial_x^2 q$ is positive definite. The following is an analogue of Theorem 3.13:

LEMMA 13.2. *Suppose that $x \mapsto q(x, Z)$ is a holomorphic quadratic form on $\mathbb{C}^n \times \mathbb{C}^N$ and put*

$$A := \frac{\partial^2 q}{\partial x^2}.$$

Suppose that

$$(13.2.8) \quad \operatorname{Im} A \text{ is positive definite.}$$

Then the equation

$$(13.2.9) \quad \frac{\partial q}{\partial x}(x, Z) = 0$$

has a unique solution $x = x(Z) \in \mathbb{C}^n$, and

$$(13.2.10) \quad \int_{\mathbb{R}^n} e^{\frac{i}{h} q(x, Z)} dx = \left(\det \frac{A}{2\pi i h} \right)^{-1/2} e^{\frac{i}{h} q(x(Z), Z)},$$

where the branch of the square root is positive on $(0, \infty)$.

Moreover, if $a(x) \in C_c^\infty(\mathbb{R}^n)$ and $x(Z) \in \mathbb{R}^n$, then

$$\int_{\mathbb{R}^n} e^{\frac{i}{h}q(x,Z)} a(x) dx = \left(\det \frac{A}{2\pi i h} \right)^{-1/2} e^{\frac{i}{h}q(x(Z),Z)} e^{h\langle A^{-1}D_x, D_x \rangle / (2i)} a(x(Z)),$$

where for each N ,

$$e^{h\langle A^{-1}D_x, D_x \rangle / (2i)} a(x) = \sum_{k=0}^{N-1} \frac{h^k}{k!} \left(\frac{1}{2i} \langle A^{-1}D_x, D_x \rangle \right)^k a(x) + O(h^N).$$

The latter expansion is valid in $S(\mathbb{R}^n)$ with constants in the remainder depending on some seminorm of $a(x)$ in $S(\mathbb{R}^n)$, similarly to Theorem 4.17(ii).

Proof. 1. We can write $q(x, Z)$ in the form (13.2.3):

$$q(x, Z) = \frac{1}{2} \langle Ax, x \rangle + \langle Bx, Z \rangle + \frac{1}{2} \langle DZ, Z \rangle.$$

The hypothesis then states that $\text{Im } A$ is positive definite and in particular A is invertible. Indeed, assume that $Ax = 0$ for some $x \in \mathbb{C}^n$. Then $\langle Ax, \bar{x} \rangle = 0 = \langle \bar{A}x, \bar{x} \rangle$ and thus $\langle (\text{Im } A)x, \bar{x} \rangle = 0$, implying $x = 0$.

The equation $\partial_x q = 0$ reads $Ax + B^T Z = 0$, with the unique solution

$$x(Z) = -A^{-1}B^T Z.$$

2. We rewrite the phase as

$$q(x, Z) = \frac{1}{2} \langle A(x - x(Z)), x - x(Z) \rangle + q(x(Z), Z)$$

and to show (13.2.10), we need to evaluate

$$\int_{\mathbb{R}^n} e^{\frac{i}{2h} \langle A(x - x(Z)), x - x(Z) \rangle} dx = \int_{\mathbb{R}^n - x(Z)} e^{\frac{i}{2h} \langle Ax, x \rangle} dx,$$

where by $\mathbb{R}^n - x(Z)$ we mean the contour $x \mapsto x - x(Z)$ in $\mathbb{C}^n$.

If $\text{Re } A = 0$, Theorem 3.1 applied with $Q = -iA$ shows that the integral is then equal to

$$(13.2.11) \quad \int_{\mathbb{R}^n} e^{\frac{i}{2h} \langle Ax, x \rangle} dx = (2\pi h)^{n/2} \det(-iA)^{-1/2},$$

where the square root is positive.

The set, H , of complex symmetric matrices, $A = A^T$, with $\text{Im } A$ positive definite, is an open convex set in the set of all complex matrices which we identify with $\mathbb{C}^{n(n+1)/2}$. Convexity of H and $\det A \neq 0$, for $A \in H$, show that there is a unique branch of $A \mapsto \det(-iA)^{1/2}$ which is positive for $\text{Re } A = 0$.

Since both sides of (13.2.11) are holomorphic in A as functions on H , the equality at the totally real subset of H , given by $\operatorname{Re} A = 0$, shows that (13.2.11) holds for all $A \in H$.

3. The stationary phase expansion for

$$\int_{\mathbb{R}^n} e^{\frac{i}{h}q(x,Z)} a(x) dx$$

is obtained as in Theorem 3.13, while the validity of this expansion in $S(\mathbb{R}^n)$ is proved similarly to Theorem 4.17. $\square$

13.2.3. Symplectic geometry. The symplectic form on $\mathbb{R}^n \times \mathbb{R}^n$,

$$\sigma = \sum_{j=1}^n d\xi_j \wedge dx_j, \quad (x, \xi) \in \mathbb{R}^n \times \mathbb{R}^n,$$

complexifies to define a complex symplectic form on $\mathbb{C}^n \times \mathbb{C}^n$,

$$\sigma_{\mathbb{C}} := \sum_{j=1}^n d\zeta_j \wedge dz_j, \quad (z, \zeta) \in \mathbb{C}^n \times \mathbb{C}^n.$$

In terms of quadratic forms, if $Z = (z, \zeta)$, $Z' = (z', \zeta') \in \mathbb{C}^n \times \mathbb{C}^n$, then

$$\sigma_{\mathbb{C}}(Z, Z') = \langle JZ, Z' \rangle, \quad J = \begin{pmatrix} O & I \\ -I & O \end{pmatrix}.$$

When we consider $\mathbb{C}^n \times \mathbb{C}^n \simeq \mathbb{R}^{2n} \times \mathbb{R}^{2n}$ as a real vector space, we have two natural real symplectic forms on $\mathbb{C}^n \times \mathbb{C}^n$: $\operatorname{Re} \sigma_{\mathbb{C}}$, $\operatorname{Im} \sigma_{\mathbb{C}}$, as they are both antisymmetric and nondegenerate. Writing

$$z = x + iy, \quad \zeta = \xi + i\eta,$$

we have

$$\begin{aligned} \operatorname{Re} \sigma_{\mathbb{C}} &= \sum_{j=1}^n d\xi_j \wedge dx_j - d\eta_j \wedge dy_j, \\ \operatorname{Im} \sigma_{\mathbb{C}} &= \sum_{j=1}^n d\xi_j \wedge dy_j + d\eta_j \wedge dx_j. \end{aligned}$$

EXAMPLE: I -Lagrangian subspaces. Let $\Lambda \subset \mathbb{C}^n \times \mathbb{C}^n \simeq \mathbb{R}^{2n} \times \mathbb{R}^{2n}$ be an $\mathbb{R}$ -linear subspace of real dimension $2n$. Then Λ is called I -Lagrangian if $\operatorname{Im} \sigma_{\mathbb{C}}$ vanishes on Λ . When $\operatorname{Re} \sigma_{\mathbb{C}}$ vanishes on Λ , we say that Λ is R -Lagrangian. For instance

$$\Lambda = \{(x, \xi) : x \in \mathbb{R}^n, \xi \in \mathbb{R}^n\} \subset \mathbb{C}^n \times \mathbb{C}^n$$

is I -Lagrangian but R -symplectic, in the sense that $\operatorname{Re} \sigma_{\mathbb{C}}$ is nondegenerate on Λ .

We can also check that if Λ is both R -Lagrangian and I -Lagrangian, then Λ is $\mathbb{C}$ -linear and Lagrangian with respect to $\sigma_{\mathbb{C}}$. When $f(z) = \frac{1}{2}\langle Az, z \rangle$ is a quadratic function of z , then

$$(13.2.12) \quad \Lambda = \{(z, \partial_z f) : z \in \mathbb{C}^n\} = \{(z, Az) : z \in \mathbb{C}^n\}$$

is Lagrangian with respect to $\sigma_{\mathbb{C}}$. When f is an arbitrary quadratic function on $\mathbb{C}^n \simeq \mathbb{R}^{2n}$, Λ will not in general be Lagrangian for $\sigma_{\mathbb{C}}$. $\square$

DEFINITION. A $\mathbb{C}$ -linear transformation $\kappa : \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}^n \times \mathbb{C}^n$ is called a *complex symplectic transformation* if $\kappa^* \sigma_{\mathbb{C}} = \sigma_{\mathbb{C}}$, that is, if

$$\kappa^T J \kappa = J.$$

EXAMPLE. Suppose $\varphi = \varphi(z, w)$ is a quadratic function of (z, w) , that is, a holomorphic quadratic function on $\mathbb{C}^n \times \mathbb{C}^n$. If

$$\det \partial_z \partial_w \varphi \neq 0,$$

then

$$(13.2.13) \quad (w, -\partial_w \varphi(z, w)) \mapsto (z, \partial_z \varphi(z, w))$$

implicitly defines a complex symplectic transformation which we will denote by κ_{φ} . Explicitly, if φ is given by (13.2.3) with $k = n$ and $\det B \neq 0$, then

$$(13.2.14) \quad \kappa_{\varphi}(w, \omega) = (-B^{-1}(Dw + \omega), (B^T - AB^{-1}D)w - AB^{-1}\omega). \quad \square$$

13.2.4. Plurisubharmonic functions. A C^2 function $u : \mathbb{C} \rightarrow \mathbb{R}$ is called subharmonic if $\Delta u \geq 0$. Writing the Laplacian as $4\partial_z \partial_{\bar{z}}$, this means that

$$\frac{\partial^2 u}{\partial z \partial \bar{z}} \geq 0.$$

This is equivalent to the statement that

$$(13.2.15) \quad r \mapsto \int_0^{2\pi} u(x + re^{i\theta}) d\theta \text{ is nondecreasing.}$$

The condition (13.2.15) assumes less regularity and can be used to give a more general definition of subharmonicity. We say that an upper semicontinuous function $u : \mathbb{C} \rightarrow [-\infty, \infty)$ is subharmonic if (13.2.15) holds.

The seemingly strange upper semicontinuity part of the definition is needed to guarantee that any two subharmonic functions equal almost everywhere are in fact equal (see [H1, Theorem 4.18]).

EXAMPLES. (i) Suppose that $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic. Then $u = \log |f|$ is subharmonic.

(ii) Suppose that $\{u_i\}_{i \in I}$ is a family of subharmonic functions on $\mathbb{C}$. If

$$u(x) := \sup_{i \in I} u_i(x)$$

takes values in $[-\infty, \infty)$ and u is upper semicontinuous, then $u(x)$ is subharmonic (see [H2, Corollary 16.1.5]). $\square$

DEFINITION. Suppose that $\Phi : \mathbb{C}^k \rightarrow \mathbb{R}$. If for all $a, b \in \mathbb{C}^k$,

$$z \mapsto \Phi(za + b), \quad z \in \mathbb{C},$$

is a subharmonic function in $\mathbb{C}$, we say that Φ is *plurisubharmonic*. This means that the restriction of Φ to any complex line is subharmonic.

When $\Phi \in C^2$, the definition is equivalent to the statement that

$$(13.2.16) \quad \sum_{i,j=1}^k \frac{\partial^2 \Phi}{\partial z_i \partial \bar{z}_j} \zeta_i \bar{\zeta}_j \geq 0, \quad \zeta \in \mathbb{C}^k.$$

In other words, the Hermitian matrix $(\partial_{z_i \bar{z}_j}^2 \Phi)_{i,j=1}^k$ is positive semidefinite.

DISCUSSION. In Section 13.1 we introduced weighted L^2 spaces of holomorphic functions, H_Φ . If $u \in H_\Phi$, then, roughly, $\log |u(z)| \leq \Phi(z)/h$. Since $\log |u(z)|$ is plurisubharmonic, optimal weights defining a space of holomorphic functions should also be plurisubharmonic.

DEFINITION. A C^2 function $\Phi : \mathbb{C}^k \rightarrow \mathbb{R}$ is called *strictly plurisubharmonic* if there exists $\lambda > 0$ such that

$$(13.2.17) \quad \sum_{i,j=1}^k \frac{\partial^2 \Phi(z)}{\partial z_i \partial \bar{z}_j} \zeta_i \bar{\zeta}_j \geq \lambda |\zeta|^2, \quad z \in \mathbb{C}^k, \quad \zeta \in \mathbb{C}^k.$$

In other words, the Hermitian matrix $(\partial_{z_i \bar{z}_j}^2 \Phi)_{i,j=1}^k$ is uniformly positive definite.

EXAMPLE. Suppose that $\Phi(z)$ is a real quadratic form on $\mathbb{C}^k$. Then if $\Phi(z)$ is strictly convex, then $\Phi(z)$ is strictly plurisubharmonic. The converse is not true, as show by the example of

$$\Phi(z) := \frac{1}{2} |\operatorname{Im} z|^2. \quad \square$$

A fundamental result which we cite from Hörmander [H2, Theorem 15.1.1] shows the importance of spaces $L_\Phi^2(\mathbb{C}^k)$, with Φ strictly plurisubharmonic. We will not use it but we record it to further motivate spaces H_Φ .

THEOREM 13.3 (Solving Cauchy–Riemann equations in L^2_Φ). *Let $\Phi : \mathbb{C}^k \rightarrow \mathbb{R}$ be a strictly plurisubharmonic function. Suppose that the $f_j \in L^2_\Phi(\mathbb{C}^k)$, $j = 1, \dots, k$, satisfy*

$$\frac{\partial f_j}{\partial \bar{z}_i} = \frac{\partial f_i}{\partial \bar{z}_j}, \quad i, j = 1, \dots, k.$$

Then there exists $u \in L^2_\Phi$ such that

$$\frac{\partial u}{\partial \bar{z}_j} = f_j, \quad j = 1, \dots, k, \quad \|u\|_{L^2_\Phi}^2 \leq \frac{h}{2\lambda} \sum_{j=1}^k \|f_j\|_{L^2_\Phi}^2,$$

where λ is given in (13.2.17).

13.3. FBI TRANSFORMS AND BERGMAN KERNELS

DEFINITION. Suppose that $\varphi(z, x)$ is a holomorphic quadratic function on $\mathbb{C}^n \times \mathbb{C}^n$ such that

$$(13.3.1) \quad \operatorname{Im} \frac{\partial^2 \varphi}{\partial x^2} \text{ is a positive definite matrix}$$

and

$$(13.3.2) \quad \det \frac{\partial^2 \varphi}{\partial x \partial z} \neq 0.$$

The *Fourier–Bros–Iagolnitzer (FBI)* transform associated to φ is the operator T defined on $\mathcal{S}(\mathbb{R}^n)$ by

$$(13.3.3) \quad \mathcal{T}_\varphi u(z) = c_\varphi h^{-3n/4} \int_{\mathbb{R}^n} e^{\frac{i}{h}\varphi(z,x)} u(x) dx,$$

where

$$c_\varphi = 2^{-n/2} \pi^{-3n/4} (\det \operatorname{Im} \partial_x^2 \varphi)^{-\frac{1}{4}} |\det \partial_x \partial z \varphi|.$$

The significance of the value of c_φ will become clear later.

EXAMPLE. When we choose

$$\varphi(z, x) = \frac{i}{2}(z - x)^2,$$

then for the function $\Phi(z)$ defined in (13.1.1) (see also (13.3.8) below),

$$\mathcal{T}_\varphi u(z) = c_\varphi h^{-3n/4} e^{\Phi(z)/h} e^{-\frac{i}{h}(\operatorname{Im} z, \operatorname{Re} z)} \mathcal{F}_h \left(e^{-\frac{1}{2h}(x - \operatorname{Re} z)^2} u(x) \right) (-\operatorname{Im} z). \quad \square$$

INTERPRETATION. The transform in the example is, up to the exponential weight $\Phi(z)$ and a factor of fixed size in z , given by

$$u \mapsto \mathcal{F}_h \left(e^{-\frac{1}{2h}(x - \operatorname{Re} z)^2} u(x) \right) (-\operatorname{Im} z).$$

This means that we first localize u in space near $\operatorname{Re} z$ using the Gaussian factor $\exp(-(x - \operatorname{Re} z)^2/2h)$ and then take the semiclassical Fourier transform at $-\operatorname{Im} z$. Hence we expect that semiclassical properties of u near a phase space point (x, ξ) should correspond to properties of $\mathcal{T}_\varphi u(z)$ near $z = x - i\xi$.

Defining $\mathcal{T}_\varphi$ with more general quadratic forms satisfying (13.3.1) and (13.3.2) provides additional flexibility and explains the geometric structure without explicit formulas. $\square$

THEOREM 13.4 (Basic mapping properties of FBI transforms).

Define

$$(13.3.4) \quad \Phi(z) := \max_{x \in \mathbb{R}^n} -\operatorname{Im} \varphi(z, x).$$

Then,

$$(i) \text{ for } u \in L^\infty, |\mathcal{T}_\varphi u(z)| \leq Ch^{-n/4} \|u\|_{L^\infty} e^{\Phi(z)/h};$$

and for u 's independent of h ,

$$(ii) \text{ for } u \in \mathcal{S}'(\mathbb{R}^n), |\mathcal{T}_\varphi u(z)| \leq Ch^{-M} (1 + |z|)^M e^{\Phi(z)/h} \text{ for some } M;$$

$$(iii) \text{ for } u \in \mathcal{S}(\mathbb{R}^n), |\mathcal{T}_\varphi u(z)| \leq C_N h^{-n/4} (1 + |z|)^{-N} e^{\Phi(z)/h} \text{ for all } N.$$

Proof. 1. The general phase φ can be written as in (13.2.3), with x taking the place of w . For proving estimates, we can make the following simplifying assumption. The term involving A contributes the factor $e^{\frac{i}{h}\langle Az, z \rangle/2}$ to $\mathcal{T}_\varphi u$ and a term $-\operatorname{Im}\langle Az, z \rangle/2$ to Φ . Hence by changing Φ we can assume that $A = 0$.

Denote

$$C = \operatorname{Im} D,$$

which gives a positive definite matrix. We check that

$$(13.3.5) \quad \Phi(z) := \max_{x \in \mathbb{R}^n} -\operatorname{Im} \varphi(z, x) = \frac{1}{2} \langle C^{-1} \operatorname{Im}(Bz), \operatorname{Im}(Bz) \rangle.$$

We denote

$$x(z) = -C^{-1} \operatorname{Im}(Bz),$$

the critical value of $-\operatorname{Im} \varphi(z, x)$,

$$\Phi(z) = -\operatorname{Im} \varphi(z, x(z)),$$

and note that

$$(13.3.6) \quad -\operatorname{Im} \varphi(z, x) \leq \Phi(z) - |x - x(z)|^2/C_0.$$

2. If $u \in L^\infty$, we use (13.3.6) to obtain

$$\begin{aligned} |\mathcal{T}_\varphi u(z)| &\leq |c_\varphi| h^{-3n/4} \|u\|_{L^\infty} \int_{\mathbb{R}^n} e^{-\operatorname{Im} \varphi(z,x)} dx \\ &\leq |c_\varphi| h^{-3n/4} \|u\|_{L^\infty} e^{\Phi(z)/h} \int_{\mathbb{R}^n} e^{-\frac{|x-x(z)|^2}{C_0 h}} dx \\ &\leq C_1 h^{-n/4} \|u\|_{L^\infty} e^{\Phi(z)/h}. \end{aligned}$$

3. If $u \in \mathcal{S}'$, then, as $e^{i\varphi(z,\cdot)/h} \in \mathcal{S}(\mathbb{R}^n)$,

$$\begin{aligned} |\mathcal{T}_\varphi u(z)| &= |c_\varphi| h^{-3n/4} |u(e^{i\varphi(z,\cdot)/h})| \\ &\leq C_0 h^{-3n/4} \sum_{|\alpha|+|\beta| \leq M} \sup_{\mathbb{R}^n} |x^\beta| |D_x^\alpha e^{i\varphi(z,\cdot)/h}| \\ &\leq C_1 h^{-3n/4} \sum_{|\alpha|+|\beta| \leq M} \sup_{\mathbb{R}^n} h^{-|\alpha|} |x^\beta| (|z| + |x|)^{|\alpha|} e^{\frac{1}{h}(\Phi(z) - |x-x(z)|^2/C_0)} \\ &\leq h^{-M-3n/4} (1 + |z|)^M e^{\Phi(z)/h}. \end{aligned}$$

Since we can change M to $M + 3n/4$, we obtain part (ii).

4. To handle the estimate in the case of $u \in \mathcal{S}$, we use the identity

$$(Bz)_j e^{i\varphi(z,x)/h} = (-(Dx)_j + hD_{x_j}) e^{i\varphi(z,x)/h}$$

to see that

$$(Bz)_j \mathcal{T}_\varphi u(z) = \mathcal{T}_\varphi ((-hD_{x_j} - (Dx)_j)u).$$

Since B is invertible, part (iii) follows from part (i). $\square$

DEFINITION. Suppose that φ satisfies (13.3.1) and (13.3.2). We define a complex linear symplectic transformation associated to φ by

$$(13.3.7) \quad \kappa_\varphi : (x, -\partial_x \varphi(z, x)) \rightarrow (z, \partial_z \varphi(x, z)), \quad x, z \in \mathbb{C}^n.$$

In a sense made precise in Theorems 13.5 and 13.7, the transformation $\mathcal{T}_\varphi$ is associated to κ_φ in the semiclassical spirit of Chapter 11.

EXAMPLES. (i) For $\varphi(z, x) = i(z - x)^2/2$ discussed in the previous example,

$$\kappa_\varphi(x, \xi) = (x - i\xi, \xi),$$

and

$$(13.3.8) \quad \Phi(z) = \frac{1}{2} |\operatorname{Im} z|^2.$$

(ii) Given Φ , a strictly plurisubharmonic quadratic function, we can find φ for which (13.3.4) holds. Let Φ be given by

$$(13.3.9) \quad \Phi(z) = \operatorname{Re}\langle A_0 z, z \rangle + \langle B_0 z, \bar{z} \rangle, \quad A_0 = A_0^T, \quad B_0 = B_0^*.$$

The condition that Φ is strictly plurisubharmonic means that B_0 is positive definite.

Suppose that φ is given by

$$\varphi(z, x) := \frac{1}{2}\langle Az, z \rangle + \langle Bz, x \rangle + \frac{it}{4}\langle x, x \rangle, \quad t > 0,$$

where $\det B \neq 0$. We made a simplifying assumption on D compared to the general form of φ given in (13.2.3).

Then, as in (13.3.5),

$$\begin{aligned} \Phi(z) &= -\frac{1}{2}\operatorname{Im}\langle Az, z \rangle + \frac{1}{t}\langle \operatorname{Im}(Bz), \operatorname{Im}(Bz) \rangle \\ &= \frac{1}{2}\operatorname{Re}\langle iAz, z \rangle - \frac{1}{4t}\langle Bz - \bar{B}\bar{z}, Bz - \bar{B}\bar{z} \rangle \\ &= \frac{1}{2}\operatorname{Re}\langle iAz, z \rangle - \frac{1}{2t}\operatorname{Re}\langle B^T Bz, z \rangle + \frac{1}{2t}\langle B^* Bz, \bar{z} \rangle. \end{aligned}$$

To match this with (13.3.9), we need

$$2tB_0 = B^* B, \quad 2tA_0 = itA - B^T B.$$

Since B_0 is positive definite such a B exists. For instance we can put

$$B = \sqrt{2ti}B_0^{\frac{1}{2}}, \quad A = -2iA_0 - \frac{i}{t}B^T B.$$

(iii) Applying the construction from example (ii) to

$$\Phi(z) = \frac{1}{2}|z|^2,$$

we first note that $B_0 = \frac{1}{2}I$ and $A_0 = 0$. Hence we can take $B = -i\sqrt{t}I$ and $A = iI$. That gives

$$(13.3.10) \quad \varphi(z, x) = \frac{i}{2}\langle z, z \rangle - i\sqrt{t}\langle z, x \rangle + \frac{it}{4}\langle x, x \rangle.$$

The corresponding symplectic transformation is

$$\kappa_\varphi : (x, i\sqrt{t}z - \frac{it}{2}x) \mapsto (z, iz - i\sqrt{t}x),$$

that is

$$\kappa_\varphi(x, \xi) = (t^{-1/2}(\frac{t}{2}x - i\xi), t^{-1/2}(\xi - \frac{it}{2}x)).$$

A nice choice, made by Bargmann, is to take $t = 2$, in which case

$$\begin{aligned}
 \varphi(z, x) &= \frac{i}{2} \langle z, z \rangle - i\sqrt{2} \langle z, x \rangle + \frac{i}{2} \langle x, x \rangle, \\
 (13.3.11) \quad \kappa_\varphi(x, \xi) &= \frac{1}{\sqrt{2}}(x - i\xi, \xi - ix), \\
 \kappa_\varphi^{-1}(z, \zeta) &= \frac{1}{\sqrt{2}}(z + i\zeta, \zeta + iz). \quad \square
 \end{aligned}$$

THEOREM 13.5 (Geometric interpretation of Φ). *The quadratic function $\Phi(z)$ given by (13.3.4) is strictly plurisubharmonic and κ_φ is a bijection of $\mathbb{R}^n \times \mathbb{R}^n \subset \mathbb{C}^n \times \mathbb{C}^n$ onto*

$$\Lambda_\Phi := \left\{ \left(z, \frac{2}{i} \frac{\partial \Phi}{\partial z} \right) : z \in \mathbb{C}^n \right\}.$$

The real linear subspace Λ_Φ is I -Lagrangian and R -symplectic.

Proof. 1. To show that

$$\kappa_\varphi(\mathbb{R}^n \times \mathbb{R}^n) = \Lambda_\Phi,$$

we need to show that if $x \in \mathbb{R}^n$ and $-\partial_x \varphi(z, x) \in \mathbb{R}^n$, then

$$(13.3.12) \quad \partial_z \varphi(z, x) = \frac{2}{i} \partial_z \Phi(z).$$

If $\partial_x \varphi(z, x)$ is real, then $\partial_x \operatorname{Im} \varphi(z, x) = 0$, which means $-\operatorname{Im} \varphi(z, x)$ has a maximum at $x = x(z)$, and $-\operatorname{Im} \varphi(z, x(z)) = \Phi(z)$. Cauchy–Riemann equations show that

$$-\partial_z \operatorname{Im} \varphi(z, x) = \frac{i}{2} \partial_z \varphi(z, x).$$

Hence, since $-\operatorname{Im} \varphi(z, x)$ is stationary at $x = x(z)$,

$$\frac{2}{i} \partial_z \Phi(z) = \partial_z \varphi(z, x(z)) - \frac{2}{i} \frac{\partial \operatorname{Im} \varphi(z, x)}{\partial x} \Big|_{x=x(z)} \frac{\partial x}{\partial z} = \partial_z \varphi(z, x),$$

and (13.3.12) follows.

2. The plane Λ_Φ is an image of an R -symplectic, I -Lagrangian plane, $\mathbb{R}^n \times \mathbb{R}^n \simeq T^*\mathbb{R}^n$ by a complex symplectic transformation, and hence it also has these properties.

3. Definition (13.3.4) gives $\Phi(z)$ as a maximum of a family of harmonic functions. Example (ii) in Section 13.2.4 shows that $\Phi(z)$ is plurisubharmonic. Hence we only need to show that

$$(13.3.13) \quad \det \frac{\partial^2 \Phi}{\partial z \partial \bar{z}} \neq 0.$$

To see this, we calculate the pull-back of the nondegenerate symplectic form on $\mathbb{R}^n \times \mathbb{R}^n$ by the inverse of κ_φ :

$$\begin{aligned} (\kappa_\varphi^{-1})^* \left(\sum_{j=1}^n d\xi_j \wedge dx_j \right) &= (\kappa_\varphi^{-1})^* \left(\sum_{j=1}^n d\zeta_j \wedge dz_j \Big|_{\mathbb{R}^n \times \mathbb{R}^n} \right) \\ &= \sum_{j=1}^n d\zeta_j \wedge dz_j \Big|_{\Lambda_\Phi} \\ &= \sum_{j=1}^n d \left(\frac{2}{i} \partial_{z_j} \Phi_j(z) \right) \wedge dz_j \\ &= \frac{2}{i} \sum_{k,j=1}^n \frac{\partial^2 \Phi}{\partial \bar{z}_k \partial z_j} d\bar{z}_k \wedge dz_j. \end{aligned}$$

As the form on the left is nondegenerate, (13.3.13) follows. □

THEOREM 13.6 (Representation of the Bergman kernel). *Let Ψ be the unique holomorphic quadratic form on $\mathbb{C}^n \times \mathbb{C}^n$ such that*

$$\Psi(z, \bar{z}) = \Phi(z).$$

Then the orthogonal projection $\Pi_\Phi : L_\Phi^2 \rightarrow H_\Phi$ is given by

$$(13.3.14) \quad \Pi_\Phi u(z) = \frac{2^n \det \partial_{z\bar{w}}^2 \Psi}{(\pi h)^n} \int_{\mathbb{C}^n} e^{2(\Psi(z, \bar{w}) - \Phi(w))/h} u(w) dm(w).$$

Proof. 1. Let Π_Φ be the operator given by (13.3.14). To show that it is the orthogonal projector onto H_Φ , we need to prove the following: (1) it is bounded $L_\Phi^2 \rightarrow H_\Phi$, (2) it is selfadjoint on L_Φ^2 , and (3) for $u \in H_\Phi$, $\Pi_\Phi u = u$.

To show that Π_Φ is bounded $L_\Phi^2 \rightarrow L_\Phi^2$, it suffices to note that

$$\begin{aligned} (13.3.15) \quad & 2\operatorname{Re} \Psi(z, \bar{w}) - \Phi(z) - \Phi(w) \\ &= \Psi(z, \bar{w}) + \Psi(w, \bar{z}) - \Psi(z, \bar{z}) - \Psi(w, \bar{w}) \\ &= \langle \partial_{\bar{z}} \Psi(z, \bar{z}) - \partial_{\bar{w}} \Psi(w, \bar{w}), \overline{w - z} \rangle \\ &= -\langle (\partial_{z\bar{z}}^2 \Phi)(z - w), \overline{z - w} \rangle \\ &\leq -|z - w|^2/C. \end{aligned}$$

The last inequality followed from Theorem 13.5, which stated that Φ is strictly plurisubharmonic.

By (13.3.14), the image of Π_Φ consists of holomorphic functions. The selfadjointness of this operator follows from (13.2.6):

$$\overline{\Psi(w, \bar{z})} = \Psi(z, \bar{w}).$$

2. It remains to show that for $u \in H_\Phi$, $\Pi_\Phi u = u$.

Using the notation reviewed in (13.2.1), we first prove that

$$(13.3.16) \quad \begin{aligned} u(z) &= \frac{1}{(2\pi h)^n} \iint_{\Gamma(z)} e^{\frac{i}{h}\langle z-w, \zeta \rangle} u(w) d\zeta \wedge dw, \\ \Gamma(z) : w &\mapsto \zeta := \frac{2}{i} \partial_z \Phi(z) + iC \overline{(z-w)}, \quad w \in \mathbb{C}^n, \end{aligned}$$

where $C > 0$ is sufficiently large. The absolute convergence of the integral for $u \in L^2_\Phi$ follows from (13.2.7):

$$(13.3.17) \quad \begin{aligned} \operatorname{Re}(i\langle z-w, \zeta \rangle) &= 2\operatorname{Re}\langle \partial_z \Phi(z), z-w \rangle - C|z-w|^2 \\ &= \Phi(z) - \Phi(w) + O(|z-w|^2) - C|z-w|^2 \\ &\leq \Phi(z) - \Phi(w) - |z-w|^2, \end{aligned}$$

if C is large enough.

To show (13.3.16), we note that its right-hand side is

$$\frac{C^n}{(\pi h)^n} \int_{\mathbb{C}^n} e^{-C|z-w|^2/h} e^{\frac{2}{h}\langle z-w, \partial_z \Phi(z) \rangle} u(w) dm(w).$$

The first factor in the product under this integral depends only on $|z-w|$, while the other two terms are holomorphic in w . We write the integral using polar coordinates centered at z and apply the mean value theorem for holomorphic functions, arriving at

$$\frac{C^n}{(\pi h)^n} \int_{\mathbb{C}^n} e^{-C|z-w|^2/h} dm(w) \cdot u(z).$$

The constant in this expression is a Gaussian integral and can be explicitly computed to be equal to 1.

3. We now perform the following contour deformation on (13.3.16). Let

$$\begin{aligned} \Gamma_t(z) : w &\mapsto (1-t) \left(\frac{2}{i} \partial_z \Phi(z) + iC \overline{(z-w)} \right) \\ &\quad + t \left(\frac{2}{i} \partial_{zw}^2 \Psi \bar{w} + \frac{1}{i} \partial_z^2 \Psi(z+w) \right), \quad 0 \leq t \leq 1. \end{aligned}$$

We have $\Gamma_0(z) = \Gamma(z)$, with $\Gamma(z)$ from (13.3.16). We also note that on $\Gamma_1(z)$,

$$i\langle z-w, \zeta \rangle = 2(\Psi(z, \bar{w}) - \Phi(w, \bar{w})).$$

Hence, (13.3.15) and (13.3.17) show that on $\Gamma_t(z)$,

$$\operatorname{Re}(i\langle z-w, \zeta \rangle) \leq \Phi(z) - \Phi(w) - |z-w|^2,$$

which provides the decay at infinity needed to justify the contour deformation. We conclude that

$$(13.3.18) \quad u(z) = \frac{1}{(2\pi h)^n} \iint_{\Gamma_1(z)} e^{\frac{i}{h}\langle z-w, \zeta \rangle} u(w) d\zeta \wedge dw.$$

4. We now make the following $\mathbb{C}$ -linear change of variables $\xi \mapsto \zeta$ in (13.3.18):

$$\zeta = \frac{2}{i} \partial_{zw}^2 \Psi \xi + \frac{1}{i} \partial_z^2 \Psi (z + w),$$

so that

$$(13.3.19) \quad i\langle z - w, \zeta \rangle = 2(\Psi(z, \xi) - \Phi(w, \xi)).$$

Hence (13.3.18) is the same as

$$u(z) = \frac{2^n \det \partial_{zw}^2 \Psi}{(\pi h)^n} \int e^{\frac{2}{h}(\Psi(z, \bar{w}) - \Phi(w))} u(w) dm(w).$$

In other words, $\Pi_\Phi u = u$, as required. $\square$

THEOREM 13.7 (Unitarity of the FBI transform). *Suppose that $\mathcal{T}_\varphi$ is defined by (13.3.3) and $\Phi = \Phi(z)$ by (13.3.4). Then*

$$\mathcal{T}_\varphi : L^2(\mathbb{R}^n) \rightarrow H_\Phi(\mathbb{C}^n)$$

is a unitary transformation. In particular, if $\mathcal{T}_\varphi^ : L_\Phi^2(\mathbb{C}^n) \rightarrow L^2(\mathbb{R}^n)$ is the adjoint of $\mathcal{T}_\varphi$, then $\Pi_\Phi = \mathcal{T}_\varphi \mathcal{T}_\varphi^*$.*

Proof. 1. We start by simplifying the function φ as in the proof of Theorem 13.4. Let φ be given by (13.2.3), with x taking the place of w . Since multiplication by $\exp(i\langle \operatorname{Re} Dx, x \rangle / (2h))$ is a unitary transformation on $L^2(\mathbb{R}^n)$ and since the multiplication operator

$$\exp\left(\frac{i}{2h}\langle Az, z \rangle\right) : H_{\Phi + \operatorname{Im}(\langle Az, z \rangle / 2)}(\mathbb{C}^n) \rightarrow H_\Phi(\mathbb{C}^n)$$

is also unitary, we may assume that $D = iC$ for some real-valued positive symmetric matrix C and that $A = 0$. That means that

$$(13.3.20) \quad \varphi(z, x) = \langle Bz, x \rangle + \frac{i}{2} \langle Cx, x \rangle.$$

2. We will first show that $\mathcal{T}_\varphi$ is an isometry $L^2(\mathbb{R}^n) \rightarrow H_\Phi(\mathbb{C}^n)$. Since $\mathcal{T}_\varphi u(z)$ is holomorphic, we only need to show that

$$(13.3.21) \quad \|u\|_{L^2(\mathbb{R}^n)} = \|\mathcal{T}_\varphi u\|_{L_\Phi^2}, \quad u \in \mathcal{S}(\mathbb{R}^n).$$

To establish this, we first note that

$$\mathcal{T}_\varphi u(B^{-1}w) = c_\varphi h^{-3n/4} \mathcal{F}_h \left(u(x) e^{-\langle Cx, x \rangle / (2h)} \right) (-w).$$

Hence by Parseval's formula (see Theorem 3.8),

$$\int_{\mathbb{R}^n} |\mathcal{T}_\varphi u(B^{-1}w)|^2 d\operatorname{Re} w = (2\pi)^n h^{-n/2} c_\varphi^2 \int_{\mathbb{R}^n} e^{-2(\langle \operatorname{Im} w, x \rangle + \langle Cx, x \rangle)/h} |u(x)|^2 dx.$$

Recalling that Φ is given by (13.3.5), we get

$$\begin{aligned} & \iint_{\mathbb{C}^n} |\mathcal{T}_\varphi u(B^{-1}w)|^2 e^{-2\Phi(B^{-1}w)/h} d\operatorname{Re} w d\operatorname{Im} w \\ &= (2\pi)^n h^{-n/2} c_\varphi^2 \iint e^{-2(\langle C^{-1}\xi, \xi \rangle + 2\langle \xi, x \rangle + \langle Cx, x \rangle)/h} |u(x)|^2 dx d\xi \\ &= (2\pi)^n h^{-n/2} c_\varphi^2 \iint e^{-\langle C^{-1}(\xi + Cx), \xi + Cx \rangle/h} |u(x)|^2 dx d\xi, \end{aligned}$$

where on the right-hand side we put $\xi = \operatorname{Im} w$. The integral in ξ can be evaluated using Theorem 3.1 and it is equal to $(\pi h)^{n/2} (\det C)^{1/2}$. A $\mathbb{C}$ -linear change of variables $z = B^{-1}w$ in the left-hand side of the integral above gives

$$\begin{aligned} & |\det B|^2 \int_{\mathbb{C}^n} |\mathcal{T}_\varphi u(z)|^2 e^{-2\Phi(z)/h} dm(z) \\ &= 2^n \pi^{3n/2} c_\varphi^2 (\det C)^{1/2} \|u\|_{L^2(\mathbb{R}^n)}^2. \end{aligned}$$

With our choice of c_φ this proves (13.3.21).

3. It remains to show that $\mathcal{T}_\varphi \mathcal{T}_\varphi^*$ is equal to the identity on H_Φ . Since it is selfadjoint on L_Φ^2 , we have to show that

$$(13.3.22) \quad \Pi_\Phi = \mathcal{T}_\varphi \mathcal{T}_\varphi^*,$$

where Π_Φ is given in (13.3.14).

The adjoint of $\mathcal{T}_\varphi : L^2 \rightarrow L_\Phi^2$ is explicitly given by

$$\mathcal{T}_\varphi^* v(x) = c_\varphi h^{-3n/4} \int_{\mathbb{C}^n} e^{-(i\overline{\varphi(w,x)} + 2\Phi(w))/h} v(w) dm(w).$$

Hence, the Schwartz kernel of $\mathcal{T}_\varphi \mathcal{T}_\varphi^*$ is equal to

$$\mathcal{T}_\varphi \mathcal{T}_\varphi^*(z, w) = c_\varphi^2 h^{-3n/2} e^{-2\Phi(w)/h} \int_{\mathbb{R}^n} e^{i(\varphi(z,x) - \overline{\varphi(w,x)})/h} dx.$$

We write $Z = (z, w)$ and

$$q(x, Z) = \varphi(z, x) - \overline{\varphi(w, x)},$$

which in view of the assumption (13.3.1) satisfies (13.2.8).

4. Hence we can apply Lemma 13.2 to obtain

$$(13.3.23) \quad \mathcal{T}_\varphi \mathcal{T}_\varphi^*(Z) = c_\varphi^2 \pi^{n/2} \det(-i\partial_x^2 q/2)^{-1/2} h^{-n} e^{-2\Phi(w)/h + iq(x(Z), Z)/h}.$$

Here we define $\bar{\varphi}(z, x) := \overline{\varphi(\bar{z}, \bar{x})}$, which is a holomorphic quadratic form in (z, x) . We can then write

$$q(x, Z) = \varphi(z, x) - \bar{\varphi}(\bar{w}, x),$$

which is now a holomorphic quadratic form in $(z, \bar{w}, x)$. If $z = w$ and x is real-valued, then $q(x, Z) = 2i\text{Im } \varphi(z, x)$. Therefore, for $z = w$ the critical point of $q(x, Z)$ coincides with the (real-valued) critical point of $\text{Im } \varphi(z, x)$ and we compute

$$Z = (z, z) \implies iq(x(Z), Z) = 2\Phi(z).$$

Since the left-hand side of the equation is a holomorphic quadratic form in $(z, \bar{w})$, we see by the definition of Ψ that

$$iq(x(Z), Z) = 2\Psi(z, \bar{w}).$$

5. In view of (13.3.23), to obtain (13.3.14), it remains to show that

$$(13.3.24) \quad c_\varphi^2 \pi^{n/2} \det(-i\partial_x^2 q/2)^{-1/2} h^{-n} = \frac{2^n \det \partial_{zw}^2 \Psi}{(\pi h)^n}.$$

In terms of φ given by (13.3.20) we have

$$c_\varphi^2 = 2^{-n} \pi^{-3n/2} (\det C)^{-\frac{1}{2}} |\det B|^2, \quad \partial_x^2 q = 2iC,$$

and, since $\Phi(z) = \frac{1}{2} \langle C^{-1} \text{Im}(Bz), \text{Im}(Bz) \rangle$,

$$\partial_{zw}^2 \Psi_{zw} = \partial_{z\bar{z}}^2 \Phi = \frac{1}{4} B^* C^{-1} B.$$

It remains to substitute those values into (13.3.24). □

13.4. QUANTIZATION AND TOEPLITZ OPERATORS

DEFINITION. Suppose $q \in L^\infty(\mathbb{C}^n)$. We define the *Toeplitz quantization* of q as

$$(13.4.1) \quad T_q u(z) := \Pi_\Phi M_q \Pi_\Phi, \quad T_q : H_\Phi \rightarrow H_\Phi.$$

Here $M_q : u \mapsto qu$ is the multiplication operator.

We will investigate the connection between the Toeplitz quantization, $q \mapsto T_q$, and the quantization corresponding to the Weyl quantization on $\mathbb{R}^n$:

$$(13.4.2) \quad a \in S(T^*\mathbb{R}^n), \quad a \mapsto \mathcal{T}_\varphi a^w(x, hD) \mathcal{T}_\varphi^*.$$

To define the Weyl quantization on the FBI transform side, we start with the following observation based on (13.2.7):

$$(13.4.3) \quad \operatorname{Re} i \langle \zeta, z - w \rangle = \Phi(z) - \Phi(w), \quad \zeta = \frac{2}{i} \partial_z \Phi \left(\frac{z + w}{2} \right).$$

We also note that boundedness of

$$Ku(z) = \iint_{\mathbb{C}^n} K(z, w) u(w) dm(w)$$

as an operator $L^2_{\Phi} \rightarrow L^2_{\Phi}$ is equivalent to the boundedness of the reduced operator

$$\tilde{K}u(z) = \iint_{\mathbb{C}^n} e^{(\Phi(w) - \Phi(z))/h} K(z, w) u(w) dm(w)$$

from L^2 to L^2 .

In what follows, we identify Λ_{Φ} with $\mathbb{C}^n$ using

$$(13.4.4) \quad z \mapsto \left(z, \frac{2}{i} \partial_z \Phi(z) \right) \in \Lambda_{\Phi}.$$

In particular, $\mathcal{S}(\Lambda_{\Phi}) \simeq \mathcal{S}(\mathbb{C}^n)$ and $S(\Lambda_{\Phi}) \simeq S(\mathbb{C}^n)$. It is nevertheless instructive to keep Λ_{Φ} in mind in the quantization procedure.

DEFINITION. For $a \in \mathcal{S}(\Lambda_{\Phi})$ we define the Weyl quantization

$$a_{\Phi}^w(z, hD_z) : L^2_{\Phi}(\mathbb{C}^n) \rightarrow L^2_{\Phi}(\mathbb{C}^n)$$

by

$$(13.4.5) \quad a_{\Phi}^w(z, hD_z)u = \frac{1}{(2\pi h)^n} \iint_{\Gamma_{\Phi}(z)} a\left(\frac{z+w}{2}\right) e^{\frac{i}{h}\langle z-w, \zeta \rangle} u(w) d\zeta \wedge dw,$$

where the contour Γ_{Φ} is given by

$$\Gamma_{\Phi}(z) : w \mapsto \zeta = \frac{2}{i} \partial_z \Phi \left(\frac{z + w}{2} \right).$$

The boundedness on L^2 follows from (13.4.3) and the comments about the reduced operator.

THEOREM 13.8 (Weyl operators on H_{Φ}). *The Weyl quantization $a \mapsto a_{\Phi}^w(z, hD)$ given by (13.4.5) extends from $\mathcal{S}(\Lambda_{\Phi})$ to $S(\Lambda_{\Phi})$ and defines a bounded operator*

$$a_{\Phi}^w(z, hD_z) : H_{\Phi}(\mathbb{C}^n) \rightarrow H_{\Phi}(\mathbb{C}^n).$$

Proof. 1. We first assume that $a \in \mathcal{S}(\Lambda_\Phi)$ so that the operator is well defined as

$$a_\Phi^w(z, hD_z) : H_\Phi(\mathbb{C}^n) \rightarrow L_\Phi^2(\mathbb{C}^n),$$

thanks to (13.4.3).

We need to show that for $u \in H_\Phi(\mathbb{C}^n)$, $a_\Phi^w(z, hD_z)u$ is holomorphic in z . For that, we compute $\partial_{\bar{z}}$ applied to the Schwartz kernel of a_Φ^w :

$$K_\Phi(z, w) = \frac{4^n \det \partial_{z\bar{z}}^2 \Phi}{(2\pi h)^n} a\left(\frac{z+w}{2}\right) \exp\left(\frac{2}{h} \left\langle z - w, \partial_z \Phi\left(\frac{z+w}{2}\right) \right\rangle\right),$$

$$\partial_{\bar{z}} K_\Phi(z, w) = \partial_{\bar{w}} K_\Phi(z, w).$$

Hence,

$$\begin{aligned} \partial_{\bar{z}} a_\Phi^w(z, hD_z)u &= \int_{\mathbb{C}^n} \partial_{\bar{z}} K_\Phi(z, w) u(w) dm(w) \\ &= \int_{\mathbb{C}^n} \partial_{\bar{w}} K_\Phi(z, w) u(w) dm(w) \\ &= \int_{\mathbb{C}^n} K_\Phi(z, w) (-\partial_{\bar{w}} u(w)) dm(w) \\ &= 0. \end{aligned}$$

2. To obtain boundedness on L_Φ^2 for $a \in S(\Lambda_\Phi)$, we first note that for

$$(13.4.6) \quad \zeta(z, w) = \frac{2}{i} \partial_z \Phi\left(\frac{z+w}{2}\right),$$

$$\begin{aligned} h \partial_{\bar{w}} e^{\frac{i}{h} \langle z-w, \zeta \rangle} &= i \langle z-w, \partial_w \zeta \rangle e^{\frac{i}{h} \langle z-w, \zeta \rangle} \\ &= \partial_{zz}^2 \Phi(z-w) e^{\frac{i}{h} \langle z-w, \zeta \rangle}. \end{aligned}$$

Since $\partial_{zz}^2 \Phi$ is nondegenerate, this means that

$$e^{\frac{i}{h} \langle z-w, \zeta \rangle} = L e^{\frac{i}{h} \langle z-w, \zeta \rangle},$$

where the differential operator $L = L(z, w, hD_{\bar{w}})$ is given by

$$L = \frac{1 + \langle (\partial_{z\bar{z}}^2 \Phi)^{-1}(\overline{z-w}), \partial_{\bar{w}} \rangle}{1 + |z-w|^2/h}.$$

This means that for $a \in \mathcal{S}(\Lambda_\Phi)$ and $v \in H_\Phi$,

$$(13.4.7) \quad a_\Phi^w(z, hD_z)v = \frac{4^n \det \partial_{z\bar{z}}^2 \Phi}{(2\pi h)^n} \int_{\mathbb{C}^n} e^{i \langle z-w, \zeta \rangle / h} (L^T)^N a\left(\frac{z+w}{2}\right) v(w) dm(w),$$

where ζ in the integrand is given by (13.4.6).

3. For $a \in S(\Lambda_\Phi)$ we have

$$(L^T)^N a\left(\frac{z+w}{2}\right) = O(\langle h^{-1/2}(z-w) \rangle^{-N})$$

and hence by Schur's inequality the formula (13.4.7) defines $a_{\Phi}^w(z, hD_z)$ as a bounded operator $H_{\Phi} \rightarrow L_{\Phi}^2$; see part 2 of the proof of Theorem 4.21. The operator norm of $a_{\Phi}^w(z, hD_z)$ on L_{Φ}^2 is bounded uniformly in h :

$$h^{-n} \int_{\mathbb{C}^n} \langle h^{-1/2}(z - w) \rangle^{-N} dm(w) = O(1).$$

□

The significance of Λ_{Φ} and of the quantization of $a \in S(\Lambda_{\Phi})$ becomes clear in the next theorem. It shows that the FBI transform, $\mathcal{T}_{\varphi}$, relates that quantization to the quantization of classical observables on $T^*\mathbb{R}^n$.

THEOREM 13.9 (Complex and real quantizations). *The FBI transform (13.3.3) provides a one-to-one correspondence between the quantizations of $S(T^*\mathbb{R}^n)$ and $S(\Lambda_{\Phi})$:*

$$(13.4.8) \quad \mathcal{T}_{\varphi}^* a_{\Phi}^w(z, hD_z) \mathcal{T}_{\varphi} = (\kappa_{\varphi}^* a)^w(x, hD_x), \quad a \in S(\Lambda_{\Phi}),$$

where by Theorem 13.5, $\kappa_{\varphi}^* : S(\Lambda_{\Phi}) \rightarrow S(T^*\mathbb{R}^n)$.

Proof. 1. We will proceed as in the proof of Theorem 11.9 and consider the case of $b = \kappa_{\varphi}^* a$ given by the exponential of a linear transformation:

$$a(z) = e^{\frac{i}{h}k(z)}, \quad b(x, \xi) = e^{\frac{i}{h}l(x, \xi)},$$

where $l(x, \xi) = \langle x^*, x \rangle + \langle \xi^*, \xi \rangle$, $x^*, \xi^* \in \mathbb{R}^n$, and $k = l \circ \kappa_{\varphi}^{-1}$ is the corresponding real-valued linear function on Λ_{Φ} . As in the proof of Theorem 11.9 the general case follows from Lemma 4.10. (Even though the S -seminorms of b are not bounded uniformly in h , the argument still goes through.)

2. We extend l to a $\mathbb{C}$ -linear function $\tilde{l}$ on $\mathbb{C}^{2n}$ and put

$$\tilde{k} = \tilde{l} \circ \kappa_{\varphi}^{-1},$$

where κ_{φ} given by (13.3.7) is considered as a $\mathbb{C}$ -linear transformation $\mathbb{C}^{2n} \rightarrow \mathbb{C}^{2n}$. Then $\tilde{k}$ is a $\mathbb{C}$ -linear function on $\mathbb{C}^{2n}$ and $\tilde{k}(z, \zeta) = k(z)$ on Λ_{Φ} . We write

$$\tilde{k}(z, \zeta) = \langle z^*, z \rangle + \langle \zeta^*, \zeta \rangle,$$

where $z^*, \zeta^* \in \mathbb{C}^n$.

Assume that $v(z)$ is a holomorphic function on $\mathbb{C}^n$ such that

$$e^{-\Phi(z)/h} v(z) \in \mathcal{S}(\mathbb{C}^n).$$

For $(w, \zeta) \in \Gamma_{\Phi}(z)$,

$$k((z + w)/2) = \tilde{k}((z + w)/2, \zeta),$$

and hence

$$a_{\Phi}^w(z, hD_z)v(z) = \frac{1}{(2\pi h)^n} \iint_{\Gamma_{\Phi}(z)} e^{\frac{i}{h}(\langle z-w, \zeta \rangle + \bar{k}((z+w)/2, \zeta))} v(w) d\zeta \wedge dw.$$

The integral converges due to (13.4.3) and the rapid decay of $e^{-\Phi(w)/h}v(w)$. We now use the method of part 2 of the proof of Theorem 13.6. Since the expression under the integral is holomorphic in w and ζ , we can deform the contour to

$$\tilde{\Gamma}(z) : w \mapsto \zeta = \frac{2}{i} \partial_z \Phi(z) + iC \overline{(z - w + \zeta^*)},$$

where C is a large enough constant so that the integral converges on the new contour. Using polar coordinates in w centered at $z + \zeta^*$ and the mean value theorem for holomorphic functions, we get

$$(13.4.9) \quad a_{\Phi}^w(z, hD_z)v(z) = e^{\frac{i}{2h}\langle z^*, \zeta^* \rangle} e^{\frac{i}{h}\langle z^*, z \rangle} v(z + \zeta^*).$$

3. Take $u \in \mathcal{S}(\mathbb{R}^n)$. By part (iii) of Theorem 13.4, the function $v = \mathcal{T}_{\varphi}u$ satisfies the assumption of step 2 of this proof. We now recall from Theorem 4.7 that

$$(13.4.10) \quad b^w(x, hD_x)u(x) = e^{\frac{i}{2h}\langle x^*, \xi^* \rangle} e^{\frac{i}{h}\langle x^*, x \rangle} u(x + \xi^*).$$

A direct calculation using (13.4.9), (13.4.10), and the definition of $\mathcal{T}_{\varphi}$ shows that

$$\mathcal{T}_{\varphi}b^w(x, hD_x)u = a_{\Phi}^w(z, hD_z)v,$$

which is the same as $\mathcal{T}_{\varphi}b^w = a_{\Phi}^w\mathcal{T}_{\varphi}$. It remains to use the unitarity of $\mathcal{T}_{\varphi}$. $\square$

REMARK. Theorems 13.7, 13.8, and 13.9 provide an alternative proof of Theorem 4.23. It is arguably simpler as it does not involve the decay estimates in Theorem 4.22 and the Cotlar–Stein Theorem (Theorem C.5).

THEOREM 13.10 (Toeplitz and Weyl quantizations). (i) Suppose that $q \in S(\mathbb{C}^n)$. Then

$$(13.4.11) \quad T_q = a_{\Phi}^w(z, hD_z),$$

for $a \in S(\Lambda_{\Phi})$ given by, under the identification (13.4.4),

$$(13.4.12) \quad a(z) = \exp\left(\frac{h}{4} \left\langle (\partial_{z\bar{z}}^2 \Phi)^{-1} \partial_z, \partial_{\bar{z}} \right\rangle\right) q(z),$$

where $\partial_{z\bar{z}}^2 \Phi$ is positive definite by Theorem 13.5.

(ii) Suppose that $a \in S(\Lambda_{\Phi})$. Then there exists $q \in S(\mathbb{C}^n)$ such that

$$(13.4.13) \quad a_{\Phi}^w(z, hD) = T_q + O_{H_{\Phi} \rightarrow H_{\Phi}}(h^{\infty}),$$

and for each N ,

$$(13.4.14) \quad q(z) = \sum_{j=0}^N \frac{h^j}{4^j j!} \left(\left\langle (\partial_{z\bar{z}}^2 \Phi)^{-1} D_z, D_{\bar{z}} \right\rangle \right)^j a(z) + O_{S(\mathbb{C}^n)}(h^{N+1}).$$

REMARK. Consider the case of $\Phi(z) = \frac{1}{2}|z|^2$. The formula (13.4.12) becomes

$$(13.4.15) \quad a(z) = \exp\left(\frac{1}{8}h\Delta\right) q(z)$$

where $\Delta := 4 \sum_{j=1}^n \partial_{z_j} \partial_{\bar{z}_j}$ is the Laplacian on $\mathbb{C}^n \simeq \mathbb{R}^{2n}$. Hence using the formula for the heat propagator, we explicitly get

$$a(z) = \left(\frac{2}{\pi h}\right)^n \int_{\mathbb{C}^n} e^{-\frac{2}{h}|z-w|^2} q(w) dm(w).$$

In particular, we see that a obtained from q is *real analytic*. We could actually define a for $q \in L^\infty$ but $a \notin S(\Lambda_\Phi)$ in general; that is, derivatives will grow as $h \rightarrow 0$.

This is true for all Φ as the formula (13.4.12) can be interpreted as the forward heat flow for an elliptic constant coefficient operator

$$\left\langle (\partial_{z\bar{z}}^2 \Phi)^{-1} \partial_z, \partial_{\bar{z}} \right\rangle. \quad \square$$

INTERPRETATION. Formula (13.4.12) has a natural interpretation. Using (13.3.14) and (13.3.19), for $u \in H_\Phi$ we can write

$$(13.4.16) \quad \Pi_\Phi(qu)(z) = \frac{1}{(2\pi h)^n} \iint_{\Gamma} e^{\frac{i}{h}\langle z-w, \xi \rangle} q(w) u(w) d\xi \wedge dw,$$

where

$$(13.4.17) \quad \Gamma : w \mapsto \xi = \frac{2}{i} \partial_{z\bar{z}}^2 \Phi \bar{w} + \frac{1}{i} \partial_z^2 \Psi(z+w).$$

Formally the expression (13.4.16) is similar to the formula for the left quantization, $\tilde{q} \mapsto \text{Op}_0(\tilde{q})$, $\tilde{q} \in S(\mathbb{R}^n \times \mathbb{R}^n)$.

If $\text{Op}_0(\tilde{q}) = \tilde{a}^w$, $\tilde{a} \in S(\mathbb{R}^n \times \mathbb{R}^n)$, then Theorem 4.13 applied with $t = 1/2$ and $s = 0$ gives

$$(13.4.18) \quad \tilde{a}(x, \xi) = \exp\left(\frac{1}{2}ih\langle D_x, D_\xi \rangle\right) \tilde{q}(x, \xi).$$

Making a formal substitution given in (13.4.17) gives

$$D_\xi = \left(\left(\frac{2}{i} \partial_{z\bar{z}}^2 \Phi \right)^{-1} \right)^T D_{\bar{z}},$$

and (13.4.18) translates to (13.4.12). □

Proof. 1. Since (ii) follows directly from (i), we only need to prove the latter. By Theorem 13.9, it suffices to show that

$$(13.4.19) \quad \mathcal{T}_\varphi^* M_q \mathcal{T}_\varphi = (\kappa_\varphi^* a)^w(x, hD_x).$$

Moreover, it suffices to consider $q \in C_c^\infty(\mathbb{C}^n)$. Indeed, every $q \in S(\mathbb{C}^n)$ is the limit in, say, $\langle z \rangle^2 L^\infty(\mathbb{C}^n)$ of a sequence $q_j \in C_c^\infty(\mathbb{C}^n)$. Then $\mathcal{T}_\varphi^* M_{q_j} \mathcal{T}_\varphi$ converges, as an operator $\mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}'(\mathbb{R}^n)$, to the left-hand side of (13.4.19) by Theorem 13.4(iii), while $a_j^w(x, hD_x)$ converges in the same topology to the right-hand side of (13.4.19) by Theorem 4.2.

For $q \in C_c^\infty(\mathbb{C}^n)$ and $\eta \in \mathbb{R}^n$ we compute the formal left symbol of the action of the Toeplitz operator on the $L^2(\mathbb{R}^n)$ side:

$$(13.4.20) \quad b_q(x, \eta) = e^{-\frac{i}{h}\langle x, \eta \rangle} (\mathcal{T}_\varphi^* M_q \mathcal{T}_\varphi (e^{\frac{i}{h}\langle \cdot, \eta \rangle}))(x),$$

$$(13.4.21) \quad \mathcal{T}_\varphi^* M_q \mathcal{T}_\varphi = b_q(x, hD).$$

Here we used Theorem 4.19.

The expression (13.4.20) is justified as $e^{\frac{i}{h}\langle y, \eta \rangle} \in \mathcal{S}'$ and we can apply part (ii) of Theorem 13.4. It shows that

$$q(z) \mathcal{T}_\varphi (e^{\frac{i}{h}\langle \cdot, \eta \rangle}) \in L_{\Phi}^2,$$

so that $\mathcal{T}_\varphi^*$ can be applied. We will later show that $b_q \in \mathcal{S}(\mathbb{R}^{2n})$ which then justifies (13.4.21).

2. The formulas for $\mathcal{T}_\varphi$ and $\mathcal{T}_\varphi^*$ give

$$(13.4.22) \quad b_q(x, \eta) = c_\varphi^2 h^{-3n/2} \int_{\mathbb{C}^n} \int_{\mathbb{R}^n} e^{\frac{i}{h}\Theta} q(z) dy dm(z),$$

where Θ is the following quadratic function of $(z, y, x, \eta) \in \mathbb{C}^n \times \mathbb{R}^{3n}$:

$$\Theta(z, y; x, \eta) = \varphi(z, y) - \overline{\varphi(z, x)} + 2i\Phi(z) + \langle y - x, \eta \rangle.$$

By the definition (13.3.4) of $\Phi(z)$, we see that $\text{Im } \Theta \geq 0$ everywhere. The critical points of the phase Θ are given by the equations

$$(13.4.23) \quad 0 = \partial_z \Theta = \partial_z \varphi(z, y) + 2i\partial_z \Phi(z) = \partial_z \varphi(z, y) - \partial_z \varphi(z, x(z)),$$

$$(13.4.24) \quad 0 = -\overline{\partial_z \Theta} = \partial_z \varphi(z, x) + 2i\partial_z \Phi(z) = \partial_z \varphi(z, x) - \partial_z \varphi(z, x(z)),$$

$$(13.4.25) \quad 0 = \partial_y \Theta = \partial_y \varphi(z, y) + \eta.$$

Here $x(z)$ is the unique point in $\mathbb{R}^n$ where the function $-\text{Im } \varphi(z, x)$ achieves its maximum. We recall from step 1 of the proof of Theorem 13.5 that if $\mathbb{C}^n$ is identified with Λ_Φ by (13.4.4), then $\partial_x \varphi(z, x(z))$ is real-valued and

$$\begin{aligned} \partial_z \varphi(z, x(z)) &= \frac{2}{i} \partial_z \Phi(z), \\ z &= \kappa_\varphi(x(z), -\partial_x \varphi(z, x(z))). \end{aligned}$$

The equations (13.4.23), (13.4.24) are equivalent to $x = y = x(z)$. Therefore, the only critical point of Θ is given by

$$z = \kappa_\varphi(x, \eta), \quad y = x.$$

The value of Θ at this critical point is 0.

3. Armed with the knowledge of critical points of Θ , we make the change of variables

$$(13.4.26) \quad z = \kappa_\varphi(x', \eta'), \quad (x', \eta') \in \mathbb{R}^{2n}.$$

We now let φ be given by (13.2.3):

$$\varphi(z, x) = \frac{1}{2} \langle Az, z \rangle + \langle Bz, x \rangle + \frac{1}{2} \langle Dx, x \rangle.$$

Put $C = \text{Im } D$. By (13.2.14),

$$\kappa_\varphi(x, \eta) = -B^{-1}(Dx + \eta).$$

Therefore, the Jacobian of the change of variables (13.4.26) is equal to $|\det B|^{-2} \det C$. Also,

$$c_\varphi^2 = 2^{-n} \pi^{-3n/2} (\det C)^{-1/2} |\det B|^2.$$

After the change of variables, (13.4.22) becomes

$$b_q(x, \eta) = 2^{-n} (\pi h)^{-3n/2} (\det C)^{1/2} \int_{\mathbb{R}^{3n}} e^{\frac{i}{h} \tilde{\Theta}} (\kappa_\varphi^* q)(x', \eta') dx' d\eta' dy,$$

where

$$\begin{aligned} \tilde{\Theta}(x', \eta', y; x, \eta) &= \Theta(\kappa_\varphi(x', \eta'), y; x, \eta) \\ &= \langle D \left(\frac{y+x}{2} - x' \right) - (\eta' - \eta), y - x \rangle + i \langle C(x' - x), x' - x \rangle. \end{aligned}$$

Now, by Lemma 13.2 (whose proof still applies despite $\partial_{(x', \eta', y)}^2 \tilde{\Theta}$ not being positive definite — this matrix is nondegenerate and its imaginary part is nonnegative, as $\text{Im } \tilde{\Theta}$ is nonnegative everywhere)

$$b_q(x, \eta) = e^{-\frac{h}{4} (\langle C^{-1} D_x, D_x \rangle + \langle C^{-1} \overline{D} D_\eta, D D_\eta \rangle - 2 \langle C^{-1} D D_\eta, D_x \rangle)} (\kappa_\varphi^* q)(x, \eta).$$

4. Passing to Weyl quantization and using (13.4.21), we get for $q \in C_c^\infty(\mathbb{C}^n)$,

$$\mathcal{T}_\varphi^* M_q \mathcal{T}_\varphi = a_q^w(x, hD_x),$$

where

$$\begin{aligned} a_q(x, \eta) &= e^{-i \frac{h}{2} \langle D_x, D_\eta \rangle} b_q(x, \eta) \\ &= e^{-\frac{h}{4} \langle C^{-1} (D_x - D D_\eta), D_x - \overline{D} D_\eta \rangle} (\kappa_\varphi^* q)(x, \eta) \\ &= (e^{-h \langle C B^{-T} D_z, \overline{B}^{-T} D_{\bar{z}} \rangle} q)(\kappa_\varphi(x, \eta)) \\ &= (e^{-\frac{h}{4} \langle (\partial_{z\bar{z}}^2 \Phi)^{-1} D_z, D_{\bar{z}} \rangle} q)(\kappa_\varphi(x, \eta)). \end{aligned}$$

In other words, $a_q(x, \eta) = a(\kappa_\varphi(x, \eta))$, with a given by (13.4.12). This proves (13.4.19). $\square$

Using the composition formula for the Weyl quantization (4.3.6), we can now get the composition formula for Toeplitz quantization.

Quadratic forms revisited. Just as in (4.3.6) and (13.4.12) quadratic forms will appear and we need to review their significance more abstractly. Suppose V and W are real vector spaces and $\kappa : V \rightarrow W$ is a linear isomorphism. We denote by V^T and W^T the corresponding dual spaces with the transpose map

$$\kappa^T : W^T \rightarrow V^T.$$

Suppose that A is a nondegenerate quadratic form on V . It defines a natural form A^{-1} on V^T as follows. First let $V^T \rightarrow V$ be the map

$$\alpha : X^* \mapsto X, \quad A(X, Y) = X^*(Y), \quad \text{for all } Y \in V,$$

where for X^* in the dual space V^T , $X^*(Y) \in \mathbb{R}$ is the action of $X^* \in V^T$ on $Y \in V$.

We then put

$$A^{-1}(X^*, Y^*) = A(\alpha(X^*), \alpha(Y^*)).$$

Since A^{-1} is defined on the dual space of V , it defines a natural differential operator acting on functions on V :

$$A^{-1}(D) : C^\infty(V) \rightarrow C^\infty(V).$$

We then have

$$(\kappa^{-1})^* A^{-1}(D) \kappa^* = (\kappa^T)^* (A^{-1})(D) = ((\kappa^{-1})^* A)^{-1}(D).$$

REMARK. In Theorem 4.11 the symplectic form $\sigma(D_z, D_w)$, $z \in \mathbb{R}^{2n}$, $w \in \mathbb{R}^{2n}$, appeared in the composition formula, but strictly speaking it should have been $\sigma^{-1}(D_z, D_w)$, which is equal to $\sigma(D_z, D_w)$. When the symplectic form is not standard, this becomes an issue as seen in the composition formula in the Toeplitz case. $\square$

THEOREM 13.11 (Composition for Toeplitz quantization). *Assume that $q_1, q_2 \in S(\mathbb{C}^n)$. Then*

$$T_{q_1} T_{q_2} = T_q + O_{H_\Phi \rightarrow H_\Phi}(h^\infty),$$

where q has asymptotic expansion in powers of h given by

$$(13.4.27) \quad q(z) = e^{\frac{h}{2} \operatorname{Re} \langle (\partial_{z\bar{z}}^2 \Phi)^{-1} D_z, D_{\bar{w}} \rangle} e^{\frac{h}{2} i \sigma_\Phi^{-1}(D_z, \bar{z}, D_w, \bar{w})} (q_1(z) q_2(w))|_{w=z}.$$

Here σ_Φ is the (real-valued, as Λ_Φ is I -Lagrangian) pull-back of $\sigma_\mathbb{C}$ by the map

$$z \in \mathbb{C}^n \mapsto \left(z, \frac{2}{i} \partial_z \Phi(z) \right) \in \Lambda_\Phi \subset \mathbb{C}^{2n},$$

$$\sigma_\Phi = \frac{2}{i} \sum_{i,j=1}^n \frac{\partial^2 \Phi}{\partial z_i \partial \bar{z}_j} dz_i \wedge d\bar{z}_j.$$

REMARK. The formula (13.4.27) can only be treated as an asymptotic series in h , as the quadratic form $\text{Re} \langle (\partial_{z\bar{z}}^2 \Phi)^{-1} D_z, D_{\bar{w}} \rangle$ has signature 0. For q_1, q_2 which have compactly supported Fourier transforms, the exponentials of the differential operators can be applied and (13.4.27) is valid not only as an asymptotic expansion but also as an exact formula. $\square$

Proof. We have

$$T_{q_j} = (a_j)_\Phi^w(z, hD_z)$$

where the $a_j \in S(\Lambda_\Phi)$ are given by (13.4.12). By Theorem 13.9 and (4.3.6),

$$(a_1)_\Phi^w(z, hD_z)(a_2)_\Phi^w(z, hD_z) = a_\Phi^w(z, hD_z),$$

where, for σ the standard symplectic form on $\mathbb{R}^{2n}$ (see the remark before the theorem),

$$\begin{aligned} a(z, \zeta) &= (e^{\frac{h}{2}i\sigma^{-1}(D_x, D_\xi, D_y, D_\eta)}(\kappa_\varphi^* a_1)(x, \xi)(\kappa_\varphi^* a_2)(y, \eta))|_{(y, \eta)=(x, \xi)=\kappa_\varphi^{-1}(z, \zeta)} \\ &= e^{\frac{h}{2}i((\kappa_\varphi^{-1})^* \sigma)^{-1}(D_{z, \zeta}, D_{w, \zeta'})}(a_1(z, \zeta)a_2(w, \zeta'))|_{\substack{w=z \\ \zeta'=\zeta}}. \end{aligned}$$

Here $(\kappa_\varphi^{-1})^* \sigma$ is a symplectic form on Λ_Φ . Since κ_φ is a symplectomorphism of $\sigma_\mathbb{C}$ and σ is the restriction of $\sigma_\mathbb{C}$ to $\mathbb{R}^{2n}$, if we parametrize Λ_Φ by z and consider a_1, a_2, a as functions of z , then

$$a(z) = e^{\frac{h}{2}i\sigma_\Phi^{-1}(D_{z, \bar{z}}, D_{w, \bar{w}})}(a_1(z)a_2(w))|_{w=z}.$$

We now see that

$$\begin{aligned} T_{q_1} T_{q_2} &= (a_1)_\Phi^w(z, hD_z)(a_2)_\Phi^w(z, hD_z) \\ &= a_\Phi^w(z, hD_z) \\ &= T_q + O_{H_\Phi \rightarrow H_\Phi}(h^\infty), \end{aligned}$$

where

$$\begin{aligned} q &:= e^{\frac{h}{4}\langle (\partial_{z\bar{z}}^2 \Phi)^{-1} D_z, D_{\bar{z}} \rangle} a(z) \\ &= e^{\frac{h}{4}\langle (\partial_{z\bar{z}}^2 \Phi)^{-1} (D_z + D_w), D_{\bar{z}} + D_{\bar{w}} \rangle} e^{\frac{h}{2}i\sigma_\Phi^{-1}(D_{z, \bar{z}}, D_{w, \bar{w}})}(a_1(z)a_2(w))|_{w=z} \\ &= e^{\frac{h}{2}\text{Re}\langle (\partial_{z\bar{z}}^2 \Phi)^{-1} D_z, D_{\bar{w}} \rangle} e^{\frac{h}{2}i\sigma_\Phi^{-1}(D_{z, \bar{z}}, D_{w, \bar{w}})}(q_1(z)q_2(w))|_{w=z}, \end{aligned}$$

as stated in (13.4.27). $\square$

13.5. APPLICATIONS

We have already seen an application in the remark after the proof of Theorem 13.9: an alternative proof of L^2 boundedness of $a^w(x, hD)$ for $a \in S$. This section gives examples of some diverse applications, including to a purely algebraic theorem of Quillen. For more on applications to PDE see Delort [De], Martinez [M], and Sjöstrand [S1].

13.5.1. Approximation by multiplication. The connection between the Weyl quantization and the Toeplitz quantization provides an answer to the motivating question of this chapter: how close can quantization be to multiplication?

THEOREM 13.12 (Approximation by multiplication). *Suppose that $a \in S(\Lambda_\Phi)$. Then, after identifying Λ_Φ with $\mathbb{C}^n$ via (13.4.4):*

(i) *Denoting by $M_a : H_\Phi \rightarrow L_\Phi^2$ the multiplication operator by $a(z)$, we have*

$$(13.5.1) \quad a_\Phi^w(z, hD_z) - M_a = O_{H_\Phi \rightarrow L_\Phi^2}(\sqrt{h}).$$

In particular, for $b \in S(T^\mathbb{R}^n)$*

$$(13.5.2) \quad \mathcal{T}_\varphi b^w(x, hD) - M_{(\kappa_\varphi^{-1})^*b} \mathcal{T}_\varphi = O_{L^2 \rightarrow L_\Phi^2}(\sqrt{h}).$$

(ii) *With the same notation,*

$$(13.5.3) \quad \begin{aligned} &\langle b^w(x, hD)u, u \rangle_{L^2(\mathbb{R}^n)} \\ &= \langle M_{(\kappa_\varphi^{-1})^*b} \mathcal{T}_\varphi u, \mathcal{T}_\varphi u \rangle_{L_\Phi^2(\mathbb{C}^n)} + O(h) \|u\|_{L^2(\mathbb{R}^n)}^2. \end{aligned}$$

REMARKS. (i) In (13.5.2) and (13.5.3) the Weyl quantization, $b \mapsto b^w$, can be replaced by any quantization $b \mapsto \text{Op}_t(b)$ as the error terms in these changes are of order $O(h)$.

(ii) The approximation in (13.5.2) does not imply (13.5.3). When $b \geq 0$, then (13.5.3) immediately gives the sharp Gårding inequality (Theorem 4.32).

Proof. 1. By Theorem 13.10,

$$a_\Phi^w(z, hD_z) = \Pi_\Phi M_a \Pi_\Phi + O_{H_\Phi \rightarrow L_\Phi^2}(h).$$

We need to show that

$$(13.5.4) \quad \Pi_\Phi M_a - M_a \Pi_\Phi = O_{H_\Phi \rightarrow L_\Phi^2}(\sqrt{h}).$$

Using (13.3.14), that means showing that the operator with the Schwartz kernel,

$$A(z, w) = \frac{2^n \det \partial_{z\bar{w}}^2 \Psi}{(\pi h)^n} (a(w) - a(z)) e^{2(\Psi(z, \bar{w}) - \Phi(w))/h},$$

is $O(h^{1/2}) : L_\Phi^2 \rightarrow L_\Phi^2$. This is the same as checking that the operator with the kernel $e^{-\Phi(z)/h} A(z, w) e^{\Phi(w)/h}$ is

$$O(h^{1/2}) : L^2 \rightarrow L^2, \quad L^2 = L^2(\mathbb{C}^n, dm).$$

From (13.3.14), (13.3.15), and the mean value theorem applied to $a \in S(\mathbb{C}^n)$ we obtain

$$\begin{aligned} e^{-\Phi(z)/h} |A(z, w)| e^{\Phi(w)/h} &\leq Ch^{-n} |a(z) - a(w)| e^{(\operatorname{Re} \Psi(z, \bar{w}) - \Phi(z) - \Phi(w))/h} \\ &\leq C' h^{-n} |z - w| e^{-2c|z-w|^2/h} \\ &\leq C'' \sqrt{h} h^{-n} e^{-c|z-w|^2/h}. \end{aligned}$$

Since

$$h^{-n} \int_{\mathbb{C}} e^{-c|z-w|^2/h} dm(z) = O(1)$$

and as we have $\sqrt{h}$ to spare, (13.5.4) follows.

2. Similarly, to establish (13.5.3), we only need

$$(13.5.5) \quad \langle \Pi_\Phi(M_a v), v \rangle_{H_\Phi} = \langle M_a v, v \rangle_{L_\Phi^2} + O(h) \|v\|_{H_\Phi}^2,$$

for $v \in H_\Phi$. However, since Π_Φ is selfadjoint on L_Φ^2 , we have

$$\begin{aligned} \langle \Pi_\Phi(av), v \rangle_{H_\Phi} &= \langle \Pi_\Phi(av), v \rangle_{L_\Phi^2} \\ &= \langle av, \Pi_\Phi v \rangle_{L_\Phi^2} \\ &= \langle av, v \rangle_{L_\Phi^2}; \end{aligned}$$

that is, (13.5.5) holds with no error term. $\square$

As an application we improve Theorem 5.1:

THEOREM 13.13 (Estimate on the operator norm). *Suppose that $b \in S$. Then*

$$(13.5.6) \quad \|b^w(x, hD)\|_{L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)} = \sup_{\mathbb{R}^{2n}} |b| + O(h).$$

Proof. 1. Let $\mathcal{T}_\varphi$ be an FBI transform defined in (13.3.3), and let Φ be the associated weight given by (13.3.4). Define $a \in S(\Lambda_\Phi)$ using the canonical transformation (13.3.7), $\kappa_\varphi^* a = b$. Theorem 13.9 shows that

$$\|b^w(x, hD)\|_{L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)} = \|a^w(z, hD)\|_{H_\Phi(\mathbb{C}^n) \rightarrow H_\Phi(\mathbb{C}^n)}.$$

Theorem 13.10 then shows that

$$\|a_{\Phi}^w(z, hD)\|_{H_{\Phi} \rightarrow H_{\Phi}} = \|T_q\|_{H_{\Phi} \rightarrow H_{\Phi}} + O_{H_{\Phi} \rightarrow H_{\Phi}}(h^{\infty}),$$

where

$$q \in S(\mathbb{C}^n), \quad q(z) = a\left(z, \frac{2}{i}\partial_z\Phi(z)\right) + O_S(h).$$

Since

$$\begin{aligned} \|T_q\|_{H_{\Phi} \rightarrow H_{\Phi}} &= \|\Pi_{\Phi} M_q\|_{H_{\Phi} \rightarrow H_{\Phi}} \leq \|M_q\|_{L_{\Phi}^2 \rightarrow L_{\Phi}^2} \\ &= \sup_{\mathbb{C}^n} |q| = \sup_{\Lambda_{\Phi}} |a| + O(h) = \sup_{\mathbb{R}^{2n}} |b| + O(h), \end{aligned}$$

the upper bound in (13.5.6) follows.

2. To prove that

$$\|b^w(x, hD)\| \geq \sup_{\mathbb{R}^{2n}} |b| - Ch,$$

we use part (ii) of Theorem 13.12; for $u \in L^2(\mathbb{R}^n)$ with $\|u\|_{L^2} = 1$,

$$\begin{aligned} (13.5.7) \quad \|b^w(x, hD)\|_{L^2 \rightarrow L^2} &\geq \langle b^w(x, hD)u, u \rangle_{L^2(\mathbb{R}^n)} \\ &\geq \langle M_a \mathcal{T}_{\varphi} u, \mathcal{T}_{\varphi} u \rangle_{L_{\Phi}^2} - C_0 h. \end{aligned}$$

We can find $v = v(h) \in L_{\Phi}^2(\mathbb{C}^n)$, $\|v\|_{L_{\Phi}^2} = 1$, such that

$$\langle M_a v, v \rangle_{L_{\Phi}^2} \geq \sup_{\mathbb{R}^{2n}} |b| - C_1 h.$$

By taking $u = \mathcal{T}_{\varphi}^* v$ in (13.5.7), we obtain

$$\|b^w(x, hD)\|_{L^2 \rightarrow L^2} \geq \sup_{\mathbb{R}^{2n}} |b| - C_2 h,$$

which provides the lower bound in (13.5.6). $\square$

13.5.2. Characterization of $\mathbf{WF}_h$. One of the original motivations for introducing the FBI transform in microlocal analysis was the characterization of wavefront sets. In our setting it is given in

THEOREM 13.14 (Wavefront set and the FBI transform). *Suppose that $u = \{u(h)\} \subset L^2(\mathbb{R}^n)$ is a tempered family, that is, $\|u(h)\|_{L^2} \leq Ch^{-N}$ for some N . Let $(x_0, \xi_0) \in \mathbb{R}^{2n}$ and put $z_0 = \kappa_{\varphi}(x_0, \xi_0)$ (we keep identifying Λ_{Φ} with $\mathbb{C}^n$ via (13.4.4)). Then the following conditions are equivalent:*

(i) $(x_0, \xi_0) \notin \mathbf{WF}_h(u)$;

(ii) *there exists a neighborhood $U_0 \subset \subset \mathbb{C}^n$ of z_0 such that*

$$(13.5.8) \quad \|\mathcal{T}_{\varphi} u\|_{L_{\Phi}^2(U_0)} = O(h^{\infty});$$

(iii) there exists a neighborhood $U_1 \subset \subset \mathbb{C}^n$ of z_0 such that

$$(13.5.9) \quad |\mathcal{T}_\varphi u(z)| \leq C_N h^N e^{\Phi(z)/h}, \quad z \in U_1,$$

for all N .

Proof. 1. We first show that (i) implies (ii). Theorem 8.13 shows that if $b \in C_c^\infty(\mathbb{R}^{2n})$ is supported in a certain fixed neighborhood V of (x_0, ξ_0) , then

$$(13.5.10) \quad \|b^w(x, hD)u\|_{L^2} = O(h^\infty).$$

Now choose $q \geq 0$, $q \in C_c^\infty(\mathbb{C}^n)$, such that

$$\text{spt } q \subset \subset \kappa_\varphi(V)$$

and $q(z) = 1$ for z in some neighborhood U_0 of z_0 . Then

$$\begin{aligned} \|\mathcal{T}_\varphi u\|_{L_\Phi^2(U_0)}^2 &\leq \langle q \mathcal{T}_\varphi u, \mathcal{T}_\varphi u \rangle_{L_\Phi^2} = \langle T_q \mathcal{T}_\varphi u, \mathcal{T}_\varphi u \rangle_{H_\Phi} \\ &= \langle a_\Phi^w(z, hD_z) \mathcal{T}_\varphi u, \mathcal{T}_\varphi u \rangle_{H_\Phi}, \end{aligned}$$

where $a \in S(\mathbb{C}^n)$ satisfying $T_q = a_\Phi^w$ is given by Theorem 13.10. Theorem 13.9 then shows that

$$\|\mathcal{T}_\varphi u\|_{L_\Phi^2(U_0)}^2 \leq \|(\kappa_\varphi^* a)^w(x, hD)u\|_{L^2} \|u\|_{L^2}.$$

The assumptions on q and (13.4.12) then give $\kappa_\varphi^* a = b + O_S(h^\infty)$ where $b \in C_c^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ is supported in V . Hence (13.5.10) implies (13.5.8):

$$\|\mathcal{T}_\varphi u\|_{L_\Phi^2(U_0)}^2 \leq \|b^w(x, hD)u\|_{L^2} \|u\|_{L^2} + O(h^\infty) = O(h^\infty).$$

2. To see that (ii) implies (iii), that is, that (13.5.8) implies (13.5.9), we take $U_1 \subset \subset U_0$ and write for $z \in U_1$,

$$\begin{aligned} e^{-\Phi(z)/h} \mathcal{T}_\varphi u(z) &= e^{-\Phi(z)/h} (\Pi_\Phi \mathcal{T}_\varphi) u(z) \\ &= \int_{U_0} e^{-\Phi(z)/h} \Pi_\Phi(z, w) e^{\Phi(w)/h} e^{-\Phi(w)/h} \mathcal{T}_\varphi u(w) dm(w) \\ &\quad + \int_{\mathbb{C}^n \setminus U_0} e^{-\Phi(z)/h} \Pi_\Phi(z, w) e^{\Phi(w)/h} e^{-\Phi(w)/h} \mathcal{T}_\varphi u(w) dm(w) \\ &= A + B. \end{aligned}$$

We recall from (13.3.14) that

$$\begin{aligned} e^{-\Phi(z)/h} |\Pi_\Phi(z, w)| e^{\Phi(w)/h} &\leq C h^{-n} e^{(\text{Re } \Psi(z, \bar{w}) - \Phi(z) - \Phi(w))/h} \\ &\leq C' h^{-n} e^{-c|z-w|^2/h}. \end{aligned}$$

This implies the estimates

$$|A| \leq C h^{-n} \left(\int_{U_0} e^{-2c|z-w|^2/h} dm(w) \right)^{\frac{1}{2}} \|\mathcal{T}_\varphi u\|_{L_\Phi^2(U_0)} = O(h^\infty)$$

and, for $z \in U_1 \subset\subset U_0$,

$$\begin{aligned} |B| &\leq Ch^{-n} \left(\int_{\mathbb{C}^n \setminus U_0} e^{-2c|z-w|^2/h} dm(w) \right)^{\frac{1}{2}} \|\mathcal{T}_\varphi u\|_{H_\Phi} \\ &\leq Ch^{-n} e^{-cd(U_1, \mathbb{C}^n \setminus U_0)^2/(2h)} \|\mathcal{T}_\varphi u\|_{H_\Phi} = O(h^\infty). \end{aligned}$$

This shows (13.5.9).

3. Finally, we show that (iii) implies (i), that is, that (13.5.9) implies (13.5.10) for some $b \in S(\mathbb{R}^n \times \mathbb{R}^n)$ with $b(x_0, \xi_0) \neq 0$. Theorem 13.9 shows that this is equivalent to proving that for some $a \in S(\Lambda_\Phi)$, $a(\kappa_\varphi(x_0, \xi_0)) \neq 0$,

$$(13.5.11) \quad \|a_\Phi^w(z, hD_z)\mathcal{T}_\varphi u\|_{H_\Phi} = O(h^\infty).$$

The action of a_Φ^w on $v = \mathcal{T}_\varphi u$ is given by (13.4.7).

Suppose that $\text{spt } a \subset B_{\mathbb{C}^n}(z_0, \delta)$ for δ small enough so that $B_{\mathbb{C}^n}(z_0, 4\delta) \subset U_1$. Let

$$\chi \in C_c^\infty(B(z_0, 3\delta)), \quad \chi|_{B_{\mathbb{C}^n}(z_0, 2\delta)} = 1.$$

We claim that

$$a_\Phi^w(z, hD_z)(1 - \chi) = O_{L_\Phi^2 \rightarrow L_\Phi^2}(h^\infty).$$

To see this, we observe that in the integrand (13.4.7) multiplied by $1 - \chi(w)$

$$\begin{aligned} |z - w| &\geq |w - z_0| - \left| \frac{z+w}{2} - z_0 + \frac{z-w}{2} \right| \\ &\geq |w - z_0| - \left| \frac{z+w}{2} - z_0 \right| - \frac{|z-w|}{2} \\ &\geq 2\delta - \delta - \frac{|z-w|}{2}, \end{aligned}$$

which implies that we have $|z-w| \geq 2\delta/3$ in the integrand. We can integrate by parts using

$$L_1 := h \frac{\langle (\partial_{z\bar{z}}^2 \Phi)^{-1}(\overline{z-w}), \partial_{\bar{w}} \rangle}{|z-w|^2}.$$

This proves the claim. Hence

$$\begin{aligned} \|a_\Phi^w(z, hD_z)\mathcal{T}_\varphi u\|_{H_\Phi} &= \|a_\Phi^w(z, hD_z)(\chi\mathcal{T}_\varphi u)\|_{L_\Phi^2} + O(h^\infty)\|\mathcal{T}_\varphi u\|_{H_\Phi} \\ &\leq \|\chi\mathcal{T}_\varphi u\|_{L_\Phi^2} + O(h^\infty)\|\mathcal{T}_\varphi u\|_{H_\Phi} \\ &= O(h^\infty). \end{aligned}$$

(We remark that (ii) could be used here in place of the seemingly stronger estimate (iii).) This is (13.5.11). $\square$

13.5.3. Sobolev spaces. When considering Toeplitz operators in Section 13.4, we took $q \in S(\mathbb{C}^n)$. To consider $q \in S(\mathbb{C}^n, m)$ where m is an order function on $\mathbb{C}^n$, we need to introduce generalized Sobolev spaces. We will consider the simplest case of polynomial order functions.

DEFINITION. Suppose that a real quadratic form $\Phi : \mathbb{C}^n \rightarrow \mathbb{R}$ is strictly plurisubharmonic. For $m \geq 0$ we define

$$(13.5.12) \quad \begin{aligned} H_\Phi^m &:= \{u \in H_\Phi : \langle z \rangle^m u \in L_\Phi^2\}, \\ \|u\|_{H_\Phi^m}^2 &:= \int_{\mathbb{C}^n} |u(z)|^2 \langle z \rangle^{2m} e^{-2\Phi(z)/h} dm(z). \end{aligned}$$

The space H_Φ^m is dense in H_Φ for any strictly plurisubharmonic Φ , as by part (iii) of Theorem 13.4 it contains the image of $\mathcal{S}(\mathbb{R}^n)$ under the FBI transform. If Φ is moreover strictly convex, we can say that any polynomial belongs to H_Φ^m :

$$P(z) \in H_\Phi^m, \quad P(z) = \sum_{|\alpha| \leq k} p_\alpha z^\alpha.$$

THEOREM 13.15 (Toeplitz operators on Sobolev spaces). *Suppose that $q_1 \in \langle z \rangle^m L^\infty(\mathbb{C}^n)$. Then*

$$T_{q_1} = O(1) : H_\Phi^m \rightarrow H_\Phi.$$

Similarly, if $q_2 \in S(\mathbb{C}^n, \langle z \rangle^{-m})$, then

$$T_{q_2} = O(1) : H_\Phi \rightarrow H_\Phi^m,$$

that is, the norms of both operators are bounded independently of h .

Proof. 1. For $u \in H_\Phi^m$ we have $q_1 u \in L_\Phi^2$ and hence $T_{q_1} u := \Pi_\Phi(q_1 u) \in H_\Phi$.

2. We need to show that $\langle z \rangle^m \Pi_\Phi \langle z \rangle^{-m}$ is uniformly bounded on L_Φ^2 . For that, we consider the reduced kernel (see part 1 of the proof of Theorem 13.12):

$$K(z, w) := \langle z \rangle^m e^{-\Phi(z)/h} \Pi_\Phi(z, w) e^{\Phi(w)/h} \langle w \rangle^{-m},$$

and we need to show that the operator defined by K is bounded on $L^2(\mathbb{C}^n)$. Using Theorem 13.6 and the inequality

$$1 + |z|^2 \leq 2(1 + |w|^2)(1 + |z - w|^2),$$

we have

$$\begin{aligned} K(z, w) &\leq C h^{-n} \frac{\langle z \rangle^m}{\langle w \rangle^m} e^{(\operatorname{Re} \Psi(z, \bar{w}) - \Phi(z) - \Phi(w))/h} \\ &\leq C' h^{-n} \frac{\langle z \rangle^m}{\langle w \rangle^m} e^{-c|z-w|^2/h} \\ &\leq C'' \langle z - w \rangle^m h^{-n} e^{-c|z-w|^2/h}, \quad c > 0. \end{aligned}$$

Since

$$\int_{\mathbb{C}^n} \langle z - w \rangle^m h^{-n} e^{-c|z-w|^2/h} dm(w) = O(1),$$

the theorem follows from Schur's inequality. $\square$

EXAMPLES. (i) Consider $q(z) = \bar{z}_j \in S(\mathbb{C}^n, \langle z \rangle)$ and $\Phi(z) = |z|^2/2$. Then

$$(13.5.13) \quad T_{\bar{z}_j} u = h \partial_{z_j} u,$$

and

$$(13.5.14) \quad T_{|z|^2} u = (h \langle z, \partial_z \rangle + hn) u.$$

(ii) Let φ be given by (13.3.11). Then $\Phi(z) = |z|^2/2$. If

$$p(x, \xi) = \xi^2 + x^2$$

is the symbol of the harmonic oscillator, then

$$\begin{aligned} a(z, \zeta) &:= (\kappa_\varphi^{-1})^* p(z, \zeta) = \frac{1}{2} ((iz + \zeta)^2 + (i\zeta + z)^2) \\ &= 2iz\zeta = 2|z|^2. \end{aligned}$$

Theorems 13.9 and 13.10 (or a direct calculation) show that

$$\mathcal{T}_\varphi(-h^2\Delta + x^2)\mathcal{T}_\varphi^* = T_q, \quad q(z) := 2|z|^2 - hn.$$

By part (i) we see that

$$\mathcal{T}_\varphi(-h^2\Delta + x^2)\mathcal{T}_\varphi^* = 2\langle z, h\partial_z \rangle + hn.$$

This simple operator has thus been identified with the harmonic oscillator. The ground state is simply the function $u_0 = 1$ and the creation and annihilation operators are M_{z_j} and $h\partial_{z_j}$, respectively. The eigenvalues are given by $(2|\alpha| + n)h$ with eigenfunctions

$$(2\langle z, h\partial_z \rangle + hn)z^\alpha = (2|\alpha| + n)hz^\alpha.$$

The eigenspace corresponding to the eigenvalue $(2M + 1)h$ is the space of homogeneous polynomials of degree M . $\square$

The next theorem describes the microlocalization properties of eigenfunctions of the harmonic oscillator, on the FBI–Bargmann transform side. It will be useful in our next application.

THEOREM 13.16. *Let $\Phi(z) = |z|^2/2$, and define*

$$(13.5.15) \quad f_\alpha(z) := \frac{1}{(\pi h)^{n/2}} \left(\frac{1}{h^{|\alpha|} \alpha!} \right)^{1/2} z^\alpha, \quad \alpha \in \mathbb{N}^n.$$

Let $\mathcal{P}_M \subset H_\Phi$ be the space of homogeneous polynomials of degree M .

Then

(i) the set of f_α 's is orthonormal:

$$(13.5.16) \quad \langle f_\alpha, f_\beta \rangle_{H_\Phi} = \delta_{\alpha\beta},$$

and $\{f_\alpha\}_{|\alpha|=M}$ forms a basis of $\mathcal{P}_M$.

(ii) Suppose $\Omega \subset \mathbb{C}^n$ is an open set satisfying

$$\Omega \cap \{z : |z| = 1\} = \emptyset.$$

If $|Mh - 1| < \epsilon$, then for ϵ small enough and for any fixed m ,

$$(13.5.17) \quad \|\langle z \rangle^m u\|_{L^2_\Phi(\Omega)} = O(h^\infty) \|u\|_{H_\Phi}, \quad u \in \mathcal{P}_M.$$

REMARKS. (i) We leave it to the reader to check that the set $\{f_\alpha\}_{\alpha \in \mathbb{N}^n}$ is also complete and hence forms a Hilbert space basis of H_Φ . One can then rederive the formula for the Bergman kernel

$$(13.5.18) \quad \begin{aligned} \Pi_\Phi(z, w) &= \sum_{\alpha \in \mathbb{N}^n} f_\alpha(z) \overline{f_\alpha(w)} e^{-|w|^2/h} \\ &= \frac{1}{(\pi h)^n} e^{\frac{1}{h}(\langle z, \bar{w} \rangle - |w|^2)} \\ &= \frac{1}{(\pi h)^n} e^{\frac{1}{h}(2\Psi(z, \bar{w}) - 2\Phi(w))}, \quad \Phi(z) = |z|^2/2. \end{aligned}$$

Conversely, this also shows that the set of f_α 's is complete.

(ii) The bound $O(h^\infty)$ in (13.5.17) can be replaced by an exponential error term $e^{-c_{m,\epsilon}/h}$. That follows from a direct analysis of integrals, as in the proofs of Stirling's formula. We give a proof using the connection with the localization of eigenfunctions of the harmonic oscillator and the methods of Section 13.4. $\square$

Proof. 1. We put $z = (z_1, \dots, z_n)$, $z_j = r_j e^{i\theta_j}$, and calculate

$$\begin{aligned} \langle z^\alpha, z^\beta \rangle_{H_\Phi} &= \int_{\mathbb{C}^n} z^\alpha \bar{z}^\beta e^{-|z|^2/h} dm(z) \\ &= \prod_{j=1}^n \int_{\mathbb{C}} z_j^{\alpha_j} \bar{z}_j^{\beta_j} e^{-|z_j|^2/h} dm(z_j) \\ &= \prod_{j=1}^n \int_0^\infty \int_0^{2\pi} r_j^{\alpha_j+\beta_j} e^{i(\alpha_j-\beta_j)\theta_j} d\theta_j e^{-r_j^2/h} r_j dr_j \\ &= \prod_{j=1}^n \pi \delta_{\alpha_j\beta_j} \int_0^\infty r_j^{2\alpha_j} e^{-r_j^2/h} 2r_j dr_j \\ &= (h\pi)^n h^{|\alpha|} \alpha! \delta_{\alpha\beta}. \end{aligned}$$

This proves (13.5.16).

2. We first reduce the case of an arbitrary fixed m in (13.5.17) to the case of $m = 0$. For that, we note that

$$(13.5.19) \quad |\langle z \rangle^m z^\alpha| \leq C_m \max_{|\beta| \leq m} |z^{\beta+\alpha}|.$$

If $1 - \epsilon < |\alpha|h < 1 + \epsilon$, then for $|\beta| \leq m$,

$$1 - 2\epsilon < 1 - \epsilon < |\alpha + \beta|h < 1 + 2\epsilon, \quad \text{provided that } h < \epsilon/m.$$

If we show that (13.5.17) holds with $m = 0$ and for all ϵ small enough, we can replace α by $\alpha + \beta$ in the statement. It then follows from (13.5.19) that (13.5.17) holds for any fixed m .

3. We now prove (13.5.17) for $m = 0$. For that, let $\chi \in C_c^\infty(\mathbb{C}^n; [0, 1])$ satisfy $\text{spt } \chi \cap \Omega = \emptyset$ and $\chi = 1$ on $|z| = 1$. Then for $u \in \mathcal{P}_M$,

$$\begin{aligned} \|u\|_{L_\Phi^2(\Omega)}^2 &\leq \langle (1 - \chi)u, u \rangle_{L_\Phi^2} = \langle \Pi_\Phi(1 - \chi)\Pi_\Phi u, u \rangle_{H_\Phi} \\ &= \langle a_\Phi^w(z, hD_z)u, u \rangle_{H_\Phi}, \quad a := 1 - \exp(\tfrac{1}{8}h\Delta)\chi, \end{aligned}$$

where we used (13.4.15). Since

$$\begin{aligned} \exp(\tfrac{1}{8}h\Delta)\chi &= \sum_{j=0}^N \frac{h^j}{j!} \left(\tfrac{1}{8}\Delta\right)^j \chi + \int_0^1 (1-t)^N \exp(\tfrac{t}{8}h\Delta) \left(\tfrac{1}{8}h\Delta\right)^{N+1} \chi dt \\ &= \sum_{j=0}^N \frac{h^j}{j!} \left(\tfrac{1}{8}\Delta\right)^j \chi + O_{\mathcal{S}}(h^{N+1}), \end{aligned}$$

$$(13.5.20) \quad a = a_0 + O_{\mathcal{S}}(h^\infty), \quad \text{dist}(\text{spt } a_0, \{|z| = 1\}) > \delta_0 > 0,$$

where a_0 depends on h but δ_0 does not.

4. Theorem 13.9 shows that for φ given by (13.3.11),

$$\|u\|_{L^2_{\Phi}(\Omega)}^2 \leq \langle (\kappa_{\varphi}^* a)^w(x, hD) \mathcal{T}_{\varphi}^* u, \mathcal{T}_{\varphi}^* u \rangle_{L^2(\mathbb{R}^n)},$$

where

$$\begin{aligned} \kappa_{\varphi}(x, \xi) &= \frac{1}{\sqrt{2}}(i\xi - x, -\xi + ix), \\ \kappa_{\varphi}(\{x^2 + \xi^2 = 2\}) &= \{(z, -i\bar{z}) : |z| = 1\}. \end{aligned}$$

From the property (13.5.20) we conclude

$$(13.5.21) \quad \kappa_{\varphi}^* a = b_0 + O_{\mathcal{S}}(h^{\infty}), \quad \text{dist}(\text{spt } b_0, \{x^2 + \xi^2 = 2\}) > \delta > 0,$$

independently of h .

Owing to example (ii) above, for $u \in \mathcal{P}_M$,

$$(-h^2 \Delta + x^2) \mathcal{T}_{\varphi}^* u = Eu, \quad E = (2M + n)h, \quad |E - 2| < 3\epsilon.$$

Hence, for ϵ small enough, (13.5.21) and Theorem 6.4 show that

$$\begin{aligned} \|u\|_{L^2_{\Phi}(\Omega)}^2 &\leq \langle (\kappa_{\varphi}^* a)^w(x, hD) \mathcal{T}_{\varphi}^* u, \mathcal{T}_{\varphi}^* u \rangle_{L^2(\mathbb{R}^n)} \\ &= O(h^{\infty}) \|\mathcal{T}_{\varphi}^* u\|_{L^2(\mathbb{R}^n)}^2 \\ &= O(h^{\infty}) \|u\|_{H_{\Phi}}^2, \end{aligned}$$

proving (13.5.17). □

13.5.4. Positive forms in several complex variables. We will use the results from Chapter 4 and Section 13.4 to prove a purely algebraic theorem.

DEFINITION. A function $f : \mathbb{C}^n \rightarrow \mathbb{C}$ given by

$$(13.5.22) \quad f(z, \bar{z}) = \sum_{|\alpha|=|\beta|=m} c_{\alpha\beta} z^{\alpha} \bar{z}^{\beta}, \quad c_{\alpha\beta} \in \mathbb{C},$$

is called a *bi-homogeneous form* of degree m on $\mathbb{C}^n$. The coefficients $c_{\alpha\beta}$ are uniquely determined by f :

$$c_{\alpha\beta} = \frac{1}{\alpha! \beta!} \partial_z^{\alpha} \partial_{\bar{z}}^{\beta} f(z, \bar{z}).$$

We start with the following observation.

LEMMA 13.17 (Sums of Hermitian squares). *A bi-homogeneous form of degree m can be written as a sum of squares of homogeneous polynomials,*

$$(13.5.23) \quad f(z, \bar{z}) = \sum_{j=1}^k |P_j(z)|^2, \quad P_j(z) = \sum_{|\alpha|=m} p_j^{\alpha} z^{\alpha},$$

if and only if for all $v_\alpha \in \mathbb{C}$, $\alpha \in \mathbb{N}^n$, $|\alpha| = m$,

$$\sum_{|\alpha|=|\beta|=m} c_{\alpha\beta} v_\alpha \bar{v}_\beta \geq 0.$$

This condition means that the matrix $(c_{\alpha\beta})$ is positive semidefinite.

Proof. 1. We recall that a Hermitian matrix A , $A = A^*$, acting on $\mathbb{C}^N$, can be factorized:

$$A = U\Lambda U^*$$

where $\Lambda = (\delta_{ij} \lambda_j)_{1 \leq i, j \leq N}$, $\lambda_j \in \mathbb{R}$, is a diagonal matrix and $U \in \mathbb{M}^{N \times N}$ is a unitary matrix whose columns, $u_j \in \mathbb{M}^{N \times 1}$, are the eigenvectors of A :

$$Au_j = \lambda_j u_j, \quad u_j^* u_i = \delta_{ij}, \quad A = \sum_{i,j=1}^n \lambda_j u_j u_i^*.$$

This means that for $w \in \mathbb{C}^N \simeq \mathbb{M}^{N \times 1}$,

$$(13.5.24) \quad \langle Aw, \bar{w} \rangle = w^* A w = w^* U \Lambda U^* w = \sum_{j=1}^N \lambda_j |u_j^* w|^2.$$

The number of negative and positive eigenvalues depends only on the quadratic form $\langle Aw, \bar{w} \rangle$. In particular $\langle Aw, \bar{w} \rangle \geq 0$ for all $w \neq 0$ if and only if all eigenvalues are nonnegative. That means that A is positive semidefinite.

2. When a bi-homogeneous form $f(z, \bar{z})$ takes only real values, then $c_{\alpha\beta} = \bar{c}_{\beta\alpha}$ and $A = (c_{\alpha\beta})_{|\alpha|=|\beta|=m}$ is a Hermitian matrix. Consequently we can apply the diagonalization procedure described above with $w = (z^\alpha)_{|\alpha|=m}$ and $u_\alpha = (u_\alpha^\beta)_{|\beta|=m}$, $|\alpha| = m$, the basis of eigenvectors:

$$f(z, \bar{z}) = w^* A w = \sum_{|\alpha|=m} \lambda_\alpha |u_\alpha^* w|^2 = \sum_{|\alpha|=m} \operatorname{sgn}(\lambda_\alpha) |P_\alpha(z)|^2,$$

where $P_\alpha(z) = |\lambda_\alpha|^{\frac{1}{2}} \sum_{|\beta|=m} \bar{u}_\alpha^\beta z^\beta$. Hence $f(z, \bar{z})$ is of the form (13.5.23) if all the eigenvalues λ_α are nonnegative. That is equivalent to the matrix $c_{\alpha\beta}$ being positive semidefinite, as claimed.

3. Conversely, if $f(z, \bar{z})$ is a sum of Hermitian squares (13.5.23), then

$$(13.5.25) \quad A := (c_{\alpha\beta})_{|\alpha|=|\beta|=m}$$

is given by

$$A = \sum_{j=1}^k p_j p_j^*, \quad p_j = (p_j^\alpha)_{|\alpha|=m},$$

that is, A is positive semidefinite. □

The next theorem is related to Hilbert's 17th problem and can be considered as its complex analogue.

THEOREM 13.18 (Quillen's Theorem). *If a bi-homogeneous form satisfies*

$$f(z, \bar{z}) > 0, \quad z \in \mathbb{C}^n \setminus \{0\},$$

then for N sufficiently large,

$$(13.5.26) \quad \langle z, \bar{z} \rangle^N f(z, \bar{z}) = \sum_{j=1}^{k_N} |P_j^N(z)|^2,$$

where the $P_j^N(z)$ are homogeneous polynomials of degree $N + m$.

REMARK. The conclusion of the theorem is not true for all N as the following example ([**Ca-D**, Example 2], [**Pu**, Chapter 10]) shows:

$$(13.5.27) \quad f(z, \bar{z}) = |z_1|^4 + |z_2|^4 - c|z_1 z_2|^2, \quad z = (z_1, z_2) \in \mathbb{C}^2, \quad 0 < c < 2.$$

In that case the matrix A in (13.5.25) is diagonal with two positive and one negative eigenvalues. Lemma 13.17 shows that f is *not* a sum of Hermitian squares.

We see that

$$\begin{aligned} \langle z, \bar{z} \rangle^N f(z, \bar{z}) &= \sum_{m=0}^N \binom{N}{m} (|z_1|^{2(m+2)} |z_2|^{2(N-m)} \\ &\quad + |z_1|^{2m} |z_2|^{2(N-m+2)} - c|z_1|^{2(m+1)} |z_2|^{2(N-m+1)}). \end{aligned}$$

Using the standard convention that $\binom{N}{m} = 0$ for $m < 0$ or $m > N$, we can rewrite this as

$$\sum_{m=-1}^{N+1} \left(\binom{N}{m-1} + \binom{N}{m+1} - c \binom{N}{m} \right) |z_1|^{2m+2} |z_2|^{2N-2m+2}.$$

Lemma 13.17 shows that to write this as a sum of Hermitian squares we need all the terms in the sum to be positive. A simple calculation shows that we need

$$\frac{m}{N-m+1} + \frac{N-m}{m+1} \geq c, \quad 0 \leq m \leq N.$$

The minimum of the left-hand side is achieved at $m = [N/2]$, which gives the condition

$$(13.5.28) \quad N \geq \frac{2c}{2-c}.$$

Hence N may have to be very large to obtain a sum of Hermitian squares. We will return to this example to compare the bound (13.5.28) with a bound obtained from a semiclassical estimate. $\square$

Proof of Quillen's Theorem (Theorem 13.18). As in the rest of this section $\Phi(z) = |z|^2/2$.

1. Since

$$\begin{aligned}
 \frac{\langle z, \bar{z} \rangle^N}{N!} &= \sum_{|\mu|=N} \frac{z^\mu \bar{z}^\mu}{\mu!}, \\
 \langle z, \bar{z} \rangle^N f(z, \bar{z}) &= \sum_{\substack{|\alpha|=|\beta|=m \\ |\mu|=N}} \frac{c_{\alpha\beta}}{\mu!} z^{\alpha+\mu} \bar{z}^{\beta+\mu} \\
 (13.5.29) \qquad &=: \sum_{|\gamma|=|\rho|=m+N} c_{\gamma\rho}^N z^\gamma \bar{z}^\rho.
 \end{aligned}$$

In view of Lemma 13.17 our goal is to show that for all N sufficiently large the matrix $(c_{\rho\gamma}^N)_{|\rho|=|\gamma|=m+N}$ is positive definite.

2. We now define the following differential operator associated to $f(z, \bar{z})$:

$$(13.5.30) \qquad P_f := \sum_{|\alpha|=|\beta|=m} c_{\alpha\beta} z^\alpha (h\partial_z)^\beta = \sum_{|\alpha|=|\beta|=m} c_{\alpha\beta} z^\alpha T_{\bar{z}\beta},$$

where the identification with Toeplitz operators follows from example (i) in Section 13.5.3 and Theorem 13.11. Since

$$(T_{\bar{z}j})^* = M_{z_j}$$

and since $\bar{c}_{\alpha\beta} = c_{\beta\alpha}$, the operator P_f is selfadjoint on H_Φ . Also,

$$P_f : \mathcal{P}_{m+N} \rightarrow \mathcal{P}_{m+N},$$

where $\mathcal{P}_{m+N}$ is the space of homogeneous holomorphic polynomials of degree $m+N$.

3. We now analyze the action of P_f on $\mathcal{P}_{m+N}$. Using the formula for the Bergman kernel (13.5.18), we write

$$\begin{aligned}
 P_f g(z) &= \sum_{\alpha, \beta} c_{\alpha\beta} z^\alpha \Pi_\Phi(\bar{z}^\beta g(z)) \\
 &= \sum_{N=0}^{\infty} \frac{1}{(\pi h)^n} \int_{\mathbb{C}^n} \frac{\langle z, \bar{w} \rangle^N}{h^N N!} f(z, \bar{w}) e^{-|w|^2/h} g(w) dm(w) \\
 &= \sum_{N=0}^{\infty} \frac{1}{(\pi h)^n} \int_{\mathbb{C}^n} \sum_{|\mu|=N} \frac{z^\mu \bar{w}^\mu}{h^N \mu!} f(z, \bar{w}) g(w) e^{-|w|^2/h} dm(w).
 \end{aligned}$$

Theorem 13.16 gave

$$(13.5.31) \quad \langle z^\alpha, z^\beta \rangle_{H_\Phi} = \delta_{\alpha\beta} (\pi h)^n h^{|\alpha|} \alpha!$$

and hence for $u \in \mathcal{P}_{m+N}$,

$$P_f u(z) = \frac{1}{(\pi h)^n} \int_{\mathbb{C}^n} \sum_{|\mu|=N} \frac{z^\mu \bar{w}^\mu}{h^N \mu!} f(z, \bar{w}) u(w) e^{-|w|^2/h} dm(w).$$

If

$$u, v \in \mathcal{P}_{m+N}, \quad u = \sum_{|\gamma|=m+N} u_\gamma z^\gamma, \quad v = \sum_{|\rho|=m+N} v_\rho z^\rho,$$

then, using (13.5.31)

$$\begin{aligned} \langle P_f u, v \rangle_{H_\Phi} &= (\pi h)^n h^{N+2m} \sum_{\substack{|\alpha|=|\beta|=m \\ |\mu|=N}} c_{\alpha\beta} \frac{(\alpha+\mu)! (\beta+\mu)!}{\mu!} u_{\beta+\mu} \bar{v}_{\alpha+\mu} \\ &= (\pi h)^n h^{N+2m} \sum_{|\gamma|=|\rho|=m+N} \rho! \gamma! c_{\rho\gamma}^N u_\gamma \bar{v}_\rho, \end{aligned}$$

where $c_{\gamma\rho}^N$ was defined in (13.5.29). Our goal is proving the positivity of the matrix $c_{\gamma\rho}^N$. That is equivalent to the positivity of the matrix $\rho! \gamma! c_{\gamma\rho}^N$ which in turn is equivalent to proving

$$(13.5.32) \quad P_f|_{\mathcal{P}_{N+m}} \geq cI, \quad c > 0, \quad N \geq N_0.$$

4. To prove (13.5.32), we use the calculus of Toeplitz operators presented in Theorem 13.11. The Toeplitz symbols are polynomials in z and $\bar{z}$ so the formulas are exact and can be justified directly. We obtain for $|\alpha| = |\beta| = m$,

$$z^\alpha T_{\bar{z}^\beta} = T_{q_{\alpha\beta}}, \quad q_{\alpha\beta} = z^\alpha \bar{z}^\beta + \sum_{j=1}^m h^j p_j^{\alpha\beta}(z, \bar{z}),$$

where $p_j^{\alpha\beta}(z, \bar{z})$ is bi-homogeneous of degree $m-j$. Consequently,

$$P_f = T_q, \quad q(z) = f(z, \bar{z}) + \sum_{j=1}^m h^j p_j(z, \bar{z}),$$

where $p_j(z, \bar{z})$ is bi-homogeneous of degree $m-j$.

For $0 < h < h_0$ there exists $c > 0$ such that

$$q(z) > 2c, \quad |z| > \frac{1}{2}.$$

Let $N > 1/h_0$ and put $h = 1/N$. Then part (ii) of Theorem 13.16 shows that for $u \in \mathcal{P}_{N+m}$,

$$\begin{aligned} \langle P_f u, u \rangle_{H_\Phi} &= \langle T_q u, u \rangle_{H_\Phi} = \langle \Pi_\Phi(qu), u \rangle_{H_\Phi} \\ &= \langle qu, u \rangle_{L_\Phi^2} = \langle qu, u \rangle_{L_\Phi^2(\{1-\epsilon < |z| < 1+\epsilon\})} - O(h^\infty) \|u\|_{H_\Phi}^2 \\ &\geq 2c \langle u, u \rangle_{L_\Phi^2(\{1-\epsilon < |z| < 1+\epsilon\})} - O(h^\infty) \|u\|_{H_\Phi}^2 \\ &= 2c \langle u, u \rangle_{L_\Phi^2} - O(h^\infty) \|u\|_{H_\Phi}^2 \\ &\geq c \|u\|_{H_\Phi}^2. \end{aligned}$$

This is (13.5.32) and the proof is completed. $\square$

DISCUSSION. The proof above involves some computations but the idea is very simple once the connection between (13.5.29) and the action of P_f on the space of homogeneous polynomials is established. Since

$$P_f u(z) = \frac{1}{(2\pi h)^n} \iint_{w \mapsto \zeta = \frac{1}{i} \bar{w}} e^{\frac{i}{h} \langle z-w, \zeta \rangle} f(z, i\zeta) u(w) dw \wedge dz,$$

we can think of P_f as the standard quantization of $f(z, i\zeta)$, $(z, \zeta) \in \Lambda_\Phi$. Its principal symbol is strictly positive away from 0, and hence on functions microlocalized away from 0, P_f is positive, provided that h is small. The space of homogeneous polynomials of order comparable to $1/h$ has that property. The next example shows that the Toeplitz calculus used in the proof has computational advantages. $\square$

EXAMPLE. We revisit the example (13.5.27) using the method of the proof of Theorem 13.18. Using example (ii) of Section 13.5.3, we calculate that for f given by (13.5.27) we have $P_f = T_q$ with

$$q(z) = |z_1|^4 + |z_2|^4 - c|z_1|^2|z_2|^2 - (4-c)h(|z_1|^2 + |z_2|^2) + (4-c)h^2.$$

To obtain a lower bound on $N = 1/h$ for which $\langle z, \bar{z} \rangle^N f(z, \bar{z})$ is a sum of Hermitian squares, we need to check the positivity of $q(z)$ for $|z| = 1$. Writing $|z_1|^2 = x$ and $|z_2|^2 = 1 - x$ gives the condition

$$\begin{aligned} 0 &< \min_{0 \leq x \leq 1} (x^2 + (1-x)^2 - cx(1-x)) - (4-c)(h-h^2) \\ &= \frac{1}{4}(2-c) - (4-c)(h-h^2). \end{aligned}$$

Hence we have the positivity for $h < \frac{2-c}{4(4-c)}$, or for

$$N > \frac{4(4-c)}{2-c},$$

which is close to the sharp inequality (13.5.28). $\square$

13.6. NOTES

For the theory of holomorphic functions in several variables see the classic text by Hörmander [H].

The space H_Φ for $\Phi(z) = |z|^2/2$ is sometimes called the Fock–Bargmann space, and the FBI transform with the phase given by (13.3.11) is called the Bargmann transform; see Bargmann [Bar] and also Folland [F].

Sections 13.3 and 13.4 are mostly based on unpublished lecture notes by Sjöstrand [S2] as presented in Sjöstrand [S3, Section 1], Hörmander [H5, Section 6] and in [S-Z4, Section 3]. This perspective on the FBI transform stresses the connection with analytic microlocal analysis and can be considered as a linear version of the theory presented in Sjöstrand [S1]. Here linear refers to the fact that the complex symplectic transformations are linear, which corresponds to quadratic phase functions. For Toeplitz operators and their connection to pseudodifferential calculus in other settings see Berezin [Ber], Berezin–Shubin [Ber-Sh], and Boutet de Monvel–Guillemin [BdM-G].

Semyon Dyatlov provided us with direct proofs of Theorems 13.10 and 13.11 given here.

To further motivate the study of Bergman kernels, we point out that a semiclassical analysis of Bergman kernels has played an important role in complex geometry since the work of Fefferman [Fe1] and Boutet de Monvel–Sjöstrand [BdM-S] (see Catlin [Ca] and Zelditch [Ze2]) and in *large N limit* quantization (see Bordemann–Meinrenken–Schlichenmaier [B-M-S], Borthwick–Paul–Uribe [B-P-U], and Charles [Cha]). A direct asymptotic construction of general Bergman kernels in the semiclassical spirit is provided in Berman–Berndtsson–Sjöstrand [Be-Ber-S]. The Bergman kernel studied here is then a model case.

A proof of the sharp Gårding inequality based on (13.5.3) was given by Córdoba–Fefferman [C-F]. For more applications of the FBI transform in PDE see Delort [De], Martinez [M], and Sjöstrand [S1].

The proof of Theorem 13.18 is essentially that of Quillen [Q] as presented in Putinar [Pu]. The theorem was rediscovered by Catlin and D’Angelo [Ca-D] whose argument used Bergman projectors.

Part 4

SEMICLASSICAL ANALYSIS ON MANIFOLDS

MANIFOLDS

- 14.1 Definitions, examples
- 14.2 Pseudodifferential operators on manifolds
- 14.3 Schrödinger operators on manifolds
- 14.4 Notes

We devote this chapter to a quick review of differential manifolds and then to an exposition of how the semiclassical calculus extends to compact manifolds. The application presented in this book is the proof of a quantum ergodicity theorem in the next chapter.

14.1. DEFINITIONS, EXAMPLES

14.1.1. Manifolds. We start with a long sequence of definitions and examples:

DEFINITION. Let M be a Hausdorff topological space with a countable basis. We say that M is a *smooth n -dimensional manifold* if there exists a family $\mathcal{F}$ of homeomorphisms between open sets

$$\gamma : U_\gamma \rightarrow V_\gamma, \quad U_\gamma \subset M, \quad V_\gamma \subset \mathbb{R}^n,$$

satisfying the following properties:

- (i) (Smooth overlaps) If $\gamma_1, \gamma_2 \in \mathcal{F}$, then

$$\gamma_2 \circ \gamma_1^{-1} \in C^\infty(\gamma_1(U_{\gamma_1} \cap U_{\gamma_2}); \gamma_2(U_{\gamma_1} \cap U_{\gamma_2})).$$

- (ii) (Covering) The open sets U_γ cover M :

$$\bigcup_{\gamma \in \mathcal{F}} U_\gamma = M.$$

We call $\{(\gamma, U_\gamma) \mid \gamma \in \mathcal{F}\}$ an *atlas* for M . The open set $U_\gamma \subset M$ is a *coordinate patch*.

Next assume that M and N are smooth manifolds of dimensions n and m , respectively. Let $\mathcal{F}$ be an atlas for M and let $\mathcal{G}$ be an atlas for N .

DEFINITIONS. (i) A function $F : M \rightarrow N$ is said to be *smooth* if for every $\gamma \in \mathcal{F}$ and $\rho \in \mathcal{G}$,

$$\rho \circ F \circ \gamma^{-1} \in C^\infty(V_\gamma; V_\rho),$$

whenever $F(U_\gamma) \cap U_\rho \neq \emptyset$.

(ii) If $F : M \rightarrow N$ is invertible and both F and F^{-1} are smooth, we call F a *diffeomorphism*.

14.1.2. Vector bundles.

DEFINITION. A C^∞ complex *vector bundle* over M with *fiber dimension* N consists of a C^∞ manifold V and a smooth map

$$\pi : V \rightarrow M,$$

defining the *fibers*

$$V_x := \pi^{-1}(\{x\})$$

for $x \in M$, each of which has the structure of a complex N -dimensional vector space. In particular, for each point $x \in M$ there exists an open neighborhood U of x and a diffeomorphism ψ such that

$$(14.1.1) \quad \psi : \pi^{-1}(U) \rightarrow U \times \mathbb{C}^N;$$

and for each point $y \in U$, ψ maps V_y onto $\{y\} \times \mathbb{C}^N$ as a linear isomorphism:

$$(14.1.2) \quad \psi(V_y) = \{y\} \times \mathbb{C}^N.$$

A *real vector bundle* is defined analogously.

REMARKS: Transition matrices. (i) We can choose a covering $\{X_i\}_{i \in \mathcal{I}}$ of M such that for each index i there exists

$$\psi_i : \pi^{-1}(X_i) \rightarrow X_i \times \mathbb{C}^N$$

with the properties (14.1.1), (14.1.2) in the definition of a vector bundle. Then

$$\gamma_{ij} := \psi_i \circ \psi_j^{-1} \in C^\infty(X_i \cap X_j; GL(N, \mathbb{C}))$$

are the associated *transition matrices*, $GL(N, \mathbb{C})$ denoting the group of invertible, complex $N \times N$ matrices.

(ii) We can recover the vector bundle V from the transition matrices. To see this, suppose that we are given functions γ_{ij} satisfying the identities

$$\begin{cases} \gamma_{ij}(x)\gamma_{ji}(x) = I & \text{for } x \in X_i \cap X_j, \\ \gamma_{ij}(x)\gamma_{jk}(x)\gamma_{ki}(x) = I & \text{for } x \in X_i \cap X_j \cap X_k. \end{cases}$$

Now define on the set $W := \mathcal{I} \times M \times \mathbb{C}^N$ the equivalence relation $(i, x, t) \sim (i', x', t')$ if and only if $x = x'$ and $t' = \gamma_{i'i}(x)t$. Then one can check that

$$V = W / \sim$$

is a vector bundle over M with the given transition functions. $\square$

DEFINITION. A *section* of the vector bundle V is a smooth map

$$u : M \rightarrow V$$

such that

$$\pi \circ u(x) = x \quad (x \in M).$$

We write

$$u \in C^\infty(M, V).$$

EXAMPLE 1: Tangent bundle. Let M be an n -dimensional smooth manifold. We define the *tangent bundle* of M , denoted

$$TM,$$

as a real vector bundle with fibers given by $\mathbb{R}^n$ and with the transition functions

$$(14.1.3) \quad \gamma_{ij}(x) := \partial(\gamma_i \circ \gamma_j^{-1})(\gamma_j(x)) \in GL(n, \mathbb{R})$$

for $x \in U_{\gamma_i} \cap U_{\gamma_j}$. Its sections $C^\infty(M, TM)$ are the smooth *vector fields* on M . The fiber

$$(14.1.4) \quad T_x M := \pi^{-1}(x)$$

is the *tangent space* at the point $x \in M$.

In local coordinates, sections of TM correspond to vector fields on $\mathbb{R}^n$ and hence act on smooth functions. Thus if $V \in C^\infty(M, TM)$, then

$$(14.1.5) \quad V : C^\infty(M) \rightarrow C^\infty(M). \quad \square$$

EXAMPLE 2: Cotangent bundle. For any vector bundle V we can define its dual

$$V^* := \bigcup_{x \in X} (V_x)^*,$$

with transition functions

$$\tilde{\gamma}_{ij} = ((\gamma_{ij})^{-1})^T,$$

γ_{ij} defined by (14.1.3).

If $V = TM$, we obtain the *cotangent bundle*, denoted

$$T^*M.$$

Its sections $C^\infty(M, T^*M)$ are the differential 1-forms on M . The fiber

$$(14.1.6) \quad T_x^*M := \pi^{-1}(x)$$

is the *cotangent space* at x .

Canonical symplectic form on T^*M . In Chapter 2 we introduced the symplectic form on $\mathbb{R}^{2n}$. The cotangent bundle is likewise equipped with a canonical symplectic form:

DEFINITIONS.

(i) The *canonical 1-form* η on T^*M is given by the rule

$$\eta_\rho(X) = \rho(\pi_*X)$$

for $\rho \in T^*M$ and tangent vectors $X \in T_\rho(T^*M)$. We note that $\rho \in T^*M$ defines a linear form on $T_{\pi(\rho)}M$. The informal but persuasive way to write the canonical 1-form is

$$(14.1.7) \quad \eta = \xi dx,$$

where ξ is the (natural) coordinate on T_xM , defined by fixing coordinate x on M .

(ii) The *canonical symplectic form* on T^*M is the 2-form

$$\sigma := d\eta.$$

This form is obviously closed, and a calculation in local coordinates shows that it is nondegenerate. We informally write

$$(14.1.8) \quad \sigma = d\xi \wedge dx.$$

NOTATION. If $f : T^*M \rightarrow \mathbb{R}$ is Lebesgue measurable and integrable, we will write

$$(14.1.9) \quad \int_{T^*M} f \, dx \, d\xi$$

to denote the integral of f over T^*M with respect to the measure induced by the symplectic form:

$$dx \, d\xi = dx_1 \wedge \cdots \wedge dx_n \wedge d\xi_1 \wedge \cdots \wedge d\xi_n = \frac{\sigma^n}{n!}. \quad \square$$

EXAMPLE 3: s-density bundles. Let M be an n -dimensional manifold and let $\{(\gamma, U_\gamma) \mid \gamma \in \mathcal{F}\}$ be an atlas.

We define the *s-density bundle* over X , denoted

$$\Omega^s(M),$$

by choosing the transition functions

$$\gamma_{ij}(x) := |\det \partial(\gamma_j \circ \gamma_i^{-1})|^s \circ \gamma_i(x)$$

for $x \in U_{\gamma_i} \cap U_{\gamma_j}$. This is a *line bundle* over M , that is, a bundle with fibers of complex dimension one.

NOTATION. As in Section 9.1 we can use an informal notation for sections of *s-density bundles*:

$$(14.1.10) \quad u(x)|dx|^s \in C^\infty(M, \Omega^s(M)).$$

Then $C^\infty(M, \Omega^1(M))$ comprises the *densities* on M , and $C^\infty(M, \Omega^{\frac{1}{2}}(M))$ the *half-densities*.

REMARK: Integrals of densities. If $u \in \Omega^1(M)$, then for any open $W \subset\subset M$, we can invariantly define

$$(14.1.11) \quad \int_W u,$$

the integral of u over W .

To see this, let $\chi \in C_c^\infty(M)$ be supported in $U = U_\gamma$ and let $\psi : \pi^{-1}(U) \rightarrow U \times \mathbb{C}$ be as in (14.1.1): $\psi(u(x)) = (x, \psi_2(x))$. Then

$$\int_{U_\gamma} \chi u := \int_{V_\gamma} \chi(\gamma^{-1}(y)) \psi_2(\gamma^{-1}(y)) dy.$$

The definition of $\Omega^1(M)$ shows that this is independent of the choice of $\gamma \in \mathcal{F}$; and a partition of unity argument then gives the definition of $\int_W u$. $\square$

DEFINITIONS. (i) If $F : M \rightarrow N$ is a C^∞ mapping between two manifolds, we define the *pull-back*

$$F^* : C^\infty(N, \mathbb{C}) \rightarrow C^\infty(M; \mathbb{C})$$

by the rule

$$u \mapsto F^*u := u \circ F$$

for $u \in C^\infty(N; \mathbb{C})$.

(ii) If M and N are manifolds of the same dimension, we can also define the pull-back for sections of the *s-density bundles*:

$$F^* : C^\infty(N; \Omega^s(N)) \rightarrow C^\infty(M; \Omega^s(M)).$$

In the informal notation of (14.1.10), we have for $F(x) = y$, $x \in M$, $y \in N$,

$$F^*(u(y)|dy|^s) = u(F(x))|\det \partial F|^s|dx|^s.$$

14.1.3. Riemannian manifolds.

DEFINITION. An n -dimensional *Riemannian manifold* (M, g) comprises a smooth, n -dimensional manifold M and a metric defined using a smooth inner product g on fibers of TM . This means that

$$g \in C^\infty(M, T^*M \otimes T^*M)$$

is symmetric and positive definite.

Note that an element of $V^* \otimes V^*$ defines a bilinear form on $V \times V$. A local construction and a partition of unity argument show that any manifold can be equipped with a Riemannian metric.

NOTATION. (i) If (M, g) is a Riemannian manifold, we write the quadratic form on $T_x M$ in local coordinates as

$$(14.1.12) \quad g(X, Y) = \sum_{1 \leq i, j \leq n} g_{ij}(x) X_i Y_j$$

and define

$$((g^{ij})) := ((g_{ij}))^{-1}, \quad \bar{g} := \det((g_{ij})).$$

(In terms of the discussion of quadratic forms preceding Theorem 13.11, g^{ij} is the quadratic form g^{-1} on the fibers of the cotangent bundle, $T_x^* M$.)

(ii) If $f : M \rightarrow \mathbb{R}$ is measurable, we will write

$$(14.1.13) \quad \int_M f \, dx$$

to denote the integral of f over M with respect to the density induced by the metric g . In local coordinates, we write

$$dx = \sqrt{\bar{g}} |dx_1 \wedge \cdots \wedge dx_n|.$$

This density is invariantly defined once a metric is fixed.

REMARKS. (i) An informal way of writing the metric g is

$$g := \sum_{i, j=1}^n g_{ij}(x) dx_i dx_j,$$

meaning that (14.1.12) holds.

(ii) We can use the metric to identify $\xi \in T_x^* M$ with $X \in T_x M$, written

$$(14.1.14) \quad \xi \sim X,$$

provided

$$\xi(Y) = g_x(Y, X)$$

for all $Y \in T_x M$.

(iii) Under the identification (14.1.14) the flow of H_p on T^*M generated by the symbol

$$(14.1.15) \quad p(x, \xi) := |\xi|_{g_x}^2 = \sum_{i,j=1}^n g^{ij}(x) \xi_i \xi_j = \sum_{i,j=1}^n g_{ij}(x) X_i X_j = g_x(X, X)$$

is the geodesic flow on TM .

(iv) Half-densities on M can be identified with functions using the Riemannian density:

$$u = u(x) |dx|^{\frac{1}{2}} = \tilde{u}(x) \left(\bar{g}^{\frac{1}{2}} |dx_1 \wedge \cdots \wedge dx_n| \right)^{\frac{1}{2}},$$

$$u \in C^\infty(M, \Omega^{\frac{1}{2}}(M)), \quad u \mapsto \tilde{u} \in C^\infty(M). \quad \square$$

14.2. PSEUDODIFFERENTIAL OPERATORS ON MANIFOLDS

We now assume that M is a *compact* Riemannian manifold, a restriction made largely for expository reasons. Our applications to spectral asymptotics and to quantum ergodicity concern this case only.

14.2.1. Differential operators on manifolds.

To define the action of vector fields and differential operators on L^2 , we first need to define distributions on M :

DEFINITION. We say that a linear map $u : C^\infty(M) \rightarrow \mathbb{C}$ is a *distribution* on M if for every $\gamma \in \mathcal{F}$ and every $\chi \in C_c^\infty(V_\gamma)$, the mapping defined for $\varphi \in \mathcal{S}(\mathbb{R}^n)$ by

$$\varphi \mapsto u(\gamma^*(\chi\varphi))$$

belongs to $\mathcal{S}'(\mathbb{R}^n)$, that is, is a distribution on $\mathbb{R}^n$.

We denote the space of distributions on M by

$$\mathcal{D}'(M).$$

DEFINITION. A *differential operator* P on M of order at most m is a finite linear combination of compositions of at most m vector fields. Thus

$$P = \sum V_{j_1} V_{j_2} \cdots V_{j_k},$$

where $V_{j_i} \in C^\infty(M, TM)$ and $1 \leq k \leq m$.

As in (14.1.5), we can interpret P as a linear mapping

$$(14.2.1) \quad \begin{aligned} P &: C^\infty(M) \rightarrow C^\infty(M), \\ P &: C_c^\infty(M) \rightarrow C_c^\infty(M), \\ P &: \mathcal{D}'(M) \rightarrow \mathcal{D}'(M). \end{aligned}$$

DEFINITIONS.

(i) We define

$$L^2(M, \Omega^{\frac{1}{2}}(M))$$

as the completion of $C^\infty(M, \Omega^{\frac{1}{2}}(M))$ with respect to the norm

$$\|u\|_{L^2} := \left(\int_M |u|^2 \right)^{\frac{1}{2}},$$

where $|u|^2 \in \Omega^1(M)$ is a density on M and hence can be integrated.

(ii) For a Riemannian compact manifold (M, g) ,

$$L^2(M) := \{u : M \rightarrow \mathbb{C} \mid u \text{ is measurable, } \int_M |u|^2 dx < \infty\}.$$

(iii) If k is a nonnegative integer, we define

$$H^k(M) := \{u \in L^2(M) \mid \text{for all } 1 \leq l \leq k, V_1 \cdots V_l u \in L^2(M) \\ \text{for all vector fields } V_1, \dots, V_l\};$$

$H^k(M)$ is a *Sobolev space*.

(iv) Choose a family of vector fields $\{W_j\}_{1 \leq j \leq J}$ that spans $T_x M$ at every $x \in M$. Since M is compact, that can always be achieved for some finite $J \geq n$. We then define the norm

$$(14.2.2) \quad \|u\|_{H^k(M)} := \left(\sum_{l=0}^k \sum_{\substack{\alpha \in \mathbb{N}^\ell \\ 1 \leq \alpha_j \leq J}} \|W_{\alpha_1} \cdots W_{\alpha_l} u\|_{L^2(M)}^2 \right)^{\frac{1}{2}},$$

with the convention that $W_{\alpha_1} \cdots W_{\alpha_\ell} u = u$ if $\ell = 0$.

We likewise define the semiclassical norms

$$(14.2.3) \quad \|u\|_{H_h^k(M)} := \left(\sum_{l=0}^k \sum_{\substack{\alpha \in \mathbb{N}^\ell \\ 1 \leq \alpha_j \leq J}} h^{2l} \|W_{\alpha_1} \cdots W_{\alpha_l} u\|_{L^2(M)}^2 \right)^{\frac{1}{2}}.$$

(v) We also define negative order Sobolev spaces of distributions:

$$H^{-k}(M) := \text{span}\{u \in \mathcal{D}'(M) \mid u = V_1 \cdots V_l f, \text{ where } f \in L^2(M), \\ 0 \leq l \leq k, V_1, \dots, V_l \in C^\infty(M, TM)\}.$$

By convention $V_1 \cdots V_l u = u$ if $l = 0$.

(vi) The norm for $H^{-k}(M)$ can be given using the fixed family of vector fields from part (iv):

(14.2.4)

$$\|u\|_{H^{-k}(M)} := \inf \left\{ \sum_{j=1}^J \|f_j\|_{L^2(M)} \mid u = \sum_{\ell=0}^k \sum_{\substack{\alpha \in \mathbb{N}^\ell \\ 1 \leq \alpha_j \leq J}} W_{\alpha_1} \cdots W_{\alpha_j} f_j \right\}.$$

The norm $\|u\|_{H_h^{-k}(M)}$ is given by replacing each W_{α_j} by hW_{α_j} .

When P is a differential operator of order m , then

$$P : H^{k+m}(M) \rightarrow H^k(M).$$

14.2.2. Pseudodifferential operators on manifolds. Recall from Section 9.3 that a symbol class for which we have invariance under coordinate changes is

$$S^m = S^m(\mathbb{R}^n) = \{a \in C^\infty(\mathbb{R}^{2n}) \mid |\partial_x^\alpha \partial_\xi^\beta a| \leq C_{\alpha\beta} \langle \xi \rangle^{m-|\beta|} \text{ for all } \alpha, \beta\}.$$

Remember that symbols in S^m can depend upon h , in which case the constants $C_{\alpha\beta}$ are uniform for $0 < h \leq h_0$.

DEFINITION. A linear operator

$$A : C^\infty(M) \rightarrow C^\infty(M)$$

is called a *pseudodifferential operator* on M if

(i) there exists an integer m such that for each coordinate patch U_γ , there exists a symbol $a_\gamma \in S^m$ such that

$$(14.2.5) \quad \varphi A(\psi u) = \varphi \gamma^* a_\gamma^w(x, hD) (\gamma^{-1})^* (\psi u)$$

for any $\varphi, \psi \in C_c^\infty(U_\gamma)$ and $u \in C^\infty(M)$;

and

(ii) for all $\varphi_1, \varphi_2 \in C^\infty(M)$ satisfying

$$\text{spt } \varphi_1 \cap \text{spt } \varphi_2 = \emptyset,$$

we have for all N

$$(14.2.6) \quad \|\varphi_1 A \varphi_2\|_{H^{-N}(M) \rightarrow H^N(M)} = O(h^\infty).$$

Condition (ii) is motivated by Theorem 9.7; we require that away from the diagonal the operator A is regularizing and negligible in h . The regularizing effect (14.2.6) means that $\varphi_1 A \varphi_2$ improves regularity by any order and in particular maps distributions to smooth functions.

NOTATION. For a pseudodifferential operator defined above we write

$$A \in \Psi^m(M)$$

and sometimes call A a *quantum observable*. We will also use the notation

$$\Psi(M) := \Psi^0(M), \quad \Psi^{-\infty}(M) := \bigcap_{m \in \mathbb{Z}} \Psi^m(M). \quad \square$$

14.2.3. Symbols of pseudodifferential operators. We can also define symbols on T^*M :

DEFINITION. We say

$$a \in S^m(T^*M)$$

if $a \in C^\infty(T^*M)$ and for any $\gamma \in \mathcal{F}$, the pull-back of a under the identification

$$V_\gamma \times \mathbb{R}^n \rightarrow T^*(U_\gamma)$$

belongs to $S^m(V_\gamma \times \mathbb{R}^n)$.

This definition does not depend on the choice of the atlas.

Our next goal is to associate with a pseudodifferential operator A a symbol a defined on T^*M .

THEOREM 14.1 (Symbols and quantizations). *There exist linear maps*

$$(14.2.7) \quad \sigma : \Psi^m(M) \rightarrow S^m(T^*M)/hS^{m-1}(T^*M)$$

and

$$(14.2.8) \quad \text{Op} : S^m(T^*M) \rightarrow \Psi^m(M)$$

such that

$$(14.2.9) \quad \sigma(A_1 A_2) = \sigma(A_1) \sigma(A_2)$$

and

$$(14.2.10) \quad \sigma(\text{Op}(a)) = [a] \in S^m(T^*M)/hS^{m-1}(T^*M).$$

NOTATION. (i) In (14.2.10) we use $[a]$ to denote the equivalence class in S^m/hS^{m-1} :

$$[a] = [\hat{a}] \quad \text{if and only if } a - \hat{a} \in hS^{m-1}(T^*M)$$

for $a, \hat{a} \in S^m(T^*M)$.

(ii) We call $a = \sigma(A)$ the *symbol* of the pseudodifferential operator A . Later we will often write

$$(14.2.11) \quad \text{Op}(a) = a^w(x, hD).$$

INTERPRETATION. We are asserting in (14.2.8) that for each symbol in $S^m(T^*M)$ we can associate a pseudodifferential operator in $\Psi^m(M)$.

Conversely, (14.2.7) says that given a pseudodifferential operator in $\Psi^m(M)$, we can associate it with a symbol in $S^m(T^*M)$, uniquely defined up to a lower-order term that is less singular and of order $O(h)$ as $h \rightarrow 0$. $\square$

Proof. 1. Let U be an open subset of $\mathbb{R}^n$. Suppose that $B : C_c^\infty(U) \rightarrow C^\infty(U)$ and that for all $\varphi, \psi \in C_c^\infty(U)$ the mapping $u \mapsto \varphi B \psi u$ belongs to $\Psi^m(\mathbb{R}^n)$.

Let us denote by $S_{\text{loc}}^m(U \times \mathbb{R}^n)$ the class of functions for which the estimates in the definition of S^m are uniformly valid on compact subsets of U .

2. We claim that there exists a symbol $a \in S_{\text{loc}}^m(U \times \mathbb{R}^n)$ such that

$$(14.2.12) \quad B = a^w(x, hD) + B_0,$$

where for all N

$$(14.2.13) \quad B_0 = O_{H^{-N}(V) \rightarrow H^N(V)}(h^\infty)$$

for each open set $V \subset\subset U$.

To see this, first choose a smooth, locally finite partition of unity $\{\psi_j\}_{j \in J}$,

$$\sum_{j \in J} \psi_j(x) \equiv 1, \quad x \in U, \quad \psi_j \in C_c^\infty(U).$$

Then

$$\psi_j B \psi_k = a_{jk}^w(x, hD),$$

where $a_{jk} \in S^m$ and $a_{jk}(x, \xi) = 0$ if $x \notin \text{spt}(\psi_j)$. Now put

$$a := \sum'_{j,k} a_{jk}(x, \xi) \in S_{\text{loc}}^m(U \times \mathbb{R}^n),$$

where we sum over those indices j, k for which $\text{spt}(\psi_j) \cap \text{spt}(\psi_k) \neq \emptyset$. We must verify (14.2.13) for

$$B_0 := B - a^w(x, hD) = \sum''_{j,k} \psi_j B \psi_k,$$

the sum now over indices j, k for which $\text{spt}(\psi_j) \cap \text{spt}(\psi_k) = \emptyset$. Since we assumed that $\psi_j B \psi_k \in \Psi^m$, Theorem 9.7 and the local finiteness in the partition of unity give the global mapping property from $H^{-N}(V)$ to $H^N(V)$, with norm bounded by $O(h^\infty)$.

3. For each coordinate chart (γ, U_γ) , we can now use (14.2.12) with $B = (\gamma^{-1})^* A \gamma^*$ to define

$$a_\gamma \in T^*(U_\gamma).$$

The second part of Theorem 9.10 shows that if $U_{\gamma_1} \cap U_{\gamma_2} \neq \emptyset$, then

$$(14.2.14) \quad (a_{\gamma_1} - a_{\gamma_2})|_{U_{\gamma_1} \cap U_{\gamma_2}} \in hS^{m-1}(T^*(U_{\gamma_1} \cap U_{\gamma_2})).$$

To define the symbol map, we first choose a covering of M by finitely many coordinate charts $\{U_\alpha\}_{\alpha \in J}$ and a corresponding finite partition of unity $\{\varphi_\alpha\}_{\alpha \in J}$, with $\text{spt}(\varphi_\alpha) \subset U_\gamma$. We then put

$$a := \sum_{\alpha \in J} \varphi_\alpha a_\alpha.$$

We see from (14.2.14) that $a \in S^m(T^*M)$ is invariantly defined up to terms in $hS^{m-1}(T^*M)$. We consequently can define

$$\sigma(A) := [a] \in S^m(T^*M)/hS^{m-1}(T^*M).$$

4. It remains to show the existence of

$$\text{Op} : S^m(T^*M) \rightarrow \Psi^m(M), \quad \sigma(\text{Op}(a)) = [a].$$

Suppose that for our covering $\{U_\alpha\}_{\alpha \in J}$ of M by coordinate charts, we choose $\{\psi_\alpha\}_{\alpha \in J}$ such that

$$\text{spt}(\psi_\alpha) \subset U_\alpha, \quad \sum_{\alpha \in J} \psi_\alpha^2(x) \equiv 1.$$

Define

$$A := \sum_{\alpha \in J} \psi_\alpha \gamma_\alpha^* \tilde{a}_\alpha^w(x, hD) (\gamma_\alpha^{-1})^* \psi_\alpha,$$

where

$$\tilde{a}_\alpha(x, \xi) := a(\gamma_\alpha^{-1}(x), (\partial \gamma_\alpha(x)^T)^{-1} \xi).$$

Theorem 9.10 demonstrates that $\sigma(A)$ equals $[a]$. □

REMARK. Theorem 14.1 is also valid when $h = 1$, in which case it is the classical result about a symbol of a pseudodifferential operator on a manifold. In this case the symbol is defined in $S^m(T^*M)/S^{m-1}(T^*M)$ and the only gain is in regularity. □

EXAMPLE: Dirichlet to Neumann map. Consider the boundary value problem

$$-\Delta u = 0 \quad \text{in } U, \quad u|_{\partial U} = g,$$

where $U \subset \mathbb{R}^n$ has a smooth boundary. We define the *Dirichlet to Neumann map*

$$N : g \rightarrow \frac{\partial u}{\partial \nu} \Big|_{\partial U}$$

where ν is the outward unit normal vector to ∂U . Then N is a pseudodifferential operator on ∂U and its symbol is

$$\sigma(N) = \sigma(\sqrt{-\Delta_{\partial U} + 1}),$$

where $\Delta_{\partial U}$ is the Laplace–Beltrami operator on ∂U for the metric induced by the Euclidean metric; see Section 14.2.6. $\square$

14.2.4. Properties of pseudodifferential operators on manifolds.

Remember that we assume that M is compact, in which case the mapping properties of operators in $\Psi^m(M)$ are easy to state. Recall from Section 14.2.1 that the Sobolev spaces $H^k(M)$ are independent of the choice of a Riemmanian metric.

THEOREM 14.2 (Mapping properties for a compact manifold).

Suppose that M is a compact manifold. Then:

(i) *If $A \in \Psi^0(M)$, then*

$$A : L^2(M) \rightarrow L^2(M) \text{ is bounded.}$$

(ii) *If $A \in \Psi^m(M)$ for $m < 0$, then*

$$A : L^2(M) \rightarrow L^2(M) \text{ is a compact operator.}$$

Proof. Since M is compact, we can find a finite partition of unity subordinate to a cover by elements of an atlas. Hence (i) and (ii) follow from Theorems 4.23 and 4.28. $\square$

REMARK. Sobolev spaces can be characterized using pseudodifferential operators:

$$H^m(M) = \{u \in \mathcal{D}'(M) \mid Au \in L^2(M) \text{ for all } A \in \Psi^m(M)\}.$$

This immediately provides a definition of $H^s(M)$ for $s \in \mathbb{R}$ and shows that if $A \in \Psi^m(M)$, then

$$A : H^s(M) \rightarrow H^{s-m}(M)$$

for each $s \in \mathbb{R}$. $\square$

14.2.5. Pseudodifferential operators and half-densities. Pseudodifferential operators on manifolds also act on half-densities:

DEFINITION. A linear operator

$$A : C^\infty(M; \Omega^{\frac{1}{2}}(M)) \rightarrow C^\infty(M; \Omega^{\frac{1}{2}}(M))$$

is called a *semiclassical pseudodifferential operator on half-densities* if there exists an integer m such that for each coordinate patch U_γ there exists a symbol $a_\gamma \in S^m$ such that for any $\varphi, \psi \in C_c^\infty(U_\gamma)$

$$(14.2.15) \quad \varphi A(\psi u) = \varphi \gamma^* a_\gamma^w(x, hD)(\gamma^{-1})^*(\psi u)$$

for each $u \in C^\infty(M; \Omega^{\frac{1}{2}}(M))$. Here

$$(\gamma^{-1})^* : C^\infty(U_\gamma, \Omega^{\frac{1}{2}}(U_\gamma)) \rightarrow C^\infty(V_\gamma, \Omega^{\frac{1}{2}}(V_\gamma))$$

with a similar statement holding for γ^* . We also require that the analogue of (14.2.6) holds.

NOTATION. We write

$$A \in \Psi^m(M; \Omega^{\frac{1}{2}}(M)).$$

By adapting the proof of Theorem 14.1 to the case of half-densities and using the first part of Theorem 9.3, we obtain

THEOREM 14.3 (Symbol on half-densities). *There exist linear maps*

$$(14.2.16) \quad \sigma : \Psi^m(M; \Omega^{1/2}(M)) \rightarrow S^m(T^*M)/h^2 S^{m-2}(T^*M)$$

and

$$(14.2.17) \quad \text{Op} : S^m(T^*M) \rightarrow \Psi^m(M; \Omega^{1/2}(M))$$

such that

$$(14.2.18) \quad \sigma(A_1 A_2) = \sigma(A_1) \sigma(A_2)$$

and

$$(14.2.19) \quad \sigma(\text{Op}(a)) = [a] \in S^m(T^*M)/h^2 S^{m-2}(T^*M).$$

14.2.6. PDE on manifolds.

In this subsection we will consider examples of linear PDE on manifolds. Here and in Section 14.3.4 we will revisit certain facts established earlier in simpler settings.

The Laplace–Beltrami operator. The operator Δ_g on a Riemannian manifold M is defined in local coordinates by

$$(14.2.20) \quad \Delta_g := \frac{1}{\sqrt{g}} \sum_{i,j=1}^n \frac{\partial}{\partial x_i} \left(g^{ij} \sqrt{g} \frac{\partial}{\partial x_j} \right).$$

One checks that (14.2.20) is independent of the choice of coordinates.

The function

$$p(x, \xi) = |\xi|_{g_x}^2,$$

defined by (14.1.15), is the symbol of the Laplace–Beltrami operator $-h^2 \Delta_g$.

The Schrödinger operator. Given a potential $V \in C^\infty(M)$, we also define

$$(14.2.21) \quad P(h) := -h^2 \Delta_g + V(x),$$

the symbol of which is

$$p(x, \xi) = |\xi|_{g_x}^2 + V(x).$$

When M is compact, then $P(h)$ has properties similar to the Schrödinger operator with a confining potential on $\mathbb{R}^n$, discussed earlier in Section 7.2.

We devote Section 14.3 to the study of (14.2.21) on a compact manifold. When $V = 0$, we can always rescale h to 1; and therefore our results also apply to the Laplace–Beltrami operator $-\Delta_g$.

The damped wave equation. We consider this initial-value problem for the wave equation:

$$(14.2.22) \quad \begin{cases} (\partial_t^2 + a \partial_t - \Delta_g)u = 0 & \text{on } M \times \mathbb{R} \\ u = 0, \quad \partial_t u = f & \text{on } M \times \{t = 0\}, \end{cases}$$

where $a = a(x) \geq 0$. As in Section 5.3 we define the energy of a solution at time t to be

$$E(t) := \frac{1}{2} \int_M (\partial_t u)^2 + |\partial_x u|^2 dx.$$

Using the pseudodifferential calculus developed in Section 14.2.3, it is straightforward to adapt the proofs in Section 5.3 to establish the following theorem.

THEOREM 14.4 (Rauch–Taylor Theorem). *Suppose u solves the damped wave equation (14.2.22) with the initial condition $f \in L^2(M)$.*

Assume also that there exists a time $T > 0$ such that each geodesic of length greater than or equal to T intersects the set $\{a > 0\}$.

Then there exist constants $C, \beta > 0$ such that

$$(14.2.23) \quad E(t) \leq Ce^{-\beta t} \|f\|_{L^2}$$

for all times $t \geq 0$.

14.3. SCHRÖDINGER OPERATORS ON MANIFOLDS

In this section we provide some basic theory for the Schrödinger operator

$$P(h) = -h^2 \Delta_g + V(x)$$

on a compact Riemannian manifold. We will in particular develop a functional calculus and use it to prove an analogue of Weyl's Law from Section 6.4.

14.3.1. Spectral theory.

LEMMA 14.5 (Smoothness of eigenfunctions). *Suppose that $z \in \mathbb{C}$ and that $u \in L^2(M)$ satisfies*

$$(P(h) - z)u = 0$$

in the sense of distributions. Then

$$u \in C^\infty(M).$$

In particular, the eigenfunctions of $P(h)$ are smooth.

Proof. 1. Choose $\chi \in C_c^\infty(\mathbb{R})$ equal to 1 for $|t| \leq T$. If $z \in K \subset \subset \mathbb{C}$, then for $T = T(K)$ sufficiently large we have

$$q(x, \xi) := (1 - \chi(|\xi|_g))(|\xi|_g^2 + V(x) - z)^{-1} \in S^{-2}(T^*M)$$

and

$$p(x, \xi)q(x, \xi) = 1 - r_1(x, \xi), \quad r_1 \in S^{-\infty}(T^*M).$$

2. Theorem 14.1 shows that

$$Q_0(P(h) - z) = I - R_1,$$

where $Q_0 := \text{Op}(q)$ and $R_1 \in \Psi^{-1}(M)$. It follows that

$$(14.3.1) \quad Q_N(P(h) - z) = I - R_{N+1},$$

for $R_{N+1} \in \Psi^{-N-1}(M)$ and

$$Q_N := \sum_{k=0}^N R_1^k Q_0.$$

3. If u solves $(P(h) - z)u = 0$, then (14.3.1) gives

$$u - R_{N+1}u = Q_N(P(h) - z)u = 0.$$

For vector fields V_j and for $1 \leq j \leq l \leq N+1$, we then have

$$h^l V_1 \cdots V_l R_{N+1} \in \Psi^{-N-1+l}(M) \subset \Psi^0(M).$$

The definition of $H^N(M)$ and Theorem 14.2 show that $R_{N+1}u \in H^{N+1}(M)$. Hence $u \in H^{N+1}$. This conclusion is valid for any N , and so $u \in C^\infty(M)$. $\square$

Next we establish a useful resolvent estimate:

LEMMA 14.6 (Resolvents on manifold). *The operator*

$$P(h) = -h^2 \Delta_g + V, \quad V \in C^\infty(M; \mathbb{R}),$$

is symmetric.

For each $z \in \mathbb{C} \setminus \mathbb{R}$, the operator

$$P(h) - z : H_h^2(M) \rightarrow L^2(M)$$

is invertible and

$$(14.3.2) \quad \|(P(h) - z)^{-1}\|_{L^2(M) \rightarrow H_h^2(M)} = O(1 + |\operatorname{Im} z|^{-1}).$$

The semiclassical Sobolev spaces $H_h^k(M)$ are defined in (14.2.3).

Proof. 1. The local coordinate definition of Δ_g in (14.2.20) and the local formula for integration on M given in (14.1.13) show that Δ_g is symmetric:

$$\begin{aligned} \int_M \Delta_g u \bar{v} \, dx &= \int_{V_\gamma} \bar{g}^{-1/2} \sum_{i,j=1}^n \partial_{y_i} \left(g^{ij} \bar{g}^{1/2} \partial_{y_j} (\gamma^{-1})^* u \right) (\gamma^{-1})^* \bar{v} \bar{g}^{1/2} \, dy \\ &= \int_{V_\gamma} \sum_{i,j=1}^n \partial_{y_i} \left(g^{ij} \bar{g}^{1/2} \partial_{y_j} (\gamma^{-1})^* u \right) (\gamma^{-1})^* \bar{v} \, dy \\ &= \int_{V_\gamma} (\gamma^{-1})^* u \sum_{i,j=1}^n \partial_{y_i} \left(g^{ij} \bar{g}^{1/2} \partial_{y_j} (\gamma^{-1})^* \bar{v} \right) \, dy \\ &= \int_M u \overline{\Delta_g v} \, dx, \end{aligned}$$

for $u, v \in C_c^\infty(U_\gamma)$.

2. We now observe that if $u \in H_h^2(M)$ and $(P(h) - z)u = 0$, then $u = 0$. Indeed, Lemma 14.5 shows that $u \in C^\infty(M)$ and the symmetry of $P(h)$ gives

$$(14.3.3) \quad 0 = \operatorname{Im} \langle (P(h) - z)u, u \rangle = -\operatorname{Im} z \|u\|_{L^2}^2.$$

Since $\operatorname{Im} z \neq 0$, we see that $u = 0$.

3. To show surjectivity, suppose that $u \in L^2(M)$ is orthogonal to the image of $P(h) - z$ on $C^\infty(M) \subset H_h^2(M)$. The symmetry of $P(h)$ implies $(P(h) - \bar{z})u = 0$ in the sense of distributions. As in step 1, we first see that $u \in C^\infty$ and then derive a contradiction.

4. The calculation in (14.3.3) shows that for $u \in C^\infty(M)$,

$$\|(P(h) - z)u\|_{L^2} \geq |\operatorname{Im} z| \|u\|_{L^2}.$$

To prove (14.3.2), we therefore need to show that

$$(14.3.4) \quad \|u\|_{H_h^2(M)} \leq C \|(P(h) - z)u\|_{L^2} + C \|u\|_{L^2}.$$

For this, we use (14.3.1) with $N = 1$; for any vector fields V_1 and V_2 , we have

$$\|h^2 V_1 V_2 u\|_{L^2} \leq \|h^2 V_1 V_2 R_2 u\|_{L^2} + \|h^2 V_1 V_2 Q_1 (P(h) - z)u\|_{L^2}.$$

Since $h^2 V_1 V_2 R_2, h^2 V_1 V_2 Q_1 \in \Psi^0(M)$, the estimate (14.3.4) follows from Theorem 14.2 and the definition (14.2.3). $\square$

REMARK. Since for a vector field V we have

$$\begin{aligned} hV(P(h) - z)^{-1} &= (P(h) - z)^{-1} hV \\ &\quad + (P(h) - z)^{-1} [P(h), hV] (P(h) - z)^{-1}, \end{aligned}$$

it also follows that

$$(14.3.5) \quad (P(h) - z)^{-1} = O_{H_h^k \rightarrow H_h^{k+2}}(|\operatorname{Im} z|^{-1-k}), \quad 0 < |\operatorname{Im} z| \leq 1,$$

for $k \geq 0$. $\square$

The operator (14.2.21) has nice properties as an unbounded operator on $L^2(M)$:

THEOREM 14.7 (Eigenvalues and eigenfunctions on manifolds).

Let M be a compact Riemannian manifold and let $V \in C^\infty(M; \mathbb{R})$.

(i) The operator $P(h) := -h^2 \Delta_g + V$ with the domain $C^\infty(M)$ is essentially selfadjoint; and the domain of the closure is $H^2(M)$.

(ii) For each $h > 0$, there exists an orthonormal basis

$$\{u_j(h)\}_{j=1}^\infty \subset C^\infty(M)$$

of $L^2(M)$ comprised of eigenfunctions of $P(h)$:

$$(14.3.6) \quad P(h)u_j(h) = E_j(h)u_j(h) \quad (j = 1, \dots),$$

where $E_j(h) \rightarrow \infty$ as $j \rightarrow \infty$.

REMARK: Eigenfunctions for the Laplace–Beltrami operator. If $V = 0$, we can take $h = 1$, since the eigenvalues can be rescaled. We then see that the Laplace–Beltrami operator $-\Delta_g$ has eigenvalues

$$0 = \lambda_0 < \lambda_1 \leq \dots \leq \lambda_j \rightarrow \infty$$

and there exists an orthonormal basis $\{\varphi_j\}_{j=1}^\infty$ of $L^2(M)$ consisting of eigenfunctions:

$$(14.3.7) \quad -\Delta_g \varphi_j = \lambda_j \varphi_j \quad (j = 1, \dots). \quad \square$$

Proof. 1. Essential selfadjointness now follows from Lemma 14.6 and Theorem C.12.

2. For $\operatorname{Im} z \neq 0$, (14.3.1) shows that

$$(P(h) - z)^{-1} = Q_1 + R_2(P(h) - z)^{-1}.$$

Since $Q_1, R_2 \in \Psi^{-2}(M)$, Theorem 14.2 implies that $(P(h) - z)^{-1}$ is a compact operator. According then to Theorem C.7, the spectrum of $(P(h) + i)^{-1}$ is discrete, with an accumulation point at 0.

An application of the Spectral Theorem (Theorem C.10) provides the orthonormal basis of eigenfunctions corresponding to $E_j(h) \rightarrow \infty$. According to Lemma 14.5, the eigenfunctions are smooth. $\square$

14.3.2. A functional calculus. Using the basis of eigenfunctions given in (14.3.6), we can write

$$(14.3.8) \quad P(h) = \sum_{j=1}^{\infty} E_j(h) u_j(h) \otimes \overline{u_j(h)},$$

where we use the notation $u \otimes v(\varphi) := u \int_M v \varphi \, dx$.

DEFINITION. For each $f \in L^\infty(\mathbb{R})$, we define the bounded operator

$$f(P(h)) : L^2(M) \rightarrow L^2(M)$$

by

$$(14.3.9) \quad f(P(h)) := \sum_{j=1}^{\infty} f(E_j(h)) u_j(h) \otimes \overline{u_j(h)}.$$

We need to rewrite (14.3.9) in a more useful form, and for this we now assume that $f \in \mathcal{S}(\mathbb{R})$ and recall the almost analytic extension $\tilde{f}$ discussed in Theorem 3.6.

THEOREM 14.8 (Helffer–Sjöstrand formula). *We have*

$$(14.3.10) \quad f(P(h)) = \frac{1}{\pi i} \int_{\mathbb{C}} \bar{\partial}_z \tilde{f}(z) (P(h) - z)^{-1} dm,$$

m denoting Lebesgue measure on $\mathbb{C}$.

We call (14.3.10) the *Helffer–Sjöstrand formula*.

Proof. Let $B(t, \epsilon)$ denote the disk in $\mathbb{C}$ with center t and radius ϵ . Using Green's formula, we calculate for $t \in \mathbb{R}$ that

$$\begin{aligned} \frac{1}{\pi i} \int_{\mathbb{C}} \bar{\partial}_z \tilde{f}(z) (t - z)^{-1} dm &= \frac{1}{\pi i} \lim_{\epsilon \rightarrow 0} \int_{\mathbb{C} \setminus B(t, \epsilon)} \bar{\partial}_z \tilde{f}(z) (t - z)^{-1} dm \\ &= \frac{1}{\pi i} \lim_{\epsilon \rightarrow 0} \int_{\mathbb{C} \setminus B(t, \epsilon)} \bar{\partial}_z \left(\tilde{f}(z) (t - z)^{-1} \right) dm \\ &= \frac{1}{2\pi i} \lim_{\epsilon \rightarrow 0} \oint_{\partial B(t, \epsilon)} \tilde{f}(z) (t - z)^{-1} dz \\ &= \frac{1}{2\pi i} \lim_{\epsilon \rightarrow 0} \oint_{\partial B(t, \epsilon)} (f(t) + O(\epsilon)) (t - z)^{-1} dz \\ &= f(t). \end{aligned}$$

We now put $t = E_j(h)$ and apply (14.3.9) to obtain (14.3.10). $\square$

We next use the Helffer–Sjöstrand formula (14.3.10) to learn that $f(P(h))$ is a pseudodifferential operator:

THEOREM 14.9 (Symbols and functional calculus). *If $f \in \mathcal{S}(\mathbb{R})$, then*

$$(14.3.11) \quad f(P(h)) \in \Psi^{-\infty}(M)$$

and has the symbol

$$(14.3.12) \quad \sigma(f(P(h))) = f(|\xi|_g^2 + V(x)).$$

Proof. 1. Write $P = P(h)$. We first demonstrate that

$$(14.3.13) \quad f(P) \in \Psi(M)$$

and start by checking part (ii) of the definition in Section 14.2.2. Let $\varphi, \psi \in C^\infty(M)$ satisfy $\text{spt } \varphi \cap \text{spt } \psi = \emptyset$. We need to show that

$$(14.3.14) \quad \varphi f(P) \psi = O_{H^{-N} \rightarrow H^N}(h^N),$$

for any N .

2. Let

$$Q_0(z) := (|\xi|_g^2 + V(x) - z)^{-1} \quad \text{for } \operatorname{Im} z \neq 0.$$

Then $Q_0(z) \in \Psi^{-2}(M)$, but with estimates that blow up polynomially as $|\operatorname{Im} z| \rightarrow 0$. A calculation based upon (14.2.9) in Theorem 14.1 demonstrates that

$$(14.3.15) \quad (P - z)Q_0(z) = I - R_1(z),$$

where $R_1(z) \in h\Psi^{-1}(M)$. Computing the symbol of $R_1(z)$ in local coordinates using Theorem 4.18 shows that

$$R_1(z) = O_{H_h^{-N} \rightarrow H_h^{-N+1}}(h|\operatorname{Im} z|^{-K}),$$

for some $K > 0$.

Iteration of (14.3.15) as in the proof of Lemma 14.5 establishes that for any M we have

$$(14.3.16) \quad (P - z)Q_M(z) = I - R_{M+1}(z),$$

where

$$R_{M+1}(z) = O_{H_h^{-N} \rightarrow H_h^{-N+M+1}}(h^M |\operatorname{Im} z|^{-K_M})$$

and $Q_M(z) \in \Psi^{-2}(M)$. The estimates for $Q_M(z)$ blow up polynomially as $\operatorname{Im} z \rightarrow 0$. The support condition on φ and ψ and the pseudodifferential calculus show that

$$\varphi Q_M(z) \psi = O_{H^{-N} \rightarrow H^N}(h^N |\operatorname{Im} z|^{-L_N}).$$

From (14.3.16) we see that

$$(P - z)^{-1} = Q_M(z) - (P - z)^{-1}R_{M+1}(z).$$

Combining the above with the estimate (14.3.5) shows that for M large enough,

$$\begin{aligned} & \varphi(P - z)^{-1} \psi \\ &= O_{H^{-N} \rightarrow H^N}(h^N |\operatorname{Im} z|^{-L_N}) + O_{H_h^{-N} \rightarrow H_h^{M-N}}(h^M |\operatorname{Im} z|^{-K'_M}) \\ &= O_{H^{-N} \rightarrow H^N}(h^N |\operatorname{Im} z|^{-M_N}), \end{aligned}$$

for some M_N . Here we replaced the semiclassical Sobolev spaces with the usual ones by taking M large enough. Since the almost analytic extension satisfies $\bar{\partial}_z \tilde{f} = O(|\operatorname{Im} z|^\infty \langle z \rangle^{-\infty})$ and $\operatorname{spt} \tilde{f} \subset \{|\operatorname{Im} z| \leq 1\}$, (14.3.10) shows that (14.3.14) holds.

3. Next, we need to check that for $\varphi \in C_c^\infty(M)$ with $\operatorname{spt} \varphi \subset U_\gamma$,

$$(14.3.17) \quad (\gamma^{-1})^* \varphi f(P) \varphi \gamma^* \in \Psi(\mathbb{R}^n).$$

A partition of unity argument and (14.3.14) then show that part (i) of the definition of $\Psi(M)$ in Section 14.2.2 holds.

In view of Theorem 9.12, to prove (14.3.17), it suffices to check that for any N and any choices of linear symbols l_j we have

$$(14.3.18) \quad \|\mathrm{ad}_{l_1(x,hD)} \cdots \mathrm{ad}_{l_N(x,hD)} (\gamma^{-1})^* \varphi f(P) \varphi \gamma^*\|_{L^2 \rightarrow L^2} = O(h^N).$$

Note that for a linear function l on $\mathbb{R}^{2n}$, we have

$$\mathrm{ad}_{l(x,hD)} ((\gamma^{-1})^* \varphi (P-z)^{-1} \varphi \gamma^*) = (\gamma^{-1})^* (\mathrm{ad}_L (\varphi (P-z)^{-1} \varphi)) \gamma^*,$$

where

$$L := \gamma^* \tilde{\varphi} l(x, hD) \tilde{\varphi} (\gamma^{-1})^* \in \Psi^1(M)$$

for a function $\tilde{\varphi} \in C^\infty(\mathbb{R}^n)$ such that $\mathrm{spt} \tilde{\varphi} \subset \gamma(U_\gamma)$, $\gamma^* \tilde{\varphi}|_{\mathrm{spt} \varphi} = 1$. Using the derivation property (8.1.9) of ad_L and Lemma 14.6, we deduce that

$$\mathrm{ad}_L (P-z)^{-1} = -(P-z)^{-1} (\mathrm{ad}_L P) (P-z)^{-1} = O_{L^2 \rightarrow L^2}(h|\mathrm{Im} z|^{-2}).$$

Therefore

$$\begin{aligned} & \|\mathrm{ad}_{l(x,hD)} (\gamma^{-1})^* \varphi f(P) \varphi \gamma^*\|_{L^2 \rightarrow L^2} \\ & \leq \int_{\mathbb{C}} |\bar{\partial}_z \tilde{f}(z)| \|\mathrm{ad}_{l(x,hD)} (\gamma^{-1})^* \varphi (P-z)^{-1} \varphi \gamma^*\|_{L^2 \rightarrow L^2} dm \\ & \leq \int_{\mathbb{C}} |\bar{\partial}_z \tilde{f}(z)| \|(\gamma^{-1})^* (\mathrm{ad}_L (\varphi (P-z)^{-1} \varphi)) \gamma^*\|_{L^2 \rightarrow L^2} dm \\ & \leq Ch \int_{\mathbb{C}} |\bar{\partial}_z \tilde{f}(z)| |\mathrm{Im} z|^{-2} dm = O(h), \end{aligned}$$

where we again used the properties of $\tilde{f}$.

This proves (14.3.18) for $N = 1$; the case $N > 1$ following similarly. We conclude that (14.3.17) holds.

4. Since

$$(P+i)^k f(P) = g_k(P)$$

for $g_k := (t+i)^k f(t) \in \mathcal{S}(\mathbb{R})$, we see using (14.3.5) that $f(P(h)) : L^2(M) \rightarrow H_h^N(M)$ for any N . Hence

$$f(P(h)) \in \Psi^{-\infty}(M).$$

5. To compute the symbol of $f(P)$, we recall that (14.3.15) gives

$$((|\xi|_g^2 + V(x) - z)^{-1})^w (P-z) = I + O_{L^2 \rightarrow L^2}(h|\mathrm{Im} z|^{-K})$$

for some K . Hence it follows that

$$((|\xi|_g^2 + V(x) - z)^{-1})^w = (P-z)^{-1} + O_{L^2 \rightarrow L^2}(h|\mathrm{Im} z|^{-K-1}).$$

The Helffer–Sjöstrand formula (14.3.10) now gives

$$\begin{aligned} f(P) &= \frac{1}{\pi i} \int_{\mathbb{C}} ((|\xi|_g^2 + V(x) - z)^{-1})^w \bar{\partial}_z \tilde{f}(z) dm + O_{L^2 \rightarrow L^2}(h) \\ &= \left(\frac{1}{\pi i} \int_{\mathbb{C}} (|\xi|_g^2 + V(x) - z)^{-1} \bar{\partial}_z \tilde{f}(z) dm \right)^w + O_{L^2 \rightarrow L^2}(h) \\ &= (f(|\xi|_g^2 + V(x)))^w + O_{L^2 \rightarrow L^2}(h). \end{aligned}$$

As we already know that $f(P) \in \Psi^{-\infty}(M)$, this calculation implies that the principal symbol of $f(P)$ is $f(|\xi|_g^2 + V(x))$. $\square$

14.3.3. Trace class operators. Trace class properties of $f(P)$ will be useful in the next section and in Section 15.3. See Section C.3 for the basic definitions and facts.

THEOREM 14.10 (Trace class properties of $f(P(h))$). *Assume that M is a compact Riemannian manifold, $V \in C^\infty(M, \mathbb{R})$, and*

$$P(h) = -h^2 \Delta_g + V(x)$$

is a Schrödinger operator.

Then for $f \in \mathcal{S}$

$$(14.3.19) \quad f(P(h)) \in \mathcal{L}_1(L^2(M))$$

and

$$(2\pi h)^n \operatorname{tr} f(P(h)) = \int_{T^*M} f(p(x, \xi)) dx d\xi = O(h).$$

Proof. This is a statement for a fixed h . Since Theorem 14.9 gave $f(P) \in \Psi^{-\infty}(M)$, $f(P)$ is a finite sum of operators of the form $\varphi \gamma_1^* a^w(x, D) (\gamma_2^{-1})^* \psi$, with $\varphi, \psi \in C^\infty(M)$ and $a \in \mathcal{S}(\mathbb{R}^{2n})$. According to (C.3.4) each of these operators is in $\mathcal{L}_1$ and hence so is $f(P(h))$. $\square$

14.3.4. Weyl's Law for compact manifolds. The symbolic calculus we developed in Section 14.3.2 gives us a quick proof of the Weyl Law for Schrödinger operators.

THEOREM 14.11 (Weyl's asymptotics on compact manifolds).

(i) *For any $a < b$, we have*

$$\begin{aligned} (14.3.20) \quad & \#\{E(h) \mid a \leq E(h) \leq b\} \\ &= \frac{1}{(2\pi h)^n} (\operatorname{Vol}_{T^*M} \{a \leq |\xi|_g^2 + V(x) \leq b\} + o(1)) \end{aligned}$$

as $h \rightarrow 0$.

(ii) If $V \equiv 0$ and $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_j \rightarrow \infty$ is the complete set of eigenvalues of the Laplace–Beltrami operator $-\Delta_g$ on M , then

$$(14.3.21) \quad \#\{j \mid \lambda_j \leq r\} \sim \frac{\alpha(n)}{\text{Vol}(M)} r^{n/2}$$

as $r \rightarrow \infty$, where $\alpha(n)$ is the volume of the unit ball in $\mathbb{R}^n$.

Proof. 1. Let f_1, f_2 be two functions satisfying the assumptions of Theorem 14.9, such that for real x

$$(14.3.22) \quad f_1(x) \leq \mathbf{1}_{[a,b]}(x) \leq f_2(x),$$

where $\mathbf{1}_{[a,b]}(x)$ is the characteristic function of the interval $[a, b]$. It follows that

$$\text{tr} f_1(P) \leq \#\{E(h) \mid a \leq E(h) \leq b\} \leq \text{tr} f_2(P).$$

The operators are of trace class by Theorem 14.10.

2. Theorem C.18 now shows that for $j = 1, 2$

$$\text{tr} f_j(P) = \frac{1}{(2\pi h)^n} \left(\int_{T^*M} f_j(|\xi|_g^2 + V(x)) dx d\xi + O(h) \right).$$

We note that since $f_j(P) \in h^\infty \Psi^{-\infty}(M)$, the errors in the symbolic computations are all $O(h\langle \xi \rangle^{-\infty})$ and hence can be integrated.

3. The next step is to construct f_1^ϵ and f_2^ϵ satisfying the hypotheses of Theorem 14.9 and (14.3.22) and such that for $j = 1, 2$, we have

$$\int_{T^*M} f_j^\epsilon(|\xi|_g^2 + V(x)) dx d\xi \rightarrow \text{Vol}_{T^*M}\{a \leq |\xi|_g^2 + V(x) \leq b\},$$

as $\epsilon \rightarrow 0$. That is easily done by regularizing the characteristic function of $a \leq E \leq b$.

4. To prove (14.3.21), we take $a = 0$, $b = 1$, $h = 1/\sqrt{r}$ and apply Theorem 14.11. The eigenvalues $-\Delta_g$ are the rescaled eigenvalues of $-h^2 \Delta_g$ and the $\alpha(n)$ term comes from integrating out the ξ variables. $\square$

14.4. NOTES

Hörmander in [H3, Section 18.2] discusses the pseudodifferential calculus on manifolds for $h = 1$, and we have adapted his presentation to the semiclassical case. The notes [G-St1] of Guillemin–Sternberg provide a more detailed and functorial exposition of invariance issues.

For more about Sobolev spaces and mapping properties of pseudodifferential operators, we refer to Hörmander [H3, Chapter 18]. For these results h can be considered as a fixed parameter.

The Helffer–Sjöstrand formula is also called the Dynkin–Droste–Helffer–Sjöstrand formula; see Dimassi–Sjöstrand [D-S] for references. Davies [Da1] provides a proof of the Spectral Theorem based on this formula. Much finer asymptotic formulas for the eigenvalue counting function are known; see Dimassi–Sjöstrand [D-S], Ivrii [I], and Safarov–Vassiliev [Sa-V].

Chapter 15

QUANTUM ERGODICITY

- 15.1 Classical ergodicity
- 15.2 A weak Egorov Theorem
- 15.3 Weyl's Law generalized
- 15.4 Quantum ergodic theorems
- 15.5 Notes

In this chapter we are given a smooth real-valued potential V on a compact Riemannian manifold (M, g) . We consider the classical Hamiltonian

$$(15.0.1) \quad p(x, \xi) = |\xi|_g^2 + V(x)$$

for $(x, \xi) \in T^*M$, the Hamiltonian flow of which is denoted

$$\varphi_t = \exp(tH_p) \quad (t \in \mathbb{R}).$$

When $V \equiv 0$ and T^*M is identified with TM using (14.1.14), φ_t gives the *geodesic flow* on TM .

We devote this chapter to proving *quantum ergodicity*, meaning that ergodicity for the classical evolution $\{\varphi_t\}_{t \in \mathbb{R}}$ implies a type of equidistribution of eigenfunctions of the associated quantum operator

$$(15.0.2) \quad P(h) = -h^2 \Delta_g + V.$$

15.1. CLASSICAL ERGODICITY

We hereafter select $a < b$ and assume that

$$(15.1.1) \quad |\partial p| \geq \gamma > 0 \quad \text{on } \{a \leq p \leq b\}.$$

Then, according to the Implicit Function Theorem, for each $a \leq c \leq b$, the set

$$\Sigma_c := p^{-1}(c)$$

is a smooth, $(2n-1)$ -dimensional hypersurface in the cotangent space T^*M . We can interpret Σ_c as an energy surface.

DEFINITION. For each $c \in [a, b]$, we denote by μ_c *Liouville measure* on the hypersurface $\Sigma_c = p^{-1}(c)$ corresponding to p . This measure is characterized by the formula

$$(15.1.2) \quad \iint_{p^{-1}[a,b]} f \, dx d\xi = \int_a^b \int_{\Sigma_c} f \, d\mu_c \, dc$$

for all $a < b$ and each continuous function $f : T^*M \rightarrow \mathbb{R}$.

Theorem 2.11 shows that the measure μ_c is invariant under the flow φ_t .

EXAMPLE. If $M = \mathbb{T}^n := \mathbb{R}^n / \mathbb{Z}^n$ is the n -dimensional torus and $T^*M = \mathbb{T}^n \times \mathbb{R}^n$, we have

$$\mu_c = \frac{1}{|\partial p|} \mathcal{H}^{2n-1} \llcorner \Sigma_c,$$

$\mathcal{H}^{2n-1}$ denoting $(2n-1)$ -dimensional Hausdorff measure. Then (15.1.2) follows from the Coarea Formula; see [E-G, Section 3.4]. $\square$

DEFINITIONS. (i) Let $z = (x, \xi) \in \Sigma_c$ and $f : T^*M \rightarrow \mathbb{C}$. For $T > 0$ we define the *time average*

$$(15.1.3) \quad \langle f \rangle_T := \frac{1}{T} \int_0^T f(\varphi_t(z)) \, dt = \oint_0^T f(\varphi_t(z)) \, dt,$$

the slash through the second integral denoting an average. Note carefully that $\langle f \rangle_T$ depends upon the starting point $z \in T^*M$.

(ii) We say that the flow φ_t is *ergodic* on $p^{-1}[a, b]$ if for each $c \in [a, b]$,

$$(15.1.4) \quad \begin{cases} \text{if } E \subset \Sigma_c \text{ is flow invariant, then} \\ \text{either } \mu_c(E) = 0 \text{ or else } \mu_c(E) = \mu_c(\Sigma_c). \end{cases}$$

In other words, we are requiring that each flow invariant subset of the energy level Σ_c have either zero measure or full measure.

THEOREM 15.1 (Mean Ergodic Theorem). *Suppose the flow is ergodic on $\Sigma_c := p^{-1}(c)$. Then for each $f \in L^2(\Sigma_c, \mu_c)$ we have*

$$(15.1.5) \quad \lim_{T \rightarrow \infty} \int_{\Sigma_c} \left(\langle f \rangle_T - \int_{\Sigma_c} f d\mu_c \right)^2 d\mu_c = 0,$$

where $\int f d\mu_c := \int f d\mu_c / \mu_c(\Sigma_c)$.

REMARK. According to Birkhoff's Ergodic Theorem, in fact

$$(15.1.6) \quad \langle f \rangle_T \rightarrow \int_{\Sigma_c} f d\mu_c \quad \text{as } T \rightarrow \infty,$$

for μ_c -a.e. point m belonging to Σ_c . But we will only need the weaker statement given in Theorem 15.1. $\square$

Proof. To simplify notation, we normalize the measure μ_c so that $\mu_c(\Sigma_c) = 1$.

1. Define

$$A := \{f \in L^2(\Sigma_c, \mu_c) \mid \varphi_t^* f = f \text{ for all times } t\},$$

$$B_0 := \{H_p g \mid g \in C^\infty(\Sigma_c)\}, \quad B := \bar{B}_0 \subset L^2(\Sigma_c, \mu_c).$$

We claim that

$$(15.1.7) \quad B_0^\perp = A,$$

where $\perp$ denotes the L^2 -orthogonal complement.

To see this, first let $h \in A$ and $f = H_p g \in B_0$. Then, using invariance of μ_c under the flow, we see that

$$\begin{aligned} \int_{\Sigma_c} h \bar{f} d\mu_c &= \int_{\Sigma_c} h \overline{H_p g} d\mu_c = \partial_t \int_{\Sigma_c} h \overline{\varphi_t^* g} d\mu_c|_{t=0} \\ &= \partial_t \int_{\Sigma_c} \varphi_{-t}^* h \bar{g} d\mu_c|_{t=0} = \partial_t \int_{\Sigma_c} h \bar{g} d\mu_c|_{t=0} = 0, \end{aligned}$$

and thus $h \in B_0^\perp$.

Conversely, suppose $h \in B_0^\perp$. Then for any $g \in C^\infty$, we have

$$0 = \int_{\Sigma_c} h \overline{H_p \varphi_{-t}^* g} d\mu_c = -\partial_t \int_{\Sigma_c} h \overline{\varphi_{-t}^* g} d\mu_c = -\partial_t \int_{\Sigma_c} \varphi_t^* h \bar{g} d\mu_c.$$

Therefore for all times t and all functions $g \in C^\infty$,

$$\int_{\Sigma_c} \varphi_t^* h \bar{g} d\mu_c = \int_{\Sigma_c} h \bar{g} d\mu_c.$$

Hence $\varphi_t^* h \equiv h$, and so $h \in A$.

2. It follows from (15.1.7) that we have the orthogonal decomposition

$$L^2(\Sigma_c, \mu_c) = A \oplus B.$$

Thus if we write $f = f_A + f_B$, for $f_A \in A$, $f_B \in B$, then

$$\langle f_A \rangle_T \equiv f_A$$

for all T .

For $g \in C^\infty$ we compute

$$\begin{aligned} \int_{\Sigma_c} |\langle H_p g \rangle_T|^2 d\mu_c &= \frac{1}{T^2} \int_{\Sigma_c} \left| \int_0^T (d/dt) \varphi_t^* g dt \right|^2 d\mu_c \\ &= \frac{1}{T^2} \int_{\Sigma_c} |\varphi_T^* g - g|^2 d\mu_c \\ &\leq \frac{4}{T^2} \int_{\Sigma_c} |g|^2 d\mu_c \rightarrow 0, \end{aligned}$$

as $T \rightarrow \infty$. Hence for $f_B \in B := \overline{B}_0$, we have $\langle f_B \rangle_T \rightarrow 0$ in $L^2(\Sigma_c, d\mu_c)$. It follows that

$$\langle f \rangle_T \rightarrow f_A \quad \text{in } L^2(\Sigma_c, d\mu_c).$$

3. The ergodicity hypothesis is equivalent to saying that A consists of constant functions. Indeed, for any $h \in A$, the set $h^{-1}([\alpha, \infty))$ is invariant under the flow and hence has either full measure or measure zero. Since the functions in $L^2(\Sigma_c, d\mu_c)$ are defined up to sets of measure zero, h is equivalent to a constant function.

Finally, observe that the orthogonal projection $f \mapsto f_A$ is just the space average with respect to μ_c . This proves (15.1.5). $\square$

15.2. A WEAK EGOROV THEOREM

We next estimate the difference between the classical and quantum evolutions governed by our symbol $p(x, \xi) = |\xi|_g^2 + V(x)$. For this, we present a simple weak variant of Egorov's Theorem (Theorem 11.12) in the setting of compact manifolds.

NOTATION. (i) We write

$$(15.2.1) \quad F(t) = e^{-itP(h)/h} \quad (t \in \mathbb{R})$$

for the unitary group on $L^2(M)$ generated by the selfadjoint operator $P(h)$. See Theorems 14.7 and C.13.

(ii) If A is a pseudodifferential operator in $\Psi^{-\infty}(M)$, we define its quantum evolution as

$$(15.2.2) \quad A(t) := F^{-1}(t) A F(t) \quad (t \in \mathbb{R}).$$

This, as mentioned earlier, accords with the Heisenberg picture of quantum mechanics.

THEOREM 15.2 (Weak form of Egorov's Theorem). *Fix a time $T > 0$ and define*

$$(15.2.3) \quad \tilde{A}(t) := a_t^w(x, hD) \quad (0 \leq t \leq T),$$

where

$$(15.2.4) \quad a_t(x, \xi) := a(\varphi_t(x, \xi)), \quad A = a^w(x, hD), \quad a \in S^{-\infty}(T^*M).$$

Then

$$(15.2.5) \quad \|A(t) - \tilde{A}(t)\|_{L^2 \rightarrow L^2} = O(h) \quad \text{uniformly for } 0 \leq t \leq T.$$

We are using the notation of Theorem 14.1.

REMARK. We need to assume that $a \in S^{-\infty}(T^*M)$ to guarantee that a_t is in the same symbol class. If $a \in S(T^*M)$, then the symbol class is not preserved by φ_t^* , due to the fact that the flow of p is faster at higher frequencies. For instance in the case of the torus and $p(x, \xi) = |\xi|^2$, $\varphi_t(x, \xi) = (x + 2t\xi, \xi)$, modulo $\mathbb{Z}^n$ in the first variable. Then for $a(x, \xi) \in S(T^*\mathbb{R}^n)$ periodic in the first variable, $a(x + t\xi, \xi) \notin S(T^*\mathbb{R}^n)$; see the definition (9.3.2).

Proof. We have

$$\partial_t a_t = \{p, a_t\},$$

where $\{p, a_t\} = H_p a_t$ is the Poisson bracket on T^*M .

If σ denotes the symbol of a pseudodifferential operator as defined in Section 14.2.3,

$$\sigma\left(\frac{i}{h}[P(h), B]\right) = \{p, \sigma(B)\}.$$

This is checked in local coordinates using Theorem 9.5 and then follows from the invariance of both sides.

Theorem 14.1 shows that

$$(15.2.6) \quad \partial_t \tilde{A}(t) = \frac{i}{h}[P(h), \tilde{A}(t)] + E(t), \quad E(t) \in h\Psi^{-\infty}(M),$$

and by Theorem 14.2, $\|E(t)\|_{L^2 \rightarrow L^2} = O(h)$. Hence

$$\begin{aligned} & \partial_t \left(e^{-\frac{itP(h)}{h}} \tilde{A}(t) e^{\frac{itP(h)}{h}} \right) \\ &= e^{-\frac{itP(h)}{h}} \left(\partial_t \tilde{A}(t) - \frac{i}{h}[P(h), \tilde{A}(t)] \right) e^{\frac{itP(h)}{h}} \\ &= e^{-\frac{itP(h)}{h}} \left(\frac{i}{h}[P(h), \tilde{A}(t)] + E(t) - \frac{i}{h}[P(h), \tilde{A}(t)] \right) e^{\frac{itP(h)}{h}} \\ &= e^{-\frac{itP(h)}{h}} E(t) e^{\frac{itP(h)}{h}} = O_{L^2 \rightarrow L^2}(h). \end{aligned}$$

Integrating, we deduce that

$$\|e^{-\frac{itP(h)}{h}} \tilde{A}(t) e^{\frac{itP(h)}{h}} - A\|_{L^2 \rightarrow L^2} = O(h);$$

and so

$$\|\tilde{A}(t) - A(t)\|_{L^2 \rightarrow L^2} = \|\tilde{A}(t) - e^{\frac{itP(h)}{h}} A e^{-\frac{itP(h)}{h}}\|_{L^2 \rightarrow L^2} = O(h),$$

uniformly for $0 \leq t \leq T$. $\square$

15.3. WEYL'S LAW GENERALIZED

NOTATION. We hereafter turn our attention to the eigenvalue problems

$$(15.3.1) \quad P(h)u_j(h) = E_j(h)u_j(h) \quad (j = 1, \dots).$$

To simplify notation, we write $u_j = u_j(h)$ and $E_j = E_j(h)$. We assume as well the normalization

$$(15.3.2) \quad \|u_j\|_{L^2(M)} = 1 \quad (j = 1, \dots).$$

We have

$$(15.3.3) \quad e^{-itP(h)/h} u_j(h) = e^{-itE_j(h)/h} u_j(h) \quad (t \in \mathbb{R})$$

for $j = 1, \dots$.

The following result generalizes Theorem 14.11, showing that we can localize the asymptotics using a quantum observable.

THEOREM 15.3 (Weyl's Theorem generalized). *Let $B \in \Psi(M)$. Then*

$$(15.3.4) \quad (2\pi h)^n \sum_{a \leq E_j \leq b} \langle B u_j, u_j \rangle \rightarrow \iint_{\{a \leq p \leq b\}} \sigma(B) dx d\xi$$

as $h \rightarrow 0$.

REMARK. If $B = I$, and so $\sigma(B) \equiv 1$, (15.3.4) reads

$$(2\pi h)^n \#\{a \leq E_j \leq b\} \rightarrow \text{Vol}(\{a \leq p \leq b\}).$$

This is the usual form of Weyl's Law, Theorem 14.11. $\square$

The proof uses the functional calculus developed in Section 14.3.2.

Proof. 1. As shown in the proof of Theorem 14.10, $A \in \Psi^{-\infty}(M)$ is of trace class. For $a \in \mathcal{S}(\mathbb{R}^{2n})$, the formula for the Schwartz kernel (4.5.1) and Theorem C.18 show that

$$\text{tr } a^w(x, hD) = \frac{1}{(2\pi h)^n} \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} a(x, \xi) dx d\xi.$$

This and the invariance of the principal symbol of A give

$$(15.3.5) \quad \operatorname{tr} A = \frac{1}{(2\pi h)^n} \left(\iint_{T^*M} \sigma(A) dx d\xi + O(h) \right).$$

2. Fix a small number $\epsilon > 0$ and construct $f_\epsilon, g_\epsilon \in C_c^\infty(\mathbb{R})$ satisfying $0 \leq f_\epsilon, g_\epsilon \leq 1$ and

$$\begin{aligned} \operatorname{spt} f_\epsilon &\subset [a + \epsilon, b - \epsilon], & f_\epsilon &\equiv 1 \text{ on } [a + 2\epsilon, b - 2\epsilon], \\ \operatorname{spt} g_\epsilon &\subset [a - 2\epsilon, b + 2\epsilon], & g_\epsilon &\equiv 1 \text{ on } [a - \epsilon, b + \epsilon]. \end{aligned}$$

Theorem 14.9 then shows that $f_\epsilon(P) \in \Psi^{-\infty}(M)$ and that

$$(15.3.6) \quad \sigma(f_\epsilon(P))(x, \xi) = f_\epsilon(p(x, \xi)).$$

Similar statements hold for g_ϵ .

Define

$$\Pi := \text{projection onto the span of } \{u_j \mid a \leq E_j \leq b\},$$

so that

$$(15.3.7) \quad f_\epsilon(P)\Pi = f_\epsilon(P), \quad g_\epsilon(P)\Pi = \Pi.$$

3. We now write

$$\begin{aligned} (15.3.8) \quad \sum_{a \leq E_j \leq b} \langle Bu_j, u_j \rangle &= \operatorname{tr}(\Pi B \Pi) \\ &= \operatorname{tr}(f_\epsilon(P)B) + \operatorname{tr}(\Pi g_\epsilon(P)(1 - f_\epsilon(P))B). \end{aligned}$$

The operator $g_\epsilon(P)(1 - f_\epsilon(P))$ is selfadjoint and, as $0 \leq g_\epsilon, f_\epsilon \leq 1$,

$$\sum_{j=1}^{\infty} s_j(g_\epsilon(P)(1 - f_\epsilon(P))) = \sum_{j=1}^{\infty} g_\epsilon(E_j(h))(1 - f_\epsilon(E_j(h))).$$

Here, s_j denote singular values, see Section C.3, which due to selfadjointness and nonnegativity are equal to eigenvalues.

The definition of the trace class norm (C.3.3) and Theorem 14.11 imply that

$$\begin{aligned} \|g_\epsilon(P)(1 - f_\epsilon(P))\|_{\mathcal{L}_1} &\leq \#\{a - 2\epsilon \leq E_j(h) \leq a + 2\epsilon\} \\ &\quad + \#\{b - 2\epsilon \leq E_j(h) \leq b + 2\epsilon\} \\ &\leq C(\alpha(\epsilon) + \beta(h, \epsilon))h^{-n}, \end{aligned}$$

where

$$\lim_{h \rightarrow 0} \beta(h, \epsilon) = 0$$

and

$$\alpha(\epsilon) := \operatorname{Vol}(\{a - 2\epsilon \leq p \leq a + 2\epsilon\} \cup \{b - 2\epsilon \leq p \leq b + 2\epsilon\})$$

satisfies

$$\lim_{\epsilon \rightarrow 0} \alpha(\epsilon) = 0.$$

We use (15.3.8) and (15.3.5) with $A = f_\epsilon(P)$ to deduce that

$$\begin{aligned} (2\pi h)^{n\text{tr}}(\Pi B \Pi) &= (2\pi h)^{n\text{tr}}(f_\epsilon(P)B) + O(h^n \|B\| \|g_\epsilon(P)(1 - f_\epsilon(P))\|_{\mathcal{L}_1}) \\ &= \iint_{T^*M} f_\epsilon(p) \sigma(B) dx d\xi + O_\epsilon(h) + O(\alpha(\epsilon) + \beta(h, \epsilon)). \end{aligned}$$

Here $O_\epsilon(h)$ indicates that the bound depends on ϵ , since the error term in (15.3.5) depends on A .

4. We conclude that

$$\begin{aligned} \limsup_{h \rightarrow 0} (2\pi h)^n \sum_{a \leq E_j \leq b} \langle B u_j, u_j \rangle &= \limsup_{h \rightarrow 0} (2\pi h)^{n\text{tr}}(\Pi B \Pi) \\ &= \iint_{T^*M} f_\epsilon(p) \sigma(B) dx d\xi + O(\alpha(\epsilon)), \end{aligned}$$

and similarly for $\liminf_{h \rightarrow 0}$. Since the left-hand side does not depend on ϵ and since

$$\lim_{\epsilon \rightarrow 0} \iint_{T^*M} f_\epsilon(p) \sigma(B) dx d\xi + O(\alpha(\epsilon)) = \iint_{\{a \leq p \leq b\}} \sigma(B) dx d\xi,$$

the theorem follows. $\square$

15.4. QUANTUM ERGODIC THEOREMS

Assume now that $A \in \Psi(M)$ has the symbol $\sigma(A)$, satisfying the condition that

$$(15.4.1) \quad \alpha := \int_{\Sigma_c} \sigma(A) d\mu_c \text{ is the same for all } c \in [a, b],$$

the slash through the integral denoting the average. In other words, we are requiring that the averages of the symbol of A over each level surface $p^{-1}(c)$ are equal.

REMARK: Symbol averages. The condition (15.4.1) on averages is easy to obtain.

Since we assume that $|\partial p| > \gamma$ on $p^{-1}([a, b])$, we also have $|\partial p| > \gamma/2$ on $p^{-1}([a - \delta, b + \delta])$ for some small $\delta > 0$. We define

$$\mathcal{T}b(x, \xi) := b(x, \xi) - \int_{\Sigma_{p(x, \xi)}} b d\mu_{p(x, \xi)}.$$

Then for $c \in [a, b]$

$$(15.4.2) \quad \int_{p^{-1}(c)} \mathcal{T}b d\mu_c = 0.$$

If $\chi \in C_c^\infty(p^{-1}((a-\delta, b+\delta)))$ is equal to 1 near $p^{-1}([a, b])$, we can define $\mathcal{T}$ by applying the above definition to χb . This defines $\mathcal{T} : C^\infty(T^*M) \rightarrow C_c^\infty(T^*M)$ such that

$$\mathcal{T}^2 = \mathcal{T}$$

and (15.4.2) holds for any b . This means that $\mathcal{T}$ is a projection onto the space of functions satisfying (15.4.1) with $\alpha = 0$.

To obtain a large class of operators A satisfying (15.4.1) with an arbitrary α , take any $B \in \Psi(M)$, $\chi \in S(T^*M)$ equal to 1 near $p^{-1}([a, b])$, and define

$$A := \text{Op}(\mathcal{T}\sigma(B)) + \alpha \text{Op}(\chi) + \text{Op}((1-\chi)\sigma(B)). \quad \square$$

We now establish our first connection between the classical ergodicity of the flow φ_t and the distribution of the eigenfunctions satisfying (15.3.1):

THEOREM 15.4 (Quantum ergodicity 1). *Assume the ergodic condition (15.1.4) and assume that $A \in \Psi(M)$ satisfies the condition (15.4.1). Then*

$$(15.4.3) \quad (2\pi h)^n \sum_{a \leq E_j \leq b} \left| \langle Au_j, u_j \rangle - \int_{\{a \leq p \leq b\}} \sigma(A) dx d\xi \right|^2 \rightarrow 0$$

as $h \rightarrow 0$.

REMARK. The condition that averages of $\sigma(A)$ are the same on different energy levels is essential for this formulation of the theorem. A result without that condition requires a finer energy localization and states that

$$(2\pi h)^{n-1} \sum_{a \leq E_j \leq a+h} \left| \langle Au_j, u_j \rangle - \int_{p^{-1}(a)} \sigma(A) d\mu_a \right|^2 \rightarrow 0;$$

see [H-M-R], [Dy-Gu], and the notes at the end of this chapter. In the homogeneous case of $-h^2\Delta_g$ the energy levels are all equivalent by rescaling and our formulation is sufficient for obtaining the standard quantum ergodicity theorems of [Sh], [CdV], and [Ze1].

Proof. 1. Write

$$(15.4.4) \quad B := \chi(P)(A - \alpha I),$$

where α is defined by (15.4.1) and $\chi \in C_c^\infty(T^*M)$ is equal to 1 near $p^{-1}([a, b])$. In view of our hypothesis (15.4.1)

$$(15.4.5) \quad \int_{\Sigma_c} \sigma(B) d\mu_c = 0 \quad \text{for each } c \in [a, b].$$

Since $|\langle Cu_j, u_j \rangle|^2 \leq \langle C^*Cu_j, u_j \rangle$, Theorem 15.3 shows that

$$\begin{aligned} (2\pi h)^n \sum_{a \leq E_j \leq b} |\langle (1 - \chi(P))Au_j, u_j \rangle|^2 \\ \leq (2\pi h)^n \sum_{a \leq E_j \leq b} \langle A^*(1 - \chi(P))^2 Au_j, u_j \rangle \rightarrow 0, \quad h \rightarrow 0. \end{aligned}$$

Hence it is enough to prove the theorem for A replaced by B .

Define

$$\epsilon(h) := (2\pi h)^n \sum_{a \leq E_j \leq b} |\langle Bu_j, u_j \rangle|^2;$$

we must show that $\epsilon(h) \rightarrow 0$.

Now

$$\langle Bu_j, u_j \rangle = \langle Be^{-\frac{itE_j}{h}} u_j, e^{-\frac{itE_j}{h}} u_j \rangle = \langle Be^{-\frac{itP(h)}{h}} u_j, e^{-\frac{itP(h)}{h}} u_j \rangle$$

according to (15.3.3). Consequently

$$(15.4.6) \quad \langle Bu_j, u_j \rangle = \langle e^{\frac{itP(h)}{h}} Be^{-\frac{itP(h)}{h}} u_j, u_j \rangle = \langle B(t)u_j, u_j \rangle$$

in the notation of (15.2.2). This identity is valid for each time $t \in \mathbb{R}$. We can therefore average:

$$(15.4.7) \quad \langle Bu_j, u_j \rangle = \left\langle \int_0^T B(t) dt u_j, u_j \right\rangle = \langle \langle B \rangle_T u_j, u_j \rangle,$$

for

$$\langle B \rangle_T := \frac{1}{T} \int_0^T B(t) dt = \int_0^T B(t) dt.$$

Now since $\|u_j\|^2 = 1$, (15.4.7) implies that

$$|\langle Bu_j, u_j \rangle| = |\langle \langle B \rangle_T u_j, u_j \rangle| \leq \|\langle B \rangle_T u_j\|^2 = \langle \langle B^* \rangle_T \langle B \rangle_T u_j, u_j \rangle.$$

Hence

$$(15.4.8) \quad \epsilon(h) \leq (2\pi h)^n \sum_{a \leq E_j \leq b} |\langle \langle B^* \rangle_T \langle B \rangle_T u_j, u_j \rangle|.$$

2. Theorem 15.2 tells us that

$$\langle B \rangle_T = \langle \tilde{B} \rangle_T + O_{L^2 \rightarrow L^2}(h), \quad \langle \tilde{B} \rangle_T := \int_0^T \tilde{B}(t) dt,$$

where $\tilde{B}(t) \in \Psi(M)$, $\sigma(\tilde{B}(t)) = \varphi_t^* \sigma(B)$, and

$$\sigma(\langle \tilde{B} \rangle_T) = \int_0^T \sigma(B) \circ \varphi_t dt = \langle \sigma(B) \rangle_T.$$

We note that the error $O_{L^2 \rightarrow L^2}(h)$ depends on T .

This means that modulo $O_T(h)$ errors we can replace $e^{itP(h)/h} B e^{-itP(h)/h}$ by $\tilde{B}(t)$. Theorem 15.3 and (15.4.8) show that

$$\begin{aligned}
 (15.4.9) \quad \limsup_{h \rightarrow 0} \epsilon(h) &\leq \limsup_{h \rightarrow 0} \left((2\pi h)^n \sum_{a \leq E_j \leq b} \langle \langle \tilde{B}^* \rangle_T \langle \tilde{B} \rangle_T u_j, u_j \rangle + O_T(h) \right) \\
 &= \iint_{\{a \leq p \leq b\}} \sigma(\langle \tilde{B}^* \rangle_T \langle \tilde{B} \rangle_T) dx d\xi \\
 &= \iint_{\{a \leq p \leq b\}} |\sigma(\langle B \rangle_T)|^2 dx d\xi,
 \end{aligned}$$

since the symbol map is multiplicative and the symbol of an adjoint is given by the complex conjugate.

3. We can now apply Theorem 15.1 with $f = \sigma(B)$, to conclude that

$$\int_{p^{-1}[a,b]} |\langle \sigma(B) \rangle_T|^2 dx d\xi \rightarrow 0,$$

as $T \rightarrow \infty$. Since the left-hand side of (15.4.9) is independent of T , this calculation shows that the limit must in fact be zero. $\square$

The pointwise limit along a sequence of density one is obtained using a density argument:

THEOREM 15.5 (Quantum ergodicity 2). *Assume the ergodic condition (15.1.4).*

Then there exists a family of subsets $\Lambda(h) \subset \{a \leq E_j \leq b\}$ such that

$$(15.4.10) \quad \lim_{h \rightarrow 0} \frac{\#\Lambda(h)}{\#\{a \leq E_j \leq b\}} = 1;$$

and for each $A \in \Psi(M)$ satisfying (15.4.1), we have

$$(15.4.11) \quad \langle A u_j, u_j \rangle \rightarrow \int_{\{a \leq p \leq b\}} \sigma(A) dx d\xi \quad \text{as } h \rightarrow 0$$

for $E_j \in \Lambda(h)$.

Proof. 1. We first show that assertion (15.4.3) implies (15.4.10) with $\Lambda(h)$ depending on A . For this, let B be given by (15.4.4) so that again

$$\int_{\{a \leq p \leq b\}} \sigma(B) dx d\xi = 0.$$

According to (15.4.3),

$$(2\pi h)^n \sum_{a \leq E_j \leq b} |\langle Bu_j, u_j \rangle|^2 =: \epsilon(h) \rightarrow 0.$$

Define

$$\Gamma(h) := \{a \leq E_j \leq b \mid |\langle Bu_j, u_j \rangle|^2 \geq \epsilon(h)^{1/2}\},$$

so that

$$(2\pi h)^n \#\Gamma(h) \leq \epsilon(h)^{1/2}.$$

Next, write

$$\Lambda(h) := \{a \leq E_j \leq b\} \setminus \Gamma(h).$$

Then if $E_j \in \Lambda(h)$,

$$|\langle Bu_j, u_j \rangle| \leq \epsilon(h)^{1/4};$$

and so

$$|\langle Au_j, u_j \rangle - \alpha| \leq \epsilon(h)^{1/4}.$$

Also,

$$\frac{\#\Lambda(h)}{\#\{a \leq E_j \leq b\}} = 1 - \frac{\#\Gamma(h)}{\#\{a \leq E_j \leq b\}}.$$

But according to Weyl's Law,

$$\frac{\#\Gamma(h)}{\#\{a \leq E_j \leq b\}} = \frac{(2\pi h)^n \#\Gamma(h)}{\text{Vol}(\{a \leq p \leq b\}) + o(1)} \leq C\epsilon(h)^{1/2} \rightarrow 0.$$

2. Now let $\{A_k\}_{k=1}^\infty \subset \Psi$ be a countable family of pseudodifferential operators each satisfying (15.4.1) for some α_k .

For each k we have $\Lambda_k(h) \subset \{a \leq E_j \leq b\}$ such that (15.4.10) and (15.4.11) hold for A_k and $\Lambda_k(h)$. Since (15.4.10) for $\Lambda(h) = \Lambda_k(h)$ and $\Lambda(h) = \Lambda_l(h)$ implies (15.4.10) for $\Lambda(h) = \Lambda_k(h) \cap \Lambda_l(h)$ (immediate by considering the zero density complements), we can assume that

$$(15.4.12) \quad \Lambda_{k+1}(h) \subset \Lambda_k(h).$$

For each k , let $h(k) > 0$ be small enough so that

$$\frac{\#\Lambda_k(h)}{\#\{a \leq E_j \leq b\}} \geq 1 - \frac{1}{k}, \quad \text{for } 0 < h < h(k),$$

which is possible in view of (15.4.10). We can take

$$h(k) > h(k+1) \rightarrow 0, \quad k \rightarrow \infty,$$

and define

$$\Lambda_\infty(h) := \Lambda_k(h), \quad h(k+1) \leq h < h(k).$$

We then have

$$\frac{\#\Lambda_\infty(h)}{\#\{a \leq E_j \leq b\}} \geq 1 - \frac{1}{k}, \quad \text{for } 0 < h < h(k),$$

and since $h(k) \rightarrow 0$,

$$\lim_{h \rightarrow 0} \frac{\#\Lambda_\infty(h)}{\#\{a \leq E_j \leq b\}} = 1.$$

3. We now claim that for any A_k

$$(15.4.13) \quad \langle A_k u_j, u_j \rangle \rightarrow \int_{\{a \leq p \leq b\}} \sigma(A_k) dx d\xi, \quad h \rightarrow 0, \quad E_j \in \Lambda_\infty(h).$$

This is true since (15.4.12) and the definition of Λ_∞ show that $\Lambda_\infty(h) \subset \Lambda_k(h)$ for $h < h(k)$ and since we assumed that (15.4.13) held for $E_j \in \Lambda_k(h)$.

4. In the last step of the proof we will choose a set $\{A_k\}_{k=1}^\infty$ dense in

$$(15.4.14) \quad \mathcal{P} := \{A \in \Psi^{-\infty}(M) \mid \int_{\Sigma_c} \sigma(A) d\mu_c = \alpha, \text{ independently of } c \in [a, b]\}.$$

The density is meant in the following sense: given $A \in \Psi$,

for any $\epsilon > 0$ there exist k and h_0 such that for $0 < h < h_0$,

$$(15.4.15) \quad \int_{\{a \leq p \leq b\}} |\sigma(A_k - A)| dx d\xi < \epsilon \quad \text{and} \quad \|A_k - A\|_{L^2 \rightarrow L^2} < \epsilon.$$

This and (15.4.13) show that for $E_j \in \Lambda_\infty(h)$,

$$\begin{aligned} \left| \limsup_{h \rightarrow 0} \left(\langle A u_j, u_j \rangle - \int_{\{a \leq p \leq b\}} \sigma(A) dx d\xi \right) \right| &< 2\epsilon, \\ \left| \liminf_{h \rightarrow 0} \left(\langle A u_j, u_j \rangle - \int_{\{a \leq p \leq b\}} \sigma(A) dx d\xi \right) \right| &< 2\epsilon, \end{aligned}$$

proving (15.4.11) for $A \in \Psi^{-\infty}(M)$. As in the proof of Theorem 15.3 we see that we only need to consider that case as A can be replaced by $f(P)A$, where $f \in C_c^\infty(\mathbb{R})$ is equal to 1 near $[a, b]$.

5. It remains to find $\{A_k\}_{k=1}^\infty \subset \mathcal{P}$, where $\mathcal{P}$ is given by (15.4.14), so that (15.4.15) holds.

Theorems 5.1 and 14.1 show that

$$\|A - A_k\|_{L^2 \rightarrow L^2} \leq \|\sigma(A) - \sigma(A_k)\|_{L^\infty(T^*M)} + C(a, a_k)h^{\frac{1}{2}},$$

and we also immediately have

$$\int_{a \leq p(x, \xi) \leq b} |\sigma(A - A_k)| dx d\xi \leq C \|\sigma(A) - \sigma(A_k)\|_{L^\infty(T^*M)}.$$

Thus we only need to find the set $\{a_k\}_{k=1}^\infty \subset S^{-\infty}(T^*M)$ satisfying

$$(15.4.16) \quad \int_{\Sigma_c} a_k(x, \xi) d\mu_c = \alpha,$$

independently of $c \in [a, b]$, such that for every $a \in S^{-\infty}(T^*M)$ and every $\epsilon > 0$ there exists k such that

$$\|a - a_k\|_{L^\infty(T^*M)} < \epsilon.$$

Hence we need to find $a_k \in C_c^\infty(T^*M)$ satisfying (15.4.16) and dense in the space $C_0(T^*M)$ of continuous functions vanishing at infinity and satisfying (15.4.16). We can assume that $\alpha = 0$ since adding $\alpha\chi$, $\chi \in C_c^\infty(\mathbb{R}^{2n})$, $\chi = 1$ near $p^{-1}([a, b])$, to a produces averages α . We can then take $\alpha \in \mathbb{Q}$.

Since $\mathcal{T}$ in (15.4.2) is continuous on $C_0(\mathbb{R}^{2n})$, we only need to select a dense set of $b_k \in C_c^\infty(\mathbb{R}^{2n})$ in $C_0(\mathbb{R}^{2n})$, as then we can take $a_k = \mathcal{T} b_k$. $\square$

REMARK. It is important that we consider $A \in \Psi^{-\infty}(M)$ in the proof above, even though the theorem applies to $A \in \Psi(M)$. This is because the space $\Psi(M)$ is not separable, just as the space of symbols S is not separable. $\square$

APPLICATION. The simplest and most striking application concerns the complete set of eigenfunctions of the Laplace–Beltrami operator on a compact Riemannian manifold:

$$-\Delta_g u_j = \lambda_j u_j \quad (j = 1, \dots),$$

normalized so that

$$\|u_j\|_{L^2(M)} = 1.$$

THEOREM 15.6 (Equidistribution of eigenfunctions). *Suppose that (M, g) is a compact Riemannian manifold with an ergodic geodesic flow.*

Then there exists a sequence $j_k \rightarrow \infty$ of density one,

$$\lim_{m \rightarrow \infty} \frac{\#\{k \mid j_k \leq m\}}{m} = 1,$$

such that

$$(15.4.17) \quad \int_M |u_{j_k}|^2 f \, dx \rightarrow \int_M f \, dx$$

for each $f \in C(M)$.

15.5. NOTES

The Quantum Ergodicity Theorem (Theorem 15.5) is from a 1974 paper of Shnirelman [Sh]. Zelditch [Ze1] gave the first complete proof. We have followed his more recent proof in [Z-Z] for Theorem 15.4 and the argument from Colin de Verdière [CdV] for Theorem 15.5; see also [Dy-Zw]. A refined version of semiclassical ergodicity can be found in Helffer–Martinez–Robert [H-M-R]. See also Dyatlov–Guillarmou [Dy-Gu, Appendix D] for a simple direct proof and a better energy localization.

Hopf showed that the geodesic flow on a negatively curved surface is ergodic, and Anosov and Sinai proved this for manifolds of negative sectional curvature in higher dimensions; see Brin [Ba, Appendix] for a self-contained presentation. This provides a large class of examples to which Theorem 15.6 applies. More examples are provided by metrics with Anosov geodesic flows. Since these are structurally stable, ergodicity is also valid for the flow of the Hamiltonian $p = |\xi|_g^2 + V(x)$ on $\{p = 1\}$ provided that V is small. Examples of Hamiltonians $p = |\xi|^2 + V(x)$ on $\mathbb{T}^2$ for which the flow is ergodic have been constructed recently by Donnay–Liverani [DoLi].

Part 5

APPENDICES

Appendix A

NOTATION

A.1. BASIC NOTATION

$\mathbb{Z}$ = integers; $\mathbb{N}$ = nonnegative integers.

$\mathbb{R}$ = real line; $\mathbb{R}_+ = (0, \infty)$.

$\mathbb{R}^n$ = n -dimensional Euclidean space.

x, y denote typical points in $\mathbb{R}^n$: $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n)$.

$\mathbb{R}^{2n} = \mathbb{R}^n \times \mathbb{R}^n$.

$z = (x, \xi), w = (y, \eta)$ denote typical points in $\mathbb{R}^n \times \mathbb{R}^n$:

$z = (x_1, \dots, x_n, \xi_1, \dots, \xi_n)$, $w = (y_1, \dots, y_n, \eta_1, \dots, \eta_n)$.

We usually write $\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ to denote a smooth diffeomorphism and $\kappa : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ to denote a smooth symplectomorphism.

$\Lambda_\kappa = \{(x, y, \xi, -\eta) \mid (x, \xi) = \kappa(y, \eta), (y, \eta) \in \mathbb{R}^{2n}\}$ = twisted graph of κ .

$\mathbb{T}^n$ = n -dimensional flat torus = $\mathbb{R}^n / \mathbb{Z}^n$.

$\mathbb{C}$ = complex plane; $\mathbb{C}^n$ = n -dimensional complex space.

$GL(n, \mathbb{C})$ = the group of invertible linear transformations on $\mathbb{C}^n$.

$\langle x, y \rangle = \sum_{i=1}^n x_i \bar{y}_i$, the inner product on $\mathbb{C}^n$, except in Chapter 13 where we use $\langle z, w \rangle := \sum_{j=1}^n z_j w_j$, which is the complexification of the inner product on $\mathbb{R}^n$.

$$|x| = \langle x, x \rangle^{1/2}; \quad \langle x \rangle = (1 + |x|^2)^{1/2}.$$

$\text{dist}(x, y)$, $\text{dist}(A, B)$ denote distances between points and sets, respectively.

$$\sigma(z, w) = \langle Jz, w \rangle = \text{symplectic inner product on } \mathbb{R}^{2n}.$$

• Sets:

$\#S$ = cardinality of the set S .

$\mathcal{L}^n$ = n -dimensional Lebesgue measure.

$|E| = \mathcal{L}^n(E)$ = Lebesgue measure of the measurable set $E \subset \mathbb{R}^n$.

$\mathcal{H}^s$ = s -dimensional Hausdorff measure.

$\mathbf{1}_E$ = indicator function of the set E .

U, V usually denote open subsets.

$U \subset\subset V$ means $\bar{U}$ is a compact subset of V .

• Matrices:

$\mathbb{M}^{m \times n}$ = $(m \times n)$ -matrices.

$\mathbb{S}^{n \times n}$ = $n \times n$ real symmetric matrices.

tr = trace; $\det$ = determinant.

A^T = transpose of the matrix A .

$$|A|^2 = \text{tr}(A^T A).$$

$[A, B] = AB - BA$ = commutator of matrices A and B .

$$\exp A = e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

$F^{-1}AF$ = conjugation of A by F .

$\text{sgn } Q$ = signature of the symmetric matrix Q .

I denotes both the identity matrix and the identity mapping.

$$J = \begin{pmatrix} O & I \\ -I & O \end{pmatrix}.$$

A.2. FUNCTIONS, DIFFERENTIATION

The support of a function is denoted “spt”, and a subscript “ c ” on a space of functions means those with compact support.

- Composition:

$$f \circ g = \text{composition of the functions } f \text{ and } g.$$

- A slash through an integral sign denotes an average; for example,

$$\oint_0^T f \, dt := \frac{1}{T} \int_0^T f \, dt.$$

- Partial derivatives:

$$\partial_{x_j} := \frac{\partial}{\partial x_j}, \quad D_{x_j} := \frac{1}{i} \frac{\partial}{\partial x_j}, \quad \partial_t = \frac{\partial}{\partial t}.$$

We will also write

$$\partial_t = \frac{d}{dt}$$

applied to functions of the single variable t .

- Multiindex notation: A multiindex is a vector $\alpha = (\alpha_1, \dots, \alpha_n)$, the entries of which are nonnegative integers: $\alpha \in \mathbb{N}^n$. The size of α is

$$|\alpha| := \alpha_1 + \dots + \alpha_n.$$

We define for $x \in \mathbb{R}^n$

$$x^\alpha := x_1^{\alpha_1} \dots x_n^{\alpha_n},$$

where $x = (x_1, \dots, x_n)$.

Also

$$\partial^\alpha := \partial_{x_1}^{\alpha_1} \dots \partial_{x_n}^{\alpha_n}$$

and

$$D^\alpha := \frac{1}{i^{|\alpha|}} \partial_{x_1}^{\alpha_1} \dots \partial_{x_n}^{\alpha_n}.$$

(WARNING: Our use of the symbols “ D ” and “ D^α ” differs from that in the textbook [E].)

- If $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$, then we write

$$\partial\varphi := (\varphi_{x_1}, \dots, \varphi_{x_n}) = \text{gradient}$$

and

$$\partial^2\varphi := \begin{pmatrix} \varphi_{x_1x_1} & \dots & \varphi_{x_1x_n} \\ & \ddots & \\ \varphi_{x_nx_1} & \dots & \varphi_{x_nx_n} \end{pmatrix} = \text{Hessian matrix}.$$

Also

$$D\varphi := \frac{1}{i} \partial \varphi.$$

- If φ depends on both of the variables $(x, y) \in \mathbb{R}^n \times \mathbb{R}^n$, we put

$$\partial_x^2 \varphi := \begin{pmatrix} \varphi_{x_1 x_1} & \cdots & \varphi_{x_1 x_n} \\ & \ddots & \\ \varphi_{x_n x_1} & \cdots & \varphi_{x_n x_n} \end{pmatrix}, \quad \partial_y^2 \varphi := \begin{pmatrix} \varphi_{y_1 y_1} & \cdots & \varphi_{y_1 y_n} \\ & \ddots & \\ \varphi_{y_n y_1} & \cdots & \varphi_{y_n y_n} \end{pmatrix},$$

and

$$\partial_{x,y}^2 \varphi := \begin{pmatrix} \varphi_{x_1 y_1} & \cdots & \varphi_{x_1 y_n} \\ & \ddots & \\ \varphi_{x_n y_1} & \cdots & \varphi_{x_n y_n} \end{pmatrix}.$$

- Jacobians: Let

$$x \mapsto y = y(x)$$

be a diffeomorphism, $y = (y^1, \dots, y^n)$. The Jacobian matrix is

$$\partial y = \partial_x y := \begin{pmatrix} \frac{\partial y^1}{\partial x_1} & \cdots & \frac{\partial y^1}{\partial x_n} \\ & \ddots & \\ \frac{\partial y^n}{\partial x_1} & \cdots & \frac{\partial y^n}{\partial x_n} \end{pmatrix}_{n \times n}.$$

- Differentiation of determinants: Suppose $A(t)$ is a function from $\mathbb{R}$ to invertible $m \times m$ matrices. Then

$$(A.2.1) \quad \partial_t \det A(t) = \operatorname{tr}(A(t)^{-1} \partial_t A(t)) \det A(t).$$

- Poisson bracket: If $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ are C^1 functions,

$$\{f, g\} := \langle \partial_\xi f, \partial_x g \rangle - \langle \partial_x f, \partial_\xi g \rangle = \sum_{j=1}^n \frac{\partial f}{\partial \xi_j} \frac{\partial g}{\partial x_j} - \frac{\partial f}{\partial x_j} \frac{\partial g}{\partial \xi_j}.$$

- The *Schwartz space* is

$$\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$$

$$:= \{\varphi \in C^\infty(\mathbb{R}^n) \mid \sup_{\mathbb{R}^n} |x^\alpha \partial^\beta \varphi| < \infty \text{ for all multiindices } \alpha, \beta\}.$$

We say

$$\varphi_j \rightarrow \varphi \quad \text{in } \mathcal{S}$$

provided

$$\sup_{\mathbb{R}^n} |x^\alpha D^\beta (\varphi_j - \varphi)| \rightarrow 0$$

for all multiindices α, β .

We write $\mathcal{S}' = \mathcal{S}'(\mathbb{R}^n)$ for the space of *tempered distributions*, which is the dual of $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$. That is, $u \in \mathcal{S}'$ provided $u : \mathcal{S} \rightarrow \mathbb{C}$ is linear and $\varphi_j \rightarrow \varphi$ in $\mathcal{S}$ implies $u(\varphi_j) \rightarrow u(\varphi)$.

We say

$$u_j \rightarrow u \quad \text{in } \mathcal{S}'$$

provided

$$u_j(\varphi) \rightarrow u(\varphi) \quad \text{for all } \varphi \in \mathcal{S}.$$

- Support of a function or distribution is denoted by $\text{spt } u$:

$$\text{spt } u := \mathbb{R}^n \setminus \bigcup \{U : U \text{ is open and } u|_U \equiv 0\}.$$

A.3. OPERATORS

Multiplication operator: $M_\lambda f(x) = \lambda f(x)$.

Translation operator: $T_\xi f(x) = f(x - \xi)$.

Reflection operator: $Rf(x) := f(-x)$.

A^* = adjoint of the operator A .

$[A, B] = AB - BA$ = commutator of operators A and B .

$\text{ad}_A B = [A, B]$.

$\sigma(A)$ = symbol of the pseudodifferential operator A .

$\text{Spec}(A)$ = spectrum of A .

$\text{tr}(A)$ = trace of A .

- If $A : X \rightarrow Y$ is a bounded linear operator, we define the operator norm

$$\|A\| := \sup\{\|Au\|_Y \mid \|u\|_X \leq 1\}.$$

We will usually write this norm as

$$\|A\|_{X \rightarrow Y},$$

to emphasize the spaces between which A maps.

- $L(X, Y)$ = the space of bounded linear operators from X to Y
 $L(X) = L(X, X)$.

A.4. ESTIMATES

A.4.1. Use of constants.

- We use “ C ” to denote a general positive constant appearing to the right of the inequality sign $\leq$ in various estimates throughout the text. The letter C will in general stand for different constants in different lines of the calculations.

- We will also employ a lower case “ c ” to denote a *positive* constant occurring in various estimates to the left of the inequality sign $\leq$ or to the right of $\geq$.

A.4.2. Order estimates.

- We write

$$f = O(h^\infty) \quad \text{as } h \rightarrow 0$$

if there exist $h_0 > 0$ and for each positive integer N a constant C_N such that

$$|f| \leq C_N h^N \quad \text{for all } 0 < h < h_0.$$

- We write

$$f = O_X(h^N)$$

to mean

$$\|f\|_X = O(h^N).$$

- If A is a bounded linear operator between the spaces X, Y , we will often write

$$A = O_{X \rightarrow Y}(h^N)$$

to mean

$$\|A\|_{X \rightarrow Y} = O(h^N).$$

- We will sometimes write

$$a = O_S(h^N)$$

to mean that for all α

$$|\partial^\alpha a| \leq C_\alpha h^N.$$

We use similar notation for other spaces with seminorms.

A.5. SYMBOL CLASSES

We record from Chapters 4 and 9 various classes of symbols:

- Given an order function m on $\mathbb{R}^{2n}$, we define the corresponding class of symbols:

$$\begin{aligned} S(m) &:= \{a \in C^\infty \mid |\partial^\alpha a| \leq C_\alpha m \text{ for all multiindices } \alpha\}, \\ S_\delta(m) &:= \{a \in C^\infty \mid |\partial^\alpha a| \leq C_\alpha h^{-\delta|\alpha|} m \text{ for all multiindices } \alpha\}, \\ S &:= S(1), \quad S_\delta = S_\delta(1). \end{aligned}$$

- Kohn–Nirenberg symbols:

$$\begin{aligned} S^m &:= \{a \in C^\infty \mid |\partial_x^\alpha \partial_\xi^\beta a| \leq C_{\alpha\beta} \langle \xi \rangle^{m-|\beta|} \text{ for all } \alpha \text{ and } \beta\}, \\ \Psi^m &:= \{a^w(x, hD) \mid a \in S^m\}. \end{aligned}$$

DIFFERENTIAL FORMS

In this section we provide a minimalist review of differential forms on $\mathbb{R}^N$. See the references for suggestions of good books that fill in the details and the motivation we omit.

B.1. DEFINITIONS

We start with a long list of algebraic and analytic concepts:

DEFINITIONS.

- (i) An m -form on $\mathbb{R}^N$ is a multilinear mapping

$$\omega : (\mathbb{R}^N)^m = \mathbb{R}^N \times \cdots \times \mathbb{R}^N \rightarrow \mathbb{R},$$

which is alternating; if $u_i = u_j$ for some $1 \leq i < j \leq m$, then

$$\omega(u_1, \dots, u_i, \dots, u_j, \dots, u_m) = 0.$$

We will write $\omega(u)$ where $u = (u_1, \dots, u_m)$ for $u_1, \dots, u_m \in \mathbb{R}^N$

- (ii) If α, β are 1-forms, their *wedge product* is the 2-form

$$(\alpha \wedge \beta)(u) := \alpha(u_1)\beta(u_2) - \alpha(u_2)\beta(u_1)$$

for $u = (u_1, u_2)$.

(iii) More generally, if $\{\alpha_j\}_{j=1}^m$ are 1-forms and $m \leq N$, we define their wedge product to be the m -form

$$(B.1.1) \quad (\alpha_1 \wedge \cdots \wedge \alpha_m)(u) = \det((\alpha_j(u_k))),$$

where $u = (u_1, \dots, u_m)$.

(iv) The 1-forms dx_j on $\mathbb{R}^N$ are defined by the rule

$$(B.1.2) \quad dx_j(u) = x_j$$

for $j = 1, \dots, N$ and $u = (x_1, \dots, x_N) \in \mathbb{R}^N$.

(v) A *differential m -form* on $\mathbb{R}^N$ is an expression

$$(B.1.3) \quad \omega = \sum_{i_1 < i_2 < \cdots < i_m} f_{i_1 \dots i_m} dx_{i_1} \wedge \cdots \wedge dx_{i_m},$$

with $f_{i_1 \dots i_m} \in C^\infty(\mathbb{R}^N)$. Its action at a point $x \in \mathbb{R}^N$ on an m -tuple of vectors u is defined using (B.1.1) and (B.1.2).

(vi) If $f : \mathbb{R}^N \rightarrow \mathbb{R}$, the *differential* of f is the differential 1-form

$$(B.1.4) \quad df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i.$$

(vii) The *differential of an m -form* is defined by induction using (B.1.3) and the rule

$$(B.1.5) \quad d(f\eta) = df \wedge \eta + f d\eta,$$

where f is a function and η is an $(m-1)$ -form. The operator d satisfies

$$(B.1.6) \quad d^2 = 0.$$

THEOREM B.1 (Alternative definition of d). Suppose w is a differential 2-form and $u \in C^\infty(\mathbb{R}^n, \mathbb{R}^3)$, $u = (u_1, u_3, u_3)$.

Then

$$(B.1.7) \quad \begin{aligned} dw(u) &= u_1 (w(u_2, u_3)) + u_2 (w(u_3, u_1)) + u_3 (w(u_1, u_2)) \\ &\quad - w([u_1, u_2], u_3) - w([u_2, u_3], u_1) - w([u_3, u_1], u_2). \end{aligned}$$

Proof. 1. Both sides of (B.1.7) are linear in w and trilinear in u .

Next observe that

$$dw(fu_1, u_2, u_3) = f dw(u).$$

The right-hand side of (B.1.7) with u_1 replaced by fu_1 equals

$$\begin{aligned} &fu_1 (w(u_2, u_3)) + u_2 (fw(u_3, u_1)) + u_3 (fw(u_1, u_2)) \\ &\quad - w([fu_1, u_2], u_3) - w([u_2, u_3], fu_1) - w([u_3, fu_1], u_2), \end{aligned}$$

and this equals the right-hand side of (B.1.7) multiplied by f . In fact,

$$[fu_1, u_2] = f[u_1, u_2] - (u_2f)u_1, \quad [u_3, fu_1] = f[u_3, u_1] + (u_3f)u_1,$$

and

$$\begin{aligned} u_2(fw(u_3, u_1)) &= fu_2(w(u_3, u_1)) + (u_2f)w(u_3, u_1), \\ u_3(fw(u_1, u_2)) &= fu_3(w(u_1, u_2)) + (u_3f)w(u_1, u_2). \end{aligned}$$

2. Hence we only need to check (B.1.7) for u constant in x and for $w = w_1dw_2 \wedge dw_3$, where $w_1 \in C^\infty$ and w_2, w_3 are coordinate functions (that is, $w_2, w_3 \in \{x_1, \dots, x_n\}$). Then (B.1.1) implies

$$\begin{aligned} dw(u) &= (dw_1 \wedge dw_2 \wedge dw_3)(u_1, u_2, u_3) \\ &= \det((dw_i(u_j))_{1 \leq i, j \leq 3}) \\ &= \det((u_jw_i)_{1 \leq i, j \leq 3}). \end{aligned}$$

The right-hand side of (B.1.7) is given by the expansion of this determinant with respect to the first row (u_1w_1, u_2w_1, u_3w_1) , since now the u_jw_i for $i = 2, 3$ are constant. $\square$

NOTATION. We write

$$\mathcal{E}^m(U)$$

to denote the space of m -forms defined on the open set $U \subseteq \mathbb{R}^N$ with smooth coefficients.

We therefore have

$$C^\infty(\mathbb{R}^N) = \mathcal{E}^0(\mathbb{R}^N) \xrightarrow{d} \mathcal{E}^1(\mathbb{R}^N) \xrightarrow{d} \dots \xrightarrow{d} \mathcal{E}^N(\mathbb{R}^N) \xrightarrow{d} 0.$$

NOTATION. We will often take $N = 2n$, in which case we write a point $z \in \mathbb{R}^{2n} = \mathbb{R}^n \times \mathbb{R}^n$ as $z = (x, \xi)$ for $x = (x_1, \dots, x_n)$, $\xi = (\xi_1, \dots, \xi_n)$.

The 1-forms $dx_j, d\xi_j$ on $\mathbb{R}^{2n}$ are then defined by the rules

$$(B.1.8) \quad dx_j(z) = x_j, \quad d\xi_j(z) = \xi_j$$

for $j = 1, \dots, n$.

We next explain how to contract an m -form with a vector field:

DEFINITION. If η is a differential m -form and V is a vector field, then the *contraction* of η by V , denoted

$$V \lrcorner \eta,$$

is the $(m-1)$ -form defined by

$$(B.1.9) \quad (V \lrcorner \eta)(u) = \eta(V, u),$$

where $u = (u_1, \dots, u_{m-1})$ for $u_1, \dots, u_{m-1} \in \mathbb{R}^N$. We will also write $V \lrcorner f = 0$ if f is a function.

It follows from (B.1.1) that

$$(B.1.10) \quad V \lrcorner (\alpha \wedge \beta) = (V \lrcorner \alpha) \wedge \beta + (-1)^k \alpha \wedge (V \lrcorner \beta)$$

if α is a k -form and β is an m -form.

B.2. PUSH-FORWARDS AND PULL-BACKS

We next record how forms transform under mappings. Let $\kappa : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be a smooth mapping. We write $\kappa = (\kappa^1, \dots, \kappa^N)$ and $y = \kappa(x)$.

DEFINITIONS.

(i) If V is a vector field on $\mathbb{R}^N$, its *push-forward* by κ is

$$(B.2.1) \quad (\kappa_* V)_y = \partial \kappa(x)(V_x), \quad \partial \kappa(x) : T_x \mathbb{R}^N \rightarrow T_y \mathbb{R}^N.$$

Here V_x is the vector in $T_x \mathbb{R}^N$ given by the value of V at x .

(ii) If η is an m -form on $\mathbb{R}^N$, its *pull-back* by κ is the m -form

$$(B.2.2) \quad (\kappa^* \eta)(u) = \eta(\kappa_* u),$$

where $\kappa_* u = (\partial \kappa u_1, \dots, \partial \kappa u_m)$ for $u = (u_1, \dots, u_m)$.

EXAMPLES.

(i) We easily check that

$$\kappa^* dy_i = \sum_{j=1}^N \frac{\partial y_i}{\partial x_j} dx_j.$$

(ii) For an m -form η and $f \in C^\infty$, $\kappa^*(f\eta) = \kappa^* f \kappa^* \eta$.

The differential commutes with the pull-back:

THEOREM B.2 (Differentials and pull-backs). *Let ω be a differential m -form. Then*

$$(B.2.3) \quad d(\kappa^* \omega) = \kappa^*(d\omega).$$

Proof. We first prove this for functions, by observing

$$\begin{aligned} d(\kappa^* f) &= \sum_{j=1}^N \frac{\partial}{\partial x_j} f(\kappa(x)) dx_j = \sum_{i,j=1}^N \frac{\partial y_i}{\partial x_j} f_{y_i}(\kappa(x)) dx_j \\ &= \sum_{i=1}^N \kappa^* f_{y_i} \kappa^* dy_i = \kappa^* \left(\sum_{i=1}^N f_{y_i} dy_i \right) \\ &= \kappa^*(df). \end{aligned}$$

The proof now follows by induction on the order of the differential form. Note that any m -form can be written as a linear combination of terms $f d\eta$, where f is a function and η is an $(m-1)$ -form. $\square$

DEFINITION. If V is a vector field generating the flow φ_t , then the *Lie derivative* of w is

$$(B.2.4) \quad \mathcal{L}_V w := \partial_t(\varphi_t^* w)|_{t=0}.$$

Here w denotes a function, a vector field, or a form and $\varphi_t = \exp(tV)$ is the flow generated by the vector field V ; see Section 2.1.

We have

$$(B.2.5) \quad d(\mathcal{L}_V \omega) = \mathcal{L}_V(d\omega)$$

and

$$(B.2.6) \quad \mathcal{L}_V(f\omega) = (\mathcal{L}_V f)\omega + f\mathcal{L}_V \omega,$$

where f is a function and ω is a differential form.

EXAMPLES.

(i) If f is a function,

$$\mathcal{L}_V f = V(f).$$

(ii) If W is a vector field,

$$\mathcal{L}_V W = [V, W].$$

$\square$

THEOREM B.3 (Cartan's formula). If ω is a differential form,

$$(B.2.7) \quad \mathcal{L}_V \omega = d(V \lrcorner \omega) + V \lrcorner d\omega.$$

Proof. 1. We proceed by induction on the order of differential forms. For 0-forms, that is, for functions, we have

$$\mathcal{L}_V f = Vf = V \lrcorner df = d(V \lrcorner f) + (V \lrcorner df),$$

by our convention that $V \lrcorner f = 0$.

2. Any m -form is a linear combination of terms $f d\eta$, where f is a function and η is an $(m-1)$ -form. Using (B.2.5), (B.2.6), $d^2 = 0$, and the induction hypothesis, we see that

$$(B.2.8) \quad \begin{aligned} \mathcal{L}_V(f d\eta) &= (\mathcal{L}_V f) d\eta + f \mathcal{L}_V d\eta \\ &= (Vf) d\eta + f d(\mathcal{L}_V \eta) \\ &= (Vf) d\eta + f d(d(V \lrcorner \eta) + V \lrcorner d\eta) \\ &= (Vf) d\eta + f d(V \lrcorner d\eta). \end{aligned}$$

3. The right-hand side of (B.2.7) for $\omega = f d\eta$ is equal to

$$(B.2.9) \quad \begin{aligned} d(V \lrcorner (f d\eta)) + V \lrcorner d(f d\eta) \\ = df \wedge (V \lrcorner d\eta) + f d(V \lrcorner d\eta) + V \lrcorner (df \wedge d\eta). \end{aligned}$$

We now use (B.1.10) to obtain

$$V \lrcorner (df \wedge d\eta) = (V f) d\eta - df \wedge (V \lrcorner d\eta).$$

Inserting this into (B.2.9) and comparing with (B.2.8) gives Cartan's formula (B.2.7) for $\omega = f d\eta$ and hence for all differential m -forms. $\square$

B.3. POINCARÉ'S LEMMA

On all of $\mathbb{R}^N$, or more generally on star-shaped regions, all closed forms are exact. This means that if $d\alpha = 0$, then there exists ω such that $\alpha = d\omega$. The construction of ω can be made explicit:

THEOREM B.4 (Poincaré's Lemma). *If α is a k -form defined in the open ball $V = B^0(0, R)$ and if*

$$d\alpha = 0,$$

then there exists a $(k-1)$ -form ω in V such that

$$d\omega = \alpha.$$

Proof. 1. Recall that $\mathcal{E}^k(V)$ denotes the space of k -forms on V . Define $A : \mathcal{E}^k(V) \rightarrow \mathcal{E}^k(V)$ by

$$A(f dx_{i_1} \wedge \cdots \wedge dx_{i_k}) = \left(\int_0^1 t^{k-1} f(tx) dt \right) dx_{i_1} \wedge \cdots \wedge dx_{i_k}.$$

Set $X := \langle x, \partial_x \rangle$. We claim that

$$(B.3.1) \quad A\mathcal{L}_X = I \quad \text{on } \mathcal{E}^k(V).$$

and that

$$(B.3.2) \quad d \circ A = A \circ d.$$

2. To prove (B.3.1), we compute using (B.2.6):

$$\begin{aligned} A\mathcal{L}_X(f dx_{i_1} \wedge \cdots \wedge dx_{i_k}) &= A \left[\left(kf + \sum_{j=1}^n x_j f_{x_j} \right) (dx_{i_1} \wedge \cdots \wedge dx_{i_k}) \right] \\ &= \int_0^1 \left(kt^{k-1} f(tx) + \sum_{j=1}^n t^k x_j f_{x_j}(tx) \right) dt dx_{i_1} \wedge \cdots \wedge dx_{i_k} \\ &= \int_0^1 \partial_t(t^k f(tx)) dt dx_{i_1} \wedge \cdots \wedge dx_{i_k} = f dx_{i_1} \wedge \cdots \wedge dx_{i_k}. \end{aligned}$$

To verify (B.3.2), note that

$$\begin{aligned}
 A \circ d(f dx_{i_1} \wedge \cdots \wedge dx_{i_k}) &= A \left(\sum_{j=1}^n f_{x_j} dx_j \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k} \right) \\
 &= \left(\int_0^1 t^k \sum_{j=1}^n f_{x_j}(tx) dx_j dt \right) dx_{i_1} \wedge \cdots \wedge dx_{i_k} \\
 &= d \left(\left(\int_0^1 t^{k-1} f(tx) dt \right) dx_{i_1} \wedge \cdots \wedge dx_{i_k} \right) \\
 &= d \circ A(f dx_{i_1} \wedge \cdots \wedge dx_{i_k}).
 \end{aligned}$$

3. Now define

$$H : \mathcal{E}^k(V) \rightarrow \mathcal{E}^{k-1}(V)$$

by $H := A \circ X \lrcorner$. By Cartan's formula (B.2.7),

$$\mathcal{L}_X = d \circ (X \lrcorner) + X \lrcorner d.$$

Thus

$$\begin{aligned}
 I = A\mathcal{L}_X &= A \circ d \circ (X \lrcorner) + A \circ X \lrcorner d \\
 &= d(A \circ X \lrcorner) + (A \circ X \lrcorner) \circ d \\
 &= d \circ H + H \circ d.
 \end{aligned}$$

Consequently

$$d \circ H + H \circ d = I;$$

and therefore

$$d(H\alpha) + Hd\alpha = \alpha.$$

So $d\omega = \alpha$ for $\omega := H\alpha$. □

B.4. DIFFERENTIAL FORMS ON MANIFOLDS

Let M be a smooth N -dimensional manifold; consult Section 14.1.1 for definitions and notation.

DEFINITION. A linear functional ω on m -tuples of vector fields

$$\omega : (C^\infty(M; TM))^m \rightarrow C^\infty(M)$$

is called a *differential form on M* if for every γ in an atlas we have

$$(\gamma^{-1})^* \omega|_{U_\gamma} \in \mathcal{E}^m(V_\gamma).$$

NOTATION. We denote by

$$\mathcal{E}^m(M)$$

the space of m -forms defined on M with smooth coefficients.

REMARKS. (i) According to Example 2 in Section 14.1.2,

$$\mathcal{E}^1(M) = C^\infty(M; T^*M).$$

(ii) The restriction of a form ω to the chart U_γ is defined by

$$\omega|_{U_\gamma} := \omega|_{C^\infty(U_\gamma; TU_\gamma)^m}.$$

(iii) The pull-back of a form is defined using the natural operations on vector fields:

$$\gamma_*^{-1} : C^\infty(V_\gamma; TV_\gamma) \rightarrow C^\infty(U_\gamma; TU_\gamma).$$

(See Example 1 in Section 14.1.2 for the definition of $C^\infty(M; TM)$.)

(iv) Theorem B.2 shows that the differential is well defined,

$$C^\infty(M) = \mathcal{E}^0(M) \xrightarrow{d} \mathcal{E}^1(M) \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{E}^N(M) \xrightarrow{d} 0,$$

and

$$d^2 = 0. \quad \square$$

A big difference with manifolds is that closed forms need not be exact, as the Poincaré Lemma holds only locally:

THEOREM B.5 (Poincaré's Lemma on manifolds). *Suppose that M is a smooth manifold. Then every $x \in M$ has a neighborhood U such that if $\alpha \in \mathcal{E}^m(M)$ is closed,*

$$d\alpha = 0,$$

then there exists $\omega \in \mathcal{E}^{m-1}(U)$ such that

$$(B.4.1) \quad \alpha|_U = d\omega.$$

Proof. Every point x belongs to U_γ for some γ and $V_\gamma = \gamma(U_\gamma) \supset B(\gamma(x), R)$ for some R . We can assume that $\gamma(x) = 0$.

We then apply Theorem B.4 to $\alpha_\gamma := (\gamma^{-1})^*(\alpha|_{U_\gamma})$ to obtain $\omega_\gamma \in \mathcal{E}^{m-1}(B(0, R))$ satisfying $d\omega_\gamma = \alpha_\gamma$ in $B(0, R)$. The theorem follows from our putting $U = \gamma^{-1}(B(0, R))$ and

$$\omega := \gamma^*\omega_\gamma \in \mathcal{E}^{m-1}(U). \quad \square$$

FUNCTIONAL ANALYSIS

C.1. OPERATOR THEORY

C.1.1. Operators on distributions.

THEOREM C.1 (Schwartz Kernel Theorem). *Let $A : \mathcal{S} \rightarrow \mathcal{S}'$ be a continuous linear operator.*

Then there exists a distribution $K_A \in \mathcal{S}'(\mathbb{R}^n \times \mathbb{R}^n)$ such that for all $u, v \in \mathcal{S}$

$$Au(v) = K_A(u \otimes v),$$

for

$$(u \otimes v)(x, y) := u(x)v(y) \in \mathcal{S}(\mathbb{R}^n \times \mathbb{R}^n).$$

NOTATION. We write this as

$$Au(x) = \int_{\mathbb{R}^n} K_A(x, y)u(y) dy$$

and call K_A the *kernel* or *Schwartz kernel* of A .

C.1.2. Operators and inverses. We next collect some results about invertibility of operators and operator norms, providing some proofs for the reader's convenience.

THEOREM C.2 (Inverse Function Theorem). *Let X, Y denote Banach spaces and assume*

$$f : X \rightarrow Y$$

is C^1 . Select a point $x_0 \in X$ and write $y_0 := f(x_0)$.

(i) *(Right inverse) If there exists $A \in L(Y, X)$ such that*

$$\partial f(x_0)A = I,$$

then there exists $g \in C^1(Y, X)$ such that

$$f \circ g = I \quad \text{near } y_0.$$

(ii) *(Left inverse) If there exists $B \in L(Y, X)$ such that*

$$B\partial f(x_0) = I,$$

then there exists $g \in C^1(Y, X)$ such that

$$g \circ f = I \quad \text{near } x_0.$$

THEOREM C.3 (Approximate inverses). *Let X, Y be Banach spaces and suppose $A : X \rightarrow Y$ is a bounded linear operator. Suppose there exist bounded linear operators $B_1, B_2 : Y \rightarrow X$ such that*

$$(C.1.1) \quad \begin{cases} AB_1 = I + R_1 & \text{on } Y \\ B_2A = I + R_2 & \text{on } X, \end{cases}$$

where

$$\|R_1\| < 1, \quad \|R_2\| < 1.$$

Then A is invertible.

Proof. The operator $I + R_1$ is invertible, with

$$(I + R_1)^{-1} = \sum_{k=0}^{\infty} (-1)^k R_1^k,$$

this series converging since $\|R_1\| < 1$. Hence

$$AC_1 = I \quad \text{for } C_1 := B_1(I + R_1)^{-1}.$$

Likewise,

$$(I + R_2)^{-1} = \sum_{k=0}^{\infty} (-1)^k R_2^k,$$

and

$$C_2A = I \quad \text{for } C_2 := (I + R_2)^{-1}B_2.$$

So A has a left and a right inverse and is consequently invertible, with $A^{-1} = C_1 = C_2$. $\square$

THEOREM C.4 (Norms of powers of operators). *For H_1, H_2 complex Hilbert spaces, let $A \in L(H_1, H_2)$ be a bounded operator.*

(i) *Then*

$$\|A\| = \sup_{\|u\|, \|v\|=1} |\langle Au, v \rangle|, \quad \|A\| = \|A^*\|, \quad \|A\|^2 = \|A^*A\|.$$

(ii) *If A is selfadjoint,*

$$\|A\|^m = \|A^m\|$$

for all $m \in \mathbb{N}$.

Proof. 1. We may assume $\|A\| > 0$. Note that

$$|\langle Au, v \rangle| \leq \|Au\| \|v\| \leq \|A\| \|u\| \|v\| = \|A\|$$

for any two unit vectors u, v . Thus $\sup_{\|u\|, \|v\|=1} |\langle Au, v \rangle| \leq \|A\|$.

Now if $u \notin \ker(A)$, we can put $v = Au/\|Au\|$. Consequently,

$$\sup_{\|u\|, \|v\|=1} |\langle Au, v \rangle| \geq \sup_{\substack{u \notin \ker(A) \\ \|u\|=1}} \frac{1}{\|Au\|} |\langle Au, Au \rangle| = \sup_{\substack{u \notin \ker(A) \\ \|u\|=1}} \|Au\| = \|A\|;$$

and therefore

$$\begin{aligned} \|A\|^2 &= \sup_{\|u\|=1} \|Au\|^2 = \sup_{\|u\|=1} |\langle A^*Au, u \rangle| \\ &\leq \sup_{\|u\|, \|v\|=1} |\langle A^*Au, v \rangle| = \|A^*A\|. \end{aligned}$$

Also, for any u, v with norm one we have

$$|\langle A^*Au, v \rangle| = |\langle Au, Av \rangle| \leq \|Au\| \|Av\| \leq \|A\|^2.$$

Taking the supremum over u, v gives us the inequality $\|A^*A\| \leq \|A\|^2$.

2. For A selfadjoint, $A = A^*$, a simple induction now yields $\|A\|^{2^k} = \|A^{2^k}\|$ for all natural numbers k . For a general m , find an n such that $m+n$ is a power of 2. Then $\|A^{m+n}\| = \|A\|^{m+n}$, and so

$$\|A\|^m \|A\|^n = \|A\|^{m+n} = \|A^{m+n}\| = \|A^m A^n\| \leq \|A\|^m \|A\|^n.$$

Therefore the inequality signs above must be equalities, and this implies $\|A^m\| = \|A\|^m$. $\square$

THEOREM C.5 (Cotlar–Stein Theorem). *Let H_1, H_2 be Hilbert spaces and let $A_j \in L(H_1, H_2)$ for $j = 1, \dots$. Assume that*

$$\sup_j \sum_{k=1}^{\infty} \|A_j^* A_k\|^{1/2} \leq C, \quad \sup_j \sum_{k=1}^{\infty} \|A_j A_k^*\|^{1/2} \leq C.$$

Then the series

$$A := \sum_{j=1}^{\infty} A_j$$

converges in strong operator topology and

$$\|A\| \leq C.$$

Proof. 1. Let us first assume that $A_j = 0$ for $j > J$ so that A is well defined. Since A^*A is selfadjoint, the previous theorem implies that

$$\|A\|^{2m} = \|(A^*A)^m\|.$$

In addition,

$$(A^*A)^m = \sum_{j_1, \dots, j_{2m}=1}^J A_{j_1}^* A_{j_2} \cdots A_{j_{2m-1}}^* A_{j_{2m}} =: \sum_{j_1, \dots, j_{2m}=1}^J a_{j_1, \dots, j_{2m}}.$$

Now

$$\|a_{j_1, \dots, j_{2m}}\| \leq \|A_{j_1}^* A_{j_2}\| \|A_{j_3}^* A_{j_4}\| \cdots \|A_{j_{2m-1}}^* A_{j_{2m}}\|,$$

and also

$$\|a_{j_1, \dots, j_{2m}}\| \leq \|A_{j_1}\| \|A_{j_2} A_{j_3}^*\| \cdots \|A_{j_{2m-2}} A_{j_{2m-1}}^*\| \|A_{j_{2m}}\|.$$

Note that $\|A_j\| = \|A_j^* A_j\|^{\frac{1}{2}} \leq C$.

Multiply these estimates and take square roots:

$$\|a_{j_1, \dots, j_{2m}}\| \leq C \|A_{j_1}^* A_{j_2}\|^{1/2} \|A_{j_2} A_{j_3}^*\|^{1/2} \cdots \|A_{j_{2m-1}}^* A_{j_{2m}}\|^{1/2}.$$

Consequently,

$$\begin{aligned} \|A\|^{2m} &= \|(A^*A)^m\| \leq \sum_{j_1, \dots, j_{2m}=1}^J \|a_{j_1, \dots, j_{2m}}\| \\ &\leq C \sum_{j_1, \dots, j_{2m}=1}^J \|A_{j_1} A_{j_2}^*\|^{1/2} \cdots \|A_{j_{2m-1}}^* A_{j_{2m}}\|^{1/2} \\ &\leq J C C^{2m}, \end{aligned}$$

where the J factor came from having $2m$ sums and only $2m - 1$ factors in the summands.

Hence

$$\|A\| \leq J^{\frac{1}{2m}} C^{\frac{2m+1}{2m}} \rightarrow C \quad \text{as } m \rightarrow \infty.$$

2. To consider the general case, take $u \in H_1$, and suppose that $u = A_k^* v$ for some k and some $v \in H_2$. Then

$$\begin{aligned} \left\| \sum_{j=1}^{\infty} A_j u \right\| &= \left\| \sum_{j=1}^{\infty} A_j A_k^* v \right\| \\ &\leq \sum_{j=1}^{\infty} \|A_j A_k^*\|^{1/2} \|A_j A_k^*\|^{1/2} \|v\| \\ &\leq C^2 \|v\|. \end{aligned}$$

Thus $\sum_{j=1}^{\infty} A_j u$ converges for $u \in \Sigma := \text{span}\{A_k^*(H_2) \mid k = 1, \dots\}$.

3. We have proved that $\|\sum_{j=1}^J A_j\| \leq C$ for any J and that $\sum_{j=1}^{\infty} A_j u$ converges for any $u \in \Sigma$. Hence, the series converges for any $u \in \bar{\Sigma}$, and

$$\left\| \sum_{j=1}^{\infty} A_j u \right\| \leq C \|u\|.$$

If u is orthogonal to $\bar{\Sigma}$, then $u \in \ker(A_k)$ for all k , in which case $\sum_{j=1}^{\infty} A_j u = 0$. Hence the series $\sum_{j=1}^{\infty} A_j u$ converges in norm for all $u \in H_1$ and defines an operator with the norm bounded by C .

□

C.2. SPECTRAL THEORY

Henceforth H denotes a complex, separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$.

C.2.1. Spectral theory for bounded operators.

DEFINITIONS. Suppose that $A : H \rightarrow H$ is a bounded linear operator.

(i) The *spectrum* of A is

$$\text{Spec}(A) := \mathbb{C} \setminus \{\lambda \in \mathbb{C} \mid (A - \lambda)^{-1} : H \rightarrow H \text{ is bounded}\}.$$

(ii) We say that $\lambda \in \text{Spec}(A)$ is an *eigenvalue* of A if there exists $u \neq 0$ such that

$$(C.2.1) \quad Au = \lambda u.$$

(iii) The operator A is called *compact* if the image of $\{u \mid \|u\| \leq 1\}$ under A is a precompact subset of H .

We recall that compact operators form a closed ideal in the space of bounded operators with operator norm topology. That means that for A

compact and B bounded, AB and BA are compact and that if $\|C_j - C\| \rightarrow 0$ and the C_j 's are compact, then C is compact.

DEFINITION. For a bounded linear operator $A : H \rightarrow H$, we define the adjoint $A^* : H \rightarrow H$ by the formula

$$\langle Au, v \rangle = \langle u, A^*v \rangle.$$

A bounded operator A is *selfadjoint* if $A^* = A$.

THEOREM C.6 (Spectral Theorem). *Let A be a bounded selfadjoint operator on H . Then there exist a measure space $(X, \mathcal{M}, \mu)$, a real-valued function $f \in L^\infty(X, \mu)$, and a unitary operator*

$$U : H \rightarrow L^2(X, \mu)$$

such that

$$(C.2.2) \quad U^* M_f U = A.$$

Here M_f is the multiplication operator:

$$M_f u = fu$$

for $u \in L^2(X, \mu)$.

THEOREM C.7 (Spectra of compact operators). *Suppose A is a compact operator on H . Then:*

- (i) *Every $\lambda \in \text{Spec}(A) \setminus \{0\}$ is an eigenvalue of A .*
- (ii) *For all nonzero $\lambda \in \text{Spec}(A) \setminus \{0\}$, there exist N such that*

$$\ker(A - \lambda)^N = \ker(A - \lambda)^{N+1}.$$

(iii) The eigenvalues can only accumulate at 0. In particular, $\text{Spec}(A)$ is countable.

- (iv) *Every $\lambda \in \text{Spec}(A) \setminus \{0\}$ is a finite rank pole of the resolvent operator*

$$\zeta \mapsto (A - \zeta)^{-1}.$$

More precisely,

$$(A - \zeta)^{-1} = \sum_{j=1}^N \frac{A_j}{(\zeta - \lambda)^j} + Q(\zeta, \lambda),$$

where $\dim A_j H < \infty$, $A_j = (A - \lambda)^{j-1} A_1$, and $\zeta \mapsto Q(\zeta, \lambda)$ is holomorphic near $\zeta = \lambda$.

(v) Suppose in addition that A is selfadjoint. Then there exists an orthonormal set $\{u_k\}_{k \in K} \subset H$ such that

$$(C.2.3) \quad Au(x) = \sum_{k \in K} \lambda_k u_k(x) \langle u, u_k \rangle,$$

where $\lambda_0 \geq \lambda_1 \geq \dots$ are the nonzero eigenvalues of A . Here the index set K is either $\{0, 1, 2, \dots, N\}$ or $\mathbb{N}$.

(vi) Conversely, if (C.2.3) holds and $\lambda_j \rightarrow 0$, then A is compact.

THEOREM C.8 (Spectra of selfadjoint operators). Suppose $A : H \rightarrow H$ is a bounded selfadjoint operator.

(i) Then $(A - \lambda)^{-1}$ exists and is a bounded linear operator on H for $\lambda \in \mathbb{C} \setminus \text{Spec}(A)$.

(ii) If $\text{Spec}(A) \subset [a, \infty)$, then

$$(C.2.4) \quad \langle Au, u \rangle \geq a \|u\|^2 \quad (u \in A).$$

C.2.2. Spectral theory for unbounded operators. We next review the more complicated theory for unbounded operators.

DEFINITIONS.

(i) An unbounded operator $A : H \rightarrow H$ is given by a subspace $\mathcal{D}(A) \subset H$ and a linear operator $A : \mathcal{D}(A) \rightarrow H$. We call $\mathcal{D}(A)$ the *domain* of A and say that A is *densely defined* if $\mathcal{D}(A)$ is dense in A .

(ii) The *graph* of A is

$$\text{graph}(A) := \{(u, Au) \mid u \in \mathcal{D}(A)\} \subset H \times H.$$

(iii) If A, B are unbounded operators on H , we say that $A \subseteq B$ if $\mathcal{D}(A) \subseteq \mathcal{D}(B)$ and $Au = Bu$ for all $u \in \mathcal{D}(A)$.

(iv) The operator A is *closed* if $\text{graph}(A)$ is a closed subspace of $H \times H$ equipped with the norm $\|(u, v)\|^2 = \|u\|^2 + \|v\|^2$.

(v) An unbounded operator A is *closable* if there exists a closed unbounded operator $\bar{A}$ such that $A \subseteq \bar{A}$. The operator $\bar{A}$ is unique and is called the *closure* of A .

THEOREM C.9 (Adjoint operator). Suppose $A : H \rightarrow H$ is an unbounded densely defined operator. Then there exists an unbounded operator $A^* : H \rightarrow H$ defined by the rule

$$(C.2.5) \quad \langle A^*v, u \rangle := \langle v, Au \rangle$$

for all $v \in \mathcal{D}(A^*)$, $u \in \mathcal{D}(A)$, where

$$(C.2.6) \quad \mathcal{D}(A^*) := \{v \in H \mid |\langle Au, v \rangle| \leq C(v)\|u\| \text{ for all } u \in \mathcal{D}(A)\}.$$

Here $C(v)$ is a constant depending on v .

The unbounded operator A^* is always closed. If A^* is densely defined, then A is closable and $\bar{A} = (A^*)^*$, $\bar{A}^* = A^*$.

DEFINITIONS.

(i) An unbounded densely defined operator A is called *symmetric* if

$$(C.2.7) \quad A \subseteq A^*.$$

Equivalently, $\langle Au, v \rangle = \langle u, Av \rangle$ for all $u, v \in \mathcal{D}(A)$.

(ii) An unbounded densely defined operator A is called *selfadjoint* if

$$(C.2.8) \quad A = A^*.$$

(iii) A symmetric operator is called *essentially selfadjoint* if

$$(C.2.9) \quad \bar{A} = A^*.$$

EXAMPLE: Quadratic symbols. Suppose

$$p(x, \xi) := \frac{1}{2} \langle Ax, x \rangle + \langle Bx, \xi \rangle + \frac{1}{2} \langle D\xi, \xi \rangle,$$

where A , B , and D are complex $n \times n$ matrices and $A^T = A$, $D^T = D$.

We define a corresponding unbounded operator N_p on $L^2(\mathbb{R}^n)$ by the rule

$$N_p u := p^w(x, D)u$$

for

$$u \in \mathcal{S} =: \mathcal{D}(N_p).$$

Since $\langle p^w(x, D)u, v \rangle = \langle u, \bar{p}^w(x, D)v \rangle$ for $u, v \in \mathcal{S}$, we see that N_p is symmetric if and only if p is real. We next define an extension M_p of N_p , by first defining $\mathcal{D}(M_p)$ to be the set of $u \in L^2$ such that $p^w(x, D)u \in L^2$, where $p^w(x, D)$ is meant in the sense of $\mathcal{S}'$. Then let $M_p u := p^w(x, D)u$ for $u \in \mathcal{D}(M_p)$.

We assert that

$$(C.2.10) \quad M_p \text{ is closed,}$$

$$(C.2.11) \quad M_p = \bar{N}_p,$$

and

$$(C.2.12) \quad N_p^* = M_p^* = M_{\bar{p}}.$$

In particular, M_p is selfadjoint if and only if p is real-valued, in which case N_p is essentially selfadjoint.

Proof of (C.2.10). Remember that $p^w(x, D) : \mathcal{S}' \rightarrow \mathcal{S}'$ is continuous. If $u_j \rightarrow u$ and $p^w(x, D)u_j \rightarrow v$ in L^2 , then $u_j \rightarrow u$ in $\mathcal{S}'$ and so $v = p^w(x, D)u \in L^2$. Therefore $u \in \mathcal{D}(M_p)$ and $M_p u = v$. $\square$

Proof of (C.2.11). To show that M_p is the closure of N_p , we have to show that for any $u \in \mathcal{D}(M_p)$ we can find $u_\epsilon \in \mathcal{S}$ such that $u_\epsilon \rightarrow u$ and $p^w(x, D)u_\epsilon \rightarrow p^w(x, D)u$ in L^2 as $\epsilon \rightarrow 0$. So take $\chi \in C_c^\infty(\mathbb{R}^{2n})$ equal to one in $B(0, 1)$ and write $\chi_\epsilon(x, \xi) := \chi(\epsilon x, \epsilon \xi)$. Set $u_\epsilon := \chi_\epsilon^w(x, D)u \in \mathcal{S}$; then

$$u_\epsilon \rightarrow u \text{ in } L^2.$$

Observe that

$$p^w(x, D)u_\epsilon = \chi_\epsilon^w(x, D)p^w(x, D)u + [p^w(x, D), \chi_\epsilon^w(x, D)]u$$

and $\chi_\epsilon^w(x, D)p^w(x, D)u \rightarrow p^w(x, D)u$ in L^2 . We therefore need to show that

$$(C.2.13) \quad [p^w(x, D), \chi_\epsilon^w(x, D)]u \rightarrow 0 \text{ in } L^2$$

as $\epsilon \rightarrow 0$. Since p is quadratic, Theorem 4.11 implies that

$$[p^w(x, D), \chi_\epsilon^w(x, D)] = \{\chi_\epsilon, p\}^w(x, D).$$

Now

$$\{\chi_\epsilon, p\}(x, \xi) = \epsilon \sum_{j=1}^n (\partial_{x_j} p(x, \xi)(\partial_{\xi_j} \chi)(\epsilon x, \epsilon \xi) - \partial_{\xi_j} p(x, \xi)(\partial_{x_j} \chi)(\epsilon x, \epsilon \xi)).$$

Hence $\{\chi_\epsilon, p\}$ is bounded in S , uniformly with respect to ϵ . Consequently Theorem 4.23 implies that the commutator is uniformly bounded on L^2 .

For $\psi \in C_c^\infty(\mathbb{R}^n)$ supported in $B(0, 1)$ and equal to one near 0, we put $\psi_\epsilon(x) = \psi(\epsilon x)$. This function vanishes on the support of $\{\chi_\epsilon, p\}$. Theorems 4.11 and 4.23 now imply that

$$[p^w(x, D), \chi_\epsilon^w(x, D)]\psi_\epsilon(x) = O_{L^2 \rightarrow L^2}(\epsilon).$$

Since $\psi_\epsilon u \rightarrow u$ in L^2 , we obtain (C.2.13). $\square$

Proof of (C.2.12). Suppose $v \in \mathcal{D}(M_p^*)$. Then there exists $C(v)$ such that

$$(C.2.14) \quad \langle M_p u, v \rangle \leq C(v) \|u\|$$

for all $u \in \mathcal{D}(M_p) \supset \mathcal{S}$. For $u \in \mathcal{S}$ we have

$$\langle M_p u, v \rangle = \langle u, \bar{p}^w(x, D)v \rangle,$$

where $p^w(x, D)v \in \mathcal{S}'$. But then (C.2.14) implies that $\bar{p}^w(x, D)v \in L^2$.

Hence $M_p^* \subset M_{\bar{p}}$. Also, since M_p^* is closed,

$$N_{\bar{p}} \subset N_p^* = \bar{N}_p^* = M_p^*.$$

Therefore $M_{\bar{p}} = \bar{N}_{\bar{p}} \subset M_p^*$ and consequently $M_p^* = M_{\bar{p}}$. $\square$

THEOREM C.10 (Spectral Theorem for unbounded operators).

Let A be an unbounded selfadjoint operator on H . Then there exist a measure space $(X, \mathcal{M}, \mu)$, a real-valued measurable function f , and a unitary operator

$$U : H \rightarrow L^2(X, \mu)$$

such that

$$(C.2.15) \quad x \in \mathcal{D}(A) \text{ if and only if } M_f(Ux) \in L^2(X, \mu)$$

and

$$(C.2.16) \quad x \in \mathcal{D}(A) \text{ implies } U(Ax) = M_f(Ux).$$

Here $M_f : x \mapsto fx$ denotes the unbounded multiplication operator on X .

As an immediate consequence we obtain this useful result valid for both bounded and unbounded operators:

THEOREM C.11 (Distance to spectrum).

(i) *If A is a selfadjoint operator, then*

$$\text{Spec}(A) = \text{ess-image}(f) \subset \mathbb{R},$$

where f is given in the Spectral Theorems (Theorems C.10 and C.6) and

$$\text{ess-image}(f) := \{t \mid \mu(f^{-1}(t - \epsilon, t + \epsilon)) > 0 \text{ for all } \epsilon > 0\}.$$

(ii) *Furthermore, if $\lambda \in \mathbb{C} \setminus \text{Spec}(A)$, then*

$$(C.2.17) \quad \|(A - \lambda)^{-1}\| = \frac{1}{\text{dist}(\lambda, \text{Spec}(A))}.$$

There are many criteria determining if an operator is essentially self-adjoint and there are many subtleties on the subject. Here we only need the simplest one:

THEOREM C.12 (Criteria for essential selfadjointness). *Suppose that $A : H \rightarrow H$ is symmetric. Then the following conditions are equivalent:*

(i) *A is essentially selfadjoint.*

(ii) *For both signs, $(A^* \pm i)x = 0$, $x \in \mathcal{D}(A^*)$, implies $x = 0$.*

(iii) *For both signs, $\{(A \pm i)x \mid x \in \mathcal{D}(A)\}$ is dense in H .*

The Schrödinger propagators of selfadjoint operators are important for quantum dynamics:

THEOREM C.13 (Stone's Theorem). *Suppose that $P : \mathcal{D}(A) \subseteq H \rightarrow H$ is a (possibly unbounded) selfadjoint operator.*

(i) *Then*

$$(C.2.18) \quad U(t) := e^{-itP} \quad (t \in \mathbb{R})$$

defines a strongly continuous unitary group:

$$(C.2.19) \quad \begin{cases} U(t)U(s) = U(t+s), & U(t)^* = U(-t), \\ \lim_{t \rightarrow 0} \|U(t)u - u\|_H = 0 & (u \in H). \end{cases}$$

In addition,

$$(C.2.20) \quad D_t(U(t)u) + U(t)Pu = 0 \quad (t \in \mathbb{R})$$

for all $u \in \mathcal{D}(P)$.

(ii) *Conversely, if $U(t)$ satisfies (C.2.19), then there exists a selfadjoint operator P such that (C.2.18) and (C.2.20) hold.*

EXAMPLE. Lemma 11.8 shows that for real quadratic functions $p = p(x, \xi)$, we can define $\exp(-itp^w(x, D)) : \mathcal{S} \rightarrow \mathcal{S}$ and extend these operators to a unitary group on L^2 .

The continuity property in (C.2.19) is easily checked for $u \in \mathcal{S}$, and owing to the density of $\mathcal{S}$ in L^2 , the continuity holds as well for $u \in L^2$. According to Stone's Theorem, $p^w(x, D)$ is essentially selfadjoint, as we have already seen in the first example of this section. $\square$

C.2.3. Minimax formulas. When the spectrum of a selfadjoint operator is discrete, tools from linear algebra are applicable.

THEOREM C.14 (Maximin and minimax principles). *Suppose that $A : H \rightarrow H$ is selfadjoint and semibounded, meaning $A \geq -c$. Assume also that the right inverse $(A + 2c)^{-1} : H \rightarrow H$ is a compact operator.*

Then the spectrum of A is discrete, $\lambda_1 \leq \lambda_2 \leq \lambda_3 \cdots$, and $\mathcal{D}(A) = (A + 2c)^{-1}H$.

Furthermore,

(i)

$$(C.2.21) \quad \lambda_j = \max_{\substack{V \subseteq H \\ \dim V < j}} \min_{\substack{w \in \mathcal{D}(A) \\ 0 \neq w \perp V}} \frac{\langle Aw, w \rangle}{\|w\|^2},$$

(ii)

$$(C.2.22) \quad \lambda_j = \min_{\substack{V \subset \mathcal{D}(A) \\ \dim V \geq j}} \max_{\substack{v \in V \\ v \neq 0}} \frac{\langle Av, v \rangle}{\|v\|^2}.$$

In these formulas, V denotes a linear subspace of H .

DEFINITIONS. (i) Let $Q : H \rightarrow H$ be a bounded linear operator. We define the *rank* of Q to be the dimension of the range $Q(H)$.

(ii) If A is an operator with real and discrete spectrum, we set

$$N(\lambda) := \#\{\lambda_j \mid \lambda_j \leq \lambda\}$$

to count the number of eigenvalues less than or equal to λ .

THEOREM C.15 (Estimating $N(\lambda)$). *Let A satisfy the assumptions of Theorem C.14.*

(i) *If*

$$(C.2.23) \quad \begin{cases} \text{there exist } \delta > 0 \text{ and an operator } Q : \mathcal{D}(A) \rightarrow H, \\ \text{with rank } Q \leq k, \text{ such that} \\ \langle Au, u \rangle \geq (\lambda + \delta)\|u\|^2 - \langle Qu, u \rangle \text{ for } u \in \mathcal{D}(A), \end{cases}$$

then

$$N(\lambda) \leq k.$$

(ii) *If*

$$(C.2.24) \quad \begin{cases} \text{for each } \delta > 0, \text{ there exists a subspace } V \subset \mathcal{D}(A) \\ \text{with } \dim V \geq k, \text{ such that} \\ \langle Au, u \rangle \leq (\lambda + \delta)\|u\|^2 \text{ for } u \in V, \end{cases}$$

then

$$N(\lambda) \geq k.$$

Proof. 1. We note that $\dim Q(H) = \text{rank } Q \leq k$. Therefore the maximin formula (C.2.21) implies

$$\begin{aligned} \lambda_{k+1} &= \max_{\substack{V \subset H \\ \dim V \leq k}} \min_{\substack{w \in \mathcal{D}(A) \\ 0 \neq w \perp V}} \frac{\langle Aw, w \rangle}{\|w\|^2} \geq \min_{\substack{w \in \mathcal{D}(A) \\ 0 \neq w \perp Q(H)}} \frac{\langle Aw, w \rangle}{\|w\|^2} \\ &\geq \min_{\substack{w \in \mathcal{D}(A) \\ 0 \neq w \perp Q(H)}} \left(\lambda + \delta - \frac{\langle Qw, w \rangle}{\|w\|^2} \right) = \lambda + \delta, \end{aligned}$$

since $\langle Qw, w \rangle = 0$ if $w \in Q(H)^\perp$. Hence $\lambda < \lambda + \delta \leq \lambda_{k+1}$, and so

$$N(\lambda) = \max\{j \mid \lambda_j \leq \lambda\} \leq k.$$

This proves assertion (i).

2. The minimax formula (C.2.22) directly implies that

$$\lambda_k \leq \max_{\substack{v \in V \\ v \neq 0}} \frac{\langle Av, v \rangle}{\|v\|^2} \leq \lambda + \delta.$$

Hence $\lambda_k \leq \lambda + \delta$. This is valid for all $\delta > 0$, and so

$$N(\lambda) = \max\{j \mid \lambda_j \leq \lambda\} \geq k.$$

This is assertion (ii). □

C.3. TRACE CLASS OPERATORS

Let $A : H \rightarrow H$ be a compact operator on a complex separable Hilbert space H . Then $A^*A : H \rightarrow H$ is a selfadjoint semidefinite compact operator, and hence it has discrete spectrum

$$\|A\|^2 = s_0(A)^2 \geq s_1(A)^2 \geq \cdots \geq s_k(A)^2 \rightarrow 0.$$

DEFINITION. The *singular values* of A are the nonnegative square roots of these eigenvalues:

$$s_j(A) \quad (j = 0, 1, \dots).$$

We note that we obtain the same singular values by considering eigenvalues of AA^* . We have

$$(C.3.1) \quad s_j(AB) \leq \|B\|s_j(A)$$

if A is compact and B is bounded.

DEFINITIONS.

(i) A compact operator $A : H \rightarrow H$ is said to be of *trace class*, written

$$A \in \mathcal{L}_1(H),$$

if

$$(C.3.2) \quad \sum_{j=1}^{\infty} s_j(A) < \infty.$$

(ii) The *trace class norm* is

$$(C.3.3) \quad \|A\|_{\mathcal{L}_1} := \sum_{j=1}^{\infty} s_j(A).$$

It is not immediate that the right-hand side of (C.3.3) in fact defines a norm.

THEOREM C.16 (Quantization and trace class). *Suppose that $a \in \mathcal{S}(\mathbb{R}^{2n})$. Then*

$$(C.3.4) \quad a^w(x, D) \in \mathcal{L}_1(L^2(\mathbb{R}^n)).$$

Proof. 1. Recalling (4.5.1), we see that $a \in \mathcal{S}(\mathbb{R}^{2n})$ is equivalent to

$$a^w(x, D)u = Au(x) := \int_{\mathbb{R}^n} K(x, y)u(y) dy$$

for $K \in \mathcal{S}(\mathbb{R}^n \times \mathbb{R}^n)$. It follows that

$$A^*Au(x) = \int_{\mathbb{R}^n} K_1(x, y)u(y) dy,$$

where

$$K_1(x, y) := \int_{\mathbb{R}^n} \overline{K(z, x)}K(z, y) dz, \quad K_1 \in \mathcal{S}(\mathbb{R}^n \times \mathbb{R}^n).$$

Let $P_0 = -\Delta + |x|^2$ be the quantum harmonic oscillator analyzed in Section 6.1. Theorem 6.2 and the discussion in Section 6.1.2 show that $P_0^{-1} : L^2 \rightarrow L^2$ exists and that the eigenvalues of the selfadjoint compact operator P_0^{-1} are $(2|\alpha| + n)^{-1}$ for $\alpha \in \mathbb{N}$. In particular,

$$(C.3.5) \quad s_j(P_0^{-N}) \leq Cj^{-N/n}$$

for $N \geq 1$.

2. Since $(-\Delta + |x|^2)^N K_1(x, y) \in \mathcal{S}(\mathbb{R}^{2n})$, we see that $P_0^N A^*A$ is bounded on L^2 for any N . We can now use (C.3.1) and (C.3.5) to estimate the singular values of A :

$$\begin{aligned} s_j(A) &= s_j(A^*A)^{\frac{1}{2}} = s_j(P_0^{-N}P_0^N A^*A)^{\frac{1}{2}} \\ &\leq s_j(P_0^{-N})^{\frac{1}{2}} \|P_0^N A^*A\|_{L^2 \rightarrow L^2}^{\frac{1}{2}} \leq Cj^{-N/2n}. \end{aligned}$$

Taking $N \geq 2n + 1$, we obtain (C.3.2) and therefore (C.3.4). $\square$

REMARK. Much finer conditions on a can be found that guarantee $a^w \in \mathcal{L}_1$. In particular,

$$(C.3.6) \quad \text{if } a \in S(m) \text{ and } m \in L^1(\mathbb{R}^{2n}), \text{ then } a^w(x, D) \in \mathcal{L}_1(L^2(\mathbb{R}^n)).$$

Consult Dimassi–Sjöstrand [D-S, Chapter 8]. $\square$

THEOREM C.17 (Definition of the trace). *Suppose A is of trace class on a Hilbert space H .*

(i) *Let $\{e_j\}_{j=0}^\infty$ be any orthonormal basis of H . Then*

$$(C.3.7) \quad \operatorname{tr} A := \sum_{j=0}^{\infty} \langle Ae_j, e_j \rangle$$

is finite and independent of the choice of $\{e_j\}_{j=0}^\infty$.

(ii) *Suppose that B is a bounded operator on H . Then AB and BA are of trace class and*

$$(C.3.8) \quad \operatorname{tr}(AB) = \operatorname{tr}(BA).$$

DEFINITION. We call $\operatorname{tr} A$ defined by (C.3.7) the *trace* of A .

THEOREM C.18 (Traces of integral operators). *Suppose that B is an operator of trace class on $L^2(M; \Omega^{\frac{1}{2}}(M))$, given by the integral kernel*

$$K \in C^\infty(M \times M; \Omega^{\frac{1}{2}}(M \times M)).$$

Then K_Δ , the restriction to the diagonal $\Delta := \{(m, m) \mid m \in M\}$, has a well-defined density; and

$$(C.3.9) \quad \operatorname{tr} B = \int_{\Delta} K_{\Delta}.$$

A next theorem relates the trace to the eigenvalues of A . It is easy for selfadjoint operators.

THEOREM C.19 (Lidskii's Theorem). *Suppose that A is of trace class and that*

$$\operatorname{Spec}(A) = \{\lambda_j\}_{j=0}^\infty \subset \mathbb{C}, \quad |\lambda_0| \geq |\lambda_1| \geq \cdots \geq |\lambda_j| \rightarrow 0.$$

Then

$$(C.3.10) \quad \operatorname{tr} A = \sum_{j=0}^{\infty} \lambda_j.$$

FREDHOLM THEORY

This appendix describes the role of the Schur complement formula in analytic Fredholm theory.

D.1. GRUSHIN PROBLEMS

Linear algebra. The *Schur complement formula* states for two-by-two systems of matrices that if

$$\begin{pmatrix} P & R_- \\ R_+ & R_0 \end{pmatrix}^{-1} = \begin{pmatrix} E & E_+ \\ E_- & E_0 \end{pmatrix},$$

then P is invertible if and only if E_0 is invertible, with

$$(D.1.1) \quad P^{-1} = E - E_+ E_0^{-1} E_-, \quad E_0^{-1} = R_0 - R_+ P^{-1} R_-.$$

Grushin problem. The Schur complement formula can be used in infinite dimensions. We will apply it to problems of the form

$$(D.1.2) \quad \begin{pmatrix} P & R_- \\ R_+ & O \end{pmatrix} \begin{pmatrix} u \\ u_- \end{pmatrix} = \begin{pmatrix} v \\ v_+ \end{pmatrix}$$

where

$$P : X_1 \rightarrow X_2, \quad R_+ : X_1 \rightarrow X_+, \quad R_- : X_- \rightarrow X_2,$$

for appropriate Banach spaces X_1, X_2, X_+, X_- .

We call (D.1.2) a *Grushin problem*. It is useful for reducing problems of infinite dimension to problems of finite dimension. In practice, we start

with an operator P and build a Grushin problem by choosing $R_{\pm}$, in which case it is normally sufficient to take $R_0 = O$, as above.

If the Grushin problem (D.1.2) is invertible, we call it *well posed* and we write its inverse as

$$(D.1.3) \quad \begin{pmatrix} u \\ u_- \end{pmatrix} = \begin{pmatrix} E & E_+ \\ E_- & E_0 \end{pmatrix} \begin{pmatrix} v \\ v_+ \end{pmatrix}$$

for operators

$$E : X_2 \rightarrow X_1, \quad E_0 : X_+ \rightarrow X_-, \quad E_+ : X_+ \rightarrow X_1, \quad E_- : X_2 \rightarrow X_-.$$

LEMMA D.1 (The operators in a Grushin problem). *If (D.1.2) is well posed, then the operators R_+ , E_- are surjective and the operators E_+ , R_- are injective.*

D.2. FREDHOLM OPERATORS

DEFINITIONS. (i) A bounded linear operator $P : X_1 \rightarrow X_2$ is called a *Fredholm operator* if the kernel of P ,

$$\ker P := \{u \in X_1 \mid Pu = 0\},$$

and the cokernel of P ,

$$\operatorname{coker} P := X_2/PX_1, \quad \text{where } PX_1 := \{Pu \mid u \in X_1\},$$

are both finite dimensional. (The quotient is defined algebraically as the set of equivalence classes for the relation $\sim$ on X_2 : $v \sim v'$ if $v - v' \in PX_1$.)

(ii) The *index* of a Fredholm operator is

$$\operatorname{ind} P := \dim \ker P - \dim \operatorname{coker} P.$$

EXAMPLE. Many important Fredholm operators have the form

$$(D.2.1) \quad P = I + K,$$

where K is a compact operator mapping a Banach space X to itself.

Theorem D.3 below shows that the index does not change under continuous deformations of Fredholm operators (with respect to operator norm topology). Hence for operators of the form (D.2.1) the index is 0:

$$\operatorname{ind} P = \operatorname{ind}(I + tK) = \operatorname{ind} I = 0 \quad (0 \leq t \leq 1). \quad \square$$

The connection between Grushin problems and Fredholm operators is this:

THEOREM D.2 (Grushin problem for Fredholm operators).

(i) Suppose that $P : X_1 \rightarrow X_2$ is a Fredholm operator.

Then there exist finite-dimensional spaces $X_{\pm}$ and operators $R_- : X_- \rightarrow X_2$, $R_+ : X_1 \rightarrow X_+$ for which the Grushin problem (D.1.2) is well posed. As a consequence $PX_1 \subset X_2$ is closed.

(ii) Conversely, suppose that for some choice of spaces $X_{\pm}$ and operators $R_{\pm}$, the Grushin problem (D.1.2) is well posed.

Then $P : X_1 \rightarrow X_2$ is a Fredholm operator if and only if $E_0 : X_+ \rightarrow X_-$ is a Fredholm operator, in which case

$$(D.2.2) \quad \text{ind } P = \text{ind } E_0.$$

Assertion (ii) is particularly useful when the spaces $X_{\pm}$ are finite dimensional.

Proof. 1. Assume that $P : X_1 \rightarrow X_2$ is Fredholm. Let $n_+ := \dim \ker P$ and $n_- := \dim \text{coker } P$, and write $X_+ := \mathbb{C}^{n_+}$, $X_- := \mathbb{C}^{n_-}$. Then select linear operators

$$R_- : X_- \rightarrow X_2, \quad R_+ : X_1 \rightarrow X_+$$

of ranks $n_{\mp}$, respectively, such that

$$R_-X_- \cap PX_1 = \{0\}, \quad \ker(R_+|_{\ker P}) = \{0\}.$$

Then the operator

$$\begin{pmatrix} P & R_- \\ R_+ & O \end{pmatrix}$$

has a trivial kernel and is onto. Hence it is invertible, and by the Open Mapping Theorem the inverse is continuous.

In particular, consider P acting on the quotient space $X_1/\ker P$, which is a Banach space since $\ker P$ is closed. We have $n_+ = 0$, and

$$PX_1 = P(X_1/\ker P) = \begin{pmatrix} P & R_- \end{pmatrix} \begin{pmatrix} X_1/\ker P \\ \{0\} \end{pmatrix}$$

is a closed subspace.

2. Conversely, suppose that Grushin problem (D.1.2) is well posed. According to Lemma D.1, the operators R_+ , E_- are surjective, and the operators E_+ , R_- are injective. We take $u_- = 0$. Then

$$(D.2.3) \quad \begin{cases} \text{the equation } Pu = v \text{ is equivalent to} \\ u = Ev + E_+v_+, \quad 0 = E_-v + E_0v_+. \end{cases}$$

This means that

$$E_- : \operatorname{Im} P \rightarrow \operatorname{Im} E_0,$$

and so we can define the induced map

$$E^\# : X_2 / \operatorname{Im} P \rightarrow X_- / \operatorname{Im} E_0.$$

Since E_- is surjective, so is $E^\#$. Also, $\ker E^\# = \{0\}$. This follows since if $E_-v \in \operatorname{Im} E_0$, we can use (D.2.3) to deduce that $v \in \operatorname{Im} P$. Hence $E^\#$ is a bijection of the cokernels $X_2 / \operatorname{Im} P$ and $X_- / \operatorname{Im} E_0$.

3. Next, we claim that

$$E_+ : \ker E_0 \rightarrow \ker P$$

is a bijection. Indeed, if $u \in \ker P$, then $u = E_+v_+$ and $E_0v_+ = 0$. Therefore E_+ is onto, and this is all we need to check since E_+ is injective.

We conclude that

$$\dim \ker P = \dim \ker E_0, \quad \dim \operatorname{coker} P = \dim \operatorname{coker} E_0.$$

In particular, the indices of P and E_0 are equal. $\square$

THEOREM D.3 (Invariance of the index under deformations). *The set of Fredholm operators is open in $L(X_1, X_2)$, and the index is constant in each component.*

Proof. When P is a Fredholm operator, we can use Theorem D.2 to obtain $E_0 : \mathbb{C}^{n_+} \rightarrow \mathbb{C}^{n_-}$, with

$$(D.2.4) \quad \operatorname{ind} E_0 = n_+ - n_-$$

according to the Rank-Nullity Theorem of linear algebra. The Grushin problem remains well posed with the same operators $R_\pm$ if P is replaced by P' , provided $\|P - P'\| < \epsilon$ for some sufficiently small $\epsilon > 0$. Hence the set of Fredholm operators is open.

Using (D.2.4), we see that the index of P' is the same as the index of P . Consequently it remains constant in each connected component of the set of Fredholm operators. $\square$

D.3. MEROMORPHIC CONTINUATION

The Grushin problem framework also provides an elegant proof of the following standard result:

THEOREM D.4 (Analytic Fredholm theory). *Suppose $U \subset \mathbb{C}$ is a connected open set and $\{A(z)\}_{z \in U}$ is a family of Fredholm operators depending holomorphically on z .*

Then if $A(z_0)^{-1}$ exists at some point $z_0 \in U$, the mapping $z \mapsto A(z)^{-1}$ is a meromorphic family of operators on U .

Proof. 1. Fix $z_1 \in U$. We form a Grushin problem for $P = A(z_1)$, as described in the proof of Theorem D.2. The same operators $R_{\pm}^{z_1}$ also provide a well-posed Grushin problem for $P = A(z)$ for z in some sufficiently small neighborhood $V(z_1)$ of z_1 .

According to Theorem D.3

$$\operatorname{ind} A(z) = \operatorname{ind} A(z_0) = 0.$$

Consequently

$$n_+ = n_- = n,$$

and $E_0^{z_1}(z)$ is an $n \times n$ matrix with holomorphic coefficients. The invertibility of $E_0^{z_1}(z)$ is equivalent to the invertibility of $A(z)$.

2. This shows that there exists a locally finite covering $\{U_j\}_{j \in J}$ of U and a family of functions f_j , holomorphic in U_j , such that if $z \in U_j$, then $A(z)$ is invertible precisely when

$$f_j(z) \neq 0.$$

Indeed, we can define $f_j := \det E_0^z$, where E_0^z exists for $z \in U_j$ by the construction in step 1. Since U is connected and since $A(z_0)$ is invertible for at least one $z_0 \in U$, none of the functions f_j is identically zero.

So $\det E_0(z)$ is a nontrivial holomorphic function in $V(z_1)$; and consequently $E_0(z)^{-1}$ is a meromorphic family of matrices. Applying (D.1.1), we conclude that

$$A(z)^{-1} = E(z) - E_+(z)E_{-+}(z)^{-1}E_-(z)$$

is a meromorphic family of operators in the neighborhood $V(z_1)$. Since z_1 was arbitrary, $A(z)^{-1}$ is in fact meromorphic in all of U . $\square$

NOTES FOR THE APPENDICES

The short book [dC] by do Carmo is a good introduction to differential forms. See also Warner [W, Chapter 2].

For a quick and elegant presentation of the Inverse Function Theorem, see Hörmander [H1, Chapter 1]; and for the Schwartz Kernel Theorem, consult Hörmander [H1, Chapter 5] or Friedlander–Joshi [F-J, Section 6.1]. For proofs of other theorems in Appendix C, and in particular the Spectral Theorem, see Reed–Simon [R-S]. Simon [Si] discusses trace class operators and singular values. A review of spectral theory, with semiclassical problems in mind, is in Dimassi–Sjöstrand [D-S, Chapter 4]; see also Hislop–Sigal [Hi-Si] and Shubin [Shu]. The first example in Section C.2 comes from Hörmander [H6] and was suggested to us by M. Hitrik. I. Hirshberg provided us with the proof of Theorem C.4.

Appendix D follows the presentation in [S-Z2]. We refer to Hörmander [H2, Section 19.1] for a comprehensive introduction to Fredholm operators. Consult Zhang [Zh] for the Schur complement formula and its applications.

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